FHIR-STU3 — Foundation Module — part 1 of 2 (sections 2 – 2.32.2)

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2 Foundation Module

URL: https://hl7.org/fhir/STU3/foundation-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group FHIR Infrastructure | Ballot Status: Informative

2.0 Foundation Module

The Foundation Module is responsible the overall infrastructure of the FHIR specification. Every implementer works with the content in the foundation module however they use FHIR.

The Foundation Module maintains most of the basic documentation for the FHIR specification. In addition, the Foundation Module includes the following resources:

Foundation Framework Resource DomainResource Basic Binary Bundle Media | Content Management Resources. Questionnaire QuestionnaireResponse List Composition DocumentReference DocumentManifest | Data Exchange Resources. OperationOutcome Parameters Subscription MessageHeader MessageDefinition

2.0.1 Relationships with other modules

* All the other modules depend on the foundation module

* The Linked Data module builds on the foundation model by adding RDF representations, and strengthening the definitions by facilitating linkage to external ontologies

* The Terminology module provides the formal basis for using Concepts defined in Code Systems in the definitions

* The Conformance module provides the basis for extending the foundation for national and local use

* The Security & Privacy provides the linking framework to external standards for security and privacy

* The Implementation Support module builds on the foundation to provide testing and reference implementations

2.0.2 Common Use Cases: Exchanging Data

FHIR supports 4 different paradigms for exchange: the RESTful API, Messaging, Documents, and Services. Each of these approaches can be used to exchange information, and each has its own strengths and weaknesses.

2.0.2.1 RESTful API

Most implementers focus on RESTful API. This is a client/server API designed to follow the principles of RESTful design for Create, Read, Update and Delete operations, along with Search and Execute (Operations) support.

The RESTful API is a general purpose interface that can be used to push and pull data between systems. Which is appropriate depends on architecture and deployment considerations.

2.0.2.2 Messaging

In addition to the RESTful API, a messaging exchange framework is documented, which supports exchange between systems by exchanging content by sending routed messages from system to system. This exchange can be implemented on the RESTful API, or using some other messaging stack. The messaging paradigm is provided to support implementers who wish to use such a messaging paradigm.

Implementers should note that the messaging framework is not provided to fill any functional deficiency in the RESTful API (or vice versa), these frameworks are provided to allow implementers to choose how to exchange content based on their own architectural and deployment considerations.

2.0.2.3 Documents

This specification also defines a document based exchange framework, where content to be exchanged is wrapped by a Composition that provides the context of the content, and that has a fixed presentation for a human reader. The document framework is provided to help with computer-assisted human to human communication uses - which are not uncommon in healthcare.

Typically, exchanging documents is associated with exchanging clinical information across clinical governance borders, while data based exchange using the RESTful API is appropriate within where there are well established clinical governance arrangements.

2.0.2.4 Services / SOA

In addition, this specification describes the use of FHIR in a services framework(e.g. a SOA). Note that any use of any of the above alternatives in production is a 'service' by some or many definitions. The services description provides context

regarding the use of FHIR (and particularly the RESTful API) in a wider enterprise architecture.

2.0.3 Developmental Roadmap

Much of the foundation module is well advanced through the maturity model process. Specifically, the following parts of the specification:

* Element / Resource + ElementDefinition

* The XML, & JSON formats

* Many of the Data Types

* The RESTful API + Search

* Binary, Bundle, OperationOutcome & Parameters

The focus over the next 18 months or so as the 4th release of FHIR is prepared is to focus on stability and move these artifacts to normative status. However, some parts of the foundation module are not as well explored, and are not as far advanced with regard to their development. Specifically Documents, Messaging and Services are areas that still need further testing with regard to interoperability. HL7 expects to focus on testing these things in connectathons over the next 18 months.

2.1 Conformance

URL: https://hl7.org/fhir/STU3/conformance-rules.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.1 Conformance

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The FHIR specification describes a set of resources, and several different frameworks for exchanging resources between different systems. Because of its general nature and wide applicability, the rules made in this specification are generally fairly loose. As a consequence, this specification allows that different applications may not be able to interoperate because of how they use optional features. As a consequence, applications claiming conformance to this specification make the claim in respect of a specific exchange framework, and in regard to specific details about their usage of those frameworks and resource contents.

Applications claim conformance to one (or more) of the following exchange frameworks:

* "RESTful FHIR": the RESTful API

* "FHIR messaging": Message based exchange

* "FHIR documents": Document based exchange

To provide details about specific usage of the frameworks and resource contents, FHIR provides a conformance layer that implementers and national/regional programs can use to provide a computable statement about how the resources and their exchange paradigms are used to solve particular use cases. The conformance layer itself is implemented using the following key resources:

Value Set | Defines a set of coded values (see "Using Codes" for more details)

StructureDefinition | Makes rules about how a resource (or type) and its data elements are used in a particular context. A structure definition references value sets for the coded elements in a resource

CapabilityStatement | A statement of the kinds of resources and operations provided and/or consumed by an application. The Capability Statement references profiles to describe specific use of resources by the application

Implementation Guide | A single coherent collection of capability statements, profiles, value set, and documentation describing a set of interoperable applications

The specification also provides a number of tools that can assist with enforcing technical conformance to this base specification and profiles on it.

Conformance with this specification does not provide any guarantee of patient or data safety. However, choosing to not conform to this specification carries additional risk in two ways:

* FHIR has been subject to a level of review and vetting unlikely to be received by any non-conformant variation; variations may result in introduction of undetected risks

* FHIR-like solutions (based on FHIR, but not conformant) may set expectations by trading partners which are not met due to the non-conformance of the system and these un-met expectations may also result in risk

Any assertion of conformance to FHIR that does not reference a CapabilityStatement instance has limited utility. System capabilities that cannot be described by a CapabilityStatement cannot be considered FHIR conformant.

2.1.1 Conformance Language

This specification uses the conformance verbs SHALL, SHOULD, and MAY as defined in RFC 2119 . Unlike RFC 2119, however, this specification allows that different applications may not be able to interoperate because of how they use optional features. In particular:

* SHALL: an absolute requirement for all implementations

* SHALL NOT: an absolute prohibition against inclusion for all implementations

* SHOULD/SHOULD NOT: A best practice or recommendation to be considered by implementers within the context of their particular implementation; there may be valid reasons to ignore an item, but the full implications must be understood and carefully weighed before choosing a different course

* MAY: This is truly optional language for an implementation; can be included or omitted as the implementer decides with no implications

2.1.2 Base Conformance Rules

The contents of a resource and the formats used to represent resources SHALL conform to the rules described in this specification, as defined in the narrative of the specification, and as controlled by the conformance properties defined below.

Data elements defined in resources and data types have 3 properties that are directly related to conformance: Cardinality, Is-Modifier, and Must-Support. These interact to place conformance requirements on implementations.

2.1.3 Cardinality

All attributes defined in FHIR have cardinality as part of their definition - a minimum number of required appearances and a maximum number. These numbers specify the number of times the attribute may appear in any instance of the resource type. This specification only defines the following cardinalities: 0..1, 0..*, 1..1, and 1..*. Profiles that describe specific use cases may use other values for cardinality within the limits of the cardinality defined by the base resource.

Note that when present, elements cannot be empty - they SHALL have a value attribute, child elements, or extensions. This means that setting an element to a minimum cardinality of 1 does not ensure that valid data will be present; specific XPath constraints are required to ensure that the required data will be present.

In this specification, very few elements have a minimum cardinality of 1. Resources are used in many contexts, often quite removed from their primary use case, and sometimes even basic information is quite incomplete. For this reason, the only elements that have a minimum cardinality of 1 are those where they are necessary to any understanding of the resource or element that contains them. The minimum cardinalities should not be taken as a guide to what elements are expected to be present in any particular use of the resource, including their normal/primary usage purpose. In some cases, this specification publishes additional profiles that define which elements are required in particular situations. Similar profiles are published by jurisdictions, vendors, or projects.

For elements that have cardinality > 1, the order in which they appear may have meaning. Unless the element definition (either in this specification or the extension) defines a meaning to the order explicitly (using ElementDefinition.orderMeaning, the meaning of the order is not defined, and implementations are allowed to reorder the elements. Note that it is not possible to define a meaning for the order of the elements in a profile using a StructureDefinition. When there is no definition of the meaning of the order, implementations that need to choose a single element from a list of elements for some use SHALL do so based on the semantics of the content of the elements that repeats. Profiles and Implementation guides may often make rules about this selection process.

Clients should not depend on servers maintaining ordering of elements, unless the retrieved resource conforms to a profile which mandates maintenance of ordering. If a server cannot maintain ordering, it must strip off known profile tags which require maintenance of ordering, and strip off unknown profiles (since they might require maintenance of ordering).

2.1.4 Is-modifier

Is-Modifier is a boolean property that is assigned when an element is defined, either as part of the base resource contents in this specification, or when extensions are defined. An element is labeled "Is-Modifier = true" if the value it contains may change the interpretation of the element that contains it (including if the element is the resource as a whole). Typical examples of elements that are labeled "Is-Modifier" are elements such as "status", "active", "refuted", or "certainty". Whether an element is a modifier cannot be changed when element usage is described in a profile (i.e. a constraining Structure Definition). When an element is labeled as Is-Modifier, the documentation must be clear about why it is a modifier.

A typical example of a modifier element is one that negates the element that contains it. For instance, in the following fragment of a resource definition:

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | | | DomainResource | Allergy or Intolerance (generally: Risk Of Adverse reaction to a substance)
onset | ? | 0..1 | dateTime | Date(/time) when manifestations showed
patient | ? | 1..1 | Reference(Patient) | Who the sensitivity is for
verificationStatus | ?! ? | 0..1 | code | unconfirmed | confirmed | refuted | entered-in-error AllergyIntoleranceVerificationStatus (Required)
criticality | ? | 0..1 | code | CRITL | CRITH | CRITU AllergyIntoleranceCriticality (Required)

The value of the 'verificationStatus' element affects the entire meaning of the resource - if it is set to refuted, the entire resource must be understood differently, and so it is not safe for implementations to ignore it. As a consequence, it is labeled as 'is modifier = true'. In this tabular representation of the resource, this shows as the flag '?!'. The JSON and XML representations of a resource definition have their own representation of 'is modifier = true' status, and it is defined directly in a ElementDefinition.

Is-Modifier elements SHALL be represented in the narrative summary of the resource.

If the value of a modifier element is not explicit in the instance, or known by the context, the resource may not be able to be safely understood. Wherever possible, elements labeled "Is-Modifier = true" also have a minimum cardinality of 1, or a default value, in order to introduce certainty in their handling. However, sometimes this is not possible - much legacy data is not well described. Implementations producing resources SHOULD ensure that appropriate values for isModifier elements are provided at all times.

Implementations processing the data in resources SHALL understand the impact of the element when using the data. Implementations are not required to "support" the element in any meaningful way - they may achieve this understanding by rejecting instances that contain values outside those they support (for instance, an application may refuse to accept observations with a reliability other than "ok"). Alternatively, implementations may be able to be sure that, due to their implementation environment, such values will never occur. However applications SHOULD always check the value irrespective of this.

Note that processing the data of a resource typically means copying or filtering data out of a resource for use in another context (display to a human, decision support, exchange in another format where not all information is included, or storing it for this kind of use). Servers and background processes that simply move whole resources around unchanged are not "processing the data of the resource", and therefore these applications are not required to check Is-Modifier elements.

Every element in the base resource has a value of "true" or "false" for the Is-Modifier flag. The value of the flag cannot be changed by profiles on the resource, in either direction. When a profile defines an extension, it labels the extension with the Is-Modifier flag, and this cannot be changed in other profiles. Note that extensions that have is-Modifier = true are represented differently in resource instances ("modifierExtension" instead of "extension"), and there are additional rules about how they are handled.

Here is a list of all the modifier elements defined in the specification:

* The status and experimental elements on the resources ActivityDefinition, CapabilityStatement, CodeSystem, CompartmentDefinition, ConceptMap, DataElement, ExpansionProfile, GraphDefinition, ImplementationGuide, Library, Measure, MessageDefinition, OperationDefinition, PlanDefinition, Questionnaire, SearchParameter, ServiceDefinition, StructureDefinition, StructureMap, TestScript and ValueSet
* Identifier.use
* Quantity.comparator
* Address.use
* ContactPoint.use
* HumanName.use
* BackboneElement.modifierExtension
* DomainResource.modifierExtension
* Resource.implicitRules
* Account.status
* AllergyIntolerance.clinicalStatus
* AllergyIntolerance.verificationStatus
* Appointment.status
* AppointmentResponse.participantStatus
* Basic.code
* BodySite.active
* CarePlan.status
* CarePlan.intent
* CarePlan.activity.detail.status
* CarePlan.activity.detail.prohibited
* CareTeam.status
* ChargeItem.status
* Claim.status
* ClaimResponse.status
* ClinicalImpression.status
* Communication.status
* Communication.notDone
* CommunicationRequest.status
* Composition.status
* Composition.confidentiality
* Composition.section.mode
* ConceptMap.group.element.target.equivalence
* Condition.clinicalStatus
* Condition.verificationStatus
* Consent.status
* Contract.status

* Coverage.status
* DetectedIssue.status
* Device.status
* DeviceRequest.status
* DeviceRequest.intent
* DeviceUseStatement.status
* DiagnosticReport.status
* DocumentManifest.status
* DocumentReference.status
* DocumentReference.relatesTo
* EligibilityRequest.status
* EligibilityResponse.status
* Encounter.status
* Endpoint.status
* EnrollmentRequest.status
* EnrollmentResponse.status
* EpisodeOfCare.status
* ExplanationOfBenefit.status
* FamilyMemberHistory.status
* FamilyMemberHistory.notDone
* FamilyMemberHistory.estimatedAge
* Flag.status
* Goal.status
* Group.characteristic.exclude
* GuidanceResponse.status
* HealthcareService.active
* Immunization.status
* Immunization.notGiven
* List.status
* List.mode
* List.entry.deleted
* Location.status
* Location.mode
* MeasureReport.status
* MedicationAdministration.status
* MedicationAdministration.notGiven
* MedicationDispense.status
* MedicationRequest.status
* MedicationRequest.intent
* MedicationRequest.substitution.allowed
* MedicationStatement.status
* MedicationStatement.taken
* NamingSystem.status
* NutritionOrder.status
* Observation.status
* OperationOutcome.issue.severity
* Organization.active
* Patient.active
* Patient.deceased[x]
* Patient.animal
* Patient.link
* PaymentNotice.status

* PaymentReconciliation.status
* Person.active
* Procedure.status
* Procedure.notDone
* ProcedureRequest.status
* ProcedureRequest.intent
* ProcedureRequest.doNotPerform
* ProcessRequest.status
* ProcessResponse.status
* Questionnaire.item.enableWhen
* QuestionnaireResponse.status
* ReferralRequest.status
* ReferralRequest.intent
* RelatedPerson.active
* RequestGroup.status
* RequestGroup.intent
* ResearchStudy.status
* ResearchSubject.status
* Schedule.active
* Specimen.status
* Subscription.status
* SupplyDelivery.status
* SupplyRequest.status
* TestReport.status
* ValueSet.compose.include.filter
* VisionPrescription.status

2.1.5 Must-Support

Labeling an element Must-Support means that implementations that produce or consume resources SHALL provide "support" for the element in some meaningful way. Because the base FHIR specification is intended to be independent of any particular implementation context, no elements are flagged as "must-support" as part of the base specification. This flag is intended for use in profiles that have a defined implementation context.

For this reason, the specification itself never labels any elements as must-support. This is done in Resource Profiles, where the profile labels an element as mustSupport=true. When a profile does this, it SHALL also make clear exactly what kind of "support" is required, as this can mean many things.

Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.

2.1.6 Constraints

All elements may have constraints attached to them (also known as 'invariants'). Constraints defined on an element have the following properties:

Key | Identifies the constraint uniquely amongst all the constraints in the context - typically, this is used to refer to the constraint in an error message
Requirements | An explanation of why the constraint has been applied - what harmful conditions are being avoided
Severity | Whether the constraint is an error, or a warning. The exact difference in meaning of these depends on context, but an error is associated with "SHALL" and systems rejecting content, whereas a warning might not be
Human Description | A human description of the rule intended to be show as the explanation for a message when the constraint is not met
Expression | An FHIRPath expression that must evaluate to true when run on the element
XPath | An XPath expression that must evaluate to true when run on the element in the XML representation

Many constraints are defined in the base specification. In addition, additional constraints may be defined in profiles that apply to resources. Systems are not required to evaluate the constraints, just as they are not required to check for conformance, or schema validity. However, systems SHOULD always ensure that all resources are valid against all applicable constraints.

Elements can also be explicitly associated with constraints defined elsewhere. This is a notification to implementers that the element is affected by the constraint. It has no meaning when the constraints are evaluated.

Profiles may define additional constraints that apply to an element, but they cannot alter or remove constraints that are already applied.

2.1.7 Other Metadata

In addition to the conformance metadata, each element has other metadata properties:

* isSummary - How the element behaves in Summary Searches

* defaultValue / meaningWhenMissing - What it means when an element is missing

* maxLength - How long the representation of the primitive value is allowed to be (not including XML or JSON escaping)

2.1.8 Examples and Reference Implementations

This specification includes many examples. While every effort has been made to ensure that the examples are fully conformant to the specification, if the examples disagree with the specification, the specification is considered correct and normative, not the examples. This same rule applies to the reference implementations.

The examples reflected in this specification do *not* represent actual people. Any resemblance to real people - alive or dead - is entirely coincidental. In some cases, examples may be drawn from real clinical data. However, if this has occurred, the content has been scrubbed to remove any identifying information.

2.2 FHIR Life Cycle Page

URL: https://hl7.org/fhir/STU3/lifecycle.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.2 FHIR Life Cycle Page

FHIR Infrastructure Work Group | Maturity Level: 3 | Ballot Status: Trial Use

This page describes several issues around lifecycle management for the resources and the content they contain. Specifically, this page describes:

* Resource Status: how resource status codes work

* Current List: issues associated with retrieving "current X list" of resources

* Entered in Error: information about how erroneous entry is handled for the resources

2.2.1 Resource Status

Many FHIR resources have a status element that represents the lifecycle state of the resource or the clinical process represented by the resource. Work groups can specify status values appropriate to the individual resource. Although consistency between resources is not the primary objective, it is helpful to users and developers to have well-crafted value sets that cover all possible states (since the value sets are typically required and non-extensible).

To understand existing status elements, and to help create extensions and resources involving resource states, we note that status value sets follow one of the following life cycles:

* Clinical workflow process life cycle

* Request/Order life cycle

* Entity status life cycle

* Clinical status life cycle

2.2.2 Clinical Workflow Process Life Cycle

Describes the lifecycle states of complex activities common in healthcare. Typically, these states follow a chronological life cycle that leads from initiation to the conclusion of the action. A characteristic (but non-exhaustive) set of states for the clinical workflow process life cycle include:

* planned - resources for the activity are being allocated but the activity has not begun

* cancelled - the planned activity did not start and will not take place

* in-progress - the activity has begun

* on-hold (suspended) - the activity has been temporarily interrupted

* stopped (aborted, failed) - the activity has not been completed but no future action is planned

* completed (finished) - the activity has been completed

Examples of the clinical workflow life cycle:

* Communication.status: preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown

* Encounter.status: planned | arrived | triaged | in-progress | onleave | finished | cancelled | entered-in-error | unknown

* Goal.status: proposed | accepted | cancelled | entered-in-error | rejected

* MedicationAdministration.status: in-progress | on-hold | completed | entered-in-error | stopped | unknown

* MedicationDispense.status: preparation | in-progress | on-hold | completed | entered-in-error | stopped

* Procedure.status: preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown

2.2.3 Request/Order Life Cycle

Some resources in FHIR represent orders or requests. The request lifecycle can be generalized in terms of four stages: creating the request, sending the request, receiving acceptance or refusal of the request, and fulfillment of the request. A characteristic (but non-exhaustive) set of states for the request/order pattern include:

* proposed: An actor (e.g. a clinical decision support system) has proposed an action to be requested

* draft: The request is in preliminary form, prior to being requested

* requested: The request has been made

* rejected: The request receiver has declined the request

* accepted: The request receiver has accepted the request

* in-progress: Work to fulfill the request has begun

* on-hold (suspended): Work on the request has been interrupted

* stopped (aborted): The activity has not been completed but no future action is planned

* completed: Work on the requested task has been completed, and no further action is required

* cancelled: The request has been withdrawn

Examples of the request/order life cycle:

* CommunicationRequest.status: draft | active | suspended | cancelled | completed | entered-in-error | unknown

* DeviceRequest.status: draft | active | suspended | cancelled | completed | entered-in-error | unknown

* MedicationRequest.status: active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown

* ProcedureRequest.status: draft | active | suspended | cancelled | completed | entered-in-error | unknown

* ReferralRequest.status: draft | active | suspended | cancelled | completed | entered-in-error | unknown

2.2.4 Entity Availability Life Cycle

The entity availability life cycle indicates if the resource, or the entity described by the resource, is ready for use, not yet ready for use, or has been retired from use. A characteristic (but non-exhaustive) set of states for the entity availability life cycle include:

* draft: The entity is being prepared but is not yet in use

* active: The entity is in use

* suspended: The entity is not in use at the moment, but may return to active status

* amended: The entity has undergone a revision but is still active

* retired (superseded): The entity is no longer in use.

Examples of the entity availability life cycle:

* DiagnosticReport.status: registered | partial | final | amended | cancelled | entered-in-error | unknown

* MedicationStatement.status: active | completed | entered-in-error | intended | stopped | on-hold. (note: in-progress and completed are states reflecting the administration of the medication)

* DocumentManifest.status: current | superseded | entered-in-error

* CapabilityStatement.status: draft | active | retired | unknown

* StructureDefinition.status: draft | active | retired | unknown

* DataElement: draft | active | retired | unknown

* Questionnaire.status: draft | active | retired | unknown

* DocumentReference.status: current | superseded | entered-in-error

* QuestionnaireResponse.status: in-progress | completed | amended | entered-in-error | stopped

* Flag.status: active | inactive | entered-in-error

* Location.status: active | suspended | inactive

* Organization.active: true | false

* Patient.active: true | false

2.2.5 Clinical Status Life Cycle

Clinical status is somewhat different than the previous status values, since it does not deal with workflow or lifecycle. Instead, it indicates how evidence is affecting a clinical interpretation. Here are two examples:

* AllergyIntolerance.clinicalStatus: active | inactive

* Condition.clinicalStatus: active | inactive

2.2.6 Current Resource Lists

Many clinical systems maintain current lists of some kind of resources for a patient. Some of the commonly maintained lists include:

* Current Problem List: a list of the problems that are of concern for care of the patient

* Current Medication List: a list of the medications that a patient is known to be on at the current time

Because of the way that resources are used, there is no simple way to determine, from examination of a resource, whether it is 'current' or not. Take, as an example, the Condition resource. In a typical EHR, condition resources might be published on the RESTful interface for the following reasons:

* to represent an item in a patient's curated problem list

* to represent a complaint or a diagnosis from an encounter record

* to represent a problem for investigation provided by a diagnostic system as part of a ProcedureRequest/DiagnosticReport pair

* the resources were received from another system as part of a referral package, and were current for that system when they were received

There is no element on the Condition resource than can convey the difference between these usages. In particular, there can be no way to differentiate between current and past resources without having to retrospectively alter resources, which is problematic with regard to integrity and digital signatures.

One consequence of this is that searching the condition resource for a given patient will return more than just the patient's current problems. Though this is somewhat counter-intuitive to some implementers, restricting searches on Condition to only include the patient's current curated problem list excludes all the other - important - uses of the Condition resource.

Determining whether a Condition is an entry on a patient's current problem list is done by checking with the Condition resource is referenced from the correct list.

On the RESTful API, this is done using the list search mechanism:

```
GET [base]/AllergyIntolerance?patient=42&_list=$current-allergies
```

This is a request to fetch all the allergies in the patient 42's "Current Problem List". Note that the server is not required to actually make a resource representation of the current allergy list available, though doing so assists clients in their audit/integrity tasks. See List Operation "Find" for further information.

In a document, current lists are determined by the code on a Composition section.

FHIR defines the following names for functional lists:

List | ResourceType | Description | Possible LOINC codes in documents / sections

$current-problems | Condition | The "Current Problem List" - A list of current and active diagnoses as well as past diagnoses relevant to the current care of the patient | 46105-3 (Problem conditions Set)

$current-medications | MedicationStatement/ MedicationRequest | A list of all medications that the patient is taking. The 'current medications list' sometimes may include a mix of prescribed and over-the-counter medications - or only some of them. The list may contain a mix of prescriptions and more general statements, or only one of the two. The list may also correspond to a formal reconciled medication administration schedule, but more often does not | 57828-6 (Prescription list), 10160-0 (History of medication)

$current-allergies | AllergyIntolerance | A list of known or suspected propensities to medications, foods, or environmental agents that is provided to help prevent reactions while care is occurring | 18716-1 (Allergy studies (set)), 52472-8 (Allergies and Adverse Drug Reactions), and 48765-2 (Allergies and adverse reactions Document)

$current-drug-allergies | AllergyIntolerance | A list of known or suspected propensities to medications that is provided to help prevent reactions while care is occurring. This list is a subset of the full allergies list | (same as above?)

2.2.7 Entered in Error Summary

The entered-in-error state indicates the resource was created accidentally, and should be ignored. This state can apply to resources created by manual entry. It is usually not associated with the Clinical Workflow Process life cycle, but can be associated with the Request/Order and the Entity Availability life cycles.

This table summarizes what is expected to happen for each resource in the case that the data it contains is subsequently found to be an erroneous entry.

Resource | Status

Account | .status = entered-in-error

ActivityDefinition | .status = retired

AdverseEvent

AllergyIntolerance | .verificationStatus = entered-in-error

Appointment | .status=entered-in-error

AppointmentResponse | .participantStatus=entered-in-error

AuditEvent | Audit Events are recorded, there is no update or delete.

Basic | There are no fixed arrangements for Basic - profiles should describe how this works as arrangements will depend on the 'type' of Basic resource

Binary | n/a (This would be handled where the binary is linked from)

BodySite | .active = false

Bundle | Depends on the type: document - see for Composition; message - see for MessageHeader; transaction / transaction-response / history / searchset - not expected to be stored; collection: just delete it if it's stored, and in error

CapabilityStatement | .status = retired

CarePlan | .status = entered-in-error

CareTeam | .status = entered-in-error

ChargeItem | .status = entered-in-error

Claim | .status = entered-in-error

ClaimResponse | .status = entered-in-error

ClinicalImpression | .status = entered-in-error

CodeSystem | .status = retired

Communication | .status = entered-in-error

CommunicationRequest | .status = entered-in-error

CompartmentDefinition | .status = retired

Composition | .status = entered-in-error

ConceptMap | .status = retired
Condition | .verificationStatus = entered-in-error
Consent | .status = entered-in-error
Contract | .status = entered-in-error
Coverage | .status = entered-in-error
DataElement | .status = retired
DetectedIssue | .status = entered-in-error
Device | .status = entered-in-error
DeviceComponent | .operationalStatus = entered-in-error
DeviceMetric | .operationalStatus = entered-in-error
DeviceRequest | .status = entered-in-error
DeviceUseStatement | .status = entered-in-error
DiagnosticReport | .status = entered-in-error
DocumentManifest | .status = entered-in-error
DocumentReference | .status = entered-in-error
EligibilityRequest | .status = entered-in-error
EligibilityResponse | .status = entered-in-error
Encounter | .status=entered-in-error
Endpoint | .status=entered-in-error
EnrollmentRequest | .status = entered-in-error
EnrollmentResponse | .status = entered-in-error
EpisodeOfCare | .status=entered-in-error
ExpansionProfile | .status=retired
ExplanationOfBenefit | .status = entered-in-error
FamilyMemberHistory | .status = entered-in-error
Flag | .status = entered-in-error
Goal | .status = entered-in-error
GraphDefinition | .status = retired
Group | .active = false
GuidanceResponse | .status = entered-in-error
HealthcareService | .active = false
ImagingManifest | Imaging Object Selection state is managed in PACS.
ImagingStudy | Imaging Study state is managed in PACS.
Immunization | .status = entered-in-error
ImmunizationRecommendation | Immunization recommendations are generated on "just in time" and typically are not retained by the system generating them so there isn't an expected workflow where a previous recommendation would be re-evaluated and found to be "in error".
ImplementationGuide | .status = retired
Library | .status = retired
Linkage | .active = false
List | .status = entered-in-error
Location | .status = inactive
Measure | Delete the report if it is
MeasureReport | Delete the report if it is
Media | n/a - this would be handled whereever the media is linked from
Medication | Medication does not have a status
MedicationAdministration | .status = entered-in-error
MedicationDispense | .status = entered-in-error
MedicationRequest | .status = entered-in-error
MedicationStatement | .status = entered-in-error
MessageDefinition | mostly n/a, but in the cases where messages are stored in error, they would simply be deleted
MessageHeader | mostly n/a, but in the cases where messages are stored in error, they would simply be deleted
NamingSystem | .status = retired
NutritionOrder | .status = entered-in-error
Observation | .status = entered-in-error
OperationDefinition | .status = retired
OperationOutcome | n/a - this resource is not expected to be stored
Organization | .active = false
Patient | .active = false
PaymentNotice | .status = entered-in-error
PaymentReconciliation | .status = entered-in-error
Person | .active = false
PlanDefinition | .status = retired
Practitioner | .active = false
PractitionerRole | .active = false
Procedure | .status = entered-in-error
ProcedureRequest | .status = entered-in-error
ProcessRequest | .status = entered-in-error
ProcessResponse | .status = entered-in-error
Provenance | Provenance are recorded, there is no update or delete.
Questionnaire | .status = retired
QuestionnaireResponse | .status = entered-in-error
ReferralRequest | .status = entered-in-error

RelatedPerson | .active = false
RequestGroup | .status = entered-in-error
ResearchStudy | .status = entered-in-error
ResearchSubject | .status = entered-in-error
RiskAssessment | .status = entered-in-error
Schedule | .active = false
SearchParameter | .status = retired
Sequence | .status = entered-in-error
ServiceDefinition | .status = retired
Slot | .status = entered-in-error
Specimen | .status = entered-in-error
StructureDefinition | .status = retired
StructureMap | .status = retired
Subscription | .status = off (just turn it off, maybe update the error message)
Substance | .status = entered-in-error
SupplyDelivery | .status = entered-in-error
SupplyRequest | .status = entered-in-error
Task | .status = entered-in-error
TestReport | .status = entered-in-error
TestScript | .status = retired
ValueSet | .status = retired
VisionPrescription | .status = entered-in-error

Note: Resources that are not listed in this table do not have any explicit documentation with regard to being entered in error.

2.3 Resource References

URL: https://hl7.org/fhir/STU3/references.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.3.0 Resource References

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

Many of the defined elements in a resource are references to other resources. Using these references, the resources combine to build a web of information about healthcare.

Resources contain two types of references:

* Internal "contained" references - references to other resources packaged inside the source resource

* External references - references to resources found elsewhere

References are always defined and represented in one particular direction - from one resource (source) to another (target). References are either provided as a literal URL, which may either be absolute or relative, or as a logical identifier. Resolving the references is discussed below.

The corresponding reverse relationship from the target to the source exists in a logical sense, but is not represented explicitly in the target resource. For external references, navigating these reverse relationships requires some external infrastructure to track the relationship between resources (the REST API provides one such infrastructure by providing the ability to search the reverse relationship by naming search parameters for the references, and by providing support for reverse includes).

Because resources are processed independently, relationships are not considered to be transitive. For example, if a Condition resource references a particular Patient as its subject, and references a Procedure resource as its cause, there is no automatic rule or implication that the procedure has the same patient for its subject. Instead, the subject of the procedure must be established directly in the Procedure resource itself. Another way to state this is that the context of the subject is not "inherited", nor does it "conduct" along the relationship to procedure. The only exception to this is the case of contained resources (see below). Note that in practice, the relationships need to describe a logical and coherent record, and in the case of the Condition and Procedure described here, they would usually be required to have the same patient for their subjects, and profiles may make rules about this.

In a resource, references are represented with a reference (literal reference), an identifier (logical reference), and a display (text description of target).

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Reference | ?I | | Element | A reference from one resource to another + SHALL have a contained resource if a local reference is provided Elements defined in Ancestors: id, extension

reference | ?I | 0..1 | string | Literal reference, Relative, internal or absolute URL

identifier | ? | 0..1 | Identifier | Logical reference, when literal reference is not known

display | ? | 0..1 | string | Text alternative for the resource

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <reference value="[string]"/><!--  0..1 Literal reference, Relative, internal or absolute URL -->
 <identifier><!-- 0..1 Identifier Logical reference, when literal reference is not known --></identifier>
 <display value="[string]"/><!-- 0..1 Text alternative for the resource -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "reference" : "<string>", // C? Literal reference, Relative, internal or absolute URL
  "identifier" : { Identifier }, // Logical reference, when literal reference is not known
  "display" : "<string>" // Text alternative for the resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Reference.reference [ string ]; # 0..1 Literal reference, Relative, internal or absolute URL
  fhir:Reference.identifier [ Identifier ]; # 0..1 Logical reference, when literal reference is not known
  fhir:Reference.display [ string ]; # 0..1 Text alternative for the resource
]
```

Changes since DSTU2

Reference

Reference.identifier | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Reference | ?I | | Element | A reference from one resource to another + SHALL have a contained resource if a local reference is provided Elements defined in Ancestors: id, extension

reference | ?I | 0..1 | string | Literal reference, Relative, internal or absolute URL

identifier | ? | 0..1 | Identifier | Logical reference, when literal reference is not known

display | ? | 0..1 | string | Text alternative for the resource

Documentation for this format

UML Diagram (Legend)

XML Template


```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <reference value="[string]"/><!--  0..1 Literal reference, Relative, internal or absolute URL -->
 <identifier><!-- 0..1 Identifier Logical reference, when literal reference is not known --></identifier>
 <display value="[string]"/><!-- 0..1 Text alternative for the resource -->
</[name]>
```


JSON Template


```
{
  // from Element: extension
  "reference" : "<string>", // C? Literal reference, Relative, internal or absolute URL
  "identifier" : { Identifier }, // Logical reference, when literal reference is not known
  "display" : "<string>" // Text alternative for the resource
}
```


Turtle Template


```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Reference.reference [ string ]; # 0..1 Literal reference, Relative, internal or absolute URL
  fhir:Reference.identifier [ Identifier ]; # 0..1 Logical reference, when literal reference is not known
  fhir:Reference.display [ string ]; # 0..1 Text alternative for the resource
]
```


Changes since DSTU2

Reference

Reference.identifier | Added Element

See the Full Difference for further information

Constraints

At least one of reference, identifier and display SHALL be present (unless an extension is provided).

* ref-1: SHALL have a contained resource if a local reference is provided (expression : reference.startsWith('#').not() or (reference.substring(1).trace('url') in %resource.contained.id.trace('ids')))

2.3.0.1 Literal References

The reference is the key element - resources are identified and addressed by their URL. It contains a url that is either

* an absolute URL

* a relative URL, which is relative to the Service Base URL, or, in a bundle, the Bundle.entry.fullUrl (see Resolving References in Bundles)

* an internal fragment reference (see "Contained Resources" below)

Notes:

* Using absolute URLs provides a stable, scalable approach suitable for a cloud/web context, while using relative/logical references provides a flexible approach suitable for use when trading across closed ecosystem boundaries. (see "Managing Resource Identity" for further discussion)

* Absolute URLs do not need to point to a , though this is the preferred approach. Whether or not the reference is to a FHIR RESTful server, the reference SHALL point to a Resource as defined by this specification. Note: This regex is true if the reference to a resource is consistent with a FHIR API: However conformance with this regex is no guarantee that the end-point is a FHIR server

FHIR RESTful server

```
((http|https)://([A-Za-z0-
9\\\.\:\%\$]\/)*)?(Account|ActivityDefinition|AdverseEvent|AllergyIntolerance|Appointment|AppointmentResponse|AuditE
vent|Basic|Binary|BodySite|Bundle|CapabilityStatement|CarePlan|CareTeam|ChargeItem|Claim|ClaimResponse|ClinicalImpre
ssion|CodeSystem|Communication|CommunicationRequest|CompartmentDefinition|Composition|ConceptMap|Condition|Consent|C
ontract|Coverage|DataElement|DetectedIssue|Device|DeviceComponent|DeviceMetric|DeviceRequest|DeviceUseStatement|Diag
nosticReport|DocumentManifest|DocumentReference|EligibilityRequest|EligibilityResponse|Encounter|Endpoint|Enrollment
Request|EnrollmentResponse|EpisodeOfCare|ExpansionProfile|ExplanationOfBenefit|FamilyMemberHistory|Flag|Goal|GraphDe
finition|Group|GuidanceResponse|HealthcareService|ImagingManifest|ImagingStudy|Immunization|ImmunizationRecommendati
on|ImplementationGuide|Library|Linkage|List|Location|Measure|MeasureReport|Media|Medication|MedicationAdministration
|MedicationDispense|MedicationRequest|MedicationStatement|MessageDefinition|MessageHeader|NamingSystem|NutritionOrde
r|Observation|OperationDefinition|OperationOutcome|Organization|Patient|PaymentNotice|PaymentReconciliation|Person|P
lanDefinition|Practitioner|PractitionerRole|Procedure|ProcedureRequest|ProcessRequest|ProcessResponse|Provenance|Que
stionnaire|QuestionnaireResponse|ReferralRequest|RelatedPerson|RequestGroup|ResearchStudy|ResearchSubject|RiskAssess
ment|Schedule|SearchParameter|Sequence|ServiceDefinition|Slot|Specimen|StructureDefinition|StructureMap|Subscription
|Substance|SupplyDelivery|SupplyRequest|Task|TestReport|TestScript|ValueSet|VisionPrescription)\/[A-Za-z0-9\-
\.]{1,64}(\/_history\/[A-Za-z0-9\-\.]{1,64})?
```

* URLs are always considered to be case-sensitive

* References SHALL be a reference to an actual FHIR resource, and SHALL be resolveable (allowing for access control, temporary unavailability, etc). Resolution can be either by retrieval from the URL, or, where applicable by resource type, by treating an absolute reference as a canonical URL (see below) and looking it up in a local registry/repository

A relative reference to the Patient "034AB16" in an element named context on a FHIR RESTful server:

```
<patient>
  <reference value="Patient/034AB16" />
</patient>
```

An absolute reference to a Structure Definition in an element named profile:

```
<profile>
  <reference value="http://fhir.hl7.org/svc/StructureDefinition/c8973a22-2b5b-4e76-9c66-00639c99e61b" />
</profile>
```

2.3.0.2 Logical References

In many contexts where FHIR is used, applications building a resource may know an identifier for the target of the reference, but there is no way for the application to convert this to a literal reference that directly references an actual resource. This situation may arise for several reasons:

* There is no server exposing any such resource. This is often the case with national identifiers (e.g. US SSN or NPI), and such identifiers are widely used

* The server that exposes the resource is not available to the source application, so it has no way to resolve an identifier to a reference

* The application is not in a RESTful environment - it is creating a message or a document

In this cases, the source application may provide the identifier as a logical reference to the entity that the target resource would describe.

A logical reference to the Patient with an SSN of 000111111:

```
<patient>
  <identifier>
    <system value="http://hl7.org/fhir/sid/us-ssn" />
    <value value="000111111" />
  </identifier>
</patient>
```

There is no requirement that a Reference.identifier point to something that is actually exposed as a FHIR instance, but it SHALL point to a business concept that would be expected to be exposed as a FHIR instance, and that instance would need to be of a FHIR resource type allowed by the reference.

When processing a resource, an application may be able to use the identifier directly, on the grounds that all it needs is the identifier, or it may be able to resolve the identifier directly. Alternatively, it may be able to use a server to resolve the logical reference to a literal reference to a resource.

Irrespective of how the resolution occurs, any system processing a logical reference will only be able to resolve the identifier to a reference if it understands the business context in which the identifier is used. Sometimes this is global (e.g. a national identifier) but often it is not.

For this reason, none of the useful mechanisms described for working with references (e.g. chaining, includes) are possible, nor should servers be expected to be able to automatically resolve the reference. Servers may accept an identifier based reference untouched, resolve it, and/or reject it - see CapabilityStatement.rest.resource.referencePolicy.

When both an identifier and a literal reference are provided, the literal reference is preferred. Applications processing the resource are allowed - but not required - to check that the identifier matches the literal reference, if they understand how to resolve the logical reference.

Applications converting a logical reference to a literal reference may choose to leave the logical reference present, or remove it.

2.3.0.3 Reference Description

Irrespective of whether a literal and/or logical reference is provided, or neither, the display element may be used to provide a very short description of the target resource.

```
<custodian>
  <reference value="Organization/123" />
  <display value="HL7, Inc" />
</custodian>
```

This text can be used by any application that cannot resolve the reference to fill out the text portion of a hyperlink referring to the target resource, for instance. It can also save time fetching a target resource, and determining how to convert it to a very short textual description.

In general, the display, if populated, does not have identical content to the Resource.text of the referenced resource. The purpose is to identify what's being referenced, not to more fully describe it.

2.3.0.4 Canonical URLs

Many resource types have a defined element "url" which is the canonical URL that always identifies the resource. These include all the conformance and knowledge resources (most of the resources not found in the Patient Compartment).

The canonical URL remains the same when the resource is copied from server to server, while the logical id of the resource - its local identifier - usually changes as the resource is copied. The canonical URL serves as a stable logical identifier for the conformance artifact, and is the preferred way to reference a conformance or knowledge resource. The canonical URL is also the location where the master copy of the artifact is found.

References to these resources may use the Reference type described above, but they can also be referenced using a uri.

When the type of the canonical reference is a uri, the URL may include a version, in order be precise about which version of the resource is being referred to. To do this, append the version to the canonical url with a '|' like this:

```
<valueSetUri value="http://hl7.org/fhir/StructureDefinition/my-profile|0.8"/>
```

This is a version specific reference to a profile. Note that this is the StructureDefinition.version not the StructureDefinition.meta.versionId. Searching for this on a FHIR server would look like this:

```
GET fhir/ValueSet?url=http://hl7.org/fhir/ValueSet/clinical-findings&version=0.8
```

Note that if a canonical reference does not have a version, and the server finds multiple versions for the value set, the system using the reference should pick the latest version of the target resource and use that. Servers SHOULD support version specific searching for canonical URLs but automatically detecting the presence of a |[version] and performing the appropriate search.

Systems resolving references to resources that might have canonical URLs SHOULD first try to resolve the reference using the canonical URL, and then fall back to direct resolution using the URL as a literal reference if a local version of the canonical resource cannot be found. This approach is safe because the local version cannot be a different artifact than the master copy, though implementations will need to make appropriate arrangements regarding the currency of their local copy of the artifact.

2.3.0.5 Contained Resources

In some circumstances, the content referred to in the resource reference does not have an independent existence apart from the resource that contains it - it cannot be identified independently, and nor can it have its own independent transaction scope. Typically, such circumstances arise where resources are being assembled by a secondary user of the source data, such as a middleware engine. If the data available when the resource is constructed does not include record keys or absolute identification information, then a properly identified resource cannot be assembled, and even if an arbitrary identification was associated with it, the resource could never be the subject of a transaction outside the context of the resource that refers to it.

For example, consider a situation where an interface engine is creating a Condition record on a patient from an HL7 v2 message, and the only information about the primary surgeon is her first name and last name (REL-7.2 & REL-7.3). In the absence of a controlled practitioner directory, this is not enough information to create an identified Practitioner resource since more than one practitioner might have the same name.

In these circumstances, the resource is placed directly in-line in the resource. This SHOULD NOT be done when the content can be identified properly, as once the identification is lost, it is extremely difficult (and context dependent) to restore it again.

An example of a contained resource:

```
<Condition xmlns="http://hl7.org/fhir">
 <contained>
   <Practitioner>
     <id value="p1"/>
     <name>
       <family value="Person"/>
       <given value="Patricia"/>
     </name>
   </Practitioner>
 </contained>
 <!-- other attributes -->
 <asserter>
   <reference value="#p1" />
 </asserter>
 <!-- other attributes -->
</Condition>
```

The same example in JSON:

```
{
  "resourceType" : "Condition",
  "contained": [
    {
      "resourceType" : "Practitioner",
      "id" : "p1",
      "name" : [{
        "family" : "Person",
        "given" : ["Patricia"]
      }]
    }],
   "asserter" : {
     "reference" : "#p1"
  }
}
```

Design Note: Contained resources are still a reference rather than being inlined directly into the element that is the reference (e.g. "custodian" above) to ensure that a single approach to resolving resource references can be used. Though direct containment would seem simpler, it would still be necessary to support internal references where the same contained resource is referenced more than once. In the end, all that it would achieve is creating additional options in the syntax. For users using XPath to process the resource, the following XPath fragment resolves the internal reference:

```
ancestor::f:*[not(parent::f:*)]/f:contained/*[@id=substring-after(current()/f:reference/@value, '#')]
```

Some notes about use and interpretation of contained resources:

* The contained element SHALL NOT have extensions on it (though contained resources can still contain extensions).

* The contained resource can be put in any resource that inherits from DomainResource. The contained element is then located at the beginning of the resource after any text narrative and before any extension.

* Contained resources share the same internal id resolution space as the parent resource (for id attributes, see below).

* When resolving references, references are resolved by looking through the 'container' resource - the one that contains the other resources. Since there are no nested contained resources, there is only one container resource.

* References to contained resources are never resolved outside the container resource. Specifically, resolution stops at the elements Bundle.entry.resource and Parameters.parameter.resource, but not at DomainResource.contained.

* Contained resources SHALL NOT contain additional contained resources.

* Contained resources SHALL NOT contain any narrative.

* A contained resource SHALL only be included in a resource if something in that resource (potentially another contained resource) has a reference to it.

Resources that are contained inline do not "inherit" context from their parent resource. For instance, if the parent resource contains a "subject", and the contained resource also has a "subject" element defined, there is no implication that the contained resource has the same subject as the parent resource.

Resources can only be contained in other resources if there is a reference from the resource to the contained resource. This is intended to ensure that the meaning of the contained resource is clear, and that there is no confusion as to its significance.

STU Note: There are some identified use cases where it would be useful to include resources that refer to the contained resource rather than the container referring to the contained resource, but this has a series of structural ramifications for the API. Whether these can be resolved is an open issue for investigation during the period of trial use.

Feedback is welcome here .

2.4 Narrative

URL: https://hl7.org/fhir/STU3/narrative.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.4.0 Narrative

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

Any resource that is a DomainResource (almost all types of resource) may include a human-readable narrative that contains a summary of the resource, and may be used to represent the content of the resource to a human.

If narrative is present with a status other than 'empty', it SHALL reflect all content needed for a human to understand the essential clinical and business information for the resource. In other words, it must be safe to render only the narrative of the resource without displaying any of the resource's discrete/encoded information. Resource definitions and/or profiles on resources may define what content should be represented in the narrative to ensure clinical safety.

The narrative for a resource can contain additional information that is not in the structured data, including human-edited content. Such additional information SHALL be in the scope of the definition of the resource, though it is common for the narrative to include additional descriptive information extracted from other referenced resources when describing references. Narrative for a resource should include summary information about any referenced resources that would be required for a consumer of the resource to be able to understand the key, essential information about a resource without retrieving any additional resources.

For example, the narrative for a MedicationRequest might include brief summary information about the referenced patient, prescriber and medication. Some resources (e.g. List, Composition) may provide specific rules about what content must (or must not) be included in the resource narrative. Consideration should be given to the fact that referenced resources may be updated without updating referencing resources, so the proportion of content of a referenced resource included in a referencing resource should be limited. Systems MAY choose how narrative is generated, including how much de-referencing to perform, but SHALL NOT assume that the resource is rendered in any particular context when generating narrative, since resources will be used in multiple contexts.

Resources SHOULD always contain narrative to support human-consumption as a fallback. Structured data SHOULD NOT generally contain information of importance to human readers that is omitted from the narrative. Creators of FHIR resources should not assume that systems will render (or that humans will see) data that is not in the narrative. However, in strictly managed trading systems where all systems share a common data model and additional text is unnecessary or even a clinical safety risk, the narrative may be omitted. Implementers should give careful consideration before doing this, as it will mean that such resources can only be understood in the limited trading environment. Closed trading partner environments are very likely to open up during the lifetime of the resources they define. Also, many workflow steps involving finding and aggregating resources are much more difficult or tedious if the resources involved do not have their own text.

In some cases, a resource may only have text with little or no additional discrete data (as long as all minOccurs=1 elements are satisfied). This may be necessary for data from legacy systems where information is captured as a "text blob" or where text is additionally entered raw or narrated and encoded in formation is added later.

Note that contained Resources SHALL NOT have a narrative of their own.

The narrative is an XHTML fragment with a flag to indicate its relationship to the data:

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Narrative | | | Element | A human-readable formatted text, including images Elements defined in Ancestors: id, extension

status | | 1..1 | code | generated | extensions | additional | empty NarrativeStatus (Required)

div | I | 1..1 | xhtml | Limited xhtml content + The narrative SHALL contain only the basic html formatting elements and attributes described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes + The narrative SHALL have some non-whitespace content

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <status value="[code]"/><!-- 1..1 generated | extensions | additional | empty -->
 <div xmlns="http://www.w3.org/1999/xhtml"> <!-- Limited xhtml content< --> </div>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "status" : "<code>", // R!  generated | extensions | additional | empty
  "div" : "(Escaped XHTML)" // R!  Limited xhtml content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Narrative.status [ code ]; # 1..1 generated | extensions | additional | empty
  fhir:Narrative.div [ fhir:value "[escaped xhtml]"^^xsd:string ]; # 1..1 Limited xhtml content
]
```

Changes since DSTU2

Narrative

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Narrative | | | Element | A human-readable formatted text, including images Elements defined in Ancestors: id, extension

status | | 1..1 | code | generated | extensions | additional | empty NarrativeStatus (Required)

div | I | 1..1 | xhtml | Limited xhtml content + The narrative SHALL contain only the basic html formatting elements and attributes described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes + The narrative SHALL have some non-whitespace content

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <status value="[code]"/><!-- 1..1 generated | extensions | additional | empty -->
 <div xmlns="http://www.w3.org/1999/xhtml"> <!-- Limited xhtml content< --> </div>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "status" : "<code>", // R!  generated | extensions | additional | empty
  "div" : "(Escaped XHTML)" // R!  Limited xhtml content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Narrative.status [ code ]; # 1..1 generated | extensions | additional | empty
  fhir:Narrative.div [ fhir:value "[escaped xhtml]"^^xsd:string ]; # 1..1 Limited xhtml content
]
```

Changes since DSTU2

Narrative

See the Full Difference for further information

The contents of the div element are an XHTML fragment that SHALL contain only the basic HTML formatting elements described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes. The XHTML content SHALL NOT contain a head, a body element, external stylesheet references, deprecated elements, scripts, forms, base/link/xlink, frames, iframes, objects or event related attributes (e.g. onClick). This is to ensure that the content of the narrative is contained within the resource and that there is no active content. Such content would introduce security issues and potentially safety issues with regard to extracting text from the XHTML.

The div element SHALL have some non-whitespace content (text or an image).

```
  <narrative>
    <div xmlns="http://www.w3.org/1999/xhtml">This is a simple
          example with only plain text</div>
  </narrative>

   <narrative>
   <div xmlns="http://www.w3.org/1999/xhtml">
     <p>
       This is an <i>example</i> with some <b>xhtml</b> formatting.
     </p>
   </div>
  </narrative>
```

The inner portion of the div content is often used for the innerHTML property in a browser. In order to simplify this kind of processing, when the narrative is represented in JSON, it SHALL be encoded so that the characters between the first '>' and

the last '<' delimiters is the content of the element; e.g.

```
  "div": "<div>text</div>"
```

is legal, but this is not:

```
  "div": "<?xml ...><div>text</div>"
```

Note that the XHTML is contained in general XML so there is no support for HTML entities like   or © etc. Unicode characters SHALL be used instead. Unicode   substitutes for .

The narrative content SHOULD be in the language of the resource, but there is no reason to expect that HTML type tooling would understand the resource language element. For this reason, a lang attribute on the SHOULD also be used (see the note in the HTML 5 specification about use of language).

2.4.0.1 Image References

Image source data (the src attribute) may refer to an image found in the resource (as a contained Media or Binary resource) by its id:

```
<Patient xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>... <img src="#pic1"/>. ....</p>
    </div>
  </text>
  <contained>
    <Binary><id value="pic1"/><contentType value="image/gif"/><content value="MEKH....SD/Z"/></Binary>
  </contained>
</Patient>
```

References between the narrative and the resource data (in either direction) are mediated by the XML id/idref attributes. in JSON, the property "id" is used which is equivalent to the XML attribute "id".

The id attribute SHALL have a unique value within the resource with regard to any other id attributes: the uniqueness and resolution scope of these id references is within the resource that contains them. Contained resources are included in the id uniqueness scope of the resource that contains them.

If multiple resources are combined into a single combined document, such as a Bundle, duplicate values of the id attribute may occur between resources. This SHALL be managed by applications reading the resources.

Since images that are not contained in the resource cannot be guaranteed to be available when the resource is presented to a user, the source for any images that are an essential part of the narrative SHOULD always be embedded as a data: url , in an attachment or a contained resource.

2.4.0.2 Styling the XHTML

The XHTML fragment in the narrative may be styled using cascading stylesheets with either external or internal styles. External styles are applied using the class and id attributes on the XHTML elements and internal styles are applied using a style attribute on the XHTML elements directly.

In order to minimise manageability and security issues, authoring systems cannot specify the CSS stylesheet to use directly. Instead, the application that displays the resource provides the stylesheets. This means that the rendering system chooses what styles can be used, but the authoring system must use them in advance. Authoring systems can use these classes, which SHALL be supported by all rendering systems:

- bold | Bold | { font-weight: bold }
- italics | Italics Text | { font-style: italic }
- underline | Underlined Text | { text-decoration: underline }
- strikethrough | Strikethrough Text | { text-decoration: line-through }
- left | Left Aligned | { text-align : left }
- right | Right Aligned | { text-align : right }
- center | Center Aligned | { text-align : center }
- justify | Justified | { text-align : justify }
- border-left | Border on the left | { border-left: 1px solid grey }
- border-right | Border on the right | { border-right: 1px solid grey }
- border-top | Border on the top | { border-top: 1px solid grey }
- border-bottom | Border on the bottom | { border-bottom: 1px solid grey }
- arabic | List is ordered using Arabic numerals: 1, 2, 3 | { list-style-type: decimal }
- little-roman | List is ordered using little Roman numerals: i, ii, iii | { list-style-type: lower-roman }
- big-roman | List is ordered using big Roman numerals: I, II, III | { list-style-type: upper-roman }
- little-alpha | List is ordered using little alpha characters: a, b, c | { list-style-type: lower-alpha }
- big-alpha | List is ordered using big alpha characters: A, B, C | { list-style-type: upper-alpha }
- disc | List bullets are simple solid discs | { list-style-type: disc }
- circle | List bullets are hollow discs | { list-style-type : circle }
- square | List bullets are solid squares | { list-style-type: square }
- unlist | List with no bullets | { list-style-type: none }

Note: for testing purposes, there is an example resource that includes all of these styles. It is also available as XHTML and a standard stylesheet that includes all of these styles. Use of styles not on this list will require an arrangement between producing and consuming systems.

Authoring systems may refer to additional classes, but cannot rely on the fact that they will be supported. If the additional classes are critical for safe rendering, trading partner agreements will be required.

Authoring systems may also use internal styles using the style attribute. This has the advantage of not depending on external interpretation, but also has the side effect of making content more difficult to manage when rendering, so applications should use this approach with care.

Authoring systems may fix the following styling aspects of the content:

* bold, italic, underline, strikethrough

* font color, family and size

* background color, text alignment

* whitespace interpretation

* ordered list number format (since it may be referred to in text)

These style properties are specified in-line using the style attribute. Rendering systems SHOULD respect any of these rendering styles when they are specified in the style attribute, although appropriate interpretation is allowed in certain contexts (e.g. a low-contrast display for dark rooms or a high-contrast display for the visually impaired may adjust colors accordingly).

Note that rendering systems can ignore or override any of the internal or external styles described above, but SHOULD be careful to ensure that this is only done in the context of well-maintained trading partner agreements, as altering the presentation of the text may create clinical safety issues.

Authors MAY specify additional styles and style properties as specified in the CSS specification, but these are extensions to this specification and renderers are not required to heed them. It SHOULD be safe to view the narrative without these additional styling features available.

Note that there are additional rules around styling for documents presentation.

2.4.0.3 Clinical Safety Concerns

Health care records are often associated with legislative and business requirements for very long retention times (up to a century), and extreme risk aversion with regards to inconsistent display across a variety of devices. Although the narrative is allowed to use the standard XHTML and CSS features as described above, implementations are encouraged to show restraint when using the features available. Even when trading partner arrangements limit the current requirements made on a system, experience shows that these trading arrangements will likely broaden over time.

In particular:

* complex layered layouts requiring careful testing of the match between the XHTML div and span elements and styles, and those that include nested tables (possibly with images) as well, make rendering consistency difficult, and implementations SHOULD avoid these

* The use of styles e.g. bolding, italics and color; SHOULD NOT be used as the sole way to convey meaning or semantics, but should be used in conjunction with other data elements to ensure consistent, long term interoperability.

2.4.1 Example Narrative

URL: https://hl7.org/fhir/STU3/narrative-example.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

2.4.1.1 Plain HTML, No Styles

2.4.1.1.1 Heading 3

2.4.1.1.1.1 Heading 4

2.4.1.1.1.1.1 Heading 5

2.4.1.1.1.1.1.1 Heading 6

Paragraph. span. Link. Bold, br: em, Italics, strong, small, big Teletype Text, small, Definition term, q, var. All provided by HL7, for FHIR (cite).

Paragraph in a blockquote, with an hr after it:

Paragraph in a div (Link Target)

* Unordered List Item

* Ordered List Item

DT Item

DD Item

```
Some Pre Text
  with a line break
```

Table:

Head Cell 1 | Head Cell 2 | Head Cell 3
Foot Cell 1 | Foot Cell 2 | Foot Cell 3
Body Cell 1 | Body Cell 2 | Body Cell 3

Code Block

Sample Block

2.4.1.2 External Styles

2.4.1.2.1 Text:

Example Text: bold, italics, underline and strikethrough

This paragraph is left aligned. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen.

This paragraph is right aligned. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen.

This paragraph is center aligned. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen.

This paragraph is justified. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen.

2.4.1.2.2 Table:

Border Left | Border Right | Border Top | Border Bottom

2.4.1.2.3 List:

* arabic (Item 1)
* Item Two
* little-roman (Item 1)
* Item Two
* big-roman (Item 1)
* Item Two
* little-alpha (Item 1)
* Item Two
* big-alpha (Item 1)
* Item Two
* disc (Item 1)
* Item Two
* unlist(Item 1)
* Item Two
* circle (Item 1)
* Item Two
* square (Item 1)
* Item Two

2.4.1.3 Internal Styles

Example Text: bold, italics, underline and strikethrough. Font-Family Serif and Sans Serif, Font-size 50% 80% 150%, Font-Color Navy Maroon Brown, Background-color Aqua Silver Pink.

Whitespace Control:

Normal Whitespace Test, long long long

Pre Whitespace Test, long long long

* Item Two

* upper-latin (Item 1)

* Item Two

* upper-roman (Item 1)

* Item Two

* upper-roman (Item 1)

* Item Two

2.5 Extensibility

URL: https://hl7.org/fhir/STU3/extensibility.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

This exchange specification is based on generally agreed common requirements across healthcare - covering many jurisdictions, domains, and different functional approaches. It is common for specific implementations to have valid requirements that are not part of these agreed common requirements. Incorporating all valid requirements would make this specification very cumbersome and difficult to implement. Instead, this specification expects that additional valid requirements will be implemented as extensions.

As such, extensibility is a fundamental part of the design of this specification. Every element in a resource can have extension child elements to represent additional information that is not part of the basic definition of the resource. Applications should not reject resources merely because they contain extensions, though they may need to reject resources because of the specific contents of the extensions.

Note that, unlike in many other specifications, there can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core simplicity for everyone.

To make the use of extensions safe and manageable, there is strict governance applied to the definition and use of extensions. Although any implementer can define and use extensions, there is a set of requirements that must be met as part of their use and definition.

2.5.0.1 Extension Element

Every element in a resource or data type includes an optional "extension" child element that may be present any number of times. This is the content model of the extension as it appears in each resource:

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Extension | I | | Element | Optional Extensions Element + Must have either extensions or value[x], not both Elements defined in Ancestors: id, extension
url | | 1..1 | uri | identifies the meaning of the extension
value[x] | | 0..1 | * | Value of extension
Documentation for this format

UML Diagram (Legend)

XML Template

```
<extension|modifierExtension xmlns="http://hl7.org/fhir" url="identifies the meaning of the extension (uri)">
 <!-- from Element: extension -->
 <value[x]><!-- 0..1 * Value of extension --></value[x]>
</extension|modifierExtension>
```

JSON Template

```
{
  // from Element: extension
  "url" : "<uri>", // R!  identifies the meaning of the extension
  // value[x]: Value of extension. One of these 23:
  "valueInteger" : <integer>
  "valueDecimal" : <decimal>
  "valueDateTime" : "<dateTime>"
  "valueDate" : "<date>"
  "valueInstant" : "<instant>"
```

```
  "valueString" : "<string>"
  "valueUri" : "<uri>"
  "valueBoolean" : <boolean>
  "valueCode" : "<code>"
  "valueBase64Binary" : "<base64Binary>"
  "valueCoding" : { Coding }
  "valueCodeableConcept" : { CodeableConcept }
  "valueAttachment" : { Attachment }
  "valueIdentifier" : { Identifier }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valuePeriod" : { Period }
  "valueRatio" : { Ratio }
  "valueHumanName" : { HumanName }
  "valueAddress" : { Address }
  "valueContactPoint" : { ContactPoint }
  "valueSchedule" : { Schedule }
  "valueReference" : { Reference }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Extension.url [ uri ]; # 1..1 identifies the meaning of the extension
  # Extension.value[x] : 0..1 Value of extension. One of these 38
    fhir:Extension.valueBase64Binary [ base64Binary ]
    fhir:Extension.valueBoolean [ boolean ]
    fhir:Extension.valueCode [ code ]
    fhir:Extension.valueDate [ date ]
    fhir:Extension.valueDateTime [ dateTime ]
    fhir:Extension.valueDecimal [ decimal ]
    fhir:Extension.valueId [ id ]
    fhir:Extension.valueInstant [ instant ]
    fhir:Extension.valueInteger [ integer ]
    fhir:Extension.valueMarkdown [ markdown ]
    fhir:Extension.valueOid [ oid ]
    fhir:Extension.valuePositiveInt [ positiveInt ]
    fhir:Extension.valueString [ string ]
    fhir:Extension.valueTime [ time ]
    fhir:Extension.valueUnsignedInt [ unsignedInt ]
    fhir:Extension.valueUri [ uri ]
    fhir:Extension.valueAddress [ Address ]
    fhir:Extension.valueAge [ Age ]
    fhir:Extension.valueAnnotation [ Annotation ]
    fhir:Extension.valueAttachment [ Attachment ]
    fhir:Extension.valueCodeableConcept [ CodeableConcept ]
    fhir:Extension.valueCoding [ Coding ]
    fhir:Extension.valueContactPoint [ ContactPoint ]
    fhir:Extension.valueCount [ Count ]
    fhir:Extension.valueDistance [ Distance ]
    fhir:Extension.valueDuration [ Duration ]
    fhir:Extension.valueHumanName [ HumanName ]
    fhir:Extension.valueIdentifier [ Identifier ]
    fhir:Extension.valueMoney [ Money ]
    fhir:Extension.valuePeriod [ Period ]
    fhir:Extension.valueQuantity [ Quantity ]
    fhir:Extension.valueRange [ Range ]
    fhir:Extension.valueRatio [ Ratio ]
    fhir:Extension.valueReference [ Reference ]
    fhir:Extension.valueSampledData [ SampledData ]
    fhir:Extension.valueSignature [ Signature ]
    fhir:Extension.valueTiming [ Timing ]
    fhir:Extension.valueMeta [ Meta ]
]
```

Changes since DSTU2

Extension

Extension.value[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Extension | I | | Element | Optional Extensions Element + Must have either extensions or value[x], not both Elements defined in Ancestors: id, extension

url | | 1..1 | uri | identifies the meaning of the extension

value[x] | | 0..1 | * | Value of extension

Documentation for this format

UML Diagram (Legend)

XML Template

```
<extension|modifierExtension xmlns="http://hl7.org/fhir" url="identifies the meaning of the extension (uri)">
 <!-- from Element: extension -->
 <value[x]><!-- 0..1 * Value of extension --></value[x]>
```

```
</extension|modifierExtension>
```

JSON Template

```
{
  // from Element: extension
  "url" : "<uri>", // R!  identifies the meaning of the extension
  // value[x]: Value of extension. One of these 23:
  "valueInteger" : <integer>
  "valueDecimal" : <decimal>
  "valueDateTime" : "<dateTime>"
  "valueDate" : "<date>"
  "valueInstant" : "<instant>"
  "valueString" : "<string>"
  "valueUri" : "<uri>"
  "valueBoolean" : <boolean>
  "valueCode" : "<code>"
  "valueBase64Binary" : "<base64Binary>"
  "valueCoding" : { Coding }
  "valueCodeableConcept" : { CodeableConcept }
  "valueAttachment" : { Attachment }
  "valueIdentifier" : { Identifier }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valuePeriod" : { Period }
  "valueRatio" : { Ratio }
  "valueHumanName" : { HumanName }
  "valueAddress" : { Address }
  "valueContactPoint" : { ContactPoint }
  "valueSchedule" : { Schedule }
  "valueReference" : { Reference }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Extension.url [ uri ]; # 1..1 identifies the meaning of the extension
  # Extension.value[x] : 0..1 Value of extension. One of these 38
    fhir:Extension.valueBase64Binary [ base64Binary ]
    fhir:Extension.valueBoolean [ boolean ]
    fhir:Extension.valueCode [ code ]
    fhir:Extension.valueDate [ date ]
    fhir:Extension.valueDateTime [ dateTime ]
    fhir:Extension.valueDecimal [ decimal ]
    fhir:Extension.valueId [ id ]
    fhir:Extension.valueInstant [ instant ]
    fhir:Extension.valueInteger [ integer ]
    fhir:Extension.valueMarkdown [ markdown ]
    fhir:Extension.valueOid [ oid ]
    fhir:Extension.valuePositiveInt [ positiveInt ]
    fhir:Extension.valueString [ string ]
    fhir:Extension.valueTime [ time ]
    fhir:Extension.valueUnsignedInt [ unsignedInt ]
    fhir:Extension.valueUri [ uri ]
    fhir:Extension.valueAddress [ Address ]
    fhir:Extension.valueAge [ Age ]
    fhir:Extension.valueAnnotation [ Annotation ]
    fhir:Extension.valueAttachment [ Attachment ]
    fhir:Extension.valueCodeableConcept [ CodeableConcept ]
    fhir:Extension.valueCoding [ Coding ]
    fhir:Extension.valueContactPoint [ ContactPoint ]
    fhir:Extension.valueCount [ Count ]
    fhir:Extension.valueDistance [ Distance ]
    fhir:Extension.valueDuration [ Duration ]
    fhir:Extension.valueHumanName [ HumanName ]
    fhir:Extension.valueIdentifier [ Identifier ]
    fhir:Extension.valueMoney [ Money ]
    fhir:Extension.valuePeriod [ Period ]
    fhir:Extension.valueQuantity [ Quantity ]
    fhir:Extension.valueRange [ Range ]
    fhir:Extension.valueRatio [ Ratio ]
    fhir:Extension.valueReference [ Reference ]
    fhir:Extension.valueSampledData [ SampledData ]
    fhir:Extension.valueSignature [ Signature ]
    fhir:Extension.valueTiming [ Timing ]
    fhir:Extension.valueMeta [ Meta ]
]
```

Changes since DSTU2

Extension
Extension.value[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money

See the Full Difference for further information

Notes:

* The url is a mandatory attribute / property and identifies a retrievable extension definition that defines the content and meaning of the extension

* The url SHALL be a URL, not a URN (e.g. not an OID or a UUID), and it SHALL be the canonical URL of a StructureDefinition that defines the extension. Except for child extensions defined within complex extensions, the URL SHALL be an absolute URL. The StructureDefinition that defines the extension SHOULD be available at the identified location so that resource processors can access the definition from the URL

* An extension SHALL have either a value (i.e. a value[x] element) or sub-extensions, but not both. If present, the value[x] element SHALL have content (value attribute or other elements)

* If it is not safe for an application processing the content of the resource to ignore the extension it must be represented differently using a Modifier Extension

* The element has an actual name of "value" and then the TitleCased name of one of these defined types, and its contents are those as defined for that type:

value[x]

* valueInteger : integer

* valueUnsignedInt : unsignedInt

* valuePositiveInt : positiveInt

* valueDecimal : decimal

* valueDateTime : dateTime

* valueDate : date

* valueTime : time

* valueInstant : instant

* valueString : string

* valueUri : uri

* valueOid : oid

* valueUuid : uuid

* valueId : id

* valueBoolean : boolean

* valueCode : code (only if the extension definition provides a fixed binding to a suitable set of codes)

* valueMarkdown : markdown

* valueBase64Binary : base64Binary

* valueCoding : Coding

* valueCodeableConcept : CodeableConcept

* valueAttachment : Attachment

* valueIdentifier : Identifier

* valueQuantity : Quantity

* valueSampledData : SampledData

* valueRange : Range

* valuePeriod : Period

* valueRatio : Ratio

* valueHumanName : HumanName

* valueAddress : Address

* valueContactPoint : ContactPoint

* valueTiming : Timing

* valueReference : Reference - a reference to another resource

* valueAnnotation : Annotation

* valueSignature : Signature

* valueMeta : A Meta heading from a resource

Here is an example of an extension in XML:

```
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso-21090-name-use" >
    <valueCode value="I" />
  </extension>
  <text value="Chief Red Cloud"/>
</name>
```

In this example the name with text = "Chief Red Cloud" is extended to have a name use code of "Indigenous" (defined in ISO 21090, but very rarely used in practice).

In JSON, extensions are represented similarly:

```
{
  "extension" : [{
```

```
    "url" : "http://hl7.org/fhir/StructureDefinition/iso-21090-name-use",
    "valueCode" : "I"
  }],
  "text" : "Chief Red Cloud"
}
```

Making the types explicit in the representation means that all systems can read and write (and therefore store and/or exchange) extensions correctly without needing to access the definition of the extension.

Note that the JSON representation for extensions on primitive data types is handled differently. See Representing primitive types in JSON for further information.

Extensions can also contain extensions, either because the extension definition itself defines complex content - that is, a nested tree of values in the extension - or because the extension is extended with an additional extension defined separately.

In the case where an extension defines complex content, the identity of the parts of the extension are local/relative to the reference to the extension definition.

As an example, consider extending a patient with an opt-in status for a clinical trial containing 3 fields: clinical trial number, period of involvement, and a reason for enrollment. In XML:

```
<Patient>
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial" >
    <extension url="NCT" >
      <valueString value="123456789" />
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="2009-03-14" />
      </valuePeriod>
    </extension>
    <extension url="reason" >
      <valueCodeableConcept>
        <coding>
          <system value="http://acme.org/codes/general" />
          <code value="tt14j" />
        </coding>
      </valueCodeableConcept>
    </extension>
  </extension>
  <!-- other data for patient -->
</Patient>
```

Or in JSON:

```
{
  "resourceType" : "Patient",
  "extension" : [{
    "url" : "http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial",
    "extension" : [{
      "url" : "NCT",
      "valueString" : "123456789"
    }, {
      "url" : "period",
      "valuePeriod" : {
         "start" : "2009-03-14"
      }
    },  {
      "url" : "reason",
      "valueCodeableConcept" : {
          "coding" : [{
             "system" : "http://acme.org/codes/general",
             "code" : "tt14j"
          }]
       }
    }]
  }]
}
```

As the URL suggests, this extension is defined as part of this specification.

This extension can be extended again, by adding a "registrar" extension:

The registrar is defined as a separate extension (e.g. by an implementing organization) rather than part of the official clinical-trial extension. The URL of the extension is thus different. In XML:

```
<Patient>
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial" >
    <extension url="NCT" >
      <valueString value="123456789" />
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="2009-03-14" />
      </valuePeriod>
    </extension>
    <extension url="reason" >
      <valueCodeableConcept>
        <coding>
          <system value="http://acme.org/codes/general" />
          <code value="tt14j" />
```

```
        </coding>
      </valueCodeableConcept>
    </extension>
  </extension>
    <extension url="http://acme.org/fhir/StructureDefinition/registrar" >
      <valueReference>
        <reference value="Practitioner/example" />
      </valueReference>
    </extension>
  <!-- other data for patient -->
</Patient>
```

or in JSON:

```
{
  "resourceType": "Patient",
  "extension": [{
    "url": "http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial",
    "extension": [{
      "url": "NCT",
      "valueString": "123456789"
    },
    {
      "url": "period",
      "valuePeriod": {
        "start": "2009-03-14"
      }
    },
    {
      "url": "reason",
      "valueCodeableConcept": {
        "coding": [{
          "system": "http://acme.org/codes/general",
          "code": "tt14j"
        }]
      }
    },
    {
      "url": "http://acme.org/fhir/StructureDefinition/registrar",
      "valueReference": {
        "reference": "Practitioner/example"
      }
    }]
  }]
}
```

2.5.0.2 Modifier Extensions

There are some cases where the information provided in an extension modifies the meaning of the element that contains it. Typically, this means information that qualifies or negates the primary meaning of the element that contains it. Some examples:

* An anti-prescription: recording an instruction not to take a medication

* Using the Condition resource to record an assertion that a patient has a family history of the condition rather than the condition itself

* Asserting that a performer was not actually involved in a Procedure

* Asserting an additional subsumption relationship on a concept in a value set

Such extensions are known as "modifier extensions". Implementers should avoid the use of modifier extensions where possible. Any use should be carefully considered against its possible downstream consequences. However, implementers are often forced into these situations by the business arrangements around the use of resources, so this specification creates a framework for handling such cases. If modifier extensions are present, an application cannot safely process the resource unless it knows what the extension means for its own use of the data.

This specification allows for such modifier elements to be included at the base of a resource or in any elements that do not have a data type (e.g. the elements that correspond to classes in the resource UML diagrams). Elements that are data types, or that are inside data types SHALL NOT have modifier extensions, and extensions SHALL NOT have modifier extensions internally.

Note that complex extensions are allowed to have elements in the complex extension that are marked Is-Modifier = true, which means that these elements modify the extension value itself.

Extensions marked "Is-Modifier" are still represented using the extension element , not modifierExtension because the impact of the modifier element is expected to be known by applications that understand the containing extension.

Any modifierExtension defined outside an extension cannot be placed in another extension.

STU Note: Is this a problem? Input from implementers identifying places where modifier extensions might be needed inside other extensions is welcome here .

In XML, these modifier elements are represented using an element named modifierExtension, which has same content as the extension element documented above:

Example: There's no element on MedicationRequest to write an "anti-prescription" - an instruction not to take a medication for a particular period. Classical clinical recording systems do not record this as a prescription - but one particular system does,

and these "anti-prescription" records need to be shared within the institution where this happens as they are an important part of the workflow. Hence, applications are allowed to extend a resource with data like this:

```
<MedicationRequest>
  <modifierExtension url="http://example.org/fhir/StructureDefinition/anti-prescription">
    <valueBoolean value="true"/>
  </modifierExtension>
  <!-- ... other content ... -->
</MedicationRequest>
```

Or in JSON:

```
{
  "resourceType" : "MedicationRequest",
  "modifierExtension" : [{
    "url" : "http://example.org/fhir/StructureDefinition/anti-prescription",
    "valueBoolean" : "true"
  }],
  .. other content ...
}
```

Implementations processing the data in resources SHALL check for modifiers anywhere they could appear, and if a modifier extension is present, SHALL do one of these things:

* understand the impact of the extension when using the data

* refuse to process the data

* carry a warning concerning the data along with any action or output that results from processing the data to inform users that it has not fully understood the source information

Processing the data of a resource typically means copying or filtering data out of a resource for use in another context (display to a human, decision support, exchange in another format where not all information is included, or storing it for this kind of use). Servers and background processes that simply move whole resources around unchanged are not "processing the data of the resource", and therefore these applications are not required to check for unknown modifier extensions.

#1: When an application understands this extension, it means that some developer has provided appropriate instructions for what to do with the data contained in it because of the existence of the modifier extension.

#2: This means that implementations are not inherently required to "support" a modifier extension in any meaningful way - they can achieve this understanding by rejecting instances that contain this extension (a server, for instance, could return a HTTP 422 status code with an OperationOutcome if a client PUTs or POSTs a modifier extension it does not know). Applications might also be able to ignore a modifier extension if they know that it is safe to do so in their own context, though this would not usually be the case.

Note that implementations might be able to be sure, due to their implementation environment (e.g. specific trading partner agreement), that modifier extensions will never occur, and can therefore meet the requirement to check for modifiers at the design stage. However, since integration and deployment options often change, applications SHOULD always check for modifier extensions when processing resources.

#3: One way to warn the user is to download the extension definition from the given URL, and then use the defined display name to present the extension to the user. An error message could look something like this:

Note that the narrative of the resource SHALL contain modifying information, so it is safe to show this to the user as an expression of the resource's content. A warning dialog box could be extended to offer the user the choice to see the original narrative.

Here is the prescription example from above with narrative:

```
<MedicationRequest xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p><b>Note: This prescription is an instruction NOT to take a medication</b></p>
      <!-- snip actual narrative -->
    </div>
  </text>
  <!-- ...data... -->
  <modifierExtension url="http://example.org/fhir/StructureDefinition/anti-prescription">
    <valueBoolean value="true"/>
  </modifierExtension>
  <!-- ...data... -->
</MedicationRequest>
```

An application only needs to concern itself with modifier extensions on elements that it processes. Take, for example, a case where a procedure resource has a modifier extension on one of the performer elements indicating that they did not participate in the procedure. If an application is not using the performer details at all, the fact that one of the performers has a modifier extension is irrelevant and the application is free to ignore it. If the application does process the performers, and it sees the modifier extension, it must act in one of the ways outlined above.

2.5.0.2.1 Summary: Conformance Rules for Modifier Extensions

* A Modifier Extension SHALL only modify the element which contains it and/or that element's children

* It SHALL always be safe to show the narrative to humans; any modifier extension SHALL be represented in the narrative

* Applications SHALL always check for modifier extensions when processing the data from any element that can carry one

* If a Modifier Extension that an application does not understand is present, the application SHALL refuse to process the resource or affected element, or SHALL provide an appropriate warning to its users

2.5.0.2.2 Special Case: Missing data

In some cases, implementers might find that they do not have appropriate data for an element with minimum cardinality = 1. In this case, the element must be present, but unless the resource or a profile on it has made the actual value of the primitive data type mandatory, it is possible to provide an extension that explains why the primitive value is not present:

```
<uri>
  <extension url="http://hl7.org/fhir/StructureDefinition/data-absent-reason">
    <valueCode="unknown"/>
  </extension>
</uri>
```

In this example, instead of a value, a data missing code is provided. Note that it is not required that this particular extension be used. This extension is not a modifier extension, because the primitive data type has no value.

It is not valid to create a fictional piece of data for the primitive value, and then add an extension indicating that the data has been constructed to meet the data rules. This would be both a bad idea, and a modifier extension, which is not allowed on data types.

2.5.0.3 Exchanging Extensions

Extensions are a way of allowing local requirements to be reflected in a resource using a common information based approach so that all systems can confidently process them using the same tools. However, when it comes to processing the information, applications will be constrained in their ability to handle extensions by the degree to which they are informed about them.

While the structured definition of an extension should always be available (see below for details), the mere availability of a definition does not automatically mean that applications know how to handle them correctly - generally, human decisions are required to determine how the data in extensions should be handled, along with the implicit obligations that surround the information.

For this reason, local requirements that manifest as extensions are an obstacle to integration and interoperability. The more the requirements are shared (i.e. regional or national scale), the less impact they will have. The consistent representation, definition and registration of extensions that this specification defines cannot resolve that problem - it only provides a framework within which such local variations can be handled more easily.

When it comes to deploying applications that support local requirements, situations will very likely arise where different applications exchanging information with each other are supporting different sets of extensions. This specification defines some basic rules that are intended to make management of these situations easier, but they cannot resolve them.

* When exchanging resources, systems SHOULD retain unknown extensions when they are capable of doing so (just as they SHOULD retain core elements when they are capable of doing so)

* If a system modifies a resource it SHOULD remove any extensions that it does not understand from the modified element and its descendants, because it cannot know whether the modifications it has made might invalidate the value of the unknown extension

* Systems that drop existing elements are considered to be "processing the resource"

* A system SHALL NOT modify a resource or element that contains "modifier" extensions it doesn't understand

* Applications SHOULD ignore extensions that they do not recognize if they are not "modifier" extensions

* Systems that do not accept unknown extensions SHALL declare so in their CapabilityStatement resource instances

The degree to which a system can retain unknown extensions is a function of the type of system it is: a general purpose FHIR server, or a middleware engine would be expected to retain all extensions, while an application that manages patient registration through a user interface can only retain extensions to the degree that the information in them is part of the set managed by the user. Other applications will fall somewhere between these two extremes.

2.5.0.3.1 Summary: Handling extensions

Use the following rules as a guideline for handling resources:

* When writing extensions, make sure they are defined and published

* When reading, navigating through or searching on elements that can have modifier extensions, check whether there are any modifier extensions present

* When reading elements, read and process the extensions you know and use, and ignore other extensions

* Retain extensions whenever you can

2.5.1 Defining Extensions

URL: https://hl7.org/fhir/STU3/defining-extensions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.5.1 Defining Extensions

All extensions used in resources require a formal published definition which can be used by application developers, or the applications themselves, to help integrate extensions into the healthcare process they support.

Every extension in a resource refers directly to its definition, which is made available as a StructureDefinition. A resource can be profiled to specify where particular extensions are used.

Whenever resources containing extensions are exchanged, the definitions of the extensions SHALL be available to all the parties that share the resources. Each extension contains a URI that references the source of the definitions as a StructureDefinition. The source SHOULD be a literal reference, such as an http: URL that refers to an end-point that responds with the contents of the definitions - preferably a FHIR RESTful server supporting the Resources Profile, or a logical reference (e.g. using a urn:) - for instance, to a national published standard. Extensions may be defined by any project or jurisdiction, up to and including international standards organizations such as HL7 itself.

Before defining a new extension, attempt to reuse existing extensions defined in one of the shared registries described below. Also consider that some concepts may be appropriate to add as part of the core specification.

2.5.1.1 Core Elements

Elements are included as part of FHIR resources and data types principally on the basis of current world-wide usage patterns. Policy is that if a significant majority of systems throughout the world that would use a resource or data type would use an element, then that element will be included as part of the resource/data type. If not, it will be left to an extension. This holds even if the element is very common or even mandatory in one or two specific jurisdictions.

Proposals suggesting a new core element can be raised by anyone. (Free registration is required.) However, given the timelines for new FHIR releases as well as the uncertainties associated with vetting the specification through a ballot process, it may still be necessary to define extensions even for elements that are likely to be supported as part of the core specification in a future release.

2.5.1.2 Context

Extensions are always defined against some particular context - the type of element that they may be used to extend. The following are possible contexts for an extension:

Code | Context type | Context format | Examples
resource | A particular element (including the root) in a single resource | The element path for that element, using the standard dotted notation | Condition Condition.code
datatype | A particular element (including the root) in a particular data type | The data type name for primitive types or the element path for complex data types. These extensions can be used anywhere the data type is used | Address.part.value string
mapping | A particular context in one of the mapped reference models | The name of the reference model followed by the mapping path. The details of the path depend on the named mapping | RIM: Act[moodCode="EVN"]
extension | Another extension | The profile URI of the extension followed by the extension code | http://myextensions.org/someExtension

Note: For type 'resource' and 'datatype', if the context is an element that can have multiple types, then use either the [x] qualified name (e.g. Observation.value[x]) if the extension works on all choice types, or an enumeration of explicitly named elements if not (e.g. Observation.valueQuantity)

In addition, the extension definition might apply additional constraints with regards to particular element values of the target that make its use appropriate. Extensions SHALL only be used on a target for which they are defined.

2.5.1.3 Cardinality

The cardinality constraints asserted by the extension definition itself apply to any contexts where the extension is used.

Minimum Cardinality

If the Extension minimum cardinality is 0, then the extension is optional anywhere it appears. A profile that defines the use of an extension may make the minimum cardinality any number up to the maximum cardinality of the extension itself. Example: Example: Patient birthplace.

If the Extension minimum cardinality is > 1, then the extension must have a minimum cardinality of at least the minimum cardinality in any profile that defines the use of the extension. The minimum cardinality may be any number up to the maximum cardinality of the extension. Even with a minimum cardinality > 0, the extension is only required to be present in instances if the instances explicitly or implicitly conform to a profile that defines the use of the extension. Example: CapabilityStatement Expectation.

Maximum Cardinality

If the Extension maximum cardinality is 1, then the extension is only allowed once on any element on which it appears. A profile that defines the use of an extension can only make the maximum cardinality 1 (or zero if the minimum cardinality is 0, and the profile constrains another profile that allows the extension). Example: Mother's Maiden Name.

If the Extension maximum cardinality is >1, then the extension is allowed up to the specified number of times on any element on which it appears. A profile that defines the use of an extension may make the maximum cardinality any value up to the specified maximum. Example: Patient Disability.

2.5.1.4 Use of ElementDefinition in Extension Definitions

An extension is a wrapper for an identifying url and either a value or other extensions. As such, some of the properties of the extension are defined on the extension itself, while others are defined on the Extension.value. This list provides guidance for the correct usage:

* Extension root element:

* Cardinality

* Short, Definition, Comments

* IsModifier

* MustSupport (is used on invocation of the extension)

* Conditions & Constraints. These SHOULD never be on url/value[x]

* Mappings. these SHALL never be on url/value[x]

* Extension.url

* Cardinality = 1...1 (fixed)

* value = canonical url (fixed)

Extension.value[x]

* Type

* Cardinality for Simple extensions (not nested): 1...1. Use 0..0 if nested. Note that the actual ED cardinality is defined by the root element

* Binding

* MaxLength, DefaultValue, Pattern, Example, MinValue, MaxValue

Note: Extensions are always sliced by their url property. Re-slicing extensions by additional properties is allowed (see Profiling/Slicing).

2.5.1.5 Publishing Extension Definitions

As well as defining the base element structure for resources, HL7 also publishes extensions, including as part of this specification. HL7 publishes such data definitions as extensions rather than as part of the base resource structure to keep the base resource structure simple and concise, and to allow implementers freedom from needing to engage with an entire world's worth of functionality up front. Note that HL7 does not generally define "modifier" extensions - if HL7 publishes an element that modifies the meaning of other elements, it will generally be part of the resource content itself, since everyone has to understand the extension anyway.

Before extensions can be used in instances, their definition SHALL be published. HL7 maintains two extension registries:

* HL7 approved extensions, approved by an appropriate part of the HL7 community following a review process, and which have formal standing

* Provided as a service to the community, where anyone can register an extension

Users are encouraged to register their extensions in the second registry, though this is not required. All that is required is that the extension is published in a context that is available for users of the extension. So, for example, if a particular extension is only used within a single institution, the definition of the extension can be placed on the institution's intranet. However since, by their nature, resources tend to travel well, it's always better to use the HL7 or other publicly accessible extension registries.

The HL7 FHIR registry can be found at http://hl7.org/fhir/registry .

HL7 extension definitions may be balloted alongside resource content as part of the FHIR specification or may be published as part of separate specifications. When HL7 publishes extension definitions as part of the FHIR specification, these extensions SHALL be used for this data whenever the data is represented in instances. Applications SHOULD use other HL7-defined extensions published to represent equivalent data in the interest of maximum interoperability.

To minimize complexity for implementers, HL7 will not elevate widely adopted extensions (defined by HL7 or other organizations) to be content defined in a core resource in future versions of the resource unless there is widespread endorsement of such a migration from the implementer community. This policy ensures that widespread adoption of an extension does not result in a forced migration to a core element. Extensions labelled as draft may be moved in either direction, but after extensions are finalised as normative they won't be moved.

In some cases, an HL7 work group or other body may publish a profile whose sole purpose is to define extensions expected to be needed by implementers in a particular context; e.g. extensions needed to map a particular set of HL7 v2 segments or a HL7 v3 model.

Implementations are encouraged to share their extensions with HL7 and register them with the HL7 extension registry. The domain committees will work to elevate the extensions into HL7 published extensions or, if adopted by a broad enough portion of the implementer community, into the base resource structure itself.

To avoid interoperability issues, extensions SHALL NOT change their definition once published. (Small clarifications to descriptions that do not affect interoperability are permitted.) Rather than modifying an existing extension, a new extension should be introduced. Revisions to an extension may extend the set of contexts in which the extension apply but may not remove or constrain any context previously listed

2.5.2 Examples

URL: https://hl7.org/fhir/STU3/extensibility-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.5.2 Extensibility Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

In order to use an extension, there is a three-step process:

* Define the extension

* Register the extension

* Use it in the instance

This page contains examples of how this process executes.

2.5.2.1 Patient Consent for Record Sharing

The basic patient resource contains no information relating to patient consent, and/or the policy under which the patient consents to their registration details. A social web provider of personal healthcare record (PHR) services might be obliged to keep track of the particular policy under which a patient has created their relationship with the PHR provider, and share this with their participants via their FHIR API. If they wish, they can extend the patient resource to represent the patient's participation agreement. Note that other approaches to this problem are possible, but it suffices to demonstrate the extension process.

For the purposes of this example, we assume that the patient agrees to a participation policy as part of their sign up, and that as the provider has to change their policy, they ask patients to agree to new participation details. Each participation agreement has a URI by which it is identified, and the patient resource will carry this URI for each policy agreement that the patient has agreed to.

2.5.2.1.1 Define the Extension

For each extension, the first thing to do is to fill out the definitional properties of the extension:

Code | "participation-agreement"
Context | This extension is used in the patient resource
Short Defn | Agreed agreement/policy
Definition | A URI that identifies a participation agreement/policy to which the patient has agreed
Comment | URI is a literal reference to agreement text (html)
Cardinality | 1..* (patient cannot participate without at least one agreement)
Type | uri
XPaths | no Invariants
is Modifier | No (The participation agreements do not affect that interpretation of the elements of the patient, though they will likely influence how the system interacts with the patient)
Binding | (No binding - not a coded value)

2.5.2.1.2 Write the Definition of the Extension

From this table, we can build a formal extension definition. In this case, it looks like this:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <url value="http://example.org/fhir/StructureDefinition/participation-agreement" />
  <name value="Example Extension Definition" />
  <!-- snip other metadata -->
  <kind value="datatype" />
  <contextType value="resource" />
  <context value="Patient" />
  <type value="Extension" />
  <baseDefinition value="http://hl7.org.fhir/StructureDefinition/Extension" />
  <derivation value="constraint" />

  <differential>
    <element>
      <path value="Extension"/>
      <short value="Agreed agreement/policy" />
      <definition value="A URI that identifies a participation agreement/policy
        to which the patient has agreed" />
      <comment value="URI is a literal reference to agreement text (html).
        Systems SHALL conform to the policies as indicated.
        For further information, see the partnership agreement..." />
      <mustSupport value="true" />
      <isModifier value="false" />
    </element>
    <element>
      <path value="Extension.valueUri"/>
      <short value="The URI value" />
      <min value="1" />
      <max value="*" />
      <type>
         <code value="uri" />
      </type>
      <mustSupport value="true" />
    </element>
  </differential>
</StructureDefinition>
```

Note that usually one would build the actual profile using a tool. This example was built by hand for this example.

2.5.2.1.3 Register the Extension

This means the profile shown above that defines the extension is placed on the web somewhere. By preference, it will be hosted at a FHIR Profile endpoint, and the best location of all is the HL7 Profile registry (yet to be implemented).

For this example, we assume that it has been uploaded to the PHR provider's own website at http://example.org/phr/documents/fhir/extensions.

2.5.2.1.4 Use it in the instance

To use the extension in an instance, the extension is placed in the root of the resource. Note that the URL of the extension refers to the registered location, with the id of the extension as a fragment identifier.

```
<Patient xmlns="http://hl7.org/fhir">
  <extension url="http://example.org/phr/documents/fhir/extensions/participation-agreement" >
    <valueUri value="http://example.org/phr/documents/patient/general/v1" />
  </extension>
  <!-- ... -->
</Patient>
```

2.5.2.1.5 Using it in a Patient Profile

The profile definition above simply defines the extension "participation-agreement", and says that it is used with patient. But the profile above doesn't say that the server actually uses it. For the PHR provider to indicate that all Patient resources will use this resource, a StructureDefinition on the patient resource is used:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <id value="patient-profile" />
  <!-- snip other metadata, narrative -->
  <differential>

    <!-- first, the patient root element
      - can be copy/paste from the base patient profile -->
    <element>
      <path value="Patient"/>
      <!-- snip definition -->
    </element>

    <!-- now, the general definition for extensions
      - can be copied/pasted from the base patient profile,
      with changes for slicing  -->
    <element>
      <path value="Patient.extension"/>
      <!-- we're going to slice the extension element, and
        one of the extensions is one we have defined -->
      <slicing>
        <!-- extension is always sliced on url -->
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <!-- we don't care what the order of any extensions is -->
        <ordered value="false"/>
        <!-- Other extensions are allowed in addition to this one -->
        <rules value="open"/>
      </slicing>
      <!-- snip definition -->
    </element>

    <!-- now, the slice that contains our extension -->
    <element>
      <path value="Patient.extension"/>
      <sliceName value="base-extension"/>
      <!-- clone information from the extension definition.
        duplicative, but this duplication makes it simpler overall -->
      <short value="Agreed agreement/policy"/>
      <definition value="A URI that identifies a participation agreement/policy
      to which the patient has agreed"/>
      <!--  min has to be 1, since the extension itself has min = 1 -->
      <min value="1"/>
      <max value="*"/>
      <type>
        <!-- obviously it has to be an extension -->
        <code value="Extension"/>
        <!-- and here is the link to the extension definition:
          this extension has to conform to the rules laid down in its definition -->
        <profile value="http://example.org/phr/documents/fhir/extensions#participation-agreement"/>
      </type>
      <isModifier value="false"/>
    </element>

  <!-- snip the rest of the profile -->

  </differential>
</StructureDefinition>
```

Note - this step is optional.

2.5.2.2 Patient Name Parts

ISO 21090 (Healthcare Data Types) defines a concept called a "name part qualifier" that contains extra information about how a particular name part should be used or interpreted. In practice, this field is used rarely, except in particular cultural contexts, where certain part qualifiers are used as a matter of practice. Following the FHIR design policy, such a field is not included in the overall definition of the core name data type, instead is it added as an extension.

In practice, for cases such as these in ISO 21090, HL7 provides common extensions, and these are defined at [location still to be finalized].

2.5.2.2.1 Define the Extension

For each extension, the first thing to do is to fill out the definitional properties of the extension:

Code | "name-qualifier"
Context | This extension can be used anywhere a HumanName.part appears
Short Defn | (one of the codes) LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN
Definition | A set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type
Comment | Used to indicate additional information about the name part and how it should be used
Cardinality | 0..* (this is always optional, but more than one can be used if required)
Type | code
XPaths | N/A
Is Modifier | No (Qualifiers do not change the fact that the part is a given or family name)
RIM Mapping | ENXP.qualifier
v2 Mapping | N/A
Binding | Bound to a subset of the codes specified for EntityNamePartQualifierR2 in ISO 21090

Not all the codes of the EntityNamePartQualifierR2 are required in this context, because prefix and suffix are explicitly part of the name types. Rather than simply refer to the OID for EntityNamePartQualifierR2 (2.16.840.1.113883.5.1122), in this case we enumerate the available codes, and set the type of the extension to code. The type of "code" is only allowed if the profile itself defines the codes that can be used. Here is a table of the codes (see the EntityNamePartQualifierR2 reference for the full definitions):

LS | Legal status | For organizations, a suffix...
AC | Academic | Indicates that a prefix like "D...
NB | Nobility | In Europe and Asia, there are s...
PR | Professional | Primarily in the British Im...
HON | Honorific | An honorific such as 'The Rig...
BR | Birth | A name that a person was given at ...
AD | Acquired | A name part a person acquired. ...
SP | Spouse | The name assumed from the partner...
MID | Middle Name | Indicates that the name par...
CL | Call me | Call me is used to indicate which...
IN | Initial | Indicates that a name part is ju...

This is all then represented formally in a profile. Such profiles do not need to include resource constraint statements; instead, they include just extension declarations and their associated bindings. In this case, the definition looks like this:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <url value="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier" />
  <name value="Iso21090NameQualifier" />
  <title value="iso-21090 Name Qualifier" />
  <!-- snip other metadata, including definition of RIM Mapping -->
  <kind value="datatype" />
  <contextType value="datatype" />
  <context value="HumanName.given" />
  <context value="HumanName.prefix" />
  <context value="HumanName.family" />
  <context value="HumanName.suffix" />
  <type value="Extension" />
  <baseDefinition value="http://hl7.org.fhir/StructureDefinition/Extension" />
  <derivation value="constraint" />
  <differential>
    <element>
      <path value="Extension"/>
      <short value="LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN" />
      <definition value="A set of codes each of which specifies a certain subcategory
          of the name part in addition to the main name part type" />
      <comment value="Used to indicate additional information about the
             name part and how it should be used" />
      <mustSupport value="false" />
      <isModifier value="false" />
    </element>
    <element>
      <path value="Extension.value"/>
      <short value="LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN" />
      <min value="0" />
      <max value="*" />
      <type>
         <code value="code" />
      </type>
      <binding>
        <strength value="required" />
        <description value="A set of codes each of which specifies a certain subcategory
            of the name part in addition to the main name part type" />
```

```
        <valueSetReference>
          <reference value="ValueSet/name-part-qualifier" />
        </valueSetReference>
      </binding>
      <mapping>
         <identity value="RIM" />
         <map value="ENXP.qualifier" />
      </mapping>
    </element>
  </differential>
</StructureDefinition>
```

Note that usually one would build the actual profile using a tool. This example was built from a spreadsheet definition by the FHIR build tooling.

2.5.2.2.2 Register the Extension

For this example, it is registered at http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier . This is the URL that will appear in the definition element when the extension is used.

2.5.2.2.3 Use it in the instance

To use the extension in an instance, the extension is nested within the attribute that is extended. Note that the URL of the extension refers to the registered location, with the id of the extension as a fragment identifier.

```
<name>
  <use value="official" />
  <given value="Östlund">
     <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier" >
        <valueCode value="MID" />
     </extension>
  </given>
</name>
```

This particular example is a Scandinavian mellannamn. See Datatypes examples for additional examples.

2.5.2.3 Complex Extension: Patient Clinical Trial

Defining complex extensions is a little different. They have the same metadata - context, etc. - but differ in the internal structure. As an example, consider enrolling a patient in a clinical trial. There are 3 data items to collect:

Code | Name | Cardinality | Type | Content
NCT | Clinical Trial number | 1..1 | string | The format for the US ClinicalTrials.gov registry number is "NCT" followed by an 8-digit number, e.g.: NCT00000419
period | trialPeriod | 0..1 | Period | The start and end times of the participation of this patient in the clinical trial
reason | reason enrolled | 0..1 | CodableConcept | Indication or reason that the patient is part of this trial

Like simple extensions, the first thing to do is to assign a URI to the extension. Since this one is part of the FHIR publication, it is assigned a URI where it will be published: http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial.

Internally, in the extension, the cardinality of the value[x] element is set to 0 since it will not (and cannot) be used. The Extension.extension element is sliced by URL, and 3 slices are defined, each with a fixed relative URI which is the code from the table above. Tthe relevant parts of the definition of this extension are as follows:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- metadata - setting up the base definition -->
  <url value="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial"/>
  <name value="ClinicalTrialParticipation"/>
  <title value="The patient's participation in clinical trials"/>
  <contextType value="resource"/>
  <context value="Patient"/>
  <type value="Extension" />
  <baseDefinition value="http://hl7.org.fhir/StructureDefinition/Extension" />
  <derivation value="constraint" />
  <snapshot>
    <element>
      <path value="Extension"/>
      <!-- etc.-->
    </element>

    <!-- set up the slicing -->
    <element>
      <path value="Extension.extension"/>
      <slicing>
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <ordered value="true"/>
        <rules value="openAtEnd"/>
      </slicing>
    </element>

    <!-- first slice, NCT -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="NCT"/>
      <short value="National Clinical Trial number"/>
      <min value="1"/>
```

```
      <max value="1"/>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/> <!-- not allowed to be used -->
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <max value="1"/>
      <fixedUri value="NCT"/>
    </element>
    <element>
      <path value="Extension.extension.valueString"/>
      <min value="1"/>
      <max value="1"/>
      <type>
        <code value="string"/>
      </type>
    </element>

    <!-- second slice, period -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="period"/>
      <short value="The period of participation in the clinical trial"/>
      <min value="0"/>
      <max value="1"/>
    </element>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/>
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <fixedUri value="period"/>
    </element>
    <element>
      <path value="Extension.extension.valuePeriod"/>
      <type>
        <code value="Period"/>
      </type>
    </element>

    <!-- third slice, reason -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="reason"/>
      <short value="The reason for participation in the clinical trial"/>
      <min value="0"/>
      <max value="1"/>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/>
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <fixedUri value="reason"/>
    </element>
    <element>
      <path value="Extension.extension.valueCodeableConcept"/>
      <type>
        <code value="CodeableConcept"/>
      </type>
    </element>

    <!-- last (for order reasons): the fixed URI -->
    <element>
      <path value="Extension.url"/>
      <fixedUri value="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial"/>
    </element>
    <!-- and no value in the root -->
    <element>
      <path value="Extension.value[x]"/>
      <min value="0"/>
      <max value="0"/>
    </element>
  </snapshot>
</StructureDefinition>
```

2.5.2.3.1 Use it in the instance

With complex extensions, only the first URL is an absolute URL:

```
<Patient xmlns="http://hl7.org/fhir">
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial" >
    <extension url="NCT" >
      <valueString value="NCT00000419" />
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="200140105" />
        <end value="20120105" />
      </valuePeriod>
    </extension>
    <extension url="reason" >
      <valueCodeableConcept>
        <text value="NCT00000419" />
      </valueCodeableConcept>
    </extension>
  </extension>
  <!-- ... -->
</Patient>
```

2.5.2.3.2 Using it in a Patient Profile

This is the same as for a simple extension:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <id value="patient-profile" />
  <!-- snip other metadata, narrative -->
  <differential>

    <!-- first, the patient root element
      - can be copy/paste from the base patient profile -->
    <element>
      <path value="Patient"/>
      <!-- snip definition -->
    </element>

    <!-- now, the general definition for extensions
      - can be copy/paste from the base patient profile,
      with changes for slicing  -->
    <element>
      <path value="Patient.extension"/>
      <!-- we're going to slice the extension element, and
        one of the extensions is one we have defined -->
      <slicing>
        <!-- extension is always sliced on url -->
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <!-- we don't care what the order of any extensions is -->
        <ordered value="false"/>
        <!-- Other extensions are allowed in addition to this one -->
        <rules value="open"/>
      </slicing>
      <!-- snip definition -->
    </element>

    <!-- now, the slice that contains our extension -->
    <element>
      <max value="1"/>
      <type>
        <!-- obviously it has to be an extension -->
        <code value="Extension"/>
        <!-- and here is the link to the extension definition:
          this extension has to conform to the rules laid down in its definition -->
        <profile value="http://hl7.org/fhir/StructureDefinition/patient-clinicalTrial"/>
      </type>
    </element>

  <!-- snip the rest of the profile -->

  </differential>
</StructureDefinition>
```

On This Page:

Patient Consent

Name Parts

Complex Extension

2.5.3 Extension - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/extensibility-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.5.3 Extension - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

Extension
Definition | Optional Extension Element - found in all resources.
Control | 1..1
Requirements | The ability to add extensions in a structured way is what keeps FHIR resources simple.
Invariants | Defined on this element ext-1: Must have either extensions or value[x], not both (expression : extension.exists() != value.exists(), xpath: exists(f:extension)!=exists(f:*[starts-with(local-name(.), 'value')]))
Extension.url
Definition | Source of the definition for the extension code - a logical name or a URL.
Control | 1..1
Type | uri
Comments | The definition may point directly to a computable or human-readable definition of the extensibility codes, or it may be a logical URI as declared in some other specification. The definition SHALL be a URI for the Structure Definition defining the extension.
Extension.value[x]
Definition | Value of extension - may be a resource or one of a constrained set of the data types (see Extensibility in the spec for list).
Control | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]

2.6 Resource Formats

URL: https://hl7.org/fhir/STU3/formats.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

This page documents how the content of the resources are described. In actual exchange, resources can be represented in the following formats: XML, JSON and Turtle. Other representations are allowed, but are not described by this specification.

2.6.0.1 Resource Definition

The resources are described in several different ways:

* a hierarchical table that presents a logical view of the content

* a UML diagram that summarizes the content graphically

* a pseudo-XML syntax that provides a visual sense of what the end resource instances will look like in XML

* a pseudo-JSON syntax that provides a visual sense of what the end resource instances will look like in JSON

* a pseudo-Turtle syntax that provides a visual sense of what the end resource instances will look like in Turtle

In addition to this descriptive syntax, other definitional forms are available, including W3C schema and Schematron, and the StructureDefinition syntax defined internally.

2.6.0.1.1 Logical table

The Logical View shows the resources as a tree structure with the following columns:

Column | Content
Name | The name of the element in the resource (manifests as XML element name, or JSON property name. Some names finish with [x] - the meaning of this is discussed below. In addition, this column contains an icon that denotes the underlying type of the content. The icons are described below
Flags | A set of information about the element that impacts how implementers handle them. The flags are described below
Card. | Cardinality: the lower and upper bounds on how many times this element is allowed to appear in the resource
Type | The type of the element (hyperlinked to the definition of the type). Note that the type of the element has one of two meanings, depending on whether the element has defined children. If the element has children, then the element has an anonymous type that specializes the given type. If the element has no children, then the element has properties and children as specified by the nominated type
Description & Constraints | A description of the element, and details about constraints that are applied to it. Particularly, for coded elements, information about which codes can be used

Here's an example:

Name | Flags | Card. | Type | Description & Constraints
Resource Name | | | Base Type | Definition
nameA | ? | 1..1 | type | description of content
nameB[x] | ?! ? | 0..1 | | | description SHALL at least have a value
nameBType1 | | 0..1 | type1
nameBType2 | I | 0..1 | type2
nameC | | 1..* | Element | Definition
nameD | | 1..1 | type | Relevant Records

Key to Type Icons and Flags

* : The base element for a resource (see Resources)

* : An element that is part of the resource and has elements within it defined in the same resource or profile

* : An element which can have one of several different types (see below)

* : An element of a data type which describes an element that has a value attribute/property

* : An element of a data type which describes an element that has other elements

* : An element that contains a reference to another resource (see references)

* : This element has the same content as another element defined within this resource or profile

* : Introduction of a set of slices (see Slicing)

* : An extension (see Extensibility)

* : A complex extension - one with nested extensions (see Extensibility)

* : An extension that has a value and no nested extensions (see Extensibility)

* : A complex modifier extension - one with nested extensions (see Extensibility)

* : A modifier extension that has a value and no nested extensions (see Extensibility)

* : The root of a logical profile

* ?!: This element is a modifying element - see Modifier Elements

* S: This element is an element that must be supported - see Must-Support Elements

* ?: This element is an element that is part of the summary set - see Summary Searches

* I: This element defines or is affected by constraints - see Constraints

* NE: This element cannot have extensions (some infrastructural elements only)

Notes:

* Resource and Element names are case-sensitive (though duplicates that differ only in case are never defined)

* Any elements that have a primitive type will have a value attribute/property to contain the actual value of the element

* This value attribute/property can never be empty. Either it is absent, or it is present with at least one character of non-whitespace content

* Elements are assigned a cardinality that specifies how many times the element may or must appear.

* Unless elements have children defined directly (as nameC does above) they are assigned one or more types (see the data types). All the type names are hyperlinked to the source definition

* Element reuse: Some data types that have children have the same set of children as some other element defined in the resource. In that case, the type of that element has a "see [name]" where [name] is the name of the element that has the defined children

* Each element name is also a hyperlink to the formal definition of the element in the data dictionary that underlies the exchange formats.

* Any of the elements may have an id attribute to serve as the target of an internal reference. The id attribute is not shown in this format. Extensions are not always shown, but may appear except where the flag NE appears

* FHIR elements can never be empty. If an element is present in the resource, it SHALL have either a value, child elements as defined for its type, or 1 or more extensions

* Infrastructural elements that are common to all resources are not shown in the logical representation. These are described in the common base classes Resource, and DomainResource

2.6.0.1.2 Choice of Data Types

A few elements have a choice of more than one data type for their content. All such elements have a name that takes the form nnn[x]. The "nnn" part of the name is constant, and the "[x]" is replaced with the title-cased name of the type that is actually used. The table view shows each of these names explicitly.

Elements that have a choice of data type cannot repeat - they must have a maximum cardinality of 1. When constructing an instance of an element with a choice of types, the authoring system must create a single element with a data type chosen from among the list of permitted data types.

Note: In object-orientated based implementations, this is naturally represented as a polymorphic property. However this is not necessary and the correct implementation varies according to the particular features of the language. In XML schema, these become an xs:choice of element.

2.6.0.1.3 UML

The UML diagrams represent the same content as a series of classes that represent the elements of a resource.

The elements and the data types are hyperlinks to the formal definitions of the parts. The UML diagrams also show the vocabulary bindings. These are hyperlinks to the value set details.

Where an element can have a choice of data types, or is a Reference these are represented by showing the common type (Reference or Type), and then showing the applicable data type names or resource types in a stereotype, separated by the | character. Type is not formally otherwise defined by this specification, but is a super type of all the data types.

The actual order of the elements in XML cannot be determined from the diagram, nor whether a UML property becomes an element or an attribute in the XML representation.

Bindings to value sets are indicated by a stereotype on the element. The stereotype has 2 parts: the value set name, and a symbol that denotes the strength of the binding:

* ??: Example Binding

* ?: Preferred Binding

* +: Extensible Binding

* !: Required Binding

2.6.0.1.4 Serialization Format Representations

This specification defines the following ways to represent resources when they are exchanged:

* XML

* JSON

* RDF (Turtle)

Clients and servers can choose what syntax(s) to implement. In the interests of interoperability, servers SHOULD support all formats (though support for RDF should best wait until it is more fully developed). Systems SHALL declare which format(s) they support in their Capability Statement. If a server receives a request for its Capability Statement in a format it does not otherwise support, it SHALL return a 415 Unsupported Media Type.

2.6.1 XML Representation of Resources

URL: https://hl7.org/fhir/STU3/xml.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.6.1 XML Representation of Resources

Implementable Technology Specifications Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The XML representation for a resource is described using this format:

```
<name xmlns="http://hl7.org/fhir" (attrA="value")>
  <nameA><!--  1..1 type description of content  --><nameA>
  <nameB[x]><!-- 0..1 type1|type1 description  --></nameB>
  <nameC> <!--  1..* -->
    <nameD><!-- 1..1 type>Relevant elements  --></nameD>
  </nameC>
<name>
```

Using this format:

* To build a valid XML instance of a resource, simply replace the contents of the elements and attributes with valid content as described by the cardinality, type rules and content description found in the comment in each element

* Resource and Element names are case-sensitive (though duplicates that differ only in case are never defined)

* Elements must always appear in the order documented

* When an element is allowed to repeat, the elements are ordered, and the technical infrastructure needs to be able to access the items in the right order (see also Cardinality Rules for a further description of elements with cardinality > 1)

* A few properties are represented as attributes: values of primitive types in a value attribute, extension URLs in the url attribute on an extension, and the id property on elements (but not on resources, where the resource id is an element)

* Any of the XML elements may have an id attribute to serve as the target of an internal reference. The id attribute is not shown in this format

* FHIR elements are always in the namespace http://hl7.org/fhir . This is usually specified as the default namespace on the root element. The only other namespace that occurs in FHIR resources is the XHTML namespace (XHTML is found in most resources)

* Infrastructural elements that are common to all resources are not shown in the xml representation. These must appear prior to any other defined child elements in the following order:

* First, the elements from the base resource, in order

* Second, the elements from the domain resource, in order

* FHIR elements are never empty. If an element is present in the resource, it SHALL have either a value attribute, child elements as defined for its type, or 1 or more extensions

* Attributes canno be empty. Either they are absent, or they are present with at least one character of non-whitespace content

* Implementers SHOULD trim leading and trailing whitespace before writing and SHOULD trim leading and trailing whitespace when reading attribute values

* The lock icon () denotes that an element defines or is affected by additional rules that control its presence and/or content

* XML comments, processing instructions and formatting are not part of the contents of a resource

* There SHALL be no DTD references in FHIR resources (because of the XXE security exploit)

* XML resources SHALL be exchanged using UTF-8 encoding. Specifying the character encoding using a <?xml encoding="UTF-8" ?> processing instruction is optional but recommended

* Other processing instructions SHOULD not be included, and SHALL NOT be required to properly understand and/or present the data or narrative of the resource. Applications MAY preserve processing instructions when handling resources, but are not required to do so

* The MIME-type for this format is application/fhir+xml.

2.6.1.1 XML Schema and Schematron

This specification provides schema definitions for all the content models it describes.

The base schema is called "fhir-base.xsd" and defines all the datatypes and base infrastructure types. In addition, there is a schema for each resource and a common schema fhir-all.xsd that includes all the resource schemas. For schema processors that do not like circular includes, there is a single schema that contains everything.

In addition to the w3c schema files, this specification also provides Schematron files that enforce the various constraints defined for the datatypes and resources. These are packaged as files for each resource.

XML that is exchanged SHALL be valid against the w3c schema and Schematron, though being valid against the schema and Schematron is not sufficient to be a conformant instance: this specification makes several rules that cannot be checked by either mechanism. Operational systems may choose to use schema tools to check validation, but are not required to do so. Exchanged content SHALL NOT specify the schema or even contain the schema instance namespace in the resource itself.

Given the way extensions work, applications reading XML resources will never encounter unknown elements. However, once an application starts trading with other applications that conform to later versions of this specification, unknown XML elements may be encountered. Applications MAY choose to ignore unknown elements to foster forwards compatibility in this regard, but may also choose not to - which would be the normal behavior for schema generated applications. Applications declare their unknown elements behavior using CapabilityStatement.acceptUnknown.

2.6.1.2 Code Generation Schema

In addition to the validation schema, this specification provides a set of schemas suitable for code generation. These schemas describe the same XML syntax, but apply less validation to create schemas that work with more code generation tooling.

Specifically, these schemas are generated without any xsd:choice elements, for code generators that don't deal with choices well. Implementers that use these schemas will need to enforce the correct usage of the choice elements without schema support.

Implementers making use of schema driven code generation tooling need to consider how to handle the decimal data type. The decimal data type is defined to be precision aware - that is, that implementers need to preserve the difference between "2.0" and "2.00" - this is ubiquitously considered important in handling observed data in healthcare. Both schemas map this data type to xsd:decimal, but the base W3C schema decimal type is specified not to be precision aware. Schema driven implementations vary as to how precision is handled. Implementers will need to determine how their generated code handles decimals and consider changing the type for decimal in the schema from xsd:decimal to xsd:string. Specifically, implementers may wish to change

```
<xs:simpleType name="decimal-primitive">
  <xs:restriction base="xs:decimal">
    <xs:pattern value="-?([0]|([1-9][0-9]*))(\.[0-9]+)?"/>
  </xs:restriction>
</xs:simpleType>
```

to this:

```
<xs:simpleType name="decimal-primitive">
  <xs:restriction base="xs:string">
    <xs:pattern value="-?([0]|([1-9][0-9]*))(\.[0-9]+)?"/>
  </xs:restriction>
</xs:simpleType>
```

Alternatively, if supported, implementers may wish to use the precisionDecimal from the XSD 1.1 framework.

Note that most code generation frameworks ignore the pattern restriction.

2.6.1.3 Canonical XML

Resources and/or Bundles may be digitally signed (see Bundle and Provenance).

This specification defines the following method for canonicalizing FHIR resources, when represented as xml:

* No whitespace other than single spaces in attribute values and in the XHTML in the Narrative

* Use default namespaces for the FHIR and XHTML namespaces

* Omit all elements that have a default value, if a default value is defined

* Omit all comments

* Always use the unicode character representation for any XML entities (e.g. ' instead of ")

* Include the XML processing instruction <?xml version="1.0" encoding="UTF-8"?>

* Using the XML canonical method Canonical XML 1.1 (http://www.w3.org/2006/12/xml-c14n11)

This canonicalization method is identified by the URL http://hl7.org/fhir/canonicalization/xml. The following additional canonicalization URLS are also defined:

http://hl7.org/fhir/canonicalization/xml#data | The narrative (Resource.text) is omitted prior to signing (note the deletion is at Resource.text, not Resource.text.div)

http://hl7.org/fhir/canonicalization/xml#static | In addition to narrative (Resource.text), the Resource.meta element is removed. This makes the signature robust as the content is moved from server to server, or workflow and access tags are added or removed

http://hl7.org/fhir/canonicalization/xml#narrative | The method only retains the Resource.id and Narrative (Resource.text

http://hl7.org/fhir/canonicalization/xml#document | The signs everything in a Bundle, except for the Bundle.id and Bundle.metadata on the root Bundle (allows for a document to copied from server to server)

These canonicalization methods allow systems the flexibility to sign the various portions of the resource that matter for the workflow the signature serves.

Note: One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

2.6.2 JSON Representation of Resources

URL: https://hl7.org/fhir/STU3/json.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.6.2 JSON Representation of Resources

Implementable Technology Specifications Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The JSON representation for a resource is described using this format:

```
{
  "resourceType" : "[Resource Type]",
  // from Source: element #1
  "property1" : "<[primitive]>", // short description
  "property2" : { [Data Type] }, // short description
  "property3" : { // Short Description
    "propertyA" : { CodeableConcept }, // Short Description (Example)
  },
  "property4" : [{ // Short Description
    "propertyB" : { Reference(ResourceType) } // R!  Short Description
  }]
}
```

Using this format:

* To build a valid JSON instance of a resource, replace the contents of the property values with valid content as described by the type rules and content description found in the property value for each element

* In this example:

* property1 has a primitive data type; the value of the property will be as described for the stated type

* property2 has a complex data type; the value of the property is an object that has the content as described for the stated type

* property3 is an object property that contains additional properties (e.g. propertyA; the allowable properties are listed (but also include extensions as appropriate)

* property4 is an array property that contains items which are objects themselves. The items may have any of the types already encountered in points 1-3

* propertyA is an example of an object property that has a binding to a value set - the Short description is a link to the value set. In addition, the binding strength is shown

* propertyB is an example of an object property that has a reference to a particular kind of resource

* Property names are case-sensitive (though duplicates that differ only in case are never defined)

* Properties can appear in any order

* XHTML is represented as an escaped string

* Objects are never empty. If an element is present in the resource, it SHALL have properties as defined for its type, or 1 or more extensions

* String property values can never be empty. Either the property is absent, or it is present with at least one character of content

* The R! denotes that an element is mandatory - it must be present (or in an array, at least one item must be present)

* In this format, // is used for comments but these can't be in the JSON instances

* The character encoding is always UTF-8

* The MIME-type for this format is application/fhir+json.

Given the way extensions work, applications reading JSON resources will never encounter unknown properties. However, once an application starts trading with other applications that conform to later versions of this specification, unknown properties may be encountered. Applications MAY choose to ignore unknown properties in order to foster forwards

compatibility in this regard, but may also choose not to. Applications declare their behavior with regard to unknown elements using CapabilityStatement.acceptUnknown.

2.6.2.1 Comparison with XML

The JSON format is similar to the XML format:

* The names for the JSON object members are the same as the names of the elements and attributes in XML, including elements that may repeat. Property names are case sensitive

* Just as in XML, JSON objects and arrays are never empty, and properties never have null values (except for a special case documented below). Omit a property if it is empty

* JSON whitespace is not part of the contents of a resource. Applications MAY preserve the whitespace when handling resources, but are not required to do so. Note that digital signatures may depend on the whitespace

There are differences to XML:

* There are no namespaces in the JSON representation

* The type of the resource is represented differently in JSON - instead of being the name of the base object (there is none in JSON), it is carried as the property resourceType

* The order of properties of an object is not significant in the JSON representation, though order within an array SHALL be maintained

* JSON does not have a notion of attributes versus elements, so attributes (e.g. id, value) are handled differently (see below)

* JSON has the array notation, which is used to represent repeating elements. Note that arrays are used when the item might repeat, even if it does not repeat in a specific instance

* The XHTML element in the Narrative datatype is represented as a single escaped string of XHTML. This is to avoid problems in JSON with mixed content, etc. The XHTML SHALL still conform to the rules described for the Narrative

The JSON format for the resources follows the standard XML format closely to make interconversion easy, and so that XPath queries can easily be mapped to query the JSON structures. However, the differences - particularly the repeating element one, which cannot be avoided - mean that generic XML --> JSON converters are not able to perform correctly. The reference platforms provide XML <--> JSON conversion functionality that accommodates these FHIR-specific characteristics.

2.6.2.2 JSON Representation for repeating elements

An element that has a maximum cardinality of >1 (e.g. x..* in the definitions) may occur more than once in the instance. In XML, this is simply done by repeating the XML element multiple times. In JSON, this is done by using an array type. Note that:

* The name of the array is singular - the same as the XML element

* An item that may repeat is represented as an array even in the case that it doesn't repeat so that the process of parsing the resource is the same either way

```
<coding>
  <system value="http://snomed.info/sct"/>
  <code value="104934005"/>
</coding/>
<coding>
  <system value="http://loinc.org"/>
  <code value="2947-0"/>
</coding/>
```

is represented in JSON like this:

```
"coding": [
  {
    "system" : "http://snomed.info/sct",
    "code" : "104934005"
  },
  {
    "system" : "http://loinc.org",
    "code" : "2947-0"
  }
]
```

2.6.2.3 JSON representation of primitive elements

FHIR elements with primitive data types are represented in two parts:

* A JSON property with the name of the element, which has a JSON type of number, boolean, or string

* a JSON property with _ prepended to the name of the element, which, if present, contains the value's id and/or extensions

The FHIR types integer and decimal are represented as a JSON number, the FHIR type boolean as a JSON boolean, and all other types are represented as a JSON string which has the same content as that specified for the relevant data type. Whitespace is always significant (i.e. no leading and trailing spaces for non-strings).

```
<code value="abc"/> <!-- code -->
<date value="1972-11-30"/> <!-- dateTime -->
<deceased value="false" /> <!-- boolean -->
<count value="23" />  <!-- integer -->
```

is represented in JSON as

```
 "code" : "abc",
 "date" : "1972-11-30",
 "deceased" : false,
 "count" : 23
```

When using a JavaScript JSON.parse() implementation, note that JavaScript natively supports only one numeric datatype, which is a floating point number. This can cause loss of precision for FHIR numbers. In particular, trailing 0s after a decimal point will be lost e.g. 2.00 will be converted to 2. The FHIR decimal data type is defined such that precision, including trailing zeros, is preserved for presentation purposes, and this is widely regard as critical for correct presentation of clinical measurements. Implementations should consider using a custom parser and big number library (e.g. https://github.com/jtobey/javascript-bignum) to meet these requirements.

If the value has an id attribute, or extensions, then this is represented as follows:

```
<birthDate id="314159" value="1970-03-30" >
  <extension url="http://example.org/fhir/StructureDefinition/text">
    <valueString value="Easter 1970"/>
  </extension>
</birthDate>
```

is represented in JSON as:

```
"birthDate": "1970-03-30",
"_birthDate": {
  "id": "314159",
  "extension" : [ {
     "url" : "http://example.org/fhir/StructureDefinition/text",
     "valueString" : "Easter 1970"
  }]
}
```

Note: If the primitive has an id attribute or extension, but no value, only the property with the _ is rendered.

In the case where the primitive element may repeat, it is represented in two arrays. JSON null values are used to fill out both arrays so that the id and/or extension are aligned with the matching value in the first array, as demonstrated in this example:

```
<code value="au"/>
<code value="nz">
  <extension url="http://hl7.org/fhir/StructureDefinition/display">
    <valueString value="New Zealand a.k.a Kiwiland"/>
  </extension>
</code>
```

is represented in JSON as:

```
"code": [ "au", "nz" ],
"_code": [
  null,
  {
    "extension" : [ {
       "url" : "http://hl7.org/fhir/StructureDefinition/display",
       "valueString" : "New Zealand a.k.a Kiwiland"
    }]
  }
 ]
```

Note: when one of the repeating elements has no value, it is represented in the first array using a null. When an element has a value but no extension/id, the second array will have a null at the position of that element.

Design Note: The representation of primitive data types has been split into two parts like this in order to simplify the representation of simple primitive values without id or extensions. This does have the cost of making the representation of the id attribute and extensions more ungainly, but these are both rarely used with primitive data types.

2.6.2.4 JSON representation of Elements, and Complex Data types

Elements, and complex datatypes (types that contain named elements of other types) are represented using a JSON object, containing a member for each element in the datatype. Composites can have id attributes, which are converted to JSON member values, in the same manner as described for primitives. For example:

```
<person>
  <text>
    <status value="generated" />
    <div xmlns="http://www.w3.org/1999/xhtml"><p>...</p></div>
  </text>
  <name>
    <use value="official" />
    <given value="Karen" />
    <family id="a2" value="Van" />
  </name>
</person>
```

is represented in JSON as:

```
{
  "person" : {
    "name" : [{
      "use" : "official" ,
      "given" : [ "Karen" ],
      "family" : [ "Van" ]
      "_family" : [ {"id" : "a2"} ]
     }],
```

```
    "text" : {
      "status" : "generated" ,
      "div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>...</p></div>"
    }
}
```

Things to note here are:

* Both given and family are repeating XML elements, so they are serialised as an Array whether or not they repeat in this instance

* In the family part of name, the id is added represented in _family as described above

* The XHTML content in the div element which is in the Narrative element text is represented as an escaped string in the value property in JSON. The xhtml's root element needs to be a in the xhtml namespace

2.6.2.5 JSON representation of Resources

A resource is a JSON object with a property resourceType which informs the parser which resource type this is:

```
{
  "resourceType" : "Patient",
  "text" : {
    "status" : "generated" ,
    "div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>...</p></div>"
  }
  etc...
}
```

Note that parsers cannot assume that the resourceType property will come first.

Design Note: This is a problem for several JSON -> Object serialisers that assume that the resourceType property does come first, including Json.NET . However, some JSON generators do not give the authoring application control of the order of the property values, and so these implementations cannot inter-operate with implementations that make assumptions about order. Given that JSON says that the property values are an unordered map of name/value pairs, this specification cannot require that properties come in any particular order, though implementers may choose to fix the property order if they are able (and the reference platforms provided with this specification do so).

There is a sample file with many edge cases to help test JSON parsers.

2.6.2.6 Canonical JSON

Resources and/or Bundles may be digitally signed (see Bundle and Provenance).

This specification defines the following method for canonicalizing FHIR resources, when represented as JSON:

* No whitespace other than single spaces in property values and in the xhtml in the Narrative

* Order properties alphabetically

* Omit all properties that have a default value, if a default value is defined

This canonicalization method is identified by the URL http://hl7.org/fhir/canonicalization/json. The following additional canonicalization URLS are also defined:

http://hl7.org/fhir/canonicalization/json#data | The narrative (Resource.text) is omitted prior to signing (note the deletion is at Resource.text, not Resource.text.div)

http://hl7.org/fhir/canonicalization/json#static | In addition to narrative (Resource.text), the Resource.meta element is removed. This makes the signature robust as the content is moved from server to server, or workflow and access tags are added or removed

http://hl7.org/fhir/canonicalization/json#narrative | This method only retains the Resource.id and Narrative elements

http://hl7.org/fhir/canonicalization/json#document | The signs everything in a Bundle, except for the Bundle.id and Bundle.metadata on the root Bundle (allows for a document to copied from server to server)

These canonicalization methods allow system the flexibility to sign the various portions of the resource that matter for the workflow the signature serves.

Note: One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

2.7 Version Management Policy

URL: https://hl7.org/fhir/STU3/versions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

2.7.0 Version Management Policy

This page documents the way version change is handled in FHIR. FHIR is a standard, so the way version change is handled is a bit different from an application API. This page describes:

* The standards development process

* The FHIR Maturity Model

* How FHIR versions work

* The rules for inter-version change once an artifact is Normative

* Strategies for handling change during the development process

* Plans for future releases

2.7.0.1 The Standards Development Process

FHIR is a standard. In order to be useful, standards need to evolve. At the same time, the evolution of standards needs to be predictable and manageable for the implementation community. This section describes how HL7 develops a standard so that implementers know what to expect as the standard evolves.

HL7 has four descriptive terms that describe the level of stability and implementation readiness associated with different aspects of the specification. They are as follows:

Standard Level | Description

Draft | This portion of the specification is not considered to be complete enough or sufficiently reviewed to be safe for implementation. It may have known issues or still be in the "in development" stage. It is included in the publication as a place-holder, to solicit feedback from the implementation community and/or to give implementers some insight as to functionality likely to be included in future versions of the specification. Content at this level should only be implemented by the brave or desperate and is very much "use at your own risk". The content that is Draft that will usually be elevated to Trial Use once review and correction is complete after it has been subjected to ballot

Trial Use | This content has been well reviewed and is considered by the authors to be ready for use in production systems. It has been subjected to ballot and approved as an official standard. However, it has not yet seen widespread use in production across the full spectrum of environments it is intended to be used in. In some cases, there may be documented known issues that require implementation experience to determine appropriate resolutions for. Future versions of FHIR may make significant changes to Trial Use content that are not compatible with previously published content.

Normative | This content has been subject to review and production implementation in a wide variety of environments. The content is considered to be stable and has been 'locked', subjecting it to FHIR Inter-version Compatibility Rules. While changes are possible, they are expected to be infrequent and are tightly constrained.

Informative | This portion of the specification is provided for implementer assistance, and does not make rules that implementers are required to follow. Typical examples of this content in the FHIR specification are tables of contents, registries, examples, and implementer advice

This release (Release 3) is an Trial Use Specification, though a little of the content (where marked specifically at the top of the page) is Draft. For Release 4, some content is planned to be Normative.

Notes:

* The above statuses can apply to both the standard overall as well as to individual components of the FHIR specification

* Between FHIR release 2 and 3, HL7 changed from using "DSTU" (Draft Standard for Trial Use) to just simply "STU" (Standard for Trial Use) to reflect the maturity of the FHIR specification: Release 2 and particularly this Release 3 are far beyond "draft" specifications, and have been and will be widely implemented.

* Most pages in the specification identify their status explicitly. The few pages that don't are Table of contents pages, and are all Informative

* Some content is labeled with the status "External", which means that the content is maintained in another standard, and the status must be found by consulting that other standard. In this case, the Maturity Model does not apply

2.7.0.2 Maturity Levels

The content of this release has been subject to significant review through ballot and other HL7 processes and many aspects of it have been implemented and subjected to interoperability testing through Connectathons and early adoption. However, the degree of testing has varied. Some resources have been well tested in a variety of environments. Others have received relatively little real-world exercise. In general, the infrastructure should be considered to be more stable than the resources themselves. In some cases, there are issues on which input is specifically requested during the Trial Use period (see the Outstanding Issue List, and known issues will arise after publication (refer to the FHIR Change Request tracker for details.) Guidance from early implementation will help address these areas.

All artifacts in this specification are assigned a "Maturity Level", known as FMM (after the well known CMM grades). The FMM level can be used by implementers to judge how advanced - and therefore stable - an artifact is. The following FMM levels are defined:

* the resource or profile (artifact) has been published on the current build. This level is synonymous with Draft.

* PLUS the artifact produces no warnings during the build process and the responsible WG has indicated that they consider the artifact substantially complete and ready for implementation

* PLUS the artifact has been tested and successfully exchanged between at least three independently developed systems leveraging at least 80% of the core data elements using semi-realistic data and scenarios based on at least one of the declared scopes of the resource (e.g. at a connectathon). These interoperability results must have been reported to and accepted by the FMG

* PLUS the artifact has been verified by the work group as meeting the Trial Use Quality Guidelines and has been subject to a round of formal balloting; has at least 10 implementer comments recorded in the tracker drawn from at least 3 organizations resulting in at least one substantive change

* PLUS the artifact has been tested across its scope (see below), published in a formal publication (e.g. a FHIR Release), and implemented in multiple prototype projects. As well, the responsible work group agrees the resource is sufficiently stable to require implementer consultation for subsequent non-backward compatible changes.

* PLUS the artifact has been published in two formal publication release cycles at FMM1+ (i.e. Trial Use level) and has been implemented in at least 5 independent production systems in more than one country

* "Normative": the artifact is now considered stable

Tested across scope means:

* The FMG has signed off on the list of "example contexts" defined for the artifact

* For each example context, the artifact has either been: reviewed and approved by a domain expert for that scope area, mapped to an existing implemented scope-area-specific standard or tested in an implementation

The Maturity level is strongly related to stability; the higher the maturity level, the more controls are enforced to restrict breaking changes to the resource. For further information, and discussion, see the FHIR Wiki .

The maturity model is significantly influenced by the degree and type of implementation activity using an artifact. For this reason, we encourage implementers to register their implementations . A detailed analysis of the basis for the maturity metrics for FHIR artifacts can be found here .

2.7.0.3 FHIR Releases and Versioning

New versions of FHIR will be published on a release cycle of approximately 18-24 months. This frequency is based on the timelines necessary to consult with implementers, and develop, review new content as well undertake the formal balloting and reconciliation processes required for ANSI-approved standards. This release cycle also ensures an opportunity to incorporate implementer feedback from earlier versions of the specification into subsequent versions. Limited-scope releases on a shorter timeline may occur occasionally where necessary to meet implementer desires.

Each new release is assigned a unique version number. The FHIR version policy is based on Semantic versioning , but with some differences due to the fact that FHIR is a specification, not a software API.

There is a single development version of FHIR. This undergoes cycles of development as managed by HL7. Each major cycle of development is concluded by a formal ballot (or more than one), and then a new specification is published. In version control terms, each published specification is a branch off the development trunk, which may then itself undergo further change as HL7 maintains the published specification (though such changes are usually minimal, limited to necessary technical corrections or security alerts).

Each FHIR version is identified by a string composed from 4 parts: publication.major.minor.revision.

publication | Incremented when HL7 publishes FHIR as an updated specification, e.g. a Trial Use or Normative version of FHIR The first Trial Use was version 0 FHIR Release 2 (DSTU) was version 1 FHIR Release 3 (STU) is version 3 (skipped '2' to align the major numbers at implementer request)

major | Increments every time a breaking change is made (see below) When a new publication is made, this is reset to 0 in the publication, and 1 in the development branch Since HL7 does not make breaking changes as technical corrections to a published specification, these versions of FHIR always have a version number X.0.n.r Because the development version is the subject of ongoing analysis, debate, ballot and repeated alterations, breaking changes are to be expected in STU content

minor | Increments every time an official snapshot release is generated that contains one or more substantive changes Resets to 0 any time the major version changes Snapshot releases are produced approximately 6 weeks in advance of the 3 annual HL7 working group meetings (and their associated connectathons), though they can also be produced for other major connectathons or to meet implementer requirements.

revision | Incremented any time a change is made to the specification At present, this is the SVN version number (this allows anyone to reconstruct the source from which the version was built) Changes are made numerous times a day, generally driven by change requests submitted by the implementation community

Additional notes:

* Changes to a formally published specification (except for minor publishing corrections, such as correcting broken external links) are only made via announced technical corrections

* The reference implementations have 2 versions - the version of the specification that they implement and their own version. Consult the reference implementation documentation for policy regarding this version number

* The current build - published by the continuous integration service (http://build.fhir.org/) - does not conform to this version policy, in that the version is not updated as changes are made. To indicate this, the revision is always "cb" e.g. 3.1.cb immediately after the publication of Release 3

* The first DSTU was published prior to these rules being agreed as v0.80-2286. This has been updated to 0.0.81.2382 as a technical correction to align with this policy on 9-May 2014

2.7.0.4 Rules for Inter-version change

The following kinds of changes may be made to the specification:

* Breaking changes are changes that mean that previously conformant applications are no longer conformant to the updated specification

* Substantive changes are changes that introduce new functionality - changes to the specification that create new capabilities - but would not render unchanged existing applications non-conformant

* Non-substantive changes do not cause changes in any conformant application. For example, section renumbering, broken links, style corrections, typos, and clarifications that do not change the meaning of the specification. In addition, some other kinds corrections may be judged not to create any expectation of change to a conformant application

Note that the following are, by definition, considered non-breaking changes, even though some implementations (including the reference implementations) may not be able to handle some consequences of these changes without error:

* Creation of new resources

* Adding new optional elements in an existing resource

* Defining new data types

* Adding new API operations

The examples provided as part of this specification are never substantive. While very effort is made to ensure that FHIR examples are correct, changes to the examples in the specification are not considered substantive.

Content with a status of Draft or Trial Use can change - including Breaking Changes - from version to version, subject to the rules described by the Maturity Process. Once an artifact achieves Normative status, the Backward compatible behavior rules will apply. These rules ensure that implementations may exercise FHIR interfaces and process the content of FHIR resources safely while exchanging data between applications using different versions of FHIR.

2.7.0.4.1 Version identification

There is no explicit version marker in the content of FHIR resource instances. Instead, the version supported by a given system is declared using the conformance framework. The resources CapabilityStatement and StructureDefinition have properties for declaring the FHIR specification version, and these may be used to determine which version of FHIR an implementation is using to aid in validation and integration.

2.7.0.4.2 Forward compatible behavior

In a typical scenario, mixed versions may need to exist, so applications SHOULD ignore elements that they do not recognize unless they are modifierExtensions. However, in a healthcare context, many application vendors are unwilling to consider this approach because of concerns about clinical risk or technical limitations in their software (e.g. schema based processing). Applications are not required to ignore unknown elements, but SHALL declare whether they will do so in their Capability Statements.

Unrecognized search criteria SHOULD always be ignored - see Handling Search Errors for further information.

Attempts to perform HTTP operations on unexpected URLs SHOULD be responded to with an appropriate error code.

2.7.0.4.3 Backward compatible behavior

Category | Allowed changes

Elements | Once Normative, subsequent versions of this specification may introduce new elements and/or content (e.g. XML attributes, etc.) at any location in the bundle, resource and/or data type structures. However, the names, path and meaning of previously existing data elements will not be changed. This means there will be no change to resource names and no changes to names assigned to slices and other elements within profiles.

Cardinality | Minimum element cardinalities will not be changed. Upper cardinality may change from 1 to * only in circumstances where all elements except for the first repetition can be safely ignored. (This may mean that an order is assigned to the repeating items or that there is no preference as to which element is retained.) Systems should follow the rules above for unexpected elements.

Descriptions | Descriptive information about a resource - short labels, definitions, usage notes, aliases, examples, rationale, mappings, etc. may be updated or revised to provide additional clarity or guidance, but not in such a manner as to invalidate a reasonable interpretation of the previously documented use of an element. (This does not preclude fixing obvious errors.)

Value Sets | The definition of any value set that is marked as immutable will never change. The expansions for immutable value sets may still change if no "stable date" is declared and the value set does not restrict code system and/or value set references to specific versions and the codes in the referenced code system(s) or value set(s) change. For non-immutable value sets: Value sets with an enumerated list of codes and having a 'fixed' binding may have additional codes introduced but will never have codes removed, though they may be deprecated. Value sets making use of filters may have filters loosened or tightened to accommodate changes to underlying code systems. StableDates and referenced code system and value set versions may be adjusted to point to newer versions. Definitions and display values for codes may change, but only in a manner that would not change the reasonable interpretation of data captured using the previous definitions or names. Abstract codes may be made concrete. Concrete codes will not be made abstract. For both immutable and non-immutable value sets, additional designations may be declared.

Terminology Bindings | Fixed bindings will remain fixed and will continue to point to the same value set. If the reference is version-specific, it will not change. Extensible bindings will remain fixed and will continue to point to the same value set. If the reference is version-specific, it will not change. Example bindings and preferred bindings may change to point to different value sets. Example bindings may be replaced with preferred bindings.

Data Types | Data types will not be removed or changed except as allowed above for elements. New data types may be introduced. Types declared on existing elements will not be removed or changed. Additional data types may be added to elements which are already expressed as a choice of data types only if those elements are optional (minimum cardinality = 0).

Value Constraints | The allowed list of Data types will not be added, removed or changed. Invariants, regular expressions, fixed values and patterns will not be added, removed or changed.

Flags | The Is Modifier and Is Summary flags will not be changed.

Slicing | Slicing rules and aggregation characteristics will not be changed.

Search Criteria | Search criteria may be added but not removed or renamed. Existing criteria will not have their type or path changed or have their description altered in any way that would invalidate the reasonable behavior of existing systems (with the exception of correcting obvious errors).

Operations | New operations may be defined but operations may not be removed or renamed. Existing parameters will not be removed or renamed, nor may their type or lower cardinality be changed. Upper cardinality may be changed from 1 to *. (Systems should ignore unexpected repetitions.) Additional optional parameters may be introduced; e.g. Operation signatures cannot change; instead, additional operation variants will be defined.

Restful interface | Existing endpoints will not be renamed or removed, nor have their expected behavior changed in a manner that would cause reasonable systems designed against prior versions to be non-interoperable. Additional endpoints and interactions may be introduced.

Profiles and extension definitions | Profile structure, extension definitions and search criteria definitions will not be removed or have their URIs changed. New profile structures, extension definitions and search criteria definitions may be introduced. Profiles may have their statuses changed to "retired". Profiles referenced by data elements for structures or data types may be replaced with a reference to a distinct profile that is "compatible" with the previously referenced profile according to these forward and backward compatibility rules.

Additional discussion on inter-versioning issues can be found here:
http://wiki.hl7.org/index.php?title=FHIR_interversion_compatibility .

2.7.0.5 Strategies for handling Changes to STU content

Regardless of the degree of prior implementation, all aspects of the FHIR specification are potentially subject to change while an artifact has a status of Draft or Trial Use. These changes may be minor (clarifications of definitions, etc.) or major (refactoring of resources, changes to serialization rules, eliminating or adding data types, etc.) There is no commitment to backward or forward compatibility during the STU process. Changes will not be made without cause, however the interests of long-term implementability will generally trump the impact on early adopters when determining what changes should be made. This balance will shift more towards early adopters as maturity levels increase. I.e. Impact on existing implementations will be weighted more highly for an FMM-level 5 artifact than they would for an FMM-level 1 artifact.

Implementers who are willing to accept the risk of change (perhaps for the benefit of early implementation experience, first mover advantage and the ability to leverage FHIR's intrinsic benefits) are encouraged to implement FHIR in real-world systems. However, those implementers should be aware that local adaptations may be necessary to meet real-world requirements. Furthermore, such implementers should architect their solutions to be tolerant of changes to the specification and, where necessary, to manage interoperability with systems that may be using different versions of the specification or different local adaptations.

The most common strategy for handling change between versions of FHIR is to use different end-points for different versions. e.g. http://acme.com/fhir/r2 and http://acme.com/fhir/r3, though other approaches are possible.

During the Trial Use period, requests for change may be submitted using the HL7 gForge tracker which can be found here . Where possible, updates to the "development" version of the specification will be made in a timely fashion. Implementers should be aware that the changes are not considered "official" until such time as they are balloted and approved as part of a subsequent Trial Use or Normative publication. Change requests might be fixes to allow implementation, clarifications or enhancements. In addition, HL7 will be developing and introducing additional resources and profiles as part of the FHIR specification.

SDOs and regulatory bodies that are interested in making use of the FHIR specification should feel free to do so, but should consider and plan for the possibility that the specification will evolve and change prior to becoming Normative.

A key objective of the Trial Use process is gaining feedback from implementers making use of the specification. As well, HL7 has a need to monitor which portions of FHIR are being implemented in what sorts of environments so as to make an informed decision on when the specification is ready to proceed to Normative status. For this reason, all FHIR implementers are encouraged to register their usage here .

This survey will capture contact and other information that will allow the FMG to perform appropriate monitoring of FHIR STU usage. Survey information will be kept confidential unless the participant authorizes inclusion of their project in an HL7-maintained wiki page of early implementers. Confidential submissions will be reported in aggregate only.

2.7.0.6 Plans for future releases

While implementation of this Trial Use release is occurring, development will be progressing on the next release. This next release will include additional resources, profiles and quality enhancements over the current release. It will also incorporate fixes for issues raised with the FHIR change tracker . It may be useful for implementers of the STU to review the development release to get a sense of what changes are likely coming and perhaps to find more robust definitions and guidance than are available in the first release. The FHIR development release can be found at build.fhir.org . Some implementers who are dependent on content that exists in a draft release may choose to implement based on a particular snapshot of the development release, though in doing so, they will limit their potential communication partners and would not be considered to be completely FHIR conformant.

The next major publication of FHIR will be Release 4. It is our hope that this release will include the transition of some of the content of the specification to Normative. This should include many of the core infrastructure resources (e.g. StructureDefinition, ValueSet as well key pages such as the XML and JSON syntaxes, RESTful protocol, etc. It should also include at least a few of the administrative resources such as Patient. Much content, including most if not all clinical and business resources, will remain at the Trial Use level as they are not expected to meet the criteria for Normative.

The anticipated timeline for Release 4 involves a finalization of scope by the end of 2017, balloting in April of 2018 and publication around October 2018. The specific timing may vary based on implementer feedback and the volume of contents received during the ballot process. More information on plans for Release 4 can be found on the HL7 product director's blog . (Subscribing to this blog is a good way to keep current on significant events in FHIR development, including ballot and publication timelines.)

There will be additional releases of FHIR with a frequency of between 18 and 24 months for the foreseeable future. These releases will include new content (e.g. in the public health, financial or clinical research spaces), revisions reflecting implementer feedback and increasing maturity on Trial Use artifacts and the migration of additional content to normative status. As well, HL7 will gradually shift focus to providing additional guidance through the publication of implementation guides and profiles where consensus can be found at the international level.

2.7.1 Inter-version Compatibility

URL: https://hl7.org/fhir/STU3/compatibility.html

test

2.7.1 Inter-version Compatibility

This content has been moved

2.7.2 FHIR Timelines

URL: https://hl7.org/fhir/STU3/timelines.html

test

2.7.2 FHIR Timelines

This content has been moved

2.8 Version History since DSTU #1

URL: https://hl7.org/fhir/STU3/history.html

This page is part of FHIR STU 3 (v3.0.2) in it's permanent home (it will always be available at this URL). It has been superceded by R4. For a full list of available versions, see the Directory of published versions .

2.8 Version History since DSTU #1

For background to this version history, see FHIR Version Management Policy.

Major Milestones:

Feb 21, 2017 | 3.0.2 | Release 3 (STU - Standard for Trial Use)
Oct 24, 2015 | 1.0.2 | DSTU2 (Second Draft Standard for Trial Use)
Sept 30, 2014 | 0.0.82 | DSTU1 (First Draft Standard for Trial Use)

This table lists substantive changes only.

This is the developmental version of FHIR. The only changes tracked here are the changes made after the publication of the first DSTU release. For earlier changes, see the DSTU #1 Version History . Note that a full archive history of everything is available through the HL7 gForge SVN archives .

Version | Changes

3.0.2 | 2nd Technical Corection to STU3: Oct 24, 2019 Technical fixes to supporting collatoral. Specifically, the following things were changed: Fix Invariants on StructureDefinition - sdf-8,12,15,20 Fix XPath for Invariant Condition on con-4 Fix Invariant for Observation(obs-7) Remove ElementDefinition Invariant eld-10 Fix date time & instant regexes Fix Invariant on Vital Signs Profile vs-2

3.0.1 | Technical Corections to STU3: April 19, 2017 Technical fixes to supporting collatoral, fixing wrong links, and correcting some erroneous documentation. Specifically, the following things were changed: Snaphots for Profiles generated by the build tooling were wrong. Corrected (no visible changes) (gForge 13172) XPath corrections for Invariants on many resources (gForge 13171) Fix problems in generated JSON Schema (gForge 13218) Rework ele-1 to work around ambiguity in FHIRPath specification (gForge 13173) Fix eld-16 to allow for multiple tokens in the name (gForge 13217) Correct long-standing erroneous narrative about JSON representation (gForge 13156) Add note about which elements are marked as summary (gForge 13124 , 13125) Change the maturity level of the RequestGroup resource (gForge 13225) Fix rendering on R2/R3 difference page (and add computable forms by request) (gForge 13153) Remove erroneous note in overview about every page having comments Broken Links: reference to resource page on resource.html points to itself, and several broken links on the terminology module page (gForge 13001 , 13158) Correctly assign section numbers to logical models (gForge 13168)

3.0.0 | Release 3: Mar 21, 2017. STU3 This is the full FHIR Release 3 (R3). R3 is a complete overhaul of the specification from R2, with over 2400 change proposals process, and applied in R3. Of those changes, 380+ are labelled as a breaking change. Given this level of change, there is no useful way to present a single change list. Users can use the R2/R3 difference comparison, the R2 <-> R3 transforms, or the "Compare to R2" link at the foot of every page to help visualise the differences between R2 and R3. Many of the changes made in this release (since 1.8.0) are in response to Quality Assurance processes in HL7, with a goal of readying the specification for Normative processes in R4: Reorder many resources and improve their definitions and bindings (Quality Assurance work) Add draft resources AdverseEvent and GraphDefinition Change to the way discriminators work in profiles (for profile/tooling authors) Finalization of the Turtle (RDF) format, along with JSON-LD Consolidation of Vresion management issues into a single Version Policy page, and update to current policy Add rules about how missing data in search Roll up: Of all the many changes (thousands) between R2 and R3, these are the most significant: Change the FHIR mime type to application/fhir+xml|json instead of application/xml|json+fhir (breaking change, for conformance to W3C+IETF rules) Rename the "Conformance" resource to "CapabilityStatement" and Deprecate use of the OPTIONS command to retrieve the CapabilityStatement Search Changes around errors in search parameters, missing data, sorting Change HumanName.family from 0..* to 0..1 CodeSystem now separated out from ValueSet

1.9.0 | FHIR Rolling Build, 2016 Dec onwards

1.8.0 | FHIR QA Release, Dec 6, 2016 STU 3 Candidate, and also the basis for several implementation guide ballots and the San Antonio Jan 2017 Connectathon Breaking Changes Summary: Change HumanName.family from 0..* to 0..1 Rename MedicationOrder and DiagnosticOrder to MedicationRequest and DiagnosticRequest and make a series of changes to align with the Request Pattern Add cross resource search + rework composite parameter expressions Harmonise resource metadata across the Terminology, Conformance, and Knowledge resources Change the way versioned references to canonical URLs are handled REST - servers must return a location header on a create Condition status elements reworked These are only the more significant changes, there were many 100s of changes made in response to ballot comments and ongoing implementation experience. These include breaking changes to may resources. Structural changes are summarised here, and Transforms between DSTU 2 and STU 3 are provided for many resources.

1.6.0 | FHIR STU3 ballot, Aug 11, 2016 STU 3 ballot version, and also the basis for the Baltimore Sept 2016 Connectathon Publication Changes: Move the US Realm implementation guides out of the specification (see http://www.fhir.org/guides/registry) Rework the home page, and introduce modules as a way to navigate the specification + add significant new implementer advice (e.g. "Getting Started") Change the Navigation menu to focus on indexes to the content in the specification Move FHIRPath expression language to its long term home Technical Changes: RESTful API Change the FHIR mime type to application/fhir+xml|json instead of application/xml|json+fhir (breaking change, for conformance to W3C+IETF rules) Add new uses for the Prefer header (return OperationOutcome, and manage behavior related to unknown / unsupported search parameters Deprecate use of the OPTIONS command to retrieve the CapabilityStatement Add support for conditional references to the transaction interaction Add reverse chaining Formats: No change to XML + JSON formats. Add Turtle Format, and related validation tools For Data Types and Resources, there is formal difference analysis from DSTU2 (also found throughout the specification), and transforms between DSTU 2 and STU 3 are provided for many resources

1.5.0 | FHIR Current Build Update, July 8 2016 Update current version to 1.5 to prevent confusion with implementations still running the May 2016 version. There are many changes in this version from 1.4.0, but no formal list is provided. A formal list will be provided for 1.6 (as a diff to 1.4)

1.4.0 | FHIR Connectathon 12 Snapshot, Mar 30 2016 Frozen base for Connectathon 12 & For Comment ballots: FHIR API and Serialization Format Enhancement The _sort parameter has been reworked New or extended search parameters for read, search or history: _type, _at, _summary, _elements Change Turtle representation - now ready for trial Conformance resources with significant breaking changes and behavior:

StructureDefinition - Type Handling: Changes to how the structure definition types work: baseType replaces constrainedType & baseDefinition replaces base CodeSystem now separated out from ValueSet: CodeSystem is now a first class resource type in order to support use throughout the FHIR eco-system, such as to support value set expansion and validation. It is intended to be used for distributing the smaller ad-hoc code systems that are ubiquitously encountered through out the healthcare process. Other resources with significant breaking changes and behavior: AllergyIntolerance Appointment AuditEvent Claim ClaimResponse Device ConceptMap Conformance (now CapabilityStatement) Contract Device DiagnosticRequest EligibilityRequest EligibilityResponse ExplanationOfBenefit ImagingManifest Library Measure Medication MedicationAdministration MedicationStatement Observation: the introduction of the Vital Signs Profile (proposed to be mandatory) OperationDefinition Patient PaymentNotice PaymentReconciliation Practitioner ProcessRequest ProcessResponse Provenance ReferralRequest Schedule Sequence Slot StructureDefinition TestScript ValueSet Add new draft resources: CareTeam CodeSystem CompartmentDefinition Linkage MeasureReport PractitionerRole Protocol StructureMap Task

1.2.0 | FHIR Connectathon 11 Snapshot, Dec 11 2015 Frozen base for Connectathon 11: Remove GuidanceRequest Add new draft resources: Sequence, ExpansionProfile Modifications to Financial Resource & TestScript resource Note: this version is temporary, and will be removed after Connectathon 11 is complete

1.1.0 | GAO Ballot + technical corrections, Dec 2 2015 A ballot publication for the GAO Ballot that also includes: Various technical corrections to the generated snapshots Populate FHIRPath expressions in extensions for all search parameters and invariants Add new draft resources: GuidanceRequest, GuidanceResponse, ModuleMetadata, ModuleDefinition, Library, DecisionSupportServiceModule, DecisionSupportRule, OrderSet, Measure Major restructure of Questionnaire

1.0.2 | Technical Correction 1, Oct 24 2015 A series of technical corrections to the specification following extensive review: Corrections to Extension cardinalities in implementation guides Corrections in the conformance resources that support the specifications Correct several erroneous invariants Various typos, broken links, and fixes in examples For a comprehensive list of corrections, see the Task list for FHIR DSTU2 Technical Correction 1

1.0.1 | DSTU 2, Sept 23 2015 Changes of significance during the QA process: Remove the Clinical Quality Improvement Framework (CQIF) from this published version made fixes to generated schematrons updated generated comformance resources (StructureDefinitions and SearchParameters) so they were consistent with the specification Many spelling / grammar / broken link fixes

1.0.0 | DSTU 2 QA Preview, Aug 31 2015 This version had extensive change as a result of the May 2015 DSTU ballot, ongoing testing, and the open change proposals (1317 gForge tasks). The extent of the changes is best illustrated by the number of the list of changes labeled 'breaking change' - 158 changes of 1317 total tasks. Below is a list of the most important changes: General: introduced the maturity framework RESTful API: add batch, several clarifications around versioning & transactional integrity, changed Bundle URL resolution rules Search: changed the way <> etc.works, added _list parameter, changed rules around contained and included resources Formats: added a note about whitespace in XML, added code generation schemas Data Types: New data types: markdown, Annotation, changed data types: Coding, Quantity, Signature, Timing, Address, ContactPoint changes to ElementDefinition: add base, make type.profile repeat, remove invariant.name and replace with invariant.requirements, remove binding.name, add min/max value Resources: New resources: Account, ImplementationGuide, TestScript renamed: Contraindication -> DetectedIssue, MedicationPrescription -> MedicationRequest, QuestionnaireAnswers -> QuestionnaireResponse removed: Supply changed: almost all resources - too many to list (1317 tasks worth of changes) - add, remove elements, change types, references, definitions & value sets, re-order elements, provide much more documentation and new examples Implementation Guide: Move Argonaut content out

0.5.0 | DSTU Ballot, May 2015 This version had extensive change as a result of the January 2015 Draft ballot, ongoing testing, and the open change proposals (over 800 gForge tasks). The list below is a summary of the major changes to resource content. It shows only a limited number of the overall changes. Enumerations All spaces removed Extensive content changes not noted here New Data Types unsignedInt positiveInt Signature Meta Changed Data Types Coding - remove valueSet property Attachment - add creation Identifier - replace label with type Timing - major rework of content ElementDefinition - add label, code, rename 'formal' to definition, rename synonym to alias, add language to mapping, remove conformance and isExtensible and replace with strength New Resources BodySite Claim ProcessRequest ProcessResponse Removed Resources CarePlan2 -> collapsed into CarePlan FamilyHistory -> broken up into FamilyMemberHistory InstitutionalClaim, OralHealthClaim, PharmacyClaim, ProfessionalClaim, VisionClaim -> collapsed into Claim Other - use Basic instead PendedRequest,Readjudicate, Reversal, StatusRequest, StatusResponse - use ProcessRequest/Response instead SupportingDocumentation - use DocumentManifest instead Renamed Resources Alert -> Flag: 'alert' made people think it was an action like an alarm SecurityEvent -> AuditEvent: it wasn't just for security purposes ClinicalAssessment -> ClinicalImpression: people got confused with 'assessment' tools like APGAR score Profile -> StructureDefinition: 'Profile' is the process, a package of statements Changes Inside Resources Parameters - allow parameter.part to contain a resource AllergyIntolerance - rename subject to patient Appointment - remove lastModifiedBy/lastModified, add location AppointmentResponse - remove lastModifiedBy/lastModified, add rename individual to actor AuditEvent - add .event.purposeOfEvent, participant.location, .policy, and .purposeOfUse Bundle - major reorganization CarePlan - pull goal out + other reorganization ClinicalImpression - add status, replace careplan & referral with trigger, rename diagnosis to finding, make plan 0..*, Composition - change .section.content to refer to List only, not any ConceptMap - change identifier to url, add useContext, change telecom to contact, Condition - rename subject to patient, rename status to clinicalStatus, change to bodySite = code or Reference(BodySite), rename .codeableConcept to .code Conformance (now CapabilityStatement) - change identifier to url, add useContext, change telecom to contact, add requirements and copyright, add support for conditional operations, Contract - extensive rewrite Coverage - add bin, subscriberId DataElement - total rewrite to use ElementDefinition Device - add status, manufactureDate DeviceMetric - rename operationalState to operationalStatus, add measurementMode, rename calibrationInfo to calibration, change color to an enumerations DeviceRequest/DeviceUseStatement - change to bodySite = code or Reference(BodySite) DiagnosticRequest- change to bodySite = code or Reference(BodySite) DiagnosticReport - add encounter DocumentManifest - add options for how content is referred to DocumentReference - add format, remove policyManager, make content : Attachment, and remove several related attributes, remove service reference and add context.practiceSetting, sourcePatientInfo, and related Encounter - add incomingReferralRequest, allow reason to repeat, rename diet to dietPreference EpisodeOfCare - rename currentStatus to status, allow referralRequest to repeat, Flag - rename subject to patient, change from note to code Goal - add targetDate, statusDate, author, priority HealthcareService - extensive rewrite ImagingObjectSelection - remove retrieveAETitle, rename retrieveUrl to url, add frames ImagingStudy - add laterality, change url to attachment Immunization - add encounter, rename subject to patient, rename refusedIndicator to wasNotGiven, rename refusalReason to reasonNotGiven ImmunizationRecommendation - rename subject to patient List - add title, status, change ordered to orderedBy, add note Location - remove status Media - remove created (-> Attachment) Medication - add batch MedicationAdministration - add reasonGiven, note, text. remove timing & maxDosePerPeriod MedicationDispense - collapse to a single dispense, add daysSupply, note and substitution, change quantity to allow range MedicationRequest - add note, change quantity to allow range, MedicationStatement - add informationSource, status, dateAsserted, replace whenGiven with effective[x], remove device, add dosage.text NamingSystem - add date, publisher, NutritionOrder - extensive rewrite Observation - change name to code, allow more types of value[x], change type of dataAbsentReason, change to bodySite = code or Reference(BodySite), allow identifier to repeat, add device, OperationDefinition - change identifier to url, add useContext, change telecom to contact, change name to title, add reuqirements, idempotent, OperationOutcome - change type of .issue.type OrderResponse - rename code to orderStatus Organization - remove location and contact.gender Patient - communication to allow 'preferred' Person - rename other to target Practitioner - change type of birthDate, allow multiple roles per practitioner Procedure - add status and category, change to bodySite = code or Reference(BodySite), allow date to be period too, add location, change followUp to code 0..*, add device tracking ProcedureRequest - change to bodySite = code or Reference(BodySite) Provenance - change integritySignature to signature & make it a type, allow reference by Reference as well as URI Questionnaire - add telecom Schedule - move lastModified SearchParameter - change telecom to contact, add status, experimental, date, Slot - move lastModified Specimen - change source to parent, change to bodySite = code or Reference(BodySite) StructureDefinition - complete rewrite Subscription - change type of tag, reanme url to endPoint, ValueSet - change identifier to url, add useContext, change telecom to contact, replace purpose with useContext, add requirements, rename stableDate to lockedDate, change type of expansion.identifier, add expansion parameters

0.4.0 | Draft For Comment, January 2015 Ballot Breaking Changes (full list): Replace atom and taglist with a native Bundle format (3728 , 3558 , 2889) (and also Binary) JSON: change how extensions are represented (3471) RESTful API: change how version specific upgrades work (3451) DataTypes: Rename Schedule to Timing (3536 , 3236) Rename Contact to ContactPoint (3533) and swap order of elements (3108)) Address - change zip to postCode (2888) Quantity: Correct schema spelling for "QuantityCompararator" (3531) Change allowable values for the id type to include capital letters, and allow up to 64 chars (3750) Restructure Profile - only one structure, and pull ExtensionDefinition out of Profile (3647, 3498), and pull SearchParameter out (3626) Profile: allow 0..* discriminator (3131), and change the way discriminators work across resource boundaries (3124) + generate multiple types properly (2856) remove _validate interaction, and replace with $validate operation (3686) Patient: separate birth time from birthDate (3731), Change Administrative Gender from a CodableConcept to a Code. Also fixed the values as male|female|other|unknown with mappings to v2 and v3 (3070) DocumentReference: change encoding of Hash to Base64 (3291) Group: rename header to title (3126) Condition: split relatedItem into two (3111) Questionnaire: drop questionnaire.group.question.remarks (3255) and move omitReason from extension to base resource (3260) QuestionnaireResponse: allow multiple answers (3146) ValueSet: replace ValueSet.compose.include.code with ValueSet.compose.include.concept (3258), added new rules about expansion content (3138) Media: Rename element 'dateTime' to 'created' (3174) and length to duration (2866) Remove DeviceObservationReport and Query Collapse AdverseReaction into AllergyIntolerance Appointment changes - individual field renamed to actor, and added mappings to HL7 v2 and HL7 v3 FamilyMemberHistory combined with List replaces FamilyHistory (with corresponding updates to related profiles) Flag replaces Alert including improved clarification of how it is used and replacement of "note" with "code" CarePlan significantly refactored including splitting Goal out as a distinct resource, moving elements between activity and detail, introduction of several new elements and supported relationship types New Resources: Appointment AppointmentResponse Basic ClaimResponse ClinicalImpression Communication CommunicationRequest Contract Contraindication Coverage DataElement DeviceComponent DeviceMetric DeviceRequest DeviceUseStatement EligibilityRequest EligibilityResponse EnrollmentRequest EnrollmentResponse EpisodeOfCare ExplanationOfBenefit StructureDefinition Goal HealthcareService ImagingObjectSelection InstitutionalClaim NamingSystem NutritionOrder OperationDefinition OralHealthClaim PaymentNotice PaymentReconciliation Person PharmacyClaim ProcedureRequest ProfessionalClaim QuestionnaireResponse ReferralRequest RiskAssessment SearchParameter Schedule Slot Subscription SupportingDocumentation VisionClaim VisionPrescription New Implementation Guides (see discussion of status) Argonaut Project Data Access Framework Structured Data Capture & Structured Data Capture - Data Element Exchange US Laboratory Order, Report & Report to Public Health EHRS Functional model - Record Lifecycle Events

0.3.0 | Renamed Namespace to NamingSystem Split QuestionnaireResponse from Questionnaire and significantly revamped the Questionnaire in response to feedback from the Connectathon. Added DataElement resource (has been previously discussed as ObservationDefinition) Defined Subscription resource for evaluation Add time data type Define RPC-type operations on the RESTful interface, and add Operation Definition Defined operations on several resources including ConceptMap, Questionnaire and ValueSet Conformance (now CapabilityStatement): Change Conformance.rest.operation to Conformance.rest.interaction and Conformance.rest.resource.operation to Conformance.rest.resource.interaction, and add Conformance.rest.operation to point to Operation Definition Profile: add Profile.url, Profile.structure.snapshot and Profile.structure.differential, and remove query definition (now in OperationDefinition) Add pages for LOINC, RxNorm, and SNOMED CT Significant rework for ConceptMap so it can be used to map between structures as well as value sets Add Contraindication and Risk Assessment and added examples to other resources in support of these Add Referral Request Add supportingInformation to DiagnosticRequest Add fulfills to Encounter Add date and age[x] to FamilyHistory Change cardinality of Location.identifier to 0..* Change cardinality of Practitioner.address to 0..* Add Observation.encounter to 0..* Temporarily added content for the Structured Data Capture implementation guide , including profiles on several resources, conformance instances, etc. This content will be moved to an external specification once the necessary tooling is in place to support that form of publication.

0.2.1 | Minor new optional elements on value set for metadata, new extensions for all the rest of the VSD project metadata, formal profile to express basic minimum metadata for value set

0.2.0 | Namespace: adjustments based on Grahame's feedback

0.1.0 | Add Appointment, Appointment Response, Schedule and Slot Add Namespace

Note: a useful tool for displaying the differences between pages is the W3C HTML Diff engine .

2.9 Differences from DSTU Release 2

URL: https://hl7.org/fhir/STU3/diff.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.9 Differences from DSTU Release 2

This is an automatically generated list of all the structural differences between FHIR DSTU2 and this version. The following possible changes are checked when preparing this list (and others throughout the specification):

* Addition or deletion of elements, or name change to elements

* Changes to minimum and maximum cardinality

* Changes to allowed types

* Changes to bindings for Extensible/Required bindings, and changes to codes if it's a required binding on a code element

* Changes to the is-modifier status of an element

* Changes to the default value for an element

This analysis only considers changes to data types and resources. Comparison of extensions, profiles and value sets will be added in a future version.

This analysis is available as XML or JSON.

Types

Address

Age | Changed from a constraint to a specialization

Annotation

Attachment

Attachment.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

BackboneElement

CodeableConcept

Coding
ContactDetail | Added Type
ContactPoint
ContactPoint.system | Add Codes url, sms
Contributor | Added Type
Count | Changed from a constraint to a specialization
DataRequirement | Added Type
Distance | Changed from a constraint to a specialization
Dosage | Added Type
Duration | Changed from a constraint to a specialization
Element
Element.id | Type changed from id to string
ElementDefinition
ElementDefinition.representation | Add Codes xmlText, typeAttr, cdaText, xhtml
ElementDefinition.sliceName | Renamed from name to sliceName
ElementDefinition.slicing.discriminator | Type changed from string to Element
ElementDefinition.slicing.discriminator.type | Added Element
ElementDefinition.slicing.discriminator.path | Added Element
ElementDefinition.comment | Renamed from comments to comment
ElementDefinition.min | Type changed from integer to unsignedInt
ElementDefinition.base.min | Type changed from integer to unsignedInt
ElementDefinition.contentReference | Renamed from nameReference to contentReference Type changed from string to uri
ElementDefinition.type.code | Type changed from code to uri Change binding strength from required to extensible
ElementDefinition.type.profile | Max Cardinality changed from * to 1
ElementDefinition.type.targetProfile | Added Element
ElementDefinition.type.versioning | Added Element
ElementDefinition.defaultValue[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.orderMeaning | Added Element
ElementDefinition.fixed[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.pattern[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.example | Renamed from example[x] to example Max Cardinality changed from 1 to * Remove boolean, Remove integer, Remove decimal, Remove base64Binary, Remove instant, Remove string, Remove uri, Remove date, Remove dateTime, Remove time, Remove code, Remove oid, Remove id, Remove unsignedInt, Remove positiveInt, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Quantity, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.example.label | Added Element
ElementDefinition.example.value[x] | Added Element
ElementDefinition.minValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.maxValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.constraint.expression | Added Element
ElementDefinition.constraint.xpath | Min Cardinality changed from 1 to 0
ElementDefinition.constraint.source | Added Element
ElementDefinition.mapping.comment | Added Element
Extension
Extension.value[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
HumanName
HumanName.family | Max Cardinality changed from * to 1
Identifier
Meta
Money | Changed from a constraint to a specialization
Narrative
ParameterDefinition | Added Type
Period
Quantity
Range
Ratio
Reference
Reference.identifier | Added Element
RelatedArtifact | Added Type
SampledData
Signature
Signature.onBehalfOf[x] | Added Element
Signature.contentType | Min Cardinality changed from 1 to 0
Signature.blob | Min Cardinality changed from 1 to 0
SimpleQuantity
Timing
Timing.repeat.bounds[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}, Add Duration

Timing.repeat.countMax | Added Element
Timing.repeat.durationUnit | Renamed from durationUnits to durationUnit
Timing.repeat.periodUnit | Renamed from periodUnits to periodUnit
Timing.repeat.dayOfWeek | Added Element
Timing.repeat.timeOfDay | Added Element
Timing.repeat.when | Max Cardinality changed from 1 to * Add Codes MORN, AFT, EVE, NIGHT, PHS
Timing.repeat.offset | Added Element
TriggerDefinition | Added Type
UsageContext | Added Type
base64Binary | No Changes
boolean | No Changes
code | No Changes
date | No Changes
dateTime | No Changes
decimal | No Changes
id | No Changes
instant | No Changes
integer | No Changes
markdown | No Changes
oid | No Changes
positiveInt | No Changes
string | No Changes
time | No Changes
unsignedInt | No Changes
uri | No Changes
uuid | No Changes
xhtml | Added Type
Resources
Account
Account.status | Add Binding http://hl7.org/fhir/ValueSet/account-status (required)
Account.period | Added Element
Account.active | Renamed from activePeriod to active
Account.balance | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Account.coverage | Added Element
Account.coverage.coverage | Added Element
Account.coverage.priority | Added Element
Account.guarantor | Added Element
Account.guarantor.party | Added Element
Account.guarantor.onHold | Added Element
Account.guarantor.period | Added Element
Account.currency | deleted
Account.coveragePeriod | deleted
ActivityDefinition | Added Resource
AdverseEvent | Added Resource
AllergyIntolerance
AllergyIntolerance.clinicalStatus | Added Element
AllergyIntolerance.verificationStatus | Added Element
AllergyIntolerance.category | Max Cardinality changed from 1 to * Remove Code other Add Code biologic
AllergyIntolerance.criticality | Remove Codes CRITL, CRITH, CRITU Add Codes low, high, unable-to-assess
AllergyIntolerance.code | Renamed from substance to code Min Cardinality changed from 1 to 0
AllergyIntolerance.onset[x] | Renamed from onset to onset[x] Add Age, Add Period, Add Range, Add string
AllergyIntolerance.assertedDate | Added Element
AllergyIntolerance.asserter | Renamed from reporter to asserter
AllergyIntolerance.lastOccurrence | Renamed from lastOccurence to lastOccurrence
AllergyIntolerance.note | Max Cardinality changed from 1 to *
AllergyIntolerance.reaction.note | Max Cardinality changed from 1 to *
AllergyIntolerance.recordedDate | deleted
AllergyIntolerance.status | deleted
AllergyIntolerance.reaction.certainty | deleted
Appointment
Appointment.status | Add Code entered-in-error
Appointment.serviceCategory | Added Element
Appointment.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *
Appointment.specialty | Added Element
Appointment.appointmentType | Added Element
Appointment.reason | Max Cardinality changed from 1 to * Remove Binding http://hl7.org/fhir/ValueSet/encounter-reason (required)
Appointment.indication | Added Element
Appointment.supportingInformation | Added Element
Appointment.created | Added Element
Appointment.incomingReferral | Added Element
Appointment.participant.type | Change binding strength from required to extensible
Appointment.requestedPeriod | Added Element

AppointmentResponse
AppointmentResponse.participantType | Change binding strength from required to extensible
AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participantstatus to http://hl7.org/fhir/ValueSet/participationstatus
AuditEvent
AuditEvent.type | Added Element
AuditEvent.subtype | Added Element
AuditEvent.action | Added Element
AuditEvent.recorded | Added Element
AuditEvent.outcome | Added Element
AuditEvent.outcomeDesc | Added Element
AuditEvent.purposeOfEvent | Added Element
AuditEvent.agent | Renamed from participant to agent
AuditEvent.agent.role | Change value set from http://hl7.org/fhir/ValueSet/dicm-402-roleid to http://hl7.org/fhir/ValueSet/security-role-type
AuditEvent.agent.purposeOfUse | Type changed from Coding to CodeableConcept
AuditEvent.entity | Renamed from object to entity
AuditEvent.entity.type | Change value set from http://hl7.org/fhir/ValueSet/object-type to http://hl7.org/fhir/ValueSet/audit-entity-type
AuditEvent.entity.lifecycle | Change value set from http://hl7.org/fhir/ValueSet/object-lifecycle to http://hl7.org/fhir/ValueSet/object-lifecycle-events
AuditEvent.event | deleted
Basic
Binary
Binary.securityContext | Added Element
BodySite
BodySite.active | Added Element
BodySite.qualifier | Renamed from modifier to qualifier
Bundle
Bundle.identifier | Added Element
Bundle.entry.response.outcome | Added Element
CapabilityStatement | Added Resource
CarePlan
CarePlan.definition | Added Element
CarePlan.basedOn | Added Element
CarePlan.replaces | Added Element
CarePlan.partOf | Added Element
CarePlan.status | Remove Code proposed Add Codes suspended, entered-in-error, unknown
CarePlan.intent | Added Element
CarePlan.title | Added Element
CarePlan.subject | Min Cardinality changed from 0 to 1
CarePlan.author | Add Reference(CareTeam)
CarePlan.careTeam | Added Element
CarePlan.supportingInfo | Renamed from support to supportingInfo
CarePlan.activity.outcomeCodeableConcept | Added Element
CarePlan.activity.outcomeReference | Renamed from actionResulting to outcomeReference
CarePlan.activity.reference | Remove Reference(DiagnosticOrder), Remove Reference(MedicationOrder), Remove Reference(Order), Remove Reference(ProcessRequest), Remove Reference(SupplyRequest), Add Reference(MedicationRequest), Add Reference(Task), Add Reference(RequestGroup)
CarePlan.activity.detail.definition | Added Element
CarePlan.activity.detail.status | Min Cardinality changed from 0 to 1 Add Code unknown
CarePlan.activity.detail.statusReason | Type changed from CodeableConcept to string
CarePlan.activity.detail.prohibited | Min Cardinality changed from 1 to 0 Default Value "false" added
CarePlan.activity.detail.performer | Add Reference(CareTeam)
CarePlan.note | Max Cardinality changed from 1 to *
CarePlan.modified | deleted
CarePlan.relatedPlan | deleted
CarePlan.participant | deleted
CareTeam | Added Resource
ChargeItem | Added Resource
Claim
Claim.status | Added Element
Claim.type | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/claim-type-link to http://hl7.org/fhir/ValueSet/claim-type
Claim.subType | Added Element
Claim.use | Change value set from http://hl7.org/fhir/ValueSet/claim-use-link to http://hl7.org/fhir/ValueSet/claim-use
Claim.patient | Min Cardinality changed from 1 to 0
Claim.billablePeriod | Added Element
Claim.insurer | Added Element
Claim.priority | Type changed from Coding to CodeableConcept
Claim.fundsReserve | Type changed from Coding to CodeableConcept
Claim.related | Added Element
Claim.related.claim | Added Element
Claim.related.relationship | Added Element
Claim.related.reference | Added Element
Claim.prescription | Remove Reference(MedicationOrder), Add Reference(MedicationRequest)

Claim.originalPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
Claim.payee.type | Min Cardinality changed from 0 to 1 Type changed from Coding to CodeableConcept
Claim.payee.resourceType | Added Element
Claim.payee.party | Added Element
Claim.careTeam | Added Element
Claim.careTeam.sequence | Added Element
Claim.careTeam.provider | Added Element
Claim.careTeam.responsible | Added Element
Claim.careTeam.role | Added Element
Claim.careTeam.qualification | Added Element
Claim.information | Added Element
Claim.information.sequence | Added Element
Claim.information.category | Added Element
Claim.information.code | Added Element
Claim.information.timing[x] | Added Element
Claim.information.value[x] | Added Element
Claim.information.reason | Added Element
Claim.diagnosis.diagnosis[x] | Renamed from diagnosis to diagnosis[x] Remove Coding, Add CodeableConcept, Add Reference(Condition)
Claim.diagnosis.type | Added Element
Claim.diagnosis.packageCode | Added Element
Claim.procedure | Added Element
Claim.procedure.sequence | Added Element
Claim.procedure.date | Added Element
Claim.procedure.procedure[x] | Added Element
Claim.insurance | Added Element
Claim.insurance.sequence | Added Element
Claim.insurance.focal | Added Element
Claim.insurance.coverage | Added Element
Claim.insurance.businessArrangement | Added Element
Claim.insurance.preAuthRef | Added Element
Claim.insurance.claimResponse | Added Element
Claim.accident | Type changed from date to BackboneElement
Claim.accident.date | Added Element
Claim.accident.type | Added Element
Claim.accident.location[x] | Added Element
Claim.employmentImpacted | Added Element
Claim.hospitalization | Added Element
Claim.item.careTeamLinkId | Added Element
Claim.item.procedureLinkId | Added Element
Claim.item.informationLinkId | Added Element
Claim.item.revenue | Added Element
Claim.item.category | Added Element
Claim.item.service | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
Claim.item.modifier | Type changed from Coding to CodeableConcept
Claim.item.programCode | Added Element
Claim.item.serviced[x] | Added Element
Claim.item.location[x] | Added Element
Claim.item.unitPrice | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.net | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.udi | Max Cardinality changed from 1 to * Type changed from Coding to Reference(Device)
Claim.item.bodySite | Type changed from Coding to CodeableConcept
Claim.item.subSite | Type changed from Coding to CodeableConcept
Claim.item.encounter | Added Element
Claim.item.detail.revenue | Added Element
Claim.item.detail.category | Added Element
Claim.item.detail.service | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
Claim.item.detail.modifier | Added Element
Claim.item.detail.programCode | Added Element
Claim.item.detail.unitPrice | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.detail.net | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.detail.udi | Max Cardinality changed from 1 to * Type changed from Coding to Reference(Device)
Claim.item.detail.subDetail.revenue | Added Element
Claim.item.detail.subDetail.category | Added Element
Claim.item.detail.subDetail.service | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
Claim.item.detail.subDetail.modifier | Added Element
Claim.item.detail.subDetail.programCode | Added Element
Claim.item.detail.subDetail.unitPrice | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.detail.subDetail.net | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Claim.item.detail.subDetail.udi | Max Cardinality changed from 1 to * Type changed from Coding to Reference(Device)
Claim.total | Added Element
Claim.ruleset | deleted
Claim.originalRuleset | deleted

Claim.target | deleted
Claim.payee.provider | deleted
Claim.payee.organization | deleted
Claim.payee.person | deleted
Claim.condition | deleted
Claim.coverage | deleted
Claim.exception | deleted
Claim.school | deleted
Claim.accidentType | deleted
Claim.interventionException | deleted
Claim.item.type | deleted
Claim.item.provider | deleted
Claim.item.serviceDate | deleted
Claim.item.points | deleted
Claim.item.detail.type | deleted
Claim.item.detail.points | deleted
Claim.item.detail.subDetail.type | deleted
Claim.item.detail.subDetail.points | deleted
Claim.item.prosthesis | deleted
Claim.additionalMaterials | deleted
Claim.missingTeeth | deleted
ClaimResponse
ClaimResponse.status | Added Element
ClaimResponse.patient | Added Element
ClaimResponse.insurer | Added Element
ClaimResponse.outcome | Type changed from code to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/remittance-outcome (required)
ClaimResponse.payeeType | Type changed from Coding to CodeableConcept
ClaimResponse.item.adjudication.category | Added Element
ClaimResponse.item.adjudication.reason | Added Element
ClaimResponse.item.adjudication.amount | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.item.detail.noteNumber | Added Element
ClaimResponse.item.detail.adjudication | Remove BackboneElement
ClaimResponse.item.detail.subDetail.noteNumber | Added Element
ClaimResponse.item.detail.subDetail.adjudication | Remove BackboneElement
ClaimResponse.addItem.revenue | Added Element
ClaimResponse.addItem.category | Added Element
ClaimResponse.addItem.service | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
ClaimResponse.addItem.modifier | Added Element
ClaimResponse.addItem.fee | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.addItem.noteNumber | Added Element
ClaimResponse.addItem.adjudication | Remove BackboneElement
ClaimResponse.addItem.detail.revenue | Added Element
ClaimResponse.addItem.detail.category | Added Element
ClaimResponse.addItem.detail.service | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
ClaimResponse.addItem.detail.modifier | Added Element
ClaimResponse.addItem.detail.fee | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.addItem.detail.noteNumber | Added Element
ClaimResponse.addItem.detail.adjudication | Remove BackboneElement
ClaimResponse.error.code | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/adjudication-error (required)
ClaimResponse.totalCost | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.unallocDeductable | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.totalBenefit | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
ClaimResponse.payment | Added Element
ClaimResponse.payment.type | Added Element
ClaimResponse.payment.adjustment | Added Element
ClaimResponse.payment.adjustmentReason | Added Element
ClaimResponse.payment.date | Added Element
ClaimResponse.payment.amount | Added Element
ClaimResponse.payment.identifier | Added Element
ClaimResponse.form | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/forms (required)
ClaimResponse.processNote | Added Element
ClaimResponse.processNote.number | Added Element
ClaimResponse.processNote.type | Added Element
ClaimResponse.processNote.text | Added Element
ClaimResponse.processNote.language | Added Element
ClaimResponse.communicationRequest | Added Element
ClaimResponse.insurance | Added Element
ClaimResponse.insurance.sequence | Added Element
ClaimResponse.insurance.focal | Added Element
ClaimResponse.insurance.coverage | Added Element
ClaimResponse.insurance.businessArrangement | Added Element

ClaimResponse.insurance.preAuthRef | Added Element
ClaimResponse.insurance.claimResponse | Added Element
ClaimResponse.ruleset | deleted
ClaimResponse.originalRuleset | deleted
ClaimResponse.organization | deleted
ClaimResponse.item.adjudication.code | deleted
ClaimResponse.item.detail.adjudication.code | deleted
ClaimResponse.item.detail.adjudication.amount | deleted
ClaimResponse.item.detail.adjudication.value | deleted
ClaimResponse.item.detail.subDetail.adjudication.code | deleted
ClaimResponse.item.detail.subDetail.adjudication.amount | deleted
ClaimResponse.item.detail.subDetail.adjudication.value | deleted
ClaimResponse.addItem.noteNumberLinkId | deleted
ClaimResponse.addItem.adjudication.code | deleted
ClaimResponse.addItem.adjudication.amount | deleted
ClaimResponse.addItem.adjudication.value | deleted
ClaimResponse.addItem.detail.adjudication.code | deleted
ClaimResponse.addItem.detail.adjudication.amount | deleted
ClaimResponse.addItem.detail.adjudication.value | deleted
ClaimResponse.paymentAdjustment | deleted
ClaimResponse.paymentAdjustmentReason | deleted
ClaimResponse.paymentDate | deleted
ClaimResponse.paymentAmount | deleted
ClaimResponse.paymentRef | deleted
ClaimResponse.note | deleted
ClaimResponse.coverage | deleted
ClinicalImpression
ClinicalImpression.identifier | Added Element
ClinicalImpression.status | Remove Code in-progress Add Code draft
ClinicalImpression.code | Added Element
ClinicalImpression.subject | Renamed from patient to subject Add Reference(Group)
ClinicalImpression.context | Added Element
ClinicalImpression.effective[x] | Added Element
ClinicalImpression.investigation | Added Element
ClinicalImpression.investigation.code | Added Element
ClinicalImpression.investigation.item | Added Element
ClinicalImpression.protocol | Max Cardinality changed from 1 to *
ClinicalImpression.finding.item[x] | Renamed from item to item[x] Add Reference(Condition), Add Reference(Observation)
ClinicalImpression.finding.basis | Added Element
ClinicalImpression.prognosisCodeableConcept | Added Element
ClinicalImpression.prognosisReference | Added Element
ClinicalImpression.action | Remove Reference(MedicationOrder), Remove Reference(DiagnosticOrder), Remove Reference(NutritionOrder), Remove Reference(SupplyRequest), Add Reference(MedicationRequest)
ClinicalImpression.note | Added Element
ClinicalImpression.trigger[x] | deleted
ClinicalImpression.investigations | deleted
ClinicalImpression.finding.cause | deleted
ClinicalImpression.resolved | deleted
ClinicalImpression.ruledOut | deleted
ClinicalImpression.prognosis | deleted
ClinicalImpression.plan | deleted
CodeSystem | Added Resource
Communication
Communication.definition | Added Element
Communication.basedOn | Renamed from requestDetail to basedOn Max Cardinality changed from 1 to * Type changed from Reference(CommunicationRequest) to Reference(Resource)
Communication.partOf | Added Element
Communication.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-status to http://hl7.org/fhir/ValueSet/event-status
Communication.notDone | Added Element
Communication.notDoneReason | Added Element
Communication.category | Max Cardinality changed from 1 to *
Communication.subject | Add Reference(Group)
Communication.topic | Added Element
Communication.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Communication.reasonCode | Renamed from reason to reasonCode
Communication.reasonReference | Added Element
Communication.note | Added Element
CommunicationRequest
CommunicationRequest.basedOn | Added Element
CommunicationRequest.replaces | Added Element
CommunicationRequest.groupIdentifier | Added Element

CommunicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-request-status to http://hl7.org/fhir/ValueSet/request-status
CommunicationRequest.category | Max Cardinality changed from 1 to *
CommunicationRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)
CommunicationRequest.subject | Add Reference(Group)
CommunicationRequest.recipient | Add Reference(Group), Add Reference(CareTeam)
CommunicationRequest.topic | Added Element
CommunicationRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
CommunicationRequest.occurrence[x] | Added Element
CommunicationRequest.authoredOn | Renamed from requestedOn to authoredOn
CommunicationRequest.requester | Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)
CommunicationRequest.requester.agent | Added Element
CommunicationRequest.requester.onBehalfOf | Added Element
CommunicationRequest.reasonCode | Renamed from reason to reasonCode
CommunicationRequest.reasonReference | Added Element
CommunicationRequest.note | Added Element
CommunicationRequest.scheduled[x] | deleted
CompartmentDefinition | Added Resource
Composition
Composition.confidentiality | Change value set from http://hl7.org/fhir/ValueSet/v3-Confidentiality to http://hl7.org/fhir/ValueSet/v3-ConfidentialityClassification
Composition.relatesTo | Added Element
Composition.relatesTo.code | Added Element
Composition.relatesTo.target[x] | Added Element
ConceptMap
ConceptMap.title | Added Element
ConceptMap.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
ConceptMap.experimental | Now marked as Modifier
ConceptMap.contact | Type changed from BackboneElement to ContactDetail
ConceptMap.description | Type changed from string to markdown
ConceptMap.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
ConceptMap.jurisdiction | Renamed from useContext to jurisdiction Change value set from http://hl7.org/fhir/ValueSet/use-context to http://hl7.org/fhir/ValueSet/jurisdiction
ConceptMap.purpose | Renamed from requirements to purpose Type changed from string to markdown
ConceptMap.copyright | Type changed from string to markdown
ConceptMap.source[x] | Min Cardinality changed from 1 to 0 Remove Reference(StructureDefinition)
ConceptMap.target[x] | Min Cardinality changed from 1 to 0 Remove Reference(StructureDefinition)
ConceptMap.group | Added Element
ConceptMap.group.source | Added Element
ConceptMap.group.sourceVersion | Added Element
ConceptMap.group.target | Added Element
ConceptMap.group.targetVersion | Added Element
ConceptMap.group.element | Min Cardinality changed from 0 to 1
ConceptMap.group.element.display | Added Element
ConceptMap.group.element.target.display | Added Element
ConceptMap.group.element.target.equivalence | Min Cardinality changed from 1 to 0 Add Code relatedto Default Value "equivalent" added
ConceptMap.group.element.target.comment | Added Element
ConceptMap.group.element.target.dependsOn.property | Added Element
ConceptMap.group.element.target.dependsOn.system | Added Element
ConceptMap.group.element.target.dependsOn.display | Added Element
ConceptMap.group.unmapped | Added Element
ConceptMap.group.unmapped.mode | Added Element
ConceptMap.group.unmapped.code | Added Element
ConceptMap.group.unmapped.display | Added Element
ConceptMap.group.unmapped.url | Added Element
ConceptMap.contact.name | deleted
ConceptMap.contact.telecom | deleted
ConceptMap.element.codeSystem | deleted
ConceptMap.element.target.codeSystem | deleted
ConceptMap.element.target.comments | deleted
ConceptMap.element.target.dependsOn.element | deleted
ConceptMap.element.target.dependsOn.codeSystem | deleted
Condition
Condition.clinicalStatus | Add Binding http://hl7.org/fhir/ValueSet/condition-clinical (required)
Condition.verificationStatus | Min Cardinality changed from 1 to 0 Default Value "unknown" added
Condition.category | Max Cardinality changed from 1 to *
Condition.code | Min Cardinality changed from 1 to 0
Condition.subject | Renamed from patient to subject Add Reference(Group)
Condition.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
Condition.abatement[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
Condition.assertedDate | Renamed from dateRecorded to assertedDate Type changed from date to dateTime

Condition.asserter | Add Reference(RelatedPerson)
Condition.evidence.code | Max Cardinality changed from 1 to *
Condition.note | Renamed from notes to note Max Cardinality changed from 1 to * Type changed from string to Annotation
Consent | Added Resource
Contract
Contract.status | Added Element
Contract.topic | Added Element
Contract.decisionType | Added Element
Contract.contentDerivative | Added Element
Contract.securityLabel | Added Element
Contract.agent | Added Element
Contract.agent.actor | Added Element
Contract.agent.role | Added Element
Contract.signer.signature | Max Cardinality changed from 1 to * Type changed from string to Signature
Contract.valuedItem.unitPrice | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Contract.valuedItem.net | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Contract.term.topic | Renamed from subject to topic Max Cardinality changed from 1 to *
Contract.term.securityLabel | Added Element
Contract.term.agent | Added Element
Contract.term.agent.actor | Added Element
Contract.term.agent.role | Added Element
Contract.term.valuedItem.unitPrice | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Contract.term.valuedItem.net | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Contract.actor | deleted
Contract.term.actor | deleted
Coverage
Coverage.status | Added Element
Coverage.type | Type changed from Coding to CodeableConcept
Coverage.policyHolder | Added Element
Coverage.subscriber | Add Reference(RelatedPerson) No longer marked as Modifier
Coverage.subscriberId | Type changed from Identifier to string
Coverage.beneficiary | Added Element
Coverage.relationship | Added Element
Coverage.payor | Added Element
Coverage.grouping | Added Element
Coverage.grouping.group | Added Element
Coverage.grouping.groupDisplay | Added Element
Coverage.grouping.subGroup | Added Element
Coverage.grouping.subGroupDisplay | Added Element
Coverage.grouping.plan | Added Element
Coverage.grouping.planDisplay | Added Element
Coverage.grouping.subPlan | Added Element
Coverage.grouping.subPlanDisplay | Added Element
Coverage.grouping.class | Added Element
Coverage.grouping.classDisplay | Added Element
Coverage.grouping.subClass | Added Element
Coverage.grouping.subClassDisplay | Added Element
Coverage.dependent | Type changed from positiveInt to string
Coverage.sequence | Type changed from positiveInt to string
Coverage.order | Added Element
Coverage.network | Type changed from Identifier to string
Coverage.issuer | deleted
Coverage.bin | deleted
Coverage.group | deleted
Coverage.plan | deleted
Coverage.subPlan | deleted
DataElement
DataElement.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
DataElement.experimental | Now marked as Modifier
DataElement.title | Added Element
DataElement.contact | Type changed from BackboneElement to ContactDetail
DataElement.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
DataElement.jurisdiction | Added Element
DataElement.copyright | Type changed from string to markdown
DataElement.mapping.comment | Renamed from comments to comment
DataElement.contact.name | deleted
DataElement.contact.telecom | deleted
DetectedIssue
DetectedIssue.status | Added Element
Device
Device.udi | Type changed from string to BackboneElement

Device.udi.deviceIdentifier | Added Element
Device.udi.name | Added Element
Device.udi.jurisdiction | Added Element
Device.udi.carrierHRF | Added Element
Device.udi.carrierAIDC | Added Element
Device.udi.issuer | Added Element
Device.udi.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/devicestatus to http://hl7.org/fhir/ValueSet/device-status
Device.type | Min Cardinality changed from 1 to 0
Device.expirationDate | Renamed from expiry to expirationDate
Device.safety | Added Element
DeviceComponent
DeviceComponent.lastSystemChange | Min Cardinality changed from 1 to 0
DeviceComponent.operationalStatus | Add Binding http://hl7.org/fhir/ValueSet/operational-status (extensible)
DeviceComponent.parameterGroup | Add Binding http://hl7.org/fhir/ValueSet/parameter-group (extensible)
DeviceComponent.productionSpecification.specType | Add Binding http://hl7.org/fhir/ValueSet/specification-type (extensible)
DeviceComponent.languageCode | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages
DeviceMetric
DeviceMetric.operationalStatus | Add Code entered-in-error
DeviceRequest
DeviceRequest.definition | Added Element
DeviceRequest.basedOn | Added Element
DeviceRequest.priorRequest | Added Element
DeviceRequest.groupIdentifier | Added Element
DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-status to http://hl7.org/fhir/ValueSet/request-status
DeviceRequest.intent | Added Element
DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/device-use-request-priority to http://hl7.org/fhir/ValueSet/request-priority Default Value "If missing, normal priority" added
DeviceRequest.code[x] | Added Element
DeviceRequest.subject | Add Reference(Group), Add Reference(Location), Add Reference(Device)
DeviceRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
DeviceRequest.occurrence[x] | Added Element
DeviceRequest.authoredOn | Added Element
DeviceRequest.requester | Added Element
DeviceRequest.requester.agent | Added Element
DeviceRequest.requester.onBehalfOf | Added Element
DeviceRequest.performerType | Added Element
DeviceRequest.performer | Added Element
DeviceRequest.reasonCode | Renamed from indication to reasonCode
DeviceRequest.reasonReference | Added Element
DeviceRequest.supportingInfo | Added Element
DeviceRequest.note | Renamed from notes to note Type changed from string to Annotation
DeviceRequest.relevantHistory | Added Element
DeviceRequest.bodySite[x] | deleted
DeviceRequest.device | deleted
DeviceRequest.prnReason | deleted
DeviceRequest.orderedOn | deleted
DeviceRequest.recordedOn | deleted
DeviceRequest.timing[x] | deleted
DeviceUseStatement
DeviceUseStatement.status | Added Element
DeviceUseStatement.subject | Add Reference(Group)
DeviceUseStatement.source | Added Element
DeviceUseStatement.bodySite | Renamed from bodySite[x] to bodySite Remove Reference(BodySite)
DeviceUseStatement.note | Added Element
DeviceUseStatement.notes | deleted
DiagnosticReport
DiagnosticReport.basedOn | Renamed from request to basedOn Remove Reference(DiagnosticOrder), Add Reference(CarePlan), Add Reference(ImmunizationRecommendation), Add Reference(MedicationRequest), Add Reference(NutritionOrder)
DiagnosticReport.status | Add Codes preliminary, amended, unknown
DiagnosticReport.subject | Min Cardinality changed from 1 to 0
DiagnosticReport.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
DiagnosticReport.effective[x] | Min Cardinality changed from 1 to 0
DiagnosticReport.issued | Min Cardinality changed from 1 to 0
DiagnosticReport.performer | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Organization)
DiagnosticReport.performer.role | Added Element
DiagnosticReport.performer.actor | Added Element
DiagnosticReport.imagingStudy | Remove Reference(ImagingObjectSelection), Add Reference(ImagingManifest)
DocumentManifest
DocumentReference
DocumentReference.docStatus | Type changed from CodeableConcept to code

DocumentReference.content.format | Max Cardinality changed from * to 1
DomainResource
EligibilityRequest
EligibilityRequest.status | Added Element
EligibilityRequest.priority | Added Element
EligibilityRequest.patient | Added Element
EligibilityRequest.serviced[x] | Added Element
EligibilityRequest.enterer | Added Element
EligibilityRequest.insurer | Added Element
EligibilityRequest.facility | Added Element
EligibilityRequest.coverage | Added Element
EligibilityRequest.businessArrangement | Added Element
EligibilityRequest.benefitCategory | Added Element
EligibilityRequest.benefitSubCategory | Added Element
EligibilityRequest.ruleset | deleted
EligibilityRequest.originalRuleset | deleted
EligibilityRequest.target | deleted
EligibilityResponse
EligibilityResponse.status | Added Element
EligibilityResponse.outcome | Type changed from code to CodeableConcept Add Code partial
EligibilityResponse.insurer | Added Element
EligibilityResponse.inforce | Added Element
EligibilityResponse.insurance | Added Element
EligibilityResponse.insurance.coverage | Added Element
EligibilityResponse.insurance.contract | Added Element
EligibilityResponse.insurance.benefitBalance | Added Element
EligibilityResponse.insurance.benefitBalance.category | Added Element
EligibilityResponse.insurance.benefitBalance.subCategory | Added Element
EligibilityResponse.insurance.benefitBalance.excluded | Added Element
EligibilityResponse.insurance.benefitBalance.name | Added Element
EligibilityResponse.insurance.benefitBalance.description | Added Element
EligibilityResponse.insurance.benefitBalance.network | Added Element
EligibilityResponse.insurance.benefitBalance.unit | Added Element
EligibilityResponse.insurance.benefitBalance.term | Added Element
EligibilityResponse.insurance.benefitBalance.financial | Added Element
EligibilityResponse.insurance.benefitBalance.financial.type | Added Element
EligibilityResponse.insurance.benefitBalance.financial.allowed[x] | Added Element
EligibilityResponse.insurance.benefitBalance.financial.used[x] | Added Element
EligibilityResponse.form | Added Element
EligibilityResponse.error | Added Element
EligibilityResponse.error.code | Added Element
EligibilityResponse.ruleset | deleted
EligibilityResponse.originalRuleset | deleted
EligibilityResponse.organization | deleted
Encounter
Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status
Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status
Encounter.class | Type changed from code to Coding Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/encounter-class to http://hl7.org/fhir/ValueSet/v3-ActEncounterCode
Encounter.classHistory | Added Element
Encounter.classHistory.class | Added Element
Encounter.classHistory.period | Added Element
Encounter.subject | Renamed from patient to subject Add Reference(Group)
Encounter.length | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration
Encounter.diagnosis | Added Element
Encounter.diagnosis.condition | Added Element
Encounter.diagnosis.role | Added Element
Encounter.diagnosis.rank | Added Element
Encounter.account | Added Element
Encounter.indication | deleted
Encounter.hospitalization.admittingDiagnosis | deleted
Encounter.hospitalization.dischargeDiagnosis | deleted
Endpoint | Added Resource
EnrollmentRequest
EnrollmentRequest.status | Added Element
EnrollmentRequest.insurer | Added Element
EnrollmentRequest.subject | Min Cardinality changed from 1 to 0
EnrollmentRequest.coverage | Min Cardinality changed from 1 to 0
EnrollmentRequest.ruleset | deleted
EnrollmentRequest.originalRuleset | deleted
EnrollmentRequest.target | deleted
EnrollmentRequest.relationship | deleted

EnrollmentResponse
EnrollmentResponse.status | Added Element
EnrollmentResponse.outcome | Type changed from code to CodeableConcept Add Code partial
EnrollmentResponse.ruleset | deleted
EnrollmentResponse.originalRuleset | deleted
EpisodeOfCare
EpisodeOfCare.status | Add Code entered-in-error
EpisodeOfCare.statusHistory.status | Add Code entered-in-error
EpisodeOfCare.diagnosis | Added Element
EpisodeOfCare.diagnosis.condition | Added Element
EpisodeOfCare.diagnosis.role | Added Element
EpisodeOfCare.diagnosis.rank | Added Element
EpisodeOfCare.team | Renamed from careTeam to team Type changed from BackboneElement to Reference(CareTeam)
EpisodeOfCare.account | Added Element
EpisodeOfCare.condition | deleted
EpisodeOfCare.careTeam.role | deleted
EpisodeOfCare.careTeam.period | deleted
EpisodeOfCare.careTeam.member | deleted
ExpansionProfile | Added Resource
ExplanationOfBenefit
ExplanationOfBenefit.status | Added Element
ExplanationOfBenefit.type | Added Element
ExplanationOfBenefit.subType | Added Element
ExplanationOfBenefit.patient | Added Element
ExplanationOfBenefit.billablePeriod | Added Element
ExplanationOfBenefit.enterer | Added Element
ExplanationOfBenefit.insurer | Added Element
ExplanationOfBenefit.provider | Added Element
ExplanationOfBenefit.referral | Added Element
ExplanationOfBenefit.facility | Added Element
ExplanationOfBenefit.claim | Added Element
ExplanationOfBenefit.claimResponse | Added Element
ExplanationOfBenefit.outcome | Type changed from code to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/remittance-outcome (required)
ExplanationOfBenefit.related | Added Element
ExplanationOfBenefit.related.claim | Added Element
ExplanationOfBenefit.related.relationship | Added Element
ExplanationOfBenefit.related.reference | Added Element
ExplanationOfBenefit.prescription | Added Element
ExplanationOfBenefit.originalPrescription | Added Element
ExplanationOfBenefit.payee | Added Element
ExplanationOfBenefit.payee.type | Added Element
ExplanationOfBenefit.payee.resourceType | Added Element
ExplanationOfBenefit.payee.party | Added Element
ExplanationOfBenefit.information | Added Element
ExplanationOfBenefit.information.sequence | Added Element
ExplanationOfBenefit.information.category | Added Element
ExplanationOfBenefit.information.code | Added Element
ExplanationOfBenefit.information.timing[x] | Added Element
ExplanationOfBenefit.information.value[x] | Added Element
ExplanationOfBenefit.information.reason | Added Element
ExplanationOfBenefit.careTeam | Added Element
ExplanationOfBenefit.careTeam.sequence | Added Element
ExplanationOfBenefit.careTeam.provider | Added Element
ExplanationOfBenefit.careTeam.responsible | Added Element
ExplanationOfBenefit.careTeam.role | Added Element
ExplanationOfBenefit.careTeam.qualification | Added Element
ExplanationOfBenefit.diagnosis | Added Element
ExplanationOfBenefit.diagnosis.sequence | Added Element
ExplanationOfBenefit.diagnosis.diagnosis[x] | Added Element
ExplanationOfBenefit.diagnosis.type | Added Element
ExplanationOfBenefit.diagnosis.packageCode | Added Element
ExplanationOfBenefit.procedure | Added Element
ExplanationOfBenefit.procedure.sequence | Added Element
ExplanationOfBenefit.procedure.date | Added Element
ExplanationOfBenefit.procedure.procedure[x] | Added Element
ExplanationOfBenefit.precedence | Added Element
ExplanationOfBenefit.insurance | Added Element
ExplanationOfBenefit.insurance.coverage | Added Element
ExplanationOfBenefit.insurance.preAuthRef | Added Element
ExplanationOfBenefit.accident | Added Element
ExplanationOfBenefit.accident.date | Added Element

ExplanationOfBenefit.accident.type | Added Element
ExplanationOfBenefit.accident.location[x] | Added Element
ExplanationOfBenefit.employmentImpacted | Added Element
ExplanationOfBenefit.hospitalization | Added Element
ExplanationOfBenefit.item | Added Element
ExplanationOfBenefit.item.sequence | Added Element
ExplanationOfBenefit.item.careTeamLinkId | Added Element
ExplanationOfBenefit.item.diagnosisLinkId | Added Element
ExplanationOfBenefit.item.procedureLinkId | Added Element
ExplanationOfBenefit.item.informationLinkId | Added Element
ExplanationOfBenefit.item.revenue | Added Element
ExplanationOfBenefit.item.category | Added Element
ExplanationOfBenefit.item.service | Added Element
ExplanationOfBenefit.item.modifier | Added Element
ExplanationOfBenefit.item.programCode | Added Element
ExplanationOfBenefit.item.serviced[x] | Added Element
ExplanationOfBenefit.item.location[x] | Added Element
ExplanationOfBenefit.item.quantity | Added Element
ExplanationOfBenefit.item.unitPrice | Added Element
ExplanationOfBenefit.item.factor | Added Element
ExplanationOfBenefit.item.net | Added Element
ExplanationOfBenefit.item.udi | Added Element
ExplanationOfBenefit.item.bodySite | Added Element
ExplanationOfBenefit.item.subSite | Added Element
ExplanationOfBenefit.item.encounter | Added Element
ExplanationOfBenefit.item.noteNumber | Added Element
ExplanationOfBenefit.item.adjudication | Added Element
ExplanationOfBenefit.item.adjudication.category | Added Element
ExplanationOfBenefit.item.adjudication.reason | Added Element
ExplanationOfBenefit.item.adjudication.amount | Added Element
ExplanationOfBenefit.item.adjudication.value | Added Element
ExplanationOfBenefit.item.detail | Added Element
ExplanationOfBenefit.item.detail.sequence | Added Element
ExplanationOfBenefit.item.detail.type | Added Element
ExplanationOfBenefit.item.detail.revenue | Added Element
ExplanationOfBenefit.item.detail.category | Added Element
ExplanationOfBenefit.item.detail.service | Added Element
ExplanationOfBenefit.item.detail.modifier | Added Element
ExplanationOfBenefit.item.detail.programCode | Added Element
ExplanationOfBenefit.item.detail.quantity | Added Element
ExplanationOfBenefit.item.detail.unitPrice | Added Element
ExplanationOfBenefit.item.detail.factor | Added Element
ExplanationOfBenefit.item.detail.net | Added Element
ExplanationOfBenefit.item.detail.udi | Added Element
ExplanationOfBenefit.item.detail.noteNumber | Added Element
ExplanationOfBenefit.item.detail.adjudication | Added Element
ExplanationOfBenefit.item.detail.subDetail | Added Element
ExplanationOfBenefit.item.detail.subDetail.sequence | Added Element
ExplanationOfBenefit.item.detail.subDetail.type | Added Element
ExplanationOfBenefit.item.detail.subDetail.revenue | Added Element
ExplanationOfBenefit.item.detail.subDetail.category | Added Element
ExplanationOfBenefit.item.detail.subDetail.service | Added Element
ExplanationOfBenefit.item.detail.subDetail.modifier | Added Element
ExplanationOfBenefit.item.detail.subDetail.programCode | Added Element
ExplanationOfBenefit.item.detail.subDetail.quantity | Added Element
ExplanationOfBenefit.item.detail.subDetail.unitPrice | Added Element
ExplanationOfBenefit.item.detail.subDetail.factor | Added Element
ExplanationOfBenefit.item.detail.subDetail.net | Added Element
ExplanationOfBenefit.item.detail.subDetail.udi | Added Element
ExplanationOfBenefit.item.detail.subDetail.noteNumber | Added Element
ExplanationOfBenefit.item.detail.subDetail.adjudication | Added Element
ExplanationOfBenefit.addItem | Added Element
ExplanationOfBenefit.addItem.sequenceLinkId | Added Element
ExplanationOfBenefit.addItem.revenue | Added Element
ExplanationOfBenefit.addItem.category | Added Element
ExplanationOfBenefit.addItem.service | Added Element
ExplanationOfBenefit.addItem.modifier | Added Element
ExplanationOfBenefit.addItem.fee | Added Element
ExplanationOfBenefit.addItem.noteNumber | Added Element
ExplanationOfBenefit.addItem.adjudication | Added Element
ExplanationOfBenefit.addItem.detail | Added Element
ExplanationOfBenefit.addItem.detail.revenue | Added Element

ExplanationOfBenefit.addItem.detail.category | Added Element
ExplanationOfBenefit.addItem.detail.service | Added Element
ExplanationOfBenefit.addItem.detail.modifier | Added Element
ExplanationOfBenefit.addItem.detail.fee | Added Element
ExplanationOfBenefit.addItem.detail.noteNumber | Added Element
ExplanationOfBenefit.addItem.detail.adjudication | Added Element
ExplanationOfBenefit.totalCost | Added Element
ExplanationOfBenefit.unallocDeductable | Added Element
ExplanationOfBenefit.totalBenefit | Added Element
ExplanationOfBenefit.payment | Added Element
ExplanationOfBenefit.payment.type | Added Element
ExplanationOfBenefit.payment.adjustment | Added Element
ExplanationOfBenefit.payment.adjustmentReason | Added Element
ExplanationOfBenefit.payment.date | Added Element
ExplanationOfBenefit.payment.amount | Added Element
ExplanationOfBenefit.payment.identifier | Added Element
ExplanationOfBenefit.form | Added Element
ExplanationOfBenefit.processNote | Added Element
ExplanationOfBenefit.processNote.number | Added Element
ExplanationOfBenefit.processNote.type | Added Element
ExplanationOfBenefit.processNote.text | Added Element
ExplanationOfBenefit.processNote.language | Added Element
ExplanationOfBenefit.benefitBalance | Added Element
ExplanationOfBenefit.benefitBalance.category | Added Element
ExplanationOfBenefit.benefitBalance.subCategory | Added Element
ExplanationOfBenefit.benefitBalance.excluded | Added Element
ExplanationOfBenefit.benefitBalance.name | Added Element
ExplanationOfBenefit.benefitBalance.description | Added Element
ExplanationOfBenefit.benefitBalance.network | Added Element
ExplanationOfBenefit.benefitBalance.unit | Added Element
ExplanationOfBenefit.benefitBalance.term | Added Element
ExplanationOfBenefit.benefitBalance.financial | Added Element
ExplanationOfBenefit.benefitBalance.financial.type | Added Element
ExplanationOfBenefit.benefitBalance.financial.allowed[x] | Added Element
ExplanationOfBenefit.benefitBalance.financial.used[x] | Added Element
ExplanationOfBenefit.request | deleted
ExplanationOfBenefit.ruleset | deleted
ExplanationOfBenefit.originalRuleset | deleted
ExplanationOfBenefit.requestProvider | deleted
ExplanationOfBenefit.requestOrganization | deleted
FamilyMemberHistory
FamilyMemberHistory.definition | Added Element
FamilyMemberHistory.notDone | Added Element
FamilyMemberHistory.notDoneReason | Added Element
FamilyMemberHistory.age[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.estimatedAge | Added Element
FamilyMemberHistory.deceased[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.reasonCode | Added Element
FamilyMemberHistory.reasonReference | Added Element
FamilyMemberHistory.note | Max Cardinality changed from 1 to *
FamilyMemberHistory.condition.onset[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Age}, Add Age
FamilyMemberHistory.condition.note | Max Cardinality changed from 1 to *
Flag
Flag.subject | Add Reference(PlanDefinition), Add Reference(Medication), Add Reference(Procedure)
Goal
Goal.status | Add Codes on-target, ahead-of-target, behind-target, entered-in-error
Goal.description | Type changed from string to CodeableConcept
Goal.target | Renamed from target[x] to target Remove date, Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}
Goal.target.measure | Added Element
Goal.target.detail[x] | Added Element
Goal.target.due[x] | Added Element
Goal.statusReason | Type changed from CodeableConcept to string
Goal.expressedBy | Renamed from author to expressedBy
Goal.outcomeCode | Added Element
Goal.outcomeReference | Added Element
Goal.outcome | deleted
GraphDefinition | Added Resource
Group
Group.active | Added Element
GuidanceResponse | Added Resource
HealthcareService
HealthcareService.active | Added Element

HealthcareService.category | Renamed from serviceCategory to category
HealthcareService.type | Added Element
HealthcareService.specialty | Added Element
HealthcareService.location | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
HealthcareService.name | Renamed from serviceName to name
HealthcareService.endpoint | Added Element
HealthcareService.serviceType | deleted
HealthcareService.publicKey | deleted
ImagingManifest | Added Resource
ImagingStudy
ImagingStudy.context | Renamed from order to context Max Cardinality changed from * to 1 Remove Reference(DiagnosticOrder), Add Reference(Encounter), Add Reference(EpisodeOfCare)
ImagingStudy.basedOn | Added Element
ImagingStudy.interpreter | Max Cardinality changed from 1 to *
ImagingStudy.endpoint | Added Element
ImagingStudy.numberOfSeries | Min Cardinality changed from 1 to 0
ImagingStudy.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.procedureReference | Renamed from procedure to procedureReference
ImagingStudy.procedureCode | Added Element
ImagingStudy.reason | Added Element
ImagingStudy.series.numberOfInstances | Min Cardinality changed from 1 to 0
ImagingStudy.series.endpoint | Added Element
ImagingStudy.series.performer | Added Element
ImagingStudy.url | deleted
ImagingStudy.series.url | deleted
ImagingStudy.series.instance.type | deleted
ImagingStudy.series.instance.content | deleted
Immunization
Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/immunization-status
Immunization.notGiven | Renamed from wasNotGiven to notGiven
Immunization.primarySource | Added Element
Immunization.reportOrigin | Added Element
Immunization.practitioner | Added Element
Immunization.practitioner.role | Added Element
Immunization.practitioner.actor | Added Element
Immunization.vaccinationProtocol.doseSequence | Min Cardinality changed from 1 to 0
Immunization.reported | deleted
Immunization.performer | deleted
Immunization.requester | deleted
ImmunizationRecommendation
ImmunizationRecommendation.recommendation.vaccineCode | Min Cardinality changed from 1 to 0
ImmunizationRecommendation.recommendation.targetDisease | Added Element
ImmunizationRecommendation.recommendation.protocol.doseSequence | Type changed from integer to positiveInt
ImplementationGuide
ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
ImplementationGuide.experimental | Now marked as Modifier
ImplementationGuide.contact | Type changed from BackboneElement to ContactDetail
ImplementationGuide.description | Type changed from string to markdown
ImplementationGuide.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
ImplementationGuide.jurisdiction | Added Element
ImplementationGuide.copyright | Type changed from string to markdown
ImplementationGuide.package | Min Cardinality changed from 1 to 0
ImplementationGuide.package.resource.example | Added Element
ImplementationGuide.global.type | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
ImplementationGuide.page | Min Cardinality changed from 1 to 0
ImplementationGuide.page.title | Renamed from name to title
ImplementationGuide.page.type | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
ImplementationGuide.contact.name | deleted
ImplementationGuide.contact.telecom | deleted
ImplementationGuide.package.resource.purpose | deleted
Library | Added Resource
Linkage | Added Resource
List

List.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
Location
Location.operationalStatus | Added Element
Location.alias | Added Element
Location.endpoint | Added Element
Measure | Added Resource
MeasureReport | Added Resource
Media
Media.basedOn | Added Element
Media.context | Added Element
Media.occurrence[x] | Added Element
Media.reasonCode | Added Element
Media.bodySite | Added Element
Media.device | Added Element
Media.note | Added Element
Media.deviceName | deleted
Medication
Medication.status | Added Element
Medication.isOverTheCounter | Added Element
Medication.form | Added Element
Medication.ingredient | Added Element
Medication.ingredient.item[x] | Added Element
Medication.ingredient.isActive | Added Element
Medication.ingredient.amount | Added Element
Medication.package.content.item[x] | Renamed from item to item[x] Add CodeableConcept
Medication.package.batch | Added Element
Medication.package.batch.lotNumber | Added Element
Medication.package.batch.expirationDate | Added Element
Medication.image | Added Element
Medication.product | deleted
MedicationAdministration
MedicationAdministration.definition | Added Element
MedicationAdministration.partOf | Added Element
MedicationAdministration.status | Add Code unknown
MedicationAdministration.category | Added Element
MedicationAdministration.subject | Renamed from patient to subject Add Reference(Group)
MedicationAdministration.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
MedicationAdministration.supportingInformation | Added Element
MedicationAdministration.effective[x] | Added Element
MedicationAdministration.performer | Renamed from practitioner to performer Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)
MedicationAdministration.performer.actor | Added Element
MedicationAdministration.performer.onBehalfOf | Added Element
MedicationAdministration.notGiven | Renamed from wasNotGiven to notGiven
MedicationAdministration.reasonCode | Added Element
MedicationAdministration.reasonReference | Added Element
MedicationAdministration.prescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationAdministration.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationAdministration.dosage.site | Renamed from site[x] to site Remove Reference(BodySite)
MedicationAdministration.dosage.dose | Renamed from quantity to dose
MedicationAdministration.dosage.rate[x] | Remove Range, Add Quantity{http://hl7.org/fhir/StructureDefinition/SimpleQuantity}
MedicationAdministration.eventHistory | Added Element
MedicationAdministration.reasonGiven | deleted
MedicationAdministration.effectiveTime[x] | deleted
MedicationDispense
MedicationDispense.identifier | Max Cardinality changed from 1 to *
MedicationDispense.partOf | Added Element
MedicationDispense.status | Add Code preparation
MedicationDispense.category | Added Element
MedicationDispense.subject | Renamed from patient to subject Add Reference(Group)
MedicationDispense.context | Added Element
MedicationDispense.supportingInformation | Added Element
MedicationDispense.performer | Added Element
MedicationDispense.performer.actor | Added Element
MedicationDispense.performer.onBehalfOf | Added Element
MedicationDispense.authorizingPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationDispense.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationDispense.dosageInstruction | Type changed from BackboneElement to Dosage
MedicationDispense.substitution.wasSubstituted | Added Element
MedicationDispense.substitution.type | Min Cardinality changed from 1 to 0
MedicationDispense.detectedIssue | Added Element
MedicationDispense.notDone | Added Element

MedicationDispense.notDoneReason[x] | Added Element
MedicationDispense.eventHistory | Added Element
MedicationDispense.dispenser | deleted
MedicationDispense.dosageInstruction.text | deleted
MedicationDispense.dosageInstruction.additionalInstructions | deleted
MedicationDispense.dosageInstruction.timing | deleted
MedicationDispense.dosageInstruction.asNeeded[x] | deleted
MedicationDispense.dosageInstruction.site[x] | deleted
MedicationDispense.dosageInstruction.route | deleted
MedicationDispense.dosageInstruction.method | deleted
MedicationDispense.dosageInstruction.dose[x] | deleted
MedicationDispense.dosageInstruction.rate[x] | deleted
MedicationDispense.dosageInstruction.maxDosePerPeriod | deleted
MedicationRequest | Name Changed from MedicationOrder to MedicationRequest
MedicationRequest.definition | Added Element
MedicationRequest.basedOn | Added Element
MedicationRequest.groupIdentifier | Added Element
MedicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/medication-order-status to http://hl7.org/fhir/ValueSet/medication-request-status
MedicationRequest.intent | Added Element
MedicationRequest.category | Added Element
MedicationRequest.priority | Added Element
MedicationRequest.subject | Added Element
MedicationRequest.context | Added Element
MedicationRequest.supportingInformation | Added Element
MedicationRequest.authoredOn | Added Element
MedicationRequest.requester | Added Element
MedicationRequest.requester.agent | Added Element
MedicationRequest.requester.onBehalfOf | Added Element
MedicationRequest.recorder | Added Element
MedicationRequest.reasonCode | Added Element
MedicationRequest.reasonReference | Added Element
MedicationRequest.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationRequest.dosageInstruction | Type changed from BackboneElement to Dosage
MedicationRequest.dispenseRequest.expectedSupplyDuration | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration
MedicationRequest.dispenseRequest.performer | Added Element
MedicationRequest.substitution.allowed | Added Element
MedicationRequest.priorPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationRequest.detectedIssue | Added Element
MedicationRequest.eventHistory | Added Element
MedicationOrder.dateWritten | deleted
MedicationOrder.dateEnded | deleted
MedicationOrder.reasonEnded | deleted
MedicationOrder.patient | deleted
MedicationOrder.prescriber | deleted
MedicationOrder.encounter | deleted
MedicationOrder.reason[x] | deleted
MedicationOrder.dosageInstruction.text | deleted
MedicationOrder.dosageInstruction.additionalInstructions | deleted
MedicationOrder.dosageInstruction.timing | deleted
MedicationOrder.dosageInstruction.asNeeded[x] | deleted
MedicationOrder.dosageInstruction.site[x] | deleted
MedicationOrder.dosageInstruction.route | deleted
MedicationOrder.dosageInstruction.method | deleted
MedicationOrder.dosageInstruction.dose[x] | deleted
MedicationOrder.dosageInstruction.rate[x] | deleted
MedicationOrder.dosageInstruction.maxDosePerPeriod | deleted
MedicationOrder.dispenseRequest.medication[x] | deleted
MedicationOrder.substitution.type | deleted
MedicationStatement
MedicationStatement.basedOn | Added Element
MedicationStatement.partOf | Added Element
MedicationStatement.context | Added Element
MedicationStatement.status | Add Codes stopped, on-hold
MedicationStatement.category | Added Element
MedicationStatement.informationSource | Add Reference(Organization)
MedicationStatement.subject | Renamed from patient to subject Add Reference(Group)
MedicationStatement.derivedFrom | Renamed from supportingInformation to derivedFrom
MedicationStatement.taken | Added Element
MedicationStatement.reasonCode | Added Element
MedicationStatement.reasonReference | Added Element

MedicationStatement.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationStatement.dosage | Type changed from BackboneElement to Dosage
MedicationStatement.wasNotTaken | deleted
MedicationStatement.reasonForUse[x] | deleted
MedicationStatement.dosage.text | deleted
MedicationStatement.dosage.timing | deleted
MedicationStatement.dosage.asNeeded[x] | deleted
MedicationStatement.dosage.site[x] | deleted
MedicationStatement.dosage.route | deleted
MedicationStatement.dosage.method | deleted
MedicationStatement.dosage.quantity[x] | deleted
MedicationStatement.dosage.rate[x] | deleted
MedicationStatement.dosage.maxDosePerPeriod | deleted
MessageDefinition | Added Resource
MessageHeader
MessageHeader.event | No longer marked as Modifier
MessageHeader.sender | Added Element
MessageHeader.response | No longer marked as Modifier
MessageHeader.response.code | No longer marked as Modifier
MessageHeader.focus | Renamed from data to focus
MetadataResource | Added Resource
NamingSystem
NamingSystem.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
NamingSystem.contact | Type changed from BackboneElement to ContactDetail
NamingSystem.description | Type changed from string to markdown
NamingSystem.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
NamingSystem.jurisdiction | Added Element
NamingSystem.uniqueId.comment | Added Element
NamingSystem.contact.name | deleted
NamingSystem.contact.telecom | deleted
NutritionOrder
NutritionOrder.status | Change value set from http://hl7.org/fhir/ValueSet/nutrition-order-status to http://hl7.org/fhir/ValueSet/nutrition-request-status
Observation
Observation.basedOn | Added Element
Observation.status | Add Code corrected
Observation.category | Max Cardinality changed from 1 to *
Observation.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Observation.value[x] | Add boolean
Observation.comment | Renamed from comments to comment
Observation.referenceRange.type | Added Element
Observation.referenceRange.appliesTo | Renamed from meaning to appliesTo Max Cardinality changed from 1 to *
Observation.related.target | Add Reference(Sequence)
Observation.component.interpretation | Added Element
OperationDefinition
OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
OperationDefinition.experimental | Now marked as Modifier
OperationDefinition.contact | Type changed from BackboneElement to ContactDetail
OperationDefinition.description | Type changed from string to markdown
OperationDefinition.useContext | Added Element
OperationDefinition.jurisdiction | Added Element
OperationDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown
OperationDefinition.comment | Renamed from notes to comment
OperationDefinition.resource | Renamed from type to resource Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
OperationDefinition.type | Min Cardinality changed from 0 to 1 Max Cardinality changed from * to 1 Type changed from code to boolean Remove Binding http://hl7.org/fhir/ValueSet/resource-types (required)
OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-type to http://hl7.org/fhir/ValueSet/all-types
OperationDefinition.parameter.searchType | Added Element
OperationDefinition.overload | Added Element
OperationDefinition.overload.parameterName | Added Element
OperationDefinition.overload.comment | Added Element
OperationDefinition.contact.name | deleted
OperationDefinition.contact.telecom | deleted
OperationOutcome
OperationOutcome.issue.expression | Added Element
Organization

Organization.type | Max Cardinality changed from 1 to *
Organization.alias | Added Element
Organization.endpoint | Added Element
Parameters
Parameters.parameter.value[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
Patient
Patient.maritalStatus | Change binding strength from required to extensible
Patient.contact.relationship | Change value set from http://hl7.org/fhir/ValueSet/patient-contact-relationship to http://hl7.org/fhir/ValueSet/v2-0131
Patient.communication.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages
Patient.generalPractitioner | Renamed from careProvider to generalPractitioner
Patient.link.other | Add Reference(RelatedPerson) No longer marked as Modifier
Patient.link.type | Remove Code replace Add Codes replaced-by, replaces No longer marked as Modifier
PaymentNotice
PaymentNotice.status | Added Element
PaymentNotice.statusDate | Added Element
PaymentNotice.paymentStatus | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
PaymentNotice.ruleset | deleted
PaymentNotice.originalRuleset | deleted
PaymentReconciliation
PaymentReconciliation.status | Added Element
PaymentReconciliation.outcome | Type changed from code to CodeableConcept Add Code partial
PaymentReconciliation.detail.type | Type changed from Coding to CodeableConcept
PaymentReconciliation.detail.response | Added Element
PaymentReconciliation.detail.amount | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
PaymentReconciliation.form | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/forms (required)
PaymentReconciliation.total | Min Cardinality changed from 1 to 0 Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
PaymentReconciliation.processNote | Added Element
PaymentReconciliation.processNote.type | Added Element
PaymentReconciliation.processNote.text | Added Element
PaymentReconciliation.ruleset | deleted
PaymentReconciliation.originalRuleset | deleted
PaymentReconciliation.detail.responce | deleted
PaymentReconciliation.note | deleted
Person
PlanDefinition | Added Resource
Practitioner
Practitioner.name | Max Cardinality changed from 1 to *
Practitioner.communication | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages
Practitioner.practitionerRole | deleted
PractitionerRole | Added Resource
Procedure
Procedure.definition | Added Element
Procedure.basedOn | Renamed from request to basedOn Max Cardinality changed from 1 to * Remove Reference(DiagnosticOrder)
Procedure.partOf | Added Element
Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/procedure-status to http://hl7.org/fhir/ValueSet/event-status
Procedure.notDone | Renamed from notPerformed to notDone
Procedure.notDoneReason | Renamed from reasonNotPerformed to notDoneReason Max Cardinality changed from * to 1
Procedure.code | Min Cardinality changed from 1 to 0
Procedure.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Procedure.performer.actor | Min Cardinality changed from 0 to 1 Add Reference(Device)
Procedure.performer.onBehalfOf | Added Element
Procedure.reasonCode | Added Element
Procedure.reasonReference | Added Element
Procedure.complicationDetail | Added Element
Procedure.note | Renamed from notes to note
Procedure.focalDevice.action | Remove Binding http://hl7.org/fhir/ValueSet/device-action (required)
Procedure.usedReference | Renamed from used to usedReference
Procedure.usedCode | Added Element
Procedure.reason[x] | deleted
ProcedureRequest
ProcedureRequest.definition | Added Element
ProcedureRequest.basedOn | Added Element
ProcedureRequest.replaces | Added Element
ProcedureRequest.requisition | Added Element
ProcedureRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/procedure-request-status to http://hl7.org/fhir/ValueSet/request-status
ProcedureRequest.intent | Added Element
ProcedureRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/procedure-request-priority to http://hl7.org/fhir/ValueSet/request-priority
ProcedureRequest.doNotPerform | Added Element

ProcedureRequest.category | Added Element
ProcedureRequest.subject | Add Reference(Location), Add Reference(Device)
ProcedureRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
ProcedureRequest.occurrence[x] | Added Element
ProcedureRequest.authoredOn | Renamed from orderedOn to authoredOn
ProcedureRequest.requester | Added Element
ProcedureRequest.requester.agent | Added Element
ProcedureRequest.requester.onBehalfOf | Added Element
ProcedureRequest.performerType | Added Element
ProcedureRequest.performer | Add Reference(Device), Add Reference(HealthcareService)
ProcedureRequest.reasonCode | Added Element
ProcedureRequest.reasonReference | Added Element
ProcedureRequest.supportingInfo | Added Element
ProcedureRequest.specimen | Added Element
ProcedureRequest.note | Renamed from notes to note
ProcedureRequest.relevantHistory | Added Element
ProcedureRequest.reason[x] | deleted
ProcedureRequest.scheduled[x] | deleted
ProcedureRequest.orderer | deleted
ProcessRequest
ProcessRequest.status | Added Element
ProcessRequest.action | Min Cardinality changed from 1 to 0
ProcessRequest.ruleset | deleted
ProcessRequest.originalRuleset | deleted
ProcessResponse
ProcessResponse.status | Added Element
ProcessResponse.outcome | Type changed from Coding to CodeableConcept
ProcessResponse.form | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/forms (required)
ProcessResponse.processNote | Added Element
ProcessResponse.processNote.type | Added Element
ProcessResponse.processNote.text | Added Element
ProcessResponse.error | Type changed from Coding to CodeableConcept Remove Binding http://hl7.org/fhir/ValueSet/adjudication-error (required)
ProcessResponse.communicationRequest | Added Element
ProcessResponse.ruleset | deleted
ProcessResponse.originalRuleset | deleted
ProcessResponse.notes | deleted
Provenance
Provenance.reason | Type changed from CodeableConcept to Coding
Provenance.activity | Type changed from CodeableConcept to Coding Change value set from http://hl7.org/fhir/ValueSet/v3-ProvenanceEventCurrentState to http://hl7.org/fhir/ValueSet/provenance-activity-type
Provenance.agent | Min Cardinality changed from 0 to 1
Provenance.agent.role | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Type changed from Coding to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/provenance-agent-role to http://hl7.org/fhir/ValueSet/security-role-type
Provenance.agent.who[x] | Added Element
Provenance.agent.onBehalfOf[x] | Added Element
Provenance.agent.relatedAgentType | Added Element
Provenance.entity.role | Add Code removal
Provenance.entity.what[x] | Added Element
Provenance.entity.agent | Max Cardinality changed from 1 to *
Provenance.agent.actor | deleted
Provenance.agent.userId | deleted
Provenance.agent.relatedAgent | deleted
Provenance.entity.type | deleted
Provenance.entity.reference | deleted
Provenance.entity.display | deleted
Questionnaire
Questionnaire.url | Added Element
Questionnaire.name | Added Element
Questionnaire.title | Added Element
Questionnaire.status | Change value set from http://hl7.org/fhir/ValueSet/questionnaire-status to http://hl7.org/fhir/ValueSet/publication-status
Questionnaire.experimental | Added Element
Questionnaire.description | Added Element
Questionnaire.purpose | Added Element
Questionnaire.approvalDate | Added Element
Questionnaire.lastReviewDate | Added Element
Questionnaire.effectivePeriod | Added Element
Questionnaire.useContext | Added Element
Questionnaire.jurisdiction | Added Element
Questionnaire.contact | Added Element
Questionnaire.copyright | Added Element
Questionnaire.code | Added Element

Questionnaire.subjectType | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
Questionnaire.item | Renamed from group to item Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
Questionnaire.item.linkId | Min Cardinality changed from 0 to 1
Questionnaire.item.definition | Added Element
Questionnaire.item.code | Added Element
Questionnaire.item.prefix | Added Element
Questionnaire.item.type | Added Element
Questionnaire.item.enableWhen | Added Element
Questionnaire.item.enableWhen.question | Added Element
Questionnaire.item.enableWhen.hasAnswer | Added Element
Questionnaire.item.enableWhen.answer[x] | Added Element
Questionnaire.item.readOnly | Added Element
Questionnaire.item.maxLength | Added Element
Questionnaire.item.options | Added Element
Questionnaire.item.option | Added Element
Questionnaire.item.option.value[x] | Added Element
Questionnaire.item.initial[x] | Added Element
Questionnaire.item.item | Added Element
Questionnaire.telecom | deleted
Questionnaire.group.title | deleted
Questionnaire.group.concept | deleted
Questionnaire.group.group | deleted
Questionnaire.group.question | deleted
QuestionnaireResponse
QuestionnaireResponse.basedOn | Added Element
QuestionnaireResponse.parent | Added Element
QuestionnaireResponse.status | Add Codes entered-in-error, stopped
QuestionnaireResponse.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
QuestionnaireResponse.item | Renamed from group to item Max Cardinality changed from 1 to *
QuestionnaireResponse.item.linkId | Min Cardinality changed from 0 to 1
QuestionnaireResponse.item.definition | Added Element
QuestionnaireResponse.item.answer | Added Element
QuestionnaireResponse.item.answer.value[x] | Added Element
QuestionnaireResponse.item.answer.item | Added Element
QuestionnaireResponse.item.item | Added Element
QuestionnaireResponse.group.title | deleted
QuestionnaireResponse.group.group | deleted
QuestionnaireResponse.group.question | deleted
ReferralRequest
ReferralRequest.definition | Added Element
ReferralRequest.basedOn | Added Element
ReferralRequest.replaces | Added Element
ReferralRequest.groupIdentifier | Added Element
ReferralRequest.status | Change value set from http://hl7.org/fhir/ValueSet/referralstatus to http://hl7.org/fhir/ValueSet/request-status
ReferralRequest.intent | Added Element
ReferralRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)
ReferralRequest.subject | Renamed from patient to subject Min Cardinality changed from 0 to 1 Add Reference(Group)
ReferralRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
ReferralRequest.occurrence[x] | Added Element
ReferralRequest.authoredOn | Renamed from date to authoredOn
ReferralRequest.requester | Remove Reference(Practitioner), Remove Reference(Organization), Remove Reference(Patient)
ReferralRequest.requester.agent | Added Element
ReferralRequest.requester.onBehalfOf | Added Element
ReferralRequest.recipient | Add Reference(HealthcareService)
ReferralRequest.reasonCode | Renamed from reason to reasonCode Max Cardinality changed from 1 to *
ReferralRequest.reasonReference | Added Element
ReferralRequest.supportingInfo | Renamed from supportingInformation to supportingInfo
ReferralRequest.note | Added Element
ReferralRequest.relevantHistory | Added Element
ReferralRequest.dateSent | deleted
ReferralRequest.fulfillmentTime | deleted
RelatedPerson
RelatedPerson.active | Added Element
RelatedPerson.relationship | Remove Binding http://hl7.org/fhir/ValueSet/relatedperson-relationshiptype (required)
RelatedPerson.name | Max Cardinality changed from 1 to *
RequestGroup | Added Resource
ResearchStudy | Added Resource
ResearchSubject | Added Resource
Resource

Resource.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages
RiskAssessment
RiskAssessment.basedOn | Added Element
RiskAssessment.parent | Added Element
RiskAssessment.status | Added Element
RiskAssessment.code | Added Element
RiskAssessment.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
RiskAssessment.occurrence[x] | Added Element
RiskAssessment.reason[x] | Added Element
RiskAssessment.prediction.probability[x] | Remove CodeableConcept
RiskAssessment.prediction.qualitativeRisk | Added Element
RiskAssessment.comment | Added Element
RiskAssessment.date | deleted
Schedule
Schedule.active | Added Element
Schedule.serviceCategory | Added Element
Schedule.serviceType | Renamed from type to serviceType
Schedule.specialty | Added Element
Schedule.actor | Max Cardinality changed from 1 to * Add Reference(PractitionerRole)
SearchParameter
SearchParameter.version | Added Element
SearchParameter.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
SearchParameter.experimental | Now marked as Modifier
SearchParameter.contact | Type changed from BackboneElement to ContactDetail
SearchParameter.useContext | Added Element
SearchParameter.jurisdiction | Added Element
SearchParameter.purpose | Renamed from requirements to purpose Type changed from string to markdown
SearchParameter.base | Max Cardinality changed from 1 to * Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
SearchParameter.derivedFrom | Added Element
SearchParameter.description | Type changed from string to markdown
SearchParameter.expression | Added Element
SearchParameter.target | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
SearchParameter.comparator | Added Element
SearchParameter.modifier | Added Element
SearchParameter.chain | Added Element
SearchParameter.component | Added Element
SearchParameter.component.definition | Added Element
SearchParameter.component.expression | Added Element
SearchParameter.contact.name | deleted
SearchParameter.contact.telecom | deleted
Sequence | Added Resource
ServiceDefinition | Added Resource
Slot
Slot.serviceCategory | Added Element
Slot.serviceType | Renamed from type to serviceType Max Cardinality changed from 1 to *
Slot.specialty | Added Element
Slot.appointmentType | Added Element
Slot.status | Renamed from freeBusyType to status Add Code entered-in-error
Specimen
Specimen.request | Added Element
Specimen.processing | Added Element
Specimen.processing.description | Added Element
Specimen.processing.procedure | Added Element
Specimen.processing.additive | Added Element
Specimen.processing.time[x] | Added Element
Specimen.note | Added Element
Specimen.collection.comment | deleted
Specimen.treatment | deleted
StructureDefinition
StructureDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
StructureDefinition.title | Renamed from display to title
StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

StructureDefinition.experimental | Now marked as Modifier
StructureDefinition.contact | Type changed from BackboneElement to ContactDetail
StructureDefinition.description | Type changed from string to markdown
StructureDefinition.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
StructureDefinition.jurisdiction | Added Element
StructureDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown
StructureDefinition.copyright | Type changed from string to markdown
StructureDefinition.keyword | Renamed from code to keyword
StructureDefinition.mapping.comment | Renamed from comments to comment
StructureDefinition.kind | Remove Code datatype Add Codes primitive-type, complex-type
StructureDefinition.contextType | Remove Code mapping
StructureDefinition.contextInvariant | Added Element
StructureDefinition.type | Renamed from constrainedType to type Min Cardinality changed from 0 to 1 Change binding strength from required to extensible
StructureDefinition.baseDefinition | Renamed from base to baseDefinition
StructureDefinition.derivation | Added Element
StructureDefinition.contact.name | deleted
StructureDefinition.contact.telecom | deleted
StructureMap | Added Resource
Subscription
Subscription.channel.payload | Min Cardinality changed from 1 to 0
Subscription.channel.header | Max Cardinality changed from 1 to *
Substance
Substance.status | Added Element
Substance.ingredient.substance[x] | Renamed from substance to substance[x] Add CodeableConcept
SupplyDelivery
SupplyDelivery.basedOn | Added Element
SupplyDelivery.partOf | Added Element
SupplyDelivery.status | Add Code entered-in-error
SupplyDelivery.suppliedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyDelivery.suppliedItem.quantity | Added Element
SupplyDelivery.suppliedItem.item[x] | Added Element
SupplyDelivery.occurrence[x] | Added Element
SupplyDelivery.supplier | Add Reference(Organization)
SupplyDelivery.quantity | deleted
SupplyDelivery.whenPrepared | deleted
SupplyDelivery.time | deleted
SupplyRequest
SupplyRequest.status | Remove Codes requested, failed Add Codes draft, active, suspended, entered-in-error, unknown
SupplyRequest.category | Renamed from kind to category Remove Binding http://hl7.org/fhir/ValueSet/supplyrequest-kind (required)
SupplyRequest.priority | Added Element
SupplyRequest.orderedItem | Remove Reference(Medication), Remove Reference(Substance), Remove Reference(Device)
SupplyRequest.orderedItem.quantity | Added Element
SupplyRequest.orderedItem.item[x] | Added Element
SupplyRequest.occurrence[x] | Added Element
SupplyRequest.authoredOn | Renamed from date to authoredOn
SupplyRequest.requester | Added Element
SupplyRequest.requester.agent | Added Element
SupplyRequest.requester.onBehalfOf | Added Element
SupplyRequest.deliverFrom | Added Element
SupplyRequest.deliverTo | Added Element
SupplyRequest.patient | deleted
SupplyRequest.source | deleted
SupplyRequest.when | deleted
Task | Added Resource
TestReport | Added Resource
TestScript
TestScript.title | Added Element
TestScript.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
TestScript.experimental | Now marked as Modifier
TestScript.contact | Type changed from BackboneElement to ContactDetail
TestScript.description | Type changed from string to markdown
TestScript.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
TestScript.jurisdiction | Added Element
TestScript.purpose | Added Element
TestScript.copyright | Type changed from string to markdown
TestScript.origin | Added Element
TestScript.origin.index | Added Element
TestScript.origin.profile | Added Element
TestScript.destination | Added Element
TestScript.destination.index | Added Element

TestScript.destination.profile | Added Element
TestScript.metadata.capability.origin | Added Element
TestScript.metadata.capability.capabilities | Added Element
TestScript.fixture.autocreate | Default Value "false" added
TestScript.fixture.autodelete | Default Value "false" added
TestScript.variable.defaultValue | Added Element
TestScript.variable.description | Added Element
TestScript.variable.expression | Added Element
TestScript.variable.hint | Added Element
TestScript.rule | Added Element
TestScript.rule.resource | Added Element
TestScript.rule.param | Added Element
TestScript.rule.param.name | Added Element
TestScript.rule.param.value | Added Element
TestScript.ruleset | Added Element
TestScript.ruleset.resource | Added Element
TestScript.ruleset.rule | Added Element
TestScript.ruleset.rule.ruleId | Added Element
TestScript.ruleset.rule.param | Added Element
TestScript.ruleset.rule.param.name | Added Element
TestScript.ruleset.rule.param.value | Added Element
TestScript.setup.action.operation.resource | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ContactDetail, Contributor, DataRequirement, Dosage, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
TestScript.setup.action.operation.accept | Add Codes ttl, none
TestScript.setup.action.operation.contentType | Add Codes ttl, none
TestScript.setup.action.operation.destination | Default Value "0" removed
TestScript.setup.action.operation.origin | Added Element
TestScript.setup.action.operation.requestId | Added Element
TestScript.setup.action.assert.compareToSourceExpression | Added Element
TestScript.setup.action.assert.contentType | Add Codes ttl, none
TestScript.setup.action.assert.expression | Added Element
TestScript.setup.action.assert.operator | Add Code eval
TestScript.setup.action.assert.requestMethod | Added Element
TestScript.setup.action.assert.requestURL | Added Element
TestScript.setup.action.assert.resource | Remove Codes Conformance, DiagnosticOrder, ImagingObjectSelection, MedicationOrder, Order, OrderResponse Add Codes ContactDetail, Contributor, DataRequirement, Dosage, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, ActivityDefinition, AdverseEvent, CapabilityStatement, CareTeam, ChargeItem, CodeSystem, CompartmentDefinition, Consent, Endpoint, ExpansionProfile, GraphDefinition, GuidanceResponse, ImagingManifest, Library, Linkage, Measure, MeasureReport, MedicationRequest, MessageDefinition, PlanDefinition, PractitionerRole, RequestGroup, ResearchStudy, ResearchSubject, Sequence, ServiceDefinition, StructureMap, Task, TestReport
TestScript.setup.action.assert.rule | Added Element
TestScript.setup.action.assert.rule.ruleId | Added Element
TestScript.setup.action.assert.rule.param | Added Element
TestScript.setup.action.assert.rule.param.name | Added Element
TestScript.setup.action.assert.rule.param.value | Added Element
TestScript.setup.action.assert.ruleset | Added Element
TestScript.setup.action.assert.ruleset.rulesetId | Added Element
TestScript.setup.action.assert.ruleset.rule | Added Element
TestScript.setup.action.assert.ruleset.rule.ruleId | Added Element
TestScript.setup.action.assert.ruleset.rule.param | Added Element
TestScript.setup.action.assert.ruleset.rule.param.name | Added Element
TestScript.setup.action.assert.ruleset.rule.param.value | Added Element
TestScript.teardown.action.operation | Min Cardinality changed from 0 to 1
TestScript.contact.name | deleted
TestScript.contact.telecom | deleted
TestScript.requirements | deleted
TestScript.metadata.capability.conformance | deleted
TestScript.multiserver | deleted
TestScript.setup.metadata | deleted
TestScript.test.metadata | deleted
ValueSet
ValueSet.identifier | Max Cardinality changed from 1 to *
ValueSet.title | Added Element
ValueSet.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
ValueSet.experimental | Now marked as Modifier
ValueSet.contact | Type changed from BackboneElement to ContactDetail
ValueSet.description | Type changed from string to markdown
ValueSet.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)
ValueSet.jurisdiction | Added Element

ValueSet.purpose | Renamed from requirements to purpose Type changed from string to markdown
ValueSet.copyright | Type changed from string to markdown
ValueSet.compose.lockedDate | Added Element
ValueSet.compose.inactive | Added Element
ValueSet.compose.include | Min Cardinality changed from 0 to 1
ValueSet.compose.include.system | Min Cardinality changed from 1 to 0
ValueSet.compose.include.concept.designation.language | Added Element
ValueSet.compose.include.concept.designation.use | Added Element
ValueSet.compose.include.concept.designation.value | Added Element
ValueSet.compose.include.filter | Now marked as Modifier
ValueSet.compose.include.filter.op | Add Codes descendent-of, generalizes, exists
ValueSet.compose.include.valueSet | Added Element
ValueSet.expansion.contains.abstract | Default Value "false" added
ValueSet.expansion.contains.inactive | Added Element
ValueSet.expansion.contains.designation | Added Element
ValueSet.contact.name | deleted
ValueSet.contact.telecom | deleted
ValueSet.lockedDate | deleted
ValueSet.codeSystem | deleted
ValueSet.compose.import | deleted
VisionPrescription
VisionPrescription.status | Added Element
VisionPrescription.dispense.product | Min Cardinality changed from 1 to 0 Type changed from Coding to CodeableConcept
VisionPrescription.dispense.note | Added Element
VisionPrescription.dispense.notes | deleted
Conformance | deleted
DiagnosticOrder | deleted
ImagingObjectSelection | deleted
MedicationOrder | deleted
Order | deleted
OrderResponse | deleted

2.10 Transforms between DSTU 2 and STU 3

URL: https://hl7.org/fhir/STU3/r2maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

2.10 Transforms between DSTU 2 and STU 3

In addition to a list of all differences between DSTU 2 and STU 3, this specification also provides transforms between DSTU and STU 3 for most resources that were in DSTU 2. These transforms are expressed using the FHIR Mapping Language, and both serve as formal documentation of the relationship between the resource structures in R2 and R3, and can also be used to convert between the formats automatically.

The following table summarizes the state of the R2:R3 maps for each resource type.

Note the following:

* Transforms are tested using the following method:

* For each example resource in the R2 release of the FHIR specification that has a transform to R3

* apply the transform to R3

* validate against the R3 specification

* apply the reverse transform back to R2

* compare the outcome with the original resource - ideally, it should be identical

* All transforms should execute - that is, they should be able to take any R2 resource, represent the content in R3, and then generate an R2 representation for the same content. Note, though, that in a few cases, the differences between R2 and R3 designs are so significant that not all examples can be successfully transformed at all.

* For some resources, the differences in design between R2 and R3 are sufficiently great that the R2 examples cannot reproduce the same output after conversion to R3 and then back to R2. For example, the R3 resource may have a different set of status codes with imperfect maps

* For some resources, the information available in the R2 examples is not enough to generate valid R3 resources, and so the resources generate R3 validation errors. (e.g. R2 resources missing an element that is required in R3)

* The transform scripts are only tested to the degree that R2 provided example data

* The Bundle tests include the large distribution files (all valuesets, structure definitions, etc) and so generate large numbers of validation errors

* For all these reasons, the maps provided in the specification are not authoritative; they are provided to help implementers who need to convert between the R2 and R3 format, but additional revision may be required in an implementation specific context. The FHIR Wiki tracks ongoing work to provide improved transforms, and other transformation options

Resource | # Tests | % Execute OK | % RoundTrip Ok | % R3 Valid | R3 Error Count
Account | 1 | 100 | 100 | 100
ActivityDefinition | No r2:r3 maps available
AdverseEvent | No r2:r3 maps available
AllergyIntolerance | 5 | 100 | 60 | 20 | 6
Appointment | 3 | 100 | 100 | 100
AppointmentResponse | 2 | 100 | 100 | 100
AuditEvent | 8 | 100 | 100 | 100
Basic | 4 | 100 | 100 | 75 | 3
Binary | 2 | 100 | 100 | 100
BodySite | 1 | 100 | 100 | 100
Bundle | 23 | 95 | 50 | 50 | 3836
CapabilityStatement | 19 | 100 | 0 | 36 | 453
CarePlan | 10 | 100 | 20 | 0 | 33
CareTeam | No r2:r3 maps available
ChargeItem | No r2:r3 maps available
Claim | No r2:r3 maps available
ClaimResponse | No r2:r3 maps available
ClinicalImpression | 1 | 100 | 0 | 100
CodeSystem | No r2:r3 maps available
Communication | 2 | 100 | 100 | 100
CommunicationRequest | 2 | 100 | 50 | 50 | 1
CompartmentDefinition | No r2:r3 maps available
Composition | 1 | 100 | 100 | 100
ConceptMap | 2 | 100 | 100 | 50 | 1
Condition | 14 | 100 | 21 | 0 | 14
Consent | No r2:r3 maps available
Contract | 1 | 100 | 100 | 100
Coverage | No r2:r3 maps available
DataElement | 4 | 100 | 75 | 50 | 4
DetectedIssue | 4 | 100 | 100 | 0 | 4
Device | 5 | 100 | 60 | 100
DeviceComponent | 2 | 100 | 100 | 100
DeviceMetric | 1 | 100 | 100 | 100
DeviceRequest | No r2:r3 maps available
DeviceUseStatement | 1 | 100 | 100 | 0 | 1
DiagnosticReport | 11 | 72 | 0 | 62 | 22
DocumentManifest | 1 | 100 | 0 | 100
DocumentReference | 1 | 100 | 0 | 100
EligibilityRequest | No r2:r3 maps available
EligibilityResponse | No r2:r3 maps available
Encounter | 10 | 100 | 90 | 80 | 2
Endpoint | No r2:r3 maps available
EnrollmentRequest | 1 | 100 | 0 | 100
EnrollmentResponse | 1 | 100 | 0 | 100
EpisodeOfCare | 1 | 100 | 100 | 100
ExpansionProfile | No r2:r3 maps available
ExplanationOfBenefit | No r2:r3 maps available
FamilyMemberHistory | 3 | 100 | 100 | 100
Flag | 3 | 100 | 100 | 100
Goal | 2 | 100 | 100 | 0 | 2
GraphDefinition | No r2:r3 maps available
Group | 2 | 100 | 100 | 100
GuidanceResponse | No r2:r3 maps available
HealthcareService | 1 | 100 | 0 | 100
ImagingManifest | No r2:r3 maps available
ImagingStudy | 2 | 100 | 0 | 50 | 2
Immunization | 2 | 100 | 0 | 0 | 2
ImmunizationRecommendation | 1 | 100 | 100 | 100
ImplementationGuide | 1 | 100 | 100 | 100
Library | No r2:r3 maps available
Linkage | No r2:r3 maps available
List | No r2:r3 maps available
Location | 6 | 100 | 100 | 100
Measure | No r2:r3 maps available
MeasureReport | No r2:r3 maps available
Media | 3 | 100 | 100 | 100
Medication | 23 | 100 | 30 | 95 | 1
MedicationAdministration | 3 | 100 | 100 | 100

MedicationDispense | 7 | 100 | 0 | 85 | 2
MedicationRequest | 18 | 94 | 100 | 70 | 6
MedicationStatement | 7 | 100 | 100 | 85 | 1
MessageDefinition | No r2:r3 maps available
MessageHeader | 1 | 100 | 100 | 100
NamingSystem | 3 | 100 | 100 | 100
NutritionOrder | No r2:r3 maps available
Observation | 34 | 100 | 100 | 55 | 58
OperationDefinition | 18 | 100 | 94 | 100
OperationOutcome | 6 | 100 | 100 | 100
Organization | 14 | 100 | 100 | 85 | 2
Patient | 18 | 100 | 100 | 66 | 12
PaymentNotice | No r2:r3 maps available
PaymentReconciliation | No r2:r3 maps available
Person | 5 | 100 | 100 | 100
PlanDefinition | No r2:r3 maps available
Practitioner | 18 | 100 | 16 | 100
PractitionerRole | No r2:r3 maps available
Procedure | 9 | 100 | 100 | 100
ProcedureRequest | 6 | 100 | 16 | 0 | 9
ProcessRequest | No r2:r3 maps available
ProcessResponse | No r2:r3 maps available
Provenance | 2 | 100 | 50 | 50 | 1
Questionnaire | 6 | 100 | 33 | 16 | 109
QuestionnaireResponse | 6 | 100 | 83 | 0 | 19
ReferralRequest | 2 | 100 | 0 | 0 | 2
RelatedPerson | 4 | 100 | 100 | 75 | 1
RequestGroup | No r2:r3 maps available
ResearchStudy | No r2:r3 maps available
ResearchSubject | No r2:r3 maps available
RiskAssessment | 4 | 100 | 100 | 0 | 4
Schedule | 1 | 100 | 100 | 100
SearchParameter | 2 | 100 | 100 | 100
Sequence | No r2:r3 maps available
ServiceDefinition | No r2:r3 maps available
Slot | 4 | 100 | 100 | 100
Specimen | 6 | 100 | 100 | 83 | 1
StructureDefinition | 248 | 100 | 75 | 41 | 441
StructureMap | No r2:r3 maps available
Subscription | 2 | 100 | 100 | 100
Substance | 6 | 100 | 100 | 100
SupplyDelivery | 1 | 100 | 100 | 100
SupplyRequest | 1 | 100 | 100 | 100
Task | No r2:r3 maps available
TestReport | No r2:r3 maps available
TestScript | 6 | 100 | 0 | 0 | 11
ValueSet | 11 | 100 | 54 | 63 | 4
VisionPrescription | 2 | 100 | 0 | 100

2.11 FHIR Overview

URL: https://hl7.org/fhir/STU3/overview.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.11 FHIR Overview

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

Welcome to the FHIR (Fast Healthcare Interoperability Resources) Specification, which is a standard for exchanging healthcare information electronically. This page provides an overview of the standard, and serves as a road map for first-time readers of the specification to help find your way around FHIR quickly.

2.11.1 Background

Healthcare records are increasingly becoming digitized. As patients move around the healthcare ecosystem, their electronic health records must be available, discoverable, and understandable. Further, to support automated clinical decision support and other machine-based processing, the data must also be structured and standardized. (See Coming digital challenges in healthcare)

HL7 has been addressing these challenges by producing healthcare data exchange and information modeling standards for over 20 years. FHIR is a new specification based on emerging industry approaches, but informed by years of lessons around requirements, successes and challenges gained through defining and implementing HL7 v2 , HL7 v3 and the RIM, and CDA .

FHIR can be used as a stand-alone data exchange standard, but can and will also be used in partnership with existing widely used standards. (See Comparing FHIR to other HL7 standards)

FHIR aims to simplify implementation without sacrificing information integrity. It leverages existing logical and theoretical models to provide a consistent, easy to implement, and rigorous mechanism for exchanging data between healthcare applications. FHIR has built-in mechanisms for traceability to the HL7 RIM and other important content models. This ensures alignment to HL7's previously defined patterns and best practices without requiring the implementer to have intimate knowledge of the RIM or any HL7 v3 derivations. (See Comparing FHIR to other HL7 standards)

2.11.2 Components

The basic building block in FHIR is a Resource. All exchangeable content is defined as a resource. Resources all share the following set of characteristics:

* A common way to define and represent them, building them from data types that define common reusable patterns of elements

* A common set of metadata

* A human readable part

2.11.3 Approach

2.11.3.1 Approach to information modeling

The philosophy behind FHIR is to build a base set of resources that, either by themselves or when combined, satisfy the majority of common use cases. FHIR resources aim to define the information contents and structure for the core information set that is shared by most implementations. There is a built-in extension mechanism to cover the remaining content as needed.

FHIR modeling uses a composition approach. In comparison, HL7 v3 modeling is based on "model by constraint" (see Comparing FHIR to other HL7 standards). With FHIR, specific use cases are usually implemented by combining resources together through the use of resource references. Although a single resource might be useful by itself for a given use case, it is more common that resources will be combined and tailored to meet use case specific requirements. Two special kinds of resources are used to describe how resources are defined and used:

* Capability Statement - describes the interfaces that an implementation exposes for exchange of data

* StructureDefinition - provide additional rules that serve to constrain the optionality, cardinality, terminology bindings, data types and extensions defined in the resources used by the implementation

2.11.4 The Specification

Broadly, the FHIR specification is broken up into a set of modules:

* Foundation: The basic definitional and exchange infrastructure on which the rest of the specification is built

* Implementer Support: Services to help implementers make use of the specification

* Security & Privacy: Documentation and services to create and maintain security, integrity and privacy

* Conformance: How to test conformance to the specification, and define implementation guides

* Terminology: Use and support of terminologies and related artifacts

* Linked Data: How to use RDF, and how ontologies are used when defining the FHIR content

* Administration: Basic resources for tracking patients, practitioners, organizations, devices, substances, etc

* Clinical: Core clinical content such as problems, allergies, and the care process (care plans, referrals) + more

* Medications: Medication management and immunization tracking

* Diagnostics: Observations, Diagnostic reports and requests + related content

* Workflow: Managing the process of care, and technical artifacts to do with obligation management

* Financial: Billing and Claiming support

* Clinical Reasoning: Clinical Decision Support and Quality Measures

Resources have a wide range of uses, from pure clinical content such as care plans and diagnostic reports to pure infrastructure such as Message Header and Capability Statements. They all share common technical characteristics (see below for a more formal definition), but they are used in totally different fashions. Note that you do not have to use REST to make use of resources.

2.11.5 Where to Start

The best place to start is to quickly read the Resources list to get a sense of what resources exist and then look at the Patient resource definition to see what resource definitions look like, and then read these background pages:

* Resource Definitions - basic background to how resources are defined

* About Resources: the Narrative they all contain, and how Resources refer to each other

* Formats: XML and JSON

* About Extensibility: a key way that the specification is kept simple

* If you are coming to FHIR with a background in another HL7 standard (HL7 v2 , HL7 v3 or CDA), The Relationship between FHIR and other HL7 Standards may also be useful

2.11.5.1 Header Tabs

These header tabs found through-out the specification are important, and many readers miss them:

Resources and the data types that they use are presented in a concise easy to read XML-like format, but they also have detailed descriptions of their contents. In addition, most resources are mapped to several different formats, including HL7 v2 , the HL7 v3 RIM, CDA , DICOM, and others. Also, all resources come with at least one example (sometimes many more) and, where appropriate, with profiles that describe their use in specific circumstances. Finally, some resources include notes that help implementers understand the design rationale underlying them.

2.11.6 Finding Additional Information and Providing Feedback

While intended to be consumable by a variety of audiences, the FHIR specification is targeted to the implementation community - those who will actually write the software that uses the specification. To help meet the needs of the implementation community, the editors have strived to keep the specification concise to reduce the amount of reading that must be done before writing useful code. For this reason, information that is not essential to the implementation process, such as considered alternatives, points of contention, future plans, etc. have been excluded from this specification. As well, it is likely that from time-to-time, implementers will encounter situations where the specification is unclear or incomplete. Finally, there may be circumstances where the specification is broken or where a change could allow it to better meet implementer needs.

HL7 has therefore provided a number of mechanisms through which additional information about FHIR can be sought and maintained and through which support and requests for change can be made.

2.11.6.1 The FHIR Wiki

The FHIR project team also maintains a wiki where development processes, methodology and design decisions are documented. Implementers and others can also contribute to the wiki to provide additional guidance and supplemental information not found in the specification. Note that wiki content is not authoritative and is not relevant for determining conformance to the FHIR specification. As well, some wiki content may not be up to date with the most recent version of the FHIR specification.

A wiki page exists for each page in the FHIR specification. These pages will generally be used to capture background rationale, decision points and other information not relevant to implementers. Additional pages defined include FHIR methodology , use of the FHIR design tools , etc. To explore the FHIR wiki, you can start at the root page

2.11.6.2 Formal Change requests

Formal requests for change can be submitted here . (There's a link at the bottom of each page as well.) These will be reviewed by the appropriate work group and a decision made on their incorporation into the specification, including which release (if any) they will be part of.

2.11.6.3 Additional Information sources/Engagement Mechanisms

In addition to the above mechanisms, HL7 provides a Stack Overflow tag, list servers and online chat system to provide various levels of implementer support and engagement. Instructions for accessing these other mechanisms (and instructions for how best to make use of them) can be found at the Support Links (and also here).

2.12 FHIR Overview - Developers

URL: https://hl7.org/fhir/STU3/overview-dev.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.12 FHIR Overview - Developers

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

FHIR (Fast Health Interoperability Resources) is designed to enable information exchange to support the provision of healthcare in a wide variety of settings. The specification builds on and adapts modern, widely used RESTful practices to enable the provision of integrated healthcare across a wide range of teams and organizations.

The intended scope of FHIR is broad, covering human and veterinary, clinical care, public health, clinical trials, administration and financial aspects. The standard is intended for global use and in a wide variety of architectures and scenarios.

2.12.1 Framework

FHIR is based on "Resources" which are the common building blocks for all exchanges. Resources are an instance-level representation of some kind of healthcare entity. All resources have the following features in common:

* A URL that identifies the resource

* Common metadata

* A human-readable XHTML summary

* A set of defined data elements - a different set for each type of resource

* An extensibility framework to support variation in healthcare

Resource instances are represented as either XML, JSON or RDF and there are currently 116 different resource types defined in the FHIR specification.

This specification describes a set of resources - that is, a set of resource types that describe the set of resource instances that can actually be exchanged. The term 'Resource' is sometimes used without clarifying whether it specifically refers to types or instances - the context of use makes this clear.

2.12.2 Example Resource Instance

This is an example of how a patient is represented as a FHIR object in JSON. An XML encoding is also defined in the specification.

```
{
  "resourceType": "Patient",
  "id" : "23434",
  "meta" : {
    "versionId" : "12",
    "lastUpdated" : "2014-08-18T15:43:30Z"
  }
  "text": {
    "status": "generated",
    "div": "<!-- Snipped for Brevity -->"
  },
  "extension": [
    {
      "url": "http://example.org/consent#trials",
      "valueCode": "renal"
    }
  ],
  "identifier": [
    {
      "use": "usual",
      "label": "MRN",
      "system": "http://www.goodhealth.org/identifiers/mrn",
      "value": "123456"
    }
  ],
  "name": [
    {
      "family": [
        "Levin"
      ],
      "given": [
        "Henry"
      ],
      "suffix": [
        "The 7th"
      ]
    }
  ],
  "gender": {
    "text": "Male"
  },
  "birthDate": "1932-09-24",
  "active": true
}
```

Each instance of a resource consists of:

* resourceType (line 2) - Required: FHIR defines many different types of resources. See the full index

* id (line 3) - The id of this resource. Always present when a resource is exchanged, except during the create operation (below)

* meta (lines 4 - 7) - Usually Present: Common use/context data to all resources and managed by the infrastructure. Missing if there is no metadata

* text (lines 8 - 11) - Recommended: XHTML that provides a human readable representation for the resource

* extension (lines 12 - 17) - Optional: Extensions defined by the extensibility framework

* data (lines 18 - 43) - Optional: data elements - a different set, defined for each type of resource

Note that although this specification always shows the JSON properties in the order that they are defined, many JSON libraries order properties by other criteria.

2.12.3 Interactions

For manipulation of resources, FHIR provides a REST API with a rich but simple set of interactions:

* Create = POST https://example.com/path/{resourceType}

* Read = GET https://example.com/path/{resourceType}/{id}

* Update = PUT https://example.com/path/{resourceType}/{id}

* Delete = DELETE https://example.com/path/{resourceType}/{id}

* Search = GET https://example.com/path/{resourceType}?search parameters...

* History = GET https://example.com/path/{resourceType}/{id}/_history

* Transaction = POST https://example.com/path/ (POST a transaction bundle to the system)

* Operation = GET https://example.com/path/{resourceType}/{id}/${opname}

The FHIR specification describes other kinds of exchanges beyond this simple RESTful API, including exchange of groups of resources as Documents, as Messages, and by using various types of Services.

2.12.4 Managing Variability

There is a wide variation between different geo-political jurisdictions and segments of the healthcare industry, and no central authority to impose common business practices. Because of this, the FHIR specification defines an extension framework and defines a framework for managing variability.

Another key aspect of the variability encountered in healthcare is that the same information may be represented differently and with different levels of detail, granularity and nesting by various parties across the system. For example, in some cases a blood pressure measurement may be just a simple observation, a vital sign measure, while in other cases can be a rich set of highly defined data that includes things like controlled vocabularies for posture, exercise, etc. The resource types defined in this specification focus on the general, common use cases. Richer and more specific content can be supported and standardized by defining "profiles" on the base resource types.

2.12.5 Managing Versions

FHIR Servers do not have to support versioning, though they are strongly encouraged to do so. There are three different levels of versioning support for FHIR servers:

* Versioning and .meta.version are not supported (and usually, .meta.lastUpdated is not supported either)

* Versioning and the VersionId meta-property are supported, but a history of old versions is not kept

* Versioning and the VersionId meta-property are supported, and a history of old versions is available

In addition, servers may require that version aware updates are used, to prevent over-writing changes, but this is not described on this page.

2.12.6 Creating a resource

To create a resource, send an HTTP POST request to the resource type's respective end point.

```
  POST https://example.com/path/{resourceType}
```

In the example below we see the creation of a Patient.

```
POST {some base path}/Patient HTTP/1.1
Authorization: Bearer 37CC0B0E-C15B-4578-9AC1-D83DCED2B2F9
Accept: application/fhir+json
Content-Type: application/fhir+json
Content-Length: 1198

{
  "resourceType": "Patient",
  ...(properties)
}
```

Submit a new patient to the server, and ask it to store the patient with an id of its own choice.

Notes:

* /Patient (line 1) - the manager for all patients - use the name of the type of resource

* Authorization (line 2) - see Security for FHIR

* Accept, Content-Type (lines 3-4) - the content type for all FHIR resources as represented in JSON (or application/fhir+xml for the XML version). FHIR resources are always represented in UTF-8

* id - The client does not need to provide an id for a resource that is being created - the server will assign one. If the client assigns one, the server will overwrite it

* Resource Content, lines 8+ - There's no meta property at this point. The rest of the resource is the same content as shown above

2.12.7 Create Response

A response will contain an HTTP code 201 to indicate that the Resource has been created successfully. A location header indicates where the resource can be fetched in subsequent requests. The server may choose to return an OperationOutcome resource, but is not required to do so.

```
HTTP/1.1 201 Created
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT
ETag: W/"1"
Location: http://example.com/Patient/347

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">The operation was successful</div>"
```

```
  }
}
```

Notes:

* HTTP/1.1 201 (line 1) - the operation was successful. Note that use of HTTP v 1.1 is strongly recommended but not required

* ETag (line 5) - used in the version aware update pattern (if the server supports versioning)

* Location (line 6) - the id the server assigned to the resource. The id in the URL must match the id in the resource when the resource is subsequently returned

* OperationOutcome (line 9) - OperationOutcome resources in this context have no id or meta element (they have no managed identity)

2.12.7.1 Error response

For a variety of reasons, servers may need to return an error. Clients should be alert to authentication related responses, but FHIR content related errors should be returned using an appropriate HTTP status code, with an OperationOutcome resource to provide additional information. Here is an example of a server rejecting a resource because of server defined business rules:

```
HTTP/1.1 422 Unprocessable Entity
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">MRN conflict
   - the MRN 123456 is already assigned to a different patient</div>"
  },
}
```

Notes:

* The server can return additional structured information using the details of the OperationOutcome

2.12.8 Read Request

Reading a resource is done by sending HTTP GET requests to the desired Resource Type end-point.

```
  GET https://example.com/path/{resourceType}/{id}
```

Here's an example.

```
GET /Patient/347?_format=xml HTTP/1.1
Host: example.com
Accept: application/fhir+xml
Cache-Control: no-cache
```

Notes:

* 347 (line 1) - The id of the resource that is being fetched

* _format=xml (line 1) - this is another method for clients to indicate the desired response format, in addition to using the accept header, and is useful for clients that don't have access to the HTTP Headers (e.g. XSLT transforms) (see Mime Types)

* cache control (line 4) - Concurrency control is important, though FHIR itself says nothing about it - see http://www.w3.org/Protocols/rfc2616/rfc2616-sec13.html or https://www.mnot.net/cache_docs/

2.12.9 Read Response

The response to a GET contains the Resource.

```
HTTP/1.1 200 OK
Content-Length: 729
Content-Type: application/fhir+xml
Last-Modified: Sun, 17 Aug 2014 15:43:30 GMT
ETag: W/"1"

<?xml version="1.0" encoding="UTF-8"?>
<Patient xmlns="http://hl7.org/fhir">
  <id value="347"/>
  <meta>
    <versionId value="1"/>
    <lastUpdated value="2014-08-17T15:43:30Z"/>
  </meta>
  <!-- content as shown above for patient -->
</Patient>
```

Notes:

* id (line 9) - The id of the resource. This must match the id in the read request

* versionId (line 11) - The current version id of the resource (if the server supports versioning). Best practice is that this value matches the ETag (see version aware update), but clients must never assume this

* Note that servers are not required to support versioning, but are strongly encouraged to do so

* lastUpdated (line 12) - if present, this must match the value in the HTTP header

2.12.10 Search Request

In addition to getting single known resources it's possible to find a collection of resources by searching the resource type end-point with a set of criteria describing the set of resources that should be retrieved, and their order. The general pattern is:

```
  GET https://example.com/path/{resourceType}?criteria
```

The criteria is a set of HTTP parameters that specify which resources to return. The search operation

```
https://example.com/base/MedicationRequest?patient=347
```

returns all the prescriptions for the patient created above.

2.12.11 Search Response

The response to a search request is a Bundle: a list of matching resources with some metadata:

```
HTTP/1.1 200 OK
Content-Length: 14523
Content-Type: application/fhir+xml
Last-Modified: Sun, 17 Aug 2014 15:49:30 GMT

{
  "resourceType": "Bundle",
  "type": "searchset",
  "id" : "eceb4882-5c7e-4ca4-af62-995dfb8cef01"
  "meta" : {
    "lastUpdated" : "2014-08-19T15:49:30Z"
  },
  "total": "3",
  "link": [
    {
      "relation" : "next",
      "url" :
    "https://example.com/base/MedicationRequest?patient=347&searchId=ff15fd40-ff71-4b48-b366-09c706bed9d0&page=2"
    }, {
      "relation" : "self",
      "url" : "https://example.com/base/MedicationRequest?patient=347"
    }
  ],
  "entry": [
    {
      "resource" : {
        "resourceType": "MedicationRequest",
        "id" : "3123",
        "meta" : {
          "versionId" : "1",
          "lastUpdated" : "2014-08-16T05:31:17Z"
        },
        ... content of resource ...
      },
    },
    ... 2 additional resources ....
  ]
}
```

Notes:

* resourceType/type (line 7/8) - the result of a search is always a bundle of type "searchset"

* id (line 9) - An identifier assigned to this particular bundle. The server should assign a unique id to this bundle that it will not be re-used

* meta.lastUpdated (line 11) - (if the server supports versioning) This should match the HTTP header, and should be the date the search was executed, or more recent, depending on how the server handles ongoing updates. The lastUpdated data SHALL be the same or more recent than the most recent resource in the results

* total (line 13) - The total number of matches in the search results. Not the number of matches in this particular bundle, which may be a paged view into the results

* link (line 14) - A set of named links that give related contexts to this bundle. Names defined in this specification: first, prev, next, last, self

* entry (line 23) - Actual resources in this set of results

* entry.resource.id (line 27) - Note that in some bundles, the entry.resource.id must be unique in the bundle

* The search operation is also able to return additional related resources as well

2.12.12 Update Request

The client sends the server a new version of the resource to replace the existing version - it PUTs it to the location of the existing resource:

```
  PUT https://example.com/path/{resourceType}/{id}
```

Note that there does not need to be a resource already existing at {id} - the server may elect to automatically create the resource at the specified address. Here is an example of updating a patient:

```
PUT /Patient/347 HTTP/1.1
Host: example.com
```

```
Content-Type: application/fhir+json
Content-Length: 1435
Accept: application/fhir+json
If-Match: 1

{
  "resourceType": "Patient",
  "id" : "347",
  "meta" : {
    "versionId" : "1",
    "lastUpdated" : "2014-08-18T15:43:30Z"
  },
  ...
}
```

Notes:

* resourceType (line 1) - "Patient" in the URL must match the resource type in the resource (line 9)

* resource id (line 1, "347") - This must match the id in the resource (line 10)

* If-Match (line 6) - if this is provided, it must match the value in meta.versionId (line 12), and the server must check the version integrity, or return 412 if it doesn't support versions

* meta.lastUpdated (line 13) - This value is ignored, and will be updated by the server (mostly, but not always, if the server does not support versioning)

* resource content (line 14) - Not shown here, the same as Patient above

2.12.13 Update Response

The response to an update request has metadata / status, and optionally an OperationOutcome:

```
HTTP/1.1 200 OK
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT
ETag: W/"2"
Location: https://example.com/Patient/347/_history/2

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">The operation was successful</div>"
  }
}
```

Notes:

* ETag (line 5) - This is the versionId of the new version, and is also found in the location header (line 6) (if the server supports versioning)

2.12.14 Base Resource Content

Here is an example that shows all the information found in all resources, fully populated:

```
{
  "resourceType" : "X",
  "id" : "12",
  "meta" : {
    "versionId" : "12",
    "lastUpdated" : "2014-08-18T15:43:30Z",
    "profile" : ["http://example-consortium.org/fhir/profile/patient"],
    "security" : [{
      "system" : "http://hl7.org/fhir/v3/ActCode",
      "code" : "EMP"
    }],
    "tag" : [{
      "system" : "http://example.com/codes/workflow",
      "code" : "needs-review"
    }]
  },
  "implicitRules" : "http://example-consortium.org/fhir/ehr-plugins",
  "language" : "X"
}
```

Implementers notes:

* resourceType (line 2) - always found in every resource. In XML, this is the name of the root element for the resource

* id (line 3) - defined when the resource is created, and never changed. Only missing when the resource is first created

* meta.versionId (line 5) - changes each time any resource contents change (except for the last 3 elements in meta - profile, security and tag)

* meta.lastUpdated (line 6) - Changes when the versionId changes. Systems that don't support versions usually don't track lastUpdated either

* meta.profile (line 7) - An assertion that the content conforms to a profile. See Extending and Restricting Resources for further discussion. Can be changed as profiles and value sets change or the system rechecks conformance

* meta.security (lines 8 - 11) - Security labels applied to this resource. These tags connect resources in specific ways to the overall security policy and infrastructure. Security tags can be updated when the resource changes, or whenever the security sub-system chooses to

* meta.tag (lines 12 - 16) - Tags applied to this resource. Tags are used to relate resources to process and workflow. Applications are not required to consider the tags when interpreting the meaning of a resource. Tags can be updated without changing the resource version

* implicitRules (line 17) - indicates that there is a custom agreement about how the resources are used that must be understood in order to safely process the resource. Use of this is discouraged because it restricts sharing, but sometimes necessary

* language (line 18) - The base language of the resource. The resource is allowed to have content from other languages; this is just the base, but should be the main language of the resource

The base properties of all resources are defined on the resource types Resource and DomainResource.

Want more information?

* Getting Started

* Resource Index

* Resource Guide

* Documentation Index

* Support Links

2.13 FHIR Overview - Clinicians

URL: https://hl7.org/fhir/STU3/overview-clinical.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.13 FHIR Overview - Clinicians

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

FHIR (Fast Health Interoperability Resources) is designed to enable the exchange of healthcare-related information. This includes clinical data as well as healthcare-related administrative, public health and research data. It covers both human and veterinary medicine and is intended to be usable world-wide in a wide variety of contexts, including in-patient, ambulatory care, acute care, long-term care, community care, allied health, etc.

The FHIR specification is targeted to individuals and organizations developing software and architecting interoperable solutions that will be using FHIR. The FHIR specification does not attempt to define good or best clinical practices, nor does it provide guidance on user interfaces or workflows. Guidance in these areas may be useful, but it is outside of FHIR's scope.

Because of FHIR's focus on implementation, many aspects of the specification deal with the technical underpinnings of the exchange of clinical information between electronic systems. This section provides an introduction to what FHIR provides, and tries to highlight those portions of the specification that are likely to be of most interest to the clinical community while skipping over some of the technical minutiae of interoperability. However, clinical readers are welcome to explore some of the more technical areas if they find them of interest.

2.13.1 Resources

From a clinical perspective, the most important parts of the FHIR specification to understand are the Resources. Think of Resources as paper "forms" reflecting different types of clinical and administrative information that can be captured and shared. The FHIR specification defines a generic "form template" for each type of clinical information - so one for allergies, one for prescriptions, one for referrals, etc.

FHIR data consists of repositories containing completed "forms" (resource instances). The resource instances describe patient-related information (such as demographics, health conditions and procedures) as well as administrative information (such as practitioners, organizations and locations). Some resources are infrastructure components used to support the technical exchange of information by describing what systems are able to do, defining allowed sets of codes, etc. FHIR repositories might be electronic health record (EHR) systems, pharmacy systems, hospital information systems (HIS), etc. Some systems, such as clinical decision support engines, may expose FHIR interfaces even though they don't actually store any patient or administrative information themselves.

Each Resource defines a small amount of highly-focused data. A single resource doesn't say very much, but a collection of Resources taken together creates a useful clinical record. Information systems map the actions that a user takes (look up patient records, make a note in their history, etc.) to operations on the relevant resources.

2.13.2 Extensibility and Profiling

The paper forms (Resources) in FHIR are somewhat generic. They have to be usable in different countries and by different types of clinicians in different contexts (human care, veterinary care, public health, research, etc.). Recognizing that a one size fits all approach is not appropriate in the healthcare space, FHIR provides the ability to adjust the forms (Resources) to be able to handle the needs of different implementation spaces by defining "extensions" as well as enforcing constraints. For example, a "prescription" form might have extension elements added to support tracking of restricted medications while also

constraining the codes that can be used to communicate types of drugs to a particular national standard. Forms are designed in such a way that these changes can be made without changing how systems pass forms around, enabling any system to consume completed forms even if they have additional elements added, whether or not those additional elements are used by the receiving system.

To keep the base forms that everyone uses from being overly complex, FHIR has a rule that, in most cases, a resource will only include data elements if there's an expectation that most implementations will use that particular data element. That doesn't mean the data must always exist. For example, most systems in the world are capable of tracking "deceased date" for a patient, even though that element will be blank for many patient records. On the other hand, not as many systems track hair color, so hair color would be omitted from the base form and those systems that need it (perhaps in a specialty clinical or research setting) can use a FHIR extension to capture it if needed.

To keep the number of resources reasonable, some of them are fairly broad. For example, the Observation resource is used for vital signs, lab results, psychological assessments and a variety of other things. To support setting rules for more narrow areas (e.g. "What should I send if I want to share a blood pressure?"), FHIR allows the creation of Profiles. There will be a great deal of clinical work involved in forming consensus around how different types of detailed clinical information should be captured and shared in particular settings. Tooling to support the creation of profiles directly by clinicians is part of the plan for FHIR, but is still in the very early stages.

2.13.3 Narrative

FHIR is intended to support sharing data in a computable manner. I.e. The information shared should be usable for computer-mediated processes such as decision support, rules triggering, trend analysis, etc. However, not every system is the same and not all systems are able to recognize all discrete data. Also, there is still considerable value in data exchange in circumstances where not all or even none of that data is captured in a discrete manner. For this reason, FHIR resources support sharing not only discrete data for computation, but also a human-readable view so that the humans on each end of a healthcare information exchange can still get a full picture of what's going on.

Narrative is expected to exist for most resource instances, although it can be omitted in a few limited circumstances. In some cases, the narrative will be generated from discrete information. For example, the narrative for a patient might look like this:

> Peter James Chalmers (OFFICIAL), Jim identifier: MRN = 12345 (USUAL) telecom: ph: (03) 5555 6473(WORK) gender: MALE birthDate: Dec 25, 1974 deceased: false address: 534 Erewhon St PleasantVille Vic 3999 (HOME)

In other cases, the narrative might be free-form text commentary entered directly by a practitioner, such as referral letters, pathology reports, etc. Certain parts of the narrative content could also later be exposed as discrete data.

2.13.4 Interfaces

In addition to defining the "forms" for data exchange (Resources), FHIR also defines a set of interfaces by which systems actually share that information. There are four primary mechanisms or "paradigms" of exchange supported by FHIR: via a REST interface, by exchanging Documents, by sending and receiving Messages and by exposing and invoking Services.

2.13.4.1 REST

REST is the simplest exchange mechanism. Continuing with the "form" metaphor, a RESTful server can be thought of as a room full of filing cabinets. Within the room is a cabinet for each "type" of form (or Resource) it supports. The cabinet contains folders where each folder has a unique number and represents one particular real-world thing: one Patient, one Encounter, one Medication, etc. Each folder (which represents a single Resource instance) contains multiple pieces of paper, with each piece of paper representing a specific "version" of that instance. Every time someone updates a record, a new piece of paper is added to the top of the file folder. To see the history of a resource, you simply have to flip through the pieces of paper in the folder.

Note that a typical medical record is generally a big "folder-of-folders" with many different types of 'forms' or 'reports' gathered together. This is convenient for someone who wants to review the whole record, but inconvenient for someone updating bits of it. There's always contention for access to it to update the right part. In the computer application the record will be decomposed to its smallest components for management purposes, and a computer will (or should) assemble the correct bits as required, by following references that exist from one piece of information to the next.

Now picture a clerk at the front door of that room. You can pass the clerk a requisition to have them do something with the information in those file cabinets. The "clerk" and the set of requisition forms on the clerk's counter make up the FHIR restful API. With that API, you can do the following:

* search: Have the clerk search through the folders for one(s) that meet a set of search criteria and give you a copy of the top piece of paper in each relevant folder

* read: Get a copy of the top piece of paper (which is the most current version) from a specific folder in one of the cabinets

* create: Add a new folder to the appropriate cabinet (with a new number)

* update: Add a new page (version) to the contents of a specific folder

* delete: Remove a folder from the cabinet (or more accurately, virtually remove it by putting a sticker on it saying "do not open")

* history: Look at all the pages in a single folder, in a particular file cabinet or possibly in the whole room. Such a broad request would be used primarily for administrative purposes rather than clinical purposes

* transaction: Give the clerk multiple pieces of paper to place in distinct folders at the same time

* operation: Ask the clerk to perform an action or procedure on papers from one or more of the folders - for example, averaging numbers across patients, producing a summary record, or perform a complex search just by ticking a box on a requisition saying "do that one"

EHRs and other systems may present a more sophisticated interface to their end users, but behind the scenes they're all making these same types of requests to the file clerk.

2.13.4.2 Documents

Documents are a familiar mechanism for sharing information in the healthcare space. They are useful whenever there's a desire to guide how a consumer of information will navigate it and there's a need to have a "frozen" view of information that can be reliably retrieved even years in the future. Examples of document-like things in healthcare include discharge summaries and lab reports.

In FHIR, there's a special resource called Composition that acts as the "cover page" for a document. It identifies the title, author date, relevant patient and the table of contents. A FHIR document can be thought of as a set of sheets (Resource instances) stacked together with a title page on top that's stapled together. That stapled collection can then be stored or passed around, conveying a complete set of information at once.

2.13.4.3 Messaging

Much healthcare information exchange happens using a messaging paradigm. In messaging, a set of information is sent from one system to another, typically triggered by an event in the sender system. For example, a patient being admitted, a lab test being ordered, a drug being administered, the clock striking 12:00 or someone pressing a button. The message serves to notify the receiver that the event occurred as well as provide details about any existing data that was modified or new data that was created. Typically receiving a message means there's an expectation that the receiving system will "do something" in response.

A message might request that a lab order be fulfilled or notify a system that two patient records have been merged or that a patient has been transferred from one bed to another. A message is similar to a document in that it collects resources together, however for a message, the "cover page" is a MessageHeader that acts as a requisition. And rather than using a staple, the resources are joined together with a paper-clip and there's no expectation that the receiving system will store the contents of the message exactly as received, if at all.

2.13.4.4 Services

Services can be thought of as a light-weight way of doing messaging. Rather than a full cover page, a small sticky note is attached to the front of a resource. And sometimes rather than sending a full piece of paper, the relevant pieces are cut out and sent as fragments. The response to a requisition is a similarly paper-clipped bundle of resource instances. Services are likely to be used for things like decision support. E.g. "Is there a problem with prescribing medication X for patient Y?" or "What's the recommended care plan for a patient with conditions A, B and C?"

2.13.5 Approaching the specification

A FHIR-based system's capabilities are defined by what the Resources can say and from a clinical perspective, these things define the clinical record:

* the kinds of Resources that are defined

* their data contents, and rules about the data such as what terminology codes are supported and/or required

* how resources reference to each other

* how you can search for information

This information can all be found in the resource definition pages. The resources most likely to be of interest can be found in the following modules:

* Clinical Summary & Care provision

* Diagnostics

* Medications

* Administrative

Instructions on how to interpret the information found on the resource pages can be found here. The Logical table or the UML views are likely to be easiest to understand. Don't forget to look at the examples tab for an idea of what kind of information can be expressed. Seeing how elements are used to convey real data is often more useful than just looking at definitions. Also, look at the Profiles tab to see examples of how different resources can be constrained for use in particular contexts.

Clinician and other domain expertise and feedback is always welcome as we continue refining the FHIR specification. At the top of each resource page is a link to the home page for the work group responsible for that particular resource. If you have feedback on resource design, consider getting involved.

2.14 FHIR Overview - Architects

URL: https://hl7.org/fhir/STU3/overview-arch.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.14 FHIR Overview - Architects

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

At its core, FHIR contains two primary components:

* Resources - a collection of information models that define the data elements, constraints and relationships for the “business objects” most relevant to healthcare. From a model-driven architecture perspective, FHIR resources are notionally equivalent to a physical model implemented in XML or JSON. See the formal definition.

* APIs – a collection of well-defined interfaces for interoperating between two applications. Although not required, the FHIR specification targets RESTful interfaces for API implementation. See details on FHIR RESTful interfaces.

In the healthcare domain, the set of “business objects” is not universally defined, but there is a notional and ongoing evolutionary, consensus-based process for standardizing on a core set of common business objects including things like “a patient”, “a procedure”, “an observation”, “an order”, etc. (see a list of defined resources). The FHIR specification provides a framework for defining these healthcare business objects (“resources”), for relating them together in a compositional manner, for implementing them in a computable form, and for sharing them across well-defined interfaces. The framework contains a verifiable and testable syntax, a set of rules and constraints, methods and interface signatures for “FHIR-aware” APIs, and specifications for the implementation of a server capable of requesting and delivering FHIR business objects.

From an operational perspective, HL7’s internal standards development and governance processes determine what constitutes a resource and which resources exist. In addition, the FHIR specification also provides a mechanism for contextualizing resources for specific needs within specific bounds (see Profiling Resources).

2.14.1 Architecture Frameworks and FHIR Alignment

FHIR resources fit firmly within the information architecture domain and the FHIR APIs for data exchange address aspects of application architecture.

From a TOGAF perspective , FHIR address aspects architecture views related information model definition and data exchange in the Information Systems Architectures portion of the TOGAF Architecture Development Method.

With regards to the Zachman Framework , FHIR fits within the What and the How dimensions of the Architect, Engineer and Technician Perspectives

When considering the HL7 Services Aware Interoperability Framework (SAIF) , FHIR resources and RESTful APIs represent the “Physical Models” and “Interface Implementations” within the Platform Specific Specifications layer of the Information Models and the Behavioral Models Viewpoints, respectively.

2.14.2 FHIR and Architectural Principles

FHIR’s primary purpose is to address interoperability with well-structured, expressive data models and simple, efficient data exchange mechanisms. In addition, FHIR aligns to the following architectural principles:

* Reuse and Composability – FHIR resources are designed with the 80/20 rule in mind – focus on the 20% of requirements that satisfy 80% of the interoperability needs. To this end, resources are designed to meet the general or common data requirements of many use cases to avoid the proliferation of numerous, overlapping and redundant resources. Extension and customizations exist (see FHIR Profiles) to allow common, somewhat generic resources to be adopted and adapted as needed for specific use case requirements. In addition, FHIR resources are highly composable in that resources commonly refer to other resources. This further promotes reuse and allows for complex structures to be built from more atomic resources.

* Scalability – Aligning FHIR APIs to the REST architectural style ensure that all transactions are stateless which reduces memory usage, eliminates the needs for “sticky” sessions within a server farm and therefore supports horizontal scalability.

* Performance – FHIR resources are lean and optimized for exchange across the network. This has the potential to improve performance in complex transactions across multiple systems connected via a shared and finite network.

* Usability – FHIR resources are understood by technical experts and non-technical people alike. Even if the details of XML or JSON syntax are not understood, non-technical people can view these in any browser or text reader and understand the contents within them.

* Data Fidelity – FHIR is strongly typed and has mechanisms built in for clinical terminology linkage and validation. In addition, XML and JSON documents can be validated syntactically as well as against a defined set of business rules. This promotes high data fidelity and goes a long way towards using FHIR to achieve semantic interoperability.

* Implementability – One of the driving forces for FHIR is the need to create a standard with high adoption across disparate developer communities. FHIR is easily understood and readily implemented using industry standards and common mark-up and data exchange technologies.

There are additional architecture principles related to consistency, granularity, referential integrity, and others that are not as well established or proven. See the section below on Outstanding Issues for details.

2.14.3 FHIR Decomposition

As discussed, FHIR’s principle components are resources and RESTful APIs. However, there is more to the FHIR specification including the components depicted below.

NOTE: The term “component” is used loosely to mean a part of something and does not intend to carry the specific meanings for this term provided by rigorous ontologies, modeling frameworks, or other architectural and organizational constructs. Diagrammatically, the components below are depicted below as UML classes. This is done purely to take advantage of the semantics afforded using this notation. FHIR is neither objected oriented in its modeling approach nor are the component that make up the FHIR specification UML classes or objects in the formal sense. Likewise, the UML packages shown below are

notional and used for organizational purposes only.

As shown in the diagram below, it is convenient to think of the FHIR specification as having components that address the following:

* Information Model – the components of FHIR related to the creation of FHIR resources

* Constraints – the components of FHIR addressing constraints and validity

* Terminology – the components of FHIR related to clinical terminologies and ontologies

* Usage – the component of FHIR addressing the use of FHIR in a run-time capacity

Component definitions:

* FHIR Information Model

* Base Clases: Element and Resource

* Definitions for Base Classes: ElementDefinition and StructureDefinition

* Data Types

* FHIR Conformance Model

* Capability Statement

* Profile

* FHIR Terminology

* Code System

* Value Set

* FHIR Usage

* REST API

2.14.4 Creating Resources – Architectural Considerations

The following list provides general guidelines that apply when FHIR resources are defined. Most of these items are not enforced programmatically requiring human due diligence and governance to ensure adherence.

* Resources should have a clear boundary; one that matches one or more logical transaction scopes

* Resources should differ from each other in meaning, not just in usage (e.g., each different way to use a lab report should not result in a different resource)

* Resources need to have a natural identity

* Most resources should be very common and used in many different business transactions

* Resources should not be specific or detailed enough to preclude support for a wide range of business transactions

* Resources should be mutually exclusive [this is a very important consideration that helps to reduce redundancy and ambiguity]

* Resources should use other resources, but they should be more than just compositions of other resources; each resource should introduce novel content

* Resources should be organized into a logical framework based on the commonality of the resource and what it links to (see resource framework below)

* Resources should be large enough to provide meaningful context; resources that contain only a few attributes are likely too small to provide meaningful business value

* Resources should reflect general usage:

* if most systems treat something as a single concept, that suggests a single resource; if most systems treat something as distinct concepts, then that suggests multiple resources

* if two different uses of a "resource" would result in wildly different interpretations of what constitutes "core" then that suggests two resources might be appropriate.

* There is a bias towards fewer resources rather than more

2.14.5 Organizing FHIR Resources

It is impractical to model the entirety of health data in a single information model. Every modeling initiative in healthcare from HL7 version 2 message specifications to FHIR resources decomposes the healthcare domain into smaller, more manageable sub-domains or information model snippets. With FHIR, each resource is essentially a snippet of the larger healthcare information domain.

When breaking down the healthcare information model into smaller chunks (or resources for FHIR), it is important to have a framework and set of guidelines to promote consistency and integrity within the resource structures and in the way resources reference each other. The framework shown below includes health information model sub-categories organized into layers based on their degree of commonness. The layers and categories are useful for identifying which parts of healthcare information are the most common and therefore need to be the most consistently defined and tightly governed. The categories at the top layers are the most common and contain the FHIR resources that support the largest number of common healthcare transactions.

Descriptions of the layers in the framework:

* Foundation Resources: Foundation resources are the most rudimentary, foundational resources. They are often used for infrastructural tasks. Although not prohibited, they are not always referenced by other resources.

* Base Resources: Layer two consists of base resources. These are often the leaf nodes of a resource graph. In other words, they are often referenced by other resources, but don't typically reference other resources themselves. These resources are typically the most commonly used, and therefore require the highest degree of consistency and architectural rigor. Governance is greatest for resources in layers one and two.

* Clinical Resources: Layer 3 includes the resources that are clinical in nature but are also very common across many use cases. This includes resources for clinical observations, clinical treatment, care provision, and medications. These resources can be used by themselves, but typically build on the resources in later two. For example, an observation resource will reference the patient resource from layer two. These resources are also frequently contextualized when they are referenced by resources in layers three, four and five.

* Financial Resources: Layer four is dedicated to financial resources. Logically, financial resources build on clinical and base resources. For example, a billing resource will reference clinical events and activities as well as base resources like a patient.

* Specialized Resources: In layer five, we find more specialized resources for less common use cases. These resources almost always reference resources in lower layers. Given that FHIR places priority on satisfying the most common use cases, there are fewer resources in this layer.

* Resource Contextualization: Layer 6 does not contain resources. However, it does extend the composition framework made up by the first five layers of resources. Layer 6 includes profiles and graphs. Profiles are used to extend, constrain, or otherwise contextualize resources for a given purpose. Graphs are compositions of resources, or webs of resource, that contain attributes of their own.

The complete set of FHIR resources organized against this framework is found on the Resources page.

The framework serves three primary purposes:

* Organize resource for navigation and identification

* Classify resources into categories based on common sense groupings or patterns describing expected structures and/or behaviors amongst resources in the same category

* Disseminate resources across layers to stratify relative common-ness with the most common resources in the top layers

Purposes 2 and 3 set the foundation for future architectural rigor and resource governance to optimize consistency, integrity and predictability of new or refined resources in the future. The actual rules and patterns will be defined and refined in future FHIR releases. However, one general guideline to state now is that resources generally reference resources in the same layer or higher. In other words, a layer 4 resource will typically only reference resources in layers 4, 3, 2 or 1. There is nothing prohibiting a layer 4 resource from referencing a layer 5 resource, but this is not as common. Given this guideline, it is possible to identify the resources that are likely to be most common across use cases and therefore demand the highest degree of consistency and governance. Further, the framework helps identify the areas where creating new FHIR resources is the highest priority. It is generally a higher priority to create FHIR resources in the higher layers (layers 1, 2 and 3) than it is to create FHIR resources in the lower layers (layers 4 and 5) because the higher layer resources will provide the greatest value across the largest number of use cases and stakeholders. This is not to say that the business transactions needed for the higher layers are not important, it’s just that they are not as common across the whole healthcare space.

The 6th layer of the framework are not actually resources. Profiles and Clusters are extensions of resources or resource compositions that continue the progression through the FHIR Composition Framework. The provide additional contextualization required to satisfy certain use cases.

There are several benefits expected from aligning the creation of FHIR resources to this framework, including::

* Organization and manageability of health domains - the framework provides a basis for decomposition and modularity

* Identifying commonality - the framework teases out the common areas from the less common areas

* FHIR resources prioritization - the framework provides a structure for determining priorities and delegating work

* Tiered governance levels - the framework separates the areas needing the most stringent and universal governance from those that require more context-specific governance

The framework is further elaborated in the FHIR Resource Considerations page .

Another useful tool for visualizing how FHIR resources are organized relative to each other can be found using the Resource Reference Visualization tool on clinFHIR .

2.14.6 FHIR Implementation – High-Level Design Considerations

2.14.6.1 FHIR Servers and the FHIR APIs

A FHIR REST server is any software that implements the FHIR APIs and uses FHIR resources to exchange data. The diagram below describes the FHIR interface definitions. The methods are classified as:

* iServeInstance – methods that perform Get, Put or Delete operations on a resource

* iServeType – methods that get type information or metadata about resources

* iServeSystem – methods that expose or enable system behaviors.

Additional details on the FHIR APIs can be found at the FHIR RESTful API and the Operations Framework.

2.14.6.2 FHIR Transactions

As mentioned, FHIR resources are optimized for stateless transactions with RESTful APIs. Although this is not the only way FHIR resources can be used, these types or transactions are the only ones with defined interfaces and behaviors in the FHIR specification.

FHIR transactions follow a simple request and response transaction pattern. The request and response can be for be for a single payload or can operate as batch. The payload or a request and response consists of a header and the content of interest. See diagram below for details.

2.14.6.3 Security

(section to be filled out) (but see Security in the meantime).

Example Use Cases Using FHIR

For illustrative purposes, the following diagram depicts a simple use case of a patient accessing their personal health record (portal) enabled by an underlying electronic medical record (EMR) system. The EMR plays the role of the FHIR server in this example.

The pre-conditions for this use case are:

* the EMR implements the necessary FHIR APIs

* the EMR implements the necessary authentication and authorization mechanisms

* the patient is successfully authenticated and authorized to access FHIR resources

The basic flow of the use case is that the patient registers (if required), logs in, enters search criteria to identify a patient or patients of interest (the patient is most like themselves in this use case), retrieves clinical documents for the patient and retrieves clinical resources for the patient. The use cases utilizes the GET methods on the iServeInstance interface and works with the following types of FHIR resources:

* The Patient resource

* One or more document resource(s)

* One or more clinical resource(s)

Although this example use case is very simple, more complex transactions using a combination of GETs, PUTs and DELETEs against resources and metadata can be envisioned. However, the exact details of these use cases including which methods are used, the orchestration of methods and the specific resources involved are outside the scope of the FHIR specification.

2.14.7 Outstanding Issues

* Resource Consistency and Granularity – there is nothing intrinsically prohibiting one resource from duplicating the same information as another resource. Further, there is nothing prohibiting resources with the same information from defining and modeling the data elements differently. HL7 has a number of processes to ensure that resources are consistently designed, but the question is when to be consistent within the specification, and when to be consistent with the real world practices of healthcare - these are sometimes in conflict with each other. Resource granularity is a related potential problem as there are variations in the size, complexity and comprehensiveness of the existing resources. Further, the degree to which the FHIR specification can impose consistency is limited to how much agreement can be gained across various communities. While the Implementers Safety Check List and the Considerations for FHIR Resource Considerations provide guidance and promote consistency, rules for achieving complete consistency of both content and granularity amongst resources are neither completely defined nor completely enforced. Considering that FHIR is still a new and emerging standard, an over-abundance of constraint and rigor has been avoided to maximize initial adoption. Further, there is a natural tension between consistency and an architectural virtue and the practicalities of supporting the real practice of health care. Considering that FHIR ultimately is a reflection of the health business processes it supports, FHIR will always carry forward some of the data discrepancies, inconsistencies and gaps that are present in the practice of healthcare across different organizations and practitioners. Nonetheless, the issues of resource consistency and granularity is a topic that gets considerable ongoing discussion, and may change as FHIR approaches a final normative standard and as FHIR adoption approaches a level where more control is warranted, or a more information/process consistency emerges in the existing healthcare systems.

* Resource References – there are currently a lack of strict rules for what resources should be referenced by other resources and under what circumstance. There is potential for ambiguity, duplication, inaccurate and/or conflicting information communicated by a resource graph (a collection of linked resources). Imagine the scenario where Resource Type A (e.g., procedure) references Resource Type B (e.g., encounter) and Resource Type C (e.g., patient), and Resource Type B (e.g., encounter) also references Resource Type C (e.g., patient). In this scenario, is a reference to Resource A to Resource C meant to provide the same information as the reference from Resource B to Resource C? If so, is this duplication of information problematic? Note that this is not unique to FHIR - it is an innate property of information systems. If an actual instances of A, and the B that a references, reference different instances of Resource C (e.g. the procedure references patient X and an encounter for patient Y), how does the system know that the references are intentionally different versus an error or data anomaly? The problem is that there is limited ability to describe the intent of the reference which leads to the possibility of ambiguity and error. The Linkage resource can be used to help with this problem, but additional capabilities may be considered in the future to allow systems to address referential integrity.

* Conditional Semantics – Currently, the constraints for element definitions including things like data types, value sets, optionality and cardinality are defined at design time with limited consideration for variable run-time semantics. Imagine the

scenario where the value of Data Element Y (e.g., “intolerance type”) is constrained differently depending on the value of Data Element X (e.g., “causative agent”) in a given instance of a resource. For example, if the instance of an Intolerance Resource has the “intolerance type” data element populated with “food intolerance”, then “causative agent” should be constrained to only valid values for this value set (e.g., valid foods instead of medications or environmental agents). Tools for addressing deep semantic consistency in this regard are only gradually developing.

* Business Rule Enforcement and Validation – As governance increases and more resource rules are defined, it may be advantageous to have a resource validation tool that checks for things like resource consistency, duplication, referential integrity, circular or non-sensical references, and other defined and approved validation rules. Once rules are agreed to, this level of automation can help address the other issues outlined above. These kinds of facilities are planned for the future.

2.15 Introducing HL7 FHIR

URL: https://hl7.org/fhir/STU3/summary.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

FHIR® – Fast Healthcare Interoperability Resources (hl7.org/fhir) – is a next generation standards framework created by HL7. FHIR combines the best features of HL7's v2 , HL7 v3 and CDA product lines while leveraging the latest web standards and applying a tight focus on implementability.

FHIR solutions are built from a set of modular components called "Resources". These resources can easily be assembled into working systems that solve real world clinical and administrative problems at a fraction of the price of existing alternatives. FHIR is suitable for use in a wide variety of contexts – mobile phone apps, cloud communications, EHR-based data sharing, server communication in large institutional healthcare providers, and much more.

2.15.1 Why FHIR is better

FHIR offers many improvements over existing standards:

* A strong focus on implementation – fast and easy to implement (multiple developers have had simple interfaces working in a single day)

* Multiple implementation libraries, many examples available to kick-start development

* Specification is free for use with no restrictions

* Interoperability out-of-the-box– base resources can be used as is, but can also be adapted for local requirements

* Evolutionary development path from HL7 Version 2 and CDA – standards can co-exist and leverage each other

* Strong foundation in Web standards– XML, JSON, HTTP, OAuth, etc.

* Support for RESTful architectures, seamless exchange of information using messages or documents, and service based architectures

* Concise and easily understood specifications

* A human-readable serialization format for ease of use by developers

* Solid ontology-based analysis with a rigorous formal mapping for correctness

2.15.2 Flexibility

A central challenge for healthcare standards is how to handle variability caused by diverse healthcare processes. Over time, more fields and optionality are added to the specification, gradually adding cost and complexity to the resulting implementations. The alternative is relying on custom extensions, but these create many implementation problems too.

FHIR solves this challenge by defining a simple framework for extending and adapting the existing resources. All systems, no matter how they are developed, can easily read these extensions and extension definitions can be retrieved using the same framework as retrieving other resources.

In addition, each resource carries a human-readable text representation using html as a fallback display option for clinical safety. This is particularly important for complex clinical information where many systems take a simple textual/document based approach.

2.15.3 Example Resource: Patient

This simple example shows the important parts of a resource: a local extension, the human readable HTML presentation, and the standard defined data content.

FHIR has resources for administrative concepts such as patient, provider, organization and device as well as a wide variety of clinical concepts covering problems, medications, diagnostics, care plans, financial concerns and more.

2.15.4 The FHIR development process

FHIR is published as a Standard for Trial Use. During the Trial Use phase, HL7 actively monitors implementations in order to continue to improve the specification, and is able to be responsive to their needs. Due to the many advantages FHIR offers, trial use is already beginning right now.

http://hl7.org/fhir . Follow us on Twitter using #FHIR

2.16 Glossary

URL: https://hl7.org/fhir/STU3/help.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.16 Glossary

This page will be the FHIR Glossary, but it has not, as yet, been created.

Looking for volunteers to take this on...

2.17 License and Legal Terms

URL: https://hl7.org/fhir/STU3/license.html

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2.17 License and Legal Terms

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

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See also the specific warnings associated with use of the STU.

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International Classification of Diseases (ICD) codes | Consult the World Health Organization (WHO)

Current Procedures Terminology (CPT) code set

2.18 Community and Credits

URL: https://hl7.org/fhir/STU3/credits.html

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2.18 Community and Credits

Maturity Level: N/A | Ballot Status: Informative

2.18.1 Community

The FHIR community meets as part of the wider HL7 community and draws on its extensive human resources, institutional memory, previous standards and corporate support. HL7 itself owns FHIR and makes it freely available and the community relies on HL7 provided infrastructure.

Te FHIR community gathers, meets, or communicates using the following infrastructure, provided by HL7 directly, or by the the HL7 FHIR Foundation :

* http://community.fhir.org - general FHIR support forum

* http://chat.fhir.org - IM based FHIR support forum

* the FHIR email list for general FHIR standards business

* the HL7 wiki : for recording informal documentation in support of the community

* gForge for task management, and also for subversion (the master FHIR source)

In addition, the community holds regular face to face connectathons and meetings as part of the HL7 Working Group meetings . The formal governance arrangements that manage FHIR development are documented on HL7's wiki .

2.18.2 Credits

FHIR is a specification produced by the HL7 Community. Many individuals and organizations contribute to the FHIR specification. Of particular note:

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* The Management group is Hans Buitendijk, Grahame Grieve, David Hay, Paul Knapp, Josh Mandel, Lloyd McKenzie, John Moehrke, Brian Pech, Brian Postlethwaite and Wayne Kubick. (Former members: Woody Beeler, Lorraine Constable, Jean Duteau, Hugh Glover, Ron Parker and John Quinn)

* The Governance board is Calvin Beebe, Woody Beeler, Lorraine Constable, Grahame Grieve, David Hay, Ewout Kramer, Wayne Kubrik, Cecil Lynch, Lloyd McKenzie, John Quinn and Dave Shaver

* The FHIR specification includes content developed by or reviewed by the following HL7 Work Groups : Application Implementation and Design , Community Based Collaborative Care , Clinical Decision Support , Clinical Genomics , Clinical Quality Information , Health Care Devices , Electronic Health Records , FHIR Infrastructure , Financial Management , Health Standards Integration , Imaging Integration , Infrastructure And Messaging , Implementable Technology Specifications , Modeling and Methodology , Orders and Observations , Patient Administration , Patient Care , Public Health and Emergency Response , Pharmacy , Regulated Clinical Research Information Management , Structured Documents , Security , US Realm Taskforce , Vocabulary

* In addition, some resources were developed in coordination with the following organizations: DICOM and IHE

* Invaluable support through review of ballots and/or participation in FHIR formal QA processes was provided by: Tomasz Adamusiak, Marla Albitz, Al Amyot, Elaine Ayres, Larry Babb, Yongjian Bao, Nagesh Bashyam, Randy Bates, Calvin Beebe, Woody Beeler, Ruth Berge, Peter Bernhardt, Andrew Berry, Bernd Blobel, Andy Bond, Keith Boone, Aziz Boxwala , Laura Bright, Helen Broberg, Hans Buitendijk, Carl Burnett, Dave Carlson, Victor Chai, Dennis Cheung, George Cole, Sheila Connolly, Kathleen Connor, Teresa Conway, Todd Cooper, Carmela Couderc, Guilherme Del Fiol, Savithri Devaraj, Gary Dickinson, Robert Dieterle, Gay Dolin, Bob Dollin, Jean Duteau, Michael Ekaireb, Sergio Enriquez, David Fallas, Sam Forouzi, Rick Geimer, Zabrina Gonzaga, William Goosen, Will Gordon, Matthew Graham, Ollie Gray, Andy Gregorowicz, Grahame Grieve, Wei Guo, Eric Haas, Stelios Halkiotis, Russel Hamm, Yegor Hanov, Smita Hastak, John Hatem, Rob Hausam, David Hay, Mike Henderson, Yan Heras, Scott Hollington, Rob Horn, Wendy Huang, Lenel James, Hozefa Jerapurwala, Emma Jones, Peter Jordan, Anthony Julian, Diego Kaminker, Naqi Khan, Bushra Khatoon, Richard Kavanagh, Kensaku Kawamoto, Thomson Khun, Ted Klein, Paul Knapp, Simon Knee, Helmut Koenig, Ewout Kramer, Mark Kramer, James Kretz, Eric Larson, Khalid Latif, Michael Lawley, Russ Leftwich, Hugo Leroux, Ben Levy, Daniel Loewenstein, Stefano Lotti, Pegeen Long, Patrick Loyd, Thomas Lukasik, Jay Lyle, Josh Mandel, Perry Mar, Brett Marquard, Rute Martins, Jason Mathews , Vince McCauley, Robert McClure, Clem McDonald, Lloyd McKenzie, Tim McNeil, Riki Merrick, Bob Milius, Michelle M Miller, Zoran Milosevic, John Moehrke, Rick Moore, Sean Moore, Saurabh Moudgil, Galen Mulrooney, Steve Munini, Stephen Munro, Ray Murakami, Ankur Nagpal, Claude Nanjo, Dale Nelson, Shamil Nizamov, Frank Oemig, Tom Oniki, Nancy Orvis, Bryn Palmer, David Parker, Brian Pech, Vadim Peretokin, Melva Peters, Vasil Peytchev, Balaji Pinjala, Fran Pivonka, Ravindran Padmanabhan, Dave Pyke, Gila Pyke, Stan Rankins, Gordon Raup, Dianne Reeves, Greg Rehwoldt, Bryn Rhodes , Wes Rishel, Mitra Rocca, Michael Rossman, Iryna Roy, Daniel Rutz, Rob Savage, Brian Scheller, Amnon Shabo, Avinash Shanbhag, Kevin Shekleton, Ioana Singureanu, Anne Smith, Rik Smithies, Harry Solomon, Corey Spears, Rene Spronk, Andy Stechishin, Howard Strasberg, Jenni Syed, David Tao, Beth Tenorio, Sylvia Thun, Leslie Tompkins, Chris Toomey, Andrew Torres, Mollie Ullman-Coliere, Cary Ussery, Michael van der Zel, Bas van Poppel, Srinivas Velamur, Serafina Versaggi, Jens Villadsen, Mead Walker, Jason Walonoski, Kathy Walsh, Yunwei Wang, , Ian Williams , Timothy Williams, Robert Worden, Kalyani Yerra and Zaib Zaveree

* Innumerable others have assisted through discussions on the Implementer's Skype chat, http://chat.fhir.org, the FHIR list server, participation at HL7 Working Group Meetings and through submission of change requests

* The basic HTML design done by studiojoyo.com - thanks

* THANK YOU all- without the support of this community, FHIR would not exist.

2.19 Outstanding Issues

URL: https://hl7.org/fhir/STU3/todo.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.19 Outstanding Issues

This specification is currently in its third round of trial use. While some parts of the specification are mature and stable, and HL7 is focusing on moving these to a formal standard, much work remains to be done in other areas. The following general areas of functionality have been deferred to a future version, or are incompletely covered in this version:

* An alarm resource to represent current issues with the patient (e.g. device created)

* Concern Tracking

* Aggregated Data Reporting including Public Health Reporting

* One or more resources for Advance Care Directive / Power of Attorney

* A full server side query framework

For some of these, some draft content is included in the specification for implementer consideration.

In addition, there are a number of specific notes in the specification requesting feedback from implementers:

* AllergyIntolerance: new codes needed for certainty?

* AllergyIntolerance: How should "No Known Allergies" be represented?

* Appointment: Values for Appointment.priority: how interoperable are they

* BodySite: Should it be an independent resource or a datatype?

* ClinicalImpression: General Questions about use

* Clinical Reasoning: Multiple questions about implementation and usage

* Composition: Is title different to type? and open questions about document signatures

* Condition: How should "No Known Problems" be represented?

* DataElement: Should DataElement, be rolled into StructureDefinition as a Logical Model?

* Markdown: Probable that will recommend/require CommonMark in release 4 - implementation consequences?

* Extensibility: Do we need to support modifier extensions on extensions?

* RESTful API: Consequence of side effects in batches and transactions?

* RESTful API: Should servers maintain ids when processing transactions?

* RESTful API: What are the documentation requirements around transaction integrity?

* Managing Resource Identity: Mandating Identification practices for cross-system interoperability?

* Observation: Request for feedback on using the lastn operation and whether it should be applied to other resources

* Operations: Do we need a way to execute operations asynchronously?

* Patient: Should linking/merging affect the RESTful API?

* Profiling Resources: Need feedback on using system profiles

* JSON-LD format: Need feedback on the JSON-LD format and it's usefulness

* References between Resources: Do we need to allow contained resources that reference the container?

* Safety: Request for comments about further development

* Search: does text search need to be standardized?

* Search: Problems with which elements are marked as 'included in summary'

* Security: Feedback about signatures on RESTful interfaces sought

* Subscription: messaging details still to be resolved

* Workflow: several implementation questions

2.20 Appendix: The Role of Informatics in the Shift from Reactive to Proactive Healthcare

URL: https://hl7.org/fhir/STU3/change.html

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2.20 Appendix: The Role of Informatics in the Shift from Reactive to Proactive Healthcare

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

FHIR offers more than a standard solving existing problems in interoperability, it provides a platform for the future. Interoperability is "all about the people"; to get past the peak of inflated expectations to the plateau of productivity on the hype cycle it is imperative that there is a shared vision of what is trying to be achieved. Healthcare is in the midst of a transformation driven by the convergence of biological and information revolutions and by economic imperative and social change. FHIR's strength lies in it being grounded in the real world which is changing rapidly. This appendix offers a high-level view of the disruption underway and helps place FHIR into a wider context.

Just as John Snow used a branch of informatics and geospatial analysis to identify the source of cholera in London and so prove the germ theory and revolutionize medicine in 1854, so too will the application of informatics and associated technology be instrumental in the next big change in healthcare.

In countries with advanced economies, health systems are facing increased demand from aging populations and increased incidence of chronic disease. In many of these countries, there is also a shortage of skilled workers. Mounting evidence points to avoidable errors causing serious harm to patients. Indeed, optimal care only occurs about half the time in even the best performing health systems (1). Small incremental improvements will not be enough to address the looming crisis in sustainable healthcare.

Convergence of trends in health care, biology, informatics and technology together with the associated social changes and economic imperative is driving a paradigm shift (2) that may be the answer. Informatics has a role in most aspects of this. The figure below provides a summary of the convergence of these trends and the likely results of disruption that will move us from reactive to proactive healthcare.

In terms of our understanding of physiological pathways, informatics is now the major tool of molecular biology. For example, the time spent computing exceeds chemical analysis for gene sequencing by 3 to 1. Informatics is being used to map neural networks and to build the models of systems biology (3) with ever increasing levels of precision and complexity that can't be done without the help of machines. Our understanding has changed so much in recent times that the American Academy of Science is now arguing that it is time for a new taxonomy (4).

Personalization is occurring both because of social change and increased biological knowledge and is being facilitated by cheap mobile computing, sensors and devices. These social forces and the enabling technologies are allowing greater participation by 'non-experts' in decision making, treatment, discovery and knowledge management.

Greater knowledge about how we think (5), advances in the information sciences and the increased availability of computing power means that our capacity to acquire knowledge and use it to predict the course of pathology has increased enormously, and that is fortuitous, because the explosion of information is impossible to deal with otherwise. These new capabilities can be used to provide better advice and to better prevent disease through discovery, monitoring and treatment.

The health system itself can also benefit from what looks like a second phase of utilization of information technology through on-line care provision, real integrated measurement of quality and integration of knowledge in work-flow. With openness and transparency there is also the prospect of co-opetition (cooperative competition) and with integrated outcome measures that pay-for-success contracting can be used as an economic lever.

Large scale change in the way healthcare is done is both essential and inevitable. It is likely that this will derive from the merging of the knowledge and machines of the biological and information revolutions facilitating a shift from reactive treatment to proactive personalized medicine. Only by such significant changes could the needed quantum improvement in the effectiveness and efficiency of healthcare be made.

The development of digital technology has disrupted other sectors, notably media, retail and manufacturing, and the health sector is unlikely to remain immune (6). Digitization of biology and health will allow machines to help, lead to a demystification of disease, the democratization of healthcare, and a move from the treatment of disease to the promotion and maintenance of wellness.

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This page maintained by Michael Legg

2.21 RESTful API

URL: https://hl7.org/fhir/STU3/http.html

2.21.0 RESTful API

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

FHIR is described as a 'RESTful' specification based on common industry level use of the term REST. In practice, FHIR only supports Level 2 of the REST Maturity model as part of the core specification, though full Level 3 conformance is possible through the use of extensions. Because FHIR is a standard, it relies on the standardization of resource structures and interfaces. This may be considered a violation of REST principles but is key to ensuring consistent interoperability across diverse systems.

Each "resource type" has the same set of interactions defined that can be used to manage the resources in a highly granular fashion. Applications claiming conformance to this framework claim to be conformant to "RESTful FHIR" (see Conformance).

Note that in this RESTful framework, transactions are performed directly on the server resource using an HTTP request/response. The API does not directly address authentication, authorization, and audit collection - for further information, see the Security Page.

The API describes the FHIR resources as a set of operations (known as "interactions") on resources where individual resource instances are managed in collections by their type. Servers can choose which of these interactions are made available and which resource types they support. Servers SHALL provide a Capability Statement that specifies which interactions and resources are supported.

The following logical interactions are defined:

Instance Level Interactions
read | Read the current state of the resource
vread | Read the state of a specific version of the resource
update | Update an existing resource by its id (or create it if it is new)
patch | Update an existing resource by posting a set of changes to it
delete | Delete a resource
history | Retrieve the change history for a particular resource
Type Level Interactions
create | Create a new resource with a server assigned id
search | Search the resource type based on some filter criteria
history | Retrieve the change history for a particular resource type
Whole System Interactions
capabilities | Get a capability statement for the system
batch/transaction | Update, create or delete a set of resources in a single interaction
history | Retrieve the change history for all resources
search | Search across all resource types based on some filter criteria

In addition to these interactions, there is an operations framework, which includes endpoints for validation, messaging and Documents.

Style Guide

The interactions on this page are defined like this:

```
VERB [base]/[type]/[id] {?_format=[mime-type]}
```

* The first word is the HTTP verb used for the interaction

* Content surrounded by [] is mandatory, and will be replaced by the string literal identified. Possible insertion values:

* base: The Service Base URL

* mime-type: The Mime Type

* type: The name of a resource type (e.g. "Patient")

* id: The Logical Id of a resource

* vid: The Version Id of a resource

* compartment: The name of a compartment

* parameters: URL parameters as defined for the particular interaction

* Content surrounded by {} is optional

Implementations constructing URLs using these patterns SHOULD conform to RFC 3986 Section 6 Appendix A which requires percent-encoding for a number of characters that occasionally appear in the URLs (mainly in search parameters).

This specification uses the underscore as a prefix to disambiguate reserved names from other names in 3 cases:

* To differentiate system wide history and search interactions from interactions on Resource Types

* To differentiate search, history and similar interactions from instances of a resource type

* To differentiate search parameters defined for all resources from those defined for specific resource types

In addition, the character $ is used as a prefix to operation names that are RPC-like additions to the base API defined either by this specification or by implementers.

2.21.0.1 Service Base URL

The Service Base URL is the address where all of the resources defined by this interface are found. The Service Base URL takes the form of

```
http(s)://server{/path}
```

The path portion is optional, and does not include a trailing slash. Each resource type defined in this specification has a manager (or "entity set") that lives at the address /[type] where the [type] is the name of the resource type. For instance, the resource manager for the type Patient will live at:

```
https://server/path/Patient
```

All the logical interactions are defined relative to the service root URL. This means that if the address of any one FHIR resource on a system is known, the address of other resources may be determined.

Note: All URLs (and ids that form part of the URL) defined by this specification are case sensitive. Clients SHOULD encode URLs using UTF-8, and servers SHOULD decode them assuming they are UTF-8 (for background, see here).

Note that a server may use a path of the form http://server/...[xx]... where the [xx] is some variable portion that identifies a particular instantiation of the FHIR API. Typically, the variable id identifies a patient or a user, and the underlying information is completely compartmented by the logical identity associated with [xx]. In this case, the FHIR API presents a patient or user centric view of a record, where authentication/authorization is explicitly granted to the URL, on the grounds that some identifiable user is associated with the logical identity. It is not necessary to explicitly embed the patient id in the URL - implementations can associate an FHIR end-point with a particular patient or provider by using an OAuth login. See Compartments for the logical underpinning.

Identity

Systems often need to compare two URLs to determine whether they refer to the same underlying object or not. For the purposes of this specification, the following rules apply:

* The query part of the URL (anything after ?) is ignored

* The comparison of the document portion of the URL (i.e. not the server/port) is case sensitive

* The protocols http: and https: SHALL NOT be used to refer to different underlying objects

* If a port is specified, then the ports must be identical or the objects are different (due to the prevalence of port mapping and/or interface engines running on different ports). Ports should only be explicit when they have explicit meaning to the server

For example: http://myserver.com/Patient/1 and https://myserver.com/Patient/1 refer to the same underlying object, while http://myserver.com:81/Patient/1 is a distinct entity from either of the above. This does not mean that the two addresses need to be treated the same, or that a server must serve both addresses, or that the content from the two addresses must be identical, but just that if these two addresses have the same identity, and if both are served, they must both represent the same underlying object. Systems are not required to check that this is true. Note: the identity comparison for protocols other than http:/https: is undefined.

2.21.0.2 Resource Metadata and Versioning

Each resource has an associated set of resource metadata elements. These map to the HTTP request and response using the following fields:

Metadata Item	Where found in HTTP
Logical Id (.id)	The Id is represented explicitly in the URL
Version Id (.meta.versionId)	The Version Id is represented in the ETag header
Last modified (.meta.lastUpdated)	HTTP Last-Modified header

Note that the Version Id is considered a "weak" ETag and ETag headers should be prefixed with W/ and enclosed in quotes, for example:

```
ETag: W/"3141"
```

2.21.0.3 Security

Using HTTPS is optional, but all production exchange of healthcare data SHOULD use SSL and additional security as appropriate. See HTTP Security for further information. Most operations will require user authentication, and all operations that do are subject to RBAC and/or ABAC , and some operations may depend on appropriate consent being granted.

See the HTTP Security guidance on how to handle access denial response.

Note: to support browser-based client applications, servers SHOULD implement cross-origin resource sharing for the interactions documented here. Experience shows that this is an area where ongoing issues may be expected as security holes are found and closed on an ongoing basis.

2.21.0.4 HTTP Status Codes

This specification makes rules about the use of specific HTTP status codes in particular circumstances where the status codes SHALL map to particular states correctly, and only where the correct status code is not obvious. Other HTTP status codes may be used for other states as appropriate, and this particularly includes various authentication related status codes and redirects. Authentication redirects should not be interpreted to change the location of the resource itself (a common web programming error).

FHIR defines an OperationOutcome resource that can be used to convey specific detailed processable error information. For some combinations of interactions and specific return codes, an OperationOutcome is required to be returned as the content

of the response. The OperationOutcome may be returned with any HTTP 4xx or 5xx response, but this is not required - many of these errors may be generated by generic server frameworks underlying a FHIR server.

2.21.0.5 Managing Return Content

In the interests of managing band-width, this specification allows clients to specify what kind of content to return.

2.21.0.5.1 conditional read

Clients may use the If-Modified-Since, or If-None-Match HTTP header on a read request. If so, they SHALL accept either a 304 Not Modified as a valid status code on the response (which means that the content is unchanged since that date) or full content (either the content has changed, or the server does not support conditional request).

Servers can return 304 Not Modified where content is unchanged because the If-Modified-Since date-time or the If-None-Match ETag was specified, or they can return the full content as normal. This optimisation is relevant in reducing bandwidth for caching purposes and servers are encouraged but not required to support this. If servers don't support conditional read, they just return the full content.

2.21.0.5.2 create/update/patch/transaction

These interactions are performed using POST, PUT or PATCH, and it may be appropriate for a server to return either only a status code, or also return the entire resource that is the outcome of the create or update (which may be different to that provided by the client). In the case of transactions this means returning a Bundle with just the Bundle.entry.response populated for each entry, and not the Bundle.entry.resource values.

The client can indicate whether the entire resource is returned using the HTTP return preference :

```
Prefer: return=minimal
Prefer: return=representation
Prefer: return=OperationOutcome
```

The first of these asks to return no body. The second asks to return the full resource. The third asks the server to return an OperationOutcome resource containing hints and warnings about the operation rather than the full resource. Servers SHOULD honor this header. In the absence of the header, servers may choose whether to return the full resource or not (but not the OperationOutcome; that should only be returned if explicitly requested). Note that this setting only applies to successful interactions. In case of failure, servers SHOULD always return a body that contains an OperationOutcome resource.

2.21.0.6 Content Types and encodings

The formal MIME-type for FHIR resources is application/fhir+xml or application/fhir+json. The correct mime type SHALL be used by clients and servers:

* XML: application/fhir+xml

* JSON: application/fhir+json

* RDF: text/turtle (only the Turtle format is supported)

Servers SHALL support server-driven content negotiation as described in section 12 of the HTTP specification.

Note: between FHIR DSTU2 and STU3, the correct mime type was changed from application/xml+fhir and application/json+fhir to application/fhir+xml and application/fhir+json. Servers MAY also support the older mime types, and are encouraged to do so to smooth the transtion process.

In order to support various implementation limitations, servers SHOULD support the optional _format parameter to specify alternative response formats by their MIME-types. This parameter allows a client to override the accept header value when it is unable to set it correctly due to internal limitations (e.g. XSLT usage). For the _format parameter, the values xml, text/xml, application/xml, and application/fhir+xml SHALL be interpreted to mean the XML format, the codes json, application/json and application/fhir+json SHALL be interpreted to mean the JSON format, and the codes ttl and text/turtle SHALL be interpreted to mean the Turtle RDF format. In addition, the values html and text/html are allowed.

FHIR uses UTF-8 for all request and response bodies. Since the HTTP specification (section 3.7.1) defines a default character encoding of ISO-8859-1, requests and responses SHALL explicitly set the character encoding to UTF-8 using the charset parameter of the MIME-type in the Content-Type header. Requests MAY also specify this charset parameter in the Accept header and/or use the Accept-Charset header.

Note: the _format parameter does not override the content-type header.

The content type application/x-www-form-urlencoded is also accepted for posting search requests.

If neither the accept header nor the _format parameter are specified, the MIME-type of the content returned by the server is undefined and may vary.

Implementation Note: If a client provides a generic mime type in the Accept header (application/xml, text/json, or application/json), the server SHOULD respond with the requested mime type, using the XML or JSON formats described in this specification as the best representation for the named mime type (though see the note on the Binary resource).

2.21.0.7 Support for Versions

Servers that support this API SHOULD provide full version support - that is, populate and track versionId correctly, support vread, and implement version aware updates. Supporting versions like this allows for related systems to track the correct version of information, and to keep integrity in clinical records. However, many current operational systems do not do this,

and cannot easily be re-engineered to do so.

For this reason, servers are allowed to not provide versioning support and this API does not enforce that versioning is supported. Clients may elect to only interact with servers that do provide full versioning support. Systems declare their support for versioning in their Capability Statements, where they can indicate one of three levels for versioning support:

* no-version: Versioning and meta.version is not supported (server) or used (client)

* versioned: Versioning and meta.version is supported (server) or used (client)

* versioned-update: Versioning and meta.version is supported, and version aware updates are used - Version ID must be correct for updates (server) or will be specified (If-match header) for updates (client)

Server should always return the default timezone for date searches in the HTTP Response headers using the Date header. Note: Servers are not required to have a default timezone.

2.21.0.8 read

The read interaction accesses the current contents of a resource. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/[type]/[id] {?_format=[mime-type]}
```

This returns a single instance with the content specified for the resource type. This url may be accessed by a browser. The possible values for the Logical Id ("id") itself are described in the id type. The returned resource SHALL have an id element with a value that is the [id]. Servers SHOULD return an ETag header with the versionId of the resource (if versioning is supported) and a Last-Modified header.

Note: Unknown resources and deleted resources are treated differently on a read: a GET for a deleted resource returns a 410 status code, whereas a GET for an unknown resource returns 404. Systems that do not track deleted records will treat deleted records as an unknown resource. Since deleted resources may be brought back to life, servers MAY include an ETag on the error response when reading a deleted record to allow version contention management when a resource is brought back to life.

In addition, the search parameter _summary can be used when reading a resource:

```
GET [base]/[type]/[id] {?_summary=text}
```

This requests that only a subset of the resource content be returned, as specified in the _summary parameter, which can have the values true, false, text & data. Note that a resource that only contains a subset of the data is not suitable for use as a base to update the resource, and may not be suitable for other uses. The same applies to the _elements parameter - both that it should be supported, and the subset implications. Servers SHOULD define a Resource.meta.tag with the SUBSETTED as a Simple Tag to explicitly mark such resources.

2.21.0.9 vread

The vread interaction preforms a version specific read of the resource. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/[type]/[id]/_history/[vid] {?_format=[mime-type]}
```

This returns a single instance with the content specified for the resource type for that version of the resource. The returned resource SHALL have an id element with a value that is the [id], and a meta.versionId element with a value of [vid]. Servers SHOULD return an ETag header with the versionId (if versioning is supported) and a Last-Modified header.

The Version Id ("vid") is an opaque identifier that conforms to the same format requirements as a Logical Id. The id may have been found by performing a history interaction (see below), by recording the version id from a content location returned from a read or from a version specific reference in a content model. If the version referred to is actually one where the resource was deleted, the server should return a 410 status code.

Servers are encouraged to support a version specific retrieval of the current version of the resource even if they do not provide access to previous versions. If a request is made for a previous version of a resource, and the server does not support accessing previous versions (either generally, or for this particular resource), it should return a 404 Not Found error, with an operation outcome explaining that history is not supported for the underlying resource type or instance.

2.21.0.10 update

The update interaction creates a new current version for an existing resource or creates an initial version if no resource already exists for the given id. The update interaction is performed by an HTTP PUT command as shown:

```
PUT [base]/[type]/[id] {?_format=[mime-type]}
```

The request body SHALL be a Resource with an id element that has an identical value to the [id] in the URL. If no id element is provided, or the value is wrong, the server SHALL respond with an HTTP 400 error code, and SHOULD provide an OperationOutcome identifying the issue. If the request body includes a meta, the server SHALL ignore the provided versionId and lastUpdated values. If the server supports versions, it SHALL populate the meta.versionId and meta.lastUpdated with the new correct values. Servers are allowed to review and alter the other metadata values, but SHOULD refrain from doing so (see metadata description for further information). Note that there is no support for updating past versions - see notes on the history interaction.

A server SHOULD accept the resource as submitted when it accepts the update, and return the same content when it is subsequently read. However systems may not be able to do this; see the note on transactional integrity for discussion. Also, see Variations between Submitted data and Retrieved data for additional discussion around update behavior. Note that

update generally updates the whole content of the resource. For partial updates, see patch below.

If the interaction is successful, the server SHALL return either a 200 OK HTTP status code if the resource was updated, or a 201 Created status code if the resource was created, with a Last-Modified header, and an ETag header which contains the new versionId of the resource. If the resource was created (i.e. the interaction resulted in a 201 Created), the server SHOULD return a Location header (this is for HTTP conformance; it's not otherwise needed).

Note: Servers MAY choose to preserve XML comments, instructions, and formatting or JSON whitespace when accepting updates, but are not required to do so. The impact of this on digital signatures may need to be considered.

Note that servers MAY choose to allow clients to PUT a resource to a location that does not yet exist on the server - effectively, allowing the client to define the id of the resource. Whether a server allows this is a deployment choice based on the nature of its relationships with the clients. While many servers will not allow clients to define their ids, there are several reasons why it may be necessary in some configurations:

* client is reproducing an existing data model on the server, and needs to keep original ids in order to retain ongoing integrity

* client is a server doing push based pub/sub (this is a special case of the first reason)

* multiple clients doing push in the context of agreed data model shared across multiple servers where ids are shared across servers

Alternatively, clients may be sharing an agreed identification model (e.g. key server, scoped identifiers, or UUIDs) where clashes do not arise.

Servers can choose whether or not to support client defined ids, and indicate such to the clients using CapabilityStatement.rest.resource.updateCreate.

2.21.0.10.1 Rejecting Updates

Servers are permitted to reject update interactions because of integrity concerns or other business rules, and return HTTP status codes accordingly (usually a 422).

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

* 400 Bad Request - resource could not be parsed or failed basic FHIR validation rules (or multiple matches were found for conditional criteria)

* 401 Not Authorized - authorization is required for the interaction that was attempted

* 404 Not Found - resource type not supported, or not a FHIR end-point

* 405 Method Not allowed - the resource did not exist prior to the update, and the server does not allow client defined ids

* 409/412 - version conflict management - see below

* 422 Unprocessable Entity - the proposed resource violated applicable FHIR profiles or server business rules

Any of these errors SHOULD be accompanied by an OperationOutcome resource providing additional detail concerning the issue.

For additional information on how systems may behave when processing updates, refer to the Variations between Submitted data and Retrieved data page.

2.21.0.10.2 Conditional update

The conditional update interaction allows a client to update an existing resource based on some identification criteria, rather than by logical id. To accomplish this, the client issues a PUT as shown:

```
PUT [base]/[type]?[search parameters]
```

When the server processes this update, it performs a search using its standard search facilities for the resource type, with the goal of resolving a single logical id for this request. The action it takes depends on how many matches are found:

* No matches: The server performs a create interaction

* One Match: The server performs the update against the matching resource

* Multiple matches: The server returns a 412 Precondition Failed error indicating the client's criteria were not selective enough

This variant can be used to allow a stateless client (such as an interface engine) to submit updated results to a server, without having to remember the logical ids that the server has assigned. For example, a client updating the status of a lab result from "preliminary" to "final" might submit the finalized result using PUT path/Observation?identifier=http://my-lab-system|123

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional update should return an HTTP 400 error and an OperationOutcome.

2.21.0.11 Managing Resource Contention

Lost Updates , where two clients update the same resource, and the second overwrites the updates of the first, can be prevented using a combination of the ETag and If-Match header. This is also known as 'Optimistic Locking'.

To support this usage, servers SHOULD always return an ETag header with each resource:

```
HTTP 200 OK
Date: Sat, 09 Feb 2013 16:09:50 GMT
Last-Modified: Sat, 02 Feb 2013 12:02:47 GMT
ETag: W/"23"
```

```
Content-Type: application/fhir+json
```

If provided, the value of the ETag SHALL match the value of the version id for the resource. Servers are allowed to generate the version id in whatever fashion that they wish, so long as they are valid according to the id data type, and are unique within the address space of all versions of the same resource. When resources are returned as part of a bundle, there is no ETag, and the versionId of the resource is used directly.

If the client wishes to request a version aware update, it submits the request with an If-Match header that quotes the ETag from the server:

```
PUT /Patient/347 HTTP/1.1
If-Match: W/"23"
```

If the version id given in the If-Match header does not match, the server returns a 409 Conflict status code instead of updating the resource.

Servers can require that clients provide an If-Match header by returning 412 Pre-condition failed status codes when no If-Match header is found.

2.21.0.12 patch

As an alternative to updating an entire resource, clients can perform a patch operation. This can be useful when a client is seeking to minimise its bandwidth utilization, or in scenarios where a client has only partial access or support for a resource. The patch interaction is performed by an HTTP PATCH command as shown:

```
PATCH [base]/[type]/[id] {?_format=[mime-type]}
```

The body of a PATCH operation SHALL be either:

* a JSON Patch document with a content type of application/json-patch+json

* an XML Patch document with a content type of application/xml-patch+xml

* a FHIRPath Patch parameters resource with FHIR Content Type

In either case, the server SHALL process its own copy of the resource in the format indicated, applying the operations specified in the document, following the relevant PATCH specification. When the operations have all been processed, the server processes the resulting document as an Update operation; all the version and error handling etc. applies as specified, as does the Prefer Header.

Processing PATCH operations may be very version sensitive. For this reason, servers SHALL support conditional PATCH, which works exactly the same as specified for update in Concurrency Management. Clients SHOULD always consider using version specific PATCH operations so that inappropriate actions are not executed. In addition, servers SHALL support Conditional PATCH, which works exactly as described for Conditional Update.

The server SHALL ensure that the narrative in a resource is not clinically unsafe after the PATCH operation is performed. Exactly how this is defined and can be achieved depends on the context, and how narrative is being maintained, but servers may wish to consider:

* If the existing narrative has a status != generated, the server could reject the PATCH operation

* The server could regenerate the narrative once the operation has been applied to the data

* In some limited circumstances, an XML PATCH operation could update the narrative

* The server could delete the narrative, on the basis that some later process will be able to populate it correctly

Processing XML Patch documents is tricky because of namespace handling. Servers SHALL handle namespaces correctly, but note that FHIR resources only contain two XML namespaces, for FHIR (http://hl7.org/fhir) and XHTML (http://www.w3.org/1999/xhtml).

For PATCH Examples, see (link todo).

Patch operations may be performed as part of Batch or Transaction Operations using the FHIRPath Patch format.

2.21.0.13 delete

The delete interaction removes an existing resource. The interaction is performed by an HTTP DELETE command as shown:

```
DELETE [base]/[type]/[id]
```

A delete interaction means that subsequent non-version specific reads of a resource return a 410 HTTP status code and that the resource is no longer found through search interactions. Upon successful deletion, or if the resource does not exist at all, the server should return either a 200 OK if the response contains a payload, or a 204 No Content with no response payload.

Whether to support delete at all, or for a particular resource type or a particular instance is at the discretion of the server based on the business rules that apply in its context. If the server refuses to delete resources of that type as a blanket policy, then it should return the 405 Method not allowed status code. If the server refuses to delete a resource because of reasons specific to that resource, such as referential integrity, it should return the 409 Conflict status code. Performing this interaction on a resource that is already deleted has no effect, and the server should return either a 200 OK if the response contains a payload, or a 204 No Content with no response payload. Resources that have been deleted may be "brought back to life" by a subsequent update interaction using an HTTP PUT.

Many resources have a status element that overlaps with the idea of deletion. Each resource type defines what the semantics of the deletion interactions are. If no documentation is provided, the deletion interaction should be understood as deleting the record of the resource, with nothing about the state of the real-world corresponding resource implied.

For servers that maintain a version history, the delete interaction does not remove a resource's version history. From a version history respect, deleting a resource is the equivalent of creating a special kind of history entry that has no content and is marked as deleted. Note that there is no support for deleting past versions - see notes on the history interaction.

Since deleted resources may be brought back to life, servers MAY include an ETag on the delete response to allow version contention management when a resource is brought back to life.

2.21.0.13.1 Conditional delete

The conditional delete interaction allows a client to delete an existing resource based on some selection criteria, rather than by a specific logical id. To accomplish this, the client issues an HTTP DELETE as shown:

```
DELETE [base]/[type]/?[search parameters]
```

When the server processes this delete, it performs a search as specified using the standard search facilities for the resource type. The action it takes depends on how many matches are found:

* No matches or One Match: The server performs an ordinary delete on the matching resource

* Multiple matches: Servers may choose to delete all the matching resources, or it may choose to return a 412 Precondition Failed error indicating the client's criteria were not selective enough. A server indicates whether it can delete multiple resources in its Capability Statement (.rest.resource.conditionalDelete). If there are multiple matches, either all must be deleted, or the server SHALL return an error

This variant can be used to allow a stateless client (such as an interface engine) to delete a resource on a server, without having to remember the logical ids that the server has assigned. For example, a client deleting a lab atomic result might delete the resource using DELETE /Observation?identifier=http://my-lab-system|123.

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional delete should return an HTTP 400 error and an OperationOutcome.

2.21.0.14 create

The create interaction creates a new resource in a server-assigned location. If the client wishes to have control over the id of a newly submitted resource, it should use the update interaction instead. The create interaction is performed by an HTTP POST command as shown:

```
POST [base]/[type] {?_format=[mime-type]}
```

The request body SHALL be a FHIR Resource. The resource does not need to have an id element (this is one of the few cases where a resource exists without an id element). If an id is provided, the server SHALL ignore it. If the request body includes a meta, the server SHALL ignore the existing versionId and lastUpdated values. The server SHALL populate the id, meta.versionId and meta.lastUpdated with the new correct values. Servers are allowed to review and alter the other metadata values, but SHOULD refrain from doing so (see metadata description for further information).

A server SHOULD otherwise accept the resource as submitted when it accepts the create, and return the same content when it is subsequently read. However some systems may not be able to do this; see the note on transactional integrity for discussion.

The server returns a 201 Created HTTP status code, and SHALL also return a Location header which contains the new Logical Id and Version Id of the created resource version:

```
Location: [base]/[type]/[id]/_history/[vid]
```

where [id] and [vid] are the newly created id and version id for the resource version. Servers SHOULD return an ETag header with the versionId (if versioning is supported) and a Last-Modified header.

When the resource syntax or data is incorrect or invalid, and cannot be used to create a new resource, the server returns a 400 Bad Request HTTP status code. When the server rejects the content of the resource because of business rules, the server returns a 422 Unprocessable Entity error HTTP status code. In either case, the server SHOULD include a response body containing an OperationOutcome with detailed error messages describing the reason for the error.

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

* 400 Bad Request - resource could not be parsed or failed basic FHIR validation rules

* 404 Not Found - resource type not supported, or not a FHIR end-point

* 422 Unprocessable Entity - the proposed resource violated applicable FHIR profiles or server business rules. This should be accompanied by an OperationOutcome resource providing additional detail

Note: Servers MAY choose to preserve XML comments, instructions, and formatting or JSON whitespace when accepting creates, but are not required to do so. The impact of this on digital signatures may need to be considered.

For additional information on how systems may behave when processing updates, refer to the Variations between Submitted data and Retrieved data page.

2.21.0.14.1 Conditional create

The conditional create interaction allows a client to create a new resource only if some equivalent resource does not already exist on the server. The client defines what equivalence means in this case by supplying a FHIR search query using an HL7 defined extension header "If-None-Exist" as shown:

```
If-None-Exist: [search parameters]
```

The parameter just contains the search parameters (what would be in the URL following the "?").

When the server processes this create, it performs a search as specified using its standard search facilities for the resource type. The action it takes depends on how many matches are found:

* No matches: The server processes the create as above

* One Match: The server ignore the post and returns 200 OK

* Multiple matches: The server returns a 412 Precondition Failed error indicating the client's criteria were not selective enough

This variant can be used to avoid the risk of two clients creating duplicate resources for the same record. For example, a client posting a new lab result might specify If-None-Exist: identifier=http://my-lab-system|123 to ensure it is does not create a duplicate record.

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional create should return an HTTP 412 error and an OperationOutcome.

2.21.0.15 search

This interaction searches a set of resources based on some filter criteria. The interaction can be performed by several different HTTP commands.

```
GET [base]/[type]{?[parameters]{&_format=[mime-type]}}
```

This searches all resources of a particular type using the criteria represented in the parameters.

Because of the way that some user agents and proxies treat GET and POST requests, in addition to the get based search method above, servers that support search SHALL also support a POST based search:

```
POST  [base]/[type]/_search{?[parameters]{&_format=[mime-type]}}
```

This has exactly the same semantics as the equivalent GET command. All these search interactions take a series of parameters that are a series of name=value pairs encoded in the URL (or as an application/x-www-form-urlencoded submission for a POST). (See W3C HTML forms).

Note: application/x-www-form-urlencoded is supported for POST so that invoking a search by GET or POST can be done from HTML forms in a browser (though considerable active content might be required in the browser), although this is not the main usage.

Searches are processed as specified for the Search handling mechanism.

If the search succeeds, the server SHALL return a 200 OK HTTP status code and the return content SHALL be a Bundle with type = searchset containing the results of the search as a collection of zero or more resources in a defined order. The result collection can be long, so servers may use paging. If they do, they SHALL use the method described below (adapted from RFC 5005 (Feed Paging and Archiving) for breaking the collection into pages if appropriate. The server MAY also return an OperationOutcome resource within the searchset Bundle entries that contains additional information about the search; if one is sent it SHALL NOT include any issues with a fatal or error severity, and it SHALL be marked with a Bundle.entry.search.mode of outcome.

If the search fails (cannot be executed, not that there is no matches), the return value is a status code 4xx or 5xx with an OperationOutcome.

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

* 400 Bad Request - search could not be processed or failed basic FHIR validation rules

* 401 Not Authorized - authorization is required for the interaction that was attempted

* 404 Not Found - resource type not supported, or not a FHIR end-point

2.21.0.15.1 Variant Searches

To search a compartment, for either all possible resources or for a particular resource type, respectively:

```
GET [base]/[Compartment]/[id]/{*?[parameters]{&_format=[mime-type]}}
GET [base]/[Compartment]/[id]/[type]{?[parameters]{&_format=[mime-type]}}
```

For example, to retrieve all the observation resources for a particular LOINC code associated with a particular encounter:

```
GET [base]/Encounter/23423445/Observation?code=2951-2  {&_format=[mime-type]}
```

Note that there are specific operations defined to support fetching an entire patient record or all record for an encounter.

It is also possible to search across multiple resource types:

```
GET [base]?_type=Condition,Observation&[parameters]{&_format=[mime-type]}
```

This is a request to search on both Condition and Observation. In this case, the only parameters that can be used are those defined for both Condition and Observation (using SearchParameter.base - see Cross-resource Search Parameters), or the parameters defined for all resources. If a search lists types not listed in SearchParameter.base for any of the parameters, this is an error, and a server SHOULD return a 400 status. It is also possible to search on all types at once:

```
GET [base]?[parameters]{&_format=[mime-type]}
```

When searching all resources at once, only the parameters defined for all resources can be used.

2.21.0.16 capabilities

The capabilities interaction retrieves the server's Capability Statement that defines how it supports resources. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/metadata {?_format=[mime-type]}
```

In addition, there is another method to get the capability statement, using the HTTP OPTIONS command:

```
OPTIONS [base] {?_format=[mime-type]}
```

However as of STU3, this method is deprecated, and will be removed in a future version. Using OPTIONS like this is not conformant with HTTP, and creates challenges for cross-origin resource sharing support.

Applications SHALL return a Capability Statement that specifies which resource types and interactions are supported for the GET command. If a 404 Unknown is returned from the GET, FHIR is not supported on the nominated service url. An ETag header SHOULD be returned with the CapabilityStatement. The value of the header SHALL change if the CapabilityStatement itself changes. Additional parameters that are required to be returned with the OPTIONS command are defined in the OMG hData RESTful Transport specification.

The Capability statement returned typically has an arbitrary id, and no meta element, though it is not prohibited. Capability statements can become quite large; servers are encouraged to support the _summary and _elements parameters on the capabilities interaction, though this is not required. In addition, servers are encouraged to implement the $subset and $implements operations to make it easy for a client to check conformance.

In addition to this capabilities interaction, a server may also choose to provide the standard set of interactions (read, search, create, update) defined on this page for the CapabilityStatement Resource end-point. This is different from the capabilities interaction:

capabilities interaction | returns a capability statement describing the server's current operational functionality
CapabilityStatement end-point | manages a repository of capability statements (e.g. the HL7 capability statement registry)

All servers are required to support the capabilities interaction, but servers may choose whether they wish to support the CapabilityStatement end-point, just like any other end-point.

Implementation Note: In DSTU 2 and earlier, the resource that this interaction returned was named "Conformance". Clients often connect to a server, and use the capabilities interaction to check whether they are version and/or feature compatible with the server. Such clients should be able to process either a Conformance or a CapabilityStatement resource.

2.21.0.17 batch/transaction

The batch and transaction interactions submit a set of actions to perform on a server in a single HTTP request/response. The actions may be performed independently as a "batch", or as a single atomic "transaction" where the entire set of changes succeed or fail as a single entity. Multiple actions on multiple resources of the same or different types may be submitted, and they may be a mix of other interactions defined on this page (e.g. read, search, create, update, delete, etc.), or using the operations framework.

The transaction mode is especially useful where one would otherwise need multiple interactions, possibly with a risk of loss of referential integrity if a later interaction fails (e.g. when storing a Provenance resource and its corresponding target resource, or, on document repositories, a document index entry and its accompanying document).

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them.

A batch or transaction interaction is performed by an HTTP POST command as shown:

```
POST [base] {?_format=[mime-type]}
```

The content of the post submission is a Bundle with Bundle.type = batch or transaction. Each entry carries request details (Bundle.entry.request) that provides the HTTP details of the action in order to inform the system processing the batch or transaction what to do for the entry (note: the request is optional, but SHOULD be present). If the HTTP command is a PUT or POST, then the entry SHALL contain a resource for the body of the action. The resources in the bundle are each processed separately as if they were an individual interaction or operation as otherwise described on this page, or the Operations framework. The actions are subject to the normal processing for each, including the meta element, verification and version aware updates, and transactional integrity.

Examples:

* Transaction Example with Matching Response

* Batch request to fetch Meds & Allergies with Response

* Batch request to fetch simple Patient Summary with Response

2.21.0.17.1 Batch Processing Rules

For a batch, there SHALL be no interdependencies between the different entries in the Bundle that cause change on the server. The success or failure of one change SHOULD not alter the success or failure or resulting content of another change. Servers SHOULD validate that this is the case. Note that it is considered that servers execute the batch in the same order as that specified below for transactions, though the order of execution should not matter given the previous rule.

Static references within a Bundle.entry.resource to another Bundle.entry.resource that is being created within the batch are considered to be non-conformant.

2.21.0.17.2 Transaction Processing Rules

For a transaction, servers SHALL either accept all actions and return a 200 OK, along with a response bundle (see below), or reject all resources and return an HTTP 400 or 500 type response. It is not an error if the submitted bundle has no resources in it. The outcome of the processing the transaction SHALL NOT depend on the order of the resources in the transaction. A resource can only appear in a transaction once (by identity).

Because of the rules that a transaction is atomic where all actions pass or fail together and the order of the entries doesn't matter, there is a particular order in which to process the actions:

* Process any DELETE interactions

* Process any POST interactions

* Process any PUT interactions

* Process any GET interactions

If any resource identities (including resolved identities from conditional update/delete) overlap in steps 1-3, then the transaction SHALL fail.

STU Note: clients are able to request that operations be executed as part of a transaction. Some transactions can cause side effects, such as the creation of new resources or other actions that may be difficult to fit into a transaction framework. Input regarding this issue is sought during the STU period.

Provide feedback here .

A transaction may include references from one resource to another in the bundle, including circular references where resources refer to each other. If the server assigns a new id to any resource in the bundle as part of the processing rules above, it SHALL also update any references to that resource in the same bundle as they are processed. References to resources that are not part of the bundle are left untouched. Version-specific references should remain as version-specific references after the references have been updated. Servers SHALL replace all matching links in the bundle, whether they are found in the resource ids, resource references, url elements, or <a href="" & <img src="" in the narrative.

When processing a "POST" (create), the full URL is treated as the id of the resource on the source, and is ignored; the server generates an id for the resource. For updates, the server performs a mapping between the fullUrl specified and the local URL the server knows that instance as, if possible. If the server does not have a mapping for the fullUrl, the server ignores the base URL and attempts an update assuming the base is the same as the server base. This allows the same transaction bundle to be sent to multiple systems without changing the fullUrls for each target.

When processing a batch or transaction, a server MAY choose to honor existing logical ids (e.g. Observation/1234 remains as Observation/1234 on the server), but since this is only safe in controlled circumstances, servers may choose to assign new ids to all submitted resources, irrespective of any claimed logical id in the resource, or fullUrl on entries in the batch/transaction.

Note: this behavior is subject to verification based on implementation experience and may change.

Provide feedback here .

Conditional References

When constructing the bundle, the client may not know the logical id of a resource, but it may know identifying information - e.g. an identifier. This situation arises commonly when building transactions from v2 messages. The client could resolve that identifier to a logical id using a search, but that would mean that the resolution to a logical id does not occur within the same transaction as the commit (as well as significantly complicating the client). Because of this, in a transaction (and only in a transaction), references to resources may be replaced by a search URI that describes how to find the correct reference:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="20160113160203" />
  <type value="transaction" />
  <entry>
    <fullUrl value="urn:uuid:c72aa430-2ddc-456e-7a09-dea8264671d8" />
    <resource>
      <Observation>
        <subject>
           <reference value="Patient?identifier=12345"/>
        </subject>
        <--! rest of resource omitted -->
      </Observation>
    </resource>
    <request>
      <method value="POST" />
    </request>
  </entry>
<Bundle>
```

The search URI is relative to the server's [base] path, and always starts with a resource type: [type]:?parameters.... Only filtering parameters are allowed; none of the parameters that control the return of resources are relevant.

When processing transactions, servers SHALL:

* check all references for search URIs

* For search URIs, use the search to locate matching resources

* if there are no matches, or multiple matches, the transaction fails, and an error is returned to the user

* if there is a single match, the server replaces the search URI with a reference to the matching resource

2.21.0.17.3 Batch/Transaction Response

For a batch, or a successful transaction, the response the server SHALL return a Bundle with type set to batch-response or transaction-response that contains one entry for each entry in the request, in the same order, with the outcome of processing the entry. For a failed transaction, the server returns a single OperationOutcome instead of a Bundle.

A client may use the returned Bundle to track the outcomes of processing the entry, and the identities assigned to the resources by the server. Each entry element SHALL contain a response element which details the outcome of processing the entry - the HTTP status code, and the location and ETag header values, which are used for identifying and versioning the resources. In addition, a resource may be included in the entry, as specified by the Prefer header.

2.21.0.17.4 Accepting other Bundle types

A server may choose to accept bundle types other than batch or transaction when POSTed to the [base] URL.

Bundles of type history inherently have the same structure as a transaction, and can be treated as either a transaction or batch, so servers SHOULD accept a history Bundle - this makes it possible to replicate data from one server to another easily using a pub/sub model. Note, however, that the original transaction boundaries may not be represented in a history list, and a resource may occur more than once in a history list, so servers processing history bundles must have some strategy to manage this.

For other Bundle types, should the server choose to accept them, there will be no request element (note that every entry will have a resource). In this case, the server treats the entry as either a create or an update interaction, depending on whether it recognises the identity of the resource - if the identity of the resource refers to a valid location on the server, it should treat it as an update to that location. Note: this option allows a client to delegate the matching process to the server.

2.21.0.18 history

The history interaction retrieves the history of either a particular resource, all resources of a given type, or all resources supported by the system. These three variations of the history interaction are performed by HTTP GET command as shown:

```
GET [base]/[type]/[id]/_history{?[parameters]&_format=[mime-type]}
GET [base]/[type]/_history{?[parameters]&_format=[mime-type]}
GET [base]/_history{?[parameters]&_format=[mime-type]}
```

The return content is a Bundle with type set to history containing the specified version history, sorted with oldest versions last, and including deleted resources. Each entry SHALL minimally contain either a resource which holds the resource as it is at the conclusion of the interaction, or a request with entry.request.method The request provides information about the interaction that occurred to cause the new version, and allows, for instance, a subscriber system to differentiate between create and update interactions. The principal reason a resource might be missing is that the resource was changed by some other channel rather than via the RESTful interface. If the entry.request.method is a PUT or a POST, the entry SHALL contain a resource.

The interactions create, update, and delete create history entries. Other interactions do not (note that these operations may produce side-effects such as new AuditEvent resources; these are represented as create interactions in their own right). New resources or updates to existing resources that are triggered by operations also appear in the history, as do updates to the resources that result from interactions outside the scope of the RESTful interface.

A create interaction is represented in a history interaction in the following way:

```
<entry>
  <resource>
    <Patient>
      <!-- the id of the created resource -->
      <id value="23424"/>
      <!-- snip -->
    </Patient>
  </resource>
  <request>
    <!-- POST: this was a create -->
    <method value="POST"/>
    <url value="Patient"/>
  </request>
</entry>
```

Note that conditional creates, updates and deletes are converted to direct updates and deletes in a history list.

In addition to the standard _format parameter, the parameters to this interaction may also include:

_count : integer | single | Number of return records requested. The server is not bound to return the number requested, but cannot return more
_since : instant | single | Only include resource versions that were created at or after the given instant in time
_at : dateTime | single | Only include resource versions that were current at some point during the time period specified in the date time value (may be more than one)

The history list can be restricted to a limited period by specifying a _since parameter which contains a full date time with time zone. Clients should be aware that due to timing imprecision, they may receive notifications of a resource update on the boundary instant more than once. Servers are not required to support a precision finer than by second.

The updates list can be long, so servers may use paging. If they do, they SHALL use the method described below for breaking the list into pages if appropriate, and maintain the specified _count across pages.

The history interaction can be used to set up a subscription from one system to another, so that resources are synchronized between them. Refer to the Subscription resource for an alternate means of system synchronization.

Additional Notes about maintaining a history of resources:

* The history is a record version history on a per-resource basis. It is not intended to support concurrent versions, or multi-branch version history

* Accordingly, there is no way to update or delete past versions of the record, except that the metadata can be modified (mainly for access control purposes)

* All past versions of a resource are considered to be superceded, and no longer active, but retained for audit/integrity purposes

* In the case that a past version of a resource needs to be explicitly documented as 'entered-in-error', use a Provenance resource pointing to the past version of the resource

* When tracing the history of a specific resource, applications should retrieve any provenance resources relating to the resource or its past versions

* If a request is made for a history that is not available (e.g. the system does not keep a history for the type, or the particular instance), the server should return a 404 Not Found along with an OperationOutcome explaining the problem

2.21.0.19 Transactional Integrity

When processing create and update interactions, a FHIR server is not obliged to accept the entire resource as it is; when the resource is retrieved through a read interaction subsequently, the resource may be different. The difference may arise for several reasons:

* The server merged updated content with existing content

* The server applied business rules and altered the content

* The server does not fully support all the features or possible values of the resource

Note that there is no general-purpose method to make merging with existing content or altering the content by business rules safe or predictable - what is possible, safe and/or required is highly context dependent. These kinds of behaviors may be driven by security considerations. With regard to incomplete support, clients can consult the server's base CapabilityStatement profile references to determine which features or values the server does not support.

To the degree that the server alters the resource for any of the 3 reasons above, the FHIR server will create implementation consequences for the eco-system that it is part of, which will need to be managed (i.e. it will cost more). For this reason, servers SHOULD change the resource as little as possible, given the constraints of the system exposing the FHIR resource. However due to the variability that exists within healthcare, this specification allows that servers MAY alter the resource on create/update.

Similarly, to the degree that an implementation context makes special rules about merging content or altering the content, that context will become more expensive to maintain.

Although these rules are stated with regard to servers, a similar concept applies to clients - to the degree that different client systems interacting with the server do not support the same feature set, the clients and/or the server will be forced to implement custom logic to prevent information from being lost or corrupted.

Some of these problems can be mitigated by following a pattern built on top of version-aware updates. In this pattern:

* The server provides a read interaction for any resource it accepts update interactions on

* Before updating, the client reads the latest version of the resource

* The client applies the changes it wants to the resource, leaving other information intact (note the extension related rules around this)

* The client writes the result back as an update interaction, and is able to handle a 409 or 412 response (usually by trying again)

If clients follow this pattern, then information from other systems that they do not understand will be maintained through the update.

Note that it is possible for a server to choose to maintain the information that would be lost, but there is no defined way for a server to determine whether the client omitted the information because it wasn't supported (perhaps in this case) or whether it wishes to delete the information.

2.21.0.19.1 Conformance

Both client and server systems SHOULD clearly document how transaction integrity is handled.

STU Note: For now, the only way to document how transaction integrity is handled is as text in the narrative portions of the CapabilityStatement resource. Feedback during the Trial use period on which - if any - of this information should be computable is welcome.

Provide feedback here .

2.21.0.20 Paging

If servers provide paging for the results of a search or history interaction, they SHALL conform to this method (adapted from RFC 5005 (Feed Paging and Archiving) for sending continuation links to the client when returning a Bundle (e.g. with history and search). If the server does not do this then there is no way to continue paging.

This example shows the third page of a search result:

```
<Bundle xmlns="http://hl7.org/fhir">
  <!-- snip metadata -->
  <!-- This Search url starts with base search, and adds the effective
    parameters, and additional parameters for search state. All searches
    SHALL return this value.

    In this case, the search continuation method is that the server
    maintains a state, with page references into the stateful list.
  -->
  <link>
    <relation value="self">
    <url value="http://example.org/Patient?name=peter&stateid=23&page=3"/>
  </link>
  <!-- 4 links for navigation in the search. All of these are optional, but recommended -->

  <link>
    <relation value="first"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=1"/>
  </link>
  <link>
    <relation value="previous"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=2"/>
  </link>
  <link>
    <relation value="next"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=4"/>
  </link>
  <link>
    <relation value="last"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=26"/>
  </link>

  <!-- then the search results... -->
</Bundle>
```

The server need not use a stateful paging method as shown in this example - it is at the discretion of the server how to best ensure that the continuation retains integrity in the context of ongoing changes to the resources. An alternative approach is to use version specific references to the records on the boundaries, but this is subject to continuity failures when records are updated.

A server MAY add additional state tracking parameters to the links, as shown in the example above. The client must use the server supplied links in order to traverse the pages. A server MAY inform the client of the total number of resources returned by the interaction for which the results are paged using the Bundle.total.

Note that for search, where _include can be used to return additional related resources, the total number of resources in the feed may exceed the number indicated in totalResults.

2.21.0.21 Intermediaries

The HTTP protocol may be routed through an HTTP proxy such as squid. Such proxies are transparent to the applications, though implementers should be alert to the effects of caching, particularly including the risk of receiving stale content. See the HTTP specification for further detail

Interface engines may also be placed between the consumer and the provider. These differ from proxies because they actively alter the content and/or destination of the HTTP exchange and are not bound the rules that apply to HTTP proxies. Such agents are allowed, but SHALL mark the HTTP header to assist with troubleshooting.

Any agent that modifies an HTTP request or response content other than under the rules for HTTP proxies SHALL add a stamp to the HTTP headers like this:

```
  request-modified-[identity]: [purpose]
  response-modified-[identity]: [purpose]
```

The identity SHALL be a single token defined by the administrator of the agent that will sufficiently identify the agent in the context of use. The header SHALL specify the agent's purpose in modifying the content. End point systems SHALL not use this header for any purpose. Its aim is to assist with system troubleshooting.

2.21.0.22 OMG hData RESTful Transport

This RESTful specification described here is based on the OMG Health RESTful specification (HData) . In this regard, FHIR functions as a Record Format Profile as described in that specification. Note the following significant factors to be aware of:

* FHIR maps the hData sections to resource types, and hData documents to resource instances. There are no subsections, and client systems are not able to create new sections, though compartments behave somewhat like sections

* Because clients cannot submit new sections (POST to service URL), POST to the service URL has been re-used for the transaction interaction (difference under review)

* FHIR does not (yet) define a root document. When defined, it will contain information about what the FHIR server has done (as opposed to a Capability Statement, which describes what it is capable of doing)

* Note that this specification does support the hData RESTful Transport OPTIONS command on the service URL

2.21.0.23 Summary

These tables present a summary of the interactions described here. Note that all requests may include an optional Accept header to indicate the format used for the response (this is even true for DELETE since an OperationOutcome may be returned).

Interaction | Path | Request
| Verb | Content-Type | Body | Prefer | Conditional
read | /[type]/[id] | GET | N/A | N/A | N/A | O: ETag, If-Modified-Since, If-None-Match
vread | /[type]/[id]/_history/[vid] | GET | N/A | N/A | N/A | N/A
update | /[type]/[id] | PUT | R | Resource | O | O: If-Match
delete | /[type]/[id] | DELETE | N/A | N/A | N/A | N/A
create | /[type] | POST | R | Resource | O | O: If-None-Exist
search | /[type]? | GET | N/A | N/A | N/A | N/A
/[type]/_search? | POST | application/x-www-form-urlencoded | form data | N/A | N/A
search-all | ? | GET | N/A | N/A | N/A | N/A
capabilities | /metadata | GET | N/A | N/A | N/A | N/A
transaction | / | POST | R | Bundle | O | N/A
history | /[type]/[id]/_history | GET | N/A | N/A | N/A | N/A
history-type | /[type]/_history | GET | N/A | N/A | N/A | N/A
history-all | /_history | GET | N/A | N/A | N/A | N/A
(operation) | /$[name], /[type]/$[name] or /[type]/[id]/$[name] | POST | R | Parameters | N/A | N/A
GET | N/A | N/A | N/A | N/A
POST | application/x-www-form-urlencoded | form data | N/A | N/A

Notes:

* N/A = not present, R = Required, O = optional

* For operations defined on all resources, including direct access to the meta element, see Resource Operations

Interaction | Response
| Content-Type | Body | Location | Versioning | Status Codes
read | R | R: Resource | N/A | R: ETag, Last-Modified | 200, 404, 410
vread | R | R: Resource | N/A | R: ETag, Last-Modified | 200, 404
update | R if body | O: Resource (Prefer) | R on create | R: ETag, Last-Modified | 200, 201, 400, 404, 405, 409, 412, 422
delete | R if body | O: OperationOutcome | N/A | N/A | 200, 204, 404, 405, 409, 412
create | R if body | O : Resource (Prefer) | R | R: ETag, Last-Modified | 201, 400, 404, 405, 422
search | R | R: Bundle | N/A | N/A | 200, 401?
search-all | R | R: Bundle | N/A | N/A | 200, 401?
capabilities | R | R: CapabilityStatement | N/A | N/A | 200, 404
transaction | R | R: Bundle | N/A | N/A | 200, 400, 404, 405, 409, 412, 422
history | R | R: Bundle | N/A | N/A | 200
history-type | R | R: Bundle | N/A | N/A | 200
history-all | R | R: Bundle | N/A | N/A | 200
(operation) | R | R: Parameters/Resource | N/A | N/A | 200

Note: this table lists the status codes described here, but other status codes are possible as described by the HTTP specification. Additional codes that are likely are server errors and various codes associated with authentication protocols.

2.21.1 Search

URL: https://hl7.org/fhir/STU3/search.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.21.1 Search

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

Searching for resources is fundamental to the mechanics of FHIR. Search operations traverse through an existing set of resources filtering by parameters supplied to the search operation. The text below describes the FHIR search framework, starting with simple cases moving to the more complex. Implementers need only implement the amount of complexity that they require for their implementations.

2.21.1.1 Summary Table

Search Parameter Types | Parameters for all resources | Search result parameters
Number Date/DateTime String Token Reference Composite Quantity URI | _id _lastUpdated _tag _profile _security _text _content _list _has _type _query | _sort _count _include _revinclude _summary _elements _contained _containedType

In addition, there is a special search parameter called _filter that allows for an alternative method of searching.

Also, there is a single page that lists all the search parameters.

2.21.1.2 Introduction

In the simplest case, a search is executed by performing a GET operation in the RESTful framework:

```
GET [base]/[type]?name=value&...{&_format=[mime-type]}}
```

For this RESTful search (see definition in RESTful API), the parameters are a series of name=[value] pairs encoded in the URL or as an application/x-www-form-urlencoded submission for a POST:

```
POST  [base]/[type]/_search{?[parameters]{&_format=[mime-type]}}
```

The server determines which of the set of resources it serves meet the specific criteria, and returns the results in the HTTP response as a bundle which includes the resources that are the results of the search. Note that the _format parameter works for search like for other interactions.

Search operations are executed in one of three defined contexts that control which set of resources are being searched:

* A specified resource type: GET [base]/[type]?parameter(s)

* A specified compartment, perhaps with a specified resource type in that compartment: GET [base]/Patient/[id]/[type]?parameter(s)

* All resource types: GET [base]?parameter(s) (parameters common to all types only)

Search operations can also be implemented in the messaging framework.

The server determines which of their resources meet the criteria contained in the search parameters as described below. However the server has the prerogative to return additional search results if it believes them to be relevant. Note: There is a special search for the most relevant context in which the search set is indeterminate: Patient MPI Search.

Search using GET may include sensitive information in the search parameters. Therefore, secure communications and endpoint management are recommended, see Security Communications

2.21.1.3 Handling Errors

If a server is unable to execute a search request, it may return an error. A HTTP status code of 403 signifies that the server refused to perform the search, while other 4xx and 5xx codes signify that some sort of error has occurred. When the search fails, a server SHOULD return an OperationOutcome detailing the cause of the failure. Note: An empty search result is not a failure.

In some cases, parameters may cause an error. For instance:

* A parameter may refer to a non-existent resource e.g. GET [base]/Observation?subject=101, where "101" does not exist

* A parameter may refer to an unknown code e.g. GET [base]/Observation?code=loinc|1234-1, where the LOINC code "1234-1" is not known to the server

* A parameter may refer to a time that is out of scope e.g. GET [base]/Condition?onset=le1995, where the system only has data going back to 2001

* A parameter may use an illegal or unacceptable modifier e.g. GET [base]/Condition?onset:text=1995, where the modifier cannot be processed by the server

* A data time parameter may have incorrect format e.g. GET [base]/Condition?onset=23%20May%202009

* A parameter may be unknown or unsupported (see below)

Where the content of the parameter is syntactically incorrect, servers SHOULD return an error. However, where the issue is a logical condition (e.g. unknown subject or code), the server SHOULD process the search, including processing the parameter - with the result of returning an empty search set, since the parameter cannot be satisfied.

In such cases, the search process MAY include an OperationOutcome in the search set that contains additional hints and warnings about the search process. This is included in the search results as an entry with search mode = outcome. Clients can use this information to improve future searches.

Unknown and unsupported parameters

Servers may receive parameters from the client that they do not recognise, or may receive parameters they recognise but do not support (either in general, or for a specific search). In general, servers SHOULD ignore unknown or unsupported parameters for the following reasons:

* Various HTTP stacks and proxies may add parameters that aren't under the control of the client

* The client can determine what parameters the server used by examining the self link in the return (see below)

Clients can specify how the server should behave, by using the prefer header

* Prefer: handling=strict: Client requests that the server return an error for any unknown or unsupported parameter

* Prefer: handling=lenient: Client requests that the server ignore any unknown or unsupported parameter

Servers SHOULD honor the client's request, but are not required to do so.

2.21.1.4 Standard Parameters

2.21.1.4.1 Parameters for all resources

The following parameters apply to all resources: _content, _id, _lastUpdated, _profile, _query, _security, _tag. In addition, the search parameter _text and _filter, (documented below) also applies to all resources (as do the search result parameters).

The search parameter _id refers to the logical id of the resource, and can be used when the search context specifies a resource type:

```
GET [base]/Patient?_id=23
```

This search finds the patient resource with the given id (there can only be one resource for a given id). Functionally, this is equivalent to a simple read operation:

```
GET [base]/Patient/23
```

However, the search with parameter _id returns a bundle with the requested resource, instead of just the resource itself. Additional parameters can be added which may provide additional functionality on top of this base read equivalence (e.g. _include). Note that although the _id parameter has a type of token, because servers SHALL use exact match with it, there is no system for the _id parameter.

The search parameter _lastUpdated can be used to select resources based on the last time they were changed:

```
GET [base]/Observation?_lastUpdated=gt2010-10-01
```

This search finds any observations changed since 1-Oct 2010. When this search parameter is used, applications should consider synchronization approaches (RESTful history or the Subscription resource).

The search parameters _tag, _profile and _security parameters search on the equivalent elements in the meta element. For example,

```
GET [base]/Condition?_tag=http://acme.org/codes|needs-review
```

searches for all Condition resources with the tag:

```
{
  "system" : "http://acme.org/codes",
  "code" : "needs-review"
}
```

In the same manner:

```
GET [base]/DiagnosticReport?_profile=http://hl7.org/fhir/StructureDefinition/lipid
GET [base]/DiagnosticReport?_profile=Profile/lipid
```

restricts the search to only DiagnosticReport resources that are tagged as conforming to a particular profile. The second reference is relative, and refers a local profile on the same server.

_tag, _profile and _security parameters are all token types (see below).

2.21.1.4.2 Parameters for each resource

In addition to the _id parameter which exists for all resources, each FHIR resource type defines its own set of search parameters with their names, types, and meanings. These search parameters are on the same page as the resource definitions, and are also published as part of the standard Capability statement (XML or JSON).

In general, the defined search parameters correspond to a single element in the resource, but this is not required, and some search parameters refer to the same type of element in multiple places, or refer to derived values.

Some search parameters defined by resources are associated with more than one path in a resource. This means that the search parameter matches if any of the paths contain matching content. If a path matches, the whole resource is returned in the search results. The client may have to examine the resource to determine which path contains the match.

Servers are not required to implement any of the standard search parameters (except for the _id parameter described above), and may define their own parameters.

2.21.1.4.3 Search Parameter Types

Each search parameter is defined by a type that specifies how the search parameter behaves. These are the defined parameter types:

number | Search parameter SHALL be a number (a whole number, or a decimal).

date | Search parameter is on a date/time. The date format is the standard XML format, though other formats may be supported.

string | Search parameter is a simple string, like a name part. Search is case-insensitive and accent-insensitive. May match just the start of a string. String parameters may contain spaces.

token | Search parameter on a coded element or identifier. May be used to search through the text, displayname, code and code/codesystem (for codes) and label, system and key (for identifier). Its value is either a string or a pair of namespace and value, separated by a "|", depending on the modifier used.

reference | A reference to another resource.

composite | A composite search parameter that combines a search on two values together.

quantity | A search parameter that searches on a quantity.

uri | A search parameter that searches on a URI (RFC 3986).

The search parameters can also append "modifiers" that control their behavior. The kinds of modifiers that available is dependent on the type of parameter being modified.

2.21.1.4.4 Modifiers

Parameters are defined per resource. Parameter names may specify a modifier as a suffix. The modifiers are separated from the parameter name by a colon. Modifiers are:

* For all parameters (except combination): :missing; e.g. gender:missing=true (or false). Searching for gender:missing=true will return all the resources that don't have a value for the gender parameter (which usually equates to not having the relevant element in the resource). Searching for gender:missing=false will return all the resources that have a value for the gender parameter.

* For string: :exact (the match needs to be exact, no partial matches, case sensitive and accent-sensitive), or :contains (case insensitive and accent-insensitive, partial match at start or end), instead of the default behavior (case insensitive and accent-insensitive, partial matches at the end of the string).

* For token: :text (the match does a partial searches on the text portion of a CodeableConcept or the display portion of a Coding), instead of the default search which uses codes. Other defined modifiers are :in, :below, :above and :not-in which are

described below.

* For reference: :[type] where [type] is the name of a type of resource

* For uri: :below, :above indicate that instead of an exact match, either the search term left matches the value, or vice-versa.

Server SHALL reject any search request that contains is suffixed by a modifier that the server does not support for that parameter. For example, if the server supports the name search param, but not the :exact modifier on the name, it should reject a search with the parameter name:exact=Bill, using an HTTP 400 error with an OperationOutcome with a clear error message.

2.21.1.4.5 Prefixes

For the ordered parameter types of number, date, and quantity, a prefix to the parameter value may be used to control the nature of the matching. To avoid URL escaping and visual confusion, the following prefixes are used:

eq | the value for the parameter in the resource is equal to the provided value | the range of the search value fully contains the range of the target value

ne | the value for the parameter in the resource is not equal to the provided value | the range of the search value does not fully contain the range of the target value

gt | the value for the parameter in the resource is greater than the provided value | the range above the search value intersects (i.e. overlaps) with the range of the target value

lt | the value for the parameter in the resource is less than the provided value | the range below the search value intersects (i.e. overlaps) with the range of the target value

ge | the value for the parameter in the resource is greater or equal to the provided value | the range above the search value intersects (i.e. overlaps) with the range of the target value, or the range of the search value fully contains the range of the target value

le | the value for the parameter in the resource is less or equal to the provided value | the range below the search value intersects (i.e. overlaps) with the range of the target value or the range of the search value fully contains the range of the target value

sa | the value for the parameter in the resource starts after the provided value | the range of the search value does not overlap with the range of the target value, and the range above the search value contains the range of the target value

eb | the value for the parameter in the resource ends before the provided value | the range of the search value does overlap not with the range of the target value, and the range below the search value contains the range of the target value

ap | the value for the parameter in the resource is approximately the same to the provided value. Note that the recommended value for the approximation is 10% of the stated value (or for a date, 10% of the gap between now and the date), but systems may choose other values where appropriate | the range of the search value overlaps with the range of the target value

If no prefix is present, the prefix eq is assumed. Note that the way search parameters operate is not the same as the way the operations on two numbers work in a mathematical sense. sa (starts-after) and eb (ends-before) are not used with integer values.

For each prefix above, two interpretations are provided - the simple intent of the prefix and the interpretation of the parameter when applied to ranges. The range interpretation is provided for decimals and dates. Searches are always performed on values that are implicitly or explicitly a range. For instance, the number 2.0 has an implicit range of 1.95 to 2.05, and the date 2015-08-12 has an implicit range of the all the time during that day. If the target value is a Range, a Period, or a Timing, then the target is explicitly a range. Three ranges are identified:

range of the value | The limits implied by the precision of the value | The number 2.0 has a range of 1.95 to 2.05 The date 2015-08-12 has a range from 00:00 to 00:00 exclusive

range below the value | Up to the specified value | The range below 2.0 includes any value less or equal to <2.00000000000000000000 The range before 2015-08-12T05:23:45 includes any time up to 2015-08-12T05:23:45.000000000000000

range above the value | The specified value and up | The range above 2.0 includes any value greater or equal to <2.00000000000000000000 The range after 2015-08-12T05:23:45 includes any time after 2015-08-12T05:23:45.000000000000000

The proper use of these ranges is discussed further below.

2.21.1.4.6 number

Searching on a simple numerical value in a resource. Examples:

[parameter]=100 | Values that equal 100, to 3 significant figures precision, so range [99.5 ... 100.5)

[parameter]=100.00 | Values that equal 100, to 5 significant figures precision, so range [99.995 ... 100.005). Whole numbers also equal 100.00, but not 100.01

[parameter]=lt100 | Values that are less than 100

[parameter]=le100 | Values that are less or equal to 100

[parameter]=gt100 | Values that are greater than 100

[parameter]=ge100 | Values that are greater or equal to 100

[parameter]=ne100 | Values that are not equal to 100

Note: Uncertainty does not factor in evaluations. The precision of the numbers is considered arbitrarily high. (The way search parameters operate in resources is not the same as whether two numbers are equal to each other in a mathematical sense).

Here are some example searches:

Search | Description

GET [base]/Encounter?length=gt20 | Search for all the encounters longer than 20 days

GET [base]/ImmunizationRecommendation?deo-number=2 | Search for any immunization recommendation recommending a second dose

2.21.1.4.7 date

A date parameter searches on a date/time or period. As is usual for date/time related functionality, while the concepts are relatively straight-forward, there are a number of subtleties involved in ensuring consistent behavior.

The date parameter format is yyyy-mm-ddThh:mm:ss[Z|(+|-)hh:mm] (the standard XML format).

Technically, this is any of the date, dateTime, and instant data types; e.g. Any degree of precision can be provided, but it SHALL be populated from the left (e.g. can't specify a month without a year), except that the minutes SHALL be present if an hour is present, and you SHOULD provide a time zone if the time part is present. Note: Time can consist of hours and minutes with no seconds, unlike the XML Schema dateTime type. Some user agents may escape the : characters in the URL, and servers SHALL handle this correctly.

Date parameters may be used with the following data types:

date | The range of the value is the day, month, or year as specified

dateTime | The range of the value as defined above; e.g. For example, the date 2013-01-10 specifies all the time from 00:00 on 10-Jan 2013 to immediately before 00:00 on 11-Jan 2013

instant | An instant is considered a fixed point in time with an interval smaller than the precision of the system, i.e. an interval with an effective width of 0

Period | Explicit, though the upper or lower bound may not actually be specified in resources.

Timing | the specified scheduling details are ignored and only the outer limits matter. For instance, a schedule that specifies every second day between 31-Jan 2013 and 24-Mar 2013 includes 1-Feb 2013, even though that is on an odd day that is not specified by the period. This is to keep the server load processing queries reasonable.

Implicitly, a missing lower boundary is "less than" any actual date. A missing upper boundary is "greater than" any actual date. The use of the prefixes:

[parameter]=eq2013-01-14 | 2013-01-14T00:00 matches (obviously) 2013-01-14T10:00 matches 2013-01-15T00:00 does not match - it's not in the range

[parameter]=ne2013-01-14 | 2013-01-15T00:00 matches - it's not in the range 2013-01-14T00:00 does not match - it's in the range 2013-01-14T10:00 does not match - it's in the range

[parameter]=lt2013-01-14T10:00 | 2013-01-14 matches, because it includes the part of 14-Jan 2013 before 10am

[parameter]=gt2013-01-14T10:00 | 2013-01-14 matches, because it includes the part of 14-Jan 2013 after 10am

[parameter]=ge2013-03-14 | "from 21-Jan 2013 onwards" is included because that period may include times after 14-Mar 2013

[parameter]=le2013-03-14 | "from 21-Jan 2013 onwards" is included because that period may include times before 14-Mar 2013

[parameter]=sa2013-03-14 | "from 15-Mar 2013 onwards" is included because that period starts after 14-Mar 2013 "from 21-Jan 2013 onwards" is not included because that period starts before 14-Mar 2013 "before and including 21-Jan 2013" is not included because that period starts (and ends) before 14-Mar 2013

[parameter]=eb2013-03-14 | "from 15-Mar 2013 onwards" is not included because that period starts after 14-Mar 2013 "from 21-Jan 2013 onwards" is not included because that period starts before 14-Mar 2013, but does not end before it "before and including 21-Jan 2013" is not included because that period ends before 14-Mar 2013

[parameter]=ap2013-03-14 | 14-Mar 2013 is included - as it exactly matches 21-Jan 2013 is not included because that is near 14-Mar 2013 15-Jun 2015 is not included - as it is not near 14-Mar 2013. Note that the exact value here is at the discretion of the system

Other notes:

* When the date parameter is not fully specified, matches against it are based on the behavior of intervals, where:

* Dates with only the year specified are equivalent to an interval that starts at the first instant of January 1st to the last instant of December 31st, e.g. 2000 is equivalent to an interval of [2000-01-01T00:00, 2000-12-31T23:59].

* Dates with the year and month are equivalent to an interval that starts at the first instant of the first day of the month and ends on the last instant of the last day of the month, e.g. 2000-04 is equivalent to an interval of [2000-04-01T00:00, 2000-04-30T23:59].

* Where possible, the system should correct for timezones when performing queries. Dates do not have time zones, and time zones should not be considered. Where both search parameters and resource element date times do not have time zones, the servers local time zone should be assumed.

To search for all the procedures in a patient compartment that occurred over a 2 year period:

```
GET [base]/Patient/23/Procedure?date=ge2010-01-01&date=le2011-12-31
```

2.21.1.4.8 string

For a simple string search, a string parameter serves as the input for a case- and accent-insensitive search against sequences of characters. By default, a field matches a string query if the value of the field equals or starts with the supplied parameter value, after both have been normalized by case and accent. The :contains modifier returns results that include the supplied parameter value anywhere within the field being searched. The :exact modifier returns results that match the entire supplied parameter, including casing and accents.

Examples:

[base]/Patient?given=eve | Any patients with a name containing a given part with "eve" at the start of the name. This would include patients with the given name "Eve", "Evelyn".

[base]/Patient?given:contains=eve | Any patients with a name with a given part containing "eve" at any position. This would include patients with the given name "Eve", "Evelyn", and also "Severine".

[base]/Patient?given:exact=Eve | Any patients with a name with a given part that is exactly "Eve". Note: This would not include patients with the given name "eve" or "EVE".

An additional modifier :text can be used to specify a search with advanced text handling (see below) though only a few servers are expected to offer this facility.

When a string search parameter refers to the types HumanName and Address, the search covers the elements of type string, and does not cover elements such as use and period. For robust search, servers should search the parts of a family name independently. E.g. Searching either Carreno or Quinones should match a family name of "Carreno Quinones". HL7 affiliates may make more specific recommendations about how search should work in their specific culture.

It is at the discretion of the server whether to pre-process names, addresses, and contact details to remove separator characters prior to matching in order to ensure more consistent behavior. For example, a server might remove all spaces and - characters from phone numbers. What is most appropriate varies depending on culture and context. A server may also use a free-text style searches on this property to achieve the best results When searching whole names and addresses (not parts), servers may also use flexible match or a free-text style searches on names to achieve the best results.

2.21.1.4.9 uri

The uri parameter refers to a URI (RFC 3986) element. Matches are precise (e.g. case, accent, and escape) sensitive, and the entire URI must match. The modifier :above or :below can be used to indicate that partial matching is used. For example:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123
GET [base]/ValueSet?url=http://acme.org/fhir/
GET [base]/ValueSet?url=urn:oid:1.2.3.4.5
```

The first line is a request to find any value set with the exact url "http://acme.org/fhir/ValueSet/123". The second line performs a search that will return any value sets that have a URL that starts with "http://acme.org/fhir/". The converse - the search for any value set above a given specific URL. This approach may be useful for searching name systems, but it is generally less useful than the :below search. The third line shows an example of searching by an OID. Note that the :above and :below modifiers only apply to URLs, and not URNS such as OIDs.

Note that for Canonical URLs, servers SHOULD support searching by Canonical URL, and SHOULD support automatically detecting a |[version] portion as part of the Canonical URL, and interpreting that portion as a search on the version.

2.21.1.4.10 token

A token type is a parameter that provides an exact match search, either on a string of characters, potentially scoped by a URI. It is mostly used against a code or identifier data type where the value may have a URI that scopes its meaning, where the search is performed against the pair from a Coding or an Identifier. Tokens are also used against other fields where exact matches are required - uris, booleans, and ContactPoints. In these cases, the URI portion is not used.

For tokens, matches are literal (e.g. not based on subsumption or other code system features), but not case sensitive. To use subsumption based logic, use the modifiers below, or list all the codes in the hierarchy. The syntax for the value is one of the following:

* [parameter]=[code]: the value of [code] matches a Coding.code or Identifier.value irrespective of the value of the system property

* [parameter]=[system]|[code]: the value of [code] matches a Coding.code or Identifier.value, and the value of [system] matches the system property of the Identifier or Coding

* [parameter]=|[code]: the value of [code] matches a Coding.code or Identifier.value, and the Coding/Identifier has no system property

* [parameter]=[system]|: any element where the value of [system] matches the system property of the Identifier or Coding

Note: The namespace URI and code both must be escaped correctly. If a system is not applicable (e.g. an element of type uri, then just the form [parameter]=[code] is used.

Token search parameters are used for the following data types:

Data Type	URI	Code	Comments
Coding	Coding.system	Coding.code	
CodeableConcept	CodeableConcept.coding.system	CodeableConcept.coding.code	Matches against any coding in the CodeableConcept
Identifier	Identifier.system	Identifier.value	
ContactPoint	ContactPoint.use	ContactPoint.value	The use is prepended by http://hl7.org/fhir/contact-point-system/
code	(implicit)	code	the system is defined in the value set (though it's not usually needed)
boolean	(implicit)	boolean	The implicit system is http://hl7.org/fhir/special-values
string	n/a	string	Token is sometimes used for string to indicate that exact matching is the correct default search stategy

Note: The use of token search parameters for boolean fields: the boolean values "true" and "false" are also represented as formal codes in the Special Values code system, which is useful when boolean values need to be represented in a Coding data type. The namespace for these codes is http://hl7.org/fhir/special-values, though there is usually no reason to use this, as a simple true or false is sufficient.

Modifiers:

Modifier	Use
:text	The search parameter is processed as a string that searches text associated with the code/value - either CodeableConcept.text, Coding.display, or Identifier.type.text.
:not	Reverse the code matching described in the paragraph above. Note that this includes resources that have no value for the parameter - e.g. ?gender:not=male includes all patients that do not have gender = male, including patients that do not have a gender at all
:above	The search parameter is a concept with the form [system]|[code], and the search parameter tests whether the coding in a resource subsumes the specified search code. For example, the search concept has an is-a relationship with the coding in the resource, and this includes the coding itself.
:below	the search parameter is a concept with the form [system]|[code], and the search parameter tests whether the coding in a resource is subsumed by the specified search code. For example, the coding in the resource has an is-a relationship with the search concept, and this includes the coding itself.
:in	The search parameter is a URI (relative or absolute) that identifies a value set, and the search parameter tests whether the coding is in the specified value set. The reference may be literal (to an address where the value set can be found) or logical (a reference to ValueSet.url). If the server can treat the reference as a literal URL, it does, else it tries to match known logical ValueSet.url values.

:not-in | The search parameter is a URI (relative or absolute) that identifies a value set, and the search parameter tests whether the coding is not in the specified value set.

Most servers will only process value sets that are already known/registered/supported internally. However, servers can elect to accept any valid reference to a value set. Servers may elect to consider concept mappings when testing for subsumption relationships.

Example searches:

Search | Description
GET [base]/Patient?identifier=http://acme.org/patient|2345 | Search for all the patients with an identifier with key = "2345" in the system "http://acme.org/patient"
GET [base]/Patient?gender=male | Search for any patient with a gender that has the code "male"
GET [base]/Patient?gender:not=male | Search for any patient with a gender that does not have the code "male"
GET [base]/Patient?active=true | Search for any patients that are active
GET [base]/Condition?code=http://acme.org/conditions/codes|ha125 | Search for any condition with a code "ha125" in the code system "http://acme.org/conditions/codes"
GET [base]/Condition?code=ha125 | Search for any condition with a code "ha125". Note that there is not often any useful overlap in literal symbols between code systems, so the previous example is generally preferred
GET [base]/Condition?code:text=headache | Search for any Condition with a code that has a text "headache" associated with it (either in the text, or a display)
GET [base]/Condition?code:in=http%3A%2F%2Fsnomed.info%2Fsct%3Ffhir_vs%3Disa%2F126851005 | Search for any condition in the SNOMED CT value set "http://snomed.info/sct?fhir_vs=isa/126851005" that includes all descendants of "Neoplasm of liver"
GET [base]/Condition?code:below=126851005 | Search for any condition that is subsumed by the SNOMED CT Code "Neoplasm of liver". Note: This is the same outcome as the previous search
GET [base]/Condition?code:in=http://acme.org/fhir/ValueSet/cardiac-conditions | Search for any condition that is in the institutions list of cardiac conditions

2.21.1.4.11 quantity

A quantity parameter searches on the Quantity data type. The syntax for the value follows the form:

* [parameter]=[prefix][number]|[system]|[code] matches a quantity with the given unit

The prefix is optional, and is as described above, both regarding how precision and comparator/range operators are interpreted. Example searches:

Search | Description
GET [base]/Observation?value=5.4|http://unitsofmeasure.org|mg | Search for all the observations with a value of 5.4 mg where mg is understood as a UCUM unit (system/code)
GET [base]/Observation?value=5.4||mg | Search for all the observations with a value of 5.4 mg where the unit - either the code or the stated human unit (unit) are "mg"
GET [base]/Observation?value=le5.4|http://unitsofmeasure.org|mg | Search for all the observations where the value of is less than 5.4 mg where mg is understood as a UCUM unit
GET [base]/Observation?value=ap5.4|http://unitsofmeasure.org|mg | Search for all the observations where the value of is about 5.4 mg where mg is understood as a UCUM unit

The search processor may choose to perform a search based on canonical units (e.g. any value where the units can be converted to a value in mg in the case above).

2.21.1.4.12 reference

A reference parameter refers to references between resources. For example, find all Conditions where the subject reference is a particular patient, where the patient is selected by name or identifier. The interpretation of a reference parameter is either:

* [parameter]=[id] the logical [id] of a resource using a local reference (i.e. a relative reference)

* [parameter]=[type]/[id] the logical [id] of a resource of a specified type using a local reference (i.e. a relative reference), for when the reference can point to different types of resources (e.g. Observation.subject)

* [parameter]=[url] where the [url] is an absolute URL - a reference to a resource by its absolute location

Note: A relative reference resolveing to the same value as a specified absolute URL, or vice versa, qualifies as a match. For example, if the search parameter value is Patient/123, then this will find references like this:

```
<patient>
  <reference value="Patient/123"/>
</patient>
```

If the server base address is http://example.org/fhir, then the full URL for that reference is http://example.org/fhir/Patient/123, which means that the search term also matches patient references like this:

```
<patient>
  <reference value="http://example.org/fhir/Patient/123"/>
</patient>
```

In addition, searching for reference=http://example.org/fhir/Patient/123 will also match both references.

Some references are allowed to point to more than one type of resource; e.g. subject : Reference(Patient|Group|Device|..). In these cases, multiple resources may have the same logical identifier. Servers SHOULD reject a search where the logical id refers to more than one matching resource across different types. In order to allow the client to perform a search in these situations the type is specified explicitly:

```
GET [base]/Observation?subject=Patient/23
```

This searches for any observations where the subject refers to the patient resource with the logical identifier "23". A modifier is also defined to to allow the client to be explicit about the intended type:

```
GET [base]/Observation?subject:Patient=23
```

This has the same effect as the previous search. The modifier becomes useful when used with chaining as explained in the next section. Note: The [type] modifier can't be used with a reference to a resource found on another server, since the server would not usually know what type that resource has. However, since these are absolute references, there can be no ambiguity about the type.

In some cases, search parameters are defined with an implicitly limited scope. For example, Observation has an element subject, which is a reference to one of a number of types. This has a matching search parameter subject, which refers to any of the possible types. In addition to this, there is another search parameter patient, which also refers to Observation.subject, but is limited to only include references of type Patient. When using the patient search parameter, there is no need to specify ":Patient" as a modifier, or "Patient/" in the search value, as this must always be true.

2.21.1.4.13 Chained parameters

In order to save a client from performing a series of search operations, reference parameters may be "chained" by appending them with a period (.) followed by the name of a search parameter defined for the target resource. This can be done recursively, following a logical path through a graph of related resources, separated by .. For instance, given that the resource DiagnosticReport has a search parameter named subject, which is usually a reference to a Patient resource, and the Patient resource includes a parameter name which searches on patient name, then the search

```
GET [base]/DiagnosticReport?subject.name=peter
```

is a request to return all the lab reports that have a subject whose name includes "peter". Because the Diagnostic Report subject can be one of a set of different resources, it's necessary to limit the search to a particular type:

```
GET [base]/DiagnosticReport?subject:Patient.name=peter
```

This request returns all the lab reports that have a subject which is a patient, whose name includes "peter".

Advanced Search Note: Where a chained parameter searches a resource reference that may have more than one type of resource as its target, the parameter chain may end up referring to search parameters with the same name on more than one kind of resource at once. Servers SHOULD reject a search where the logical id refers to more than one matching resource across different types. For example, the client has to specify the type explicitly using the syntax in the second example above.

2.21.1.4.14 Reverse Chaining

The _has parameter provides limited support for reverse chaining - that is, selecting resources based on the properties of resources that refer to them (instead of chaining, above, where resources can be selected based on the properties of resources that they refer to). Here is an example of the _has parameter:

```
GET [base]/Patient?_has:Observation:patient:code=1234-5
```

This requests the server to return Patient resources, where the patient resource is referred to by at least one Observation where the observation has a code of 1234, and where the Observation refers to the patient resource in the patient search parameter.

"Or" searches are allowed (e.g. GET [base]/Patient?_has:Observation:subject:code=123,456), and multiple _has parameters are allowed (e.g. GET [base]/Patient?_has:Observation:subject:code=123&_has:Observation:subject:code=456). Note that each _has parameter is processed independently of other _has parameters.

The _has parameter can be chained, like this:

```
GET [base]/Patient?_has:Observation:patient:_has:AuditEvent:entity:user=MyUserId
```

Fetch all the patients that have an Observation where the observation has an audit event from a specific user.

2.21.1.4.15 Composite Search Parameters

Composite search parameters support joining single values with a $. For example, the result of the search operation is the intersection of the resources that match the criteria specified by each individual search parameter. If a parameter repeats, such as /Patient?language=FR&language=NL, then this matches a patient who speaks both languages. This is known as an AND search parameter, since the server is expected to respond only with results which match both values.

If, instead, the search is to find patients that speak either language, then this is a single parameter with multiple values, separated by a ,. For example, /Patient?language=FR,NL. This is known as an OR search parameter, since the server is expected to respond with results which match either value.

AND parameters and OR parameters may also be combined, for example: /Patient?language=FR,NL&language=EN would refer to any patient who speaks English, as well as either French or Dutch.

This approach allows for simple combinations of and/or values, but doesn't allow a search based on a pair of values, such as all observations with a sodium value >150 mmol/L (particularly as the end criteria of a chained search), or searching on Group.characteristic where you need find a combination of key/value, not an intersection of separate matches on key and value. Another example is spatial coordinates when doing geographical searches.

To allow these searches, a resource may also specify composite parameters that take sequences of single values that match other defined parameters as an argument. The matching parameter of each component in such a sequence is documented in the definition of the parameter. These sequences are formed by joining the single values with a $. Note: This sequence is a single value and itself can be composed into a set of values, so that, for example, multiple matching state-on-date parameters

can be specified as state-on-date=new$2013-05-04,active$2013-05-05.

Note: Modifiers are not used on composite parameters.

Examples of using composite parameters:

Search	Description
GET [base]/DiagnosticReport?result.code-value-quantity=http://loinc.org|2823-3$gt5.4|http://unitsofmeasure.org|mmol/L	Search for all diagnostic reports that contain on observation with a potassium value of >5.4 mmol/L (UCUM)
GET [base]/Observation?component-code-value-quantity=http://loinc.org|8480-6$lt60	Search for all the observations with a systolic blood pressure < 60. Note that in this case, the unit is assumed (everyone uses mmHg)
GET [base]/Group?characteristic-value=gender$mixed	Search for all groups that have a characteristic "gender" with a text value of "mixed"

2.21.1.4.16 Handling Missing Data

Consider the case of searching for all AllergyIntolerance resources:

```
GET [base]/AllergyIntolerance?clinical-status=active
```

This search will only return resources that have a value for clinicalStatus:

```
{
  "resourceType" : "AllergyIntolerance",
  "clinicalStatus" : "active"
}
```

Resources missing a clinicalStatus will not be returned. This is probably unsafe - it would not usually be appropriate to ignore AllergyIntolerance warnings with an unknown clinical status, and only return resources with an explicit clinicalStatus. Instead, it might be desired to return AllergyIntolerance resources with either an explicit value for clinicalStatus, or none:

```
GET [base]/AllergyIntolerance?clinical-status=active
GET [base]/AllergyIntolerance?clinical-status:exists=false
```

Note that this is 2 separate queries. They can be combined in a batch, but not in a single operation. This query will always return an empty list, as no resource can satisfy both criteria at once:

```
GET [base]/AllergyIntolerance?clinical-status=active&clinical-status:exists=false
```

There is no way to use the :exists modifier and mix with a value using the comma syntax documented above for for composite search parameters.

An alternative approach is to use the _filter parameter, for servers that support this parameter.

2.21.1.4.17 Escaping Search Parameters

In the rules described above, special rules are defined for the characters $, ,, and |. As a consequence, if these characters appear in an actual parameter value, they must be differentiated from their use as separator characters. When any of these characters appear in an actual parameter value, they must be prepended by the character \, which also must be used to prepend itself. Therefore, param=xxx$xxx indicates that it is a composite parameter, while param=xx\$xx indicates that the parameter has the literal value xx$xx. The parameter value xx\xx is illegal, and the parameter value param=xx\\xx indicates a literal value of xx\xx.

This specification defines this additional form of escape for a reason. The classic %xx escaping which is part of normal HTTP URLs ensures that the character appears at the FHIR server correctly, while the , versus \ becomes important once it has reached the server and the query is parsed. Therefore:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124%2CValueSet/125
```

uses url escaping to make sure the FHIR server received:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124,125
```

This request will compare the URL against three values: the last one being a relative and incorrect url, which is likely not the actual intent. However:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124\,125
```

is equivalent to:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124\%2C125
```

which would mean: url = http://.....123 OR http://....124,125.

2.21.1.4.18 Text Search Parameters

The special text search parameters, _text and _content, search on the narrative of the resource, and the entire content of the resource respectively. These parameters SHOULD support a sophisticated search functionality of the type offered by typical text indexing services. The value of the parameter is a text based search, which may involve searching multiple words with thesaurus and proximity considerations, and logical operations such as AND, OR etc. For example:

```
 GET [base]/Condition?_text=(bone OR liver) and metastases
```

This request returns all Condition resources with the word "metastases" and either "bone" or "liver" in the narrative. The server MAY choose to search for related words as well.

STU Note: The issues around standardizing text search are not fully resolved. During the trial use period for this specification, we recommend that systems use the rules specified by the OData specification for the $search parameter . Typical implementations would use Lucene, an sql-based full text search, or some indexing service.

Feedback is welcome here .

2.21.1.4.19 Searching by list

The _list parameter allows for the retrieval of resources that are referenced by a List resource.

```
GET [base]/Patient?_list=42
```

This request returns all Patient resources that are referenced from the list found at [base]/List/42) in List.entry.item. While it is possible to retrieve the list, and then iterate the entries in the list fetching each patient, using a list as a search criteria allows for additional search criteria to be specified. For instance:

```
GET [base]/Patient?_list=42&gender=female
```

This request will return all female patients in the list. The server can return the list referred to in the search parameter as an included resource, but is not required to do so. In addition, a system can support searching by lists by their logical function. For example:

```
GET [base]/AllergyIntolerance?patient=42&_list=$current-allergies
```

This request will return all allergies in patient 42's "Current Allergy List". The server returns all relevant AllergyIntolerance resources, and can also choose to return the list. For further information, refer to the definition of "$current-allergies", and the List Operation "Find". Note: Servers are not required to make these lists available to the clients as list resources, but may choose to do so.

2.21.1.4.20 Advanced filtering

The search mechanism described above is flexible, and easy to implement for simple cases, but is limited in its ability to express combination queries. To complement this mechanism, a the "_filter" search expression parameter can be used.

For example, "Find all the observations for patient with a name including peter that have a LOINC code 1234-5":

```
GET [base]/Observation?code=http://loinc.org|1234-5&subject.name=peter
```

Using the _filter parameter, the search would be expressed like this:

```
GET [base]/Observation?_filter=name eq http://loinc.org|1234-5 and subject.name co "peter"
```

The _filter parameter is described in detail on the "_Filter Parameter" page.

2.21.1.4.21 Specifying the type of Resource

Normally, a search is initiated against a known type of resource, e.g.

```
GET [base]/Observation?params...
```

However, in some circumstances, a search is executed where there is no fixed type of resource:

* Using search across all resource types (GET [base]?params...)

* Using search with messaging

* Some search specifications inside other services e.g. decision support usage

In these circumstances, the search criteria may need to specify one or more resource types that the search applies to. This can be done by using the _type parameter:

```
GET [base]/?_type=Observation,Condition&other params...
```

Note that the only search parameters that be can be used in global search like this are the base parameters that apply to all resources. In other contexts, searches on multiple types may allow resource specific search parameters, but implementation experience will be required to determine the correct behaviour in these cases.

Technically, the _type parameter is a token parameter on the Resource Types Value Set.

2.21.1.5 Managing Returned Resources

2.21.1.5.1 Sorting

The client can indicate which order to return the results by using the parameter _sort, which can contains a comma-separated list of sort rules in priority order:

```
GET [base]/Observation?_sort=status,-date,category
```

Each item in the comma separated list is a search parameter, optionally with a '-' prefix. The prefix indicates decreasing order; in its absence, the parameter is applied in increasing order.

Notes:

* When sorting, the actual sort value used is not returned explicitly by the server for each resource, just the resource contents.

* To sort by relevance, use _score.

* The server returns the sort it performs as part of the returned search parameters (see below).

* A search parameter can refer to an element that repeats, and therefore there can be multiple values for a given search parameter for a single resource. In this case, the sort is based on the item in the set of multiple parameters that comes earliest in the specified sort order when ordering the returned resources.

* When sorting on string search parameters, sorting SHOULD be performed on a case-insensitive basis. Accents may either be ignored or sorted as per realm convention.

* This specification does not specify exacts rules for consistency of sorting across servers. In general, this is deemed to be not as essential as consistency of filtering (though even that is a little variable). The purpose of sorting is to provide data in a "reasonable" order for end-users. "Reasonable" may vary by use case and realm, particularly for accented characters.

2.21.1.5.2 Page Count

In order to keep the load on clients, servers and the network minimized, the server may choose to return the results in a series of pages. The search result set contains the URLs that the client uses to request additional pages from the search set. For a simple RESTful search, the page links are contained in the returned bundle as links.

Typically, a server will provide its own parameters in the links that it uses to manage the state of the search as pages are retrieved. These parameters do not need to be understood or processed by the client.

The parameter _count is defined as a hint to the server regarding how many resources should be returned in a single page. Servers SHALL NOT return more resources than requested, even if they don't support paging, but are allowed to return less than the client requested. The server should repeat the original _count parameter in its returned page links so that subsequent paging requests honor the original _count. Note: It is at the discretion of the search engine as to how to handle ongoing updates to the resources while the search is proceeding.

Note: The combination of _sort and _count can be used to return only the latest resource that meets a particular criteria - set the criteria, and then sort by date in descending order, with _count=1. This way, the last matching resource will be returned.

if _count has the value 0, this shall be treated the same as _summary=count: the server resturns a bundle that reports the total number of resources that match in Bundle.total, but with no entries, and no prev/next/last links. Note that the Bundle.total only include the total number of matching resources. It does not count extra resources such as OperationOutcome or included resources that may also be returned.

2.21.1.5.3 Including other resources in result (_include and _revinclude)

Clients may request that the engine return resources related to the search results, in order to reduce the overall network delay of repeated retrievals of related resources. This is useful when the client is searching on a clinical resource, but for every such resource returned, the client will also need the subject (patient) resource that the clinical resource refers to. The client can use the _include parameter to indicate that the subject resources be included in the results. An alternative scenario is where the client wishes to fetch a particular resource, and any resources that refer to it. For example, the client may wish to fetch a MedicationRequest, and any provenance resources that refer to the prescription. This is known as a reverse include, and is specified by providing a _revinclude parameter.

Both _include and _revinclude are based on search parameters, rather than paths in the resource, since joins, such as chaining, are already done by search parameter.

Each _include parameter specifies a search parameter to join on:

```
GET [base]/MedicationRequest?_include=MedicationRequest:patient&criteria...
GET [base]/MedicationRequest?_revinclude=Provenance:target&criteria...
```

The first search requests all matching MedicationRequests, to include any patient that the medication prescriptions in the result set refer to. The second search requests all matching prescriptions, return all the provenance resources that refer to them.

Parameter values for both _include and _revinclude have three parts, separated by a : character:

* The name of the source resource from which the join comes

* The name of the search parameter which must be of type reference

* (Optional) A specific of type of target resource (for when the search parameter refers to multiple possible target types)

_include and _revInclude parameters do not include multiple values. Instead, the parameters are repeated for each different include criteria.

For each returned resource, the server identifies the resources that meet the criteria expressed in the join, and adds to the results, with the entry.search.mode set to "include" (in some searches, it is not obvious which resources are matches, and which are includes). If there is no reference, or no matching resource, the resource cannot be retrieved (e.g. on a different server), then the resource is omitted, and no error is returned.

The inclusion process can be recursive, if the modifier :recurse is included. For example, this search returns all Medication Request resources and their prescribing Practitioner Resources for the matching Medication Dispense resources:

```
GET [base]/MedicationDispense?_include=MedicationDispense:authorizingPrescription
    &_include:recurse=MedicationRequest:prescriber&criteria...
```

This technique applies to circular relationships as well. For example, the first of these two searches includes any related observations to the target relationships, but only those directly related. The second search asks for the _include based on related parameter to be executed recursively, so it will retrieve observations that are directly related, and also any related observations to any other included observation.

```
GET [base]/Observation?_include=Observation:related-target&criteria...
GET [base]/Observation?_include:recurse=Observation:related-target&criteria...
```

Both _include and _revInclude use the wild card "*" for the search parameter name, indicating that any search parameter of type=reference be included. Though both clients and servers need to take care not to request or return too many resources when doing this. Most notably, using recursive inclusions might lead to the retrieval of the full patient's record, or even more: resources are organized into an interlinked network and broad _include paths may eventually traverse all possible paths on the server. For servers, these recursive and wildcard _includes are demanding and may slow the search response time significantly.

It is at the server's discretion how deep to recursively evaluate the inclusions. Servers are expected to limit the number of iterations done to an appropriate level and are not obliged to honor requests to include additional resources in the search

results.

When search results are paged, each page of search results should include the matching includes for the resources in each page, so that each page stands alone as a coherent package.

2.21.1.5.4 Contained Resources

By default, search results only include resources that are not contained in other resources. A chained condition will be evaluated inside contained resources. To illustrate this, consider a MedicationRequest resource that has a contained Medication resource specifying a custom formulation that has ingredient with a itemCodeableConcept "abc" in "http://acme.com./medications". In this case, a search:

```
GET MedicationRequest?medication.ingredient-code=abc
```

will include the MedicationRequest resource in the results. However, this search:

```
GET Medication?ingredient-code=abc
```

will not include the contained Medication resource in the results, since either the wrong type of resource would be returned, or the contained resource would be returned without its container resource, which provides context to the contained resource.

Clients are able to modify this behavior using the _contained parameter, which can have one of the following values:

* false (default): Do not return contained resources

* true: return only contained resources

* both: return both contained and non-contained (normal) resources

When contained resources are being returned, the server should return either the container resource, or the contained resource alone. The client can specify which by using the _containedType parameter, which can have one of the following values:

* container (default): Return the container resources

* contained: return only the contained resource

When returning a container resource, the server simply puts this in the search results:

```
<Bundle>
  ...
  <entry>
    <resource>
      <MedicationRequest>
        <id value="23">
        ....
        <contained>
          <Medication>
            <id value="m1">
            ...
          </Medication>
        <contained>

      </MedicationRequest>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

In the case of returning container resources, the server SHALL populate the entry.search.mode element, as shown, so that the client can pick apart matches and includes (since the usual approach of doing it by type may not work).

If the return type is the contained resource, this must be done slightly differently:

```
<Bundle>
  ...
  <entry>
    <fullUrl value="http://example.com/fhir/MedicationRequest/23#m1"/>
    <resource>
      <Medication>
        <id value="m1">
        ...
      </Medication>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

In this case, the fullUrl informs the client that this is a contained resource, along with indicating the identity of the containing resource.

2.21.1.5.5 External References

If the _include path selects a reference that refers to a resource on another server, the server can elect to include that resource in the search results for the convenience of the client.

If the _include path selects a reference that refers to an entity that is not a Resource, such as an image attachment, the server may also elect to include this in the returned results as a Binary resource. For example, the include path may point to

an attachment which is by reference, like this:

```
<content>
  <contentType>image/jpeg</contentType>
  <url>http://example.org/images/2343434/234234.jpg</url>
</content>
```

The server can retrieve the target of this reference, and add it to the results for the convenience of the client.

2.21.1.5.6 Paging

When returning paged results for a search with _include resources, all _include resources that are related to the primary resources returned for the page SHOULD also be returned as part of that same page, even if some of those resource instances have previously been returned on previous pages. This approach allows both sender and receiver to avoid caching results of other pages.

2.21.1.5.7 Summary

The client can request the server to return only a portion of the resources by using the parameter _summary:

```
GET [base]/ValueSet?_summary=true
```

The _summary parameter requests the server to return a subset of the resource. It can contain one of the following values:

true | Return only those elements marked as "summary" in the base definition of the resource(s) (see ElementDefinition.isSummary)
text | Return only the "text" element, the 'id' element, the 'meta' element, and only top-level mandatory elements
data | Remove the text element
count | Search only: just return a count of the matching resources, without returning the actual matches
false | Return all parts of the resource(s)

The intent of the _summary parameter is to reduce the total processing load on server, client, and resources between them such as the network. It is most useful for resources that are large, particularly ones that include images or elements that may repeat many times. The purpose of the summary form is to allow a client to quickly retrieve a large set of resources, and let a user pick the appropriate one. The summary for an element is defined to allow a user to quickly sort and filter the resources, and typically omit important content on the basis that the entire resource will be retrieved when the user selects a resource.

Servers are not obliged to return just a summary as requested. There are only a limited number of summary forms defined for resources in order to allow servers to store the summarized form(s) in advance. Servers SHOULD mark the resources with the tag SUBSETTED to ensure that the incomplete resource is not acidentally used to overwrite a complete resource.

Note that the _include and _revinclude parameters cannot be mixed with _summary=text.

STU Note: There are open issues around the definitions of which elements are in the summary for each resource. There are 2 kinds of problems in the existing definitions:

* Elements are marked as 'include in summary', but their parent element is not

* Elements have minimum cardinality > 0, but are not included in the summary (and their parent element is) - so sumarised resources will not be schema valid

These issues will be resolved for FHIR Release 4. In the meantime, implementers may choose to override which elements are marked as 'included in summary', and implementations may vary.

Feedback on which elements should be in the summaries is welcome here .

2.21.1.5.8 Elements

If one of the summary views defined above is not appropriate, a client can request a specific set of elements be returned as part of a resource using the _elements parameter:

```
GET [base]/Patient?_elements=identifier,active,link
```

The _elements parameter consists of a comma separated list of base element names such as, elements defined at the root level in the resource. Only elements that are listed are to be returned. Clients SHOULD list all mandatory and modifier elements in a resource as part of the list of elements. The list of elements does not apply to included resources.

Servers are not obliged to return just the requested elements. Servers SHOULD always return mandatory elements whether they are requested or not. Servers SHOULD mark the resources with the tag SUBSETTED to ensure that the incomplete resource is not actually used to overwrite a complete resource.

2.21.1.5.9 Relevance

Where a search specifies a non-deterministic sort, the search algorithm may generate some kind of ranking score to indicate which resources meet the specified criteria better than others. The server can return this score in entry.score:

```
<entry>
  <score value=".45"/>
  <Patient>
    ... patient data ...
  </Patient>
</entry>
```

The score is a decimal number with a value between (and including) 0 and 1, where 1 is best match, and 0 is least match.

2.21.1.6 Server Conformance

In order to allow the client to be confident about what search parameters were used as criteria by the server, the server SHALL return the parameters that were actually used to process the search. Applications processing search results SHALL

check these returned values where necessary. For example, if the server did not support some of the filters specified in the search, a client might manually apply those filters to the retrieved result set, display a warning message to the user or take some other action.

In the case of a RESTful search, these parameters are encoded in the self link in the bundle that is returned:

```
<link>
  <relation value="self"/>
  <url value="http://example.org/Patient?name=peter"/>
</link>
```

In other respects, servers have considerable discretion with regards to supporting search:

* Servers can choose which parameters to support (other than _id above).

* Servers can choose when and where to implement parameter chaining, and when and where they support the _include parameter.

* Servers are able to declare additional parameters in the profiles referenced from their Capability statements. Servers should define search parameters starting with a "-" character to ensure that the names they choose do not clash with future parameters defined by this specification.

* Servers are not required to enforce case sensitivity on parameter names, though the names are case sensitive (and URLs are generally case-sensitive).

* Servers may choose how many results to return, though the client can use _count as above

* Servers can choose how to sort the return results, though they SHOULD honor the _sort parameter.

2.21.1.7 Advanced Search

The search framework described above is a useful framework for providing a simple search based on indexed criteria, but more sophisticated query capability is needed to handle precise queries, complex decision support based requests, and direct queries that have human resolution.

More advanced search operations are specified by the _query parameter:

```
GET [base]/Patient?_query=name&parameters...
```

The _query parameter names a custom search profile that describes a specific query operation. The named query may define additional named parameters that are used with that particular named query. Servers can define their own additional named queries to meet their own uses using an OperationDefinition.

There can only ever be one _query parameter in a set of search parameters. Servers processing search requests SHALL refuse to process a search request if they do not recognize the _query parameter value.

2.21.1.8 Search Result Currency

The results of a search operation are only guaranteed to be current at the moment the operation is executed. After the operation is executed, ongoing actions performed on the resources against which the search was executed will render the results increasingly stale. The significance of this depends on the nature of the search, and the kind of use that is being made of the results.

This is particularly relevant when the server is returning the results in a series of pages. It is at the discretion of the search engine of how to handle ongoing updates to the resources while the search is proceeding.

Note: Performing a search operation does not change the set of resources on the server, with the exception of the creation of Audit Event resources auditing the search itself.

2.21.1.9 Summary Tables

Common Parameters defined for all resources:

Name | Type | Description | Paths
_id | token | Resource id (not a full URL) | Resource.id
_lastUpdated | date | Date last updated. Server has discretion on the boundary precision | Resource.meta.lastUpdated
_tag | token | Search by a resource tag | Resource.meta.tag
_profile | uri | Search for all resources tagged with a profile | Resource.meta.profile
_security | token | Search by a security label | Resource.meta.security
_text | string | Text search against the narrative
_content | string | Text search against the entire resource
_list | string | All resources in nominated list (by id, not a full URL)
_query | string | Custom named query

Search Control Parameters:

Name | Type | Description | Allowable Content
_sort | string | Order to sort results in (can repeat for inner sort orders) | Name of a valid search parameter
_count | number | Number of results per page | Whole number
_include | string | Other resources to include in the search results that search matches point to | SourceType:searchParam(:targetType)
_revinclude | string | Other resources to include in the search results when they refer to search matches | SourceType:searchParam(:targetType)
_summary | string | Just return the summary elements (for resources where this is defined) | true | false (false is default)
_contained | string | Whether to return resources contained in other resources in the search matches | true | false | both (false is default)
_containedType | string | If returning contained resources, whether to return the contained or container resources | container | contained

Cross-map between search parameter types and Data types:

Data Type | number | date | string | token | reference | quantity | uri

boolean | N | N | N | Y . true|false (System = http://hl7.org/fhir/special-values but not usually needed) | N | N | N
code | N | N | N | Y . (System, if desired, is defined in the underlying value set for eeach code) | N | N | N
date | N | Y | N | N | N | N | N
dateTime | N | Y | N | N | N | N | N
decimal | Y | N | N | N | N | N | N
instant | N | Y | N | N | N | N | N
integer | Y | N | N | N | N | N | N
string | N | N | Y | Y | N | N | N
uri | N | N | N | N | Y | N | Y
Address | N | N | Y search on any string element in the address | N | N | N | N
Annotation | N | N | N | N | N | N | N
CodeableConcept | N | N | N | Y | N | N | N
Coding | N | N | N | Y | N | N | N
ContactPoint | N | N | N | Y | N | N | N
Duration | Y | N | N | N | N | N | N
HumanName | N | N | Y Search on any string element in the name | N | N | N | N
Identifier | N | N | N | Y | N | N | N
Period | N | Y | N | N | N | N | N
Quantity | N | N | N | N | N | Y | N
Range | N | N | N | N | N | N | N
Reference | N | N | N | N | Y | N | N
SampledData | N | N | N | N | N | N | N
Timing | N | Y | N | N | N | N | N

2.21.2 _filter Parameter

URL: https://hl7.org/fhir/STU3/search_filter.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.21.2 _filter Parameter

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

2.21.2.1 Introduction

_filter is a parameter that can be used with the Search Operation. It uses the same set of standard parameters defined for the resources, and provides a syntax for expressing a set of query expressions on the underlying resources.

Examples of filters:

* Patient: name co "pet" - all patients with the characters "pet" in a given or family name

* Patient: given eq "peter" and birthdate ge 2014-10-10 - all patients with a given name of peter, born on or after 10-Oct 2014

* Observation: name eq http://loinc.org|1234-5 - all observations with the loinc code "1234-5"

* Observation: subject.name co "pet" - all observations on a patient with the characters "pet" in a given or family name

* Observation: related[type eq "has-component"].target pr true - all observations that have component observations (note: this uses one of the search parameters defined for this mechanism, see below)

* Observation: related[type eq has-component].target re Observation/4 - all observations that have Observation/v as a component

The _filter syntax has the following features:

* A filter can be a logical one (x or x, or x and x, or not x).

* A filter can contain other filters in a set of parentheses : "()".

* A filter can be a test - path operation value, where operation is taken from the table below, and value is either a "true", "false", a JSON string, or a token (any sequence of non-whitespace characters, excluding ")" and "]". Values are never case sensitive.

* A 'path' is a name, with chained searches done by name.name etc.as per existing source. There can also be a filter: name[filter].name.

* The name is one of the defined search parameters that are used with the other search mechanism, with some special exemptions defined below.

Note: The only difference between a "string" value and a "token" value is that a string can contain spaces and) and]. There is otherwise no significant difference between them.

Formal grammar for the syntax:

```
filter       = paramExp / logExp / ("not") "(" filter ")"
logExp       = filter ("and" / "or" filter)+
paramExp     = paramPath SP compareOp SP compValue
compareOp    = (see table below)
compValue    = string / numberOrDate / token
string       = json string
token        = any sequence of non-whitespace characters (by Unicode rules) except "]" and ")"
paramPath    = paramName (("[" filter "]") "." paramPath)
```

```
paramName    = ALPHA (nameChar)*
nameChar     = "_" / "-" / DIGIT / ALPHA
numberOrDate = DIGIT (DateChar)*
dateChar     = DIGIT / "T" / "-" / "." / "+"
```

Notes about using the syntax:

* Logical expressions are evaluated left to right, with no precedence between "and" and "or". If there is ambiguity, use parentheses to be explicit.

* Rhe compareOp is always evaluated against the set of values produced by evaluating the param path.

* Rhe parameter names are those defined by the specification for search parameters, except for those defined below.

* Rhe date format is a standard XML (i.e. XSD) dateTime (including timezone).

2.21.2.2 Operators

This table summarizes the comparison operations available:

Operation | Definition
eq | an item in the set has an equal value
ne | An item in the set has an unequal value
co | An item in the set contains this value
sw | An item in the set starts with this value
ew | An item in the set ends with this value
gt / lt / ge / le | A value in the set is (greater than, less than, greater or equal, less or equal) the given value
ap | A value in the set isis approximately the same as this value. Note that the recommended value for the approximation is 10% of the stated value (or for a date, 10% of the gap between now and the date), but systems may choose other values where appropriate
sa | The value starts after the specified value
eb | The value ends before the specified value
pr | The set is empty or not (value is false or true)
po | True if a (implied) date period in the set overlaps with the implied period in the value
ss | True if the value subsumes a concept in the set
sb | True if the value is subsumed by a concept in the set
in | True if one of the concepts is in the nominated value set by URI, either a relative, literal or logical vs
ni | True if none of the concepts are in the nominated value set by URI, either a relative, literal or logical vs
re | True if one of the references in set points to the given URL

For detailed rules about the operators eq, ne, le, ge, lt, gt, sa, and eb see Search Prefixes.

The interpretation of the operation depends on the type of the search parameter it is being evaulated against. This table contains those details:

Operation | String | Number | Date | Token | Reference | Quantity
Eq | Character sequence is the same (case insensitive) | Number is the same incl same precision | Date is the same including same precision and timezone if provided | Token is the same, including namespace if specified (case insensitive) | n/a | Unit and value are the same
Ne | (same)
Co | Character sequence matches somewhere (case insensitive) | An item in the set's implicit imprecision includes the stated value | An item in the set's implicit period includes the stated value | n/a | n/a | n/a?
Sw | Character sequence matches from first digit (left most, when L->R) (case insensitive) | n/a | n/a | n/a | n/a | n/a
ew | Character sequence matches up to last digit (right most, when L->R) (case insensitive) | n/a | n/a | n/a | n/a | n/a
gt / lt / ge / le | Based on Integer comparison of Unicode code points of starting character (trimmed) (case insensitive) | Based on numerical comparison | Based on date period comparison per 2.2.2.3 | n/a | n/a | Based on numerical comparison if units are the same (or are canonicalised)
pr
po | n/a | n/a | Based on date period comparison per 2.2.2.3 | | n/a | n/a
ss | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
sb | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
in | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
re | n/a | n/a | n/a | n/a | Relative or absolute url | n/a

Note:

* For token, the format is the same as the existing search parameter.

* For convenience, the codes "loinc", "snomed", "rxnorm" and "ucum" are predefined and can be used in place of their associated full namespace.

2.21.2.3 Additional Parameters

Some additional parameters are defined for the filter parameter (to do: move these into the standard parameters):

Resource Type | Parameter Name | Children
Observation | related | target = related-target Type = related-type
Group | characteristic | value = value code = characteristic
DocumentReference | relatesTo | code = relation target = relatesTo
ProcedureRequest | event | status = event-status date = event-date
ProcedureRequest | item | status = item-status code = item-code site = bodysite event = item-event
ProcedureRequest | item-event | status = item-past-status date = item-date actor = actor

Note:

* Any time these names are used in a parameter, they must have a filter and a chained name under them.

* The first column is the resource type against which this name can be used.

* The second column is the parameter name that is used.

* The third column defines the names that can be used in the chained parameter, and in the filter, and shows which existing search parameters they equate to.

* For example, you could search on Observation for '_filter=related[type eq has-component].target re url'. "type" refers to the search parameter "related-type", and "target" to the search parameter "related-target".

2.21.3 Defined Search Parameters

URL: https://hl7.org/fhir/STU3/searchparameter-registry.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.21.3 Defined Search Parameters

This page lists all search parameters defined as part of this specification. For further details regarding using search parameters, see searching on the RESTful API.

Parameter | Type | Description | Paths

Resource

_content | string | Search on the entire content of the resource

_id | token | Logical id of this artifact | Resource.id

_lastUpdated | date | When the resource version last changed | Resource.meta.lastUpdated

_profile | uri | Profiles this resource claims to conform to | Resource.meta.profile

_query | token | A custom search profile that describes a specific defined query operation

_security | token | Security Labels applied to this resource | Resource.meta.security

_tag | token | Tags applied to this resource | Resource.meta.tag

DomainResource

_text | string | Search on the narrative of the resource

Common Search Parameters

address | string | Multiple Resources: RelatedPerson: A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text Practitioner: A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text Person: A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text Patient: A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | RelatedPerson.address | Practitioner.address | Person.address | Patient.address

address-city | string | Multiple Resources: RelatedPerson: A city specified in an address Practitioner: A city specified in an address Person: A city specified in an address Patient: A city specified in an address | RelatedPerson.address.city | Practitioner.address.city | Person.address.city | Patient.address.city

address-country | string | Multiple Resources: RelatedPerson: A country specified in an address Practitioner: A country specified in an address Person: A country specified in an address Patient: A country specified in an address | RelatedPerson.address.country | Practitioner.address.country | Person.address.country | Patient.address.country

address-postalcode | string | Multiple Resources: RelatedPerson: A postal code specified in an address Practitioner: A postalCode specified in an address Person: A postal code specified in an address Patient: A postalCode specified in an address | RelatedPerson.address.postalCode | Practitioner.address.postalCode | Person.address.postalCode | Patient.address.postalCode

address-state | string | Multiple Resources: RelatedPerson: A state specified in an address Practitioner: A state specified in an address Person: A state specified in an address Patient: A state specified in an address | RelatedPerson.address.state | Practitioner.address.state | Person.address.state | Patient.address.state

address-use | token | Multiple Resources: RelatedPerson: A use code specified in an address Practitioner: A use code specified in an address Person: A use code specified in an address Patient: A use code specified in an address | RelatedPerson.address.use | Practitioner.address.use | Person.address.use | Patient.address.use

birthdate | date | Multiple Resources: RelatedPerson: The Related Person's date of birth Person: The person's date of birth Patient: The patient's date of birth | RelatedPerson.birthDate | Person.birthDate | Patient.birthDate

code | token | Multiple Resources: Medication: Codes that identify this medication MedicationRequest: Return prescriptions of this medication code MedicationAdministration: Return administrations of this medication code MedicationStatement: Return statements of this medication code MedicationDispense: Return dispenses of this medicine code | Medication.code | MedicationRequest.medication.as(CodeableConcept) | MedicationAdministration.medication.as(CodeableConcept) | MedicationStatement.medication.as(CodeableConcept) | MedicationDispense.medication.as(CodeableConcept)

code | token | Multiple Resources: FamilyMemberHistory: A search by a condition code DeviceRequest: Code for what is being requested/ordered AllergyIntolerance: Code that identifies the allergy or intolerance Procedure: A code to identify a procedure List: What the purpose of this list is ProcedureRequest: What is being requested/ordered Observation: The code of the observation type DiagnosticReport: The code for the report as a whole, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result Condition: Code for the condition | FamilyMemberHistory.condition.code | DeviceRequest.code.as(CodeableConcept) | AllergyIntolerance.code | AllergyIntolerance.reaction.substance | Procedure.code | List.code | ProcedureRequest.code | Observation.code | DiagnosticReport.code | Condition.code

date | date | Multiple Resources: Consent: When this Consent was created or indexed SupplyRequest: When the request was made RiskAssessment: When was assessment made? CareTeam: Time period team covers FamilyMemberHistory: When history was captured/updated Encounter: A date within the period the Encounter lasted AllergyIntolerance: Date record was believed accurate CarePlan: Time period plan covers EpisodeOfCare: The provided date search value falls within the episode of care's period Procedure: Date/Period the procedure was performed List: When the list was prepared Immunization: Vaccination (non)-Administration Date Flag: Time period when flag is active Observation: Obtained date/time. If the obtained element is a period, a date that falls in the period DiagnosticReport: The clinically relevant time of the report Composition: Composition editing time DetectedIssue: When identified ClinicalImpression: When the assessment was documented | Consent.dateTime | SupplyRequest.authoredOn | RiskAssessment.occurrence.as(DateTime) | CareTeam.period | FamilyMemberHistory.date | Encounter.period | AllergyIntolerance.assertedDate | CarePlan.period | EpisodeOfCare.period | Procedure.performed | List.date | Immunization.date | Flag.period | Observation.effective | DiagnosticReport.effective | Composition.date | DetectedIssue.date | ClinicalImpression.date

email | token | Multiple Resources: PractitionerRole: A value in an email contact RelatedPerson: A value in an email contact Practitioner: A value in an email contact Person: A value in an email contact Patient: A value in an email contact | PractitionerRole.telecom.where(system='email') |

RelatedPerson.telecom.where(system='email') | Practitioner.telecom.where(system='email') | Person.telecom.where(system='email') | Patient.telecom.where(system='email')

encounter | reference | Multiple Resources: DocumentReference: Context of the document content RiskAssessment: Where was assessment performed? DeviceRequest: Encounter or Episode during which request was created Procedure: Search by encounter List: Context in which list created VisionPrescription: Return prescriptions with this encounter identifier ProcedureRequest: An encounter in which this request is made Flag: Alert relevant during encounter Observation: Encounter related to the observation DiagnosticReport: The Encounter when the order was made NutritionOrder: Return nutrition orders with this encounter identifier Composition: Context of the Composition | DocumentReference.context.encounter | RiskAssessment.context | DeviceRequest.context | Procedure.context | List.encounter | VisionPrescription.encounter | ProcedureRequest.context | Flag.encounter | Observation.context | DiagnosticReport.context | NutritionOrder.encounter | Composition.encounter

family | string | Multiple Resources: Practitioner: A portion of the family name Patient: A portion of the family name of the patient | Practitioner.name.family | Patient.name.family

gender | token | Multiple Resources: RelatedPerson: Gender of the related person Practitioner: Gender of the practitioner Person: The gender of the person Patient: Gender of the patient | RelatedPerson.gender | Practitioner.gender | Person.gender | Patient.gender

given | string | Multiple Resources: Practitioner: A portion of the given name Patient: A portion of the given name of the patient | Practitioner.name.given | Patient.name.given

identifier | token | Multiple Resources: DocumentManifest: Unique Identifier for the set of documents Goal: External Ids for this goal Consent: Identifier for this record (external references) DocumentReference: Master Version Specific Identifier SupplyRequest: Unique identifier RiskAssessment: Unique identifier for the assessment CareTeam: External Ids for this team ImagingStudy: Other identifiers for the Study FamilyMemberHistory: A search by a record identifier Encounter: Identifier(s) by which this encounter is known DeviceRequest: Business identifier for request/order AllergyIntolerance: External ids for this item CarePlan: External Ids for this plan EpisodeOfCare: Business Identifier(s) relevant for this EpisodeOfCare Procedure: A unique identifier for a procedure List: Business identifier Immunization: Business identifier VisionPrescription: Return prescriptions with this external identifier ProcedureRequest: Identifiers assigned to this order Observation: The unique id for a particular observation DiagnosticReport: An identifier for the report NutritionOrder: Return nutrition orders with this external identifier Condition: A unique identifier of the condition record Composition: Logical identifier of composition (version-independent) DetectedIssue: Unique id for the detected issue SupplyDelivery: External identifier | DocumentManifest.masterIdentifier | DocumentManifest.identifier | Goal.identifier | Consent.identifier | DocumentReference.masterIdentifier | DocumentReference.identifier | SupplyRequest.identifier | RiskAssessment.identifier | CareTeam.identifier | ImagingStudy.identifier | FamilyMemberHistory.identifier | Encounter.identifier | DeviceRequest.identifier | AllergyIntolerance.identifier | CarePlan.identifier | EpisodeOfCare.identifier | Procedure.identifier | List.identifier | Immunization.identifier | VisionPrescription.identifier | ProcedureRequest.identifier | Observation.identifier | DiagnosticReport.identifier | NutritionOrder.identifier | Condition.identifier | Composition.identifier | DetectedIssue.identifier | SupplyDelivery.identifier

identifier | token | Multiple Resources: MedicationRequest: Return prescriptions with this external identifier MedicationAdministration: Return administrations with this external identifier MedicationStatement: Return statements with this external identifier MedicationDispense: Return dispenses with this external identifier | MedicationRequest.identifier | MedicationAdministration.identifier | MedicationStatement.identifier | MedicationDispense.identifier

medication | reference | Multiple Resources: MedicationRequest: Return prescriptions of this medication reference MedicationAdministration: Return administrations of this medication resource MedicationStatement: Return statements of this medication reference MedicationDispense: Return dispenses of this medicine resource | MedicationRequest.medication.as(Reference) | MedicationAdministration.medication.as(Reference) | MedicationStatement.medication.as(Reference) | MedicationDispense.medication.as(Reference)

patient | reference | Multiple Resources: ReferralRequest: Who the referral is about DocumentManifest: The subject of the set of documents Goal: Who this goal is intended for Consent: Who the consent applies to DocumentReference: Who/what is the subject of the document ImagingManifest: Subject of the ImagingManifest (or a DICOM Key Object Selection which it represents) RiskAssessment: Who/what does assessment apply to? CareTeam: Who care team is for ImagingStudy: Who the study is about FamilyMemberHistory: The identity of a subject to list family member history items for Encounter: The patient ro group present at the encounter DeviceUseStatement: Search by subject - a patient DeviceRequest: Individual the service is ordered for AllergyIntolerance: Who the sensitivity is for CarePlan: Who care plan is for EpisodeOfCare: The patient who is the focus of this episode of care Procedure: Search by subject - a patient List: If all resources have the same subject Immunization: The patient for the vaccination record VisionPrescription: The identity of a patient to list dispenses for ProcedureRequest: Search by subject - a patient Flag: The identity of a subject to list flags for Observation: The subject that the observation is about (if patient) DiagnosticReport: The subject of the report if a patient NutritionOrder: The identity of the person who requires the diet, formula or nutritional supplement Condition: Who has the condition? Composition: Who and/or what the composition is about DetectedIssue: Associated patient SupplyDelivery: Patient for whom the item is supplied ClinicalImpression: Patient or group assessed | ReferralRequest.subject | DocumentManifest.subject | Goal.subject | Consent.patient | DocumentReference.subject | ImagingManifest.patient | RiskAssessment.subject | CareTeam.subject | ImagingStudy.patient | FamilyMemberHistory.patient | Encounter.subject | DeviceUseStatement.subject | DeviceRequest.subject | AllergyIntolerance.patient | CarePlan.subject | EpisodeOfCare.patient | Procedure.subject | List.subject | Immunization.patient | VisionPrescription.patient | ProcedureRequest.subject | Flag.subject | Observation.subject | DiagnosticReport.subject | NutritionOrder.patient | Condition.subject | Composition.subject | DetectedIssue.patient | SupplyDelivery.patient | ClinicalImpression.subject

patient | reference | Multiple Resources: MedicationRequest: Returns prescriptions for a specific patient MedicationAdministration: The identity of a patient to list administrations for MedicationStatement: Returns statements for a specific patient. MedicationDispense: The identity of a patient to list dispenses for | MedicationRequest.subject | MedicationAdministration.subject | MedicationStatement.subject | MedicationDispense.subject

phone | token | Multiple Resources: PractitionerRole: A value in a phone contact RelatedPerson: A value in a phone contact Practitioner: A value in a phone contact Person: A value in a phone contact Patient: A value in a phone contact | PractitionerRole.telecom.where(system='phone') | RelatedPerson.telecom.where(system='phone') | Practitioner.telecom.where(system='phone') | Person.telecom.where(system='phone') | Patient.telecom.where(system='phone')

phonetic | string | Multiple Resources: RelatedPerson: A portion of name using some kind of phonetic matching algorithm Practitioner: A portion of either family or given name using some kind of phonetic matching algorithm Person: A portion of name using some kind of phonetic matching algorithm Patient: A portion of either family or given name using some kind of phonetic matching algorithm | RelatedPerson.name | Practitioner.name | Person.name | Patient.name

prescription | reference | Multiple Resources: MedicationAdministration: The identity of a prescription to list administrations from MedicationDispense: The identity of a prescription to list dispenses from | MedicationAdministration.prescription | MedicationDispense.authorizingPrescription

status | token | Multiple Resources: MedicationRequest: Status of the prescription MedicationAdministration: MedicationAdministration event status (for example one of active/paused/completed/nullified) MedicationStatement: Return statements that match the given status MedicationDispense: Return dispenses with a specified dispense status | MedicationRequest.status | MedicationAdministration.status | MedicationStatement.status | MedicationDispense.status

telecom | token | Multiple Resources: PractitionerRole: The value in any kind of contact RelatedPerson: The value in any kind of contact Practitioner: The value in any kind of contact Person: The value in any kind of contact Patient: The value in any kind of telecom details of the patient | PractitionerRole.telecom | RelatedPerson.telecom | Practitioner.telecom | Person.telecom | Patient.telecom

type | token | Multiple Resources: ReferralRequest: The type of the referral DocumentManifest: Kind of document set DocumentReference: Kind of document (LOINC if possible) Encounter: Specific type of encounter AllergyIntolerance: allergy | intolerance - Underlying mechanism (if known)

EpisodeOfCare: Type/class - e.g. specialist referral, disease management Composition: Kind of composition (LOINC if possible) | ReferralRequest.type | DocumentManifest.type | DocumentReference.type | Encounter.type | AllergyIntolerance.type | EpisodeOfCare.type | Composition.type

Account

balance | quantity | How much is in account? | Account.balance

identifier | token | Account number | Account.identifier

name | string | Human-readable label | Account.name

owner | reference | Who is responsible? | Account.owner

patient | reference | What is account tied to? | Account.subject

period | date | Transaction window | Account.period

status | token | active | inactive | entered-in-error | Account.status

subject | reference | What is account tied to? | Account.subject

type | token | E.g. patient, expense, depreciation | Account.type

ActivityDefinition

composed-of | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='composed-of').resource

date | date | The activity definition publication date | ActivityDefinition.date

depends-on | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='depends-on').resource | ActivityDefinition.library

derived-from | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='derived-from').resource

description | string | The description of the activity definition | ActivityDefinition.description

effective | date | The time during which the activity definition is intended to be in use | ActivityDefinition.effectivePeriod

identifier | token | External identifier for the activity definition | ActivityDefinition.identifier

jurisdiction | token | Intended jurisdiction for the activity definition | ActivityDefinition.jurisdiction

name | string | Computationally friendly name of the activity definition | ActivityDefinition.name

predecessor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='predecessor').resource

publisher | string | Name of the publisher of the activity definition | ActivityDefinition.publisher

status | token | The current status of the activity definition | ActivityDefinition.status

successor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='successor').resource

title | string | The human-friendly name of the activity definition | ActivityDefinition.title

topic | token | Topics associated with the module | ActivityDefinition.topic

url | uri | The uri that identifies the activity definition | ActivityDefinition.url

version | token | The business version of the activity definition | ActivityDefinition.version

AdverseEvent

category | token | AE | PAE An adverse event is an event that caused harm to a patient, an adverse reaction is a something that is a subject-specific event that is a result of an exposure to a medication, food, device or environmental substance, a potential adverse event is something that occurred and that could have caused harm to a patient but did not | AdverseEvent.category

date | date | When the event occurred | AdverseEvent.date

location | reference | Location where adverse event occurred | AdverseEvent.location

reaction | reference | Adverse Reaction Events linked to exposure to substance | AdverseEvent.reaction

recorder | reference | Who recorded the adverse event | AdverseEvent.recorder

seriousness | token | Mild | Moderate | Severe | AdverseEvent.seriousness

study | reference | AdverseEvent.study | AdverseEvent.study

subject | reference | Subject or group impacted by event | AdverseEvent.subject

substance | reference | Refers to the specific entity that caused the adverse event | AdverseEvent.suspectEntity.instance

type | token | actual | potential | AdverseEvent.type

AllergyIntolerance

asserter | reference | Source of the information about the allergy | AllergyIntolerance.asserter

category | token | food | medication | environment | biologic | AllergyIntolerance.category

clinical-status | token | active | inactive | resolved | AllergyIntolerance.clinicalStatus

code | token | Code that identifies the allergy or intolerance | AllergyIntolerance.code | AllergyIntolerance.reaction.substance

criticality | token | low | high | unable-to-assess | AllergyIntolerance.criticality

date | date | Date record was believed accurate | AllergyIntolerance.assertedDate

identifier | token | External ids for this item | AllergyIntolerance.identifier

last-date | date | Date(/time) of last known occurrence of a reaction | AllergyIntolerance.lastOccurrence

manifestation | token | Clinical symptoms/signs associated with the Event | AllergyIntolerance.reaction.manifestation

onset | date | Date(/time) when manifestations showed | AllergyIntolerance.reaction.onset

patient | reference | Who the sensitivity is for | AllergyIntolerance.patient

recorder | reference | Who recorded the sensitivity | AllergyIntolerance.recorder

route | token | How the subject was exposed to the substance | AllergyIntolerance.reaction.exposureRoute

severity | token | mild | moderate | severe (of event as a whole) | AllergyIntolerance.reaction.severity

type | token | allergy | intolerance - Underlying mechanism (if known) | AllergyIntolerance.type

verification-status | token | unconfirmed | confirmed | refuted | entered-in-error | AllergyIntolerance.verificationStatus

Appointment

actor | reference | Any one of the individuals participating in the appointment | Appointment.participant.actor

appointment-type | token | The style of appointment or patient that has been booked in the slot (not service type) | Appointment.appointmentType

date | date | Appointment date/time. | Appointment.start

identifier | token | An Identifier of the Appointment | Appointment.identifier

incomingreferral | reference | The ReferralRequest provided as information to allocate to the Encounter | Appointment.incomingReferral

location | reference | This location is listed in the participants of the appointment | Appointment.participant.actor

part-status | token | The Participation status of the subject, or other participant on the appointment. Can be used to locate participants that have not responded to meeting requests. | Appointment.participant.status

patient | reference | One of the individuals of the appointment is this patient | Appointment.participant.actor

practitioner | reference | One of the individuals of the appointment is this practitioner | Appointment.participant.actor

service-type | token | The specific service that is to be performed during this appointment | Appointment.serviceType
status | token | The overall status of the appointment | Appointment.status
AppointmentResponse
actor | reference | The Person, Location/HealthcareService or Device that this appointment response replies for | AppointmentResponse.actor
appointment | reference | The appointment that the response is attached to | AppointmentResponse.appointment
identifier | token | An Identifier in this appointment response | AppointmentResponse.identifier
location | reference | This Response is for this Location | AppointmentResponse.actor
part-status | token | The participants acceptance status for this appointment | AppointmentResponse.participantStatus
patient | reference | This Response is for this Patient | AppointmentResponse.actor
practitioner | reference | This Response is for this Practitioner | AppointmentResponse.actor
AuditEvent
action | token | Type of action performed during the event | AuditEvent.action
address | string | Identifier for the network access point of the user device | AuditEvent.agent.network.address
agent | reference | Direct reference to resource | AuditEvent.agent.reference
agent-name | string | Human-meaningful name for the agent | AuditEvent.agent.name
agent-role | token | Agent role in the event | AuditEvent.agent.role
altid | token | Alternative User id e.g. authentication | AuditEvent.agent.altId
date | date | Time when the event occurred on source | AuditEvent.recorded
entity | reference | Specific instance of resource | AuditEvent.entity.reference
entity-id | token | Specific instance of object | AuditEvent.entity.identifier
entity-name | string | Descriptor for entity | AuditEvent.entity.name
entity-role | token | What role the entity played | AuditEvent.entity.role
entity-type | token | Type of entity involved | AuditEvent.entity.type
outcome | token | Whether the event succeeded or failed | AuditEvent.outcome
patient | reference | Direct reference to resource | AuditEvent.agent.reference | AuditEvent.entity.reference
policy | uri | Policy that authorized event | AuditEvent.agent.policy
site | token | Logical source location within the enterprise | AuditEvent.source.site
source | token | The identity of source detecting the event | AuditEvent.source.identifier
subtype | token | More specific type/id for the event | AuditEvent.subtype
type | token | Type/identifier of event | AuditEvent.type
user | token | Unique identifier for the user | AuditEvent.agent.userId
Basic
author | reference | Who created | Basic.author
code | token | Kind of Resource | Basic.code
created | date | When created | Basic.created
identifier | token | Business identifier | Basic.identifier
patient | reference | Identifies the focus of this resource | Basic.subject
subject | reference | Identifies the focus of this resource | Basic.subject
Binary
contenttype | token | MimeType of the binary content | Binary.contentType
BodySite
code | token | Named anatomical location | BodySite.code
identifier | token | Identifier for this instance of the anatomical location | BodySite.identifier
patient | reference | Patient to whom bodysite belongs | BodySite.patient
Bundle
composition | reference | The first resource in the bundle, if the bundle type is "document" - this is a composition, and this parameter provides access to searches its contents | Bundle.entry[0].resource
identifier | token | Persistent identifier for the bundle | Bundle.identifier
message | reference | The first resource in the bundle, if the bundle type is "message" - this is a message header, and this parameter provides access to search its contents | Bundle.entry[0].resource
type | token | document | message | transaction | transaction-response | batch | batch-response | history | searchset | collection | Bundle.type
CapabilityStatement
date | date | The capability statement publication date | CapabilityStatement.date
description | string | The description of the capability statement | CapabilityStatement.description
event | token | Event code in a capability statement | CapabilityStatement.messaging.event.code
fhirversion | token | The version of FHIR | CapabilityStatement.version
format | token | formats supported (xml | json | ttl | mime type) | CapabilityStatement.format
guide | uri | Implementation guides supported | CapabilityStatement.implementationGuide
jurisdiction | token | Intended jurisdiction for the capability statement | CapabilityStatement.jurisdiction
mode | token | Mode - restful (server/client) or messaging (sender/receiver) | CapabilityStatement.rest.mode
name | string | Computationally friendly name of the capability statement | CapabilityStatement.name
publisher | string | Name of the publisher of the capability statement | CapabilityStatement.publisher
resource | token | Name of a resource mentioned in a capability statement | CapabilityStatement.rest.resource.type
resource-profile | reference | A profile id invoked in a capability statement | CapabilityStatement.rest.resource.profile
security-service | token | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates | CapabilityStatement.rest.security.service
software | string | Part of a the name of a software application | CapabilityStatement.software.name
status | token | The current status of the capability statement | CapabilityStatement.status
supported-profile | reference | Profiles for use cases supported | CapabilityStatement.profile
title | string | The human-friendly name of the capability statement | CapabilityStatement.title
url | uri | The uri that identifies the capability statement | CapabilityStatement.url
version | token | The business version of the capability statement | CapabilityStatement.version
CarePlan

activity-code | token | Detail type of activity | CarePlan.activity.detail.code
activity-date | date | Specified date occurs within period specified by CarePlan.activity.timingSchedule | CarePlan.activity.detail.scheduled
activity-reference | reference | Activity details defined in specific resource | CarePlan.activity.reference
based-on | reference | Fulfills care plan | CarePlan.basedOn
care-team | reference | Who's involved in plan? | CarePlan.careTeam
category | token | Type of plan | CarePlan.category
condition | reference | Health issues this plan addresses | CarePlan.addresses
context | reference | Created in context of | CarePlan.context
date | date | Time period plan covers | CarePlan.period
definition | reference | Protocol or definition | CarePlan.definition
encounter | reference | Created in context of | CarePlan.context
goal | reference | Desired outcome of plan | CarePlan.goal
identifier | token | External Ids for this plan | CarePlan.identifier
intent | token | proposal | plan | order | option | CarePlan.intent
part-of | reference | Part of referenced CarePlan | CarePlan.partOf
patient | reference | Who care plan is for | CarePlan.subject
performer | reference | Matches if the practitioner is listed as a performer in any of the "simple" activities. (For performers of the detailed activities, chain through the activitydetail search parameter.) | CarePlan.activity.detail.performer
replaces | reference | CarePlan replaced by this CarePlan | CarePlan.replaces
status | token | draft | active | suspended | completed | entered-in-error | cancelled | unknown | CarePlan.status
subject | reference | Who care plan is for | CarePlan.subject
CareTeam
category | token | Type of team | CareTeam.category
context | reference | Encounter or episode associated with CareTeam | CareTeam.context
date | date | Time period team covers | CareTeam.period
encounter | reference | Encounter or episode associated with CareTeam | CareTeam.context
identifier | token | External Ids for this team | CareTeam.identifier
participant | reference | Who is involved | CareTeam.participant.member
patient | reference | Who care team is for | CareTeam.subject
status | token | proposed | active | suspended | inactive | entered-in-error | CareTeam.status
subject | reference | Who care team is for | CareTeam.subject
ChargeItem
account | reference | Account to place this charge | ChargeItem.account
code | token | A code that identifies the charge, like a billing code | ChargeItem.code
context | reference | Encounter / Episode associated with event | ChargeItem.context
entered-date | date | Date the charge item was entered | ChargeItem.enteredDate
enterer | reference | Individual who was entering | ChargeItem.enterer
factor-override | number | Factor overriding the associated rules | ChargeItem.factorOverride
identifier | token | Business Identifier for item | ChargeItem.identifier
occurrence | date | When the charged service was applied | ChargeItem.occurrence
participant-actor | reference | Individual who was performing | ChargeItem.participant.actor
participant-role | token | What type of performance was done | ChargeItem.participant.role
patient | reference | Individual service was done for/to | ChargeItem.subject
performing-organization | reference | Organization providing the charged sevice | ChargeItem.performingOrganization
price-override | quantity | Price overriding the associated rules | ChargeItem.priceOverride
quantity | quantity | Quantity of which the charge item has been serviced | ChargeItem.quantity
requesting-organization | reference | Organization requesting the charged service | ChargeItem.requestingOrganization
service | reference | Which rendered service is being charged? | ChargeItem.service
subject | reference | Individual service was done for/to | ChargeItem.subject
Claim
care-team | reference | Member of the CareTeam | Claim.careTeam.provider
created | date | The creation date for the Claim | Claim.created
encounter | reference | Encounters associated with a billed line item | Claim.item.encounter
enterer | reference | The party responsible for the entry of the Claim | Claim.enterer
facility | reference | Facility responsible for the goods and services | Claim.facility
identifier | token | The primary identifier of the financial resource | Claim.identifier
insurer | reference | The target payor/insurer for the Claim | Claim.insurer
organization | reference | The reference to the providing organization | Claim.organization
patient | reference | Patient receiving the services | Claim.patient
payee | reference | The party receiving any payment for the Claim | Claim.payee.party
priority | token | Processing priority requested | Claim.priority
provider | reference | Provider responsible for the Claim | Claim.provider
use | token | The kind of financial resource | Claim.use
ClaimResponse
created | date | The creation date | ClaimResponse.created
disposition | string | The contents of the disposition message | ClaimResponse.disposition
identifier | token | The identity of the claimresponse | ClaimResponse.identifier
insurer | reference | The organization who generated this resource | ClaimResponse.insurer
outcome | token | The processing outcome | ClaimResponse.outcome
patient | reference | The subject of care. | ClaimResponse.patient
payment-date | date | The expected paymentDate | ClaimResponse.payment.date
request | reference | The claim reference | ClaimResponse.request

request-provider | reference | The Provider of the claim | ClaimResponse.requestProvider
ClinicalImpression
action | reference | Action taken as part of assessment procedure | ClinicalImpression.action
assessor | reference | The clinician performing the assessment | ClinicalImpression.assessor
context | reference | Encounter or Episode created from | ClinicalImpression.context
date | date | When the assessment was documented | ClinicalImpression.date
finding-code | token | What was found | ClinicalImpression.finding.item.as(CodeableConcept)
finding-ref | reference | What was found | ClinicalImpression.finding.item.as(Reference)
identifier | token | Business identifier | ClinicalImpression.identifier
investigation | reference | Record of a specific investigation | ClinicalImpression.investigation.item
patient | reference | Patient or group assessed | ClinicalImpression.subject
previous | reference | Reference to last assessment | ClinicalImpression.previous
problem | reference | Relevant impressions of patient state | ClinicalImpression.problem
status | token | draft | completed | entered-in-error | ClinicalImpression.status
subject | reference | Patient or group assessed | ClinicalImpression.subject
CodeSystem
code | token | A code defined in the code system | CodeSystem.concept.code
content-mode | token | not-present | example | fragment | complete | CodeSystem.content
date | date | The code system publication date | CodeSystem.date
description | string | The description of the code system | CodeSystem.description
identifier | token | External identifier for the code system | CodeSystem.identifier
jurisdiction | token | Intended jurisdiction for the code system | CodeSystem.jurisdiction
language | token | A language in which a designation is provided | CodeSystem.concept.designation.language
name | string | Computationally friendly name of the code system | CodeSystem.name
publisher | string | Name of the publisher of the code system | CodeSystem.publisher
status | token | The current status of the code system | CodeSystem.status
system | uri | The system for any codes defined by this code system (same as 'url') | CodeSystem.url
title | string | The human-friendly name of the code system | CodeSystem.title
url | uri | The uri that identifies the code system | CodeSystem.url
version | token | The business version of the code system | CodeSystem.version
Communication
based-on | reference | Request fulfilled by this communication | Communication.basedOn
category | token | Message category | Communication.category
context | reference | Encounter or episode leading to message | Communication.context
definition | reference | Instantiates protocol or definition | Communication.definition
encounter | reference | Encounter leading to message | Communication.context
identifier | token | Unique identifier | Communication.identifier
medium | token | A channel of communication | Communication.medium
part-of | reference | Part of this action | Communication.partOf
patient | reference | Focus of message | Communication.subject
received | date | When received | Communication.received
recipient | reference | Message recipient | Communication.recipient
sender | reference | Message sender | Communication.sender
sent | date | When sent | Communication.sent
status | token | preparation | in-progress | suspended | aborted | completed | entered-in-error | Communication.status
subject | reference | Focus of message | Communication.subject
CommunicationRequest
authored | date | When request transitioned to being actionable | CommunicationRequest.authoredOn
based-on | reference | Fulfills plan or proposal | CommunicationRequest.basedOn
category | token | Message category | CommunicationRequest.category
context | reference | Encounter or episode leading to message | CommunicationRequest.context
encounter | reference | Encounter leading to message | CommunicationRequest.context
group-identifier | token | Composite request this is part of | CommunicationRequest.groupIdentifier
identifier | token | Unique identifier | CommunicationRequest.identifier
medium | token | A channel of communication | CommunicationRequest.medium
occurrence | date | When scheduled | CommunicationRequest.occurrence.as(DateTime)
patient | reference | Focus of message | CommunicationRequest.subject
priority | token | Message urgency | CommunicationRequest.priority
recipient | reference | Message recipient | CommunicationRequest.recipient
replaces | reference | Request(s) replaced by this request | CommunicationRequest.replaces
requester | reference | Individual making the request | CommunicationRequest.requester.agent
sender | reference | Message sender | CommunicationRequest.sender
status | token | draft | active | suspended | cancelled | completed | entered-in-error | unknown | CommunicationRequest.status
subject | reference | Focus of message | CommunicationRequest.subject
CompartmentDefinition
code | token | Patient | Encounter | RelatedPerson | Practitioner | Device | CompartmentDefinition.code
date | date | The compartment definition publication date | CompartmentDefinition.date
description | string | The description of the compartment definition | CompartmentDefinition.description
jurisdiction | token | Intended jurisdiction for the compartment definition | CompartmentDefinition.jurisdiction
name | string | Computationally friendly name of the compartment definition | CompartmentDefinition.name
publisher | string | Name of the publisher of the compartment definition | CompartmentDefinition.publisher
resource | token | Name of resource type | CompartmentDefinition.resource.code

status | token | The current status of the compartment definition | CompartmentDefinition.status
title | string | The human-friendly name of the compartment definition | CompartmentDefinition.title
url | uri | The uri that identifies the compartment definition | CompartmentDefinition.url
Composition
attester | reference | Who attested the composition | Composition.attester.party
author | reference | Who and/or what authored the composition | Composition.author
class | token | Categorization of Composition | Composition.class
confidentiality | token | As defined by affinity domain | Composition.confidentiality
context | token | Code(s) that apply to the event being documented | Composition.event.code
date | date | Composition editing time | Composition.date
encounter | reference | Context of the Composition | Composition.encounter
entry | reference | A reference to data that supports this section | Composition.section.entry
identifier | token | Logical identifier of composition (version-independent) | Composition.identifier
patient | reference | Who and/or what the composition is about | Composition.subject
period | date | The period covered by the documentation | Composition.event.period
related-id | token | Target of the relationship | Composition.relatesTo.target.as(Identifier)
related-ref | reference | Target of the relationship | Composition.relatesTo.target.as(Reference)
section | token | Classification of section (recommended) | Composition.section.code
status | token | preliminary | final | amended | entered-in-error | Composition.status
subject | reference | Who and/or what the composition is about | Composition.subject
title | string | Human Readable name/title | Composition.title
type | token | Kind of composition (LOINC if possible) | Composition.type
ConceptMap
date | date | The concept map publication date | ConceptMap.date
dependson | uri | Reference to property mapping depends on | ConceptMap.group.element.target.dependsOn.property
description | string | The description of the concept map | ConceptMap.description
identifier | token | External identifier for the concept map | ConceptMap.identifier
jurisdiction | token | Intended jurisdiction for the concept map | ConceptMap.jurisdiction
name | string | Computationally friendly name of the concept map | ConceptMap.name
other | uri | Canonical URL for other concept map | ConceptMap.group.unmapped.url
product | uri | Reference to property mapping depends on | ConceptMap.group.element.target.product.property
publisher | string | Name of the publisher of the concept map | ConceptMap.publisher
source | reference | Identifies the source of the concepts which are being mapped | ConceptMap.source.as(Reference)
source-code | token | Identifies element being mapped | ConceptMap.group.element.code
source-system | uri | Code System (if value set crosses code systems) | ConceptMap.group.source
source-uri | reference | Identifies the source of the concepts which are being mapped | ConceptMap.source.as(Uri)
status | token | The current status of the concept map | ConceptMap.status
target | reference | Provides context to the mappings | ConceptMap.target.as(Reference)
target-code | token | Code that identifies the target element | ConceptMap.group.element.target.code
target-system | uri | System of the target (if necessary) | ConceptMap.group.target
target-uri | reference | Provides context to the mappings | ConceptMap.target.as(Uri)
title | string | The human-friendly name of the concept map | ConceptMap.title
url | uri | The uri that identifies the concept map | ConceptMap.url
version | token | The business version of the concept map | ConceptMap.version
Condition
abatement-age | quantity | Abatement as age or age range | Condition.abatement.as(Age) | Condition.abatement.as(Range) | Condition.abatement.as(Age)
abatement-boolean | token | Abatement boolean (boolean is true or non-boolean values are present) | Condition.abatement.as(boolean) | Condition.abatement.is(dateTime) | Condition.abatement.is(Age) | Condition.abatement.is(Period) | Condition.abatement.is(Range) | Condition.abatement.is(string)
abatement-date | date | Date-related abatements (dateTime and period) | Condition.abatement.as(dateTime) | Condition.abatement.as(Period)
abatement-string | string | Abatement as a string | Condition.abatement.as(string)
asserted-date | date | Date record was believed accurate | Condition.assertedDate
asserter | reference | Person who asserts this condition | Condition.asserter
body-site | token | Anatomical location, if relevant | Condition.bodySite
category | token | The category of the condition | Condition.category
clinical-status | token | The clinical status of the condition | Condition.clinicalStatus
code | token | Code for the condition | Condition.code
context | reference | Encounter or episode when condition first asserted | Condition.context
encounter | reference | Encounter when condition first asserted | Condition.context
evidence | token | Manifestation/symptom | Condition.evidence.code
evidence-detail | reference | Supporting information found elsewhere | Condition.evidence.detail
identifier | token | A unique identifier of the condition record | Condition.identifier
onset-age | quantity | Onsets as age or age range | Condition.onset.as(Age) | Condition.onset.as(Range)
onset-date | date | Date related onsets (dateTime and Period) | Condition.onset.as(dateTime) | Condition.onset.as(Period)
onset-info | string | Onsets as a string | Condition.onset.as(string)
patient | reference | Who has the condition? | Condition.subject
severity | token | The severity of the condition | Condition.severity
stage | token | Simple summary (disease specific) | Condition.stage.summary
subject | reference | Who has the condition? | Condition.subject
verification-status | token | provisional | differential | confirmed | refuted | entered-in-error | unknown | Condition.verificationStatus
Consent

action | token | Actions controlled by this consent | Consent.action | Consent.except.action
actor | reference | Resource for the actor (or group, by role) | Consent.actor.reference | Consent.except.actor.reference
category | token | Classification of the consent statement - for indexing/retrieval | Consent.category
consentor | reference | Who is agreeing to the policy and exceptions | Consent.consentingParty
data | reference | The actual data reference | Consent.data.reference | Consent.except.data.reference
date | date | When this Consent was created or indexed | Consent.dateTime
identifier | token | Identifier for this record (external references) | Consent.identifier
organization | reference | Custodian of the consent | Consent.organization
patient | reference | Who the consent applies to | Consent.patient
period | date | Period that this consent applies | Consent.period
purpose | token | Context of activities for which the agreement is made | Consent.purpose | Consent.except.purpose
securitylabel | token | Security Labels that define affected resources | Consent.securityLabel | Consent.except.securityLabel
source | reference | Source from which this consent is taken | Consent.source
status | token | draft | proposed | active | rejected | inactive | entered-in-error | Consent.status
Contract
agent | reference | Agent to the Contact | Contract.agent.actor
authority | reference | The authority of the contract | Contract.authority
domain | reference | The domain of the contract | Contract.domain
identifier | token | The identity of the contract | Contract.identifier
issued | date | The date/time the contract was issued | Contract.issued
patient | reference | The identity of the subject of the contract (if a patient) | Contract.subject
signer | reference | Contract Signatory Party | Contract.signer.party
subject | reference | The identity of the subject of the contract | Contract.subject
term-topic | reference | The identity of the topic of the contract terms | Contract.term.topic
Coverage
beneficiary | reference | Covered party | Coverage.beneficiary
class | string | Class identifier | Coverage.grouping.class
dependent | string | Dependent number | Coverage.dependent
group | string | Group identifier | Coverage.grouping.group
identifier | token | The primary identifier of the insured and the coverage | Coverage.identifier
payor | reference | The identity of the insurer or party paying for services | Coverage.payor
plan | string | A plan or policy identifier | Coverage.grouping.plan
policy-holder | reference | Reference to the policyholder | Coverage.policyHolder
sequence | string | Sequence number | Coverage.sequence
subclass | string | Sub-class identifier | Coverage.grouping.subClass
subgroup | string | Sub-group identifier | Coverage.grouping.subGroup
subplan | string | Sub-plan identifier | Coverage.grouping.subPlan
subscriber | reference | Reference to the subscriber | Coverage.subscriber
type | token | The kind of coverage (health plan, auto, Workers Compensation) | Coverage.type
DataElement
code | token | A code for the data element (server may choose to do subsumption) | DataElement.element.code
date | date | The data element publication date | DataElement.date
description | string | Text search in the description of the data element. This corresponds to the definition of the first DataElement.element. | DataElement.element.definition
identifier | token | External identifier for the data element | DataElement.identifier
jurisdiction | token | Intended jurisdiction for the data element | DataElement.jurisdiction
name | string | Computationally friendly name of the data element | DataElement.name
objectClass | token | Matches on the 11179-objectClass extension value
objectClassProperty | token | Matches on the 11179-objectClassProperty extension value
publisher | string | Name of the publisher of the data element | DataElement.publisher
status | token | The current status of the data element | DataElement.status
stringency | token | The stringency of the data element definition | DataElement.stringency
title | string | The human-friendly name of the data element | DataElement.title
url | uri | The uri that identifies the data element | DataElement.url
version | token | The business version of the data element | DataElement.version
DetectedIssue
author | reference | The provider or device that identified the issue | DetectedIssue.author
category | token | Issue Category, e.g. drug-drug, duplicate therapy, etc. | DetectedIssue.category
date | date | When identified | DetectedIssue.date
identifier | token | Unique id for the detected issue | DetectedIssue.identifier
implicated | reference | Problem resource | DetectedIssue.implicated
patient | reference | Associated patient | DetectedIssue.patient
Device
device-name | string | A server defined search that may match any of the string fields in the Device.udi.name or Device.type.coding.display or Device.type.text | Device.udi.name | Device.type.text | Device.type.coding.display
identifier | token | Instance id from manufacturer, owner, and others | Device.identifier
location | reference | A location, where the resource is found | Device.location
manufacturer | string | The manufacturer of the device | Device.manufacturer
model | string | The model of the device | Device.model
organization | reference | The organization responsible for the device | Device.owner
patient | reference | Patient information, if the resource is affixed to a person | Device.patient
status | token | active | inactive | entered-in-error | unknown | Device.status

type | token | The type of the device | Device.type
udi-carrier | string | UDI Barcode (RFID or other technology) string either in HRF format or AIDC format converted to base64 string. | Device.udi.carrierHRF | Device.udi.carrierAIDC
udi-di | string | The udi Device Identifier (DI) | Device.udi.deviceIdentifier
url | uri | Network address to contact device | Device.url
DeviceComponent
identifier | token | The identifier of the component | DeviceComponent.identifier
parent | reference | The parent DeviceComponent resource | DeviceComponent.parent
source | reference | The device source | DeviceComponent.source
type | token | The device component type | DeviceComponent.type
DeviceMetric
category | token | The category of the metric | DeviceMetric.category
identifier | token | The identifier of the metric | DeviceMetric.identifier
parent | reference | The parent DeviceMetric resource | DeviceMetric.parent
source | reference | The device resource | DeviceMetric.source
type | token | The component type | DeviceMetric.type
DeviceRequest
authored-on | date | When the request transitioned to being actionable | DeviceRequest.authoredOn
based-on | reference | Plan/proposal/order fulfilled by this request | DeviceRequest.basedOn
code | token | Code for what is being requested/ordered | DeviceRequest.code.as(CodeableConcept)
definition | reference | Protocol or definition followed by this request | DeviceRequest.definition
device | reference | Reference to resource that is being requested/ordered | DeviceRequest.code.as(Reference)
encounter | reference | Encounter or Episode during which request was created | DeviceRequest.context
event-date | date | When service should occur | DeviceRequest.occurrence.as(DateTime) | DeviceRequest.occurrence.as(Period)
group-identifier | token | Composite request this is part of | DeviceRequest.groupIdentifier
identifier | token | Business identifier for request/order | DeviceRequest.identifier
intent | token | proposal | plan | original-order |reflex-order | DeviceRequest.intent
patient | reference | Individual the service is ordered for | DeviceRequest.subject
performer | reference | Desired performer for service | DeviceRequest.performer
priorrequest | reference | Request takes the place of referenced completed or terminated requests | DeviceRequest.priorRequest
requester | reference | Who/what is requesting service | DeviceRequest.requester.agent
status | token | entered-in-error | draft | active |suspended | completed | DeviceRequest.status
subject | reference | Individual the service is ordered for | DeviceRequest.subject
DeviceUseStatement
device | reference | Search by device | DeviceUseStatement.device
identifier | token | Search by identifier | DeviceUseStatement.identifier
patient | reference | Search by subject - a patient | DeviceUseStatement.subject
subject | reference | Search by subject | DeviceUseStatement.subject
DiagnosticReport
based-on | reference | Reference to the procedure request. | DiagnosticReport.basedOn
category | token | Which diagnostic discipline/department created the report | DiagnosticReport.category
code | token | The code for the report as a whole, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result | DiagnosticReport.code
context | reference | Healthcare event (Episode of Care or Encounter) related to the report | DiagnosticReport.context
date | date | The clinically relevant time of the report | DiagnosticReport.effective
diagnosis | token | A coded diagnosis on the report | DiagnosticReport.codedDiagnosis
encounter | reference | The Encounter when the order was made | DiagnosticReport.context
identifier | token | An identifier for the report | DiagnosticReport.identifier
image | reference | A reference to the image source. | DiagnosticReport.image.link
issued | date | When the report was issued | DiagnosticReport.issued
patient | reference | The subject of the report if a patient | DiagnosticReport.subject
performer | reference | Who was the source of the report (organization) | DiagnosticReport.performer.actor
result | reference | Link to an atomic result (observation resource) | DiagnosticReport.result
specimen | reference | The specimen details | DiagnosticReport.specimen
status | token | The status of the report | DiagnosticReport.status
subject | reference | The subject of the report | DiagnosticReport.subject
DocumentManifest
author | reference | Who and/or what authored the manifest | DocumentManifest.author
content-ref | reference | Contents of this set of documents | DocumentManifest.content.p.as(Reference)
created | date | When this document manifest created | DocumentManifest.created
description | string | Human-readable description (title) | DocumentManifest.description
identifier | token | Unique Identifier for the set of documents | DocumentManifest.masterIdentifier | DocumentManifest.identifier
patient | reference | The subject of the set of documents | DocumentManifest.subject
recipient | reference | Intended to get notified about this set of documents | DocumentManifest.recipient
related-id | token | Identifiers of things that are related | DocumentManifest.related.identifier
related-ref | reference | Related Resource | DocumentManifest.related.ref
source | uri | The source system/application/software | DocumentManifest.source
status | token | current | superseded | entered-in-error | DocumentManifest.status
subject | reference | The subject of the set of documents | DocumentManifest.subject
type | token | Kind of document set | DocumentManifest.type
DocumentReference
authenticator | reference | Who/what authenticated the document | DocumentReference.authenticator

author | reference | Who and/or what authored the document | DocumentReference.author
class | token | Categorization of document | DocumentReference.class
created | date | Document creation time | DocumentReference.created
custodian | reference | Organization which maintains the document | DocumentReference.custodian
description | string | Human-readable description (title) | DocumentReference.description
encounter | reference | Context of the document content | DocumentReference.context.encounter
event | token | Main clinical acts documented | DocumentReference.context.event
facility | token | Kind of facility where patient was seen | DocumentReference.context.facilityType
format | token | Format/content rules for the document | DocumentReference.content.format
identifier | token | Master Version Specific Identifier | DocumentReference.masterIdentifier | DocumentReference.identifier
indexed | date | When this document reference was created | DocumentReference.indexed
language | token | Human language of the content (BCP-47) | DocumentReference.content.attachment.language
location | uri | Uri where the data can be found | DocumentReference.content.attachment.url
patient | reference | Who/what is the subject of the document | DocumentReference.subject
period | date | Time of service that is being documented | DocumentReference.context.period
related-id | token | Identifier of related objects or events | DocumentReference.context.related.identifier
related-ref | reference | Related Resource | DocumentReference.context.related.ref
relatesto | reference | Target of the relationship | DocumentReference.relatesTo.target
relation | token | replaces | transforms | signs | appends | DocumentReference.relatesTo.code
relationship | composite | Combination of relation and relatesTo | DocumentReference.relatesTo
securitylabel | token | Document security-tags | DocumentReference.securityLabel
setting | token | Additional details about where the content was created (e.g. clinical specialty) | DocumentReference.context.practiceSetting
status | token | current | superseded | entered-in-error | DocumentReference.status
subject | reference | Who/what is the subject of the document | DocumentReference.subject
type | token | Kind of document (LOINC if possible) | DocumentReference.type
EligibilityRequest
created | date | The creation date for the EOB | EligibilityRequest.created
enterer | reference | The party who is responsible for the request | EligibilityRequest.enterer
facility | reference | Facility responsible for the goods and services | EligibilityRequest.facility
identifier | token | The business identifier of the Eligibility | EligibilityRequest.identifier
organization | reference | The reference to the providing organization | EligibilityRequest.organization
patient | reference | The reference to the patient | EligibilityRequest.patient
provider | reference | The reference to the provider | EligibilityRequest.provider
EligibilityResponse
created | date | The creation date | EligibilityResponse.created
disposition | string | The contents of the disposition message | EligibilityResponse.disposition
identifier | token | The business identifier | EligibilityResponse.identifier
insurer | reference | The organization which generated this resource | EligibilityResponse.insurer
outcome | token | The processing outcome | EligibilityResponse.outcome
request | reference | The EligibilityRequest reference | EligibilityResponse.request
request-organization | reference | The EligibilityRequest organization | EligibilityResponse.requestOrganization
request-provider | reference | The EligibilityRequest provider | EligibilityResponse.requestProvider
Encounter
appointment | reference | The appointment that scheduled this encounter | Encounter.appointment
class | token | inpatient | outpatient | ambulatory | emergency + | Encounter.class
date | date | A date within the period the Encounter lasted | Encounter.period
diagnosis | reference | Reason the encounter takes place (resource) | Encounter.diagnosis.condition
episodeofcare | reference | Episode(s) of care that this encounter should be recorded against | Encounter.episodeOfCare
identifier | token | Identifier(s) by which this encounter is known | Encounter.identifier
incomingreferral | reference | The ReferralRequest that initiated this encounter | Encounter.incomingReferral
length | number | Length of encounter in days | Encounter.length
location | reference | Location the encounter takes place | Encounter.location.location
location-period | date | Time period during which the patient was present at the location | Encounter.location.period
part-of | reference | Another Encounter this encounter is part of | Encounter.partOf
participant | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual
participant-type | token | Role of participant in encounter | Encounter.participant.type
patient | reference | The patient ro group present at the encounter | Encounter.subject
practitioner | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual
reason | token | Reason the encounter takes place (code) | Encounter.reason
service-provider | reference | The custodian organization of this Encounter record | Encounter.serviceProvider
special-arrangement | token | Wheelchair, translator, stretcher, etc. | Encounter.hospitalization.specialArrangement
status | token | planned | arrived | triaged | in-progress | onleave | finished | cancelled + | Encounter.status
subject | reference | The patient ro group present at the encounter | Encounter.subject
type | token | Specific type of encounter | Encounter.type
Endpoint
connection-type | token | Protocol/Profile/Standard to be used with this endpoint connection | Endpoint.connectionType
identifier | token | Identifies this endpoint across multiple systems | Endpoint.identifier
name | string | A name that this endpoint can be identified by | Endpoint.name
organization | reference | The organization that is managing the endpoint | Endpoint.managingOrganization
payload-type | token | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) | Endpoint.payloadType
status | token | The current status of the Endpoint (usually expected to be active) | Endpoint.status
EnrollmentRequest

identifier | token | The business identifier of the Enrollment | EnrollmentRequest.identifier
organization | reference | The organization who generated this resource | EnrollmentRequest.organization
patient | reference | The party to be enrolled | EnrollmentRequest.subject
subject | reference | The party to be enrolled | EnrollmentRequest.subject
EnrollmentResponse
identifier | token | The business identifier of the EnrollmentResponse | EnrollmentResponse.identifier
organization | reference | The organization who generated this resource | EnrollmentResponse.organization
request | reference | The reference to the claim | EnrollmentResponse.request
EpisodeOfCare
care-manager | reference | Care manager/care co-ordinator for the patient | EpisodeOfCare.careManager
condition | reference | Conditions/problems/diagnoses this episode of care is for | EpisodeOfCare.diagnosis.condition
date | date | The provided date search value falls within the episode of care's period | EpisodeOfCare.period
identifier | token | Business Identifier(s) relevant for this EpisodeOfCare | EpisodeOfCare.identifier
incomingreferral | reference | Incoming Referral Request | EpisodeOfCare.referralRequest
organization | reference | The organization that has assumed the specific responsibilities of this EpisodeOfCare | EpisodeOfCare.managingOrganization
patient | reference | The patient who is the focus of this episode of care | EpisodeOfCare.patient
status | token | The current status of the Episode of Care as provided (does not check the status history collection) | EpisodeOfCare.status
type | token | Type/class - e.g. specialist referral, disease management | EpisodeOfCare.type
ExpansionProfile
date | date | The expansion profile publication date | ExpansionProfile.date
description | string | The description of the expansion profile | ExpansionProfile.description
identifier | token | External identifier for the expansion profile | ExpansionProfile.identifier
jurisdiction | token | Intended jurisdiction for the expansion profile | ExpansionProfile.jurisdiction
name | string | Computationally friendly name of the expansion profile | ExpansionProfile.name
publisher | string | Name of the publisher of the expansion profile | ExpansionProfile.publisher
status | token | The current status of the expansion profile | ExpansionProfile.status
url | uri | The uri that identifies the expansion profile | ExpansionProfile.url
version | token | The business version of the expansion profile | ExpansionProfile.version
ExplanationOfBenefit
care-team | reference | Member of the CareTeam | ExplanationOfBenefit.careTeam.provider
claim | reference | The reference to the claim | ExplanationOfBenefit.claim
coverage | reference | The plan under which the claim was adjudicated | ExplanationOfBenefit.insurance.coverage
created | date | The creation date for the EOB | ExplanationOfBenefit.created
disposition | string | The contents of the disposition message | ExplanationOfBenefit.disposition
encounter | reference | Encounters associated with a billed line item | ExplanationOfBenefit.item.encounter
enterer | reference | The party responsible for the entry of the Claim | ExplanationOfBenefit.enterer
facility | reference | Facility responsible for the goods and services | ExplanationOfBenefit.facility
identifier | token | The business identifier of the Explanation of Benefit | ExplanationOfBenefit.identifier
organization | reference | The reference to the providing organization | ExplanationOfBenefit.organization
patient | reference | The reference to the patient | ExplanationOfBenefit.patient
payee | reference | The party receiving any payment for the Claim | ExplanationOfBenefit.payee.party
provider | reference | The reference to the provider | ExplanationOfBenefit.provider
FamilyMemberHistory
code | token | A search by a condition code | FamilyMemberHistory.condition.code
date | date | When history was captured/updated | FamilyMemberHistory.date
definition | reference | Instantiates protocol or definition | FamilyMemberHistory.definition
gender | token | A search by a gender code of a family member | FamilyMemberHistory.gender
identifier | token | A search by a record identifier | FamilyMemberHistory.identifier
patient | reference | The identity of a subject to list family member history items for | FamilyMemberHistory.patient
relationship | token | A search by a relationship type | FamilyMemberHistory.relationship
status | token | partial | completed | entered-in-error | health-unknown | FamilyMemberHistory.status
Flag
author | reference | Flag creator | Flag.author
date | date | Time period when flag is active | Flag.period
encounter | reference | Alert relevant during encounter | Flag.encounter
identifier | token | Business identifier | Flag.identifier
patient | reference | The identity of a subject to list flags for | Flag.subject
subject | reference | The identity of a subject to list flags for | Flag.subject
Goal
category | token | E.g. Treatment, dietary, behavioral, etc. | Goal.category
identifier | token | External Ids for this goal | Goal.identifier
patient | reference | Who this goal is intended for | Goal.subject
start-date | date | When goal pursuit begins | Goal.start.as(Date)
status | token | proposed | accepted | planned | in-progress | on-target | ahead-of-target | behind-target | sustaining | achieved | on-hold | cancelled | entered-in-error | rejected | Goal.status
subject | reference | Who this goal is intended for | Goal.subject
target-date | date | Reach goal on or before | Goal.target.due.as(Date)
GraphDefinition
date | date | The graph definition publication date | GraphDefinition.date
description | string | The description of the graph definition | GraphDefinition.description
jurisdiction | token | Intended jurisdiction for the graph definition | GraphDefinition.jurisdiction

name | string | Computationally friendly name of the graph definition | GraphDefinition.name
publisher | string | Name of the publisher of the graph definition | GraphDefinition.publisher
start | token | Type of resource at which the graph starts | GraphDefinition.start
status | token | The current status of the graph definition | GraphDefinition.status
url | uri | The uri that identifies the graph definition | GraphDefinition.url
version | token | The business version of the graph definition | GraphDefinition.version
Group
actual | token | Descriptive or actual | Group.actual
characteristic | token | Kind of characteristic | Group.characteristic.code
characteristic-value | composite | A composite of both characteristic and value | Group.characteristic
code | token | The kind of resources contained | Group.code
exclude | token | Group includes or excludes | Group.characteristic.exclude
identifier | token | Unique id | Group.identifier
member | reference | Reference to the group member | Group.member.entity
type | token | The type of resources the group contains | Group.type
value | token | Value held by characteristic | Group.characteristic.value
GuidanceResponse
identifier | token | The identifier of the guidance response | GuidanceResponse.identifier
patient | reference | The identity of a patient to search for guidance response results | GuidanceResponse.subject
request | token | The identifier of the request associated with the response | GuidanceResponse.requestId
subject | reference | The subject that the guidance response is about | GuidanceResponse.subject
HealthcareService
active | token | The Healthcare Service is currently marked as active | HealthcareService.active
category | token | Service Category of the Healthcare Service | HealthcareService.category
characteristic | token | One of the HealthcareService's characteristics | HealthcareService.characteristic
endpoint | reference | Technical endpoints providing access to services operated for the location | HealthcareService.endpoint
identifier | token | External identifiers for this item | HealthcareService.identifier
location | reference | The location of the Healthcare Service | HealthcareService.location
name | string | A portion of the Healthcare service name | HealthcareService.name
organization | reference | The organization that provides this Healthcare Service | HealthcareService.providedBy
programname | string | One of the Program Names serviced by this HealthcareService | HealthcareService.programName
type | token | The type of service provided by this healthcare service | HealthcareService.type
ImagingManifest
author | reference | Author of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.author
authoring-time | date | Time of the ImagingManifest (or a DICOM Key Object Selection which it represents) authoring | ImagingManifest.authoringTime
endpoint | reference | The endpoint for the study or series | ImagingManifest.study.endpoint | ImagingManifest.study.series.endpoint
identifier | token | UID of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.identifier
imaging-study | reference | ImagingStudy resource selected in the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.study.imagingStudy
patient | reference | Subject of the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.patient
selected-study | uri | Study selected in the ImagingManifest (or a DICOM Key Object Selection which it represents) | ImagingManifest.study.uid
ImagingStudy
accession | token | The accession identifier for the study | ImagingStudy.accession
basedon | reference | The order for the image | ImagingStudy.basedOn
bodysite | token | The body site studied | ImagingStudy.series.bodySite
context | reference | The context of the study | ImagingStudy.context
dicom-class | uri | The type of the instance | ImagingStudy.series.instance.sopClass
endpoint | reference | The endpoint for te study or series | ImagingStudy.endpoint | ImagingStudy.series.endpoint
identifier | token | Other identifiers for the Study | ImagingStudy.identifier
modality | token | The modality of the series | ImagingStudy.series.modality
patient | reference | Who the study is about | ImagingStudy.patient
performer | reference | The person who performed the study | ImagingStudy.series.performer
reason | token | The reason for the study | ImagingStudy.reason
series | uri | The identifier of the series of images | ImagingStudy.series.uid
started | date | When the study was started | ImagingStudy.started
study | uri | The study identifier for the image | ImagingStudy.uid
uid | uri | The instance unique identifier | ImagingStudy.series.instance.uid
Immunization
date | date | Vaccination (non)-Administration Date | Immunization.date
dose-sequence | number | Dose number within series | Immunization.vaccinationProtocol.doseSequence
identifier | token | Business identifier | Immunization.identifier
location | reference | The service delivery location or facility in which the vaccine was / was to be administered | Immunization.location
lot-number | string | Vaccine Lot Number | Immunization.lotNumber
manufacturer | reference | Vaccine Manufacturer | Immunization.manufacturer
notgiven | token | Administrations which were not given | Immunization.notGiven
patient | reference | The patient for the vaccination record | Immunization.patient
practitioner | reference | The practitioner who played a role in the vaccination | Immunization.practitioner.actor
reaction | reference | Additional information on reaction | Immunization.reaction.detail
reaction-date | date | When reaction started | Immunization.reaction.date
reason | token | Why immunization occurred | Immunization.explanation.reason
reason-not-given | token | Explanation of reason vaccination was not administered | Immunization.explanation.reasonNotGiven

status | token | Immunization event status | Immunization.status
vaccine-code | token | Vaccine Product Administered | Immunization.vaccineCode
ImmunizationRecommendation
date | date | Date recommendation created | ImmunizationRecommendation.recommendation.date
dose-number | number | Recommended dose number | ImmunizationRecommendation.recommendation.doseNumber
dose-sequence | number | Dose number within sequence | ImmunizationRecommendation.recommendation.protocol.doseSequence
identifier | token | Business identifier | ImmunizationRecommendation.identifier
information | reference | Patient observations supporting recommendation |
ImmunizationRecommendation.recommendation.supportingPatientInformation
patient | reference | Who this profile is for | ImmunizationRecommendation.patient
status | token | Vaccine administration status | ImmunizationRecommendation.recommendation.forecastStatus
support | reference | Past immunizations supporting recommendation | ImmunizationRecommendation.recommendation.supportingImmunization
target-disease | token | Disease to be immunized against | ImmunizationRecommendation.recommendation.targetDisease
vaccine-type | token | Vaccine recommendation applies to | ImmunizationRecommendation.recommendation.vaccineCode
ImplementationGuide
date | date | The implementation guide publication date | ImplementationGuide.date
dependency | uri | Where to find dependency | ImplementationGuide.dependency.uri
description | string | The description of the implementation guide | ImplementationGuide.description
experimental | token | For testing purposes, not real usage | ImplementationGuide.experimental
jurisdiction | token | Intended jurisdiction for the implementation guide | ImplementationGuide.jurisdiction
name | string | Computationally friendly name of the implementation guide | ImplementationGuide.name
publisher | string | Name of the publisher of the implementation guide | ImplementationGuide.publisher
resource | reference | Location of the resource | ImplementationGuide.package.resource.source
status | token | The current status of the implementation guide | ImplementationGuide.status
url | uri | The uri that identifies the implementation guide | ImplementationGuide.url
version | token | The business version of the implementation guide | ImplementationGuide.version
Library
composed-of | reference | What resource is being referenced | Library.relatedArtifact.where(type='composed-of').resource
date | date | The library publication date | Library.date
depends-on | reference | What resource is being referenced | Library.relatedArtifact.where(type='depends-on').resource
derived-from | reference | What resource is being referenced | Library.relatedArtifact.where(type='derived-from').resource
description | string | The description of the library | Library.description
effective | date | The time during which the library is intended to be in use | Library.effectivePeriod
identifier | token | External identifier for the library | Library.identifier
jurisdiction | token | Intended jurisdiction for the library | Library.jurisdiction
name | string | Computationally friendly name of the library | Library.name
predecessor | reference | What resource is being referenced | Library.relatedArtifact.where(type='predecessor').resource
publisher | string | Name of the publisher of the library | Library.publisher
status | token | The current status of the library | Library.status
successor | reference | What resource is being referenced | Library.relatedArtifact.where(type='successor').resource
title | string | The human-friendly name of the library | Library.title
topic | token | Topics associated with the module | Library.topic
url | uri | The uri that identifies the library | Library.url
version | token | The business version of the library | Library.version
Linkage
author | reference | Author of the Linkage | Linkage.author
item | reference | Matches on any item in the Linkage | Linkage.item.resource
source | reference | Matches on any item in the Linkage with a type of 'source' | Linkage.item.resource
List
code | token | What the purpose of this list is | List.code
date | date | When the list was prepared | List.date
empty-reason | token | Why list is empty | List.emptyReason
encounter | reference | Context in which list created | List.encounter
identifier | token | Business identifier | List.identifier
item | reference | Actual entry | List.entry.item
notes | string | The annotation - text content | List.note.text
patient | reference | If all resources have the same subject | List.subject
source | reference | Who and/or what defined the list contents (aka Author) | List.source
status | token | current | retired | entered-in-error | List.status
subject | reference | If all resources have the same subject | List.subject
title | string | Descriptive name for the list | List.title
Location
address | string | A (part of the) address of the location | Location.address
address-city | string | A city specified in an address | Location.address.city
address-country | string | A country specified in an address | Location.address.country
address-postalcode | string | A postal code specified in an address | Location.address.postalCode
address-state | string | A state specified in an address | Location.address.state
address-use | token | A use code specified in an address | Location.address.use
endpoint | reference | Technical endpoints providing access to services operated for the location | Location.endpoint
identifier | token | An identifier for the location | Location.identifier
name | string | A portion of the location's name or alias | Location.name | Location.alias

near | token | The coordinates expressed as [latitude]:[longitude] (using the WGS84 datum, see notes) to find locations near to (servers may search using a square rather than a circle for efficiency) Requires the near-distance parameter to be provided also | Location.position
near-distance | quantity | A distance quantity to limit the near search to locations within a specific distance Requires the near parameter to also be included | Location.position
operational-status | token | Searches for locations (typically bed/room) that have an operational status (e.g. contaminated, housekeeping) | Location.operationalStatus
organization | reference | Searches for locations that are managed by the provided organization | Location.managingOrganization
partof | reference | A location of which this location is a part | Location.partOf
status | token | Searches for locations with a specific kind of status | Location.status
type | token | A code for the type of location | Location.type
Measure
composed-of | reference | What resource is being referenced | Measure.relatedArtifact.where(type='composed-of').resource
date | date | The measure publication date | Measure.date
depends-on | reference | What resource is being referenced | Measure.relatedArtifact.where(type='depends-on').resource | Measure.library
derived-from | reference | What resource is being referenced | Measure.relatedArtifact.where(type='derived-from').resource
description | string | The description of the measure | Measure.description
effective | date | The time during which the measure is intended to be in use | Measure.effectivePeriod
identifier | token | External identifier for the measure | Measure.identifier
jurisdiction | token | Intended jurisdiction for the measure | Measure.jurisdiction
name | string | Computationally friendly name of the measure | Measure.name
predecessor | reference | What resource is being referenced | Measure.relatedArtifact.where(type='predecessor').resource
publisher | string | Name of the publisher of the measure | Measure.publisher
status | token | The current status of the measure | Measure.status
successor | reference | What resource is being referenced | Measure.relatedArtifact.where(type='successor').resource
title | string | The human-friendly name of the measure | Measure.title
topic | token | Topics associated with the module | Measure.topic
url | uri | The uri that identifies the measure | Measure.url
version | token | The business version of the measure | Measure.version
MeasureReport
identifier | token | External identifier of the measure report to be returned | MeasureReport.identifier
patient | reference | The identity of a patient to search for individual measure report results for | MeasureReport.patient
status | token | The status of the measure report | MeasureReport.status
Media
based-on | reference | Procedure that caused this media to be created | Media.basedOn
context | reference | Encounter / Episode associated with media | Media.context
created | date | Date attachment was first created | Media.content.creation
date | date | When Media was collected | Media.occurrence
device | reference | Observing Device | Media.device
identifier | token | Identifier(s) for the image | Media.identifier
operator | reference | The person who generated the image | Media.operator
patient | reference | Who/What this Media is a record of | Media.subject
site | token | Body part in media | Media.bodySite
subject | reference | Who/What this Media is a record of | Media.subject
subtype | token | The type of acquisition equipment/process | Media.subtype
type | token | photo | video | audio | Media.type
view | token | Imaging view, e.g. Lateral or Antero-posterior | Media.view
Medication
code | token | Codes that identify this medication | Medication.code
container | token | E.g. box, vial, blister-pack | Medication.package.container
form | token | powder | tablets | capsule + | Medication.form
ingredient | reference | The product contained | Medication.ingredient.item.as(Reference)
ingredient-code | token | The product contained | Medication.ingredient.item.as(CodeableConcept)
manufacturer | reference | Manufacturer of the item | Medication.manufacturer
over-the-counter | token | True if medication does not require a prescription | Medication.isOverTheCounter
package-item | reference | The item in the package | Medication.package.content.item.as(Reference)
package-item-code | token | The item in the package | Medication.package.content.item.as(CodeableConcept)
status | token | active | inactive | entered-in-error | Medication.status
MedicationAdministration
code | token | Return administrations of this medication code | MedicationAdministration.medication.as(CodeableConcept)
context | reference | Return administrations that share this encounter or episode of care | MedicationAdministration.context
device | reference | Return administrations with this administration device identity | MedicationAdministration.device
effective-time | date | Date administration happened (or did not happen) | MedicationAdministration.effective
identifier | token | Return administrations with this external identifier | MedicationAdministration.identifier
medication | reference | Return administrations of this medication resource | MedicationAdministration.medication.as(Reference)
not-given | token | Administrations that were not made | MedicationAdministration.notGiven
patient | reference | The identity of a patient to list administrations for | MedicationAdministration.subject
performer | reference | The identify of the individual who administered the medication | MedicationAdministration.performer.actor
prescription | reference | The identity of a prescription to list administrations from | MedicationAdministration.prescription
reason-given | token | Reasons for administering the medication | MedicationAdministration.reasonCode
reason-not-given | token | Reasons for not administering the medication | MedicationAdministration.reasonNotGiven
status | token | MedicationAdministration event status (for example one of active/paused/completed/nullified) | MedicationAdministration.status
subject | reference | The identify of the individual or group to list administrations for | MedicationAdministration.subject

MedicationDispense
code | token | Return dispenses of this medicine code | MedicationDispense.medication.as(CodeableConcept)
context | reference | Returns dispenses with a specific context (episode or episode of care) | MedicationDispense.context
destination | reference | Return dispenses that should be sent to a specific destination | MedicationDispense.destination
identifier | token | Return dispenses with this external identifier | MedicationDispense.identifier
medication | reference | Return dispenses of this medicine resource | MedicationDispense.medication.as(Reference)
patient | reference | The identity of a patient to list dispenses for | MedicationDispense.subject
performer | reference | Return dispenses performed by a specific individual | MedicationDispense.performer.actor
prescription | reference | The identity of a prescription to list dispenses from | MedicationDispense.authorizingPrescription
receiver | reference | The identity of a receiver to list dispenses for | MedicationDispense.receiver
responsibleparty | reference | Return dispenses with the specified responsible party | MedicationDispense.substitution.responsibleParty
status | token | Return dispenses with a specified dispense status | MedicationDispense.status
subject | reference | The identity of a patient to list dispenses for | MedicationDispense.subject
type | token | Return dispenses of a specific type | MedicationDispense.type
whenhandedover | date | Returns dispenses handed over on this date | MedicationDispense.whenHandedOver
whenprepared | date | Returns dispenses prepared on this date | MedicationDispense.whenPrepared
MedicationRequest
authoredon | date | Return prescriptions written on this date | MedicationRequest.authoredOn
category | token | Returns prescriptions with different categories | MedicationRequest.category
code | token | Return prescriptions of this medication code | MedicationRequest.medication.as(CodeableConcept)
context | reference | Return prescriptions with this encounter or episode of care identifier | MedicationRequest.context
date | date | Multiple Resources: MedicationRequest: Returns medication request to be administered on a specific date | MedicationRequest.dosageInstruction.timing.event
identifier | token | Return prescriptions with this external identifier | MedicationRequest.identifier
intended-dispenser | reference | Returns prescriptions intended to be dispensed by this Organization | MedicationRequest.dispenseRequest.performer
intent | token | Returns prescriptions with different intents | MedicationRequest.intent
medication | reference | Return prescriptions of this medication reference | MedicationRequest.medication.as(Reference)
patient | reference | Returns prescriptions for a specific patient | MedicationRequest.subject
priority | token | Returns prescriptions with different priorities | MedicationRequest.priority
requester | reference | Returns prescriptions prescribed by this prescriber | MedicationRequest.requester.agent
status | token | Status of the prescription | MedicationRequest.status
subject | reference | The identity of a patient to list orders for | MedicationRequest.subject
MedicationStatement
category | token | Returns statements of this category of medicationstatement | MedicationStatement.category
code | token | Return statements of this medication code | MedicationStatement.medication.as(CodeableConcept)
context | reference | Returns statements for a specific context (episode or episode of Care). | MedicationStatement.context
effective | date | Date when patient was taking (or not taking) the medication | MedicationStatement.effective
identifier | token | Return statements with this external identifier | MedicationStatement.identifier
medication | reference | Return statements of this medication reference | MedicationStatement.medication.as(Reference)
part-of | reference | Returns statements that are part of another event. | MedicationStatement.partOf
patient | reference | Returns statements for a specific patient. | MedicationStatement.subject
source | reference | Who or where the information in the statement came from | MedicationStatement.informationSource
status | token | Return statements that match the given status | MedicationStatement.status
subject | reference | The identity of a patient, animal or group to list statements for | MedicationStatement.subject
MessageDefinition
category | token | The behavior associated with the message | MessageDefinition.category
date | date | The message definition publication date | MessageDefinition.date
description | string | The description of the message definition | MessageDefinition.description
event | token | The event that triggers the message | MessageDefinition.event
focus | token | A resource that is a permitted focus of the message | MessageDefinition.focus.code
identifier | token | External identifier for the message definition | MessageDefinition.identifier
jurisdiction | token | Intended jurisdiction for the message definition | MessageDefinition.jurisdiction
name | string | Computationally friendly name of the message definition | MessageDefinition.name
publisher | string | Name of the publisher of the message definition | MessageDefinition.publisher
status | token | The current status of the message definition | MessageDefinition.status
title | string | The human-friendly name of the message definition | MessageDefinition.title
url | uri | The uri that identifies the message definition | MessageDefinition.url
version | token | The business version of the message definition | MessageDefinition.version
MessageHeader
author | reference | The source of the decision | MessageHeader.author
code | token | ok | transient-error | fatal-error | MessageHeader.response.code
destination | string | Name of system | MessageHeader.destination.name
destination-uri | uri | Actual destination address or id | MessageHeader.destination.endpoint
enterer | reference | The source of the data entry | MessageHeader.enterer
event | token | Code for the event this message represents | MessageHeader.event
focus | reference | The actual content of the message | MessageHeader.focus
receiver | reference | Intended "real-world" recipient for the data | MessageHeader.receiver
response-id | token | Id of original message | MessageHeader.response.identifier
responsible | reference | Final responsibility for event | MessageHeader.responsible
sender | reference | Real world sender of the message | MessageHeader.sender
source | string | Name of system | MessageHeader.source.name

source-uri | uri | Actual message source address or id | MessageHeader.source.endpoint
target | reference | Particular delivery destination within the destination | MessageHeader.destination.target
timestamp | date | Time that the message was sent | MessageHeader.timestamp
NamingSystem
contact | string | Name of an individual to contact | NamingSystem.contact.name
date | date | The naming system publication date | NamingSystem.date
description | string | The description of the naming system | NamingSystem.description
id-type | token | oid | uuid | uri | other | NamingSystem.uniqueId.type
jurisdiction | token | Intended jurisdiction for the naming system | NamingSystem.jurisdiction
kind | token | codesystem | identifier | root | NamingSystem.kind
name | string | Computationally friendly name of the naming system | NamingSystem.name
period | date | When is identifier valid? | NamingSystem.uniqueId.period
publisher | string | Name of the publisher of the naming system | NamingSystem.publisher
replaced-by | reference | Use this instead | NamingSystem.replacedBy
responsible | string | Who maintains system namespace? | NamingSystem.responsible
status | token | The current status of the naming system | NamingSystem.status
telecom | token | Contact details for individual or organization | NamingSystem.contact.telecom
type | token | e.g. driver, provider, patient, bank etc. | NamingSystem.type
value | string | The unique identifier | NamingSystem.uniqueId.value
NutritionOrder
additive | token | Type of module component to add to the feeding | NutritionOrder.enteralFormula.additiveType
datetime | date | Return nutrition orders requested on this date | NutritionOrder.dateTime
encounter | reference | Return nutrition orders with this encounter identifier | NutritionOrder.encounter
formula | token | Type of enteral or infant formula | NutritionOrder.enteralFormula.baseFormulaType
identifier | token | Return nutrition orders with this external identifier | NutritionOrder.identifier
oraldiet | token | Type of diet that can be consumed orally (i.e., take via the mouth). | NutritionOrder.oralDiet.type
patient | reference | The identity of the person who requires the diet, formula or nutritional supplement | NutritionOrder.patient
provider | reference | The identify of the provider who placed the nutrition order | NutritionOrder.orderer
status | token | Status of the nutrition order. | NutritionOrder.status
supplement | token | Type of supplement product requested | NutritionOrder.supplement.type
Observation
based-on | reference | Reference to the test or procedure request. | Observation.basedOn
category | token | The classification of the type of observation | Observation.category
code | token | The code of the observation type | Observation.code
code-value-concept | composite | Code and coded value parameter pair | Observation
code-value-date | composite | Code and date/time value parameter pair | Observation
code-value-quantity | composite | Code and quantity value parameter pair | Observation
code-value-string | composite | Code and string value parameter pair | Observation
combo-code | token | The code of the observation type or component type | Observation.code | Observation.component.code
combo-code-value-concept | composite | Code and coded value parameter pair, including in components | Observation | Observation.component
combo-code-value-quantity | composite | Code and quantity value parameter pair, including in components | Observation | Observation.component
combo-data-absent-reason | token | The reason why the expected value in the element Observation.value[x] or Observation.component.value[x] is missing. | Observation.dataAbsentReason | Observation.component.dataAbsentReason
combo-value-concept | token | The value or component value of the observation, if the value is a CodeableConcept | Observation.value.as(CodeableConcept) | Observation.component.value.as(CodeableConcept)
combo-value-quantity | quantity | The value or component value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.value.as(Quantity) | Observation.component.value.as(Quantity)
component-code | token | The component code of the observation type | Observation.component.code
component-code-value-concept | composite | Component code and component coded value parameter pair | Observation.component
component-code-value-quantity | composite | Component code and component quantity value parameter pair | Observation.component
component-data-absent-reason | token | The reason why the expected value in the element Observation.component.value[x] is missing. | Observation.component.dataAbsentReason
component-value-concept | token | The value of the component observation, if the value is a CodeableConcept | Observation.component.value.as(CodeableConcept)
component-value-quantity | quantity | The value of the component observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.component.value.as(Quantity)
context | reference | Healthcare event (Episode-of-care or Encounter) related to the observation | Observation.context
data-absent-reason | token | The reason why the expected value in the element Observation.value[x] is missing. | Observation.dataAbsentReason
date | date | Obtained date/time. If the obtained element is a period, a date that falls in the period | Observation.effective
device | reference | The Device that generated the observation data. | Observation.device
encounter | reference | Encounter related to the observation | Observation.context
identifier | token | The unique id for a particular observation | Observation.identifier
method | token | The method used for the observation | Observation.method
patient | reference | The subject that the observation is about (if patient) | Observation.subject
performer | reference | Who performed the observation | Observation.performer
related | composite | Related Observations - search on related-type and related-target together | Observation.related
related-target | reference | Resource that is related to this one | Observation.related.target
related-type | token | has-member | derived-from | sequel-to | replaces | qualified-by | interfered-by | Observation.related.type
specimen | reference | Specimen used for this observation | Observation.specimen
status | token | The status of the observation | Observation.status
subject | reference | The subject that the observation is about | Observation.subject
value-concept | token | The value of the observation, if the value is a CodeableConcept | Observation.value.as(CodeableConcept)

value-date | date | The value of the observation, if the value is a date or period of time | Observation.value.as(DateTime) | Observation.value.as(Period)
value-quantity | quantity | The value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | Observation.value.as(Quantity)
value-string | string | The value of the observation, if the value is a string, and also searches in CodeableConcept.text | Observation.value.as(String)
OperationDefinition
base | reference | Marks this as a profile of the base | OperationDefinition.base
code | token | Name used to invoke the operation | OperationDefinition.code
date | date | The operation definition publication date | OperationDefinition.date
description | string | The description of the operation definition | OperationDefinition.description
instance | token | Invoke on an instance? | OperationDefinition.instance
jurisdiction | token | Intended jurisdiction for the operation definition | OperationDefinition.jurisdiction
kind | token | operation | query | OperationDefinition.kind
name | string | Computationally friendly name of the operation definition | OperationDefinition.name
param-profile | reference | Profile on the type | OperationDefinition.parameter.profile
publisher | string | Name of the publisher of the operation definition | OperationDefinition.publisher
status | token | The current status of the operation definition | OperationDefinition.status
system | token | Invoke at the system level? | OperationDefinition.system
type | token | Invole at the type level? | OperationDefinition.type
url | uri | The uri that identifies the operation definition | OperationDefinition.url
version | token | The business version of the operation definition | OperationDefinition.version
Organization
active | token | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Organization.active
address | string | A (part of the) address of the organization | Organization.address
address-city | string | A city specified in an address | Organization.address.city
address-country | string | A country specified in an address | Organization.address.country
address-postalcode | string | A postal code specified in an address | Organization.address.postalCode
address-state | string | A state specified in an address | Organization.address.state
address-use | token | A use code specified in an address | Organization.address.use
endpoint | reference | Technical endpoints providing access to services operated for the organization | Organization.endpoint
identifier | token | Any identifier for the organization (not the accreditation issuer's identifier) | Organization.identifier
name | string | A portion of the organization's name or alias | Organization.name | Organization.alias
partof | reference | An organization of which this organization forms a part | Organization.partOf
phonetic | string | A portion of the organization's name using some kind of phonetic matching algorithm | Organization.name
type | token | A code for the type of organization | Organization.type
Patient
active | token | Whether the patient record is active | Patient.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Patient.address
address-city | string | A city specified in an address | Patient.address.city
address-country | string | A country specified in an address | Patient.address.country
address-postalcode | string | A postalCode specified in an address | Patient.address.postalCode
address-state | string | A state specified in an address | Patient.address.state
address-use | token | A use code specified in an address | Patient.address.use
animal-breed | token | The breed for animal patients | Patient.animal.breed
animal-species | token | The species for animal patients | Patient.animal.species
birthdate | date | The patient's date of birth | Patient.birthDate
death-date | date | The date of death has been provided and satisfies this search value | Patient.deceased.as(DateTime)
deceased | token | This patient has been marked as deceased, or as a death date entered | Patient.deceased.exists()
email | token | A value in an email contact | Patient.telecom.where(system='email')
family | string | A portion of the family name of the patient | Patient.name.family
gender | token | Gender of the patient | Patient.gender
general-practitioner | reference | Patient's nominated general practitioner, not the organization that manages the record | Patient.generalPractitioner
given | string | A portion of the given name of the patient | Patient.name.given
identifier | token | A patient identifier | Patient.identifier
language | token | Language code (irrespective of use value) | Patient.communication.language
link | reference | All patients linked to the given patient | Patient.link.other
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Patient.name
organization | reference | The organization at which this person is a patient | Patient.managingOrganization
phone | token | A value in a phone contact | Patient.telecom.where(system='phone')
phonetic | string | A portion of either family or given name using some kind of phonetic matching algorithm | Patient.name
telecom | token | The value in any kind of telecom details of the patient | Patient.telecom
PaymentNotice
created | date | Creation date fro the notice | PaymentNotice.created
identifier | token | The business identifier of the notice | PaymentNotice.identifier
organization | reference | The organization who generated this resource | PaymentNotice.organization
payment-status | token | The type of payment notice | PaymentNotice.paymentStatus
provider | reference | The reference to the provider | PaymentNotice.provider
request | reference | The Claim | PaymentNotice.request
response | reference | The ClaimResponse | PaymentNotice.response
statusdate | date | The date of the payment action | PaymentNotice.statusDate

PaymentReconciliation
created | date | The creation date | PaymentReconciliation.created
disposition | string | The contents of the disposition message | PaymentReconciliation.disposition
identifier | token | The business identifier of the Explanation of Benefit | PaymentReconciliation.identifier
organization | reference | The organization who generated this resource | PaymentReconciliation.organization
outcome | token | The processing outcome | PaymentReconciliation.outcome
request | reference | The reference to the claim | PaymentReconciliation.request
request-organization | reference | The organization who generated this resource | PaymentReconciliation.requestOrganization
request-provider | reference | The reference to the provider who sumbitted the claim | PaymentReconciliation.requestProvider
Person
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Person.address
address-city | string | A city specified in an address | Person.address.city
address-country | string | A country specified in an address | Person.address.country
address-postalcode | string | A postal code specified in an address | Person.address.postalCode
address-state | string | A state specified in an address | Person.address.state
address-use | token | A use code specified in an address | Person.address.use
birthdate | date | The person's date of birth | Person.birthDate
email | token | A value in an email contact | Person.telecom.where(system='email')
gender | token | The gender of the person | Person.gender
identifier | token | A person Identifier | Person.identifier
link | reference | Any link has this Patient, Person, RelatedPerson or Practitioner reference | Person.link.target
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Person.name
organization | reference | The organization at which this person record is being managed | Person.managingOrganization
patient | reference | The Person links to this Patient | Person.link.target
phone | token | A value in a phone contact | Person.telecom.where(system='phone')
phonetic | string | A portion of name using some kind of phonetic matching algorithm | Person.name
practitioner | reference | The Person links to this Practitioner | Person.link.target
relatedperson | reference | The Person links to this RelatedPerson | Person.link.target
telecom | token | The value in any kind of contact | Person.telecom
PlanDefinition
composed-of | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='composed-of').resource
date | date | The plan definition publication date | PlanDefinition.date
depends-on | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='depends-on').resource | PlanDefinition.library
derived-from | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='derived-from').resource
description | string | The description of the plan definition | PlanDefinition.description
effective | date | The time during which the plan definition is intended to be in use | PlanDefinition.effectivePeriod
identifier | token | External identifier for the plan definition | PlanDefinition.identifier
jurisdiction | token | Intended jurisdiction for the plan definition | PlanDefinition.jurisdiction
name | string | Computationally friendly name of the plan definition | PlanDefinition.name
predecessor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | Name of the publisher of the plan definition | PlanDefinition.publisher
status | token | The current status of the plan definition | PlanDefinition.status
successor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='successor').resource
title | string | The human-friendly name of the plan definition | PlanDefinition.title
topic | token | Topics associated with the module | PlanDefinition.topic
url | uri | The uri that identifies the plan definition | PlanDefinition.url
version | token | The business version of the plan definition | PlanDefinition.version
Practitioner
active | token | Whether the practitioner record is active | Practitioner.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Practitioner.address
address-city | string | A city specified in an address | Practitioner.address.city
address-country | string | A country specified in an address | Practitioner.address.country
address-postalcode | string | A postalCode specified in an address | Practitioner.address.postalCode
address-state | string | A state specified in an address | Practitioner.address.state
address-use | token | A use code specified in an address | Practitioner.address.use
communication | token | One of the languages that the practitioner can communicate with | Practitioner.communication
email | token | A value in an email contact | Practitioner.telecom.where(system='email')
family | string | A portion of the family name | Practitioner.name.family
gender | token | Gender of the practitioner | Practitioner.gender
given | string | A portion of the given name | Practitioner.name.given
identifier | token | A practitioner's Identifier | Practitioner.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Practitioner.name
phone | token | A value in a phone contact | Practitioner.telecom.where(system='phone')
phonetic | string | A portion of either family or given name using some kind of phonetic matching algorithm | Practitioner.name
telecom | token | The value in any kind of contact | Practitioner.telecom
PractitionerRole
active | token | Whether this practitioner's record is in active use | PractitionerRole.active
date | date | The period during which the practitioner is authorized to perform in these role(s) | PractitionerRole.period

email | token | A value in an email contact | PractitionerRole.telecom.where(system='email')
endpoint | reference | Technical endpoints providing access to services operated for the practitioner with this role | PractitionerRole.endpoint
identifier | token | A practitioner's Identifier | PractitionerRole.identifier
location | reference | One of the locations at which this practitioner provides care | PractitionerRole.location
organization | reference | The identity of the organization the practitioner represents / acts on behalf of | PractitionerRole.organization
phone | token | A value in a phone contact | PractitionerRole.telecom.where(system='phone')
practitioner | reference | Practitioner that is able to provide the defined services for the organation | PractitionerRole.practitioner
role | token | The practitioner can perform this role at for the organization | PractitionerRole.code
service | reference | The list of healthcare services that this worker provides for this role's Organization/Location(s) | PractitionerRole.healthcareService
specialty | token | The practitioner has this specialty at an organization | PractitionerRole.specialty
telecom | token | The value in any kind of contact | PractitionerRole.telecom
Procedure
based-on | reference | A request for this procedure | Procedure.basedOn
category | token | Classification of the procedure | Procedure.category
code | token | A code to identify a procedure | Procedure.code
context | reference | Encounter or episode associated with the procedure | Procedure.context
date | date | Date/Period the procedure was performed | Procedure.performed
definition | reference | Instantiates protocol or definition | Procedure.definition
encounter | reference | Search by encounter | Procedure.context
identifier | token | A unique identifier for a procedure | Procedure.identifier
location | reference | Where the procedure happened | Procedure.location
part-of | reference | Part of referenced event | Procedure.partOf
patient | reference | Search by subject - a patient | Procedure.subject
performer | reference | The reference to the practitioner | Procedure.performer.actor
status | token | preparation | in-progress | suspended | aborted | completed | entered-in-error | unknown | Procedure.status
subject | reference | Search by subject | Procedure.subject
ProcedureRequest
authored | date | Date request signed | ProcedureRequest.authoredOn
based-on | reference | What request fulfills | ProcedureRequest.basedOn
body-site | token | Where procedure is going to be done | ProcedureRequest.bodySite
code | token | What is being requested/ordered | ProcedureRequest.code
context | reference | Encounter or Episode during which request was created | ProcedureRequest.context
definition | reference | Protocol or definition | ProcedureRequest.definition
encounter | reference | An encounter in which this request is made | ProcedureRequest.context
identifier | token | Identifiers assigned to this order | ProcedureRequest.identifier
intent | token | proposal | plan | order + | ProcedureRequest.intent
occurrence | date | When procedure should occur | ProcedureRequest.occurrence
patient | reference | Search by subject - a patient | ProcedureRequest.subject
performer | reference | Requested perfomer | ProcedureRequest.performer
performer-type | token | Performer role | ProcedureRequest.performerType
priority | token | routine | urgent | asap | stat | ProcedureRequest.priority
replaces | reference | What request replaces | ProcedureRequest.replaces
requester | reference | Individual making the request | ProcedureRequest.requester.agent
requisition | token | Composite Request ID | ProcedureRequest.requisition
specimen | reference | Specimen to be tested | ProcedureRequest.specimen
status | token | draft | active | suspended | completed | entered-in-error | cancelled | ProcedureRequest.status
subject | reference | Search by subject | ProcedureRequest.subject
ProcessRequest
action | token | The action requested by this resource | ProcessRequest.action
identifier | token | The business identifier of the ProcessRequest | ProcessRequest.identifier
organization | reference | The organization who generated this request | ProcessRequest.organization
provider | reference | The provider who regenerated this request | ProcessRequest.provider
ProcessResponse
identifier | token | The business identifier of the Explanation of Benefit | ProcessResponse.identifier
organization | reference | The organization who generated this resource | ProcessResponse.organization
request | reference | The reference to the claim | ProcessResponse.request
request-organization | reference | The Organization who is responsible the request transaction | ProcessResponse.requestOrganization
request-provider | reference | The Provider who is responsible the request transaction | ProcessResponse.requestProvider
Provenance
agent | reference | Who participated | Provenance.agent.who
agent-role | token | What the agents role was | Provenance.agent.role
end | date | End time with inclusive boundary, if not ongoing | Provenance.period.end
entity-id | token | Identity of entity | Provenance.entity.what.as(Identifier)
entity-ref | reference | Identity of entity | Provenance.entity.what.as(Reference)
location | reference | Where the activity occurred, if relevant | Provenance.location
patient | reference | Target Reference(s) (usually version specific) | Provenance.target
recorded | date | When the activity was recorded / updated | Provenance.recorded
signature-type | token | Indication of the reason the entity signed the object(s) | Provenance.signature.type
start | date | Starting time with inclusive boundary | Provenance.period.start
target | reference | Target Reference(s) (usually version specific) | Provenance.target
Questionnaire

code | token | A code that corresponds to one of its items in the questionnaire | Questionnaire.item.code
date | date | The questionnaire publication date | Questionnaire.date
description | string | The description of the questionnaire | Questionnaire.description
effective | date | The time during which the questionnaire is intended to be in use | Questionnaire.effectivePeriod
identifier | token | External identifier for the questionnaire | Questionnaire.identifier
jurisdiction | token | Intended jurisdiction for the questionnaire | Questionnaire.jurisdiction
name | string | Computationally friendly name of the questionnaire | Questionnaire.name
publisher | string | Name of the publisher of the questionnaire | Questionnaire.publisher
status | token | The current status of the questionnaire | Questionnaire.status
title | string | The human-friendly name of the questionnaire | Questionnaire.title
url | uri | The uri that identifies the questionnaire | Questionnaire.url
version | token | The business version of the questionnaire | Questionnaire.version
QuestionnaireResponse
author | reference | The author of the questionnaire response | QuestionnaireResponse.author
authored | date | When the questionnaire response was last changed | QuestionnaireResponse.authored
based-on | reference | Plan/proposal/order fulfilled by this questionnaire response | QuestionnaireResponse.basedOn
context | reference | Encounter or episode associated with the questionnaire response | QuestionnaireResponse.context
identifier | token | The unique identifier for the questionnaire response | QuestionnaireResponse.identifier
parent | reference | Procedure or observation this questionnaire response was performed as a part of | QuestionnaireResponse.parent
patient | reference | The patient that is the subject of the questionnaire response | QuestionnaireResponse.subject
questionnaire | reference | The questionnaire the answers are provided for | QuestionnaireResponse.questionnaire
source | reference | The individual providing the information reflected in the questionnaire respose | QuestionnaireResponse.source
status | token | The status of the questionnaire response | QuestionnaireResponse.status
subject | reference | The subject of the questionnaire response | QuestionnaireResponse.subject
ReferralRequest
authored-on | date | Creation or activation date | ReferralRequest.authoredOn
based-on | reference | Request being fulfilled | ReferralRequest.basedOn
context | reference | Part of encounter or episode of care | ReferralRequest.context
definition | reference | Instantiates protocol or definition | ReferralRequest.definition
encounter | reference | Originating encounter | ReferralRequest.context
group-identifier | token | Part of common request | ReferralRequest.groupIdentifier
identifier | token | Business identifier | ReferralRequest.identifier
intent | token | Proposal, plan or order | ReferralRequest.intent
occurrence-date | date | When the service(s) requested in the referral should occur | ReferralRequest.occurrence
patient | reference | Who the referral is about | ReferralRequest.subject
priority | token | The priority assigned to the referral | ReferralRequest.priority
recipient | reference | The person that the referral was sent to | ReferralRequest.recipient
replaces | reference | Request(s) replaced by this request | ReferralRequest.replaces
requester | reference | Individual making the request | ReferralRequest.requester.agent
service | token | Actions requested as part of the referral | ReferralRequest.serviceRequested
specialty | token | The specialty that the referral is for | ReferralRequest.specialty
status | token | The status of the referral | ReferralRequest.status
subject | reference | Patient referred to care or transfer | ReferralRequest.subject
type | token | The type of the referral | ReferralRequest.type
RelatedPerson
active | token | Indicates if the related person record is active | RelatedPerson.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | RelatedPerson.address
address-city | string | A city specified in an address | RelatedPerson.address.city
address-country | string | A country specified in an address | RelatedPerson.address.country
address-postalcode | string | A postal code specified in an address | RelatedPerson.address.postalCode
address-state | string | A state specified in an address | RelatedPerson.address.state
address-use | token | A use code specified in an address | RelatedPerson.address.use
birthdate | date | The Related Person's date of birth | RelatedPerson.birthDate
email | token | A value in an email contact | RelatedPerson.telecom.where(system='email')
gender | token | Gender of the related person | RelatedPerson.gender
identifier | token | An Identifier of the RelatedPerson | RelatedPerson.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | RelatedPerson.name
patient | reference | The patient this related person is related to | RelatedPerson.patient
phone | token | A value in a phone contact | RelatedPerson.telecom.where(system='phone')
phonetic | string | A portion of name using some kind of phonetic matching algorithm | RelatedPerson.name
telecom | token | The value in any kind of contact | RelatedPerson.telecom
RequestGroup
author | reference | The author of the request group | RequestGroup.author
authored | date | The date the request group was authored | RequestGroup.authoredOn
context | reference | The context the request group applies to | RequestGroup.context
definition | reference | The definition from which the request group is realized | RequestGroup.definition
encounter | reference | The encounter the request group applies to | RequestGroup.context
group-identifier | token | The group identifier for the request group | RequestGroup.groupIdentifier
identifier | token | External identifiers for the request group | RequestGroup.identifier
intent | token | The intent of the request group | RequestGroup.intent

participant | reference | The participant in the requests in the group | RequestGroup.action.participant
patient | reference | The identity of a patient to search for request groups | RequestGroup.subject
priority | token | The priority of the request group | RequestGroup.priority
status | token | The status of the request group | RequestGroup.status
subject | reference | The subject that the request group is about | RequestGroup.subject
ResearchStudy
category | token | Classifications for the study | ResearchStudy.category
date | date | When the study began and ended | ResearchStudy.period
focus | token | Drugs, devices, conditions, etc. under study | ResearchStudy.focus
identifier | token | Business Identifier for study | ResearchStudy.identifier
jurisdiction | token | Geographic region(s) for study | ResearchStudy.jurisdiction
keyword | token | Used to search for the study | ResearchStudy.keyword
partof | reference | Part of larger study | ResearchStudy.partOf
principalinvestigator | reference | The individual responsible for the study | ResearchStudy.principalInvestigator
protocol | reference | Steps followed in executing study | ResearchStudy.protocol
site | reference | Location involved in study execution | ResearchStudy.site
sponsor | reference | Organization responsible for the study | ResearchStudy.sponsor
status | token | draft | in-progress | suspended | stopped | completed | entered-in-error | ResearchStudy.status
title | string | Name for this study | ResearchStudy.title
ResearchSubject
date | date | Start and end of participation | ResearchSubject.period
identifier | token | Business Identifier for research subject | ResearchSubject.identifier
individual | reference | Who is part of study | ResearchSubject.individual
patient | reference | Who is part of study | ResearchSubject.individual
status | token | candidate | enrolled | active | suspended | withdrawn | completed | ResearchSubject.status
RiskAssessment
condition | reference | Condition assessed | RiskAssessment.condition
date | date | When was assessment made? | RiskAssessment.occurrence.as(DateTime)
encounter | reference | Where was assessment performed? | RiskAssessment.context
identifier | token | Unique identifier for the assessment | RiskAssessment.identifier
method | token | Evaluation mechanism | RiskAssessment.method
patient | reference | Who/what does assessment apply to? | RiskAssessment.subject
performer | reference | Who did assessment? | RiskAssessment.performer
probability | number | Likelihood of specified outcome | RiskAssessment.prediction.probability
risk | token | Likelihood of specified outcome as a qualitative value | RiskAssessment.prediction.qualitativeRisk
subject | reference | Who/what does assessment apply to? | RiskAssessment.subject
Schedule
active | token | Is the schedule in active use | Schedule.active
actor | reference | The individual(HealthcareService, Practitioner, Location, ...) to find a Schedule for | Schedule.actor
date | date | Search for Schedule resources that have a period that contains this date specified | Schedule.planningHorizon
identifier | token | A Schedule Identifier | Schedule.identifier
type | token | The type of appointments that can be booked into associated slot(s) | Schedule.serviceType
SearchParameter
base | token | The resource type(s) this search parameter applies to | SearchParameter.base
code | token | Code used in URL | SearchParameter.code
component | reference | Defines how the part works | SearchParameter.component.definition
date | date | The search parameter publication date | SearchParameter.date
derived-from | uri | Original Definition for the search parameter | SearchParameter.derivedFrom
description | string | The description of the search parameter | SearchParameter.description
jurisdiction | token | Intended jurisdiction for the search parameter | SearchParameter.jurisdiction
name | string | Computationally friendly name of the search parameter | SearchParameter.name
publisher | string | Name of the publisher of the search parameter | SearchParameter.publisher
status | token | The current status of the search parameter | SearchParameter.status
target | token | Types of resource (if a resource reference) | SearchParameter.target
type | token | number | date | string | token | reference | composite | quantity | uri | SearchParameter.type
url | uri | The uri that identifies the search parameter | SearchParameter.url
version | token | The business version of the search parameter | SearchParameter.version
Sequence
chromosome | token | Chromosome number of the reference sequence | Sequence.referenceSeq.chromosome
coordinate | composite | Search parameter for region of the reference DNA sequence string. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as coordinate=1$lt345$gt123, this means it will search for the Sequence resource on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | Sequence.variant
end | number | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the reference sequence. | Sequence.referenceSeq.windowEnd
identifier | token | The unique identity for a particular sequence | Sequence.identifier
patient | reference | The subject that the observation is about | Sequence.patient
start | number | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the reference sequence. | Sequence.referenceSeq.windowStart
type | token | Amino Acid Sequence/ DNA Sequence / RNA Sequence | Sequence.type
ServiceDefinition

composed-of | reference | What resource is being referenced | ServiceDefinition.relatedArtifact.where(type='composed-of').resource
date | date | The service definition publication date | ServiceDefinition.date
depends-on | reference | What resource is being referenced | ServiceDefinition.relatedArtifact.where(type='depends-on').resource
derived-from | reference | What resource is being referenced | ServiceDefinition.relatedArtifact.where(type='derived-from').resource
description | string | The description of the service definition | ServiceDefinition.description
effective | date | The time during which the service definition is intended to be in use | ServiceDefinition.effectivePeriod
identifier | token | External identifier for the service definition | ServiceDefinition.identifier
jurisdiction | token | Intended jurisdiction for the service definition | ServiceDefinition.jurisdiction
name | string | Computationally friendly name of the service definition | ServiceDefinition.name
predecessor | reference | What resource is being referenced | ServiceDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | Name of the publisher of the service definition | ServiceDefinition.publisher
status | token | The current status of the service definition | ServiceDefinition.status
successor | reference | What resource is being referenced | ServiceDefinition.relatedArtifact.where(type='successor').resource
title | string | The human-friendly name of the service definition | ServiceDefinition.title
topic | token | Topics associated with the module | ServiceDefinition.topic
url | uri | The uri that identifies the service definition | ServiceDefinition.url
version | token | The business version of the service definition | ServiceDefinition.version
Slot
identifier | token | A Slot Identifier | Slot.identifier
schedule | reference | The Schedule Resource that we are seeking a slot within | Slot.schedule
slot-type | token | The type of appointments that can be booked into the slot | Slot.serviceType
start | date | Appointment date/time. | Slot.start
status | token | The free/busy status of the appointment | Slot.status
Specimen
accession | token | The accession number associated with the specimen | Specimen.accessionIdentifier
bodysite | token | The code for the body site from where the specimen originated | Specimen.collection.bodySite
collected | date | The date the specimen was collected | Specimen.collection.collected
collector | reference | Who collected the specimen | Specimen.collection.collector
container | token | The kind of specimen container | Specimen.container.type
container-id | token | The unique identifier associated with the specimen container | Specimen.container.identifier
identifier | token | The unique identifier associated with the specimen | Specimen.identifier
parent | reference | The parent of the specimen | Specimen.parent
patient | reference | The patient the specimen comes from | Specimen.subject
status | token | available | unavailable | unsatisfactory | entered-in-error | Specimen.status
subject | reference | The subject of the specimen | Specimen.subject
type | token | The specimen type | Specimen.type
StructureDefinition
abstract | token | Whether the structure is abstract | StructureDefinition.abstract
base | uri | Definition that this type is constrained/specialized from | StructureDefinition.baseDefinition
base-path | token | Path that identifies the base element | StructureDefinition.snapshot.element.base.path | StructureDefinition.differential.element.base.path
context-type | token | resource | datatype | extension | StructureDefinition.contextType
date | date | The structure definition publication date | StructureDefinition.date
derivation | token | specialization | constraint - How relates to base definition | StructureDefinition.derivation
description | string | The description of the structure definition | StructureDefinition.description
experimental | token | For testing purposes, not real usage | StructureDefinition.experimental
ext-context | string | Where the extension can be used in instances | StructureDefinition.context
identifier | token | External identifier for the structure definition | StructureDefinition.identifier
jurisdiction | token | Intended jurisdiction for the structure definition | StructureDefinition.jurisdiction
keyword | token | A code for the profile | StructureDefinition.keyword
kind | token | primitive-type | complex-type | resource | logical | StructureDefinition.kind
name | string | Computationally friendly name of the structure definition | StructureDefinition.name
path | token | A path that is constrained in the profile | StructureDefinition.snapshot.element.path | StructureDefinition.differential.element.path
publisher | string | Name of the publisher of the structure definition | StructureDefinition.publisher
status | token | The current status of the structure definition | StructureDefinition.status
title | string | The human-friendly name of the structure definition | StructureDefinition.title
type | token | Type defined or constrained by this structure | StructureDefinition.type
url | uri | The uri that identifies the structure definition | StructureDefinition.url
valueset | reference | A vocabulary binding reference | StructureDefinition.snapshot.element.binding.valueSet
version | token | The business version of the structure definition | StructureDefinition.version
StructureMap
date | date | The structure map publication date | StructureMap.date
description | string | The description of the structure map | StructureMap.description
identifier | token | External identifier for the structure map | StructureMap.identifier
jurisdiction | token | Intended jurisdiction for the structure map | StructureMap.jurisdiction
name | string | Computationally friendly name of the structure map | StructureMap.name
publisher | string | Name of the publisher of the structure map | StructureMap.publisher
status | token | The current status of the structure map | StructureMap.status
title | string | The human-friendly name of the structure map | StructureMap.title
url | uri | The uri that identifies the structure map | StructureMap.url
version | token | The business version of the structure map | StructureMap.version
Subscription

add-tag | token | A tag to be added to the resource matching the criteria | Subscription.tag
contact | token | Contact details for the subscription | Subscription.contact
criteria | string | The search rules used to determine when to send a notification | Subscription.criteria
payload | string | The mime-type of the notification payload | Subscription.channel.payload
status | token | The current state of the subscription | Subscription.status
type | token | The type of channel for the sent notifications | Subscription.channel.type
url | uri | The uri that will receive the notifications | Subscription.channel.endpoint
Substance
category | token | The category of the substance | Substance.category
code | token | The code of the substance or ingredient | Substance.code | Substance.ingredient.substance.as(CodeableConcept)
container-identifier | token | Identifier of the package/container | Substance.instance.identifier
expiry | date | Expiry date of package or container of substance | Substance.instance.expiry
identifier | token | Unique identifier for the substance | Substance.identifier
quantity | quantity | Amount of substance in the package | Substance.instance.quantity
status | token | active | inactive | entered-in-error | Substance.status
substance-reference | reference | A component of the substance | Substance.ingredient.substance.as(Reference)
SupplyDelivery
identifier | token | External identifier | SupplyDelivery.identifier
patient | reference | Patient for whom the item is supplied | SupplyDelivery.patient
receiver | reference | Who collected the Supply | SupplyDelivery.receiver
status | token | in-progress | completed | abandoned | entered-in-error | SupplyDelivery.status
supplier | reference | Dispenser | SupplyDelivery.supplier
SupplyRequest
category | token | The kind of supply (central, non-stock, etc.) | SupplyRequest.category
date | date | When the request was made | SupplyRequest.authoredOn
identifier | token | Unique identifier | SupplyRequest.identifier
requester | reference | Individual making the request | SupplyRequest.requester.agent
status | token | draft | active | suspended + | SupplyRequest.status
supplier | reference | Who is intended to fulfill the request | SupplyRequest.supplier
Task
authored-on | date | Search by creation date | Task.authoredOn
based-on | reference | Search by requests this task is based on | Task.basedOn
business-status | token | Search by business status | Task.businessStatus
code | token | Search by task code | Task.code
context | reference | Search by encounter or episode | Task.context
focus | reference | Search by task focus | Task.focus
group-identifier | token | Search by group identifier | Task.groupIdentifier
identifier | token | Search for a task instance by its business identifier | Task.identifier
intent | token | Search by task intent | Task.intent
modified | date | Search by last modification date | Task.lastModified
organization | reference | Search by responsible organization | Task.requester.onBehalfOf
owner | reference | Search by task owner | Task.owner
part-of | reference | Search by task this task is part of | Task.partOf
patient | reference | Search by patient | Task.for
performer | token | Search by recommended type of performer (e.g., Requester, Performer, Scheduler). | Task.performerType
period | date | Search by period Task is/was underway | Task.executionPeriod
priority | token | Search by task priority | Task.priority
requester | reference | Search by task requester | Task.requester.agent
status | token | Search by task status | Task.status
subject | reference | Search by subject | Task.for
TestReport
identifier | token | An external identifier for the test report | TestReport.identifier
issued | date | The test report generation date | TestReport.issued
participant | uri | The reference to a participant in the test execution | TestReport.participant.uri
result | token | The result disposition of the test execution | TestReport.result
tester | string | The name of the testing organization | TestReport.tester
testscript | reference | The test script executed to produce this report | TestReport.testScript
TestScript
date | date | The test script publication date | TestScript.date
description | string | The description of the test script | TestScript.description
identifier | token | External identifier for the test script | TestScript.identifier
jurisdiction | token | Intended jurisdiction for the test script | TestScript.jurisdiction
name | string | Computationally friendly name of the test script | TestScript.name
publisher | string | Name of the publisher of the test script | TestScript.publisher
status | token | The current status of the test script | TestScript.status
testscript-capability | string | TestScript required and validated capability | TestScript.metadata.capability.description
title | string | The human-friendly name of the test script | TestScript.title
url | uri | The uri that identifies the test script | TestScript.url
version | token | The business version of the test script | TestScript.version
ValueSet
date | date | The value set publication date | ValueSet.date
description | string | The description of the value set | ValueSet.description

expansion | uri | Uniquely identifies this expansion | ValueSet.expansion.identifier
identifier | token | External identifier for the value set | ValueSet.identifier
jurisdiction | token | Intended jurisdiction for the value set | ValueSet.jurisdiction
name | string | Computationally friendly name of the value set | ValueSet.name
publisher | string | Name of the publisher of the value set | ValueSet.publisher
reference | uri | A code system included or excluded in the value set or an imported value set | ValueSet.compose.include.system
status | token | The current status of the value set | ValueSet.status
title | string | The human-friendly name of the value set | ValueSet.title
url | uri | The uri that identifies the value set | ValueSet.url
version | token | The business version of the value set | ValueSet.version
VisionPrescription
datewritten | date | Return prescriptions written on this date | VisionPrescription.dateWritten
encounter | reference | Return prescriptions with this encounter identifier | VisionPrescription.encounter
identifier | token | Return prescriptions with this external identifier | VisionPrescription.identifier
patient | reference | The identity of a patient to list dispenses for | VisionPrescription.patient
prescriber | reference | Who authorizes the vision product | VisionPrescription.prescriber

2.21.4 FHIRPath Patch

URL: https://hl7.org/fhir/STU3/fhirpatch.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.21.4 FHIRPath Patch

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

This page documents how the Parameters resource is used to define a set of FHIRPath based patch operations.

2.21.4.1 Conceptual FHIRPath Operations

This table documents the FHIRPath operations that may be used to specify changes to a resource:

Type | Path | Name | Value | Index | source | destination | Details
add | Path at which to add the content | Name of the property to add | Data type to add at nominated place | | | | The content will be appended to the element identified in the path, using the name specified
insert | Path of the collection in which to insert the content | | value (Data type) to add at nominated place | index at which to insert | | | The content will be inserted into the nominated list at the index specified (0 based). The index must be equal or less than the number of elements in the list
delete | Path of the element to delete | | | | | Only a single element can be deleted
replace | Path of the element to replace | | value (Data type) to replace it with
move | Path of the collection in which to move the content | | | | list index to move from | list index to move to | Move an element within a single list

There are a few base rules that apply for all operations:

* The FHIRPath statement must return elements from the resource being modified. e.g. a FHIRPath could cross resources like Observation.subject.resolve().identifier. Servers SHALL not allow patch operations to alter other resources than the nominated target, and SHOULD return an error if the patch operation tries

* The type of the value must be correct for the place at which it will be added/inserted. Servers SHALL return an error if the type is wrong

* Servers SHALL return an error if the outcome of the patch operation is a not a valid resource

2.21.4.2 Parameters Format

The FHIRPath patch operations are encoded in a Parameters resource as follows:

* Each operation is a Parameter named "operation"

* Each operation has a series of parts, using the following parameter names from the table above with these types:

Parameter | Type
Type | code
Path | string
Name | string
Value | *
index | integer
source | integer
destination | integer

Here is an example of adding an element:

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="operation"/>
    <part>
      <name value="type"/>
      <valueCode value="add"/>
    </part>
    <part>
      <name value="path"/>
      <valueString value="Patient"/>
    </part>
```

```
    <part>
      <name value="name"/>
      <valueString value="birthDate"/>
    </part>
    <part>
      <name value="value"/>
      <valueDate value="1930-01-01"/>
    </part>
  </parameter>
</Parameters>
```

2.21.4.2.1 Anonymous Types

Only some named data types (see the list) are allowed to be used directly in parameters. In order to add or insert other kinds of types - including anonymous elements (e.g. Observation.component, Timing.repeat), the content is defined by defining the name as described above, and instead of providing a value, a set of parts that are values are provided. Here is an example:

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="operation"/>
    <part>
      <name value="type"/>
      <valueCode value="add"/>
    </part>
    <part>
      <name value="path"/>
      <valueString value="Patient"/>
    </part>
    <part>
      <name value="name"/>
      <valueString value="contact"/>
    </part>
    <part>
      <name value="value"/>
      <part>
        <name value="name"/>
        <valueHumanName>
          <text value="a name"/>
        </valueHumanName>
      </part>
    </part>
  </parameter>
</Parameters>
```

This pattern repeats as deep as necessary.

There is a set of test cases for implementers as part of the Test Case Downloads.

2.22 Extended Operations on the RESTful API

URL: https://hl7.org/fhir/STU3/operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.22.0 Extended Operations on the RESTful API

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The RESTful API defines a set of common interactions (read, update, search, etc.) performed on a repository of typed resources. These interactions follow the RESTful paradigm of managing state by Create/Read/Update/Delete actions on a set of identified resources. While this approach solves many use cases, there is some specific functionality that can be met more efficiently using an RPC-like paradigm, where named operations are performed with inputs and outputs (Execute). Operations are used (a) where the server needs to play an active role in formulating the content of the response, not merely return existing information, or (b)where the intended purpose is to cause side effects such as the modification of existing resources, or creation of new resources. This specification describes a lightweight operation framework that seamlessly extends the RESTful API.

Operations have the following general properties:

* Each operation has a name

* Each operation has a list of 'in' and 'out' parameters

* Parameters are either resources, data types, or search parameters

* Operations are subject to the same security constraints and requirements as the RESTful API

* The URIs for the operation end-points are based on the existing RESTful API address scheme

* Operations may make use of the existing repository of resources in their definitions

* Operations may be performed on a specific resource, a resource type, or a whole system

2.22.0.1 Executing an Operation

Operations are executed using a URL derived from the FHIR endpoint, where the name of the operations is prefixed by a "dollar sign" ('$') character. For example:

```
POST http://fhir.someserver.org/fhir/Patient/1/$everything
```

When an operation is idempotent, and the parameters are all primitive data types with no extensions (as is the case with the example above), it may be invoked using GET as well.

Operations can be invoked on four types of FHIR endpoints:

* The "base" FHIR service endpoint (e.g. http://fhir.someserver.org/fhir): These are operations that operate on the full scale of the server. For example, "return me all extensions known by this server"

* A Resource type (e.g. http://fhir.someserver.org/fhir/Patient): These are operations that operate across all instances of a given resource type

* A Resource instance (e.g. http://fhir.someserver.org/fhir/Patient/1): These are operations that involve only a single instance of a Resource, like the $everything operation above does

* A specific version of a resource instance (http://fhir.someserver.org/fhir/Patient/1/_history/4): These operations involve only a specific version of a single instance of a FHIR Resource and exists only to allow manipulation of profile and tag metadata of past versions

The body of the invocation contains a special infrastructure resource called Parameters, which represents a collection of named parameters as <key,value> pairs, where the value may be any primitive or complex datatype or even a full Resource. It may also include strings formatted as search parameter types.

Upon completion, the operation returns another Parameters resource, containing one or more output parameters. This means that a FHIR operation can take a set of zero or more parameters in and return a set of zero or more result parameters out. Both the body of the POST and the returned result are always a Resource.

Some Operations with primitive input types and a single Resource output parameter named 'return' can be invoked using a GET directly, with parameters as HTTP URL parameters. In this case, the response is simply the resource that is the return value, with no Parameters resource. These kinds of usage are discussed further below.

2.22.0.2 FHIR defined Operations

See the list of defined operations.

2.22.0.3 Implementation Defined Operations

Implementations are able to define their own operations in addition to those defined here. Name clashes between operations defined by different implementers can be resolved by the use of the server's Capability Statement.

Also, the definition of these or additional run time operations does not prevent the use of other kinds of operations that are not dependent on and/or not integrated with the RESTful API, provided that their addressing scheme does not clash with the scheme defined here.

2.22.0.4 Defining an Operation

Each Operation is defined by:

* A context for the Operation - system, resource type, or resource instance

* A name for the Operation

* A list of parameters along with their definitions

For each parameter, the following information is needed:

* Name - the name of the parameter. For implementer convenience, the name should be a valid token (see below)

* Use - In | Out | Both

* Type - a data type or a Resource type

* Search Type - for string search parameters, what kind of search parameter they are (& and what kind of modifiers they have)

* Profile - a StructureDefinition that applies additional restrictions about the resource

* Documentation - a description of the parameter's use

* (Optional) Search Type - if the type is a string, and the parameter is being used like a search parameter, which kind of search type applies

Parameters may be nested into multi-part parameters. Each part has the same information as a parameter, except for use, which is taken from the parameter it is part of.

The resource Operation Definition is used to provide a computable definition of the Operation.

2.22.0.5 Extending an Operation

Implementations are able to extend an operation by defining new named parameters. Implementations can publish their own extended definitions using the Operation Definition resource, and this variant definition can use OperationDefinition.base to refer to the underlying definition.

Note that the FHIR specification will never define any parameter names starting with "x-".

2.22.0.6 Executing an Operation Synchronously

Operations are typically executed synchronously: a client sends a request to a server that includes the operation's in parameters and the server replies with the operation's out parameters.

The URL for an operation end-point depends on its context:

* system: the URL is [base]/$[name]

* resource type: the URL is [base]/[type]/$[name]

* resource instance: the URL is [base]/[type]/[id]/$[name]

2.22.0.6.1 Operation Request

An operation is generally invoked by performing an HTTP POST to the operation's end-point. The submitted content is the special Parameters format (the "in" parameters) - a list of named parameters. For an example, see the value set expansion request example. Note that when parameters have a search type, the search modifiers are available, and are used on the parameter name in the Parameters resource (e.g. "code:in").

Note that the same arrangement as for the RESTful interface applies with respect to content types.

If all the parameters for the operation are primitive types, and the operation is idempotent (see HTTP specification definition of idempotent), the operation may be invoked by performing an HTTP GET operation where all of the values of the parameters are appended to the URL in the search portion of the URL (e.g. after the '?' character). Servers SHALL support this method of invocation. E.g.

```
GET [base]/ValueSet/$expand?url=http://hl7.org/fhir/ValueSet/body-sit&filter=abdo
```

When using the HTTP GET operation, if there is a repeating parameter for the extended operation the values for that parameter are repeated by repeating the named parameter. E.g. Observation $stats statistic parameter

```
GET [base]/Observation/$stats?subject=Patient/123&code=55284-
    4&system=http://loinc.org&duration=1&statistic=average&statistic=min&statistic=max&statistic=count
```

If the only parameter to a specific invocation of the operation is a resource, then the operation can also be executed by a POST with that resource as the body of the request.

Servers MAY choose to support submission of the parameters represented in multi-part/form-data format as well, which can be useful when testing an operation using HTML forms.

2.22.0.6.2 Operation Response

If an operation succeeds, an HTTP Status success code is returned. This will usually be a 2xx code, though it may also be a 303 See Other. Other kinds of 3xx codes should be understood to indicate that the operation did not proceed, and the client will need to re-issue the operation if it can perform the redirection (e.g. may get redirected to an authentication step). User agents should note that servers may issue redirects, etc. to authenticate the client in response to an operation request. An HTTP status code of 4xx or 5xx indicates an error, and an OperationOutcome SHOULD be returned with details.

In general, an operation response uses the same Parameters format whether there is only one or there are multiple named out parameters.

If there is only one out parameter, which is a Resource with the parameter name "return" then the parameter format is not used, and the response is simply the resource itself.

The resources that are returned by the operation may be retained and made available in the resource repository on the operation server. In that case, the server will provide the identity of the resource in the returned resources. When resources that are not persisted are returned in the response, they will have no id property.

2.22.0.7 Executing an Operation Asynchronously

STU Note: there is presently no mechanism to execute operations asynchronously in a RESTful manner. However, the messaging page describes a way to execute operations asynchronously using messages.

Provide feedback/discussion here .

2.22.1 Operation Request Example

URL: https://hl7.org/fhir/STU3/op-example-request.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.22.1 Operation Request Example

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

```
POST [base]/ValueSet/$expand HTTP/1.1
Content-Type: application/fhir+xml

<!--
   This is an example of a operation request for a value set
   expansion where the value set is submitted on the fly
-->
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="filter"/>
    <valueString name="abdo"/>
  </parameter>
```

```
  <parameter>
    <name value="valueset"/>
    <resource>
      <ValueSet>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml"><!-- Snipped for brevity --></div>
        </text>
        <identifier value="http://hl7.org/fhir/ValueSet/body-site"/>
        <name value="SNOMED CT Body Structures"/>
        <publisher value="FHIR Project team"/>
        <telecom>
          <system value="url"/>
          <value value="http://hl7.org/fhir"/>
        </telecom>
        <description value="This value set includes all the &quot;Clinical finding&quot; SNOMED CT codes (i.e. codes
         with an is-a relationship with 91723000: Anatomical structure)"/>
        <status value="draft"/>
        <compose>
          <include><!--   all the descendants of clinical finding, not include itself   -->
            <system value="http://snomed.info/sct"/>
            <filter><!--   todo: work this over. what this means is any concepts where they have is-a with 91723000.
               how should this be done?   -->
              <property value="concept"/>
              <op value="is-a"/>
              <value value="91723000"/>
            </filter>
          </include>
        </compose>
      </ValueSet>
    </resource>
  </parameter>
</Parameters>
```

2.23 FHIR Documents

URL: https://hl7.org/fhir/STU3/documents.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 3 | Ballot Status: Trial Use

FHIR resources can be used to build documents that represent a composition: a set of coherent information that is a statement of healthcare information, including clinical observations and services. A document is an immutable set of resources with a fixed presentation that is authored and/or attested by humans, organizations and devices.

Documents built in this fashion may be exchanged between systems and persisted in document storage and management systems, including systems such as IHE XDS.

Applications claiming conformance to this framework claim to be conformant to "FHIR documents" (see Conformance).

FHIR documents may be 'clinical' (focused on patient healthcare information) but may also serve non-clinical purposes (e.g. FHIR Implementation guides, practice guidelines, patient handouts, etc.) HL7 will develop profiles in the future giving additional guidance on appropriate representation of clinical documents in general as well as specific types of clinical documents (e.g. Consolidated CDA).

Note that FHIR defines both this document format and a document reference resource. FHIR documents are for documents that are authored and assembled in FHIR, while the document reference resource is for general references to pre-existing documents.

* Example discharge summary: XML or JSON

2.23.1 Document Content

All documents have the same structure: a Bundle of resources of type "document" that has a Composition resource as the first resource in the bundle, followed by a series of other resources, referenced from the Composition resource, that provide supporting evidence for the document. The bundle gathers all the content of the document into a single XML or JSON document which may be signed and managed as required. The resources include both human readable and computer processable portions. In addition, the bundle may include CSS stylesheets , provenance statements and a signature.

The composition resource is the foundation of the clinical document. It:

* provides identity and its purpose, and sets the context of the document

* carries key information such as the subject and author, and who attests to the document

* divides the document up into a series of sections, each with their own narrative

Any resource referenced directly in the Composition SHALL be included in the bundle when the document is assembled. Specifically, this means the following resource references:

* Composition.subject

* Composition.encounter

* Composition.author

* Composition.attester.party

* Composition.custodian

* Composition.event.detail

* Composition.section.entry

Other resources that these referenced resources refer to may also be included in the bundle if the document construction system chooses to do so. Including these additional resources will make the document bigger, but will save applications from needing to retrieve the linked resources if they need them while processing the document. Thus, whether these linked resources should be included or not depends on the implementation environment.

The document bundle SHALL include only:

* The Composition resource, and any resources directly or indirectly (e.g. recursively) referenced from it

* A Binary resource containing a stylesheet (as described below)

* Provenance Resources that have a target which is a resource included in item #1

There are two key identifiers in the document:

* The document identifier (mandatory). This is found in Bundle.identifier and is globally unique for this instance of the document, and is never re-used, including for other documents derived from the same composition

* The Composition identifier (optional). This is found in Composition.identifier, and is the same for all documents that are derived from this composition

The document has several dates in it:

* The document date (mandatory). This is found in Bundle.meta.lastUpdated and identifies when the document bundle was assembled from the underlying resources

* The Composition date (mandatory). This is found in Composition.date, which is when the author wrote the document logically

* The Attestation dates (optional). This is found in Composition.attester.time and is when the document was witnessed by the attesters. This would usually be at the same time as the composition date or afterwards

* The Composition last modified time (optional). This is found in Composition.meta.lastUpdated for the composition, and is the last date of change of the composition. This must be >= the composition date

Document Bundles may be signed using digital signatures following the rules laid out in the digital signatures page. The signature SHOULD be provided by a listed attester of the document and the signature SHOULD contain a KeyInfo element that contains a KeyName element whose value is a URI that matches the fullUri for the matching attester resource.

Once assembled into a bundle, the document is immutable - its content can never be changed, and the document id can never be reused. Note that the document may be represented in either XML or JSON, and interconverted between these or have its character encoding changed, all the while remaining the same document. However, the directly referenced content within the document and the presentation of the document cannot change substantially (such that it changes the clinical meaning of the content). Any additional documents derived from the same composition SHALL have a different document id.

2.23.1.1 Document Presentation

When the document is presented for human consumption, applications SHOULD present the collated narrative portions in order:

* The subject resource Narrative

* The Composition resource Narrative

* The section.text Narratives

The presentation of the document is called the 'attested content' of the document. Additional resources can be included in the bundle (e.g. resources referenced from the List that represent the section.content SHOULD be in the bundle, and other additional resources they reference can be included), but these (and any narrative) are not attested content. Specifically, the Composition.attester attests to the presented form of the document.

The Composition resource narrative should summarize the important parts of the document header that are required to establish clinical context for the document (other than the subject, which is displayed in its own right). To actually build the combined narrative, simply append all the narrative fragments together.

If the document is presented in a different order from that given above, it may not represent the original attested content. Implementation Guides may restrict document narrative and display behaviour further.

The XML Tools reference implementation includes a XSLT transform that converts an XML document into browser-ready XHTML.

In addition to the basic style rules about Narratives, which must be followed, a document can reference or contain one or more stylesheets that contains additional styles that apply to the collated narrative. This is done by asserting stylesheet links on the feed:

```
<Bundle xmlns="http://hl7.org/fhir">
  <!-- metadata and type -->
  <link>
    <relation value="stylesheet"/>
    <url value="[uri]"/>
```

```
  </link>
</Bundle>
```

The url can be an absolute reference to a CSS stylesheet or a relative reference to a Binary resource that carries a CSS stylesheet. Stylesheet references can only refer to a CSS stylesheet - other forms of stylesheet are not acceptable.

Relative (internal) references SHOULD be used for stylesheets, because the viewer may be unable to resolve external content at the time of viewing, due to technical problems or local policy decisions.

Any stylesheet referenced or used SHALL NOT alter the presentation in such a way that it changes the clinical meaning of the content.

Unless otherwise agreed in local trading partner agreements, applications displaying the collated narrative SHOULD use the stylesheets specified by the document (see security note). Parties entering into a trading agreement to do otherwise should consider the implications this action will have on their long-term scope for document exchange very carefully. If the parties agree to use stylesheets that are not contained in the document, then it may be that they will never be able to share their documents safely in a more general context, such as a regional or national EHR or a global personal health record.

2.23.2 Document Profiles

Document profiles are used to describe documents for a particular purpose. Document profiles may make rules about:

* The content of the Composition resource, including

* The structure of the sections in the composition

* Which resources are to be included in the bundle along with the resources that are directly referenced in the Document resource

Applications should consider publishing Capability Statements that identify particular documents they support. Documents can identify a profile that they conform to by placing a profile identifier in the Bundle.meta.profile element - see Profile Tags for a discussion of the utility of this.

2.23.3 Document Handling Obligations

The authors/constructors and processors of Clinical Documents, whether human or software, have obligations that they must satisfy.

2.23.3.1 Author/Constructor Obligations

A document constructor is an application that creates a document. An author is a human, organization or device that uses the constructor to create a document. Between them, the constructor and the author may create new content resources and/or assemble already existing content resources while performing their tasks. They also have the following responsibilities:

* To assure that the document SHALL contain valid composition that conforms to the rules described here and that only links to other valid resources

* To assure that the content of document SHALL conform to any declared Profiles (see below).

* Ensure that the attesters are properly aware of the presentation of the document to which they are attesting

2.23.3.2 Processor Obligations

A document processor is an application and/or human user that receives documents and extracts data from them, or makes decisions because of them. The documents may be received directly from a document constructor, accessed via a document management system or forwarded by a third party. The document processor is responsible for ensuring that received documents are processed and/or rendered in accordance to this specification. A document processor has the obligation to assure that the following rules are followed:

* When storing/transmitting a document, any method may be used as long as the bundled document can be (re-)assembled with sufficient integrity to validate a digital signature

* When presenting the narrative of the document, the rules described above SHALL be followed

* Resources or data from the document may be extracted for additional uses, but such data is no longer considered to be attested by the document author

* Wherever the data from the document is displayed to a user, there SHOULD always be a way for the user to access a presentation of the original document

In addition to these obligations, document receivers SHOULD carefully track the source of documents for new documents that supersede existing documents, particularly when the documents represent compositions that have been retracted. When documents have been replaced, they SHOULD either withdraw data extracted from superseded documents or warn users when they view the document or data taken from it.

2.23.4 Document End-Points

There are several different RESTful end-points used when working with documents. The use of the various end-points can be best be described by considering the consequences of posting to them:

End-Point | Type of Content | Description

[baseurl]/Bundle | Document Bundle | This works like a normal end-point for managing a type of resource, but it works with whole document bundles - i.e. a read operation returns a bundle, an update gets a bundle and a search returns a bundle of bundles. Note that if documents are POSTed using a create interaction the Bundle.id will change, but the Bundle.identifier will not

[baseurl]/Composition | Composition Resource | The normal end-point for managing composition resources. This can be used while building a document or after breaking a document up into its constituent resources or when using compositions separately from documents

[baseurl]/Binary | Document Bundle | Just store the entire document as a sequence of bytes and return exactly that sequence when requested. There is no way to find content in the /Binary end-point, so usually this would be associated with a Document Reference so that applications can find and process the document, though this is not required

[baseurl] (e.g. a transaction) | Document Bundle | Ignore the fact that the bundle is a document and process all of the resources that it contains as individual resources. Clients SHOULD not expect that a server that receives a document submitted using this method will be able to reassemble the document exactly. (Even if the server can reassemble the document (see below), the result cannot be expected to be in the same order, etc. Thus a document signature will very likely be invalid.)

Note: While these end-points are defined for use with document resources and document bundles, it is not necessary to use them. Documents may be transferred between systems using any method desired. In addition, servers and/or specifications may define additional operations for handling documents beyond the options described above.

2.23.4.1 Generating a Document

A client can ask a server to generate a fully bundled document from a composition resource. For details, see Generate Document Operation.

2.24 Messaging using FHIR Resources

URL: https://hl7.org/fhir/STU3/messaging.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 3 | Ballot Status: Trial Use

FHIR Resources can be used in a traditional messaging context, much like HL7 v2 (see detailed comparison). Applications asserting conformance to this framework claim to be conformant to "FHIR messaging" (see Conformance).

In FHIR messaging, a "request message" is sent from a source application to a destination application when an event happens. Events mostly correspond to things that happen in the real world. The request message consists of a Bundle identified by the type "message", with the first resource in the bundle being a MessageHeader resource. The MessageHeader resource has a code - the message event - that identifies the nature of the request message, and it also carries additional request metadata. The other resources in the bundle depend on the type of the request.

The events supported in FHIR, along with the resources that are included in them, are defined below.

The destination application processes the request and returns one or more response messages which are also a bundle of resources identified by the type "message", with the first resource in each bundle being a MessageHeader resource with a response section that reports the outcome of processing the message and any additional response resources required.

* Example Request Message:

* Example Response Message:

2.24.1 Basic Messaging Assumptions

This specification assumes that content will be delivered from one application to another by some delivery mechanism, and then one or more responses will be returned to the source application. The exact mechanism of transfer is irrelevant to this specification, but may include file transfer, HTTP based transfer, MLLP (HL7 minimal lower layer protocol), MQ series messaging or anything else. The only requirement for the transfer layer is that requests are sent to a known location and responses are returned to the source of the request. This specification considers the source and destination applications as logical entities, and the mapping from logical source and destination to implementation specific addresses is outside the scope of this specification, though this specification does provide a direct delivery mechanism below.

The agreements around the content of the messages and the behavior of the two applications form the "contract" that describes the exchange. The contract will add regional and local agreements to the rules defined in this specification.

This specification ignores the existence of interface engines and message transfer agents that exist between the source and destination. Either they are transparent to the message/transaction content and irrelevant to this specification, or they are actively involved in manipulating the message content (in particular, the source and destination headers are often changed). If these middleware agents are modifying the message content, then they become responsible for honoring the contract that applies (including applicable profiles) in both directions.

A key aspect of a message is the impact of its content:

2.24.1.1 The impact of the content of a message.

Consequence | The message represents/requests a change that should not be processed more than once; e.g., making a booking for an appointment.

Currency | The message represents a response to query for current information. Retrospective processing is wrong and/or wasteful.

Notification | The content is not necessarily intended to be current, and it can be reprocessed, though there may be version issues created by processing old notifications.

Some Events defined by FHIR are assigned to one of these categories, but others are not able to be categorized in advance, and the category must be determined by the content, or the context.

2.24.1.2 Message Exchange Patterns

Each FHIR request message has one or more response messages. There must be at least one response message so that the sender can know that the message was properly received. Multiple response messages SHALL NOT be returned for messages of consequence, and SHOULD not be returned for notifications.

In principle, source applications are not required to wait for a response to a transaction before issuing a new transaction. However, in many cases, the messages in a given stream are dependent on each other, and must be sent and processed in order. In addition, some transfer methods may require sequential delivery of messages.

For this reason, a synchronous exchange pattern - where the sender sends a message, and waits on the same channel for a single response, and then sends the next message - is the easiest to understand and manage. The $process-message operation described below works in this fashion.

However synchronous message exchange does not cater for multiple response messages, which may arise when processing queries, and also imposes through-put limitations which may become relevant at high volumes. Additionally, it may not be practical or appropriate to wait for response messages. In these cases, the asynchronous message pattern described below should be used.

2.24.1.3 MessageHeader Identifiers

An incoming message contains two identifiers: the Bundle.id and the MessageHeader.id. Each time a new message is created, it SHALL be assigned an identifier (MessageHeader.id) that is unique within that message stream. Note that since message streams are often merged with other streams, it is recommended that the identifier should be globally unique. This can be achieved by using a UUID or an OID. Each time a message is sent, the Bundle.id should be changed to a new value.

When a receiver receives and processes the message, it responds with a new message with a new identifier, wrapped in a bundle which also has a new id. The response message also quotes the request MessageHeader.id in MessageHeader.response.identifier so that the source system can relate the response to its request.

2.24.1.4 Absence of Reliable Messaging

Some of the message delivery mechanisms mentioned above are reliable delivery systems - the message is always delivered, or an appropriate error is returned to the source. However most implementations use methods which do not provide reliable messaging, and either the request or the response can get lost in transit. FHIR messaging describes a simple approach that receivers SHOULD conform to in order to handle the absence of reliable messaging that maintains predictable functionality.

If the sender of the message implements reliable messaging, it SHALL do the following when it receives no response to a message within a configured timeout period based on the value specified in the CapabilityStatement messaging.event.category for the event associated with the message:

Consequence | Resend the same message (with the same MessageHeader.id) with the same Bundle.id
Currency | Resend the same message (with the same MessageHeader.id) with a different Bundle.id
Notification | Resend the same message (with the same MessageHeader.id) with a different Bundle.id

When a receiver implements reliable messaging, it SHALL check the incoming Bundle.id and MessageHeader.id against a cache of previously received messages. The correct action to take depends on what is received:

Both the Bundle.id and MessageHeader.id have not been received | This is the normal case, and the message should be processed
Both envelope and message already received | The original response has been lost (failed to return to the request issuer), and the original response SHALL be resent
The MessageHeader.id has already been received, but the Bundle.id is new | A previously seen message has been resubmitted for processing again. The server may either reprocess the message, or reject the message
The Bundle.id has already been received, but the MessageHeader.id is new | This is an error - Bundle.id values should never be reused

The duration period for caching does generally not need to be very long. At a minimum, it could be 1 minute longer than the timeout of the sending system, though it may need to be longer depending on the re-sending policies of the sending system.

Applications that implement reliable messaging declare their reliable cache period in their Capability Statement.

2.24.1.4.1 Example: Consequence

In the first example, a Clinical EHR issues an order for a particular imaging examination to be performed on a patient. This is considered to be a message of Consequence: multiple orders should not be created (in practice there are usually human review processes that catch multiple orders, but repeat orders create entropy in the system that is harmful). The EHR sends a message where the Bundle.id is UUID 1 (72edc4e0-6708-42ab-9734-f56721882c10), with a MessageHeader.id of UUID 2 (dad53a57-dcb4-4f18-b066-7239eb4b5229).

The EHR system never receives a response to the message; it does not know whether the request message got lost, or the imaging management systems was unable to process the request, or whether it successfully processed the message and the response was lost. In this case, the EHR system resends the message with same two identifiers.

In this case, the imaging system successfully received the message, and processed it. Because it receives the resent order after 1 minute (which is within its 15 minute cache time), and the two UUIDs 1 and 2 match a message it has already processed, it knows that it already processed the order, and simply returns the previous response. In the case of additional resent queries, the application keeps sending the original response, though it may also alert system administrators that the same original message keeps being resent, since lost messages should be a rare occurrence.

When the EHR system finally receives the message, it knows how the imaging management system responded; it can be sure because the message id from the original request is echoed in the response portion of the returned message.

2.24.1.4.2 Example: Currency

In this second example, a Clinical EHR needs to know what appointment slots are available for a particular imaging procedure. This is a message of Currency: available slots are ever disappearing, and ordering a slot that has become unavailable is a waste of time for the humans and systems involved. The EHR sends a message where the Bundle.id is UUID 3 (4c7f5cb2-5964-4d42-b719-e0227461818c), with a MessageHeader.id is UUID 4 (63ed7d68-b2cc-421d-ba1c-a6c7785581f2).

The EHR system never receives a response to the message; it does not know whether the request message got lost, or the imaging management systems was unable to process the request, or whether it successfully processed the message and the response was lost. In this case, the EHR system resends the message with same MessageHeader.id (UUID 4), but creates a new Bundle.id (c7c17fe4-9560-49c7-b2ae-42636476fb86).

In this case, the imaging system successfully received the message, and processed it. When it receives the resent order after 1 minute (which is within its 15 minute cache time), it sees that although the message id is the same, the Bundle.id has changed, and it reprocesses the message again, and sends a new response.

When the EHR system finally receives the message, it knows the current slot availability on the imaging management system responded.

Note that the existence of active intermediaries (or "middleware") creates the need for this protocol - the original sender matches the response to the request based on the MessageHeader.id, and so an active intermediary that choose the re-initiate a query that it previously relayed cannot change the MessageHeader.id. This protocol avoids the need for the MessageHeader.id to change, and only requires change to the Bundle.id which is never the basis for context linking outside the immediate message exchange protocol described here.

2.24.2 Capability Statement

Applications may only assert conformance to "FHIR messaging" if they publish a Capability statement so the claim may be verified. A Capability statement lists all the message events supported (either as sender or receiver) and for each event, a profile that states which resources are bundled (sender), or are required to be bundled (receiver), and any rules about the information content of the individual resources.

2.24.3 $process-message

The simplest way to handle messages where there are also RESTful interactions occurring is to use the $process-message. This operation accepts a message, processes it according to the definition of the event in the message header, and returns a one or more response messages. For example, in addition to processing the message event, a server may choose to retain all or some the resources and make them available on a RESTful interface, but is not required to do so.

When processing messages, a server may return one of several status codes:

* 200 OK: Indicates that the message has been fully processed. If an application-level response is expected for the submitted message, that response SHALL be returned as the body of the 200 response.

* 202 Accepted: Indicates that the receiving system has accepted custody of the message

* 204 No Content: Indicates that the message has been fully processed and would normally have had an application-level response, but because of instructions from the sender (e.g. the messageheader-response-request extension), no response is being provided

* 300+: Indicates that the message was not successfully processed. The server MAY return an OperationOutcome with additional information, and SHOULD do so if the response code is 400 or greater. The client SHALL interpret a 4xx response to indicate that there is no point resubmitting the unaltered message, and a 5xx response to indicate an unexpected error occurred on the part of the server, with the implication that it may be appropriate to resubmit the original message. Doing so SHOULD NOT result in a duplicate message response. Repeated failures indicate either a fatal problem with the submission or a problem with the receiving application.

The following rules apply when using $process-message:

* The operation only accepts POST transactions - any other HTTP method will result in an HTTP error

* The request content type submitted is always a Bundle with type "message" containing a Message Header resource as the first resource

* The response content type returned is always a Bundle with type "message" containing a Message Header resource as the first resource, or an HTTP error

* If the response is an error, the body SHOULD be an Errors & Warning resource with full details

* The mailbox may be authenticated using standard HTTP authentication methods, including OAuth

The $process-message operation can be used by any HTTP end-point that accepts FHIR messages, not just FHIR RESTful servers.

In order to ensure consistency of processing, the logical rules regarding processing of Bundle.id and message id described above SHALL be followed when messages are processed using this operation.

The $process-message operation may be used synchronously, or asynchronously.

2.24.3.1 Synchronous Operation

Synchronous messaging is the easiest to understand; the sender sends a message to the receiver (the server), the server processes it, and then returns a response. Usually (though not always) the sender waits for the response to the current message before sending the next message. This kind of messaging exchange is the most common because it's the simplest to understand.

The following rules apply when using the $process-message operation synchronously:

* The URL (http://server/base/$process-message) has no parameters

* It is an error if the sender POSTs a message that requires multiple response messages

* Servers SHALL accept multiple concurrent message submissions and process them correctly (they are allowed to process them sequentially internally, but multiple concurrent submissions is not an error in its own right)

2.24.3.2 Asynchronous Operation

In Asynchronous messaging, the server acknowledges receipt of the message immediately, and responds to the sender separately. The server may respond more than once to any given message.

The following rules apply when using the $process-message operation synchronously:

* The URL has at least one parameter: http://server/base/$process-message?async=true

* The server acknowledges the message with a 200 OK with no body, or returns an HTTP error if the message cannot be processed. An OperationOutcome SHOULD be returned in such a case

* Accepting the message means that the server has understood the message enough to know where to respond

* By default, the server responds by invoking the $process-message using the sender's stated end-point in the message: POST [MessageHeader.source.endpoint]/$process-messages]

* Since the source end-point may be manipulated by message transfer engines, an alternative response address may be specified using the parameter "response-url": http://server/base/$process-message?async=true&response-url=http://server2.com/base/anything. The endpoint at the specified URL SHALL implement the signature of the $process-message operation (parameter async=true, accept a Bundle, return a 200 OK or an error)

* The server submits response messages to the appropriate end-point with the parameter async=true. There is no response message for the response messages

When a message is received, a receiver can determine from the content of the message header whether it's a new message to process, or a response to a message that has already been sent. Note that asynchronous messaging is less reliable than synchronous messaging; more can go wrong. This specification does not dictate any particular error handling protocols or responsibilities; these are left to trading partner agreements between implementers.

2.24.4 Relationship between Messaging and REST

As well as this messaging framework documented here, FHIR also defines a RESTful API. The messaging and RESTful frameworks are related in that both share the same set of resources on which they operate. In fact, the basic MessageHeader resource that the messaging framework is implemented is itself a resource that can treated in a RESTful approach.

The kinds of functionality that the RESTful API and the messaging framework offer are very similar; their primary difference is architectural in nature.

For instance, the messaging framework defines an event for notifying that a administration resource has been created or updated; the REST API offers similar services (history and Subscription). On the other hand, there are differences in the capabilities offered - while a patient merge can be implemented as a series of RESTful operations performed by the client that update all resources linked to the patient, when a message command to merge patient records is processed, the server will do all the work, and is also able to merge in areas not exposed on the RESTful API. The REST API, however, provides a set of basic operations on all resources that would need special definitions in the messaging framework - definitions that are not provided.

There is no expectation that RESTful systems will need to offer messaging support, or vice versa, though systems may find it useful to support both sets of functionality in order to satisfy a wider range of implementers.

As a resource that can be used with the RESTful framework, the MessageHeader resource has the normal resource end-point (/MessageHeader), which is used to manage a set of static message resources. This could be used to make an archive of past messages available. Creating or updating MessageHeader resources in this fashion does not represent the actual occurrence of any event, nor can it trigger any logic associated with the actual event. It is just for managing a set of message header resources.

2.24.4.1 Asynchronous Messaging using the RESTful API

It is possible to exchange messages using the RESTful end-point as a central point of exchange. This is not particularly efficient compared to other methods, but is useful for low-volume asynchronous exchange.

To send a message, a sender posts the message bundle to the /Bundle end-point, with a uri that identifies the receiver at MessageHeader.destination.endpoint. The RESTful server accepts the bundle, stores it as a single bundle, and indexes it on the MessageHeader.

To receive messages, a receiver searches for all messages destined for itself, since its last check:

```
GET [base]/Bundle?message.destination-uri=[rcv]&_lastUpdated=>2015-03-01T02:00:02+01:00
```

The receiver works through the response, processing each message. As each message is processed, the receiver creates a response message, reversing the source and destination, and posts it back to the server.

To check for responses, the original sender searches for response messages destined for itself, since its last check:

```
GET [base]/Bundle?message.destination-uri=[snd]&message.response-id:missing=false
     &_lastUpdated=>2015-03-03T06:03:522+01:00
```

This lightweight protocol needs ongoing administration to ensure that multiple parties do not interfere with each other by re-using the same system identifier (and against malicious attack).

2.24.5 Event List

The message.code element carries a Coding that identifies the event that the message conveys. This table lists the message event codes defined in this specification (the system value for these is "http://hl7.org/fhir/message-events"):

Code | Category | Description | Request Resources | Response Resources | Notes
CodeSystem-expand | Currency | The definition of a code system is used to create a simple collection of codes suitable for use for data entry or validation. An expanded code system will be returned, or an error message. | CodeSystem | CodeSystem
MedicationAdministration-Complete | Consequence | Change the status of a Medication Administration to show that it is complete. | MedicationAdministration | MedicationAdministration
MedicationAdministration-Nullification | Consequence | Someone wishes to record that the record of administration of a medication is in error and should be ignored. | MedicationAdministration | MedicationAdministration
MedicationAdministration-Recording | Consequence | Indicates that a medication has been recorded against the patient's record. | MedicationAdministration | MedicationAdministration
MedicationAdministration-Update | Consequence | Update a Medication Administration record. | MedicationAdministration | MedicationAdministration
admin-notify | Notification of a change to an administrative resource (either create or update). Note that there is no delete, though some administrative resources have status or period elements for this use. | Device | --
Device | --
Group | --
Location | --
-- | Organization | (see Patient)
Patient | --
Practitioner | --
Practitioner | --
-- | RelatedPerson
Person | --
communication-request | Notification | Notification to convey information. | CommunicationRequest | --
diagnosticreport-provide | Notification | Provide a diagnostic report, or update a previously provided diagnostic report. | DiagnosticReport DiagnosticReport.patient .perfomer .results.specimen .results.result .image | --
observation-provide | Provide a simple observation or update a previously provided simple observation. | Observation Observation.subjectPatient .subjectPatient.person .subjectGroup .subjectDevice .subjectAnimal .performerAgent .performerAgent.person .performerPatient .performerPerson | --
Observation Observation.subjectPatient .subjectPatient.person .subjectGroup .subjectDevice .subjectAnimal .performerAgent .performerAgent.person .performerPatient .performerPerson | --
Observation Observation.subjectPatient .subjectPatient.person .subjectGroup .subjectDevice .subjectAnimal .performerAgent .performerAgent.person .performerPatient .performerPerson | --
patient-link | Notification | Notification that two patient records actually identify the same patient. | Patient,Patient | -- | Follow ups: patient-unlink?
patient-unlink | Notification | Notification that previous advice that two patient records concern the same patient is now considered incorrect. | Patient,Patient | --
valueset-expand | Currency | The definition of a value set is used to create a simple collection of codes suitable for use for data entry or validation. An expanded value set will be returned, or an error message. | ValueSet | ValueSet

The request and response details: The column values are either a resource that is included as part of the response, or an element that refers to another resource, which means that the target of these references SHALL also be in the message. In this table, the request and response columns list the focus resource for the event, along with other resources that should also be carried in the message directly (if they exist).

Additional events may be defined using the MessageDefinition resource.

2.24.6 Invoking Operations via Messages

A message can be used to invoke an operation as defined for a RESTful interface using an operation definition. To invoke an operation using a message:

* The requester sends a message (a bundle with type = message, and a message header resource)

* The message header has an event.system of urn:ietf:rfc:3986

* The event.code is the URL from the operation definition OperationDefinition.url

* The MessageHeader.data refers to a Parameters resource

* The parameters resource is populated appropriately as specified by the nominated operation definition

The recipient executes the operation as specified, and then:

* The receiver sends a message (a bundle with type = message, and a message header resource)

* The message header has the same event as the original message

* The MessageHeader contains a response that refers to the original request message, and a code for the outcome, with details if the operation failed

* The MessageHeader.data refers to a Parameters resource

* The parameters resource is populated appropriately as specified for the response by the nominated operation definition

* If the operation definition specifies a single return, then this is returned as the target of the MesssageHeader.data directly

Here's an example:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="urn:uuid:77831928-2a35-4c08-9496-8232323bf48c"/>
  <!-- normal bundle stuff -->
  <entry>
    <fullUrl value="urn:uuid:6080d4a7-5e05-45dc-96d5-f75329564d1f"/>
    <resource>
      <MessageHeader>
        <id value="cac8143e-6138-4f45-b086-bb8ebf976aae">
        <!-- normal message header stuff -->
        <event>
          <system value="urn:ietf:rfc:3986"/>
          <!-- value set expansion -->
          <code value="http://hl7.org/fhir/OperationDefinition/ValueSet-expand"/>
        </event>
        <!-- more normal message header stuff -->
        <data>
          <reference value="urn:uuid:00213637-dc7c-40d2-a7de-f4ef1eea5685"/>
        </data>
      </MessageHeader>
    </resource>
  </entry>
  <entry>
    <fullUrl value="urn:uuid:00213637-dc7c-40d2-a7de-f4ef1eea5685"/>
    <resource>
      <Parameters>
        <parameter>
          <name value="identifier"/>
          <valueUri value="http://hl7.org/fhir/ValueSet/identifier-type"/>
        </parameter>
      </Parameters>
    </resource>
  </entry>
</Bundle>
```

Note that there's no way to anchor the execution of the operation against a URL. The only operations that can be executed in this way are defined to be executed at the System or Resource level for a particular resource.

2.24.6.1 Invoking Search via Messages

In the same way that a defined operation can be invoked, a regular search operation can be invoked. This also uses the Parameters resource, with the following rules:

* The event code is "search-type" or "search-system" in the system http://hl7.org/fhir/restful-interaction

* If the event type is "search-type" there SHALL be a parameter "resourceType" with specifies the type of resource being searched

* The search parameters are converted to FHIR data types according to the following table

Search Parameter Type	Data Type
number	integer
date	dateTime
string	string
token	string or Coding (split the system and code apart)
reference	uri
composite	string
quantity	string or Quantity (split the syntax out)
uri	uri

Here's an example:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="urn:uuid:77831928-2a35-4c08-9496-8232323bf48c"/>
  <!-- normal bundle stuff -->
  <entry>
    <fullUrl value="urn:uuid:c466754c-09c0-4f59-9f76-a48bd0ea27c9"/>
    <resource>
      <MessageHeader>
        <!-- normal message header stuff -->
        <event>
          <system value="http://hl7.org/fhir/restful-interaction"/>
          <!-- Search against Patient -->
          <code value="search-type"/>
        </event>
        <!-- more normal message header stuff -->
        <data>
          <reference value="urn:uuid:59a17a19-46eb-42d9-821a-f93a0c530cac"/>
        </data>
      </MessageHeader>
    </resource>
  </entry>
  <entry>
```

```
    <fullUrl value="urn:uuid:59a17a19-46eb-42d9-821a-f93a0c530cac"/>
    <resource>
      <Parameters>
        <parameter>
          <name value="resourceType"/>
          <valueString value="Patient"/>
        </parameter>
        <parameter>
          <name value="gender"/>
          <valueString value="m"/>
        </parameter>
      </Parameters>
    </resource>
  </entry>
</Bundle>
```

2.25 Using Resources with Services and Service-oriented Architecture

URL: https://hl7.org/fhir/STU3/services.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.25 Using Resources with Services and Service-oriented Architecture

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

Though FHIR resources are primarily designed for the RESTful HTTP-based implementation, it is not necessary to use a RESTful interface when exchanging the resources; one way that they can be exchanged is in the context of services. Service Oriented Architecture (SOA) is an architecture pattern using services to encapsulate and provide discrete pieces of application functionality to each other. Services communicate by invoking public interfaces and exchanging information (as parameters and outputs) in accordance with a well-defined service contract.

Generally speaking, services are cohesive sets of functions that maintain responsibility for both data and "state" for the scope of their responsibility. Services have a unity of function, such as Terminology Management, Identity Management, or so on, working with other services in collaboration as part of an orchestrated workflow. Data within a service typically follows a "black box" model and is not classically exposed, instead using pre-defined interfaces to make specific behavioral requests for the service to fill.

To relate this to FHIR, resources in a service-oriented implementation can serve two discrete roles: either as a payload parameter, specifying information flowing into or out of a service; or in a behavioral sense, using FHIR APIs as the invocation mechanism for engaging a service. Service APIs are often realized using either RESTful or SOAP-based interfaces.

Of note, there are potential benefits in considering usage of FHIR and SOA together. FHIR allows for ease of implementation and ready access to data payload in an open way. SOA has maturity around transactional integrity across distributed systems instances, providing a framework for loose-coupling and addressing pre-conditions, exception handling, and other implementation considerations of complex distributed systems.

This section illustrates different contexts to allow implementers to make informed decisions at implementation time, considering potential impacts and solution patterns to benefit specific situations. In doing so, this section will clarify what is meant to apply SOA principles and specifications within a FHIR environment/implementation, and the converse. It will identify the situational factors to consider, and the elements of the SOA discipline that can be applied for more effective FHIR implementation, providing a roadmap on how to optimize your implementation toolkit to make you and your team most effective.

This section is being included as FHIR offers a highly flexible framework for interoperability that can be deployed that can be used in a variety of different ways, and in ways that have the potential to hinder interoperability if not done thoughtfully or consistently. Use of SOA patterns and approaches can mitigate this risk, limiting the potential liability resulting from the extreme flexibility of FHIR and fostering an overall framework that leads to deployment and consistency. SOA provides guidance for how components interact, how to partition responsibilities, and how to manage workflows among different parts of systems, all of which have potential utility in FHIR implementation settings.

2.25.1 Service References and Patterns

Generally, the definition of particular services is a domain or context specific task, and it is anticipated that this would be done as separate specifications that make use of the underlying facilities defined in this specification. HL7 has already defined a portfolio of services that take advantage of REST- and SOAP-based interfaces, but the parameters passed to and from those services are not at present resource-based. These are potential extension points that may lend well for FHIR implementation. Services defined like this are able to build on the common underlying platform features defined in this specification such as REST or messaging, and add specific interactions where appropriate. Alternatively, the service interfaces can build an entirely separate implementation.

Note that for each of the items below, there are paired sets of standards, an HL7 Standard defining the functional specification, and an OMG Technical Standard containing SOAP bindings (for all of the items below) and REST bindings (for about half of the items below).

The portfolio of healthcare SOA service specifications includes:

* IXS (Identification and Cross-Reference Services) that identifies the characteristic of a generic identification management services (www.omg.org/spec/IXS)

* RLUS (Retrieve, Locate, Update Service) that supports the generic management resource in a SOA environment. (www.omg.org/spec/RLUS)

* ServD (Service directory) that supports the registration and discovery of health and human-related services (www.omg.org/spec/ServD)

* CDSS (Clinical Decision Support Service) that defines a standard access interface to a clinical decision support system (www.omg.org/spec/CDSS)

* CTS2 (Common Terminology Service V 2) that defines the capabilities, responsibilities, inputs, outputs, and expected behavior to support the management, maintenance, and interaction with ontologies and medical vocabulary systems (www.omg.org/spec/CTS2)

* CCS (Care Coordination Service) that provides the capabilities to support the coordination of patient care across the care continuum spanning multiple organization [emerging specification]

* OS (Order Service) that provides functional capabilities for ordering pharmacy, laboratory, radiology, consult and nutritional services individually or part of an order set [emerging specification].

* EPSS (Event Publication & Subscription Service) that provides a Service Functional Model (SFM) for services, components and systems to subscribe to clinical events of interest and receive notice when new data are available [emerging specification].

* UCS (Unified Communications Service) facilitates the bi-directional communication between participants (both SOA services and human participants) that is so essential for the efficient delivery and coordination of patient care [emerging specification].

Note: FHIR-enablement of the services above is already occurring on a limited basis. The Clinical Quality Improvement Framework (CQIF) portion of the FHIR specification is an example of a technical interface and resources that have been influenced by prior SOA work. Similarly, this specification defines a Terminology Service which is tightly integrated with the RESTful API (note: this terminology service has different goals than the CTS2 service, so they are not functionally equivalent). FHIR itself is a framework for the instantiation of an RLUS service. Definition of other additional services based on the service interface definitions provided by the HL7 SOA work group will be considered if there is sufficient interest in this.

2.25.2 Implementation Approaches

Given that there are a set of available services that can be applied directly or used as a pattern in conjunction with FHIR, it merits consideration that there are alternative ways to bring together FHIR and SOA, each of which has benefits and drawbacks. Three specific styles of implementation have been identified:

* FHIR + REST ("RESTful FHIR"), the predominant approach for implementing FHIR today

* FHIR + WS* represents FHIR implementations using the web services stack as the communication protocol instead of REST. This would include use of FHIR resources as payload parameters in SOAP calls, for example

* FHIR + SOA Pattern illustrates the impact of applying interaction patterns, exception handing and role definition, guided by SOA practices, atop an implementation technology (SOAP, REST, or others)

Determining which of the above implementation styles is best suited to a particular situation is context-dependent. The following table presents these alternatives, deliberately using qualitative measures to help navigate to the best-fit based upon influencing factors. "Harvey Balls" in the table below indicate relative strengths and weaknesses of the implementation alternatives.

Solution Quality	Short Description	#1:FHIR + REST (RESTful FHIR)	#2 FHIR + WS*	#3 FHIR + SOA Pattern	Comments
Support of transactional integrity	Transactions may involve multiple steps composed together into a single unit	?	?	*	Transactional integrity is not assured in #1 or #2 but can be hand-coded. #3 assures that this is addressed
Support of stateless transactions	Transactions are independent and do not require supporting context	*	*	*	Each of the implementation approaches are capable of supporting stateless processing
Support of "loose coupling"	Minimum of dependencies between the calling and responding system; reduces burden on client to manage context of interaction with service	?	?	*	The notion of "loose coupling" is fundamental to SOA and inherent in that architecture. Loose coupling can be supported by FHIR, but it does not innately do so
Use of FHIR Resources as Data Payload	Representation of FHIR resources for use as query or return parameters	*	*	*	Data payload includes input data as well as data returned
Support of Dynamic Service Discovery	Provides for service registration, search, discovery, and late binding	?	?	*	This is part of SOA specification and customary in implementations. For the others, it is up to the approach
Provides resource-oriented operations	Supports ability to create, read, update, delete resources;	*	?	?	Direct access to resources/data revisions in a strength of REST and FHIR. Direct access to fine-grained transactional operations are generally not supported within SOA
Suitability for atomic transactions	Fitness to support fine-grained transactions, such as data access or targeted update	*	?	?	REST is ideally suited for point access or updates of specific data elements/resources - functions discouraged within SOA and WS*
Suitability for composite transactions	Fitness to support complex transactions: context-sensitivity, multi-step workflows, etc.	?	*	*	SOA provides for complex event processing, multi-step sequencing, orchestration

2.25.3 Architectural Considerations

FHIR resources are designed to be used in a wide variety of contexts. In particular, FHIR resources are required to be suitable for use in a REST environment. This means that there are number of design requirements and choices that impact how suitable resources are for use with services.

If service use was the only consideration, different decisions would be made, and resources would be more suitable for use with services. However, this would curtail their usefulness and reusability in other contexts.

API/Interface Design. Services' API design is explicit and exposes service capabilities which are a part of a service contract to consuming applications. This contrasts with what is the default interface into a FHIR server of a REST-based interface exposing CRUD operations and stateless access. Good SOA design practice indicates loosely-coupled access (e.g., external clients do not have visibility into the inner workings of a service) improves implementation flexibility over time, permitting design and performance improvements with the maturation of the implementation. As such, interfaces are typically exposed and aligned with business functionality (such as "Register Patient" or "Validate Identity").

Bridging what is not necessarily a natural-fit between SOA operations and RESTful operations are key to interface design. Implementers may elect to apply finer grained FHIR interactions to be encapsulated in coarse-grained functions, delegating those responsibilities to either implementations within a FHIR instance or to external code not visible to the service consumer.

Principal within the design objective is to minimize or eliminate responsibilities on behalf of the service consumer to understand or apply knowledge of the service implementation. This frees up the service provider to be able to evolve and improve the inner workings of the service without a "ripple effect" adversely impacting service consumers. In doing so, technical dependencies are minimized. Several good sources of SOA Design Patterns are available from HL7 and other public sources.

Data Storage and Coherence. The approach to and complexities associated with data storage and overall data coherence will significantly depend upon whether an implementation is subject to enterprise policies and whether data consistency is the responsibility of the service provider or consumer. In Enterprise settings, these issues are typically governed by organizational policy. FHIR servers and their data persistence would need to fit within that policy, with data access being in compliance with that policy and services interacting with "policy enforcement points" to assure that appropriate permissions are in place.

Where data consistency is the responsibility of the service (or server) use of SOA-friendly interface protocols, such as SOAP and potentially REST, create the access channel for integration. This may necessitate defining minimum sets of data that are required to comply with data coherence expectations (in other words, a service contract that specifies the collection of resources that must be updated in tandem, and the scope of what constitutes a ‘single transaction’ - more on this in the Transactional Integrity section).

In alternative implementations data storage is managed within the FHIR server, consistent with non-SOA FHIR implementations. Data consistency is the responsibility of the Server but responsibility may carry to consuming applications (if they are provided with sufficient update rights so as to make that a concern). Data validation also has a shared responsibility, first with the FHIR Server, and additionally with the authorized updating applications. Data visibility is a product of the REST interface, typically allowing CRUD operations to the data element level.

Transactional Integrity. When encapsulating functionality within a SOA-type service, data and transactional integrity are essential in making sure that data received and stored is accurate and consistent with what was received. An issue particularly with update functions, the ability to update multiple different resources and maintain harmony among them as part of a broader transaction is how this is typically realized.

Should one part of the update ‘fail’, one would expect all of the other updates to be "rolled back" to a prior, stable state. Rollback is not expressly specified as part of the FHIR. Data update and integrity becomes the responsibility of each service. The FHIR specification does not inherently provide for concurrency management (e.g., managing of "deadlock" conditions where two resources each await update permission from the other). As a result, within the FHIR server itself it is assumed that fault tolerance and server availability is being managed (beyond the scope of the specification).

Applied use of SOA patterns have benefits in terms of distributed transactional integrity, as SOA has substantially removed burdens from calling (client) applications, instead levying these responsibilities upon the service provider. This has relevance because it is customary in larger scale implementations (such as large enterprises) that service consumers are varied and not necessarily know in advance to the service. As a result, these implementation patterns make for more robust infrastructure, mitigating potential implementation risk and ultimately improving data quality. We note that these elements can be manually accommodated in FHIR implementations, absent use of SOA patterns, though that can lead to inconsistent interpretation, delegated responsibility to client application(s), and the potential for incompatible implementations.

Modularity. The modularity of resources - which resources are defined, and why - is driven by a wide set of considerations around how they are used. Resources are a platform on which a set of business/clinical services of various kinds are provided. Accordingly, the resources are defined for general use, and they can be expected to be less suitable for a particular service than custom defined structures. The pay back is wider re-use of the information that the service deals with.

Effective "service-orientation" establishes modularity at both a macro and micro-level. At the macro-level, services themselves are modular, componentized, and composable. Since each service has a singularity of purpose with a well-defined interface, they are naturally suited to working in tandem with other services and consistent with modular design. At a more fine-grained level, the interfaces into and out of services also allow for modularity, and the ability to leverage the structure of FHIR resources as a payload descriptor for carrying data structures and corresponding semantics across APIs is modularly based.

Explicit State. All resources represent the various states of the record and real world entities involved in a transaction explicitly. This is necessary for use in RESTful and document contexts, where there is no explicit transaction. Service interactions are typically associated with implicit semantics, such as a request to change the status of a particular resource to something else, for example. The fact that resources that carry this state explicitly as well as the transaction fixing state implicitly creates duplication between the two, and this will need to be managed.

Error Handling. SOA provides an implementation approach providing consistency in error handling, escalation, and error management. These tools can be leveraged to identify likely error conditions and exceptions based upon prior SOA work, helping to put into place infrastructure within FHIR to manage those exceptions. Moreover, this alignment creates future opportunity to more effectively integrate FHIR resources into an enterprise SOA fabric should that become a need. It is not anticipated that there is a need to change error management for FHIR, except to address payload-specific errors that may arise.

The FHIR specification does not specifically address error handling between and across servers and clients. When implementing within larger or more complex environments, particularly in situations where multiple FHIR servers may be involved (especially if they are provided by different vendors/implementations), error handing and management can quickly become complex and untenable.

Resource References. The most obvious impact is that resources refer to each other using full URL based references, and there are a number of rules around how these references are resolved. In the context of a service, this means that the references between modules carry this extra weight of choice and obligation, even when it might not be necessary.

FHIR resources or bundles may be used as the parameters or outputs of service interfaces.

2.25.4 Other Considerations

Orchestration. Orchestration is a term typically used within SOA to describe the steps, sequencing, and dynamic adjustment of workflow to meet a process need. Orchestrations may be entirely automated and fulfilled in a short timeline, or longer and multistep in fulfillment of a business process that may involve manual steps and human intervention. In the context of FHIR implementation, orchestration would refer to the sequencing of collections of FHIR (or service) calls that are used in tandem to fulfil a specific need.

Orchestration is neither natively supported nor unsupported in FHIR, save the availability of the FHIR Batch mode which allows for some degree of compounding of operations. In a SOA environment, orchestration is typically realized by documenting in some formalism, such as Business Process Modeling Notation (BPMN), a sequence of steps and flows, and inherent logic or decision-points affecting that flow. An engine is then capable of executing the process flow, receiving inputs during execution to adjust or adapt those processes based upon situational need as part of delivery fulfillment.

There are many examples where this approach is advantageous. For example, in a Care Management situation, test results, current problems, chief complaint, and potentially even resource availability might affect subsequent steps in fulfilling a care plan. Based upon changes to any of these factors, the sequence of calls and ultimately the systems or FHIR servers involved would vary. The role of automated clinical decision support is another example that naturally ties to workflow orchestration, adjusting care pathways based upon patient evaluation and affecting process flows within a health system.

At present, to support this complexity within a FHIR setting without the use of SOA tools, these flows would need to be manually coded, either by having a directed sequence of calls, or by creating an independent capability effectively acting as an orchestrator. This can be supported within FHIR exclusively, but all of the steps and corresponding state management would need to be done by hand. FHIR Batch provides some of this capability, allowing for aggregation of multiple steps, but does not necessarily support the event processing needed in complex.

Security. Security is inherently a dimension of any enterprise SOA architecture, meaning that the responsibility within a service implementation is to provide the "hooks" to interact with that architecture. In other words, the service does not need to create or enforce security, it needs to interact with those enterprise components that have that responsibility. This includes topics such as identity management, access control, or other dimensions of a secure solution. SOA guidance can foster an effective design involving FHIR in support of authentication, policy enforcement, role-based access controls, and a host of other provisions and protections. It is important to note that within SOA, security is an established, mainstream, and mature offering. Reuse of these concepts can help prevent inadequate, incomplete, or ineffective security measures within FHIR, and eliminate the need to re-invent solutions exclusively for FHIR.

Enterprises typically have security architecture into which a SOA environment will have already integrated. Service implementations would rely upon this existing infrastructure as part of authentication, access control, etc. Policy enforcement is a function of the architecture and not a specific service itself. Services would rely upon policy enforcement points to govern access to information.

While technical dimensions of security would be addressed by the security architecture directly, FHIR implementers should anticipate that data-specific access control policies would need to be captured and formalized so that they are enforceable within the architecture. For example, if a new service is handling protected health information, the policy enforcement point within the architecture would need to know that the service has that nature of data.

Resource Identity. All resources have a single identity (the full URL), and a logical id which may be maintained as the resource moves from server to server (see Managing Resource Identity for further discussion). Since this identity is used by any reference to the resource, it must be maintained when the resource is exchanged so that references from other resources to the one being exchanged can still be resolved. Any use of resources in a service environment needs to address how these references can be resolved. This can be achieved by delegating the reference resolution to a RESTful framework, by ensuring that all the relevant resources are contained in the service call, or by making some service based arrangement by which additional resources can be retrieved. Services that exchange resources SHOULD maintain resource identity. Further, as explicit version tracking is not a guaranteed capability of a service call, services that exchange resources should include version information associated with resources being exchanged.

Capability Statement. When using RESTful exchange, messaging, and document based exchange, the Capability statement allows authoring and reading applications to describe how they use a resource. The Capability Statement supports trading

partner negotiation from specification time through to run-time discovery. The Capability statement doesn't provide any equivalent way to make declarations about services, though this might be added in the future if common requirements emerge. Services are expected to make appropriate arrangements around discovery and compatibility, though it is expected that these will vary considerably.

Services that exchange resources MAY choose to provide support for describing and changing descriptions of service implementations.

2.25.5 Relating SOA and FHIR: Rationale

At its core, FHIR is based upon the elemental capabilities of Create, Read, Update, Delete (CRUD), allowing access to resources where interaction is primarily based upon these operations. Of note, multi-step or complex processing levies requirements upon the calling, client application. Sequencing of events, particularly when those events involve orchestration of complex or dynamic processes, can benefit from the application of SOA techniques and patterns. This is common where multi-step processing is involved, and particularly important when adjustments to a process flow can be made based upon situational context.

As these areas are both commonplace within health settings and not inherent strengths of FHIR, the marriage of FHIR and SOA techniques provides a viable and beneficial path forward to improving implementation and mitigating risks by leveraging proven and established industry best-practices.

SOA design patterns are frequently used to govern interactions for dynamic processes, non-sequential workflows, or dynamic workflows involving human intervention and/or consideration of external factors as part of processing. The ability to model interaction patterns using languages such as SOAml, and to define roles and role interactions, are tools that may be useful to FHIR development in these circumstances, and would be indicators signaling that SOA has the potential to add value to a FHIR implementation.

SOA design principles can provide guidance to a FHIR implementer resulting in reduction of co-dependencies between components, promoting "loose coupling" and minimizing potential impacts resulting from changes to inner workings of one component adversely affecting others. (For those familiar with the term, this encourages "black box" implementation).

SOA provides guidance around data persistence, durability, and expectations in support of the data lifecycle.

2.25.6 Summary

FHIR has proven itself as beneficial in that it provides an easy-to-implement and coherent approach to accessing healthcare resources using a modern protocol stack. SOA has proven itself over time in multiple vertical market segments as a way to divide responsibilities and authoritatively manage information across distributed systems. Marrying the discipline of SOA with the implementation ease of FHIR is a winning combination when the situation warrants. Implementers should consider evaluation of the contextual landscape to determine where FHIR and SOA have the opportunity to benefit from complementary implementation leverage points.

Recognizing that there are situations where the coming-together of FHIR implementation and SOA techniques are advantageous, there are several implementation approaches available. Note that the selection of which approach is advantageous for any given situation will depend upon a variety of factors: existing legacy implementation and available APIs (particularly for interfacing applications), the strategic direction/IT roadmap of the organization, and so on. Noting that there are different styles of realizing FHIR and SOA together, a recap of the principal alternatives would include:

* "Full Support in FHIR" Approach. This alternative involves simplification of the exposed behaviors of the service/server so as to be supported within the REST protocol, thus allowing full support within FHIR. Behind a FHIR technology stack, SOA design principles such as loose coupling and "black box" API design would be coded to create service autonomy.

* "Classic SOA with FHIR Payload" Approach. This alternative is attractive particularly in enterprises that have or are making investments in SOA infrastructure. This alternative relies upon traditional SOA service definition, leveraging use of FHIR Resources as the parameters into and out of the service.

It is important to note that electing to apply SOA patterns to FHIR implementation does not fundamentally change the nature of what is being implemented. Applied correctly, it results in minor adaptation to coding and interface design resulting in enhanced consistency across implementations as well as enhanced interoperability and robustness.

These considerations are summarized in this table:

Capability | #1:FHIR + REST (RESTful FHIR) | #2 FHIR + WS* | #3 FHIR + SOA Pattern | Comments
Supports dynamic adjustment in workflows | ? | ? | * | In a native FHIR environment, this must be done by hand. The WS* stack provides for some interaction patterns, but limited. Use of formal notation (BPMN) and service orchestration is most robust option
Supports ability to batch multiple operations | ? | ? | * | FHIR Batch allows for concatenation of multiple process steps. SOA Orchestration allows for fine grained control, grouping and sequencing of multiple operations
Provides for management of data coherence (e.g. Deadlocks, transactions) | ? | ? | * | SOA principles define data governance, assigning responsibility for all data management within a service contract
Provide support for orchestration languages (BPMN , SOAml) | ? | ? | * | SOA applies a systematic framework for orchestration using industry accepted formalisms, avoiding costs and complexities associated with an ad-hoc approach

SOA in a FHIR Environment considers the benefits of applying SOA patterns and best-practices to a FHIR implementation community. In these cases, developers are faced with implementation decisions that have the potential to benefit from existing case studies, design patterns, or guidance that may either help provide consistency among FHIR implementations, or which may address gaps - either known or unidentified - resulting from implicit assumptions around a FHIR implementation.

FHIR in a SOA Environment considers the implications of using FHIR in a large, established enterprise that has or is making investments in SOA infrastructure. In these cases, shared services, enterprise policies, and existing infrastructure is common,

and FHIR implementations would need to fit within the fabric of that environment.

Finally, it merits mention that not every FHIR implementation benefits from SOA, and vice-versa. The goal of this section is to help implementers navigate based upon their specific considerations to help determine if and to what extent these approaches provide benefits.

2.26 Data Types

URL: https://hl7.org/fhir/STU3/datatypes.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

The FHIR specification defines a set of data types that are used for the resource elements. There are four categories of data types:

* Simple / primitive types, which are single elements with a primitive value (below)

* General purpose complex types, which are re-usable clusters of elements (below)

* Complex data types for metadata

* Special purpose data types: Reference, Narrative, Extension, Meta, and Dosage

This page describes the general purpose data types.

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Primitive Types

Complex Types

Legend: White = abstract type. Light blue = concrete types. Light Pink = Profile on Type.

2.26.0.1 Primitive Types

The following table describes the primitive types that are used in this specification. Primitive types are those with only a value, and no additional elements as children (though, like all types, they have extensions). See also the Examples.

Primitive Types

FHIR Name | Value Domain | XML Representation | JSON representation

boolean | true | false | xs:boolean, except that 0 and 1 are not valid values | JSON boolean (true or false)

integer | A signed 32-bit integer (for larger values, use decimal) | xs:int, except that leading 0 digits are not allowed | JSON number (with no decimal point)

| Regex: [0]|[-+]?[1-9][0-9]*

string | A sequence of Unicode characters | xs:string | JSON String

| Note that strings SHALL NOT exceed 1MB in size. String should not contain Unicode character points below 32, except for u0009 (horizontal tab), u0010 (carriage return) and u0013 (line feed)

| Todo: is there any reason not to make this a SHALL?

decimal | Rational numbers that have a decimal representation. See below about the precision of the number | xs:decimal, except that decimals may not use exponents, and leading 0 digits are not allowed | A JSON number, except that exponents are not allowed

uri | A Uniform Resource Identifier Reference (RFC 3986). Note: URIs are case sensitive. For UUID (urn:uuid:53fefa32-fcbb-4ff8-8a92-55ee120877b7) use all lowercase | xs:anyURI | A JSON string - a URI

| URIs can be absolute or relative, and may have an optional fragment identifier

base64Binary | A stream of bytes, base64 encoded (RFC 4648) | xs:base64Binary | A JSON string - base64 content

| Todo: is it possible to impose an upper absolute limit on a base64Binary (for denial of service reasons, like on string)?

instant | An instant in time - known at least to the second and always includes a time zone. Note: This is intended for precisely observed times (typically system logs etc.), and not human-reported times - for them, use date and dateTime. instant is a more constrained dateTime | xs:dateTime | A JSON string - an xs:dateTime | Regex: -?([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?

| Note: This type is for system times, not human times (see date and dateTime below).

date | A date, or partial date (e.g. just year or year + month) as used in human communication. There is no time zone. Dates SHALL be valid dates | union of xs:date, xs:gYearMonth, xs:gYear | A JSON string - a union of xs:date, xs:gYearMonth, xs:gYear

| Regex: -?[0-9]{4}(-(0[1-9]|1[0-2])(-(0[0-9]|[1-2][0-9]|3[0-1]))?)?

dateTime | A date, date-time or partial date (e.g. just year or year + month) as used in human communication. If hours and minutes are specified, a time zone SHALL be populated. Seconds must be provided due to schema type constraints but may be zero-filled and may be ignored. Dates SHALL be valid dates. The time "24:00" is not allowed | union of xs:dateTime, xs:date, xs:gYearMonth, xs:gYear | A JSON string - a union of xs:dateTime, xs:date, xs:gYearMonth, xs:gYear

| Regex: -?([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?

time | A time during the day, with no date specified (can be converted to a Duration since midnight). Seconds must be provided due to schema type constraints but may be zero-filled and may be ignored. The time "24:00" is not allowed, and neither is a time zone | xs:time | A JSON string - an xs:time

| Regex: ([01][0-9]|2[0-3]):[0-5][0-9]:[0-5][0-9](\.[0-9]+)?

code | Indicates that the value is taken from a set of controlled strings defined elsewhere (see Using codes for further discussion). Technically, a code is restricted to a string which has at least one character and no leading or trailing whitespace, and where there is no whitespace other than single spaces in the contents | xs:token | JSON string

| Regex: [^\s]+([\s]?[^\s]+)*

oid | An OID represented as a URI (RFC 3001); e.g. urn:oid:1.2.3.4.5 | xs:anyURI | JSON string - uri

| Regex: urn:oid:[0-2](\.[1-9]\d*)+

id | Any combination of upper or lower case ASCII letters ('A'..'Z', and 'a'..'z', numerals ('0'..'9'), '-' and '.', with a length limit of 64 characters. (This might be an integer, an un-prefixed OID, UUID or any other identifier pattern that meets these constraints.) | xs:string | JSON string | Regex: [A-Za-z0-9\-\.]{1,64}
markdown | A string that may contain markdown syntax for optional processing by a markdown presentation engine | xs:string | JSON string
unsignedInt | Any non-negative integer (e.g. >= 0) | xs:nonNegativeInteger | JSON number | Regex: [0]|([1-9][0-9]*)
positiveInt | Any positive integer (e.g. >= 1) | xs:positiveInteger | JSON number | Regex: +?[1-9][0-9]*

Notes:

* For all the types, the XML, JSON and Turtle representations of the primitive values are the same except for different escaping in XML and JSON

* Boolean values can also be represented using coded values (such as HL7 v2 Table 0136). See Observation for one such use

* The regexes may allow a broader set of values than are actually valid (e.g. leap years) so additional validation is needed

* The regexes should be qualified with start of string and end of string anchors based on the regex implementation used (e.g. caret '^' and dollar-sign '$' for JavaScript, POSIX, XML and XPath; '\A' and '\Z' for .NET, Java, Python and others; please verify these definitions with the regex implementation used).

* The data types code, string, and uri can be bound to a value set

* The precision of the decimal value has significance:

* e.g. 0.010 is regarded as different to 0.01, and the original precision should be preserved

* Implementations SHALL handle decimal values in ways that preserve and respect the precision of the value as represented for presentation purposes

* Implementations are not required to perform calculations with these numbers differently, though they may choose to do so (i.e. preserve significance)

* See implementation comments for XML, JSON and RDF

* In object code, implementations that might meet this constraint are GMP implementations or equivalents to Java BigDecimal that implement arbitrary precision, or a combination of a (64 bit) floating point value with a precision field

* Note that there is no absolute limit to the magnitude of the value, though large and/or highly precise values are extremely rare in medicine. One element where highly precise decimals may be encountered is the Location coordinates

* About the id datatype:

* Ids are case sensitive. UUIDs SHALL be sent using lowercase letters

* The ID type includes identifiers consistent with ISO 18232 , but also includes other identifier formats as well, and is not case insensitive like ISO 18232.

* In a typical FHIR URL, like http://example.com/fhir/Patient/1234, the last part "1234" (highlighted in red) is the part that is an id datatype

* A full UUID is a uri, not an id. UUIDs in URIs SHALL also be represented in lowercase (urn:uuid:59bf0ef4-e89c-4628-9b51-12ae3fdbe22b)

* About the markdown datatype:

* Systems are not required to have markdown support, and there is considerable variation in markdown syntax, so the content of a string should be readable without markdown processing, per markdown philosophy

* The preferred markdown syntax is described at http://daringfireball.net/projects/markdown/syntax (see tests here: http://daringfireball.net/projects/downloads/MarkdownTest_1.0.zip)

* Markdown content should not contain Unicode character points below 32, except for u0009 (horizontal tab), u0010 (carriage return) and u0013 (line feed). Todo: is there any reason not to make this a SHALL?

STU Note: It is likely that FHIR will change to require/recommend CommonMark in Release 4.

Feedback is welcome here .

2.26.0.1.1 Representations in XML, JSON, and Turtle

All elements using these primitive types may have one or more of a value as described above, an internal identity (e.g. xml:id), and extensions. For an example, take an element of name "count" and type "integer".

XML

The value is represented in XML as an attribute named "value":

```
<count value="2"/>
```

The full representation, with id, extensions and value:

```
<count id="a1" value="2">
  <extension url="...">
    <valueXX.../>
  </extension>
</count>
```

JSON

In JSON, for convenience, the value is represented as the property itself:

```
"count" : 2
```

The full representation, with id, extensions and value, showing the id and extensions in the sibling property:

```
"count" : 2
"count_" : {
  "id" : "a1",
  "extension" : [{
    "url" : "...",
    "valueXXX" : "...."
  }]
}
```

RDF

The value is represented in RDF as an relationship with the URI "http://h;7.org/fhir/value". Using the normal prefix, this becomes:

```
fhir:Type.count [ fhir:value "2"^^xsd:integer ]
```

For the types date and DateTime, the type must be specified explicitly. For all other types, it is optional. The full representation, with id, extensions and value:

```
fhir:Type.count [
  Element.id "a1";
  fhir:value "2"^^xsd:integer;
  Element.extension [
    fhir:Extension.url "..";
    fhir:Extension.valueXX...
  ]
]
```

For additional details, see the XML, JSON and Turtle format definitions. When the value is missing, and there are no extensions, the element is not represented at all. This means that in xml, attributes are never present with a length of 0 (value=""), and properties are never a 0 length string or null in JSON ("name" : "" is not valid). (note: there is one specific use of the null in the JSON representation).

According to XML schema, leading and trailing whitespace in the value attribute is ignored for the types boolean, integer, decimal, base64Binary, instant, uri, date, dateTime, oid, and uri. Note that this means that the schema aware XML libraries give different attribute values to non-schema aware libraries when reading the XML instances. For this reason, the value attribute for these types SHOULD not have leading and trailing spaces. String values should only have leading and trailing spaces if they are part of the content of the value. In JSON and Turtle whitespace in string values is always significant. Primitive types other than string SHALL NOT have leading or trailing whitespace.

2.26.0.2 Complex Types

In XML, these types are represented as XML Elements with child elements with the name of the defined elements of the type. The name of the element is defined where the type is used. In JSON, the data type is represented by an object with properties named the same as the XML elements. Since the JSON representation is almost exactly the same, only the first example has an additional explicit JSON representation.

Complex data types may be "profiled". A Structure Definition or type "constraint" makes a set of rules about which elements SHALL have values and what the possible values are.

UML Diagrams of the Data types

2.26.0.3 Attachment

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

This type is for containing or referencing attachments - additional data content defined in other formats. The most common use of this type is to include images or reports in some report format such as PDF. However it can be used for any data that has a MIME type.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Attachment | I | | Element | Content in a format defined elsewhere + It the Attachment has data, it SHALL have a contentType Elements defined in Ancestors: id, extension
contentType | ? | 0..1 | code | Mime type of the content, with charset etc. MimeType (Required)
language | ? | 0..1 | code | Human language of the content (BCP-47) Common Languages (Extensible but limited to All Languages)
data | | 0..1 | base64Binary | Data inline, base64ed
url | ? | 0..1 | uri | Uri where the data can be found

size | ? | 0..1 | unsignedInt | Number of bytes of content (if url provided)
hash | ? | 0..1 | base64Binary | Hash of the data (sha-1, base64ed)
title | ? | 0..1 | string | Label to display in place of the data
creation | ? | 0..1 | dateTime | Date attachment was first created
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <contentType value="[code]"/><!-- 0..1 Mime type of the content, with charset etc.  -->
 <language value="[code]"/><!-- 0..1 Human language of the content (BCP-47) -->
 <data value="[base64Binary]"/><!-- 0..1 Data inline, base64ed -->
 <url value="[uri]"/><!-- 0..1 Uri where the data can be found -->
 <size value="[unsignedInt]"/><!-- 0..1 Number of bytes of content (if url provided) -->
 <hash value="[base64Binary]"/><!-- 0..1 Hash of the data (sha-1, base64ed) -->
 <title value="[string]"/><!-- 0..1 Label to display in place of the data -->
 <creation value="[dateTime]"/><!-- 0..1 Date attachment was first created -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "contentType" : "<code>", // Mime type of the content, with charset etc.
  "language" : "<code>", // Human language of the content (BCP-47)
  "data" : "<base64Binary>", // Data inline, base64ed
  "url" : "<uri>", // Uri where the data can be found
  "size" : "<unsignedInt>", // Number of bytes of content (if url provided)
  "hash" : "<base64Binary>", // Hash of the data (sha-1, base64ed)
  "title" : "<string>", // Label to display in place of the data
  "creation" : "<dateTime>" // Date attachment was first created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Attachment.contentType [ code ]; # 0..1 Mime type of the content, with charset etc.
  fhir:Attachment.language [ code ]; # 0..1 Human language of the content (BCP-47)
  fhir:Attachment.data [ base64Binary ]; # 0..1 Data inline, base64ed
  fhir:Attachment.url [ uri ]; # 0..1 Uri where the data can be found
  fhir:Attachment.size [ unsignedInt ]; # 0..1 Number of bytes of content (if url provided)
  fhir:Attachment.hash [ base64Binary ]; # 0..1 Hash of the data (sha-1, base64ed)
  fhir:Attachment.title [ string ]; # 0..1 Label to display in place of the data
  fhir:Attachment.creation [ dateTime ]; # 0..1 Date attachment was first created
]
```

Changes since DSTU2

Attachment
Attachment.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Attachment | I | | Element | Content in a format defined elsewhere + It the Attachment has data, it SHALL have a contentType Elements defined in Ancestors: id, extension
contentType | ? | 0..1 | code | Mime type of the content, with charset etc. MimeType (Required)
language | ? | 0..1 | code | Human language of the content (BCP-47) Common Languages (Extensible but limited to All Languages)
data | | 0..1 | base64Binary | Data inline, base64ed
url | ? | 0..1 | uri | Uri where the data can be found
size | ? | 0..1 | unsignedInt | Number of bytes of content (if url provided)
hash | ? | 0..1 | base64Binary | Hash of the data (sha-1, base64ed)
title | ? | 0..1 | string | Label to display in place of the data
creation | ? | 0..1 | dateTime | Date attachment was first created
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <contentType value="[code]"/><!-- 0..1 Mime type of the content, with charset etc.  -->
 <language value="[code]"/><!-- 0..1 Human language of the content (BCP-47) -->
 <data value="[base64Binary]"/><!-- 0..1 Data inline, base64ed -->
 <url value="[uri]"/><!-- 0..1 Uri where the data can be found -->
 <size value="[unsignedInt]"/><!-- 0..1 Number of bytes of content (if url provided) -->
 <hash value="[base64Binary]"/><!-- 0..1 Hash of the data (sha-1, base64ed) -->
 <title value="[string]"/><!-- 0..1 Label to display in place of the data -->
 <creation value="[dateTime]"/><!-- 0..1 Date attachment was first created -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "contentType" : "<code>", // Mime type of the content, with charset etc.
  "language" : "<code>", // Human language of the content (BCP-47)
  "data" : "<base64Binary>", // Data inline, base64ed
  "url" : "<uri>", // Uri where the data can be found
  "size" : "<unsignedInt>", // Number of bytes of content (if url provided)
  "hash" : "<base64Binary>", // Hash of the data (sha-1, base64ed)
  "title" : "<string>", // Label to display in place of the data
  "creation" : "<dateTime>" // Date attachment was first created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Attachment.contentType [ code ]; # 0..1 Mime type of the content, with charset etc.
  fhir:Attachment.language [ code ]; # 0..1 Human language of the content (BCP-47)
  fhir:Attachment.data [ base64Binary ]; # 0..1 Data inline, base64ed
  fhir:Attachment.url [ uri ]; # 0..1 Uri where the data can be found
  fhir:Attachment.size [ unsignedInt ]; # 0..1 Number of bytes of content (if url provided)
  fhir:Attachment.hash [ base64Binary ]; # 0..1 Hash of the data (sha-1, base64ed)
  fhir:Attachment.title [ string ]; # 0..1 Label to display in place of the data
  fhir:Attachment.creation [ dateTime ]; # 0..1 Date attachment was first created
]
```

Changes since DSTU2

Attachment

Attachment.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

See the Full Difference for further information

The actual content of an Attachment can be conveyed directly using the data element or a URL reference can be provided. If both are provided, the reference SHALL point to the same content as found in the data. The reference can never be reused to point to some different data (i.e. the reference is version specific). The URL reference SHALL point to a location that resolves to actual data; some URIs such as cid: meet this requirement. If the URL is a relative reference, it is interpreted in the same way as a resource reference.

The contentType element SHALL always be populated when an Attachment contains data, and MAY be populated when there is a url. It can include charset information and other mime type extensions as appropriate. If there is no character set in the contentType then the correct course of action is undefined, though some media types may define a default character set and/or the correct character set may be able to be determined by inspection of the content.

The hash is included so that applications can verify that the content returned by the URL has not changed. The hash and size relate to the data before it is represented in base64 form.

In many cases where Attachment is used, the cardinality is >1. A valid use of repeats is to convey the same content in different mime types and languages. Guidance on the meaning of repeating elements SHALL be provided in the definition of the repeating resource element or extension that references this type. The language element describes the language of the attachment using the codes defined in BCP 47 .

Constraints

* att-1: It the Attachment has data, it SHALL have a contentType (expression : data.empty() or contentType.exists())

If neither data nor a URL is provided, the value should be understood as an assertion that no content for the specified mimeType and/or language is available for the combination of language and contentType.

The context of use may frequently make rules about the kind of attachment (and therefore, the kind of mime types) that can be used.

Attachment is used in the following places: BodySite, Claim, Communication, CommunicationRequest, Consent, Contract, DiagnosticReport, DocumentManifest, DocumentReference, ExplanationOfBenefit, HealthcareService, Library, Media, Medication, Observation, Patient, Person, Practitioner, Questionnaire, QuestionnaireResponse and RelatedPerson

2.26.0.4 Coding

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Coding is a representation of a defined concept using a symbol from a defined "code system" - see Using Codes in resources for more details.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Coding | ? | | Element | A reference to a code defined by a terminology system Elements defined in Ancestors: id, extension
system | ? | 0..1 | uri | Identity of the terminology system
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
 <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
 <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
 <display value="[string]"/><!-- 0..1 Representation defined by the system -->
 <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<uri>", // Identity of the terminology system
  "version" : "<string>", // Version of the system - if relevant
  "code" : "<code>", // Symbol in syntax defined by the system
  "display" : "<string>", // Representation defined by the system
  "userSelected" : <boolean> // If this coding was chosen directly by the user
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Coding.system [ uri ]; # 0..1 Identity of the terminology system
  fhir:Coding.version [ string ]; # 0..1 Version of the system - if relevant
  fhir:Coding.code [ code ]; # 0..1 Symbol in syntax defined by the system
  fhir:Coding.display [ string ]; # 0..1 Representation defined by the system
  fhir:Coding.userSelected [ boolean ]; # 0..1 If this coding was chosen directly by the user
]
```

Changes since DSTU2

Coding

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Coding | ? | | Element | A reference to a code defined by a terminology system Elements defined in Ancestors: id, extension
system | ? | 0..1 | uri | Identity of the terminology system
version | ? | 0..1 | string | Version of the system - if relevant
code | ? | 0..1 | code | Symbol in syntax defined by the system
display | ? | 0..1 | string | Representation defined by the system
userSelected | ? | 0..1 | boolean | If this coding was chosen directly by the user
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
 <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
 <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
 <display value="[string]"/><!-- 0..1 Representation defined by the system -->
 <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<uri>", // Identity of the terminology system
  "version" : "<string>", // Version of the system - if relevant
  "code" : "<code>", // Symbol in syntax defined by the system
  "display" : "<string>", // Representation defined by the system
  "userSelected" : <boolean> // If this coding was chosen directly by the user
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[
 # from Element: Element.extension
  fhir:Coding.system [ uri ]; # 0..1 Identity of the terminology system
  fhir:Coding.version [ string ]; # 0..1 Version of the system - if relevant
  fhir:Coding.code [ code ]; # 0..1 Symbol in syntax defined by the system
  fhir:Coding.display [ string ]; # 0..1 Representation defined by the system
  fhir:Coding.userSelected [ boolean ]; # 0..1 If this coding was chosen directly by the user
]
```

Changes since DSTU2

Coding

See the Full Difference for further information

The meaning of the Coding is defined by the code. The system provides the source of the definition of the code, along with an optional version reference. The display is a human display for the text defined by the system - it is not intended for computation.

The system is a URI that references the code system that defines the code. Choosing the correct system is important; for more information about the code system URI, read Managing Terminology System URIs. The system URI SHALL NOT contain a reference to a value set (e.g. ValueSet.url). If the code is taken from a CodeSystem resource, CodeSystem.url is the correct value for the system uri. Resolvable URLs are generally preferred by implementers over non-resolvable URNs, particularly opaque URNs such as OIDs (urn:oid:) or UUIDs (urn:uuid:).

A code system version may also be supplied. If the meaning of codes within the code system is consistent across releases, this is not required. The version SHOULD be exchanged when the system does not maintain consistent definitions across versions. Note that the following systems SHOULD always have a version specified:

* National releases of SNOMED CT (consistency of definitions varies amongst jurisdictions, and some jurisdictions may make their own rules on this)

* Various versions of ICD (note: the major releases are labeled as different code systems altogether, but there is variation within versions)

More generally, any classification (e.g. a code system that includes concepts with relative definitions such as "not otherwise coded" will require a version. See the discussion of code system versions in the Code System resource for further discussion on versioning.

If present, the code SHALL be a syntactically correct symbol as defined by the system. In some code systems such as SNOMED CT, the symbol may be an expression composed of other predefined symbol (e.g. post-coordination). Note that codes are case sensitive unless specified otherwise by the code system. The display is a text representation of the code defined by the system and is used to display the meaning of the code by an application that is not aware of the system.

Where the code system defines multiple possible display strings, one of these SHALL be used in display. If one is labeled as preferred, it SHOULD be used. If the code system does not define a text representation (e.g. SNOMED CT Expressions) then display cannot be populated, and the meaning of the code won't be accessible to systems that don't understand the code expression.

In some cases, the system may not be known - only the code is known. In this case, no useful processing of the code may be performed unless the system can be safely inferred by the context. This practice should be avoided where possible, as information sharing in a wider context is very likely to arise eventually, and codes cannot be used in the absence of a known system.

If the system is present, and there is no code, then this is understood to mean that there is no suitable code in the system in which to represent the code.

If two codings have the same system, version and code then they have the same meaning. If the version information is missing, or the system, version or the code elements differ, then how the codes are related can only be determined by consulting the definitions of the system(s) and any mappings available.

A coding may be marked as a "userSelected" if a user selected the particular coded value in a user interface (e.g. the user selects an item in a pick-list). If a user selected coding exists, it is the preferred choice for performing translations etc.

Constraints

The context of use (as defined in the resource or applicable profile) usually makes rules about what codes and systems are allowed or required in a particular context by binding the element to a value set.

Coding is used in the following places: Signature, CodeableConcept, AuditEvent, CapabilityStatement, Claim, ClaimResponse, CodeSystem, Consent, Contract, DocumentReference, Encounter, Endpoint, ExpansionProfile, ExplanationOfBenefit, ImagingStudy, Location, MessageDefinition, MessageHeader, PlanDefinition, Provenance, Questionnaire, QuestionnaireResponse, RequestGroup, StructureDefinition, Subscription, TestScript and ValueSet

Design Note: This specification defines two types for representing coded values:

* Coding: a simple direct reference to a code defined by a code system

* CodeableConcept: a text description and/or a list of Codings (i.e. a list of references to codes defined by code systems)

The Coding data type corresponds to the simple case of selecting a single code from a code list. However this type is rarely used in the FHIR specifications; long experience with exchanging coded values in HL7 shows that in the general case, systems need to able to exchange multiple translation codes, and/or an original text.

The Coding data type is used directly when there is certainty that the value must be selected directly from one of the available codes, and the list of possible codes is agreed to by all participants. This is not usually the case in the context of FHIR - general interoperability - so Coding is mostly used in extensions, which are usually intended to be defined for a well-controlled context of use.

2.26.0.5 CodeableConcept

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A CodeableConcept represents a value that is usually supplied by providing a reference to one or more terminologies or ontologies, but may also be defined by the provision of text. This is a common pattern in healthcare data.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CodeableConcept | ? | | Element | Concept - reference to a terminology or just text Elements defined in Ancestors: id, extension
coding | ? | 0..* | Coding | Code defined by a terminology system
text | ? | 0..1 | string | Plain text representation of the concept
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <coding><!-- 0..* Coding Code defined by a terminology system --></coding>
 <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "coding" : [{ Coding }], // Code defined by a terminology system
  "text" : "<string>" // Plain text representation of the concept
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:CodeableConcept.coding [ Coding ], ... ; # 0..* Code defined by a terminology system
  fhir:CodeableConcept.text [ string ]; # 0..1 Plain text representation of the concept
]
```

Changes since DSTU2

CodeableConcept

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
CodeableConcept | ? | | Element | Concept - reference to a terminology or just text Elements defined in Ancestors: id, extension
coding | ? | 0..* | Coding | Code defined by a terminology system
text | ? | 0..1 | string | Plain text representation of the concept
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <coding><!-- 0..* Coding Code defined by a terminology system --></coding>
 <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "coding" : [{ Coding }], // Code defined by a terminology system
  "text" : "<string>" // Plain text representation of the concept
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:CodeableConcept.coding [ Coding ], ... ; # 0..* Code defined by a terminology system
  fhir:CodeableConcept.text [ string ]; # 0..1 Plain text representation of the concept
]
```

Changes since DSTU2

CodeableConcept

See the Full Difference for further information

Each coding is a representation of the concept as described above. The concept may be coded multiple times in different code systems (or even multiple times in the same code systems, where multiple forms are possible, such as with SNOMED CT). The different codings may have slightly different granularity due to the differences in the definitions of the underlying codes. There is no meaning associated with the ordering of coding within a CodeableConcept. A typical use of CodeableConcept is to send the local code that the concept was coded with, and also one or more translations to publicly defined code systems such as LOINC or SNOMED CT. Sending local codes is useful and important for the purposes of debugging and integrity auditing.

Whether or not coding elements are present, the text is the representation of the concept as entered or chosen by the user, and which most closely represents the intended meaning of the user or concept. Very often the text is the same as a display of one of the codings. One of the codings may be flagged as the userSelected - the code or concept that the user actually selected directly. When none of the coding elements is marked as userSelected, the text (if present) is the preferred source of meaning.

Constraints

The context of use usually makes rules about what codes and systems are allowed or required in a particular context by binding the element to a value set.

CodeableConcept is used in the following places: Identifier, Account, ActivityDefinition, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BodySite, CapabilityStatement, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, ClinicalImpression, CodeSystem, Communication, CommunicationRequest, CompartmentDefinition, Composition, ConceptMap, Condition, Consent, Contract, Coverage, DataElement, DetectedIssue, Device, DeviceComponent, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EligibilityRequest, EligibilityResponse, Encounter, Endpoint, EnrollmentResponse, EpisodeOfCare, ExpansionProfile, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, GraphDefinition, Group, GuidanceResponse, HealthcareService, ImagingStudy, Immunization, ImmunizationRecommendation, ImplementationGuide, Library, List, Location, Measure, MeasureReport, Media, Medication, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MessageDefinition, MessageHeader, NamingSystem, NutritionOrder, Observation, OperationDefinition, OperationOutcome, Organization, Patient, PaymentNotice, PaymentReconciliation, PlanDefinition, Practitioner, PractitionerRole, Procedure, ProcedureRequest, ProcessResponse, Provenance, Questionnaire, ReferralRequest, RelatedPerson, RequestGroup, ResearchStudy, RiskAssessment, Schedule, SearchParameter, Sequence, ServiceDefinition, Slot, Specimen, StructureDefinition, StructureMap, Substance, SupplyDelivery, SupplyRequest, Task, TestScript, ValueSet and VisionPrescription

2.26.0.6 Quantity

See also Examples, Detailed Descriptions and Mappings, Profiles & Extensions and and R2 Conversions.

A measured amount (or an amount that can potentially be measured).

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Quantity | ?I | | Element | A measured or measurable amount + If a code for the unit is present, the system SHALL also be present Elements defined in Ancestors: id, extension
value | ? | 0..1 | decimal | Numerical value (with implicit precision)
comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value QuantityComparator (Required)
unit | ? | 0..1 | string | Unit representation
system | ?I | 0..1 | uri | System that defines coded unit form
code | ? | 0..1 | code | Coded form of the unit
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value -->
 <unit value="[string]"/><!-- 0..1 Unit representation -->
 <system value="[uri]"/><!--  0..1 System that defines coded unit form -->
 <code value="[code]"/><!-- 0..1 Coded form of the unit -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "comparator" : "<code>", // < | <= | >= | > - how to understand the value
  "unit" : "<string>", // Unit representation
  "system" : "<uri>", // C? System that defines coded unit form
  "code" : "<code>" // Coded form of the unit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Quantity.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Quantity.comparator [ code ]; # 0..1 < | <= | >= | > - how to understand the value
  fhir:Quantity.unit [ string ]; # 0..1 Unit representation
  fhir:Quantity.system [ uri ]; # 0..1 System that defines coded unit form
  fhir:Quantity.code [ code ]; # 0..1 Coded form of the unit
]
```

Changes since DSTU2

Quantity

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Quantity | ?I | | Element | A measured or measurable amount + If a code for the unit is present, the system SHALL also be present Elements defined in Ancestors: id, extension

value | ? | 0..1 | decimal | Numerical value (with implicit precision)

comparator | ?!? | 0..1 | code | < | <= | >= | > - how to understand the value QuantityComparator (Required)

unit | ? | 0..1 | string | Unit representation

system | ?I | 0..1 | uri | System that defines coded unit form

code | ? | 0..1 | code | Coded form of the unit

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value -->
 <unit value="[string]"/><!-- 0..1 Unit representation -->
 <system value="[uri]"/><!--  0..1 System that defines coded unit form -->
 <code value="[code]"/><!-- 0..1 Coded form of the unit -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "comparator" : "<code>", // < | <= | >= | > - how to understand the value
  "unit" : "<string>", // Unit representation
  "system" : "<uri>", // C? System that defines coded unit form
  "code" : "<code>" // Coded form of the unit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Quantity.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Quantity.comparator [ code ]; # 0..1 < | <= | >= | > - how to understand the value
  fhir:Quantity.unit [ string ]; # 0..1 Unit representation
  fhir:Quantity.system [ uri ]; # 0..1 System that defines coded unit form
  fhir:Quantity.code [ code ]; # 0..1 Coded form of the unit
]
```

Changes since DSTU2

Quantity

See the Full Difference for further information

The value contains the numerical value of the quantity, including an implicit precision. If no comparator is specified, the value is a point value (i.e. '='). The comparator element can never be ignored.

The unit element contains a displayable unit that defines what is measured. The unit may additionally be coded in some formal way using the code and the system (see Coding for further information about how to use the system element).

If the unit can be coded in UCUM and a code is provided, it SHOULD be a UCUM code. If a UCUM unit is provided in the code, then a canonical value can be generated for purposes of comparison between quantities. Note that the unit element will often contain text that is a valid UCUM unit, but it cannot be assumed that the unit actually contains a valid UCUM unit.

Constraints

* qty-3: If a code for the unit is present, the system SHALL also be present (expression : code.empty() or system.exists())

The context of use may frequently define what kind of measured quantity this is and therefore what kind of unit can be used. The context of use may additionally require a code from a particular system. The context of use may also restrict the values for the value or range.

Quantity is used in the following places: Duration, Count, Money, Ratio, Distance, Age, ChargeItem, Claim, ExplanationOfBenefit, Goal, Group, Observation, PlanDefinition, Questionnaire, QuestionnaireResponse, Sequence and SupplyRequest

2.26.0.6.1 Defined Variations on Quantity

There are several additional data types that are specializations of Quantity that only introduce new restrictions on the existing elements defined as part of the Quantity data type:

Type Name | Rules | Formal Definitions

Age | age-1: There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. If value is present, it SHALL be positive. (expression : (code or value.empty()) and (system.empty() or system = %ucum) and (value.empty() or value > 0)) | XML, JSON

| Usage: AllergyIntolerance, Condition and FamilyMemberHistory

Count | cnt-3: There SHALL be a code with a value of "1" if there is a value and it SHALL be an expression of length. If system is present, it SHALL be UCUM. If present, the value SHALL a whole number. (expression : (code or value.empty()) and (system.empty() or system = %ucum) and (code.empty() or code = '1') and (value.empty() or value.toString().contains('.').not())) | XML, JSON

| Usage: (not used as yet)

Distance | dis-1: There SHALL be a code if there is a value and it SHALL be an expression of length. If system is present, it SHALL be UCUM. (expression : (code or value.empty()) and (system.empty() or system = %ucum)) | XML, JSON

| Usage: (not used as yet)

Duration | drt-1: There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. (expression : code.exists() implies ((system = %ucum) and value.exists())) | XML, JSON

| Usage: Encounter, Goal, MedicationRequest, PlanDefinition and RequestGroup

Money | mny-1: There SHALL be a code if there is a value and it SHALL be an expression of currency. If system is present, it SHALL be ISO 4217 (system = "urn:iso:std:iso:4217" - currency). (expression : (code or value.empty()) and (system.empty() or system = 'urn:iso:std:iso:4217')) | XML, JSON

| Usage: Account, ChargeItem, Claim, ClaimResponse, Contract, EligibilityResponse, ExplanationOfBenefit and PaymentReconciliation

In addition to the specializations, there is one profile on Quantity used in several resources:

Profile Name | Rules | Formal Definitions

Simple Quantity | sqty-1: The comparator is not used on a SimpleQuantity (expression : comparator.empty()) | XML, JSON

| Usage: Range, SampledData, ActivityDefinition, CarePlan, Claim, Contract, ExplanationOfBenefit, Immunization, Medication, MedicationAdministration, MedicationDispense, MedicationRequest, NutritionOrder, Observation, Specimen, Substance, SupplyDelivery and VisionPrescription

Note that the profile is different from the other specializations because it is not a type, just rules applied where the Quantity type is used.

2.26.0.7 Range

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A set of ordered Quantity values defined by a low and high limit.

A Range specifies a set of possible values; usually, one value from the range applies (e.g. "give the patient between 2 and 4 tablets"). Ranges are typically used in instructions.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Range | ?I | | Element | Set of values bounded by low and high + If present, low SHALL have a lower value than high Elements defined in Ancestors: id, extension

low | ?I | 0..1 | SimpleQuantity | Low limit

high | ?I | 0..1 | SimpleQuantity | High limit

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <low><!--  0..1 Quantity(SimpleQuantity) Low limit --></low>
 <high><!--  0..1 Quantity(SimpleQuantity) High limit --></high>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "low" : { Quantity(SimpleQuantity) }, // C? Low limit
  "high" : { Quantity(SimpleQuantity) } // C? High limit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Range.low [ Quantity(SimpleQuantity) ]; # 0..1 Low limit
  fhir:Range.high [ Quantity(SimpleQuantity) ]; # 0..1 High limit
]
```

Changes since DSTU2

Range

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Range | ?I | | Element | Set of values bounded by low and high + If present, low SHALL have a lower value than high Elements defined in Ancestors: id, extension

low | ?I | 0..1 | SimpleQuantity | Low limit

high | ?I | 0..1 | SimpleQuantity | High limit

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <low><!--  0..1 Quantity(SimpleQuantity) Low limit --></low>
 <high><!--  0..1 Quantity(SimpleQuantity) High limit --></high>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "low" : { Quantity(SimpleQuantity) }, // C? Low limit
  "high" : { Quantity(SimpleQuantity) } // C? High limit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Range.low [ Quantity(SimpleQuantity) ]; # 0..1 Low limit
  fhir:Range.high [ Quantity(SimpleQuantity) ]; # 0..1 High limit
]
```

Changes since DSTU2

Range

See the Full Difference for further information

The unit and code/system elements of the low or high elements SHALL match. If the low or high elements are missing, the meaning is that the low or high boundaries are not known and therefore neither is the complete range.

The comparator flag on the low or high elements cannot be present. Note that the Range type should not be used to represent out of range measurements: A quantity type with the comparator element should be used instead.

The low and the high values are inclusive, and are assumed to have arbitrarily high precision; e.g. the range 1.5 to 2.5 includes 1.50, and 2.50 but not 1.49 or 2.51.

Constraints

* rng-2: If present, low SHALL have a lower value than high (expression : low.empty() or high.empty() or (low <= high))

Range is used in the following places: ActivityDefinition, AllergyIntolerance, Condition, FamilyMemberHistory, Goal, Group, Observation, PlanDefinition, RequestGroup and RiskAssessment

2.26.0.8 Ratio

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A relationship between two Quantity values expressed as a numerator and a denominator.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Ratio | ?I | | Element | A ratio of two Quantity values - a numerator and a denominator + Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present Elements defined in Ancestors: id, extension

numerator | ? | 0..1 | Quantity | Numerator value

denominator | ? | 0..1 | Quantity | Denominator value

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <numerator><!-- 0..1 Quantity Numerator value --></numerator>
 <denominator><!-- 0..1 Quantity Denominator value --></denominator>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "numerator" : { Quantity }, // Numerator value
  "denominator" : { Quantity } // Denominator value
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Ratio.numerator [ Quantity ]; # 0..1 Numerator value
  fhir:Ratio.denominator [ Quantity ]; # 0..1 Denominator value
]
```

Changes since DSTU2

Ratio

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Ratio | ?I | | Element | A ratio of two Quantity values - a numerator and a denominator + Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present Elements defined in Ancestors: id, extension

numerator | ? | 0..1 | Quantity | Numerator value

denominator | ? | 0..1 | Quantity | Denominator value

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <numerator><!-- 0..1 Quantity Numerator value --></numerator>
 <denominator><!-- 0..1 Quantity Denominator value --></denominator>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "numerator" : { Quantity }, // Numerator value
  "denominator" : { Quantity } // Denominator value
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Ratio.numerator [ Quantity ]; # 0..1 Numerator value
```

```
  fhir:Ratio.denominator [ Quantity ]; # 0..1 Denominator value
]
```

Changes since DSTU2

Ratio

See the Full Difference for further information

Common factors in the numerator and denominator are not automatically cancelled out. The Ratio data type is used for titers (e.g. "1:128") and other quantities produced by laboratories that truly represent ratios. Ratios are not simply "structured numbers" - for example blood pressure measurements (e.g. "120/60") are not ratios. In addition, ratios are used where common factors in the numerator and denominator do not cancel out. The most common example of this is where the ratio represents a unit cost, and the numerator is a currency (e.g. 50/$10).

A proper ratio has both a numerator and a denominator; however these are not mandatory in order to allow an invalid ratio with an extension with further information.

Constraints

* rat-1: Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present (expression : (numerator.empty() xor denominator.exists()) and (numerator.exists() or extension.exists()))

The context of use may require particular types of Quantity for the numerator or denominator.

Ratio is used in the following places: Medication, MedicationAdministration, NutritionOrder, Observation and Substance

2.26.0.9 Period

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A time period defined by a start and end date/time.

A period specifies a range of times. The context of use will specify whether the entire range applies (e.g. "the patient was an inpatient of the hospital for this time range") or one value from the period applies (e.g. "give to the patient between 2 and 4 pm on 24-Jun 2013").

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Period | ?I | | Element | Time range defined by start and end date/time + If present, start SHALL have a lower value than end Elements defined in Ancestors: id, extension

start | ?I | 0..1 | dateTime | Starting time with inclusive boundary

end | ?I | 0..1 | dateTime | End time with inclusive boundary, if not ongoing

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <start value="[dateTime]"/><!--  0..1 Starting time with inclusive boundary -->
 <end value="[dateTime]"/><!--  0..1 End time with inclusive boundary, if not ongoing -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "start" : "<dateTime>", // C? Starting time with inclusive boundary
  "end" : "<dateTime>" // C? End time with inclusive boundary, if not ongoing
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Period.start [ dateTime ]; # 0..1 Starting time with inclusive boundary
  fhir:Period.end [ dateTime ]; # 0..1 End time with inclusive boundary, if not ongoing
]
```

Changes since DSTU2

Period

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Period | ?I | | Element | Time range defined by start and end date/time + If present, start SHALL have a lower value than end Elements defined in Ancestors: id, extension
start | ?I | 0..1 | dateTime | Starting time with inclusive boundary
end | ?I | 0..1 | dateTime | End time with inclusive boundary, if not ongoing
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <start value="[dateTime]"/><!--  0..1 Starting time with inclusive boundary -->
 <end value="[dateTime]"/><!--  0..1 End time with inclusive boundary, if not ongoing -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "start" : "<dateTime>", // C? Starting time with inclusive boundary
  "end" : "<dateTime>" // C? End time with inclusive boundary, if not ongoing
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Period.start [ dateTime ]; # 0..1 Starting time with inclusive boundary
  fhir:Period.end [ dateTime ]; # 0..1 End time with inclusive boundary, if not ongoing
]
```

Changes since DSTU2

Period

See the Full Difference for further information

If the start element is missing, the start of the period is not known. If the end element is missing, it means that the period is ongoing, or the start may be in the past, and the end date in the future, which means that period is expected/planned to end at the specified time

The end value includes any matching date/time. For example, the period 2011-05-23 to 2011-05-27 includes all the times from the start of the 23rd May through to the end of the 27th of May.

Period is used in the following places: Identifier, Account, ActivityDefinition, AllergyIntolerance, Appointment, CarePlan, CareTeam, ChargeItem, Claim, ClinicalImpression, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentReference, EligibilityRequest, Encounter, Endpoint, EpisodeOfCare, ExplanationOfBenefit, FamilyMemberHistory, Flag, Group, HealthcareService, Library, Measure, MeasureReport, Media, MedicationAdministration, MedicationRequest, MedicationStatement, NamingSystem, Observation, Patient, PaymentReconciliation, PlanDefinition, Practitioner, PractitionerRole, Procedure, ProcedureRequest, ProcessRequest, Provenance, Questionnaire, ReferralRequest, RelatedPerson, RequestGroup, ResearchStudy, ResearchSubject, RiskAssessment, Schedule, ServiceDefinition, Specimen, SupplyDelivery, SupplyRequest and Task

2.26.0.10 SampledData

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Data that comes from a series of measurements taken by a device, which may have upper and lower limits. The data type also supports more than one dimension in the data.

A SampledData provides a concise way to handle the data produced by devices that sample a particular physical state at a high frequency. A typical use for this is for the output of an ECG or EKG device. The data type includes a series of raw decimal values (which are mostly simple integers), along with adjustments for scale and factor. These are interpreted such that

```
original measured value[i] = SampledData.data[i] * SampledData.scaleFactor + SampledData.origin.value
```

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
SampledData | | | Element | A series of measurements taken by a device Elements defined in Ancestors: id, extension
origin | ? | 1..1 | SimpleQuantity | Zero value and units

period | ? | 1..1 | decimal | Number of milliseconds between samples
factor | ? | 0..1 | decimal | Multiply data by this before adding to origin
lowerLimit | ? | 0..1 | decimal | Lower limit of detection
upperLimit | ? | 0..1 | decimal | Upper limit of detection
dimensions | ? | 1..1 | positiveInt | Number of sample points at each time point
data | | 1..1 | string | Decimal values with spaces, or "E" | "U" | "L"
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <origin><!-- 1..1 Quantity(SimpleQuantity) Zero value and units --></origin>
 <period value="[decimal]"/><!-- 1..1 Number of milliseconds between samples -->
 <factor value="[decimal]"/><!-- 0..1 Multiply data by this before adding to origin -->
 <lowerLimit value="[decimal]"/><!-- 0..1 Lower limit of detection -->
 <upperLimit value="[decimal]"/><!-- 0..1 Upper limit of detection -->
 <dimensions value="[positiveInt]"/><!-- 1..1 Number of sample points at each time point -->
 <data value="[string]"/><!-- 1..1 Decimal values with spaces, or "E" | "U" | "L" -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "origin" : { Quantity(SimpleQuantity) }, // R!  Zero value and units
  "period" : <decimal>, // R!  Number of milliseconds between samples
  "factor" : <decimal>, // Multiply data by this before adding to origin
  "lowerLimit" : <decimal>, // Lower limit of detection
  "upperLimit" : <decimal>, // Upper limit of detection
  "dimensions" : "<positiveInt>", // R!  Number of sample points at each time point
  "data" : "<string>" // R!  Decimal values with spaces, or "E" | "U" | "L"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:SampledData.origin [ Quantity(SimpleQuantity) ]; # 1..1 Zero value and units
  fhir:SampledData.period [ decimal ]; # 1..1 Number of milliseconds between samples
  fhir:SampledData.factor [ decimal ]; # 0..1 Multiply data by this before adding to origin
  fhir:SampledData.lowerLimit [ decimal ]; # 0..1 Lower limit of detection
  fhir:SampledData.upperLimit [ decimal ]; # 0..1 Upper limit of detection
  fhir:SampledData.dimensions [ positiveInt ]; # 1..1 Number of sample points at each time point
  fhir:SampledData.data [ string ]; # 1..1 Decimal values with spaces, or "E" | "U" | "L"
]
```

Changes since DSTU2

SampledData

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
SampledData | | | Element | A series of measurements taken by a device Elements defined in Ancestors: id, extension
origin | ? | 1..1 | SimpleQuantity | Zero value and units
period | ? | 1..1 | decimal | Number of milliseconds between samples
factor | ? | 0..1 | decimal | Multiply data by this before adding to origin
lowerLimit | ? | 0..1 | decimal | Lower limit of detection
upperLimit | ? | 0..1 | decimal | Upper limit of detection
dimensions | ? | 1..1 | positiveInt | Number of sample points at each time point
data | | 1..1 | string | Decimal values with spaces, or "E" | "U" | "L"
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <origin><!-- 1..1 Quantity(SimpleQuantity) Zero value and units --></origin>
 <period value="[decimal]"/><!-- 1..1 Number of milliseconds between samples -->
 <factor value="[decimal]"/><!-- 0..1 Multiply data by this before adding to origin -->
 <lowerLimit value="[decimal]"/><!-- 0..1 Lower limit of detection -->
 <upperLimit value="[decimal]"/><!-- 0..1 Upper limit of detection -->
 <dimensions value="[positiveInt]"/><!-- 1..1 Number of sample points at each time point -->
 <data value="[string]"/><!-- 1..1 Decimal values with spaces, or "E" | "U" | "L" -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "origin" : { Quantity(SimpleQuantity) }, // R!  Zero value and units
  "period" : <decimal>, // R!  Number of milliseconds between samples
  "factor" : <decimal>, // Multiply data by this before adding to origin
```

```
  "lowerLimit" : <decimal>, // Lower limit of detection
  "upperLimit" : <decimal>, // Upper limit of detection
  "dimensions" : "<positiveInt>", // R!  Number of sample points at each time point
  "data" : "<string>" // R!  Decimal values with spaces, or "E" | "U" | "L"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:SampledData.origin [ Quantity(SimpleQuantity) ]; # 1..1 Zero value and units
  fhir:SampledData.period [ decimal ]; # 1..1 Number of milliseconds between samples
  fhir:SampledData.factor [ decimal ]; # 0..1 Multiply data by this before adding to origin
  fhir:SampledData.lowerLimit [ decimal ]; # 0..1 Lower limit of detection
  fhir:SampledData.upperLimit [ decimal ]; # 0..1 Upper limit of detection
  fhir:SampledData.dimensions [ positiveInt ]; # 1..1 Number of sample points at each time point
  fhir:SampledData.data [ string ]; # 1..1 Decimal values with spaces, or "E" | "U" | "L"
]
```

Changes since DSTU2

SampledData

See the Full Difference for further information

The digits are a set of decimal values separated by a single space (Unicode character u20). In addition to decimal values, the special values "E" (error), "L" (below detection limit) and "U" (above detection limit) can also be used. If there is more than one dimension, the different dimensions are interlaced - all the data points for a particular time are represented together. The default value for factor is 1.

SampledData is used in the following places: Observation

2.26.0.11 Identifier

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A numeric or alphanumeric string that is associated with a single object or entity within a given system. Typically, identifiers are used to connect content in resources to external content available in other frameworks or protocols. Identifiers are associated with objects, and may be changed or retired due to human or system process and errors.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Identifier | ? | | Element | An identifier intended for computation Elements defined in Ancestors: id, extension
use | ?!? | 0..1 | code | usual | official | temp | secondary (If known) IdentifierUse (Required)
type | ? | 0..1 | CodeableConcept | Description of identifier Identifier Type Codes (Extensible)
system | ? | 0..1 | uri | The namespace for the identifier value
value | ? | 0..1 | string | The value that is unique
period | ? | 0..1 | Period | Time period when id is/was valid for use
assigner | ? | 0..1 | Reference(Organization) | Organization that issued id (may be just text)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | secondary (If known) -->
 <type><!-- 0..1 CodeableConcept Description of identifier --></type>
 <system value="[uri]"/><!-- 0..1 The namespace for the identifier value -->
 <value value="[string]"/><!-- 0..1 The value that is unique -->
 <period><!-- 0..1 Period Time period when id is/was valid for use --></period>
 <assigner><!-- 0..1 Reference(Organization) Organization that issued id (may be just text) --></assigner>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | secondary (If known)
  "type" : { CodeableConcept }, // Description of identifier
  "system" : "<uri>", // The namespace for the identifier value
  "value" : "<string>", // The value that is unique
  "period" : { Period }, // Time period when id is/was valid for use
  "assigner" : { Reference(Organization) } // Organization that issued id (may be just text)
```

```
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Identifier.use [ code ]; # 0..1 usual | official | temp | secondary (If known)
  fhir:Identifier.type [ CodeableConcept ]; # 0..1 Description of identifier
  fhir:Identifier.system [ uri ]; # 0..1 The namespace for the identifier value
  fhir:Identifier.value [ string ]; # 0..1 The value that is unique
  fhir:Identifier.period [ Period ]; # 0..1 Time period when id is/was valid for use
  fhir:Identifier.assigner [ Reference(Organization) ]; # 0..1 Organization that issued id (may be just text)
]
```

Changes since DSTU2

Identifier

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Identifier | ? | | Element | An identifier intended for computation Elements defined in Ancestors: id, extension

use | ?!? | 0..1 | code | usual | official | temp | secondary (If known) IdentifierUse (Required)

type | ? | 0..1 | CodeableConcept | Description of identifier Identifier Type Codes (Extensible)

system | ? | 0..1 | uri | The namespace for the identifier value

value | ? | 0..1 | string | The value that is unique

period | ? | 0..1 | Period | Time period when id is/was valid for use

assigner | ? | 0..1 | Reference(Organization) | Organization that issued id (may be just text)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | secondary (If known) -->
 <type><!-- 0..1 CodeableConcept Description of identifier --></type>
 <system value="[uri]"/><!-- 0..1 The namespace for the identifier value -->
 <value value="[string]"/><!-- 0..1 The value that is unique -->
 <period><!-- 0..1 Period Time period when id is/was valid for use --></period>
 <assigner><!-- 0..1 Reference(Organization) Organization that issued id (may be just text) --></assigner>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | secondary (If known)
  "type" : { CodeableConcept }, // Description of identifier
  "system" : "<uri>", // The namespace for the identifier value
  "value" : "<string>", // The value that is unique
  "period" : { Period }, // Time period when id is/was valid for use
  "assigner" : { Reference(Organization) } // Organization that issued id (may be just text)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Identifier.use [ code ]; # 0..1 usual | official | temp | secondary (If known)
  fhir:Identifier.type [ CodeableConcept ]; # 0..1 Description of identifier
  fhir:Identifier.system [ uri ]; # 0..1 The namespace for the identifier value
  fhir:Identifier.value [ string ]; # 0..1 The value that is unique
  fhir:Identifier.period [ Period ]; # 0..1 Time period when id is/was valid for use
  fhir:Identifier.assigner [ Reference(Organization) ]; # 0..1 Organization that issued id (may be just text)
]
```

Changes since DSTU2

Identifier

See the Full Difference for further information

The value SHALL be unique within the defined system and have a consistent meaning wherever it appears. Both system and value are always case sensitive.

The system is a URI that defines a set of identifiers (i.e. how the value is made unique). It might be a specific application or a recognized standard/specification for a set or identifiers or a way of making identifiers unique. FHIR defines some useful or important system URIs directly. Here are some example identifier namespaces:

* http://hl7.org/fhir/sid/us-ssn for United States Social Security Number (SSN) values

* http://ns.electronichealth.net.au/id/hi/ihi/1.0 for Australian Individual Healthcare Identifier (IHI) numbers

* urn:ietf:rfc:3986 for when the value of the identifier is itself a globally unique URI

If the system is a URL, it SHOULD resolve. Resolution might be to a web page that describes the identifier system and/or supports look-up of identifiers. Alternatively, it could be to a NamingSystem resource instance. Resolvable URLs are generally preferred by implementers over non-resolvable URNs, particularly opaque URNs such as OIDs (urn:oid:) or UUIDs (urn:uuid:). If used, OIDs and UUIDs may be registered in the HL7 OID registry and SHOULD be registered if the content is shared or exchanged across institutional boundaries.

It is up to the implementer organization to determine an appropriate URL or URN structure that will avoid collisions and to manage that space (and the resolvability of URLs) over time.

Note that the scope of a given identifier system may extend beyond identifiers that might be captured by a single resource. For example, some systems might draw all "order" identifiers from a single namespace, though some might be used on MedicationRequest while others would appear on ProcedureRequest.

If the identifier value itself is naturally a globally unique URI (e.g. an OID, a UUID, or a URI with no trailing local part), then the system SHALL be "urn:ietf:rfc:3986", and the URI is in the value (OIDs and UUIDs using urn:oid: and urn:uuid: - see examples).

In some cases, the system may not be known - only the value is known (e.g. a simple device that scans a barcode), or the system is known implicitly (simple exchange in a limited context, often driven by barcode readers). In this case, no useful matching may be performed using the value unless the system can be safely inferred by the context. Applications should provide a system wherever possible, as information sharing in a wider context is very likely to arise eventually, and values without a system are inherently limited in use.

In addition to the system (which provides a uniqueness scope) and the value, identifiers may also have a type, which may be useful when a system encounters identifiers with unknown system values. Note, however, that the type of an identifier is not a well-controlled vocabulary with wide variations in practice. The type deals only with general categories of identifiers and SHOULD not be used for codes that correspond 1..1 with the Identifier.system. Some identifiers may fall into multiple categories due to variations in common usage.

The assigner is used to indicate what registry/state/facility/etc. assigned the identifier. As a Reference, the assigner can include just a text description in the display.

Identifier is used in the following places: Reference, Account, ActivityDefinition, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BodySite, Bundle, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, ClinicalImpression, CodeSystem, Communication, CommunicationRequest, Composition, ConceptMap, Condition, Consent, Contract, Coverage, DataElement, DetectedIssue, Device, DeviceComponent, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EligibilityRequest, EligibilityResponse, Encounter, Endpoint, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExpansionProfile, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, Group, GuidanceResponse, HealthcareService, ImagingManifest, ImagingStudy, Immunization, ImmunizationRecommendation, Library, List, Location, Measure, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MessageDefinition, NutritionOrder, Observation, Organization, Patient, PaymentNotice, PaymentReconciliation, Person, PlanDefinition, Practitioner, PractitionerRole, Procedure, ProcedureRequest, ProcessRequest, ProcessResponse, Provenance, Questionnaire, QuestionnaireResponse, ReferralRequest, RelatedPerson, RequestGroup, ResearchStudy, ResearchSubject, RiskAssessment, Schedule, Sequence, ServiceDefinition, Slot, Specimen, StructureDefinition, StructureMap, Substance, SupplyDelivery, SupplyRequest, Task, TestReport, TestScript, ValueSet and VisionPrescription

2.26.0.12 HumanName

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A name of a human with text, parts and usage information.

Names may be changed or repudiated. People may have different names in different contexts. Names may be divided into parts of different type that have variable significance depending on context, though the division into parts is not always significant. With personal names, the different parts may or may not be imbued with some implicit meaning; various cultures associate different importance with the name parts and the degree to which systems SHALL care about name parts around the world varies widely.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
HumanName | ? | | Element | Name of a human - parts and usage Elements defined in Ancestors: id, extension
use | ?!? | 0..1 | code | usual | official | temp | nickname | anonymous | old | maiden NameUse (Required)
text | ? | 0..1 | string | Text representation of the full name
family | ? | 0..1 | string | Family name (often called 'Surname')

given | ? | 0..* | string | Given names (not always 'first'). Includes middle names This repeating element order: Given Names appear in the correct order for presenting the name
prefix | ? | 0..* | string | Parts that come before the name This repeating element order: Prefixes appear in the correct order for presenting the name
suffix | ? | 0..* | string | Parts that come after the name This repeating element order: Suffixes appear in the correct order for presenting the name
period | ? | 0..1 | Period | Time period when name was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | nickname | anonymous | old | maiden -->
 <text value="[string]"/><!-- 0..1 Text representation of the full name -->
 <family value="[string]"/><!-- 0..1 Family name (often called 'Surname') -->
 <given value="[string]"/><!-- 0..* Given names (not always 'first'). Includes middle names -->
 <prefix value="[string]"/><!-- 0..* Parts that come before the name -->
 <suffix value="[string]"/><!-- 0..* Parts that come after the name -->
 <period><!-- 0..1 Period Time period when name was/is in use --></period>
</[name]>
```

JSON Template

```
{
  "resourceType" : "HumanName",
  // from Element: extension
  "use" : "<code>", // usual | official | temp | nickname | anonymous | old | maiden
  "text" : "<string>", // Text representation of the full name
  "family" : "<string>", // Family name (often called 'Surname')
  "given" : ["<string>"], // Given names (not always 'first'). Includes middle names
  "prefix" : ["<string>"], // Parts that come before the name
  "suffix" : ["<string>"], // Parts that come after the name
  "period" : { Period } // Time period when name was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:HumanName.use [ code ]; # 0..1 usual | official | temp | nickname | anonymous | old | maiden
  fhir:HumanName.text [ string ]; # 0..1 Text representation of the full name
  fhir:HumanName.family [ string ]; # 0..1 Family name (often called 'Surname')
  fhir:HumanName.given [ string ], ... ; # 0..* Given names (not always 'first'). Includes middle names
  fhir:HumanName.prefix [ string ], ... ; # 0..* Parts that come before the name
  fhir:HumanName.suffix [ string ], ... ; # 0..* Parts that come after the name
  fhir:HumanName.period [ Period ]; # 0..1 Time period when name was/is in use
]
```

Changes since DSTU2

HumanName
HumanName.family | Max Cardinality changed from * to 1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
HumanName | ? | | Element | Name of a human - parts and usage Elements defined in Ancestors: id, extension
use | ?!? | 0..1 | code | usual | official | temp | nickname | anonymous | old | maiden NameUse (Required)
text | ? | 0..1 | string | Text representation of the full name
family | ? | 0..1 | string | Family name (often called 'Surname')
given | ? | 0..* | string | Given names (not always 'first'). Includes middle names This repeating element order: Given Names appear in the correct order for presenting the name
prefix | ? | 0..* | string | Parts that come before the name This repeating element order: Prefixes appear in the correct order for presenting the name
suffix | ? | 0..* | string | Parts that come after the name This repeating element order: Suffixes appear in the correct order for presenting the name
period | ? | 0..1 | Period | Time period when name was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | nickname | anonymous | old | maiden -->
 <text value="[string]"/><!-- 0..1 Text representation of the full name -->
 <family value="[string]"/><!-- 0..1 Family name (often called 'Surname') -->
 <given value="[string]"/><!-- 0..* Given names (not always 'first'). Includes middle names -->
 <prefix value="[string]"/><!-- 0..* Parts that come before the name -->
 <suffix value="[string]"/><!-- 0..* Parts that come after the name -->
 <period><!-- 0..1 Period Time period when name was/is in use --></period>
</[name]>
```

JSON Template

```
{
  "resourceType" : "HumanName",
  // from Element: extension
```

```
  "use" : "<code>", // usual | official | temp | nickname | anonymous | old | maiden
  "text" : "<string>", // Text representation of the full name
  "family" : "<string>", // Family name (often called 'Surname')
  "given" : ["<string>"], // Given names (not always 'first'). Includes middle names
  "prefix" : ["<string>"], // Parts that come before the name
  "suffix" : ["<string>"], // Parts that come after the name
  "period" : { Period } // Time period when name was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:HumanName.use [ code ]; # 0..1 usual | official | temp | nickname | anonymous | old | maiden
  fhir:HumanName.text [ string ]; # 0..1 Text representation of the full name
  fhir:HumanName.family [ string ]; # 0..1 Family name (often called 'Surname')
  fhir:HumanName.given [ string ], ... ; # 0..* Given names (not always 'first'). Includes middle names
  fhir:HumanName.prefix [ string ], ... ; # 0..* Parts that come before the name
  fhir:HumanName.suffix [ string ], ... ; # 0..* Parts that come after the name
  fhir:HumanName.period [ Period ]; # 0..1 Time period when name was/is in use
]
```

Changes since DSTU2

HumanName

HumanName.family | Max Cardinality changed from * to 1

See the Full Difference for further information

This table summarizes where common parts of a person's name are found.

Name | Example | Destination / Comments

Surname | Smith | Family Name

First name | John | Given Name

Title | Mr | Prefix

Middle Name | Samuel | Subsequent Given Names

Patronymic | bin Osman | Family Name

Multiple family names | Carreño Quiñones | Family Name. See note below about decomposition of family name

Initials | Q. | Given Name as initial ("." recommended)

Nick Name | Jock | Given name, with Use = common

Qualifications | PhD | Suffix

Honorifics | Senior | Suffix

Voorvoegsel / Nobility | van Beethoven | Family Name. See note below about decomposition of family name

For further information, including all W3C International Examples , consult the examples. Note: Implementers should read the name examples for a full understanding of how name works.

The multiple given parts and family name combine to form a single name. Where a person has alternate names that may be used in place of each other (e.g. Nicknames, Aliases), these are different instances of HumanName.

The text element specifies the entire name as it should be represented. This may be provided instead of or as well as specific parts, and can be built from the parts, though the correct order of assembly of the parts is culture dependent: the order of the parts within a given part type has significance and SHALL be observed. The appropriate order between family name and given names depends on culture and context of use. Note that there is an extension for the few times name assembly order is not fixed by the culture.

The given name parts may contain whitespace, though generally they don't. Initials may be used in place of the full name if that is all that is recorded. Systems that operate across cultures should generally rely on the text form for presentation, and use the parts for index/search functionality. For this reasons, applications SHOULD populate the text element for future robustness.

In some cultures (e.g. German, Dutch, Spanish, Portuguese), family names are complex and composed of various parts that may need to be managed separately, e.g. they have differing significance for searching. In these cases, the full family name is populated in family, and a decomposition of the name can be provided using the family extensions own-name, own-prefix, partner-name, partner-prefix, fathers-family and mothers-family.

For robust search, servers should search the parts of a family name independently. E.g. Searching either Carreno or Quinones should match a family name of "Carreno Quinones". HL7 affiliates may make more specific recommendations about how search should work in their specific culture.

HumanName is used in the following places: Organization, Patient, Person, Practitioner and RelatedPerson

2.26.0.13 Address

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

An address expressed using postal conventions (as opposed to GPS or other location definition formats). This data type may be used to convey addresses for use in delivering mail as well as for visiting locations which might not be valid for mail delivery. There are a variety of postal address formats defined around the world.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Address | ? | | Element | An address expressed using postal conventions (as opposed to GPS or other location definition formats) Elements defined in Ancestors: id, extension

use | ?!? | 0..1 | code | home | work | temp | old - purpose of this address AddressUse (Required)

type | ? | 0..1 | code | postal | physical | both AddressType (Required)

text | ? | 0..1 | string | Text representation of the address

line | ? | 0..* | string | Street name, number, direction & P.O. Box etc. This repeating element order: The order in which lines should appear in an address label

city | ? | 0..1 | string | Name of city, town etc.

district | ? | 0..1 | string | District name (aka county)

state | ? | 0..1 | string | Sub-unit of country (abbreviations ok)

postalCode | ? | 0..1 | string | Postal code for area

country | ? | 0..1 | string | Country (e.g. can be ISO 3166 2 or 3 letter code)

period | ? | 0..1 | Period | Time period when address was/is in use

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old - purpose of this address -->
 <type value="[code]"/><!-- 0..1 postal | physical | both -->
 <text value="[string]"/><!-- 0..1 Text representation of the address -->
 <line value="[string]"/><!-- 0..* Street name, number, direction & P.O. Box etc. -->
 <city value="[string]"/><!-- 0..1 Name of city, town etc. -->
 <district value="[string]"/><!-- 0..1 District name (aka county) -->
 <state value="[string]"/><!-- 0..1 Sub-unit of country (abbreviations ok) -->
 <postalCode value="[string]"/><!-- 0..1 Postal code for area -->
 <country value="[string]"/><!-- 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code) -->
 <period><!-- 0..1 Period Time period when address was/is in use --></period>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Address",
  // from Element: extension
  "use" : "<code>", // home | work | temp | old - purpose of this address
  "type" : "<code>", // postal | physical | both
  "text" : "<string>", // Text representation of the address
  "line" : ["<string>"], // Street name, number, direction & P.O. Box etc.
  "city" : "<string>", // Name of city, town etc.
  "district" : "<string>", // District name (aka county)
  "state" : "<string>", // Sub-unit of country (abbreviations ok)
  "postalCode" : "<string>", // Postal code for area
  "country" : "<string>", // Country (e.g. can be ISO 3166 2 or 3 letter code)
  "period" : { Period } // Time period when address was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Address.use [ code ]; # 0..1 home | work | temp | old - purpose of this address
  fhir:Address.type [ code ]; # 0..1 postal | physical | both
  fhir:Address.text [ string ]; # 0..1 Text representation of the address
  fhir:Address.line [ string ], ... ; # 0..* Street name, number, direction & P.O. Box etc.
  fhir:Address.city [ string ]; # 0..1 Name of city, town etc.
  fhir:Address.district [ string ]; # 0..1 District name (aka county)
  fhir:Address.state [ string ]; # 0..1 Sub-unit of country (abbreviations ok)
  fhir:Address.postalCode [ string ]; # 0..1 Postal code for area
  fhir:Address.country [ string ]; # 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code)
  fhir:Address.period [ Period ]; # 0..1 Time period when address was/is in use
]
```

Changes since DSTU2

Address

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Address | ? | | Element | An address expressed using postal conventions (as opposed to GPS or other location definition formats) Elements defined in Ancestors: id, extension

use | ?!? | 0..1 | code | home | work | temp | old - purpose of this address AddressUse (Required)
type | ? | 0..1 | code | postal | physical | both AddressType (Required)
text | ? | 0..1 | string | Text representation of the address
line | ? | 0..* | string | Street name, number, direction & P.O. Box etc. This repeating element order: The order in which lines should appear in an address label
city | ? | 0..1 | string | Name of city, town etc.
district | ? | 0..1 | string | District name (aka county)
state | ? | 0..1 | string | Sub-unit of country (abbreviations ok)
postalCode | ? | 0..1 | string | Postal code for area
country | ? | 0..1 | string | Country (e.g. can be ISO 3166 2 or 3 letter code)
period | ? | 0..1 | Period | Time period when address was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old - purpose of this address -->
 <type value="[code]"/><!-- 0..1 postal | physical | both -->
 <text value="[string]"/><!-- 0..1 Text representation of the address -->
 <line value="[string]"/><!-- 0..* Street name, number, direction & P.O. Box etc. -->
 <city value="[string]"/><!-- 0..1 Name of city, town etc. -->
 <district value="[string]"/><!-- 0..1 District name (aka county) -->
 <state value="[string]"/><!-- 0..1 Sub-unit of country (abbreviations ok) -->
 <postalCode value="[string]"/><!-- 0..1 Postal code for area -->
 <country value="[string]"/><!-- 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code) -->
 <period><!-- 0..1 Period Time period when address was/is in use --></period>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Address",
  // from Element: extension
  "use" : "<code>", // home | work | temp | old - purpose of this address
  "type" : "<code>", // postal | physical | both
  "text" : "<string>", // Text representation of the address
  "line" : ["<string>"], // Street name, number, direction & P.O. Box etc.
  "city" : "<string>", // Name of city, town etc.
  "district" : "<string>", // District name (aka county)
  "state" : "<string>", // Sub-unit of country (abbreviations ok)
  "postalCode" : "<string>", // Postal code for area
  "country" : "<string>", // Country (e.g. can be ISO 3166 2 or 3 letter code)
  "period" : { Period } // Time period when address was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Address.use [ code ]; # 0..1 home | work | temp | old - purpose of this address
  fhir:Address.type [ code ]; # 0..1 postal | physical | both
  fhir:Address.text [ string ]; # 0..1 Text representation of the address
  fhir:Address.line [ string ], ... ; # 0..* Street name, number, direction & P.O. Box etc.
  fhir:Address.city [ string ]; # 0..1 Name of city, town etc.
  fhir:Address.district [ string ]; # 0..1 District name (aka county)
  fhir:Address.state [ string ]; # 0..1 Sub-unit of country (abbreviations ok)
  fhir:Address.postalCode [ string ]; # 0..1 Postal code for area
  fhir:Address.country [ string ]; # 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code)
  fhir:Address.period [ Period ]; # 0..1 Time period when address was/is in use
]
```

Changes since DSTU2

Address

See the Full Difference for further information

The text element specifies the entire address as it should be represented. This may be provided instead of or as well as the specific parts. Applications updating an address SHALL ensure either that the text and the parts are in agreement, or that only one of the two is present.

Constraints

Address is used in the following places: Claim, ExplanationOfBenefit, Location, Organization, Patient, Person, Practitioner and RelatedPerson

2.26.0.14 ContactPoint

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Details for all kinds of technology-mediated contact points for a person or organization, including telephone, email, etc.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

ContactPoint | ?I | | Element | Details of a Technology mediated contact point (phone, fax, email, etc.) + A system is required if a value is provided.

Elements defined in Ancestors: id, extension

system | ?I | 0..1 | code | phone | fax | email | pager | url | sms | other ContactPointSystem (Required)

value | ? | 0..1 | string | The actual contact point details

use | ?!? | 0..1 | code | home | work | temp | old | mobile - purpose of this contact point ContactPointUse (Required)

rank | ? | 0..1 | positiveInt | Specify preferred order of use (1 = highest)

period | ? | 0..1 | Period | Time period when the contact point was/is in use

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[code]"/><!--  0..1 phone | fax | email | pager | url | sms | other -->
 <value value="[string]"/><!-- 0..1 The actual contact point details -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | mobile - purpose of this contact point -->
 <rank value="[positiveInt]"/><!-- 0..1 Specify preferred order of use (1 = highest) -->
 <period><!-- 0..1 Period Time period when the contact point was/is in use --></period>
</[name]>
```

JSON Template

```
{
  "resourceType" : "ContactPoint",
  // from Element: extension
  "system" : "<code>", // C? phone | fax | email | pager | url | sms | other
  "value" : "<string>", // The actual contact point details
  "use" : "<code>", // home | work | temp | old | mobile - purpose of this contact point
  "rank" : "<positiveInt>", // Specify preferred order of use (1 = highest)
  "period" : { Period } // Time period when the contact point was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactPoint.system [ code ]; # 0..1 phone | fax | email | pager | url | sms | other
  fhir:ContactPoint.value [ string ]; # 0..1 The actual contact point details
  fhir:ContactPoint.use [ code ]; # 0..1 home | work | temp | old | mobile - purpose of this contact point
  fhir:ContactPoint.rank [ positiveInt ]; # 0..1 Specify preferred order of use (1 = highest)
  fhir:ContactPoint.period [ Period ]; # 0..1 Time period when the contact point was/is in use
]
```

Changes since DSTU2

ContactPoint

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

ContactPoint | ?I | | Element | Details of a Technology mediated contact point (phone, fax, email, etc.) + A system is required if a value is provided.

Elements defined in Ancestors: id, extension

system | ?I | 0..1 | code | phone | fax | email | pager | url | sms | other ContactPointSystem (Required)

value | ? | 0..1 | string | The actual contact point details

use | ?!? | 0..1 | code | home | work | temp | old | mobile - purpose of this contact point ContactPointUse (Required)

rank | ? | 0..1 | positiveInt | Specify preferred order of use (1 = highest)

period | ? | 0..1 | Period | Time period when the contact point was/is in use

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[code]"/><!--  0..1 phone | fax | email | pager | url | sms | other -->
 <value value="[string]"/><!-- 0..1 The actual contact point details -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | mobile - purpose of this contact point -->
 <rank value="[positiveInt]"/><!-- 0..1 Specify preferred order of use (1 = highest) -->
 <period><!-- 0..1 Period Time period when the contact point was/is in use --></period>
</[name]>
```

JSON Template

```
{
```

```
  "resourceType" : "ContactPoint",
  // from Element: extension
  "system" : "<code>", // C? phone | fax | email | pager | url | sms | other
  "value" : "<string>", // The actual contact point details
  "use" : "<code>", // home | work | temp | old | mobile - purpose of this contact point
  "rank" : "<positiveInt>", // Specify preferred order of use (1 = highest)
  "period" : { Period } // Time period when the contact point was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactPoint.system [ code ]; # 0..1 phone | fax | email | pager | url | sms | other
  fhir:ContactPoint.value [ string ]; # 0..1 The actual contact point details
  fhir:ContactPoint.use [ code ]; # 0..1 home | work | temp | old | mobile - purpose of this contact point
  fhir:ContactPoint.rank [ positiveInt ]; # 0..1 Specify preferred order of use (1 = highest)
  fhir:ContactPoint.period [ Period ]; # 0..1 Time period when the contact point was/is in use
]
```

Changes since DSTU2

ContactPoint

See the Full Difference for further information

If capturing a phone, fax or similar contact point, the value should be a properly formatted telephone number according to ITU-T E.123 . However, this is frequently not possible due to legacy data and/or clerical practices when recording contact details. For this reason, phone, fax, pager, and email addresses are not handled as formal URLs. For other kinds of contact points, the system is "other" and the value SHOULD be a URL so that its use can be determined automatically. Typical URL schemes used in the value are http(s): for web addresses, and URL schemes for various kinds of messaging systems. If the value is not a URL, then human interpretation will be required.

The rank element can be used to specify a preference for the order in which a set of contacts is used. Contacts are ranked with lower values coming before higher values. Note that rank does not necessarily follow the order in which the contacts are represented in the instance.

Constraints

* cpt-2: A system is required if a value is provided. (expression : value.empty() or system.exists())

ContactPoint is used in the following places: Device, Endpoint, HealthcareService, Location, MessageHeader, Organization, Patient, Person, Practitioner, PractitionerRole, RelatedPerson and Subscription

2.26.0.15 Timing

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Specifies an event that may occur multiple times. Timing schedules are used for specifying when events are expected or requested to occur, and may also be used to represent a schedule that was followed for past events. A Timing schedule can be a list of events and/or criteria for when the event happens, which can be expressed in a structured form and/or as a code. When both event and a repeating specification are provided, the list of events should be understood as an interpretation of the information in the repeat structure.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Timing | ? | | Element | A timing schedule that specifies an event that may occur multiple times Elements defined in Ancestors: id, extension

event | ? | 0..* | dateTime | When the event occurs

repeat | ?I | 0..1 | Element | When the event is to occur + If there's an offset, there must be a when (and not C, CM, CD, CV) + period SHALL be a non-negative value + If there's a periodMax, there must be a period + If there's a durationMax, there must be a duration + If there's a countMax, there must be a count + if there's a duration, there needs to be duration units + If there's a timeOfDay, there cannot be be a when, or vice versa + if there's a period, there needs to be period units + duration SHALL be a non-negative value

bounds[x] | ? | 0..1 | | Length/Range of lengths, or (Start and/or end) limits

boundsDuration | | | Duration

boundsRange | | | Range

boundsPeriod | | | Period

count | ? | 0..1 | integer | Number of times to repeat

countMax | ? | 0..1 | integer | Maximum number of times to repeat

duration | ? | 0..1 | decimal | How long when it happens

durationMax | ? | 0..1 | decimal | How long when it happens (Max)

durationUnit | ? | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)

frequency | ? | 0..1 | integer | Event occurs frequency times per period
frequencyMax | ? | 0..1 | integer | Event occurs up to frequencyMax times per period
period | ? | 0..1 | decimal | Event occurs frequency times per period
periodMax | ? | 0..1 | decimal | Upper limit of period (3-4 hours)
periodUnit | ? | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
dayOfWeek | ? | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
timeOfDay | ? | 0..* | time | Time of day for action
when | ? | 0..* | code | Regular life events the event is tied to EventTiming (Required)
offset | ? | 0..1 | unsignedInt | Minutes from event (before or after)
code | ? | 0..1 | CodeableConcept | BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H + TimingAbbreviation (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <event value="[dateTime]"/><!-- 0..* When the event occurs -->
 <repeat>  <!-- 0..1 When the event is to occur -->
  <bounds[x]><!-- 0..1 Duration|Range|Period Length/Range of lengths, or (Start and/or end) limits --></bounds[x]>
  <count value="[integer]"/><!-- 0..1 Number of times to repeat -->
  <countMax value="[integer]"/><!-- 0..1 Maximum number of times to repeat -->
  <duration value="[decimal]"/><!-- 0..1 How long when it happens -->
  <durationMax value="[decimal]"/><!-- 0..1 How long when it happens (Max) -->
  <durationUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <frequency value="[integer]"/><!-- 0..1 Event occurs frequency times per period -->
  <frequencyMax value="[integer]"/><!-- 0..1 Event occurs up to frequencyMax times per period -->
  <period value="[decimal]"/><!-- 0..1 Event occurs frequency times per period -->
  <periodMax value="[decimal]"/><!-- 0..1 Upper limit of period (3-4 hours) -->
  <periodUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <dayOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <timeOfDay value="[time]"/><!-- 0..* Time of day for action -->
  <when value="[code]"/><!-- 0..* Regular life events the event is tied to -->
  <offset value="[unsignedInt]"/><!-- 0..1 Minutes from event (before or after) -->
 </repeat>
 <code><!-- 0..1 CodeableConcept BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H + --></code>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Timing",
  // from Element: extension
  "event" : ["<dateTime>"], // When the event occurs
  "repeat" : { // When the event is to occur
    // bounds[x]: Length/Range of lengths, or (Start and/or end) limits. One of these 3:
    "boundsDuration" : { Duration },
    "boundsRange" : { Range },
    "boundsPeriod" : { Period },
    "count" : <integer>, // Number of times to repeat
    "countMax" : <integer>, // Maximum number of times to repeat
    "duration" : <decimal>, // How long when it happens
    "durationMax" : <decimal>, // How long when it happens (Max)
    "durationUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "frequency" : <integer>, // Event occurs frequency times per period
    "frequencyMax" : <integer>, // Event occurs up to frequencyMax times per period
    "period" : <decimal>, // Event occurs frequency times per period
    "periodMax" : <decimal>, // Upper limit of period (3-4 hours)
    "periodUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "dayOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "timeOfDay" : ["<time>"], // Time of day for action
    "when" : ["<code>"], // Regular life events the event is tied to
    "offset" : "<unsignedInt>" // Minutes from event (before or after)
  },
  "code" : { CodeableConcept } // BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H +
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Timing.event [ dateTime ], ... ; # 0..* When the event occurs
  fhir:Timing.repeat [ # 0..1 When the event is to occur
    # Timing.repeat.bounds[x] : 0..1 Length/Range of lengths, or (Start and/or end) limits. One of these 3
      fhir:Timing.repeat.boundsDuration [ Duration ]
      fhir:Timing.repeat.boundsRange [ Range ]
      fhir:Timing.repeat.boundsPeriod [ Period ]
    fhir:Timing.repeat.count [ integer ]; # 0..1 Number of times to repeat
    fhir:Timing.repeat.countMax [ integer ]; # 0..1 Maximum number of times to repeat
    fhir:Timing.repeat.duration [ decimal ]; # 0..1 How long when it happens
    fhir:Timing.repeat.durationMax [ decimal ]; # 0..1 How long when it happens (Max)
    fhir:Timing.repeat.durationUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.frequency [ integer ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.frequencyMax [ integer ]; # 0..1 Event occurs up to frequencyMax times per period
    fhir:Timing.repeat.period [ decimal ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.periodMax [ decimal ]; # 0..1 Upper limit of period (3-4 hours)
```

```
    fhir:Timing.repeat.periodUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.dayOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Timing.repeat.timeOfDay [ time ], ... ; # 0..* Time of day for action
    fhir:Timing.repeat.when [ code ], ... ; # 0..* Regular life events the event is tied to
    fhir:Timing.repeat.offset [ unsignedInt ]; # 0..1 Minutes from event (before or after)
  ];
  fhir:Timing.code [ CodeableConcept ]; # 0..1 BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H +
]
```

Changes since DSTU2

Timing
Timing.repeat.bounds[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}, Add Duration
Timing.repeat.countMax | Added Element
Timing.repeat.durationUnit | Renamed from durationUnits to durationUnit
Timing.repeat.periodUnit | Renamed from periodUnits to periodUnit
Timing.repeat.dayOfWeek | Added Element
Timing.repeat.timeOfDay | Added Element
Timing.repeat.when | Max Cardinality changed from 1 to *
Timing.repeat.offset | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Timing | ? | | Element | A timing schedule that specifies an event that may occur multiple times Elements defined in Ancestors: id, extension
event | ? | 0..* | dateTime | When the event occurs
repeat | ?I | 0..1 | Element | When the event is to occur + If there's an offset, there must be a when (and not C, CM, CD, CV) + period SHALL be a non-negative value + If there's a periodMax, there must be a period + If there's a durationMax, there must be a duration + If there's a countMax, there must be a count + if there's a duration, there needs to be duration units + If there's a timeOfDay, there cannot be be a when, or vice versa + if there's a period, there needs to be period units + duration SHALL be a non-negative value
bounds[x] | ? | 0..1 | | Length/Range of lengths, or (Start and/or end) limits
boundsDuration | | | Duration
boundsRange | | | Range
boundsPeriod | | | Period
count | ? | 0..1 | integer | Number of times to repeat
countMax | ? | 0..1 | integer | Maximum number of times to repeat
duration | ? | 0..1 | decimal | How long when it happens
durationMax | ? | 0..1 | decimal | How long when it happens (Max)
durationUnit | ? | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
frequency | ? | 0..1 | integer | Event occurs frequency times per period
frequencyMax | ? | 0..1 | integer | Event occurs up to frequencyMax times per period
period | ? | 0..1 | decimal | Event occurs frequency times per period
periodMax | ? | 0..1 | decimal | Upper limit of period (3-4 hours)
periodUnit | ? | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
dayOfWeek | ? | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
timeOfDay | ? | 0..* | time | Time of day for action
when | ? | 0..* | code | Regular life events the event is tied to EventTiming (Required)
offset | ? | 0..1 | unsignedInt | Minutes from event (before or after)
code | ? | 0..1 | CodeableConcept | BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H + TimingAbbreviation (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <event value="[dateTime]"/><!-- 0..* When the event occurs -->
 <repeat>  <!-- 0..1 When the event is to occur -->
  <bounds[x]><!-- 0..1 Duration|Range|Period Length/Range of lengths, or (Start and/or end) limits --></bounds[x]>
  <count value="[integer]"/><!-- 0..1 Number of times to repeat -->
  <countMax value="[integer]"/><!-- 0..1 Maximum number of times to repeat -->
  <duration value="[decimal]"/><!-- 0..1 How long when it happens -->
  <durationMax value="[decimal]"/><!-- 0..1 How long when it happens (Max) -->
  <durationUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <frequency value="[integer]"/><!-- 0..1 Event occurs frequency times per period -->
  <frequencyMax value="[integer]"/><!-- 0..1 Event occurs up to frequencyMax times per period -->
  <period value="[decimal]"/><!-- 0..1 Event occurs frequency times per period -->
  <periodMax value="[decimal]"/><!-- 0..1 Upper limit of period (3-4 hours) -->
  <periodUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <dayOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <timeOfDay value="[time]"/><!-- 0..* Time of day for action -->
  <when value="[code]"/><!-- 0..* Regular life events the event is tied to -->
  <offset value="[unsignedInt]"/><!-- 0..1 Minutes from event (before or after) -->
 </repeat>
 <code><!-- 0..1 CodeableConcept BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H + --></code>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Timing",
  // from Element: extension
```

```
  "event" : ["<dateTime>"], // When the event occurs
  "repeat" : { // When the event is to occur
    // bounds[x]: Length/Range of lengths, or (Start and/or end) limits. One of these 3:
    "boundsDuration" : { Duration },
    "boundsRange" : { Range },
    "boundsPeriod" : { Period },
    "count" : <integer>, // Number of times to repeat
    "countMax" : <integer>, // Maximum number of times to repeat
    "duration" : <decimal>, // How long when it happens
    "durationMax" : <decimal>, // How long when it happens (Max)
    "durationUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "frequency" : <integer>, // Event occurs frequency times per period
    "frequencyMax" : <integer>, // Event occurs up to frequencyMax times per period
    "period" : <decimal>, // Event occurs frequency times per period
    "periodMax" : <decimal>, // Upper limit of period (3-4 hours)
    "periodUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "dayOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "timeOfDay" : ["<time>"], // Time of day for action
    "when" : ["<code>"], // Regular life events the event is tied to
    "offset" : "<unsignedInt>" // Minutes from event (before or after)
  },
  "code" : { CodeableConcept } // BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H +
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Timing.event [ dateTime ], ... ; # 0..* When the event occurs
  fhir:Timing.repeat [ # 0..1 When the event is to occur
    # Timing.repeat.bounds[x] : 0..1 Length/Range of lengths, or (Start and/or end) limits. One of these 3
      fhir:Timing.repeat.boundsDuration [ Duration ]
      fhir:Timing.repeat.boundsRange [ Range ]
      fhir:Timing.repeat.boundsPeriod [ Period ]
    fhir:Timing.repeat.count [ integer ]; # 0..1 Number of times to repeat
    fhir:Timing.repeat.countMax [ integer ]; # 0..1 Maximum number of times to repeat
    fhir:Timing.repeat.duration [ decimal ]; # 0..1 How long when it happens
    fhir:Timing.repeat.durationMax [ decimal ]; # 0..1 How long when it happens (Max)
    fhir:Timing.repeat.durationUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.frequency [ integer ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.frequencyMax [ integer ]; # 0..1 Event occurs up to frequencyMax times per period
    fhir:Timing.repeat.period [ decimal ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.periodMax [ decimal ]; # 0..1 Upper limit of period (3-4 hours)
    fhir:Timing.repeat.periodUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.dayOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Timing.repeat.timeOfDay [ time ], ... ; # 0..* Time of day for action
    fhir:Timing.repeat.when [ code ], ... ; # 0..* Regular life events the event is tied to
    fhir:Timing.repeat.offset [ unsignedInt ]; # 0..1 Minutes from event (before or after)
  ];
  fhir:Timing.code [ CodeableConcept ]; # 0..1 BID | TID | QID | AM | PM | QD | QOD | Q4H | Q6H +
]
```

Changes since DSTU2

Timing
Timing.repeat.bounds[x] | Remove Quantity{http://hl7.org/fhir/StructureDefinition/Duration}, Add Duration
Timing.repeat.countMax | Added Element
Timing.repeat.durationUnit | Renamed from durationUnits to durationUnit
Timing.repeat.periodUnit | Renamed from periodUnits to periodUnit
Timing.repeat.dayOfWeek | Added Element
Timing.repeat.timeOfDay | Added Element
Timing.repeat.when | Max Cardinality changed from 1 to *
Timing.repeat.offset | Added Element

See the Full Difference for further information

If the timing schedule has repeating criteria, the repeat can occur a given number of times per the specified duration or in relation to some repeating real world event. If no end condition is specified, the schedule will terminate on some criteria that are expressed elsewhere.

This table summarizes some common uses of the Timing Data Type criteria.

description | duration | durationUnit | frequency | frequencyMax | period | periodUnit | periodMax | Day of Week | Time Of Day | when | offset | bounds[x] | count
Every 8 hours | | | 1 | | 8 | h
Every 7 days | | | 1 | | 7 | d
3 times a day | | | 3 | | 1 | d
3-4 times a day | | | 3 | 4 | 1 | d
Every 4-6 hours | | | 1 | | 4 | h | 6
Every 21 days for 1 hour | 1 | hr | 1 | | 21 | d
Three times a week for ½ hour | 0.5 | hr | 3 | | 1 | wk
With breakfast | | | | | | | | | | CM
For 5 minutes, 10 minutes before meals | 5 | min | | | | | | AC | 10
1 tablet 3 times daily, 30 minutes before meals | | | 3 | | 1 | d | | | AC | 30
BID, 30 mins before meal, for next 10 days | | | 2 | | 1 | d | | | AC | 30 | Duration = 10 days

TID, for 14 days | | | 3 | | 1 | d | | | | | | Duration = 14 days
BID, start on 7/1/2015 at 1:00 PM | | | 2 | | 1 | d | | | | | | Period.start = 2015-07-01T13:00:00
Mon, Wed, Fri Morning | | | 1 | | 1 | d | | mon | wed | fri | | MORN
Every day at 10am | | | 1 | | 1 | d | | | 10:00
Take once, at any time | | | | | | | | | | | | | | 1

Many systems avoid the complexity of the Timing structure by using a text field for these things (e.g. "Dosage Text"). Those systems do not use the Timing data type. Other systems use a set of 'common' codes - including, but usually not limited to, widely understood acronyms such as "BID". If a code is provided, the code is understood to be a complete statement of whatever is specified in the structured timing data (except for Timing.repeat.bounds, which applies to the code), and either the code or the data may be used to interpret the Timing. A structured timing specification SHOULD be provided whenever possible, unless the code is BID, TID, QID, AM or PM, which have a ubiquitous meaning.

This table shows the relationship between the codes provided as part of the base specification, and the structured data portions of the Timing type:

description | duration | durationUnit | frequency | frequencyMax | period | periodUnit | periodMax | when | bounds[x]
QOD | | | 1 | | 2 | d
QD | | | 1 | | 1 | d
BID | | | 2 | | 1 | d
TID | | | 3 | | 1 | d
QID | | | 4 | | 1 | d
Q4H | | | 1 | | 4 | h
Q6H | | | 1 | | 6 | h
AM | | | 1 | | 1 | d | | MORN
PM | | | 1 | | 1 | d | | AFT or EVE

These codes SHALL be understood as having the formal meanings documented in this table. Note that BID, etc. are defined as 'at institutionally specified times'. For example, an institution may choose that BID is "always at 7am and 6pm". If it is inappropriate for this choice to be made, the code BID should not be used. Instead, a distinct organization-specific code should be used in place of the HL7-defined BID code and/or a structured representation should be used (in this case, timeOfDay).

Constraints

* tim-1: On Timing.repeat: if there's a duration, there needs to be duration units (expression on Timing.repeat: duration.empty() or durationUnit.exists())

* tim-2: On Timing.repeat: if there's a period, there needs to be period units (expression on Timing.repeat: period.empty() or periodUnit.exists())

* tim-4: On Timing.repeat: duration SHALL be a non-negative value (expression on Timing.repeat: duration.exists() implies duration >= 0)

* tim-5: On Timing.repeat: period SHALL be a non-negative value (expression on Timing.repeat: period.exists() implies period >= 0)

* tim-6: On Timing.repeat: If there's a periodMax, there must be a period (expression on Timing.repeat: periodMax.empty() or period.exists())

* tim-7: On Timing.repeat: If there's a durationMax, there must be a duration (expression on Timing.repeat: durationMax.empty() or duration.exists())

* tim-8: On Timing.repeat: If there's a countMax, there must be a count (expression on Timing.repeat: countMax.empty() or count.exists())

* tim-9: On Timing.repeat: If there's an offset, there must be a when (and not C, CM, CD, CV) (expression on Timing.repeat: offset.empty() or (when.exists() and ((when in ('C' | 'CM' | 'CD' | 'CV')).not())))

* tim-10: On Timing.repeat: If there's a timeOfDay, there cannot be be a when, or vice versa (expression on Timing.repeat: timeOfDay.empty() or when.empty())

Note that these constraints still allow for non-sensical timing specifications such as "Once per day at 2:00 and 4:00" or "every 3 days on Friday". Implementers must take care to ensure that their configuration and data collection designs do not lead to these non-interpretable timing specifications. The elements dayOfWeek, timeOfDay, and when are particularly likely to be at issue here.

Timing is used in the following places: ActivityDefinition, CarePlan, ChargeItem, DeviceMetric, DeviceRequest, DeviceUseStatement, NutritionOrder, PlanDefinition, ProcedureRequest, RequestGroup, SupplyDelivery and SupplyRequest

2.26.0.16 Signature

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Signature holds an electronic representation of a signature and its supporting context in a FHIR accessible form. The signature may either be a cryptographic type (XML DigSig or a JWT), which is able to provide non-repudiation proof, or it may be a graphical image that represents a signature or a signature process.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Signature | | | Element | A digital Signature - XML DigSig, JWT, Graphical image of signature, etc. Elements defined in Ancestors: id, extension
type | ? | 1..* | Coding | Indication of the reason the entity signed the object(s) Signature Type Codes (Preferred)
when | ? | 1..1 | instant | When the signature was created
who[x] | ? | 1..1 | | Who signed
whoUri | | | uri
whoReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
onBehalfOf[x] | ? | 0..1 | | The party represented
onBehalfOfUri | | | uri
onBehalfOfReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
contentType | ? | 0..1 | code | The technical format of the signature MimeType (Required)
blob | | 0..1 | base64Binary | The actual signature content (XML DigSig. JWT, picture, etc.)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type><!-- 1..* Coding Indication of the reason the entity signed the object(s) --></type>
 <when value="[instant]"/><!-- 1..1 When the signature was created -->
 <who[x]><!-- 1..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) Who signed --></who[x]>
 <onBehalfOf[x]><!-- 0..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) The party represented --></onBehalfOf[x]>
 <contentType value="[code]"/><!-- 0..1 The technical format of the signature  -->
 <blob value="[base64Binary]"/><!-- 0..1 The actual signature content (XML DigSig. JWT, picture, etc.) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : [{ Coding }], // R!  Indication of the reason the entity signed the object(s)
  "when" : "<instant>", // R!  When the signature was created
  // who[x]: Who signed. One of these 2:
  "whoUri" : "<uri>",
  "whoReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) },
  // onBehalfOf[x]: The party represented. One of these 2:
  "onBehalfOfUri" : "<uri>",
  "onBehalfOfReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) },
  "contentType" : "<code>", // The technical format of the signature
  "blob" : "<base64Binary>" // The actual signature content (XML DigSig. JWT, picture, etc.)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Signature.type [ Coding ], ... ; # 1..* Indication of the reason the entity signed the object(s)
  fhir:Signature.when [ instant ]; # 1..1 When the signature was created
  # Signature.who[x] : 1..1 Who signed. One of these 2
    fhir:Signature.whoUri [ uri ]
    fhir:Signature.whoReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
  # Signature.onBehalfOf[x] : 0..1 The party represented. One of these 2
    fhir:Signature.onBehalfOfUri [ uri ]
    fhir:Signature.onBehalfOfReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
  fhir:Signature.contentType [ code ]; # 0..1 The technical format of the signature
  fhir:Signature.blob [ base64Binary ]; # 0..1 The actual signature content (XML DigSig. JWT, picture, etc.)
]
```

Changes since DSTU2

Signature
Signature.onBehalfOf[x] | Added Element
Signature.contentType | Min Cardinality changed from 1 to 0
Signature.blob | Min Cardinality changed from 1 to 0

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Signature | | | Element | A digital Signature - XML DigSig, JWT, Graphical image of signature, etc. Elements defined in Ancestors: id, extension
type | ? | 1..* | Coding | Indication of the reason the entity signed the object(s) Signature Type Codes (Preferred)
when | ? | 1..1 | instant | When the signature was created

who[x] | ? | 1..1 | | Who signed
whoUri | | | uri
whoReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
onBehalfOf[x] | ? | 0..1 | | The party represented
onBehalfOfUri | | | uri
onBehalfOfReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
contentType | ? | 0..1 | code | The technical format of the signature MimeType (Required)
blob | | 0..1 | base64Binary | The actual signature content (XML DigSig. JWT, picture, etc.)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type><!-- 1..* Coding Indication of the reason the entity signed the object(s) --></type>
 <when value="[instant]"/><!-- 1..1 When the signature was created -->
 <who[x]><!-- 1..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) Who signed --></who[x]>
 <onBehalfOf[x]><!-- 0..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) The party represented --></onBehalfOf[x]>
 <contentType value="[code]"/><!-- 0..1 The technical format of the signature  -->
 <blob value="[base64Binary]"/><!-- 0..1 The actual signature content (XML DigSig. JWT, picture, etc.) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : [{ Coding }], // R!  Indication of the reason the entity signed the object(s)
  "when" : "<instant>", // R!  When the signature was created
  // who[x]: Who signed. One of these 2:
  "whoUri" : "<uri>",
  "whoReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) },
  // onBehalfOf[x]: The party represented. One of these 2:
  "onBehalfOfUri" : "<uri>",
  "onBehalfOfReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
   Organization) },
  "contentType" : "<code>", // The technical format of the signature
  "blob" : "<base64Binary>" // The actual signature content (XML DigSig. JWT, picture, etc.)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Signature.type [ Coding ], ... ; # 1..* Indication of the reason the entity signed the object(s)
  fhir:Signature.when [ instant ]; # 1..1 When the signature was created
  # Signature.who[x] : 1..1 Who signed. One of these 2
    fhir:Signature.whoUri [ uri ]
    fhir:Signature.whoReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
  # Signature.onBehalfOf[x] : 0..1 The party represented. One of these 2
    fhir:Signature.onBehalfOfUri [ uri ]
    fhir:Signature.onBehalfOfReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
  fhir:Signature.contentType [ code ]; # 0..1 The technical format of the signature
  fhir:Signature.blob [ base64Binary ]; # 0..1 The actual signature content (XML DigSig. JWT, picture, etc.)
]
```

Changes since DSTU2

Signature
Signature.onBehalfOf[x] | Added Element
Signature.contentType | Min Cardinality changed from 1 to 0
Signature.blob | Min Cardinality changed from 1 to 0

See the Full Difference for further information

Constraints

Note: One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

2.26.0.16.1 XML Signature rules

When the signature is an XML Digital Signature (contentType = application/signature+xml), the following rules apply:

* The Signature.blob is base64 encoded XML-Signature

* The XML-Signature is a Detached Signature (where the content that is signed is separate from the signature itself)

* The Signature SHOULD conform to XAdES-X-L for support of Long Term signatures. The XAdES-X-L specification adds the timestamp of the signing, inclusion of the signing certificate, and statement of revocation

* When FHIR Resources are signed, the signature is across the Canonical XML form of the resource(s)

* The Signature SHOULD use the hashing algorithm sha256. Signature validation policy will apply to the signature and determine acceptability

* The Signature SHALL include a "CommitmentTypeIndication" element for the Purpose(s) of Signature. The Purpose can be the action being attested to, or the role associated with the signature. The value shall come from ASTM E1762-95(2013). The Signature.type shall contain the same values as the CommitmentTypeIndication element.

There are three levels of signature verification:

* Verifying that the Digital Signature block itself has integrity through verifying the signature across the XML-Signature.

* Confirming that the signer was authentic, not revoked, and appropriate to the signature purpose.

* Confirming that the signed content of interest is unmodified using the hash algorithm.

Deviations from these guidelines would need to be expressed in site policy, and would be enumerated in the XML-Signature block. For example, some environments may choose a different XAdES profile, hashing algorithm, policy identifier, or signature purpose vocabulary.

Signature is used in the following places: Bundle, Contract and Provenance

2.26.0.17 Annotation

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A text note which also contains information about who made the statement and when.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Annotation | | | Element | Text node with attribution Elements defined in Ancestors: id, extension
author[x] | ? | 0..1 | | Individual responsible for the annotation
authorReference | | | Reference(Practitioner | Patient | RelatedPerson)
authorString | | | string
time | ? | 0..1 | dateTime | When the annotation was made
text | | 1..1 | string | The annotation - text content
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <author[x]><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson)|string Individual responsible for the annotation
    --></author[x]>
 <time value="[dateTime]"/><!-- 0..1 When the annotation was made -->
 <text value="[string]"/><!-- 1..1 The annotation  - text content -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  // author[x]: Individual responsible for the annotation. One of these 2:
  "authorReference" : { Reference(Practitioner|Patient|RelatedPerson) },
  "authorString" : "<string>",
  "time" : "<dateTime>", // When the annotation was made
  "text" : "<string>" // R!  The annotation  - text content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  # Annotation.author[x] : 0..1 Individual responsible for the annotation. One of these 2
    fhir:Annotation.authorReference [ Reference(Practitioner|Patient|RelatedPerson) ]
    fhir:Annotation.authorString [ string ]
  fhir:Annotation.time [ dateTime ]; # 0..1 When the annotation was made
  fhir:Annotation.text [ string ]; # 1..1 The annotation  - text content
]
```

Changes since DSTU2

Annotation

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Annotation | | | Element | Text node with attribution Elements defined in Ancestors: id, extension
author[x] | ? | 0..1 | | Individual responsible for the annotation
authorReference | | | Reference(Practitioner | Patient | RelatedPerson)
authorString | | | string
time | ? | 0..1 | dateTime | When the annotation was made
text | | 1..1 | string | The annotation - text content
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <author[x]><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson)|string Individual responsible for the annotation
    --></author[x]>
 <time value="[dateTime]"/><!-- 0..1 When the annotation was made -->
 <text value="[string]"/><!-- 1..1 The annotation  - text content -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  // author[x]: Individual responsible for the annotation. One of these 2:
  "authorReference" : { Reference(Practitioner|Patient|RelatedPerson) },
  "authorString" : "<string>",
  "time" : "<dateTime>", // When the annotation was made
  "text" : "<string>" // R!  The annotation  - text content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  # Annotation.author[x] : 0..1 Individual responsible for the annotation. One of these 2
    fhir:Annotation.authorReference [ Reference(Practitioner|Patient|RelatedPerson) ]
    fhir:Annotation.authorString [ string ]
  fhir:Annotation.time [ dateTime ]; # 0..1 When the annotation was made
  fhir:Annotation.text [ string ]; # 1..1 The annotation  - text content
]
```

Changes since DSTU2

Annotation

See the Full Difference for further information

Systems that do not have structured annotations simply communicate a single annotation with no author or time.

This element may need to be included in narrative because of the potential for modifying information.

Annotations SHOULD NOT be used to communicate "modifying" information that could be computable (this is a SHOULD because enforcing user behavior is nearly impossible).

Annotation is used in the following places: AllergyIntolerance, CarePlan, CareTeam, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Device, DeviceRequest, DeviceUseStatement, FamilyMemberHistory, Goal, GuidanceResponse, Immunization, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Procedure, ProcedureRequest, ReferralRequest, RequestGroup, ResearchStudy, Specimen, Task and VisionPrescription

2.26.0.18 Open Type Element

Some elements do not have a specified type. The type is represented by the wildcard symbol "*". In these cases, the element type may be one of the following:

* boolean

* integer

* decimal

* base64Binary

* instant

* string

* uri

* date

* dateTime

* time

* code
* oid
* id
* unsignedInt
* positiveInt
* markdown
* Annotation
* Attachment
* Identifier
* CodeableConcept
* Coding
* Quantity
* Range
* Period
* Ratio
* SampledData
* Signature
* HumanName
* Address
* ContactPoint
* Timing
* Reference - a reference to another resource
* Meta

The element name ends with "[x]", and this is replaced with the Title cased name of the data type.

Open references are used in the following places: Parameters, StructureMap and Task

2.26.0.19 Other Types

The following types are defined as part of the data types, but are documented elsewhere in the specification:

* Resource - the conceptual base class for all resources
* Reference - for references from one resource to another
* Extension - used to convey additional data in a resource
* Narrative - conveys a human-readable representation of the content of a resource

2.26.1 Data Type Examples

URL: https://hl7.org/fhir/STU3/datatypes-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.26.1 Data Type Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This page includes additional examples of the data types, based on common usages and questions

Table of Contents

Primitive Types

Complex Types

2.26.1.1 Primitive Types

A boolean true value:

```
<active value="true" />
```

A negative integer value:

```
<score value="-14" />
```

A high-precision decimal value:

```
<pi value="3.14159265358979323846264338327950288419716939937510" />
```

A stream of bytes, base64 encoded:

```
<data value="/9j/4...KAP//Z" /> <!-- covers many lines -->
```

A unicode string:

```
<caption value="Noodles are called ?? in Chinese" />
```

A URI that points to a website:

```
<reference value="http://hl7.org/fhir" />
```

A URI that is a urn:

```
<id value="urn:isbn:0451450523" />
```

A date of birth:

```
<date value="1951-06-04" />
```

An approximate date of birth:

```
<date value="1951-06" />
```

The instant a document was created, including time zone:

```
<instant value="2013-06-08T10:57:34+01:00" />
```

The instant a document was created, expressed in UTC, with milliseconds:

```
<instant value="2013-06-08T09:57:34.2112Z" />
```

2:35pm in the afternoon:

```
<time value="14:35" />
```

2.26.1.2 String Patterns

A URI that is the root oid of HL7:

```
<root value="urn:oid:2.16.840.1.113883" />
```

A URI that is a uuid:

```
<id value="urn:uuid:a5afddf4-e880-459b-876e-e4591b0acc11" />
```

A code:

```
<code value="acq4+acq5" />
```

A code with single internal space:

```
<code value="Question 4b" />
```

A numeric id:

```
<id value="314" />
```

An alphanumeric id:

```
<id value="alpha-gamma-14" />
```

2.26.1.3 Attachment

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A PDF document:

```
  <document>
    <contentType value="application/pdf" />
    <language value="en" />
    <data value="/9j/4...KAP//Z" /> <!-- covers many lines -->
    <title value="Definition of Procedure" />
  </document>
```

```
  document : {
    contentType :  { value : "application/pdf" },
    language : { value : "en" },
    data :  { value : "/9j/4...KAP//Z"},
    title :  { value : "Definition of Procedure" }
  }
```

Since the JSON examples have the same structure as the XML, only XML is shown for the rest of the examples.

A reference to a DICOM image via WADO:

```
  <image>
    <contentType value="application/dicom" />
    <url value="http://10.1.2.3:1000/wado?requestType=WADO&amp;wado_details..." />
    <hash value="EQH/..AgME" />
  </image>
```

2.26.1.4 Identifier

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A primary key from an application table (an OID in the space allocated by HL7 to some organization to further sub-allocate):

```
  <identifier>
    <use value="official" />
    <system value="urn:oid:2.16.840.1.113883.16.4.3.2.5" />
    <value value="123" />
  </identifier>
```

A patient identifier defined by a hospital:

```
  <identifier>
    <use value="official" />
    <system value="http://www.acmehosp.com/patients" />
```

```
    <value value="44552" />
    <period>
      <start value="2003-05-03" />
    </period>
  </identifier>
```

In this case, the period is used to track when the identifier was first assigned to the patient.

An identifier that refers to a patient FHIR resource on a particular system:

```
  <identifier>
   <system value="urn:ietf:rfc:3986" />
   <value value="http://pas-server/xxx/Patient/443556" />
  </identifier>
```

This is not a resource reference - it's a logical reference by the patient identifier.

A UUID:

```
  <identifier>
    <use value="temp" />
    <system value="urn:ietf:rfc:3986" />
    <value value="urn:uuid:a76d9bbf-f293-4fb7-ad4c-2851cac77162" />
  </identifier>
```

UUIDs are often used for temporary identifiers, though this is not necessary.

A US SSN:

```
  <identifier>
    <use value="usual" />
    <type>
      <coding>
        <system value="http://hl7.org/fhir/v2/0203"/>
        <code value="SSN"/>
      </coding>
    </type>
    <system value="http://hl7.org/fhir/sid/us-ssn" />
    <value value="000111111" />
  </identifier>
```

Notes:

* US SSNs are often presented like this: 000-11-1111, the dashes are for presentation and should be removed, as specified in the definition of ssn-us

* The use of "usual" means that this institution prefers to use SSN when identifying the patient

A medical record number assigned on 5-July 2009:

```
  <identifier>
    <use value="usual" />
    <type>
      <coding>
        <system value="http://hl7.org/fhir/v2/0203"/>
        <code value="MRN"/>
      </coding>
    </type>
    <system value="urn:oid:0.1.2.3.4.5.6.7" />
    <value value="2356" />
    <period>
      <start value="2009-07-05" />
    </period>
  </identifier>
```

2.26.1.5 Coding

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A simple code for headache, in ICD-10:

```
  <code>
    <system value="http://hl7.org/fhir/sid/icd-10" />
    <code value="G44.1" />
  </code>
```

A SNOMED CT expression:

```
  <problem>
    <system value="http://snomed.info/sct" />
    <code value="128045006:{363698007=56459004}" />
  </problem>
```

2.26.1.6 CodeableConcept

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A simple code for headache initially coded in SNOMED CT (by picking the SNOMED CT code from a pick-list), and then translated to ICD-10:

```
  <concept>
    <coding>
      <system value="http://hl7.org/fhir/sid/icd-10" />
```

```
    <code value="R51" />
  </coding>
  <coding>
    <system value="http://snomed.info/sct" />
    <code value="25064002" />
    <display value="Headache" />
    <userSelected value="true" />
  </coding>
  <text value="general headache" />
</concept>
```

A concept represented in an institution's local coding systems for unit for which no UCUM equivalent exists:

```
<unit>
  <coding>
    <system value="urn:oid:2.16.840.1.113883.19.5.2" />
    <code value="tab" />
    <display value="Tablet" />
  </coding>
  <coding>
    <system value="http://unitsofmeasure.org" />
  </coding>
</unit>
```

A SNOMED CT expression:

```
<diagnosis>
  <coding>
    <system value="http://snomed.info/sct" />
    <code value="128045006:{363698007=56459004}" />
  </coding>
  <text value="Cellulitis of the foot" />
</diagnosis>
```

In this case, there is no display element, because no display is defined for SNOMED CT expressions.

Using the value set:

The results on a urinalysis strip:

```
<valueCoding>
  <system value="http://example.org/codes/simple-grades" />
  <code value="+" />
</valueCoding>
```

And where the value set would be something like this:

```
<ValueSet xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Possible Clinistix codes: neg, trace, +, ++, and +++</p>
    </div>
  </text>
  <url value="http://hl7.org/fhir/ValueSet/clinistix"/>
  <name value="Codes for Clinistix"/>
  <publisher value="HL7"/>
  <contact>
      <name value="FHIR project team"/>
      <telecom>
        <system value="url"/>
        <value value="http://hl7.org/fhir"/>
      </telecom>
  </contact>
  <description value="Clinistix Codes"/>
  <status value="draft"/>
  <experimental value="true"/>
  <date value="2013-10-01"/>
  <compose>
    <include>
      <system value="http://example.org/codes/simple-grades"/>
      <concept>
        <code value="neg"/>
      </concept>
      <concept>
        <code value="trace"/>
      </concept>
      <concept>
        <code value="+"/>
      </concept>
      <concept>
        <code value="+"/>
      </concept>
      <concept>
        <code value="++"/>
      </concept>
      <concept>
        <code value="+++"/>
      </concept>
    </include>
  </compose>
</ValueSet>
```

2.26.1.7 Quantity

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A duration:

```
<time>
  <value value="25" />
  <unit value="sec" />
  <system value="http://unitsofmeasure.org" />
  <code value="s" />
</time>
```

A concentration where the value was out of range:

```
<result>
  <value value="40000" />
  <comparator value="&gt;" />
  <unit value="mcg/L" />
  <system value="http://unitsofmeasure.org" />
  <code value="ug" />
</result>
```

An amount of prescribed medication:

```
<dose>
  <value value="3" />
  <unit value="capsules" />
  <system value="http://snomed.info/sct" />
  <code value="385049006" />
</dose>
```

A price (coded using currency codes defined in ISO 4217):

```
<cost>
  <value value="25.45" />
  <unit value="US$" />
  <system value="urn:iso:std:iso:4217" />
  <code value="USD" />
</cost>
```

2.26.1.8 Range

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

Range of Quantity (distance):

```
<estimate>
 <low>
   <value value="1.6" />
   <unit value="m" />
 </low>
 <high>
   <value value="1.9" />
   <unit value="m" />
 </high>
</estimate>
```

2.26.1.9 Ratio

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

Titer (Ratio of integer:integer)

```
<result>
 <numerator>
   <value value="1" />
 </numerator>
 <denominator>
   <value value="128" />
 </denominator>
</result>
```

Unit cost (Ratio of Money:Quantity):

```
<charge>
 <numerator>
   <value value="103.50" />
   <unit value="US$" />
   <code value="USD" />
   <system value="urn:iso:std:iso:4217" />
 </numerator>
 <denominator>
   <value value="1" />
   <unit value="day" />
   <code value="day" />
   <system value="http://unitsofmeasure.org" />
 </denominator>
</charge>
```

2.26.1.10 Period

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

23rd May 2011 to 27th May, including 27th May:

```
  <coverage>
   <start value="2011-05-23" />
   <end value="2011-05-27" />
  </coverage>
```

2.26.1.11 SampledData

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

The output from an EKG device:

```
 <sampledData>
  <origin>
   <value value="0"/>
   <unit value="?V"/>
   <system value="http://unitsofmeasure.org"/>
   <code value="uV"/>
  </origin>
  <period value="2"/>
  <factor value="2.5"/>
  <dimensions value="1"/>
  <data value="-4 -13 -18 -18 -18 -17 -16 -16 -16 -16 -16 -17 -18 -18 -18 ...."/>
 </sampledData>
```

2.26.1.12 HumanName

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Simple example

```
<name>
  <family value="Everyman" />
  <given value="Adam" />
  <given value="A." />
</name>
```

Composite names

```
<name>
  <family value="Contrata" />
  <given value="Mary Jane" />
</name>
```

These cases can be quite ambiguous - is "Mary Jane" one name, or two? Different systems, and data enterers may treat this differently, and the person themselves may not know. Parts are allowed to contain spaces, but systems should consider how to treat these cases. Composite names separated by "-" should be treated as a single name part.

A common pattern: a person is called by a name other than that expected from their official name (first given name in most cultures).

```
<name>
  <use value="official" />
  <family value="Chalmers" />
  <given value="Peter" />
  <given value="James" />
</name>
<name>
  <use value="usual" />
  <given value="Jim" />
</name>
```

This same pattern is often encountered with immigrants, who retain their real name for official use, but adopt a localized name for everyday use:

```
<name>
  <use value="official" />
  <family value="Sczypinski" />
  <given value="Piotr" />
  <given value="Andre" />
</name>
<name>
  <use value="usual" />
  <family value="Skipper" />
  <given value="Jim" />
</name>
```

Karen van Hentenryck is of Dutch origin, and the "van" is a voorvoegsel.

```
<name>
  <use value="official" />
  <family value="van Hentenryck" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix" >
     <valueString value="van" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-name">
```

```
      <valueString value="Hentenryck" />
    </extension>
  </family>
  <given value="Karen" />
</name>
```

See the Extensibility Example for more information about the use of extensions.

Complex example from Germany: Dr.phil. Regina Johanna Maria von Hochheim-Weilenfels, NCFSA. This example shows extensive use of multiple given names, prefixes, suffixes, for academic degrees, nobility titles, and professional designations.

```
<name>
  <use value="official" />
  <family value="von Hochheim-Weilenfels">
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix" >
      <valueString value="von" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-name">
      <valueString value="Hochheim-Weilenfels" />
    </extension>
  </family>
  <given value="Regina" />
  <given value="Johanna" />
  <given value="Maria" />
  <prefix value="Dr. phil.">
    <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier" >
      <valueCode value="AC" />
    </extension>
  </prefix>
  <suffix value="NCFSA" />
</name>
<name>
  <use value="maiden" />
  <family value="Hochheim" />
</name>
```

This example makes use of the ISO 21090 extensions to carry the rarely used ISO 21090 qualifier attribute "AC".

Japanese example in the three forms: ideographic (Kanji), syllabic (Hiragana) and alphabetic (Romaji).

```
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="IDE" />
  </extension>
  <family value="??" />
  <given value="??" />
  </name>
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="SYL" />
  </extension>
 <family value="???" />
 <given value="???" />
</name>
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="ABC" />
  </extension>
  <family value="KIMURA" />
  <given value="MICHIO" />
</name>
```

The three forms may be differentiated by the character subset each contains, but some systems require the differentiation to be made manually, which can be done using the ED representation extension.

Russian example in the two forms: cyrillic, and latin:

```
<name>
  <family value="??????" />
  <given value="????" />
  <given value="????????????" />
</name>
<name>
  <family value="EMELIN" />
  <given value="IVAN" />
  <given value="VLADIMIROVICH" />
</name>
```

In Russian usage, these names are known as the domestic and foreign names respectively. The two forms are differentiated by the character subset each contains.

Scandinavian example: Erikson is the family name. Jan Erik are the given names, and Östlund the family name of the mother, which is taken as a Mellannamn.

```
<name>
  <use value="official" />
  <family value="Erikson" />
  <given value="Jan" />
  <given value="Erik" />
  <given value="Östlund">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
```

```
        <code value="MID" />
        <system value="http://hl7.org/fhir/v3/EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
```

This example makes use of the ISO 21090 extension to carry the culture specific ISO 21090 qualifier attribute "MID" for the Mellannamn.

Then Jan Erikson has a daughter, Karin, with his wife Margrete Hansen. The first communications of the new born name is "Margrete Jente" (Margrete's Girl) and the mother's family name, not the given name (Karin). The father's Family name is not used at all. This is a known temporary name assigned directly after the birth of the child.

```
<name>
  <use value="temp" />
  <!-- use could be OR+OLD, depends how record keeping is done -->
  <family value="Hansen" />
  <given value="Margrete Jente" />
</name>
```

The baby's name is subsequently changed to the fathers' family name, and to use the mother's name as mellomnamn.

```
<name>
  <use value="official" />
  <family value="Erikson" />
  <given value="Karin" />
  <given value="Hansen">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
        <code value="MID" />
        <system value="http://hl7.org/fhir/v3/EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
```

Later, Karin gets married to Per Berg, and decides to adopt Berg as her family name, and also decides to use Erikson as the mellom navn. (Note: Karin could have chosen to use another mellom navn, e.g. the family name of her mother, her father or other family names as specified by naming laws of the country in question).

```
<name>
  <use value="old" />
  <family value="Erikson" />
  <given value="Karin" />
  <given value="Hansen">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
        <code value="MID" />
        <system value="http://hl7.org/fhir/v3/EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
<name>
  <use value="official" />
  <family value="Berg" />
  <given value="Karin" />
  <given value="Erikson">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
        <code value="MID" />
        <system value="http://hl7.org/fhir/v3/EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
<name>
  <use value="usual" />
  <family value="Berg" />
  <given value="Karin" />
</name>
```

2.26.1.12.1 W3C International Examples

These examples are taken from the W3C International Examples , which should be consulted for further information.

A patronymic is "The part of a name that links to the genealogy":

```
<name>
  <text value="Björk Guðmundsdóttir"/>
  <family value="Guðmundsdóttir"/>
  <given value="Björk"/>
</name>
```

A patronymic with a "son/daughter of" appellation:

```
<name>
  <text value="Isa bin Osman"/>
  <family value="bin Osman"/>
  <given value="Isa"/>
</name>
```

A Chinese name with a generational name:

```
<name>
  <text value="???"/> <!-- left to right -->
  <family value="?"/>
  <given value="??"/>
</name>
<name>
  <text value="Mao Ze Dong"/> <!-- left to right -->
  <family value="Mao"/>
  <given value="Ze Dong"/>
</name>
```

Note that many systems in China do not store family and given names separately, and just use text.

Additional Western name (see also example above):

```
<name>
  <use value="official" />
  <family value="Yao" />
  <given value="Ming" />
</name>
<name>
  <use value="usual" />
  <given value="Fred" />
</name>
```

Composite Family name:

```
<name>
  <family value="Carreño Quiñones" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Carreño" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Quiñones" />
    </extension>
  </family>
  <given value="María-Jose" />
</name>
```

Note that it is optional whether to break down the family name to mother's and father's parts, and not always of value, as in this Brazilian Example:

```
<name>
  <family value="Santos Tavares Melo Silva" />
  <given value="José" />
  <given value="Eduardo" />
</name>
```

Note that this naming pattern can become quite extreme ("Pedro de Alcântara João Carlos Leopoldo Salvador Bibiano Francisco Xavier de Paula Leocádio Miguel Gabriel Rafael Gonzaga"). A more practical example is

```
<name>
  <family value="Costa Teixeira" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Costa" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Teixeira" />
    </extension>
  </family>
  <given value="Manuel" />
</name>
```

After marriage, this may change to

```
<name>
  <family value="Costa Teixeira Sanches" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Costa" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Teixeira" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-partners-name">
        <valueString value="Sanches" />
    </extension>
  </family>
  <given value="Manuel" />
</name>
```

Russian Examples (using Cyrillic):

```
<name>
  <family value="??????" />
  <given value="?????" />
  <given value="??????????" />
</name>

<name>
  <family value="???????" />
  <given value="?????" />
  <given value="?????????" />
```

```
</name>
```

Example with Initial:

```
<name>
  <family value="Public" />
  <given value="John" />
  <given value="Q." />
</name>
```

Other Examples:

```
<name>
  <text value="Velikkakathu Sankaran Achuthanandan"/>
  <family value="Velikkakathu" />
  <given value="Sankaran" />
  <given value="Achuthanandan" />
</name>
<name>
  <text value="Kogaddu Birappa Timappa Nair"/>
  <family value="Nair" />
  <given value="Birappa" />
  <given value="Timappa" />
  <prefix value="Kogaddu" />
</name>
<name>
  <text value="Aditya Pratap Singh Chauhan"/>
  <family value="Singh" />
  <given value="Aditya" />
  <given value="Pratap" />
  <suffix value="Chauhan" />
</name>
<name>
  <text value="Madurai Mani Iyer"/>
  <given value="Mani" />
  <prefix value="Madurai" />
  <suffix value="Iyer" />
</name>
<name>
  <text value="Abu Karim Muhammad al-Jamil ibn Nidal ibn Abdulaziz al-Filistini"/>
  <family value="ibn Nidal ibn Abdulaziz" />
  <given value="Muhammad" />
  <given value="al-Jamil" />
  <prefix value="Abu Karim" />
  <suffix value="al-Filistini" />
</name>
```

Todo: need to discuss this with Indian / Arabic implementers. Note that collecting and storing the text element makes the primary purpose of the structured parts for index/searching, and fidelity of the name parts is not critical.

2.26.1.13 Address

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Note about these examples: the most important element in an address is the 'text' element - this defines what is printed on the envelope, the actual postal address. The other data elements are provided to support either data analysis based on a patient's address, or for the many systems that exchange fully structured addresses specific to a particular culture. For this reason, these examples focus on the structured data elements of the address.

Example

HL7 office's address.

```
  <address>
   <use value="work" />
   <text value="1050 W Wishard Blvd
RG
         5th floor
Indianapolis, IN 46240" />
   <line value="1050 W Wishard Blvd" />
   <line value="RG 5th floor" />
   <city value="Indianapolis" />
   <state value="IN" />
   <postalCode value="46240" />
  </address>
```

A UK example address, with the county 'HUDDERSFIELD'.

```
  <address>
    <text value="1 Back Lane&#13;&#10;Holmfirth&#13;&#10;HUDDERSFIELD&#13;&#10;HD7 1HQ"/>
    <line value="1 Back Lane"/>
    <city value="Holmfirth"/>
    <district value="HUDDERSFIELD"/>
    <postalCode value="HD7 1HQ"/>
  </address>
```

A temporary postal address - i.e. an address that it doesn't make sense to try and visit.

```
  <address>
    <use value="temp"/>
    <type value="postal"/>
    <line value="PO Box 31445"/>
    <city value="Erewhon"/>
```

```
  <postalCode value="0001"/>
</address>
```

The next set of examples are taken from the official international postal union examples.

Rue Lougoraïa 12, app. 10 | thoroughfare type, name and number, apartment
Korolevo | locality
223016 NOVY DVOR | postcode + post office name
Minsk | District name of district
Minsk | Region name of region
BELARUS | Country

```
<address>
  <line value="Rue Lougoraïa 12, app. 10"/>
  <city value="Korolevo"/>
  <district value="Minsk"/>
  <state value="Minsk" />
  <country value="BELARUS" />
</address>
```

Protea Apt 12 | apartment number
22 Ally Hassan Mwinyi | premise + street
14111 MSASANI | postcode + locality
DAR ES SALAM | province
TANZANIA | country

```
<address>
  <line value="Protea Apt 12"/>
  <line value="22 Ally Hassan Mwinyi"/>
  <city value="MSASANI"/>
  <state value="DAR ES SALAM" />
  <postalCode value="14111"/>
  <country value="TANZANIA" />
</address>
```

15 Shiri | premise + sub-locality 2
Kimashuku Village | sub-locality
25204 MACHAME KUSINI | postcode + locality
HAI | sub-province
KILIMANJARO | province
TANZANIA | country

```
<address>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-ADXP-precinct">
    <valueCode value="Kimashuku Village"/>
  </extension>
  <line value="15 Shiri"/>
  <city value="MACHAME KUSINI"/>
  <district value="HAI"/>
  <state value="KILIMANJARO" />
  <postalCode value="25204"/>
  <country value="TANZANIA" />
</address>
```

705-1104 | building number + door
56, Dalgubeol-daero 323beon-gil, Suseong-gu | street
Daegu | city
Rep. OF KOREA | country
706-907 | postcode

```
<address>
  <line value="705-1104"/>
  <line value="56, Dalgubeol-daero 323beon-gil, Suseong-gu"/>
  <city value="Daegu"/>
  <country value="Rep. OF KOREA" />
  <postalCode value="706-907"/>
</address>
```

3F | building floor
42, Toegye-ro 77beon-gil | street
Chuncheon-si | city
Gangwon-do | province
Rep. OF KOREA | country
200-066 | postcode

```
<address>
  <line value="3F"/>
  <line value="42, Toegye-ro 77beon-gil"/>
  <city value="Chuncheon-si"/>
  <state value="Gangwon-do" />
  <country value="Rep. OF KOREA" />
  <postalCode value="200-066"/>
</address>
```

Ga-B101 | building number + door
136, Sesil-ro | street
Busan | city
Rep. OF KOREA | country

612-837 | postcode

```
<address>
  <line value="Ga-B101"/>
  <line value="136, Sesil-ro"/>
  <city value="Busan"/>
  <country value="Rep. OF KOREA" />
  <postalCode value="612-837"/>
</address>
```

2.26.1.14 ContactPoint

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

Home phone number:

```
<telecom>
 <system value="phone" />
 <value value="(555) 675 5745" />
 <use value="home" />
</telecom>
```

In ISO 21090, CDA and other HL7 v3 specifications, the TEL data type requires that the telephone number be represented as a formal URI in RFC 3166 syntax. Many CDA documents violate this rule, since the telephone numbers are not collected from the user in conformance with the underlying rules (particularly around extensions, notes about use, etc.), so the base FHIR type does not require the phone number to be conformant. A ISO 21090 extension can be used to convey this if desired:

Example

Formally correct URI:

```
<telecom>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address" >
    <valueUri value="tel:+15556755745" />
  </extension>
  <system value="phone" />
  <value value="(555) 675 5745" />
  <use value="home" />
</telecom>
```

2.26.1.15 Timing

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

A series of appointments for radiotherapy:

```
<schedule>
  <event value="2012-01-07T09:00:00+10:00" />
  <event value="2012-01-14T09:00:00+10:00" />
  <event value="2012-01-22T11:00:00+10:00" />
</schedule>
```

BID (twice a day) (no start or end specified):

```
<schedule>
 <repeat>
   <frequency value="2" />
   <period value="1" />
   <periodUnit value="d" />
 </repeat>
</schedule>
```

1/2 an hour before breakfast for 10 days from 23-Dec 2011:

```
<schedule>
  <repeat>
    <boundsPeriod>
      <start value="2011-12-23" />
      <end value="2012-01-02" />
    </boundsPeriod>
    <when value="ACM" />
    <offset value="30" />
  </repeat>
</schedule>
```

Note that the end date is inclusive like the end date of a Period.

TID, for 14 days:

```
<schedule>
  <repeat>
    <boundsDuration>
      <value value="14" />
      <unit value="d" />
      <system value="http://unitsofmeasure.org" />
      <code value="d" />
    </boundsDuration>
    <frequency value="3" />
    <period value="1" />
    <periodUnit value="d" />
  </repeat>
```

```
  </schedule>
```

BID, start on 7/1/2015 at 1:00 PM:

```
<schedule>
  <repeat>
    <boundsPeriod>
      <start value="2015-07-01T13:00:00" />
    </boundsPeriod>
    <frequency value="2" />
    <period value="1" />
    <periodUnit value="d" />
  </repeat>
</schedule>
```

Take just once, with no specified time

```
<schedule>
  <repeat>
    <count value="1" />
  </repeat>
</schedule>
```

2.26.1.16 Signature

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

todo

```
<signature>
  <!-- todo -->
</signature>
```

2.26.1.17 Annotation

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

todo

```
<Annotation>
  <!-- todo -->
</Annotation>
```

2.26.2 Data Type Detailed Descriptions

URL: https://hl7.org/fhir/STU3/datatypes-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.26.2 Data Type Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

This page provides the detailed descriptions for the data types

Table of Contents

Primitive Types | Complex Types

2.26.2.1 Primitive Types

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

todo

2.26.2.2 Attachment

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Attachment

Definition | For referring to data content defined in other formats.

Control | 1..1

Requirements | Many models need to include data defined in other specifications that is complex and opaque to the healthcare model. This includes documents, media recordings, structured data, etc.

Comments | When providing a summary view (for example with Observation.value[x]) Attachment should be represented with a brief display text such as "Attachment".

Invariants | Defined on this element att-1: It the Attachment has data, it SHALL have a contentType (expression : data.empty() or contentType.exists(), xpath: not(exists(f:data)) or exists(f:contentType))

Attachment.contentType

Definition | Identifies the type of the data in the attachment and allows a method to be chosen to interpret or render the data. Includes mime type parameters such as charset where appropriate.

Control | 0..1

Terminology Binding | MimeType :

Type | code

Requirements | Processors of the data need to be able to know how to interpret the data.

Summary | true

Attachment.language
Definition | The human language of the content. The value can be any valid value according to BCP 47.
Control | 0..1
Terminology Binding | Common Languages (Extensible but limited to ??)
Type | code
Requirements | Users need to be able to choose between the languages in a set of attachments.
Summary | true
Attachment.data
Definition | The actual data of the attachment - a sequence of bytes. In XML, represented using base64.
Control | 0..1
Type | base64Binary
Requirements | The data needs to able to be transmitted inline.
Comments | The base64-encoded data SHALL be expressed in the same character set as the base resource XML or JSON.
Attachment.url
Definition | An alternative location where the data can be accessed.
Control | 0..1
Type | uri
Requirements | The data needs to be transmitted by reference.
Summary | true
Comments | If both data and url are provided, the url SHALL point to the same content as the data contains. Urls may be relative references or may reference transient locations such as a wrapping envelope using cid: though this has ramifications for using signatures. Relative URLs are interpreted relative to the service url, like a resource reference, rather than relative to the resource itself. If a URL is provided, it SHALL resolve to actual data.
Attachment.size
Definition | The number of bytes of data that make up this attachment (before base64 encoding, if that is done).
Control | 0..1
Type | unsignedInt
Requirements | Representing the size allows applications to determine whether they should fetch the content automatically in advance, or refuse to fetch it at all.
Summary | true
Comments | The number of bytes is redundant if the data is provided as a base64binary, but is useful if the data is provided as a url reference.
Attachment.hash
Definition | The calculated hash of the data using SHA-1. Represented using base64.
Control | 0..1
Type | base64Binary
Requirements | Included so that applications can verify that the contents of a location have not changed and so that a signature of the content can implicitly sign the content of an image without having to include the data in the instance or reference the url in the signature.
Summary | true
Comments | The hash is calculated on the data prior to base64 encoding, if the data is based64 encoded.
Attachment.title
Definition | A label or set of text to display in place of the data.
Control | 0..1
Type | string
Requirements | Applications need a label to display to a human user in place of the actual data if the data cannot be rendered or perceived by the viewer.
Summary | true
Attachment.creation
Definition | The date that the attachment was first created.
Control | 0..1
Type | dateTime
Requirements | This is often tracked as an integrity issue for use of the attachment.
Summary | true

2.26.2.3 Identifier

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Identifier
Definition | A technical identifier - identifies some entity uniquely and unambiguously.
Control | 1..1
Requirements | Need to be able to identify things with confidence and be sure that the identification is not subject to misinterpretation.
Summary | true
Identifier.use
Definition | The purpose of this identifier.
Control | 0..1
Terminology Binding | IdentifierUse (Required)
Type | code
Is Modifier | true
Requirements | Allows the appropriate identifier for a particular context of use to be selected from among a set of identifiers.
Summary | true
Comments | This is labeled as "Is Modifier" because applications should not mistake a temporary id for a permanent one. Applications can assume that an identifier is permanent unless it explicitly says that it is temporary.
Identifier.type
Definition | A coded type for the identifier that can be used to determine which identifier to use for a specific purpose.

Control | 0..1
Terminology Binding | Identifier Type Codes (Extensible)
Type | CodeableConcept
Requirements | Allows users to make use of identifiers when the identifier system is not known.
Summary | true
Comments | This element deals only with general categories of identifiers. It SHOULD not be used for codes that correspond 1..1 with the Identifier.system. Some identifiers may fall into multiple categories due to common usage. Where the system is known, a type is unnecessary because the type is always part of the system definition. However systems often need to handle identifiers where the system is not known. There is not a 1:1 relationship between type and system, since many different systems have the same type.
Identifier.system
Definition | Establishes the namespace for the value - that is, a URL that describes a set values that are unique.
Control | 0..1
Type | uri
Requirements | There are many sets of identifiers. To perform matching of two identifiers, we need to know what set we're dealing with. The system identifies a particular set of unique identifiers.
Summary | true
Identifier.value
Definition | The portion of the identifier typically relevant to the user and which is unique within the context of the system.
Control | 0..1
Type | string
Summary | true
Comments | If the value is a full URI, then the system SHALL be urn:ietf:rfc:3986. The value's primary purpose is computational mapping. As a result, it may be normalized for comparison purposes (e.g. removing non-significant whitespace, dashes, etc.) A value formatted for human display can be conveyed using the Rendered Value extension.
Identifier.period
Definition | Time period during which identifier is/was valid for use.
Control | 0..1
Type | Period
Summary | true
Identifier.assigner
Definition | Organization that issued/manages the identifier.
Control | 0..1
Type | Reference(Organization)
Summary | true
Comments | The Identifier.assigner may omit the .reference element and only contain a .display element reflecting the name or other textual information about the assigning organization.

2.26.2.4 Coding

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Coding
Definition | A reference to a code defined by a terminology system.
Control | 1..1
Requirements | References to codes are very common in healthcare models.
Summary | true
Comments | Codes may be defined very casually in enumerations or code lists, up to very formal definitions such as SNOMED CT - see the HL7 v3 Core Principles for more information.
Coding.system
Definition | The identification of the code system that defines the meaning of the symbol in the code.
Control | 0..1
Type | uri
Requirements | Need to be unambiguous about the source of the definition of the symbol.
Summary | true
Comments | The URI may be an OID (urn:oid:...) or a UUID (urn:uuid:...). OIDs and UUIDs SHALL be references to the HL7 OID registry. Otherwise, the URI should come from HL7's list of FHIR defined special URIs or it should de-reference to some definition that establish the system clearly and unambiguously.
Coding.version
Definition | The version of the code system which was used when choosing this code. Note that a well-maintained code system does not need the version reported, because the meaning of codes is consistent across versions. However this cannot consistently be assured. and when the meaning is not guaranteed to be consistent, the version SHOULD be exchanged.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | Where the terminology does not clearly define what string should be used to identify code system versions, the recommendation is to use the date (expressed in FHIR date format) on which that version was officially published as the version date.
Coding.code
Definition | A symbol in syntax defined by the system. The symbol may be a predefined code or an expression in a syntax defined by the coding system (e.g. post-coordination).
Control | 0..1
Type | code
Requirements | Need to refer to a particular code in the system.
Summary | true
Coding.display

Definition | A representation of the meaning of the code in the system, following the rules of the system.
Control | 0..1
Type | string
Requirements | Need to be able to carry a human-readable meaning of the code for readers that do not know the system.
Summary | true
Coding.userSelected
Definition | Indicates that this coding was chosen by a user directly - i.e. off a pick list of available items (codes or displays).
Control | 0..1
Type | boolean
Requirements | This has been identified as a clinical safety criterium - that this exact system/code pair was chosen explicitly, rather than inferred by the system based on some rules or language processing.
Summary | true
Comments | Amongst a set of alternatives, a directly chosen code is the most appropriate starting point for new translations. There is some ambiguity about what exactly 'directly chosen' implies, and trading partner agreement may be needed to clarify the use of this element and its consequences more completely.

2.26.2.5 CodeableConcept

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

CodeableConcept
Definition | A concept that may be defined by a formal reference to a terminology or ontology or may be provided by text.
Control | 1..1
Requirements | This is a common pattern in healthcare - a concept that may be defined by one or more codes from formal definitions including LOINC and SNOMED CT, and/or defined by the provision of text that captures a human sense of the concept.
Summary | true
Comments | Not all terminology uses fit this general pattern. In some cases, models should not use CodeableConcept and use Coding directly and provide their own structure for managing text, codings, translations and the relationship between elements and pre- and post-coordination.
CodeableConcept.coding
Definition | A reference to a code defined by a terminology system.
Control | 0..*
Type | Coding
Requirements | Allows for translations and alternate encodings within a code system. Also supports communication of the same instance to systems requiring different encodings.
Summary | true
Comments | Codes may be defined very casually in enumerations, or code lists, up to very formal definitions such as SNOMED CT - see the HL7 v3 Core Principles for more information. Ordering of codings is undefined and SHALL NOT be used to infer meaning. Generally, at most only one of the coding values will be labeled as UserSelected = true.
CodeableConcept.text
Definition | A human language representation of the concept as seen/selected/uttered by the user who entered the data and/or which represents the intended meaning of the user.
Control | 0..1
Type | string
Requirements | The codes from the terminologies do not always capture the correct meaning with all the nuances of the human using them, or sometimes there is no appropriate code at all. In these cases, the text is used to capture the full meaning of the source.
Summary | true
Comments | Very often the text is the same as a displayName of one of the codings.

2.26.2.6 Quantity

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Quantity
Definition | A measured amount (or an amount that can potentially be measured). Note that measured amounts include amounts that are not precisely quantified, including amounts involving arbitrary units and floating currencies.
Control | 1..1
Requirements | Need to able to capture all sorts of measured values, even if the measured value are not precisely quantified. Values include exact measures such as 3.51g, customary units such as 3 tablets, and currencies such as $100.32USD.
Summary | true
Comments | The context of use may frequently define what kind of quantity this is and therefore what kind of units can be used. The context of use may also restrict the values for the comparator.
Invariants | Defined on this element qty-3: If a code for the unit is present, the system SHALL also be present (expression : code.empty() or system.exists(), xpath: not(exists(f:code)) or exists(f:system))
Quantity.value
Definition | The value of the measured amount. The value includes an implicit precision in the presentation of the value.
Control | 0..1
Type | decimal
Requirements | Precision is handled implicitly in almost all cases of measurement.
Summary | true
Comments | The implicit precision in the value should always be honored. Monetary values have their own rules for handling precision (refer to standard accounting text books).
Quantity.comparator
Definition | How the value should be understood and represented - whether the actual value is greater or less than the stated value due to measurement issues; e.g. if the comparator is "<" , then the real value is < stated value.
Control | 0..1
Terminology Binding | QuantityComparator (Required)

Type | code
Is Modifier | true
Meaning if Missing | If there is no comparator, then there is no modification of the value
Requirements | Need a framework for handling measures where the value is <5ug/L or >400mg/L due to the limitations of measuring methodology.
Summary | true
Comments | This is labeled as "Is Modifier" because the comparator modifies the interpretation of the value significantly. If there is no comparator, then there is no modification of the value.
Quantity.unit
Definition | A human-readable form of the unit.
Control | 0..1
Type | string
Requirements | There are many representations for units of measure and in many contexts, particular representations are fixed and required. I.e. mcg for micrograms.
Summary | true
Quantity.system
Definition | The identification of the system that provides the coded form of the unit.
Control | 0..1
Type | uri
Requirements | Need to know the system that defines the coded form of the unit.
Summary | true
Invariants | Affect this element qty-3: If a code for the unit is present, the system SHALL also be present (expression : code.empty() or system.exists(), xpath: not(exists(f:code)) or exists(f:system))
Quantity.code
Definition | A computer processable form of the unit in some unit representation system.
Control | 0..1
Type | code
Requirements | Need a computable form of the unit that is fixed across all forms. UCUM provides this for quantities, but SNOMED CT provides many units of interest.
Summary | true
Comments | The preferred system is UCUM, but SNOMED CT can also be used (for customary units) or ISO 4217 for currency. The context of use may additionally require a code from a particular system.

2.26.2.7 Range

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Range
Definition | A set of ordered Quantities defined by a low and high limit.
Control | 1..1
Requirements | Need to be able to specify ranges of values.
Summary | true
Comments | The stated low and high value are assumed to have arbitrarily high precision when it comes to determining which values are in the range. I.e. 1.99 is not in the range 2 -> 3.
Invariants | Defined on this element rng-2: If present, low SHALL have a lower value than high (expression : low.empty() or high.empty() or (low <= high), xpath: not(exists(f:low/f:value/@value)) or not(exists(f:high/f:value/@value)) or (number(f:low/f:value/@value) <= number(f:high/f:value/@value)))
Range.low
Definition | The low limit. The boundary is inclusive.
Control | 0..1
Type | SimpleQuantity
Summary | true
Comments | If the low element is missing, the low boundary is not known.
Invariants | Affect this element rng-2: If present, low SHALL have a lower value than high (expression : low.empty() or high.empty() or (low <= high), xpath: not(exists(f:low/f:value/@value)) or not(exists(f:high/f:value/@value)) or (number(f:low/f:value/@value) <= number(f:high/f:value/@value)))
Range.high
Definition | The high limit. The boundary is inclusive.
Control | 0..1
Type | SimpleQuantity
Summary | true
Comments | If the high element is missing, the high boundary is not known.
Invariants | Affect this element rng-2: If present, low SHALL have a lower value than high (expression : low.empty() or high.empty() or (low <= high), xpath: not(exists(f:low/f:value/@value)) or not(exists(f:high/f:value/@value)) or (number(f:low/f:value/@value) <= number(f:high/f:value/@value)))

2.26.2.8 Ratio

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Ratio
Definition | A relationship of two Quantity values - expressed as a numerator and a denominator.
Control | 1..1
Requirements | Need to able to capture ratios for some measurements (titers) and some rates (costs).
Summary | true
Invariants | Defined on this element rat-1: Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present (expression : (numerator.empty() xor denominator.exists()) and (numerator.exists() or extension.exists()), xpath: (count(f:numerator) = count(f:denominator)) and ((count(f:numerator) > 0) or (count(f:extension) > 0)))
Ratio.numerator

Definition | The value of the numerator.
Control | 0..1
Type | Quantity
Summary | true
Ratio.denominator
Definition | The value of the denominator.
Control | 0..1
Type | Quantity
Summary | true

2.26.2.9 Period

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Period
Definition | A time period defined by a start and end date and optionally time.
Control | 1..1
Summary | true
Comments | This is not a duration - that's a measure of time (a separate type), but a duration that occurs at a fixed value of time. A Period specifies a range of time; the context of use will specify whether the entire range applies (e.g. "the patient was an inpatient of the hospital for this time range") or one value from the range applies (e.g. "give to the patient between these two times"). If duration is required, specify the type as Interval|Duration.
Invariants | Defined on this element per-1: If present, start SHALL have a lower value than end (expression : start.empty() or end.empty() or (start <= end), xpath: not(exists(f:start)) or not(exists(f:end)) or (f:start/@value <= f:end/@value))
Period.start
Definition | The start of the period. The boundary is inclusive.
Control | 0..1
Type | dateTime
Summary | true
Comments | If the low element is missing, the meaning is that the low boundary is not known.
Invariants | Affect this element per-1: If present, start SHALL have a lower value than end (expression : start.empty() or end.empty() or (start <= end), xpath: not(exists(f:start)) or not(exists(f:end)) or (f:start/@value <= f:end/@value))
Period.end
Definition | The end of the period. If the end of the period is missing, it means that the period is ongoing. The start may be in the past, and the end date in the future, which means that period is expected/planned to end at that time.
Control | 0..1
Type | dateTime
Meaning if Missing | If the end of the period is missing, it means that the period is ongoing
Summary | true
Comments | The high value includes any matching date/time. i.e. 2012-02-03T10:00:00 is in a period that has a end value of 2012-02-03.
Invariants | Affect this element per-1: If present, start SHALL have a lower value than end (expression : start.empty() or end.empty() or (start <= end), xpath: not(exists(f:start)) or not(exists(f:end)) or (f:start/@value <= f:end/@value))

2.26.2.10 SampledData

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

SampledData
Definition | A series of measurements taken by a device, with upper and lower limits. There may be more than one dimension in the data.
Control | 1..1
Requirements | There is a need for a concise way to handle the data produced by devices that sample a physical state at a high frequency.
Comments | The data is not interpretable without at least origin, period, and dimensions, but these are optional to allow a separation between the template of measurement and the actual measurement, such as between DeviceCapabilities and DeviceLog. When providing a summary view (for example with Observation.value[x]) SampledData should be represented with a brief display text such as "Sampled Data".
SampledData.origin
Definition | The base quantity that a measured value of zero represents. In addition, this provides the units of the entire measurement series.
Control | 1..1
Type | SimpleQuantity
Summary | true
SampledData.period
Definition | The length of time between sampling times, measured in milliseconds.
Control | 1..1
Type | decimal
Summary | true
Comments | This is usually a whole number.
SampledData.factor
Definition | A correction factor that is applied to the sampled data points before they are added to the origin.
Control | 0..1
Type | decimal
Default Value | 1
Summary | true
SampledData.lowerLimit
Definition | The lower limit of detection of the measured points. This is needed if any of the data points have the value "L" (lower than detection limit).
Control | 0..1
Type | decimal
Summary | true

SampledData.upperLimit
Definition | The upper limit of detection of the measured points. This is needed if any of the data points have the value "U" (higher than detection limit).
Control | 0..1
Type | decimal
Summary | true
SampledData.dimensions
Definition | The number of sample points at each time point. If this value is greater than one, then the dimensions will be interlaced - all the sample points for a point in time will be recorded at once.
Control | 1..1
Type | positiveInt
Summary | true
Comments | If there is more than one dimension, the code for the type of data will define the meaning of the dimensions (typically ECG data).
SampledData.data
Definition | A series of data points which are decimal values separated by a single space (character u20). The special values "E" (error), "L" (below detection limit) and "U" (above detection limit) can also be used in place of a decimal value.
Control | 1..1
Type | string

2.26.2.11 HumanName

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

HumanName
Definition | A human's name with the ability to identify parts and usage.
Control | 1..1
Requirements | Need to be able to record names, along with notes about their use.
Summary | true
Comments | Names may be changed, or repudiated, or people may have different names in different contexts. Names may be divided into parts of different type that have variable significance depending on context, though the division into parts does not always matter. With personal names, the different parts may or may not be imbued with some implicit meaning; various cultures associate different importance with the name parts and the degree to which systems must care about name parts around the world varies widely.
HumanName.use
Definition | Identifies the purpose for this name.
Control | 0..1
Terminology Binding | NameUse (Required)
Type | code
Is Modifier | true
Requirements | Allows the appropriate name for a particular context of use to be selected from among a set of names.
Summary | true
Comments | This is labeled as "Is Modifier" because applications should not mistake a temporary or old name etc.for a current/permanent one. Applications can assume that a name is current unless it explicitly says that it is temporary or old.
HumanName.text
Definition | A full text representation of the name.
Control | 0..1
Type | string
Requirements | A renderable, unencoded form.
Summary | true
Comments | Can provide both a text representation and structured parts.
HumanName.family
Definition | The part of a name that links to the genealogy. In some cultures (e.g. Eritrea) the family name of a son is the first name of his father.
Control | 0..1
Type | string
Alternate Names | surname
Summary | true
Comments | Family Name may be decomposed into specific parts using extensions (de, nl, es related cultures).
HumanName.given
Definition | Given name.
Control | 0..*
Type | string
Element Order Meaning | Given Names appear in the correct order for presenting the name
Alternate Names | first name; middle name
Summary | true
Comments | If only initials are recorded, they may be used in place of the full name. Not called "first name" since given names do not always come first.
HumanName.prefix
Definition | Part of the name that is acquired as a title due to academic, legal, employment or nobility status, etc. and that appears at the start of the name.
Control | 0..*
Type | string
Element Order Meaning | Prefixes appear in the correct order for presenting the name
Summary | true
HumanName.suffix

Definition | Part of the name that is acquired as a title due to academic, legal, employment or nobility status, etc. and that appears at the end of the name.
Control | 0..*
Type | string
Element Order Meaning | Suffixes appear in the correct order for presenting the name
Summary | true
HumanName.period
Definition | Indicates the period of time when this name was valid for the named person.
Control | 0..1
Type | Period
Requirements | Allows names to be placed in historical context.
Summary | true

2.26.2.12 Address

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Address
Definition | An address expressed using postal conventions (as opposed to GPS or other location definition formats). This data type may be used to convey addresses for use in delivering mail as well as for visiting locations which might not be valid for mail delivery. There are a variety of postal address formats defined around the world.
Control | 1..1
Requirements | Need to be able to record postal addresses, along with notes about their use.
Summary | true
Comments | Note: address is for postal addresses, not physical locations.
Address.use
Definition | The purpose of this address.
Control | 0..1
Terminology Binding | AddressUse (Required)
Type | code
Is Modifier | true
Requirements | Allows an appropriate address to be chosen from a list of many.
Summary | true
Comments | This is labeled as "Is Modifier" because applications should not mistake a temporary or old address etc.for a current/permanent one. Applications can assume that an address is current unless it explicitly says that it is temporary or old.
Address.type
Definition | Distinguishes between physical addresses (those you can visit) and mailing addresses (e.g. PO Boxes and care-of addresses). Most addresses are both.
Control | 0..1
Terminology Binding | AddressType (Required)
Type | code
Summary | true
Address.text
Definition | A full text representation of the address.
Control | 0..1
Type | string
Requirements | A renderable, unencoded form.
Summary | true
Comments | Can provide both a text representation and parts.
Address.line
Definition | This component contains the house number, apartment number, street name, street direction, P.O. Box number, delivery hints, and similar address information.
Control | 0..*
Type | string
Element Order Meaning | The order in which lines should appear in an address label
Requirements | home | work | temp | old - purpose of this address.
Summary | true
Address.city
Definition | The name of the city, town, village or other community or delivery center.
Control | 0..1
Type | string
Alternate Names | Municpality
Summary | true
Address.district
Definition | The name of the administrative area (county).
Control | 0..1
Type | string
Alternate Names | County
Summary | true
Comments | District is sometimes known as county, but in some regions 'county' is used in place of city (municipality), so county name should be conveyed in city instead.
Address.state
Definition | Sub-unit of a country with limited sovereignty in a federally organized country. A code may be used if codes are in common use (i.e. US 2 letter state codes).

Control | 0..1
Type | string
Alternate Names | Province; Territory
Summary | true
Address.postalCode
Definition | A postal code designating a region defined by the postal service.
Control | 0..1
Type | string
Alternate Names | Zip
Summary | true
Address.country
Definition | Country - a nation as commonly understood or generally accepted.
Control | 0..1
Type | string
Summary | true
Comments | ISO 3166 3 letter codes can be used in place of a full country name.
Address.period
Definition | Time period when address was/is in use.
Control | 0..1
Type | Period
Requirements | Allows addresses to be placed in historical context.
Summary | true

2.26.2.13 ContactPoint

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

ContactPoint
Definition | Details for all kinds of technology mediated contact points for a person or organization, including telephone, email, etc.
Control | 1..1
Requirements | Need to track phone, fax, mobile, sms numbers, email addresses, twitter tags, etc.
Summary | true
Invariants | Defined on this element cpt-2: A system is required if a value is provided. (expression : value.empty() or system.exists(), xpath: not(exists(f:value)) or exists(f:system))
ContactPoint.system
Definition | Telecommunications form for contact point - what communications system is required to make use of the contact.
Control | 0..1
Terminology Binding | ContactPointSystem (Required)
Type | code
Summary | true
Invariants | Affect this element cpt-2: A system is required if a value is provided. (expression : value.empty() or system.exists(), xpath: not(exists(f:value)) or exists(f:system))
ContactPoint.value
Definition | The actual contact point details, in a form that is meaningful to the designated communication system (i.e. phone number or email address).
Control | 0..1
Type | string
Requirements | Need to support legacy numbers that are not in a tightly controlled format.
Summary | true
Comments | Additional text data such as phone extension numbers, or notes about use of the contact are sometimes included in the value.
ContactPoint.use
Definition | Identifies the purpose for the contact point.
Control | 0..1
Terminology Binding | ContactPointUse (Required)
Type | code
Is Modifier | true
Requirements | Need to track the way a person uses this contact, so a user can choose which is appropriate for their purpose.
Summary | true
Comments | This is labeled as "Is Modifier" because applications should not mistake a temporary or old contact etc.for a current/permanent one. Applications can assume that a contact is current unless it explicitly says that it is temporary or old.
ContactPoint.rank
Definition | Specifies a preferred order in which to use a set of contacts. Contacts are ranked with lower values coming before higher values.
Control | 0..1
Type | positiveInt
Summary | true
Comments | Note that rank does not necessarily follow the order in which the contacts are represented in the instance.
ContactPoint.period
Definition | Time period when the contact point was/is in use.
Control | 0..1
Type | Period
Summary | true

2.26.2.14 Timing

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Timing

Definition | Specifies an event that may occur multiple times. Timing schedules are used to record when things are planned, expected or requested to occur. The most common usage is in dosage instructions for medications. They are also used when planning care of various kinds, and may be used for reporting the schedule to which past regular activities were carried out.

Control | 1..1

Requirements | Need to able to track proposed timing schedules. There are several different ways to do this: one or more specified times, a simple rules like three times a day, or before/after meals.

Summary | true

Comments | A timing schedule can be either a list of events - intervals on which the event occurs, or a single event with repeating criteria or just repeating criteria with no actual event. When both event and a repeating specification are provided, the list of events should be understood as an interpretation of the information in the repeat structure.

Timing.event

Definition | Identifies specific times when the event occurs.

Control | 0..*

Type | dateTime

Requirements | In an MAR, for instance, you need to take a general specification, and turn it into a precise specification.

Summary | true

Timing.repeat

Definition | A set of rules that describe when the event is scheduled.

Control | 0..1

Requirements | Many timing schedules are determined by regular repetitions.

Summary | true

Invariants | Defined on this element tim-1: if there's a duration, there needs to be duration units (expression : duration.empty() or durationUnit.exists(), xpath: not(exists(f:duration)) or exists(f:durationUnit)) tim-10: If there's a timeOfDay, there cannot be be a when, or vice versa (expression : timeOfDay.empty() or when.empty(), xpath: not(exists(f:timeOfDay)) or not(exists(f:when))) tim-2: if there's a period, there needs to be period units (expression : period.empty() or periodUnit.exists(), xpath: not(exists(f:period)) or exists(f:periodUnit)) tim-4: duration SHALL be a non-negative value (expression : duration.exists() implies duration >= 0, xpath: f:duration/@value >= 0 or not(f:duration/@value)) tim-5: period SHALL be a non-negative value (expression : period.exists() implies period >= 0, xpath: f:period/@value >= 0 or not(f:period/@value)) tim-6: If there's a periodMax, there must be a period (expression : periodMax.empty() or period.exists(), xpath: not(exists(f:periodMax)) or exists(f:period)) tim-7: If there's a durationMax, there must be a duration (expression : durationMax.empty() or duration.exists(), xpath: not(exists(f:durationMax)) or exists(f:duration)) tim-8: If there's a countMax, there must be a count (expression : countMax.empty() or count.exists(), xpath: not(exists(f:countMax)) or exists(f:count)) tim-9: If there's an offset, there must be a when (and not C, CM, CD, CV) (expression : offset.empty() or (when.exists() and ((when in ('C' | 'CM' | 'CD' | 'CV')).not())), xpath: not(exists(f:offset)) or exists(f:when))

Timing.repeat.bounds[x]

Definition | Either a duration for the length of the timing schedule, a range of possible length, or outer bounds for start and/or end limits of the timing schedule.

Control | 0..1

Type | Duration|Range|Period

[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true

Timing.repeat.count

Definition | A total count of the desired number of repetitions.

Control | 0..1

Type | integer

Requirements | Repetitions may be limited by end time or total occurrences.

Summary | true

Comments | If you have both bounds and count, then this should be understood as within the bounds period, until count times happens.

Timing.repeat.countMax

Definition | A maximum value for the count of the desired repetitions (e.g. do something 6-8 times).

Control | 0..1

Type | integer

Summary | true

Timing.repeat.duration

Definition | How long this thing happens for when it happens.

Control | 0..1

Type | decimal

Requirements | Some activities are not instantaneous and need to be maintained for a period of time.

Summary | true

Comments | For some events the duration is part of the definition of the event (e.g. IV infusions, where the duration is implicit in the specified quantity and rate). For others, it's part of the timing specification (e.g. exercise).

Timing.repeat.durationMax

Definition | The upper limit of how long this thing happens for when it happens.

Control | 0..1

Type | decimal

Requirements | Some activities are not instantaneous and need to be maintained for a period of time.

Summary | true

Comments | For some events the duration is part of the definition of the event (e.g. IV infusions, where the duration is implicit in the specified quantity and rate). For others, it's part of the timing specification (e.g. exercise).

Timing.repeat.durationUnit

Definition | The units of time for the duration, in UCUM units.

Control | 0..1

Terminology Binding | UnitsOfTime (Required)

Type | code
Summary | true
Timing.repeat.frequency
Definition | The number of times to repeat the action within the specified period / period range (i.e. both period and periodMax provided).
Control | 0..1
Type | integer
Default Value | 1
Summary | true
Timing.repeat.frequencyMax
Definition | If present, indicates that the frequency is a range - so to repeat between [frequency] and [frequencyMax] times within the period or period range.
Control | 0..1
Type | integer
Summary | true
Timing.repeat.period
Definition | Indicates the duration of time over which repetitions are to occur; e.g. to express "3 times per day", 3 would be the frequency and "1 day" would be the period.
Control | 0..1
Type | decimal
Summary | true
Timing.repeat.periodMax
Definition | If present, indicates that the period is a range from [period] to [periodMax], allowing expressing concepts such as "do this once every 3-5 days.
Control | 0..1
Type | decimal
Summary | true
Timing.repeat.periodUnit
Definition | The units of time for the period in UCUM units.
Control | 0..1
Terminology Binding | UnitsOfTime (Required)
Type | code
Summary | true
Timing.repeat.dayOfWeek
Definition | If one or more days of week is provided, then the action happens only on the specified day(s).
Control | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
Summary | true
Comments | If no days are specified, the action is assumed to happen every day as otherwise specified. The elements frequency and period cannot be used as well as dayOfWeek.
Timing.repeat.timeOfDay
Definition | Specified time of day for action to take place.
Control | 0..*
Type | time
Summary | true
Comments | When time of day is specified, it is inferred that the action happens every day (as filtered by dayofWeek) on the specified times. The elements when, frequency and period cannot be used as well as timeOfDay.
Timing.repeat.when
Definition | Real world events that the occurrence of the event should be tied to.
Control | 0..*
Terminology Binding | EventTiming (Required)
Type | code
Requirements | Timings are frequently determined by occurrences such as waking, eating and sleep.
Summary | true
Comments | When more than one event is listed, the event is tied to the union of the specified events.
Timing.repeat.offset
Definition | The number of minutes from the event. If the event code does not indicate whether the minutes is before or after the event, then the offset is assumed to be after the event.
Control | 0..1
Type | unsignedInt
Summary | true
Timing.code
Definition | A code for the timing schedule. Some codes such as BID are ubiquitous, but many institutions define their own additional codes. If a code is provided, the code is understood to be a complete statement of whatever is specified in the structured timing data, and either the code or the data may be used to interpret the Timing, with the exception that .repeat.bounds still applies over the code (and is not contained in the code).
Control | 0..1
Terminology Binding | TimingAbbreviation (Preferred)
Type | CodeableConcept
Summary | true
Comments | BID etc are defined as 'at institutionally specified times'. For example, an institution may choose that BID is "always at 7am and 6pm". If it is inappropriate for this choice to be made, the code BID should not be used. Instead, a distinct organization-specific code should be used in place of the HL7-defined BID code and/or the a structured representation should be used (in this case, specifying the two event times).

2.26.2.15 Signature

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Signature
Definition | A digital signature along with supporting context. The signature may be electronic/cryptographic in nature, or a graphical image representing a hand-written signature, or a signature process. Different signature approaches have different utilities.
Control | 1..1
Requirements | There are a number of places where content must be signed in healthcare.
Comments | The elements of the Signature Resource are for ease of access of these elements. Foro digital signatures (Xml DigSig, JWT), the non-repudiation proof comes from the Signature validation, which includes validation of the referenced objects (e.g. Resources) (a.k.a., Content) in the XML-Signature Detached form.
Signature.type
Definition | An indication of the reason that the entity signed this document. This may be explicitly included as part of the signature information and can be used when determining accountability for various actions concerning the document.
Control | 1..*
Terminology Binding | Signature Type Codes (Preferred)
Type | Coding
Summary | true
Comments | Examples include attesting to: authorship, correct transcription, and witness of specific event. Also known as a "Commitment Type Indication".
Signature.when
Definition | When the digital signature was signed.
Control | 1..1
Type | instant
Summary | true
Comments | This should agree with the information in the signature.
Signature.who[x]
Definition | A reference to an application-usable description of the identity that signed (e.g. the signature used their private key).
Control | 1..1
Type | uri|Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This should agree with the information in the signature.
Signature.onBehalfOf[x]
Definition | A reference to an application-usable description of the identity that is represented by the signature.
Control | 0..1
Type | uri|Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | used when the signature is on behalf of a non-signer.
Summary | true
Comments | The party that can't sign. For example a child.
Signature.contentType
Definition | A mime type that indicates the technical format of the signature. Important mime types are application/signature+xml for X ML DigSig, application/jwt for JWT, and image/* for a graphical image of a signature, etc.
Control | 0..1
Terminology Binding | MimeType :
Type | code
Summary | true
Signature.blob
Definition | The base64 encoding of the Signature content. When signature is not recorded electronically this element would be empty.
Control | 0..1
Type | base64Binary
Comments | Where the signature type is an XML DigSig, the signed content is a FHIR Resource(s), the signature is of the XML form of the Resource(s) using XML-Signature (XMLDIG) "Detached Signature" form.

2.26.2.16 Annotation

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Annotation
Definition | A text note which also contains information about who made the statement and when.
Control | 1..1
Comments | For systems that do not have structured annotations, they can simply communicate a single annotation with no author or time. This element may need to be included in narrative because of the potential for modifying information. Annotations SHOULD NOT be used to communicate "modifying" information that could be computable. (This is a SHOULD because enforcing user behavior is nearly impossible).
Annotation.author[x]
Definition | The individual responsible for making the annotation.
Control | 0..1
Type | Reference(Practitioner | Patient | RelatedPerson)|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Annotation.time
Definition | Indicates when this particular annotation was made.
Control | 0..1

Type | dateTime
Summary | true
Annotation.text
Definition | The text of the annotation.
Control | 1..1
Type | string

2.26.3 Data Type Mappings

URL: https://hl7.org/fhir/STU3/datatypes-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.26.3 Data Type Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This page provides mappings for the data types. There are mappings to HL7 v2 , HL7 v3 , and (where appropriate) vCard.

Table of Contents

Primitive Types | Complex Types

2.26.3.1 Primitive Types

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

Data Type | V2 | RIM
base64Binary | ~ED | BIN
boolean | n/a | BL
date | DT | TS
dateTime | DTM | TS
decimal | ~NM | REAL
instant | DTM | TS
integer | ~NM/~SI | INT
string | ST | ST
time | TM | n/a
uri | n/a | URL

2.26.3.2 Attachment

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.3 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Attachment | ED/RP
contentType | ED.2+ED.3/RP.2+RP.3. Note conversion may be needed if old style values are being used
language
data | ED.5
url | RP.1+RP.2 - if they refer to a URL (see v2.6)
size
hash
title
creation

2.26.3.4 Mappings for RIM Mapping (http://hl7.org/v3)

Attachment | ED
contentType | ./mediaType, ./charset
language | ./language
data | ./data
url | ./reference/literal
size | N/A (needs data type R3 proposal)
hash | .integrityCheck[parent::ED/integrityCheckAlgorithm="SHA-1"]
title | ./title/data
creation | N/A (needs data type R3 proposal)

2.26.3.5 Identifier

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.6 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Identifier | CX / EI (occasionally, more often EI maps to a resource id or a URL)
use | N/A
type | CX.5
system | CX.4 / EI-2-4
value | CX.1 / EI.1
period | CX.7 + CX.8

assigner | CX.4 / (CX.4,CX.9,CX.10)

2.26.3.7 Mappings for RIM Mapping (http://hl7.org/v3)

Identifier | II - see see identifier pattern at http://wiki.hl7.org/index.php?title=Common_Design_Patterns#Identifier_Pattern for relevant discussion. The Identifier class is a little looser than the v3 type II because it allows URIs as well as registered OIDs or GUIDs. Also maps to Role[classCode=IDENT]
use | Role.code or implied by context
type | Role.code or implied by context
system | II.root or Role.id.root
value | II.extension or II.root if system indicates OID or GUID (Or Role.id.extension or root)
period | Role.effectiveTime or implied by context
assigner | II.assigningAuthorityName but note that this is an improper use by the definition of the field. Also Role.scoper

2.26.3.8 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

Identifier | Identifier
use
type
system | ./IdentifierType
value | ./Value
period | ./StartDate and ./EndDate
assigner | ./IdentifierIssuingAuthority

2.26.3.9 Coding

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.10 Mappings for Ontological RIM Mapping (http://hl7.org/orim)

Coding | fhir:Coding rdfs:subClassOf dt:CDCoding
system | fhir:Coding.system rdfs:subPropertyOf dt:CDCoding.codeSystem
version | fhir:Coding.version rdfs:subPropertyOf dt:CDCoding.codeSystemVersion
code | fhir:Coding.code rdfs:subPropertyOf dt:CDCoding.code
display | fhir:Coding.display rdfs:subPropertyOf dt:CDCoding.displayName
userSelected | fhir:Coding.userSelected fhir:mapsTo dt:CDCoding.codingRationale. fhir:Coding.userSelected fhir:hasMap fhir:Coding.userSelected.map. fhir:Coding.userSelected.map a fhir:Map; fhir:target dt:CDCoding.codingRationale. fhir:Coding.userSelected\#true a [fhir:source "true"; fhir:target dt:CDCoding.codingRationale\#O]

2.26.3.11 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Coding | CE/CNE/CWE subset one of the sets of component 1-3 or 4-6
system | C*E.3
version | C*E.7
code | C*E.1
display | C*E.2 - but note this is not well followed
userSelected | Sometimes implied by being first

2.26.3.12 Mappings for RIM Mapping (http://hl7.org/v3)

Coding | CV
system | ./codeSystem
version | ./codeSystemVersion
code | ./code
display | CV.displayName
userSelected | CD.codingRationale

2.26.3.13 CodeableConcept

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.14 Mappings for Ontological RIM Mapping (http://hl7.org/orim)

CodeableConcept | fhir:CodeableConcept rdfs:subClassOf dt:CD
coding | fhir:CodeableConcept.coding rdfs:subPropertyOf dt:CD.coding
text | fhir:CodeableConcept.text rdfs:subPropertyOf dt:CD.originalText

2.26.3.15 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

CodeableConcept | CE/CNE/CWE
coding | C*E.1-8, C*E.10-22
text | C*E.9. But note many systems use C*E.2 for this

2.26.3.16 Mappings for RIM Mapping (http://hl7.org/v3)

CodeableConcept | CD
coding | union(., ./translation)
text | ./originalText[mediaType/code="text/plain"]/data

2.26.3.17 Quantity

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.18 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Quantity | SN (see also Range) or CQ
value | SN.2 / CQ - N/A
comparator | SN.1 / CQ.1
unit | (see OBX.6 etc.) / CQ.2
system | (see OBX.6 etc.) / CQ.2
code | (see OBX.6 etc.) / CQ.2

2.26.3.19 Mappings for RIM Mapping (http://hl7.org/v3)

Quantity | PQ, IVL<PQ>, MO, CO, depending on the values
value | PQ.value, CO.value, MO.value, IVL.high or IVL.low depending on the value
comparator | IVL properties
unit | PQ.unit
system | CO.codeSystem, PQ.translation.codeSystem
code | PQ.code, MO.currency, PQ.translation.code

2.26.3.20 Range

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.21 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Range | NR and also possibly SN (but see also quantity)
low | NR.1
high | NR.2

2.26.3.22 Mappings for RIM Mapping (http://hl7.org/v3)

Range | IVL<QTY[not(type="TS")]> [lowClosed="true" and highClosed="true"]or URG<QTY[not(type="TS")]>
low | ./low
high | ./high

2.26.3.23 Ratio

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.24 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Ratio | N/A

2.26.3.25 Mappings for RIM Mapping (http://hl7.org/v3)

Ratio | RTO
numerator | .numerator
denominator | .denominator

2.26.3.26 Period

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.27 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Period | DR
start | DR.1
end | DR.2

2.26.3.28 Mappings for RIM Mapping (http://hl7.org/v3)

Period | IVL<TS>[lowClosed="true" and highClosed="true"] or URG<TS>[lowClosed="true" and highClosed="true"]
start | ./low
end | ./high

2.26.3.29 SampledData

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.30 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

SampledData | N/A (but see section 7.14 in v2.7)
origin
period
factor
lowerLimit
upperLimit
dimensions
data

2.26.3.31 Mappings for RIM Mapping (http://hl7.org/v3)

SampledData | SLIST + GLIST
origin | SLIST.origin
period | GLIST.increment
factor | SLIST.factor
lowerLimit | N/A (use SLIST.digits.nullFlavor)
upperLimit | N/A (use SLIST.digits.nullFlavor)
dimensions | N/A (fixed to 1)
data | SLIST.digits

2.26.3.32 HumanName

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.33 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

HumanName | XPN
use | XPN.7, but often indicated by which field contains the name
text | implied by XPN.11
family | XPN.1/FN.1
given | XPN.2 + XPN.3
prefix | XPN.5
suffix | XPN/4
period | XPN.13 + XPN.14

2.26.3.34 Mappings for RIM Mapping (http://hl7.org/v3)

HumanName | EN (actually, PN)
use | unique(./use)
text | ./formatted
family | ./part[partType = FAM]
given | ./part[partType = GIV]
prefix | ./part[partType = PFX]
suffix | ./part[partType = SFX]
period | ./usablePeriod[type="IVL<TS>"]

2.26.3.35 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

HumanName | ProviderName
use | ./NamePurpose
text
family | ./FamilyName
given | ./GivenNames
prefix | ./TitleCode
suffix
period | ./StartDate and ./EndDate

vCard Mappings

* HumanName.text = vCard "FN" field

* HumanName.use = use of the vCard "TYPE" parameter

* HumanName.family, .given, .prefix, .suffix = parts of vCard "N" field. Note that there is no FHIR equivalent for the poorly defined "additional" name field. In FHIR, given names go in "middle" names

* The vCard nickname corresponds to a name with the use "nickname"

2.26.3.36 Address

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.37 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Address | XAD
use | XAD.7
type | XAD.18
text | XAD.1 + XAD.2 + XAD.3 + XAD.4 + XAD.5 + XAD.6
line | XAD.1 + XAD.2 (note: XAD.1 and XAD.2 have different meanings for a company address than for a person address)
city | XAD.3
district | XAD.9
state | XAD.4
postalCode | XAD.5
country | XAD.6
period | XAD.12 / XAD.13 + XAD.14

2.26.3.38 Mappings for RIM Mapping (http://hl7.org/v3)

Address | AD
use | unique(./use)

type | unique(./use)
text | ./formatted
line | AD.part[parttype = AL]
city | AD.part[parttype = CTY]
district | AD.part[parttype = CNT | CPA]
state | AD.part[parttype = STA]
postalCode | AD.part[parttype = ZIP]
country | AD.part[parttype = CNT]
period | ./usablePeriod[type="IVL<TS>"]

2.26.3.39 Mappings for vCard Mapping (http://w3.org/vcard)

Address
use
type | address type parameter
text | address label parameter
line | street
city | locality
district
state | region
postalCode | code
country | country
period

2.26.3.40 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

Address | Address
use | ./AddressPurpose
type
text
line | ./StreetAddress (newline delimitted)
city | ./Jurisdiction
district
state | ./Region
postalCode | ./PostalIdentificationCode
country | ./Country
period | ./StartDate and ./EndDate

2.26.3.41 ContactPoint

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.42 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ContactPoint | XTN
system | XTN.3
value | XTN.1 (or XTN.12)
use | XTN.2 - but often indicated by field
rank | n/a
period | N/A

2.26.3.43 Mappings for RIM Mapping (http://hl7.org/v3)

ContactPoint | TEL
system | ./scheme
value | ./url
use | unique(./use)
rank | n/a
period | ./usablePeriod[type="IVL<TS>"]

2.26.3.44 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

ContactPoint | ContactPoint
system | ./ContactPointType
value | ./Value
use | ./ContactPointPurpose
rank
period | ./StartDate and ./EndDate

2.26.3.45 Timing

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.46 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Timing | N/A

2.26.3.47 Mappings for RIM Mapping (http://hl7.org/v3)

Timing | QSET<TS> (GTS)
event | QLIST<TS>
repeat | Implies PIVL or EIVL
bounds[x] | IVL(TS) used in a QSI
count | PIVL.count
countMax | PIVL.count
duration | PIVL.phase
durationMax | PIVL.phase
durationUnit | PIVL.phase.unit
frequency | PIVL.phase
frequencyMax | PIVL.phase
period | PIVL.phase
periodMax | PIVL.phase
periodUnit | PIVL.phase.unit
dayOfWeek | n/a
timeOfDay | n/a
when | EIVL.event
offset | EIVL.offset
code | QSC.code

2.26.3.48 Signature

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.49 Mappings for RIM Mapping (http://hl7.org/v3)

Signature | N/A

2.26.3.50 Annotation

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.26.3.51 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Annotation | N/A

2.26.3.52 Mappings for RIM Mapping (http://hl7.org/v3)

Annotation | Act
author[x] | Act.participant[typeCode=AUT].role
time | Act.effectiveTime
text | Act.text

2.26.4 Data Type Profiles and Extensions

URL: https://hl7.org/fhir/STU3/datatypes-extras.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.26.4 Data Type Profiles and Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This page lists profiles and extensions defined in this specification itself for the data types. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

Table of Contents

Primitive Types | Complex Types

2.26.4.1 Primitive Types

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
iso21090-SC-coding | SC-coding | for ISO 21090 Data Type Extensions
rendered-value | Rendered Value | for General Extensions for use by FHIR Implementers
rendering-markdown | markdown | for Extensions for rendering data elements
rendering-xhtml | xhtml | for Extensions for rendering data elements
translation | Translation | for General Extensions for use by FHIR Implementers
Profiles
No Profiles defined for for primitive types

2.26.4.2 Attachment

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
cqif-citation | citation | for CQIF Extensions

cqif-qualityOfEvidence | qualityOfEvidence | for CQIF Extensions
cqif-strengthOfRecommendation | strengthOfRecommendation | for CQIF Extensions
Profiles
No Profiles defined for for this type

2.26.4.3 Identifier

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
identifier-validDate | validDate | for Common extensions for Identifier data type
rendered-value | Rendered Value | for General Extensions for use by FHIR Implementers
Profiles
No Profiles defined for for this type

2.26.4.4 Coding

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
codesystem-reference | reference | for CodeSystem HL7 Extensions
coding-sctdescid | sctdescid | for Common extensions for Coding data type
cqif-sourceValueSet | sourceValueSet | for CQIF Extensions
iso21090-CO-value | CO-value | for ISO 21090 Data Type Extensions
valueset-reference | reference | for ValueSet HL7 Extensions
Profiles
No Profiles defined for for this type

2.26.4.5 CodeableConcept

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.6 Quantity

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
iso21090-uncertainty | uncertainty | for ISO 21090 Data Type Extensions
iso21090-uncertaintyType | uncertaintyType | for ISO 21090 Data Type Extensions
Profiles
No Profiles defined for for this type

2.26.4.7 Range

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.8 Ratio

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.9 Period

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.10 SampledData

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.11 HumanName

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
humanname-assembly-order | assembly-order | for Common extensions for HumanName data type
humanname-fathers-family | fathers-family | for Common extensions for HumanName data type
humanname-mothers-family | mothers-family | for Common extensions for HumanName data type
humanname-own-name | own-name | for Common extensions for HumanName data type
humanname-own-prefix | own-prefix | for Common extensions for HumanName data type
humanname-partner-name | partner-name | for Common extensions for HumanName data type
humanname-partner-prefix | partner-prefix | for Common extensions for HumanName data type
iso21090-EN-qualifier | EN-qualifier | for ISO 21090 Data Type Extensions
iso21090-EN-representation | EN-representation | for ISO 21090 Data Type Extensions
Profiles
No Profiles defined for for this type

2.26.4.12 Address

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
geolocation | Geolocation | for General Extensions for use by FHIR Implementers
iso21090-AD-use | AD-use | for ISO 21090 Data Type Extensions
iso21090-ADXP-additionalLocator | ADXP-additionalLocator | for ISO 21090 Data Type Extensions
iso21090-ADXP-buildingNumberSuffix | ADXP-buildingNumberSuffix | for ISO 21090 Data Type Extensions
iso21090-ADXP-careOf | ADXP-careOf | for ISO 21090 Data Type Extensions
iso21090-ADXP-censusTract | ADXP-censusTract | for ISO 21090 Data Type Extensions
iso21090-ADXP-delimiter | ADXP-delimiter | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryAddressLine | ADXP-deliveryAddressLine | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationArea | ADXP-deliveryInstallationArea | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationQualifier | ADXP-deliveryInstallationQualifier | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationType | ADXP-deliveryInstallationType | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryMode | ADXP-deliveryMode | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryModeIdentifier | ADXP-deliveryModeIdentifier | for ISO 21090 Data Type Extensions
iso21090-ADXP-direction | ADXP-direction | for ISO 21090 Data Type Extensions
iso21090-ADXP-houseNumber | ADXP-houseNumber | for ISO 21090 Data Type Extensions
iso21090-ADXP-houseNumberNumeric | ADXP-houseNumberNumeric | for ISO 21090 Data Type Extensions
iso21090-ADXP-postBox | ADXP-postBox | for ISO 21090 Data Type Extensions
iso21090-ADXP-precinct | ADXP-precinct | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetAddressLine | ADXP-streetAddressLine | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetName | ADXP-streetName | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetNameBase | ADXP-streetNameBase | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetNameType | ADXP-streetNameType | for ISO 21090 Data Type Extensions
iso21090-ADXP-unitID | ADXP-unitID | for ISO 21090 Data Type Extensions
iso21090-ADXP-unitType | ADXP-unitType | for ISO 21090 Data Type Extensions
Profiles
No Profiles defined for for this type

2.26.4.13 ContactPoint

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
iso21090-TEL-address | TEL-address | for ISO 21090 Data Type Extensions
Profiles
No Profiles defined for for this type

2.26.4.14 Timing

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
timing-exact | exact | for Common extensions for Timing data type
Profiles
No Profiles defined for for this type

2.26.4.15 Signature

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.4.16 Annotation

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions
No Extensions defined for this type
Profiles
No Profiles defined for for this type

2.26.5 Data Type Detailed Descriptions

URL: https://hl7.org/fhir/STU3/datatypes-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.26.5 Data Type Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This page provides the detailed descriptions for the data types

Table of Contents

Primitive Types | Complex Types

2.26.5.1 Primitive Types

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.1.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/primitives2to3" = "R2 to R3 Primitive Conversions"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/base64Binary" alias base64BinaryR2 as source
uses "http://hl7.org/fhir/StructureDefinition/base64Binary" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/boolean" alias booleanR2 as source
uses "http://hl7.org/fhir/StructureDefinition/boolean" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/code" alias codeR2 as source
uses "http://hl7.org/fhir/StructureDefinition/code" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/date" alias dateR2 as source
uses "http://hl7.org/fhir/StructureDefinition/date" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/dateTime" alias dateTimeR2 as source
uses "http://hl7.org/fhir/StructureDefinition/dateTime" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/decimal" alias decimalR2 as source
uses "http://hl7.org/fhir/StructureDefinition/decimal" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/id" alias idR2 as source
uses "http://hl7.org/fhir/StructureDefinition/id" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/instant" alias instantR2 as source
uses "http://hl7.org/fhir/StructureDefinition/instant" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/integer" alias integerR2 as source
uses "http://hl7.org/fhir/StructureDefinition/integer" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/markdown" alias markdownR2 as source
uses "http://hl7.org/fhir/StructureDefinition/markdown" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/oid" alias oidR2 as source
uses "http://hl7.org/fhir/StructureDefinition/oid" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/positiveInt" alias positiveIntR2 as source
uses "http://hl7.org/fhir/StructureDefinition/positiveInt" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/string" alias stringR2 as source
uses "http://hl7.org/fhir/StructureDefinition/string" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/time" alias timeR2 as source
uses "http://hl7.org/fhir/StructureDefinition/time" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/unsignedInt" alias unsignedIntR2 as source
uses "http://hl7.org/fhir/StructureDefinition/unsignedInt" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/uri" alias uriR2 as source
uses "http://hl7.org/fhir/StructureDefinition/uri" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/uuid" alias uuidR2 as source
uses "http://hl7.org/fhir/StructureDefinition/uuid" as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/xhtml" alias xhtmlR2 as source
uses "http://hl7.org/fhir/StructureDefinition/xhtml" as target

imports "http://hl7.org/fhir/StructureMap/Element2to3"

group for type+types base64Binary extends Element
  input src : base64BinaryR2 as source
  input tgt : base64Binary as target

  "base64Binary-value" : for src.value as v make tgt.value = v

endgroup

group for type+types boolean extends Element
  input src : booleanR2 as source
  input tgt : boolean as target

  "boolean-value" : for src.value as v make tgt.value = v
endgroup

group for type+types code extends Element
  input src : codeR2 as source
```

```
  input tgt : code as target

  "code-value" : for src.value as v make tgt.value = v
endgroup

group for type+types date extends Element
  input src : dateR2 as source
  input tgt : date as target

  "date-value" : for src.value as v make tgt.value = v
endgroup

group for type+types dateTime extends Element
  input src : dateTimeR2 as source
  input tgt : dateTime as target

  "dateTime-value" : for src.value as v make tgt.value = v
endgroup

group for type+types decimal extends Element
  input src : decimalR2 as source
  input tgt : decimal as target

  "decimal-value" : for src.value as v make tgt.value = v
endgroup

group for type+types id extends Element
  input src : idR2 as source
  input tgt : id as target

  "id-value" : for src.value as v make tgt.value = v
endgroup

group for type+types instant extends Element
  input src : instantR2 as source
  input tgt : instant as target

  "instant-value" : for src.value as v make tgt.value = v
endgroup

group for type+types integer extends Element
  input src : integerR2 as source
  input tgt : integer as target

  "integer-value" : for src.value as v make tgt.value = v
endgroup

group for type+types markdown extends Element
  input src : markdownR2 as source
  input tgt : markdown as target

  "markdown-value" : for src.value as v make tgt.value = v
endgroup

group for type+types oid extends Element
  input src : oidR2 as source
  input tgt : oid as target

  "oid-value" : for src.value as v make tgt.value = v
endgroup

group for type+types positiveInt extends Element
  input src : positiveIntR2 as source
  input tgt : positiveInt as target

  "positiveInt-value" : for src.value as v make tgt.value = v
endgroup

group for type+types string extends Element
  input src : stringR2 as source
  input tgt : string as target

  "string-value" : for src.value as v make tgt.value = v
endgroup

group for type+types time extends Element
  input src : timeR2 as source
  input tgt : time as target

  "time-value" : for src.value as v make tgt.value = v
endgroup

group for type+types unsignedInt extends Element
  input src : unsignedIntR2 as source
  input tgt : unsignedInt as target

  "unsignedInt-value" : for src.value as v make tgt.value = v
endgroup

group for type+types uri extends Element
```

```
  input src : uriR2 as source
  input tgt : uri as target

  "uri-value" : for src.value as v make tgt.value = v
endgroup

group for type+types uuid extends Element
  input src : uuidR2 as source
  input tgt : uuid as target

  "uuid-value" : for src.value as v make tgt.value = v
endgroup

group for type+types xhtml extends Element
  input src : xhtmlR2 as source
  input tgt : xhtml as target

  "xhtml-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToMarkdown extends Element
  input src : stringR2 as source
  input tgt : markdown as target

  "stringToMarkdown-value" : for src.value as v make tgt.value = v
endgroup

group for types codeToUri extends Element
  input src : codeR2 as source
  input tgt : uri as target

  "codeToUri-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToId extends Element
  input src : stringR2 as source
  input tgt : id as target

  "stringToId-value" : for src.value as v make tgt.value = v
endgroup

group for types idToString extends Element
  input src : idR2 as source
  input tgt : string as target

  "stringToId-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToUri extends Element
  input src : stringR2 as source
  input tgt : uri as target

  "stringToUri-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToCode extends Element
  input src : stringR2 as source
  input tgt : code as target

  "stringToCode-value" : for src.value as v make tgt.value = v
endgroup

group for types codeToString extends Element
  input src : codeR2 as source
  input tgt : string as target

  "stringToCode-value" : for src.value as v make tgt.value = v
endgroup

group for types integerToPositiveInt extends Element
  input src : integerR2 as source
  input tgt : positiveInt as target

  "integerToPositiveInt-value" : for src.value as v make tgt.value = v
endgroup

group for types integerToUnsignedInt extends Element
  input src : integerR2 as source
  input tgt : unsignedInt as target

  "integerToUnsignedInt-value" : for src.value as v make tgt.value = v
endgroup

group for types dateToDateTime extends Element
  input src : dateR2 as source
  input tgt : dateTime as target

  "dateToDateTime-value" : for src.value as v make tgt.value = v
endgroup
```

```
group for types dateTimeToDate extends Element
  input src : dateTimeR2 as source
  input tgt : date as target

  "dateTimeToDate-value" : for src.value as v make tgt.value = (v.substring(0, 8))
endgroup
```

2.26.5.1.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/primitives3to2" = "R3 to R2 Primitive Conversions"

uses "http://hl7.org/fhir/StructureDefinition/base64Binary" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/base64Binary" alias base64BinaryR2 as target
uses "http://hl7.org/fhir/StructureDefinition/boolean" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/boolean" alias booleanR2 as target
uses "http://hl7.org/fhir/StructureDefinition/code" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/code" alias codeR2 as target
uses "http://hl7.org/fhir/StructureDefinition/date" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/date" alias dateR2 as target
uses "http://hl7.org/fhir/StructureDefinition/dateTime" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/dateTime" alias dateTimeR2 as target
uses "http://hl7.org/fhir/StructureDefinition/decimal" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/decimal" alias decimalR2 as target
uses "http://hl7.org/fhir/StructureDefinition/id" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/id" alias idR2 as target
uses "http://hl7.org/fhir/StructureDefinition/instant" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/instant" alias instantR2 as target
uses "http://hl7.org/fhir/StructureDefinition/integer" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/integer" alias integerR2 as target
uses "http://hl7.org/fhir/StructureDefinition/markdown" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/markdown" alias markdownR2 as target
uses "http://hl7.org/fhir/StructureDefinition/oid" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/oid" alias oidR2 as target
uses "http://hl7.org/fhir/StructureDefinition/positiveInt" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/positiveInt" alias positiveIntR2 as target
uses "http://hl7.org/fhir/StructureDefinition/string" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/string" alias stringR2 as target
uses "http://hl7.org/fhir/StructureDefinition/time" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/time" alias timeR2 as target
uses "http://hl7.org/fhir/StructureDefinition/unsignedInt" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/unsignedInt" alias unsignedIntR2 as target
uses "http://hl7.org/fhir/StructureDefinition/uri" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/uri" alias uriR2 as target
uses "http://hl7.org/fhir/StructureDefinition/uuid" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/uuid" alias uuidR2 as target
uses "http://hl7.org/fhir/StructureDefinition/xhtml" as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/xhtml" alias xhtmlR2 as target

imports "http://hl7.org/fhir/StructureMap/Element3to2"

group for type+types base64Binary extends Element
  input src : base64Binary as source
  input tgt : base64BinaryR2 as target

  "base64Binary-value" : for src.value as v make tgt.value = v
endgroup

group for type+types boolean extends Element
  input src : boolean as source
  input tgt : booleanR2 as target

  "boolean-value" : for src.value as v make tgt.value = v
endgroup

group for type+types code extends Element
  input src : code as source
  input tgt : codeR2 as target

  "code-value" : for src.value as v make tgt.value = v
endgroup

group for type+types date extends Element
  input src : date as source
  input tgt : dateR2 as target

  "date-value" : for src.value as v make tgt.value = v
endgroup

group for type+types dateTime extends Element
  input src : dateTime as source
  input tgt : dateTimeR2 as target

  "dateTime-value" : for src.value as v make tgt.value = v
endgroup

group for type+types decimal extends Element
  input src : decimal as source
  input tgt : decimalR2 as target

  "decimal-value" : for src.value as v make tgt.value = v
```

```
endgroup

group for type+types id extends Element
  input src : id as source
  input tgt : idR2 as target

  "id-value" : for src.value as v make tgt.value = v
endgroup

group for type+types instant extends Element
  input src : instant as source
  input tgt : instantR2 as target

  "instant-value" : for src.value as v make tgt.value = v
endgroup

group for type+types integer extends Element
  input src : integer as source
  input tgt : integerR2 as target

  "integer-value" : for src.value as v make tgt.value = v
endgroup

group for type+types markdown extends Element
  input src : markdown as source
  input tgt : markdownR2 as target

  "markdown-value" : for src.value as v make tgt.value = v
endgroup

group for type+types oid extends Element
  input src : oid as source
  input tgt : oidR2 as target

  "oid-value" : for src.value as v make tgt.value = v
endgroup

group for type+types positiveInt extends Element
  input src : positiveInt as source
  input tgt : positiveIntR2 as target

  "positiveInt-value" : for src.value as v make tgt.value = v
endgroup

group for type+types string extends Element
  input src : string as source
  input tgt : stringR2 as target

  "string-value" : for src.value as v make tgt.value = v
endgroup

group for type+types time extends Element
  input src : time as source
  input tgt : timeR2 as target

  "time-value" : for src.value as v make tgt.value = v
endgroup

group for type+types unsignedInt extends Element
  input src : unsignedInt as source
  input tgt : unsignedIntR2 as target

  "unsignedInt-value" : for src.value as v make tgt.value = v
endgroup

group for type+types uri extends Element
  input src : uri as source
  input tgt : uriR2 as target

  "uri-value" : for src.value as v make tgt.value = v
endgroup

group for type+types uuid extends Element
  input src : uuid as source
  input tgt : uuidR2 as target

  "uuid-value" : for src.value as v make tgt.value = v
endgroup

group for type+types xhtml extends Element
  input src : xhtml as source
  input tgt : xhtmlR2 as target

  "xhtml-value" : for src.value as v make tgt.value = v
endgroup

group for types markdownToString extends Element
  input src : markdown as source
  input tgt : stringR2 as target
```

```
  "markdownToString-value" : for src.value as v make tgt.value = v
endgroup

group for types uriToCode extends Element
  input src : uri as source
  input tgt : codeR2 as target

  "uriToCode-value" : for src.value as v make tgt.value = v
endgroup

group for types uriToString extends Element
  input src : uri as source
  input tgt : stringR2 as target

  "uriToString-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToXhtml extends Element
  input src : string as source
  input tgt : xhtmlR2 as target

  "stringToXhtml-value" : for src.value as v make tgt.value = v
endgroup

group for types codeToString extends Element
  input src : code as source
  input tgt : stringR2 as target

  "uriToCode-value" : for src.value as v make tgt.value = v
endgroup

group for types stringToId extends Element
  input src : string as source
  input tgt : idR2 as target

  "stringToId-value" : for src.value as v make tgt.value = v
endgroup

group for types idToString extends Element
  input src : id as source
  input tgt : stringR2 as target

  "stringToId-value" : for src.value as v make tgt.value = v
endgroup

group for types positiveIntToInteger extends Element
  input src : positiveIntR2 as source
  input tgt : integer as target

  "positiveIntToInteger-value" : for src.value as v make tgt.value = v
endgroup

group for types unsignedIntToInteger extends Element
  input src : unsignedIntR2 as source
  input tgt : integer as target

  "unsignedIntToInteger-value" : for src.value as v make tgt.value = v
endgroup

group for types dateToDateTime extends Element
  input src : date as source
  input tgt : dateTimeR2 as target

  "dateToDateTime-value" : for src.value as v make tgt.value = v
endgroup

group for types dateTimeToDate extends Element
  input src : dateTime as source
  input tgt : dateR2 as target

  "dateTimeToDate-value" : for src.value as v make tgt.value = (v.substring(0, 8))
endgroup
```

2.26.5.2 Attachment

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.2.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Attachment2to3" = "R2 to R3 Conversions for Attachment"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Attachment" alias AttachmentR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Attachment" alias Attachment as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"

group for type+types  Attachment extends Element
```

```
  input src : AttachmentR2 as source
  input tgt : Attachment as target

  "Attachment-contentType" : for src.contentType make tgt.contentType
  "Attachment-language" : for src.language make tgt.language
  "Attachment-data" : for src.data make tgt.data
  "Attachment-url" : for src.url make tgt.url
  "Attachment-size" : for src.size make tgt.size
  "Attachment-hash" : for src.hash make tgt.hash
  "Attachment-title" : for src.title make tgt.title
  "Attachment-creation" : for src.creation make tgt.creation

endgroup
```

2.26.5.2.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Attachment3to2" = "R3 to R2 Conversion for Attachment"

uses "http://hl7.org/fhir/StructureDefinition/Attachment" alias Attachment as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Attachment" alias AttachmentR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"

group for type+types  Attachment extends Element
  input src : Attachment as source
  input tgt : AttachmentR2 as target

  "Attachment-contentType" : for src.contentType make tgt.contentType
  "Attachment-language" : for src.language make tgt.language
  "Attachment-data" : for src.data make tgt.data
  "Attachment-url" : for src.url make tgt.url
  "Attachment-size" : for src.size make tgt.size
  "Attachment-hash" : for src.hash make tgt.hash
  "Attachment-title" : for src.title make tgt.title
  "Attachment-creation" : for src.creation make tgt.creation

endgroup
```

2.26.5.3 Identifier

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.3.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Identifier2to3" = "R2 to R3 Conversions for Identifier"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Identifier" alias IdentifierR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Identifier" alias Identifier as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Reference2to3"
imports "http://hl7.org/fhir/StructureMap/Period2to3"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept2to3"

group for type+types  Identifier extends Element
  input src : IdentifierR2 as source
  input tgt : Identifier as target

  "Identifier-use" : for src.use make tgt.use
  "Identifier-type" : for src.type make tgt.type
  "Identifier-system" : for src.system make tgt.system
  "Identifier-value" : for src.value make tgt.value
  "Identifier-period" : for src.period make tgt.period
  "Identifier-assigner" : for src.assigner make tgt.assigner

endgroup
```

2.26.5.3.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Identifier3to2" = "R3 to R2 Conversion for Identifier"

uses "http://hl7.org/fhir/StructureDefinition/Identifier" alias Identifier as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Identifier" alias IdentifierR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"
imports "http://hl7.org/fhir/StructureMap/Period3to2"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept3to2"

group for type+types  Identifier extends Element
  input src : Identifier as source
  input tgt : IdentifierR2 as target

  "Identifier-use" : for src.use make tgt.use
  "Identifier-type" : for src.type make tgt.type
  "Identifier-system" : for src.system make tgt.system
```

```
  "Identifier-value" : for src.value make tgt.value
  "Identifier-period" : for src.period make tgt.period
  "Identifier-assigner" : for src.assigner make tgt.assigner

endgroup
```

2.26.5.4 Coding

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.4.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Coding2to3" = "R2 to R3 Conversions for Coding"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Coding" alias CodingR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Coding" alias Coding as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"

group for type+types  Coding extends Element
  input src : CodingR2 as source
  input tgt : Coding as target

  "Coding-system" : for src.system make tgt.system
  "Coding-version" : for src.version make tgt.version
  "Coding-code" : for src.code make tgt.code
  "Coding-display" : for src.display make tgt.display
  "Coding-userSelected" : for src.userSelected make tgt.userSelected
endgroup
```

2.26.5.4.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Coding3to2" = "R3 to R2 Conversion for Coding"

uses "http://hl7.org/fhir/StructureDefinition/Coding" alias Coding as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Coding" alias CodingR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"

group for type+types  Coding extends Element
  input src : Coding as source
  input tgt : CodingR2 as target

  "Coding-system" : for src.system make tgt.system
  "Coding-version" : for src.version make tgt.version
  "Coding-code" : for src.code make tgt.code
  "Coding-display" : for src.display make tgt.display
  "Coding-userSelected" : for src.userSelected make tgt.userSelected

endgroup
```

2.26.5.5 CodeableConcept

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.5.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/CodeableConcept2to3" = "R2 to R3 Conversions for CodeableConcept"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CodeableConcept" alias CodeableConceptR2 as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Coding" alias CodingR2 as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/string" alias stringR2 as source
uses "http://hl7.org/fhir/StructureDefinition/CodeableConcept" alias CodeableConcept as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Coding2to3"

group for type+types  CodeableConcept extends Element
  input src : CodeableConceptR2 as source
  input tgt : CodeableConcept as target

  "CodeableConcept-coding" : for src.coding make tgt.coding
  "CodeableConcept-text" : for src.text make tgt.text
endgroup

group for types CodingToCodeableConcept extends Element
  input src : CodingR2 as source
  input tgt : CodeableConcept as target

  "CodingToCodeableConcept-coding" : for src make tgt.coding as vt then Coding(src, vt)

endgroup

group for types CodeableConceptToCoding extends Element
```

```
  input src : CodeableConceptR2 as source
  input tgt : Coding as target

  "CodeableConcept-coding" : for src.coding only_one as vs then Coding(vs, tgt)
endgroup

group for types stringToCodeableConcept extends Element
  input src : stringR2 as source
  input tgt : CodeableConcept as target

  "stringToCodeableConcept-text" : for src make tgt.text as vt then string(src, vt)
endgroup
```

2.26.5.5.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/CodeableConcept3to2" = "R3 to R2 Conversion for CodeableConcept"

uses "http://hl7.org/fhir/StructureDefinition/CodeableConcept" alias CodeableConcept as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CodeableConcept" alias CodeableConceptR2 as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Coding" alias CodingR2 as target
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/string" alias stringR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Coding3to2"

group for type+types  CodeableConcept extends Element
  input src : CodeableConcept as source
  input tgt : CodeableConceptR2 as target

  "CodeableConcept-coding" : for src.coding make tgt.coding
  "CodeableConcept-text" : for src.text make tgt.text

endgroup

group for types CodeableConceptToCoding extends Element
  input src : CodeableConcept as source
  input tgt : CodingR2 as target

  "CodeableConcept-coding" : for src.coding only_one as vs then Coding(vs, tgt)
endgroup

group for types CodingToCodeableConcept extends Element
  input src : Coding as source
  input tgt : CodeableConceptR2 as target

  "CodingToCodeableConcept-coding" : for src make tgt.coding as vt then Coding(src, vt)
endgroup


group for types CodeableConceptToString extends Element
  input src : CodeableConcept as source
  input tgt : stringR2 as target

  "CodeableConceptToString-text" : for src.text as vs then string(vs, tgt)
endgroup
```

2.26.5.6 Quantity

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.6.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Quantity2to3" = "R2 to R3 Conversions for Quantity"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Quantity" alias QuantityR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Quantity" alias Quantity as target
uses "http://hl7.org/fhir/StructureDefinition/Duration" alias Duration as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"

group for type+types  Quantity extends Element
  input src : QuantityR2 as source
  input tgt : Quantity as target

  "Quantity-value" : for src.value make tgt.value
  "Quantity-comparator" : for src.comparator make tgt.comparator
  "Quantity-unit" : for src.unit make tgt.unit
  "Quantity-system" : for src.system make tgt.system
  "Quantity-code" : for src.code make tgt.code

endgroup

group QuantityToDuration extends Element
  input src : QuantityR2 as source
  input tgt : Duration as target

  "Duration-value" : for src.value make tgt.value
```

```
  "Duration-comparator" : for src.comparator make tgt.comparator
  "Duration-unit" : for src.unit make tgt.unit
  "Duration-system" : for src.system make tgt.system
  "Duration-code" : for src.code make tgt.code

endgroup
```

2.26.5.6.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Quantity3to2" = "R3 to R2 Conversion for Quantity"

uses "http://hl7.org/fhir/StructureDefinition/Quantity" alias Quantity as source
uses "http://hl7.org/fhir/StructureDefinition/Duration" alias Duration as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Quantity" alias QuantityR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"

group for type+types  Quantity extends Element
  input src : Quantity as source
  input tgt : QuantityR2 as target

  "Quantity-value" : for src.value make tgt.value
  "Quantity-comparator" : for src.comparator make tgt.comparator
  "Quantity-unit" : for src.unit make tgt.unit
  "Quantity-system" : for src.system make tgt.system
  "Quantity-code" : for src.code make tgt.code

endgroup

group DurationToQuantity extends Element
  input src : Duration as source
  input tgt : QuantityR2 as target

  "Duration-value" : for src.value make tgt.value
  "Duration-comparator" : for src.comparator make tgt.comparator
  "Duration-unit" : for src.unit make tgt.unit
  "Duration-system" : for src.system make tgt.system
  "Duration-code" : for src.code make tgt.code
endgroup
```

2.26.5.7 Range

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.7.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Range2to3" = "R2 to R3 Conversions for Range"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Range" alias RangeR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Range" alias Range as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Quantity2to3"

group for type+types  Range extends Element
  input src : RangeR2 as source
  input tgt : Range as target

  "Range-low" : for src.low make tgt.low
  "Range-high" : for src.high make tgt.high

endgroup
```

2.26.5.7.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Range3to2" = "R3 to R2 Conversion for Range"

uses "http://hl7.org/fhir/StructureDefinition/Range" alias Range as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Range" alias RangeR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Quantity3to2"

group for type+types  Range extends Element
  input src : Range as source
  input tgt : RangeR2 as target

  "Range-low" : for src.low make tgt.low
  "Range-high" : for src.high make tgt.high

endgroup
```

2.26.5.8 Ratio

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.8.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Ratio2to3" = "R2 to R3 Conversions for Ratio"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Ratio" alias RatioR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Ratio" alias Ratio as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Quantity2to3"

group for type+types  Ratio extends Element
  input src : RatioR2 as source
  input tgt : Ratio as target

  "Ratio-numerator" : for src.numerator make tgt.numerator
  "Ratio-denominator" : for src.denominator make tgt.denominator

endgroup
```

2.26.5.8.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Ratio3to2" = "R3 to R2 Conversion for Ratio"

uses "http://hl7.org/fhir/StructureDefinition/Ratio" alias Ratio as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Ratio" alias RatioR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Quantity3to2"

group for type+types  Ratio extends Element
  input src : Ratio as source
  input tgt : RatioR2 as target

  "Ratio-numerator" : for src.numerator make tgt.numerator
  "Ratio-denominator" : for src.denominator make tgt.denominator

endgroup
```

2.26.5.9 Period

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.9.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Period2to3" = "R2 to R3 Conversions for Period"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Period" alias PeriodR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Period" alias Period as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"

group for type+types  Period extends Element
  input src : PeriodR2 as source
  input tgt : Period as target

  "Period-start" : for src.start make tgt.start
  "Period-end" : for src.end make tgt.end

endgroup
```

2.26.5.9.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Period3to2" = "R3 to R2 Conversion for Period"

uses "http://hl7.org/fhir/StructureDefinition/Period" alias Period as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Period" alias PeriodR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"

group for type+types  Period extends Element
  input src : Period as source
  input tgt : PeriodR2 as target

  "Period-start" : for src.start make tgt.start
  "Period-end" : for src.end make tgt.end

endgroup
```

2.26.5.10 SampledData

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/SampledData2to3" = "R2 to R3 Conversions for SampledData"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SampledData" alias SampledDataR2 as source
uses "http://hl7.org/fhir/StructureDefinition/SampledData" alias SampledData as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Quantity2to3"

group for type+types  SampledData extends Element
  input src : SampledDataR2 as source
  input tgt : SampledData as target

  "SampledData-origin" : for src.origin make tgt.origin
  "SampledData-period" : for src.period make tgt.period
  "SampledData-factor" : for src.factor make tgt.factor
  "SampledData-lowerLimit" : for src.lowerLimit make tgt.lowerLimit
  "SampledData-upperLimit" : for src.upperLimit make tgt.upperLimit
  "SampledData-dimensions" : for src.dimensions make tgt.dimensions
  "SampledData-data" : for src.data make tgt.data

endgroup
```

2.26.5.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/SampledData3to2" = "R3 to R2 Conversion for SampledData"

uses "http://hl7.org/fhir/StructureDefinition/SampledData" alias SampledData as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SampledData" alias SampledDataR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Quantity3to2"

group for type+types  SampledData extends Element
  input src : SampledData as source
  input tgt : SampledDataR2 as target

  "SampledData-origin" : for src.origin make tgt.origin
  "SampledData-period" : for src.period make tgt.period
  "SampledData-factor" : for src.factor make tgt.factor
  "SampledData-lowerLimit" : for src.lowerLimit make tgt.lowerLimit
  "SampledData-upperLimit" : for src.upperLimit make tgt.upperLimit
  "SampledData-dimensions" : for src.dimensions make tgt.dimensions
  "SampledData-data" : for src.data make tgt.data

endgroup
```

2.26.5.11 HumanName

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/HumanName2to3" = "R2 to R3 Conversions for HumanName"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/HumanName" alias HumanNameR2 as source
uses "http://hl7.org/fhir/StructureDefinition/HumanName" alias HumanName as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  HumanName extends Element
  input src : HumanNameR2 as source
  input tgt : HumanName as target

  "HumanName-use" : for src.use make tgt.use
  "HumanName-text" : for src.text make tgt.text
  "HumanName-family" : for src.family first make tgt.family
  "HumanName-family-rpt" : for src.family not_first as f make tgt.extension as ex, ex.url =
    "http://hl7.org/fhir/StructureDefinition/HumanName-familyRepeat", ex.value = f
  "HumanName-given" : for src.given make tgt.given
  "HumanName-prefix" : for src.prefix make tgt.prefix
  "HumanName-suffix" : for src.suffix make tgt.suffix
  "HumanName-period" : for src.period make tgt.period

endgroup
```

2.26.5.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/HumanName3to2" = "R3 to R2 Conversion for HumanName"

uses "http://hl7.org/fhir/StructureDefinition/HumanName" alias HumanName as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/HumanName" alias HumanNameR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  HumanName
```

```
  input src : HumanName as source
  input tgt : HumanNameR2 as target

  "HumanName-id" : for src.id make tgt.id
  "HumanName-extensions" : for src.extension where url != 'http://hl7.org/fhir/StructureDefinition/HumanName-
    familyRepeat' make tgt.extension

  "HumanName-use" : for src.use make tgt.use
  "HumanName-text" : for src.text make tgt.text
  "HumanName-family" : for src.family first make tgt.family
  "HumanName-family-rpt-ex" : for src.extension as ex where url = 'http://hl7.org/fhir/StructureDefinition/HumanName-
    familyRepeat' make tgt.family as f then nameRepeat(ex, f)
  "HumanName-given" : for src.given make tgt.given
  "HumanName-prefix" : for src.prefix make tgt.prefix
  "HumanName-suffix" : for src.suffix make tgt.suffix
  "HumanName-period" : for src.period make tgt.period
endgroup

group nameRepeat
  input src as source
  input tgt as target

  "HumanName-family-rpt-2" : for src.value as v then string(v, tgt)
endgroup
```

2.26.5.12 Address

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Address2to3" = "R2 to R3 Conversions for Address"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Address" alias AddressR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Address" alias Address as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Period2to3"

group for type+types  Address extends Element
  input src : AddressR2 as source
  input tgt : Address as target

  "Address-use" : for src.use make tgt.use
  "Address-type" : for src.type make tgt.type
  "Address-text" : for src.text make tgt.text
  "Address-line" : for src.line make tgt.line
  "Address-city" : for src.city make tgt.city
  "Address-district" : for src.district make tgt.district
  "Address-state" : for src.state make tgt.state
  "Address-postalCode" : for src.postalCode make tgt.postalCode
  "Address-country" : for src.country make tgt.country
  "Address-period" : for src.period make tgt.period

endgroup
```

2.26.5.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Address3to2" = "R3 to R2 Conversion for Address"

uses "http://hl7.org/fhir/StructureDefinition/Address" alias Address as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Address" alias AddressR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Period3to2"

group for type+types  Address extends Element
  input src : Address as source
  input tgt : AddressR2 as target

  "Address-use" : for src.use make tgt.use
  "Address-type" : for src.type make tgt.type
  "Address-text" : for src.text make tgt.text
  "Address-line" : for src.line make tgt.line
  "Address-city" : for src.city make tgt.city
  "Address-district" : for src.district make tgt.district
  "Address-state" : for src.state make tgt.state
  "Address-postalCode" : for src.postalCode make tgt.postalCode
  "Address-country" : for src.country make tgt.country
  "Address-period" : for src.period make tgt.period

endgroup
```

2.26.5.13 ContactPoint

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ContactPoint2to3" = "R2 to R3 Conversions for ContactPoint"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ContactPoint" alias ContactPointR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ContactPoint" alias ContactPoint as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Period2to3"

group for type+types  ContactPoint extends Element
  input src : ContactPointR2 as source
  input tgt : ContactPoint as target

  "ContactPoint-system" : for src.system make tgt.system
  "ContactPoint-value" : for src.value make tgt.value
  "ContactPoint-use" : for src.use make tgt.use
  "ContactPoint-rank" : for src.rank make tgt.rank
  "ContactPoint-period" : for src.period make tgt.period

endgroup
```

2.26.5.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ContactPoint3to2" = "R3 to R2 Conversion for ContactPoint"

uses "http://hl7.org/fhir/StructureDefinition/ContactPoint" alias ContactPoint as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ContactPoint" alias ContactPointR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Period3to2"

group for type+types  ContactPoint extends Element
  input src : ContactPoint as source
  input tgt : ContactPointR2 as target

  "ContactPoint-system" : for src.system make tgt.system
  "ContactPoint-value" : for src.value make tgt.value
  "ContactPoint-use" : for src.use make tgt.use
  "ContactPoint-rank" : for src.rank make tgt.rank
  "ContactPoint-period" : for src.period make tgt.period

endgroup
```

2.26.5.14 Timing

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.14.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Timing2to3" = "R2 to R3 Conversions for Timing"

conceptmap "EventTiming" {
  prefix s = "http://hl7.org/fhir/v3/TimingEvent"
  prefix t = "http://hl7.org/fhir/event-timing"

  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
  s:HS ~ t:MORN
}

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Timing" alias TimingR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Timing" alias Timing as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Period2to3"
imports "http://hl7.org/fhir/StructureMap/Range2to3"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept2to3"

group for type+types  Timing extends Element
  input src : TimingR2 as source
  input tgt : Timing as target
```

```
  "Timing-event" : for src.event make tgt.event
  "Timing-repeat" : for src.repeat as vs0 make tgt.repeat as vt0 then {
    "Timing.repeat-bounds-Range" : for vs0.bounds : Range as vs make vt0.bounds = create("Range") as vt then Range(vs,
    vt)
    "Timing.repeat-bounds-Period" : for vs0.bounds : Period as vs make vt0.bounds = create("Period") as vt then
    Period(vs, vt)
    "Timing.repeat-count" : for vs0.count make vt0.count
    "Timing.repeat-duration" : for vs0.duration make vt0.duration
    "Timing.repeat-durationMax" : for vs0.durationMax make vt0.durationMax
    "Timing.repeat-durationUnit" : for vs0.durationUnits make vt0.durationUnit
    "Timing.repeat-frequency" : for vs0.frequency make vt0.frequency
    "Timing.repeat-frequencyMax" : for vs0.frequencyMax make vt0.frequencyMax
    "Timing.repeat-period" : for vs0.period make vt0.period
    "Timing.repeat-periodMax" : for vs0.periodMax make vt0.periodMax
    "Timing.repeat-periodUnit" : for vs0.periodUnits make vt0.periodUnit
    "Timing.repeat-when" : for vs0.when as v make vt0.when = translate(v, "#EventTiming", "code")
  }

  "Timing-code" : for src.code make tgt.code

endgroup
```

2.26.5.14.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Timing3to2" = "R3 to R2 Conversion for Timing"

conceptmap "EventTiming" {
  prefix s = "http://hl7.org/fhir/event-timing"
  prefix t = "http://hl7.org/fhir/v3/TimingEvent"

  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
  s:MORN ~ t:HS
}

uses "http://hl7.org/fhir/StructureDefinition/Timing" alias Timing as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Timing" alias TimingR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Period3to2"
imports "http://hl7.org/fhir/StructureMap/Range3to2"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept3to2"

group for type+types  Timing extends Element
  input src : Timing as source
  input tgt : TimingR2 as target

  "Timing-event" : for src.event make tgt.event
  "Timing-repeat" : for src.repeat as vs0 make tgt.repeat as vt0 then {
    "Timing.repeat-bounds-Range" : for vs0.bounds : Range as vs make vt0.bounds = create("Range") as vt then Range(vs,
    vt)
    "Timing.repeat-bounds-Period" : for vs0.bounds : Period as vs make vt0.bounds = create("Period") as vt then
    Period(vs, vt)
    "Timing.repeat-count" : for vs0.count make vt0.count
    "Timing.repeat-duration" : for vs0.duration make vt0.duration
    "Timing.repeat-durationMax" : for vs0.durationMax make vt0.durationMax
    "Timing.repeat-durationUnits" : for vs0.durationUnit make vt0.durationUnits
    "Timing.repeat-frequency" : for vs0.frequency make vt0.frequency
    "Timing.repeat-frequencyMax" : for vs0.frequencyMax make vt0.frequencyMax
    "Timing.repeat-period" : for vs0.period make vt0.period
    "Timing.repeat-periodMax" : for vs0.periodMax make vt0.periodMax
    "Timing.repeat-periodUnits" : for vs0.periodUnit make vt0.periodUnits
    "Timing.repeat-when" : for vs0.when as v make vt0.when = translate(v, "#EventTiming", "code")
  }

  "Timing-code" : for src.code make tgt.code

endgroup
```

2.26.5.15 Signature

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.15.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Signature2to3" = "R2 to R3 Conversions for Signature"
```

```
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Signature" alias SignatureR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Signature" alias Signature as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Element2to3"
imports "http://hl7.org/fhir/StructureMap/Coding2to3"
imports "http://hl7.org/fhir/StructureMap/Reference2to3"

group for type+types  Signature extends Element
  input src : SignatureR2 as source
  input tgt : Signature as target

  "Signature-type" : for src.type make tgt.type
  "Signature-when" : for src.when make tgt.when
  "Signature-who-uri" : for src.who : uri as vs make tgt.who = create("uri") as vt then uri(vs, vt)
  "Signature-who-Reference" : for src.who : Reference as vs make tgt.who = create("Reference") as vt then Reference(vs,
    vt)
  "Signature-contentType" : for src.contentType make tgt.contentType
  "Signature-blob" : for src.blob make tgt.blob

endgroup
```

2.26.5.15.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Signature3to2" = "R3 to R2 Conversion for Signature"

uses "http://hl7.org/fhir/StructureDefinition/Signature" alias Signature as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Signature" alias SignatureR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Coding3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"

group for type+types  Signature extends Element
  input src : Signature as source
  input tgt : SignatureR2 as target

  "Signature-type" : for src.type make tgt.type
  "Signature-when" : for src.when make tgt.when
  "Signature-who-uri" : for src.who : uri as vs make tgt.who = create("uri") as vt then uri(vs, vt)
  "Signature-who-Reference" : for src.who : Reference as vs make tgt.who = create("Reference") as vt then Reference(vs,
    vt)
  "Signature-contentType" : for src.contentType make tgt.contentType
  "Signature-blob" : for src.blob make tgt.blob

endgroup
```

2.26.5.16 Annotation

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.26.5.16.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Annotation2to3" = "R2 to R3 Conversions for Annotation"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Annotation" alias AnnotationR2 as source
  uses "http://hl7.org/fhir/DSTU2/StructureDefinition/string" alias stringR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Annotation" alias Annotation as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Annotation extends Element
  input src : AnnotationR2 as source
  input tgt : Annotation as target

  "Annotation-author-Reference" : for src.author : Reference as vs make tgt.author = create("Reference") as vt then
    Reference(vs, vt)
  "Annotation-author-string" : for src.author : string as vs make tgt.author = create("string") as vt then string(vs,
    vt)
  "Annotation-time" : for src.time make tgt.time
  "Annotation-text" : for src.text make tgt.text

endgroup

group for types stringToAnnotation extends Element
  input src : stringR2 as source
  input tgt : Annotation as target

  "Annotation-text1" : for src make tgt.text = src
endgroup
```

2.26.5.16.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Annotation3to2" = "R3 to R2 Conversion for Annotation"

uses "http://hl7.org/fhir/StructureDefinition/Annotation" alias Annotation as source
uses "http://hl7.org/fhir/StructureDefinition/string" alias string as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Annotation" alias AnnotationR2 as target
```

```
imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Element3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"

group for type+types  Annotation extends Element
  input src : Annotation as source
  input tgt : AnnotationR2 as target

  "Annotation-author-Reference" : for src.author : Reference as vs make tgt.author = create("Reference") as vt then
    Reference(vs, vt)
  "Annotation-author-string" : for src.author : string as vs make tgt.author = create("string") as vt then string(vs,
    vt)
  "Annotation-time" : for src.time make tgt.time
  "Annotation-text" : for src.text make tgt.text

endgroup

group for type+types  AnnotationToString extends Element
  input src : AnnotationR2 as source
  input tgt : string as target

  "Annotation-text1" : for src.text as v make tgt.value = v

endgroup
```

2.27 MetaData Types

URL: https://hl7.org/fhir/STU3/metadatatypes.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

The FHIR specification defines a set of data types that are used for the resource elements. There are four categories of data types:

* simple / primitive types, which are single elements with a primitive value

* General purpose complex types, which are re-usable clusters of elements

* Complex data types for metadata

* Special purpose data types: Reference, Narrative, Extension, Meta, and Dosage

This page describes the metadata types.

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | UsageContext

2.27.0.0.1 ContactDetail

See also Examples, Detailed Descriptions and Mappings.

The ContactDetail structure defines general contact details.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ContactDetail | ? | | Element | Contact information Elements defined in Ancestors: id, extension
name | ? | 0..1 | string | Name of an individual to contact
telecom | ? | 0..* | ContactPoint | Contact details for individual or organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[string]"/><!-- 0..1 Name of an individual to contact -->
 <telecom><!-- 0..* ContactPoint Contact details for individual or organization --></telecom>
</[name]>
```

JSON Template

```
{
  "resourceType" : "ContactDetail",
  // from Element: extension
```

```
  "name" : "<string>", // Name of an individual to contact
  "telecom" : [{ ContactPoint }] // Contact details for individual or organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactDetail.name [ string ]; # 0..1 Name of an individual to contact
  fhir:ContactDetail.telecom [ ContactPoint ], ... ; # 0..* Contact details for individual or organization
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name	Flags	Card.	Type	Description & Constraints
ContactDetail	?		Element	Contact information Elements defined in Ancestors: id, extension
name	?	0..1	string	Name of an individual to contact
telecom	?	0..*	ContactPoint	Contact details for individual or organization

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[string]"/><!-- 0..1 Name of an individual to contact -->
 <telecom><!-- 0..* ContactPoint Contact details for individual or organization --></telecom>
</[name]>
```

JSON Template

```
{
  "resourceType" : "ContactDetail",
  // from Element: extension
  "name" : "<string>", // Name of an individual to contact
  "telecom" : [{ ContactPoint }] // Contact details for individual or organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactDetail.name [ string ]; # 0..1 Name of an individual to contact
  fhir:ContactDetail.telecom [ ContactPoint ], ... ; # 0..* Contact details for individual or organization
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

ContactDetail is used in the following places: ActivityDefinition, CapabilityStatement, CodeSystem, CompartmentDefinition, ConceptMap, DataElement, ExpansionProfile, GraphDefinition, ImplementationGuide, Library, Measure, MessageDefinition, NamingSystem, OperationDefinition, PlanDefinition, Questionnaire, ResearchStudy, SearchParameter, ServiceDefinition, StructureDefinition, StructureMap, TestScript and ValueSet

2.27.0.0.1.1 Interpretation of ContactDetail in different contexts

2.27.0.0.2 Contributor

See also Examples, Detailed Descriptions and Mappings.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name	Flags	Card.	Type	Description & Constraints
Contributor	?		Element	Contributor information Elements defined in Ancestors: id, extension
type	?	1..1	code	author | editor | reviewer | endorser ContributorType (Required)
name	?	1..1	string	Who contributed the content
contact	?	0..*	ContactDetail	Contact details of the contributor

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 author | editor | reviewer | endorser -->
 <name value="[string]"/><!-- 1..1 Who contributed the content -->
 <contact><!-- 0..* ContactDetail Contact details of the contributor --></contact>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Contributor",
  // from Element: extension
  "type" : "<code>", // R!  author | editor | reviewer | endorser
  "name" : "<string>", // R!  Who contributed the content
  "contact" : [{ ContactDetail }] // Contact details of the contributor
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Contributor.type [ code ]; # 1..1 author | editor | reviewer | endorser
  fhir:Contributor.name [ string ]; # 1..1 Who contributed the content
  fhir:Contributor.contact [ ContactDetail ], ... ; # 0..* Contact details of the contributor
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

Contributor | ? | | Element | Contributor information Elements defined in Ancestors: id, extension

type | ? | 1..1 | code | author | editor | reviewer | endorser ContributorType (Required)

name | ? | 1..1 | string | Who contributed the content

contact | ? | 0..* | ContactDetail | Contact details of the contributor

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 author | editor | reviewer | endorser -->
 <name value="[string]"/><!-- 1..1 Who contributed the content -->
 <contact><!-- 0..* ContactDetail Contact details of the contributor --></contact>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Contributor",
  // from Element: extension
  "type" : "<code>", // R!  author | editor | reviewer | endorser
  "name" : "<string>", // R!  Who contributed the content
  "contact" : [{ ContactDetail }] // Contact details of the contributor
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Contributor.type [ code ]; # 1..1 author | editor | reviewer | endorser
  fhir:Contributor.name [ string ]; # 1..1 Who contributed the content
  fhir:Contributor.contact [ ContactDetail ], ... ; # 0..* Contact details of the contributor
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

Contributor is used in the following places: ActivityDefinition, Library, Measure, PlanDefinition and ServiceDefinition

2.27.0.0.2.1 Interpretation of Contributor in different contexts

2.27.0.0.3 DataRequirement

See also Examples, Detailed Descriptions and Mappings.

The DataRequirement structure defines a general data requirement for a knowledge asset such as a decision support rule or quality measure.

* Structure

* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DataRequirement | ? | | Element | Describes a required data item Elements defined in Ancestors: id, extension
type | ? | 1..1 | code | The type of the required data FHIRAllTypes (Required)
profile | ? | 0..* | uri | The profile of the required data
mustSupport | ? | 0..* | string | Indicates that specific structure elements are referenced by the knowledge module
codeFilter | ? | 0..* | Element | What codes are expected
path | ? | 1..1 | string | The code-valued attribute of the filter
valueSet[x] | ? | 0..1 | | Valueset for the filter
valueSetString | | | string
valueSetReference | | | Reference(ValueSet)
valueCode | ? | 0..* | code | What code is expected
valueCoding | ? | 0..* | Coding | What Coding is expected
valueCodeableConcept | ? | 0..* | CodeableConcept | What CodeableConcept is expected
dateFilter | ? | 0..* | Element | What dates/date ranges are expected
path | ? | 1..1 | string | The date-valued attribute of the filter
value[x] | ? | 0..1 | | The value of the filter, as a Period, DateTime, or Duration value
valueDateTime | | | dateTime
valuePeriod | | | Period
valueDuration | | | Duration
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 The type of the required data -->
 <profile value="[uri]"/><!-- 0..* The profile of the required data -->
 <mustSupport value="[string]"/><!-- 0..* Indicates that specific structure elements are referenced by the knowledge
    module -->
 <codeFilter>  <!-- 0..* What codes are expected -->
  <path value="[string]"/><!-- 1..1 The code-valued attribute of the filter -->
  <valueSet[x]><!-- 0..1 string|Reference(ValueSet) Valueset for the filter --></valueSet[x]>
  <valueCode value="[code]"/><!-- 0..* What code is expected -->
  <valueCoding><!-- 0..* Coding What Coding is expected --></valueCoding>
  <valueCodeableConcept><!-- 0..* CodeableConcept What CodeableConcept is expected --></valueCodeableConcept>
 </codeFilter>
 <dateFilter>  <!-- 0..* What dates/date ranges are expected -->
  <path value="[string]"/><!-- 1..1 The date-valued attribute of the filter -->
  <value[x]><!-- 0..1 dateTime|Period|Duration The value of the filter, as a Period, DateTime, or Duration value
    --></value[x]>
 </dateFilter>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  The type of the required data
  "profile" : ["<uri>"], // The profile of the required data
  "mustSupport" : ["<string>"], // Indicates that specific structure elements are referenced by the knowledge module
  "codeFilter" : [{ // What codes are expected
    "path" : "<string>", // R!  The code-valued attribute of the filter
    // valueSet[x]: Valueset for the filter. One of these 2:
    "valueSetString" : "<string>",
    "valueSetReference" : { Reference(ValueSet) },
    "valueCode" : ["<code>"], // What code is expected
    "valueCoding" : [{ Coding }], // What Coding is expected
    "valueCodeableConcept" : [{ CodeableConcept }] // What CodeableConcept is expected
  }],
  "dateFilter" : [{ // What dates/date ranges are expected
    "path" : "<string>", // R!  The date-valued attribute of the filter
    // value[x]: The value of the filter, as a Period, DateTime, or Duration value. One of these 3:
    "valueDateTime" : "<dateTime>"
    "valuePeriod" : { Period }
    "valueDuration" : { Duration }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
```

```
 # from Element: Element.extension
  fhir:DataRequirement.type [ code ]; # 1..1 The type of the required data
  fhir:DataRequirement.profile [ uri ], ... ; # 0..* The profile of the required data
  fhir:DataRequirement.mustSupport [ string ], ... ; # 0..* Indicates that specific structure elements are referenced by
     the knowledge module
  fhir:DataRequirement.codeFilter [ # 0..* What codes are expected
    fhir:DataRequirement.codeFilter.path [ string ]; # 1..1 The code-valued attribute of the filter
    # DataRequirement.codeFilter.valueSet[x] : 0..1 Valueset for the filter. One of these 2
      fhir:DataRequirement.codeFilter.valueSetString [ string ]
      fhir:DataRequirement.codeFilter.valueSetReference [ Reference(ValueSet) ]
    fhir:DataRequirement.codeFilter.valueCode [ code ], ... ; # 0..* What code is expected
    fhir:DataRequirement.codeFilter.valueCoding [ Coding ], ... ; # 0..* What Coding is expected
    fhir:DataRequirement.codeFilter.valueCodeableConcept [ CodeableConcept ], ... ; # 0..* What CodeableConcept is
    expected
  ], ...;
  fhir:DataRequirement.dateFilter [ # 0..* What dates/date ranges are expected
    fhir:DataRequirement.dateFilter.path [ string ]; # 1..1 The date-valued attribute of the filter
    # DataRequirement.dateFilter.value[x] : 0..1 The value of the filter, as a Period, DateTime, or Duration value. One
    of these 3
      fhir:DataRequirement.dateFilter.valueDateTime [ dateTime ]
      fhir:DataRequirement.dateFilter.valuePeriod [ Period ]
      fhir:DataRequirement.dateFilter.valueDuration [ Duration ]
  ], ...;
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name	Flags	Card.	Type	Description & Constraints
DataRequirement	?		Element	Describes a required data item Elements defined in Ancestors: id, extension
type	?	1..1	code	The type of the required data FHIRAllTypes (Required)
profile	?	0..*	uri	The profile of the required data
mustSupport	?	0..*	string	Indicates that specific structure elements are referenced by the knowledge module
codeFilter	?	0..*	Element	What codes are expected
path	?	1..1	string	The code-valued attribute of the filter
valueSet[x]	?	0..1		Valueset for the filter
valueSetString			string	
valueSetReference			Reference(ValueSet)	
valueCode	?	0..*	code	What code is expected
valueCoding	?	0..*	Coding	What Coding is expected
valueCodeableConcept	?	0..*	CodeableConcept	What CodeableConcept is expected
dateFilter	?	0..*	Element	What dates/date ranges are expected
path	?	1..1	string	The date-valued attribute of the filter
value[x]	?	0..1		The value of the filter, as a Period, DateTime, or Duration value
valueDateTime			dateTime	
valuePeriod			Period	
valueDuration			Duration	
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 The type of the required data -->
 <profile value="[uri]"/><!-- 0..* The profile of the required data -->
 <mustSupport value="[string]"/><!-- 0..* Indicates that specific structure elements are referenced by the knowledge
    module -->
 <codeFilter>  <!-- 0..* What codes are expected -->
  <path value="[string]"/><!-- 1..1 The code-valued attribute of the filter -->
  <valueSet[x]><!-- 0..1 string|Reference(ValueSet) Valueset for the filter --></valueSet[x]>
  <valueCode value="[code]"/><!-- 0..* What code is expected -->
  <valueCoding><!-- 0..* Coding What Coding is expected --></valueCoding>
  <valueCodeableConcept><!-- 0..* CodeableConcept What CodeableConcept is expected --></valueCodeableConcept>
 </codeFilter>
 <dateFilter>  <!-- 0..* What dates/date ranges are expected -->
  <path value="[string]"/><!-- 1..1 The date-valued attribute of the filter -->
  <value[x]><!-- 0..1 dateTime|Period|Duration The value of the filter, as a Period, DateTime, or Duration value
    --></value[x]>
 </dateFilter>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  The type of the required data
  "profile" : ["<uri>"], // The profile of the required data
  "mustSupport" : ["<string>"], // Indicates that specific structure elements are referenced by the knowledge module
  "codeFilter" : [{ // What codes are expected
    "path" : "<string>", // R!  The code-valued attribute of the filter
    // valueSet[x]: Valueset for the filter. One of these 2:
    "valueSetString" : "<string>",
    "valueSetReference" : { Reference(ValueSet) },
```

```
    "valueCode" : ["<code>"], // What code is expected
    "valueCoding" : [{ Coding }], // What Coding is expected
    "valueCodeableConcept" : [{ CodeableConcept }] // What CodeableConcept is expected
  }],
  "dateFilter" : [{ // What dates/date ranges are expected
    "path" : "<string>", // R!  The date-valued attribute of the filter
    // value[x]: The value of the filter, as a Period, DateTime, or Duration value. One of these 3:
    "valueDateTime" : "<dateTime>"
    "valuePeriod" : { Period }
    "valueDuration" : { Duration }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:DataRequirement.type [ code ]; # 1..1 The type of the required data
  fhir:DataRequirement.profile [ uri ], ... ; # 0..* The profile of the required data
  fhir:DataRequirement.mustSupport [ string ], ... ; # 0..* Indicates that specific structure elements are referenced by
     the knowledge module
  fhir:DataRequirement.codeFilter [ # 0..* What codes are expected
    fhir:DataRequirement.codeFilter.path [ string ]; # 1..1 The code-valued attribute of the filter
    # DataRequirement.codeFilter.valueSet[x] : 0..1 Valueset for the filter. One of these 2
      fhir:DataRequirement.codeFilter.valueSetString [ string ]
      fhir:DataRequirement.codeFilter.valueSetReference [ Reference(ValueSet) ]
    fhir:DataRequirement.codeFilter.valueCode [ code ], ... ; # 0..* What code is expected
    fhir:DataRequirement.codeFilter.valueCoding [ Coding ], ... ; # 0..* What Coding is expected
    fhir:DataRequirement.codeFilter.valueCodeableConcept [ CodeableConcept ], ... ; # 0..* What CodeableConcept is
    expected
  ], ...;
  fhir:DataRequirement.dateFilter [ # 0..* What dates/date ranges are expected
    fhir:DataRequirement.dateFilter.path [ string ]; # 1..1 The date-valued attribute of the filter
    # DataRequirement.dateFilter.value[x] : 0..1 The value of the filter, as a Period, DateTime, or Duration value. One
    of these 3
      fhir:DataRequirement.dateFilter.valueDateTime [ dateTime ]
      fhir:DataRequirement.dateFilter.valuePeriod [ Period ]
      fhir:DataRequirement.dateFilter.valueDuration [ Duration ]
  ], ...;
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

A DataRequirement differs from a parameter in that it specifies the data to be provided in terms of the type of data, and specific filters on code-valued and/or date-valued attributes. Data requirements are not named because they are referenced by type within the evaluation context.

Data requirements are used by knowledge modules to communicate the set of required data to a consumer in a way that is computable (as opposed to a set of named parameters which must be integrated by hand based on the meaning of the parameter as communicated through the documentation).

DataRequirements are typically used to communicate patient-dependent information such as MedicationStatements and Encounters, whereas Parameters are typically used to communicate patient-independent information such as configuration values.

For example, consider the following CQL expression:

```
define "Total Colectomy Procedures":
  [Procedure: "Total Colectomy Value Set"] P
    where P.performedPeriod during "Measurement Period"
```

The criteria is looking for procedures matching the "Total Colectomy Value Set" that were performed during the "Measurement Period". In this case, "Measurement Period" is a parameter, referenced by name, whereas the reference to Procedure uses the name of the resource type, and so constitutes a data requirement of the criteria:

```
<dataRequirement>
  <type value="Procedure"/>
  <codeFilter>
    <path value="code"/>
    <valueSetString value="Total Colectomy Value Set"/>
  </codeFilter>
  <dateFilter>
    <path value="performedPeriod"/>
    <valuePeriod>
      <start value="2016-01-01"/>
      <end value="2016-12-31"/>
    </valuePeriod>
  </dateFilter>
</dataRequirement>
```

If a resource type has multiple date attributes, then it may be necessary to include multiple date criteria. For instance, in the example above the Procedure resource is using the performedPeriod date attribute. However, the Procedure resource also provides a performedDateTime (specific date and/or time when procedure was performed) attribute. Therefore, if the date criteria for the Procedure in the example was recorded using the performedDateTime attribute, the date criteria would be

incomplete. To account for this additional date attribute, the example could be expanded as follows:

```
define "Total Colectomy Procedures":
  [Procedure: "Total Colectomy Value Set"] P
    where exists (P.performedPeriod during "Measurement Period")
    or exists (P.performedDateTime during "Measurement Period")
```

The data requirement for the expanded criteria:

```
<dataRequirement>
  <type value="Procedure"/>
  <codeFilter>
    <path value="code"/>
    <valueSetString value="Total Colectomy Value Set"/>
  </codeFilter>
  <dateFilter>
    <path value="performedPeriod"/>
    <path value="performedDateTime"/>
  </dateFilter>
</dataRequirement>
```

DataRequirement is used in the following places: GuidanceResponse, Library, PlanDefinition and ServiceDefinition

2.27.0.0.4 ParameterDefinition

See also Examples, Detailed Descriptions and Mappings.

The ParameterDefinition structure defines a parameter to a knowledge asset such as a decision support rule or quality measure.

Parameters are typically used to communicate patient-independent information such as configuration values, whereas DataRequirements are typically used to communicate patient-dependent information such as MedicationStatements and Encounters.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ParameterDefinition | ? | | Element | Definition of a parameter to a module Elements defined in Ancestors: id, extension
name | ? | 0..1 | code | Name used to access the parameter value
use | ? | 1..1 | code | in | out OperationParameterUse (Required)
min | ? | 0..1 | integer | Minimum cardinality
max | ? | 0..1 | string | Maximum cardinality (a number of *)
documentation | ? | 0..1 | string | A brief description of the parameter
type | ? | 1..1 | code | What type of value FHIRAllTypes (Required)
profile | ? | 0..1 | Reference(StructureDefinition) | What profile the value is expected to be
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[code]"/><!-- 0..1 Name used to access the parameter value -->
 <use value="[code]"/><!-- 1..1 in | out -->
 <min value="[integer]"/><!-- 0..1 Minimum cardinality -->
 <max value="[string]"/><!-- 0..1 Maximum cardinality (a number of *) -->
 <documentation value="[string]"/><!-- 0..1 A brief description of the parameter -->
 <type value="[code]"/><!-- 1..1 What type of value -->
 <profile><!-- 0..1 Reference(StructureDefinition) What profile the value is expected to be --></profile>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<code>", // Name used to access the parameter value
  "use" : "<code>", // R!  in | out
  "min" : <integer>, // Minimum cardinality
  "max" : "<string>", // Maximum cardinality (a number of *)
  "documentation" : "<string>", // A brief description of the parameter
  "type" : "<code>", // R!  What type of value
  "profile" : { Reference(StructureDefinition) } // What profile the value is expected to be
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
```

```
 # from Element: Element.extension
  fhir:ParameterDefinition.name [ code ]; # 0..1 Name used to access the parameter value
  fhir:ParameterDefinition.use [ code ]; # 1..1 in | out
  fhir:ParameterDefinition.min [ integer ]; # 0..1 Minimum cardinality
  fhir:ParameterDefinition.max [ string ]; # 0..1 Maximum cardinality (a number of *)
  fhir:ParameterDefinition.documentation [ string ]; # 0..1 A brief description of the parameter
  fhir:ParameterDefinition.type [ code ]; # 1..1 What type of value
  fhir:ParameterDefinition.profile [ Reference(StructureDefinition) ]; # 0..1 What profile the value is expected to be
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints
ParameterDefinition | ? | | Element | Definition of a parameter to a module Elements defined in Ancestors: id, extension
name | ? | 0..1 | code | Name used to access the parameter value
use | ? | 1..1 | code | in | out OperationParameterUse (Required)
min | ? | 0..1 | integer | Minimum cardinality
max | ? | 0..1 | string | Maximum cardinality (a number of *)
documentation | ? | 0..1 | string | A brief description of the parameter
type | ? | 1..1 | code | What type of value FHIRAllTypes (Required)
profile | ? | 0..1 | Reference(StructureDefinition) | What profile the value is expected to be
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[code]"/><!-- 0..1 Name used to access the parameter value -->
 <use value="[code]"/><!-- 1..1 in | out -->
 <min value="[integer]"/><!-- 0..1 Minimum cardinality -->
 <max value="[string]"/><!-- 0..1 Maximum cardinality (a number of *) -->
 <documentation value="[string]"/><!-- 0..1 A brief description of the parameter -->
 <type value="[code]"/><!-- 1..1 What type of value -->
 <profile><!-- 0..1 Reference(StructureDefinition) What profile the value is expected to be --></profile>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<code>", // Name used to access the parameter value
  "use" : "<code>", // R!  in | out
  "min" : <integer>, // Minimum cardinality
  "max" : "<string>", // Maximum cardinality (a number of *)
  "documentation" : "<string>", // A brief description of the parameter
  "type" : "<code>", // R!  What type of value
  "profile" : { Reference(StructureDefinition) } // What profile the value is expected to be
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ParameterDefinition.name [ code ]; # 0..1 Name used to access the parameter value
  fhir:ParameterDefinition.use [ code ]; # 1..1 in | out
  fhir:ParameterDefinition.min [ integer ]; # 0..1 Minimum cardinality
  fhir:ParameterDefinition.max [ string ]; # 0..1 Maximum cardinality (a number of *)
  fhir:ParameterDefinition.documentation [ string ]; # 0..1 A brief description of the parameter
  fhir:ParameterDefinition.type [ code ]; # 1..1 What type of value
  fhir:ParameterDefinition.profile [ Reference(StructureDefinition) ]; # 0..1 What profile the value is expected to be
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

2.27.0.0.5 RelatedArtifact

See also Examples, Detailed Descriptions and Mappings.

The RelatedArtifact structure defines resources related to a module such as previous and next versions of documents, documentation, citations, etc. Note that the name resource here is being used in a more general sense than the FHIR-specific Resource. The related resource may be a FHIR resource, or it may be another type of resource, represented using the Attachment data type.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

RelatedArtifact | ? | | Element | Related artifacts for a knowledge resource Elements defined in Ancestors: id, extension

type | ? | 1..1 | code | documentation | justification | citation | predecessor | successor | derived-from | depends-on | composed-of RelatedArtifactType (Required)

display | ? | 0..1 | string | Brief description of the related artifact

citation | ? | 0..1 | string | Bibliographic citation for the artifact

url | ? | 0..1 | uri | Where the artifact can be accessed

document | ? | 0..1 | Attachment | What document is being referenced

resource | ? | 0..1 | Reference(Any) | What resource is being referenced

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 documentation | justification | citation | predecessor | successor | derived-from |
    depends-on | composed-of -->
 <display value="[string]"/><!-- 0..1 Brief description of the related artifact -->
 <citation value="[string]"/><!-- 0..1 Bibliographic citation for the artifact -->
 <url value="[uri]"/><!-- 0..1 Where the artifact can be accessed -->
 <document><!-- 0..1 Attachment What document is being referenced --></document>
 <resource><!-- 0..1 Reference(Any) What resource is being referenced --></resource>
</[name]>
```

JSON Template

```
{
  "resourceType" : "RelatedArtifact",
  // from Element: extension
  "type" : "<code>", // R!  documentation | justification | citation | predecessor | successor | derived-from | depends-
    on | composed-of
  "display" : "<string>", // Brief description of the related artifact
  "citation" : "<string>", // Bibliographic citation for the artifact
  "url" : "<uri>", // Where the artifact can be accessed
  "document" : { Attachment }, // What document is being referenced
  "resource" : { Reference(Any) } // What resource is being referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:RelatedArtifact.type [ code ]; # 1..1 documentation | justification | citation | predecessor | successor |
    derived-from | depends-on | composed-of
  fhir:RelatedArtifact.display [ string ]; # 0..1 Brief description of the related artifact
  fhir:RelatedArtifact.citation [ string ]; # 0..1 Bibliographic citation for the artifact
  fhir:RelatedArtifact.url [ uri ]; # 0..1 Where the artifact can be accessed
  fhir:RelatedArtifact.document [ Attachment ]; # 0..1 What document is being referenced
  fhir:RelatedArtifact.resource [ Reference(Any) ]; # 0..1 What resource is being referenced
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

RelatedArtifact | ? | | Element | Related artifacts for a knowledge resource Elements defined in Ancestors: id, extension

type | ? | 1..1 | code | documentation | justification | citation | predecessor | successor | derived-from | depends-on | composed-of RelatedArtifactType (Required)

display | ? | 0..1 | string | Brief description of the related artifact

citation | ? | 0..1 | string | Bibliographic citation for the artifact

url | ? | 0..1 | uri | Where the artifact can be accessed

document | ? | 0..1 | Attachment | What document is being referenced

resource | ? | 0..1 | Reference(Any) | What resource is being referenced

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 documentation | justification | citation | predecessor | successor | derived-from |
    depends-on | composed-of -->
 <display value="[string]"/><!-- 0..1 Brief description of the related artifact -->
 <citation value="[string]"/><!-- 0..1 Bibliographic citation for the artifact -->
 <url value="[uri]"/><!-- 0..1 Where the artifact can be accessed -->
```

```
 <document><!-- 0..1 Attachment What document is being referenced --></document>
 <resource><!-- 0..1 Reference(Any) What resource is being referenced --></resource>
</[name]>
```

JSON Template

```
{
  "resourceType" : "RelatedArtifact",
  // from Element: extension
  "type" : "<code>", // R!  documentation | justification | citation | predecessor | successor | derived-from | depends-
    on | composed-of
  "display" : "<string>", // Brief description of the related artifact
  "citation" : "<string>", // Bibliographic citation for the artifact
  "url" : "<uri>", // Where the artifact can be accessed
  "document" : { Attachment }, // What document is being referenced
  "resource" : { Reference(Any) } // What resource is being referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:RelatedArtifact.type [ code ]; # 1..1 documentation | justification | citation | predecessor | successor |
    derived-from | depends-on | composed-of
  fhir:RelatedArtifact.display [ string ]; # 0..1 Brief description of the related artifact
  fhir:RelatedArtifact.citation [ string ]; # 0..1 Bibliographic citation for the artifact
  fhir:RelatedArtifact.url [ uri ]; # 0..1 Where the artifact can be accessed
  fhir:RelatedArtifact.document [ Attachment ]; # 0..1 What document is being referenced
  fhir:RelatedArtifact.resource [ Reference(Any) ]; # 0..1 What resource is being referenced
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

RelatedArtifact is used in the following places: ActivityDefinition, Library, Measure, PlanDefinition, RequestGroup, ResearchStudy and ServiceDefinition

2.27.0.0.5.1 Interpretation of RelatedArtifact in different contexts

2.27.0.0.6 TriggerDefinition

See also Examples, Detailed Descriptions and Mappings.

The TriggerDefinition structure defines when a knowledge artifact is expected to be evaluated. The structure can represent three main kinds of triggering events:

* Named Event

* Scheduled Event

* Data Event

A named event is an event identified by the implementation environment. This allows any event generated within the implementation environment to be used as a trigger, but it requires pre-coordination of the names involved with the consuming environments.

A scheduled event occurs on a fixed or periodic schedule.

And finally, a data event occurs in response to some data-related event in the integrated environment such as a record being added or updated. The data-of-interest for a data event is described using a DataRequirement. This allows for systems to automatically invoke based on data activity occurring within the system.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name	Flags	Card.	Type	Description & Constraints
TriggerDefinition	?		Element	Defines an expected trigger for a module Elements defined in Ancestors: id, extension
type	?	1..1	code	named-event | periodic | data-added | data-modified | data-removed | data-accessed | data-access-ended TriggerType (Required)
eventName	?	0..1	string	Triggering event name
eventTiming[x]	?	0..1		Timing of the event
eventTimingTiming			Timing	
eventTimingReference			Reference(Schedule)	
eventTimingDate			date	
eventTimingDateTime			dateTime	

eventData | ? | 0..1 | DataRequirement | Triggering data of the event

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 named-event | periodic | data-added | data-modified | data-removed | data-accessed |
    data-access-ended -->
 <eventName value="[string]"/><!-- 0..1 Triggering event name -->
 <eventTiming[x]><!-- 0..1 Timing|Reference(Schedule)|date|dateTime Timing of the event --></eventTiming[x]>
 <eventData><!-- 0..1 DataRequirement Triggering data of the event --></eventData>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  named-event | periodic | data-added | data-modified | data-removed | data-accessed | data-
    access-ended
  "eventName" : "<string>", // Triggering event name
  // eventTiming[x]: Timing of the event. One of these 4:
  "eventTimingTiming" : { Timing },
  "eventTimingReference" : { Reference(Schedule) },
  "eventTimingDate" : "<date>",
  "eventTimingDateTime" : "<dateTime>",
  "eventData" : { DataRequirement } // Triggering data of the event
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:TriggerDefinition.type [ code ]; # 1..1 named-event | periodic | data-added | data-modified | data-removed |
    data-accessed | data-access-ended
  fhir:TriggerDefinition.eventName [ string ]; # 0..1 Triggering event name
  # TriggerDefinition.eventTiming[x] : 0..1 Timing of the event. One of these 4
    fhir:TriggerDefinition.eventTimingTiming [ Timing ]
    fhir:TriggerDefinition.eventTimingReference [ Reference(Schedule) ]
    fhir:TriggerDefinition.eventTimingDate [ date ]
    fhir:TriggerDefinition.eventTimingDateTime [ dateTime ]
  fhir:TriggerDefinition.eventData [ DataRequirement ]; # 0..1 Triggering data of the event
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

TriggerDefinition | ? | | Element | Defines an expected trigger for a module Elements defined in Ancestors: id, extension

type | ? | 1..1 | code | named-event | periodic | data-added | data-modified | data-removed | data-accessed | data-access-ended TriggerType (Required)

eventName | ? | 0..1 | string | Triggering event name

eventTiming[x] | ? | 0..1 | | Timing of the event

eventTimingTiming | | | Timing

eventTimingReference | | | Reference(Schedule)

eventTimingDate | | | date

eventTimingDateTime | | | dateTime

eventData | ? | 0..1 | DataRequirement | Triggering data of the event

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 named-event | periodic | data-added | data-modified | data-removed | data-accessed |
    data-access-ended -->
 <eventName value="[string]"/><!-- 0..1 Triggering event name -->
 <eventTiming[x]><!-- 0..1 Timing|Reference(Schedule)|date|dateTime Timing of the event --></eventTiming[x]>
 <eventData><!-- 0..1 DataRequirement Triggering data of the event --></eventData>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  named-event | periodic | data-added | data-modified | data-removed | data-accessed | data-
    access-ended
  "eventName" : "<string>", // Triggering event name
  // eventTiming[x]: Timing of the event. One of these 4:
  "eventTimingTiming" : { Timing },
  "eventTimingReference" : { Reference(Schedule) },
  "eventTimingDate" : "<date>",
  "eventTimingDateTime" : "<dateTime>",
```

```
  "eventData" : { DataRequirement } // Triggering data of the event
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:TriggerDefinition.type [ code ]; # 1..1 named-event | periodic | data-added | data-modified | data-removed |
    data-accessed | data-access-ended
  fhir:TriggerDefinition.eventName [ string ]; # 0..1 Triggering event name
  # TriggerDefinition.eventTiming[x] : 0..1 Timing of the event. One of these 4
    fhir:TriggerDefinition.eventTimingTiming [ Timing ]
    fhir:TriggerDefinition.eventTimingReference [ Reference(Schedule) ]
    fhir:TriggerDefinition.eventTimingDate [ date ]
    fhir:TriggerDefinition.eventTimingDateTime [ dateTime ]
  fhir:TriggerDefinition.eventData [ DataRequirement ]; # 0..1 Triggering data of the event
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

2.27.0.0.7 UsageContext

See also Examples, Detailed Descriptions and Mappings.

The UsageContext structure defines the context of use for a module.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
UsageContext | ? | | Element | Describes the context of use for a conformance or knowledge resource Elements defined in Ancestors: id, extension
code | ? | 1..1 | Coding | Type of context being specified UsageContextType (Extensible)
value[x] | ? | 1..1 | | Value that defines the context Context of Use ValueSet (Example)
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <code><!-- 1..1 Coding Type of context being specified --></code>
 <value[x]><!-- 1..1 CodeableConcept|Quantity|Range Value that defines the context --></value[x]>
</[name]>
```

JSON Template

```
{
  "resourceType" : "UsageContext",
  // from Element: extension
  "code" : { Coding }, // R!  Type of context being specified
  // value[x]: Value that defines the context. One of these 3:
  "valueCodeableConcept" : { CodeableConcept }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:UsageContext.code [ Coding ]; # 1..1 Type of context being specified
  # UsageContext.value[x] : 1..1 Value that defines the context. One of these 3
    fhir:UsageContext.valueCodeableConcept [ CodeableConcept ]
    fhir:UsageContext.valueQuantity [ Quantity ]
    fhir:UsageContext.valueRange [ Range ]
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints
UsageContext | ? | | Element | Describes the context of use for a conformance or knowledge resource Elements defined in Ancestors: id, extension
code | ? | 1..1 | Coding | Type of context being specified UsageContextType (Extensible)
value[x] | ? | 1..1 | | Value that defines the context Context of Use ValueSet (Example)
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <code><!-- 1..1 Coding Type of context being specified --></code>
 <value[x]><!-- 1..1 CodeableConcept|Quantity|Range Value that defines the context --></value[x]>
</[name]>
```

JSON Template

```
{
  "resourceType" : "UsageContext",
  // from Element: extension
  "code" : { Coding }, // R!  Type of context being specified
  // value[x]: Value that defines the context. One of these 3:
  "valueCodeableConcept" : { CodeableConcept }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:UsageContext.code [ Coding ]; # 1..1 Type of context being specified
  # UsageContext.value[x] : 1..1 Value that defines the context. One of these 3
    fhir:UsageContext.valueCodeableConcept [ CodeableConcept ]
    fhir:UsageContext.valueQuantity [ Quantity ]
    fhir:UsageContext.valueRange [ Range ]
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

UsageContext is used in the following places: ActivityDefinition, CapabilityStatement, CodeSystem, CompartmentDefinition, ConceptMap, DataElement, ExpansionProfile, GraphDefinition, ImplementationGuide, Library, Measure, MessageDefinition, NamingSystem, OperationDefinition, PlanDefinition, Questionnaire, SearchParameter, ServiceDefinition, StructureDefinition, StructureMap, TestScript and ValueSet

2.27.0.1 Interpretation of UsageContext in different contexts

2.27.1 Examples

URL: https://hl7.org/fhir/STU3/metadatatypes-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.27.1 MetaData Types - Examples

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | UsageContext

2.27.1.1 ContactDetail

See also Base Definition, Detailed Descriptions and Mappings.

For ContactDetail examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.1.2 Contributor

See also Base Definition, Detailed Descriptions and Mappings.

For Contributor examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.1.3 DataRequirement

See also Base Definition, Detailed Descriptions and Mappings.

For DataRequirement examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.1.4 ParameterDefinition

See also Base Definition, Detailed Descriptions and Mappings.

For ParameterDefinition examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.1.5 RelatedArtifact

See also Base Definition, Detailed Descriptions and Mappings.

For RelatedArtifact examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.1.6 TriggerDefinition

See also Base Definition, Detailed Descriptions and Mappings.

For TriggerDefinition examples, see the examples in ServiceDefinition and PlanDefinition.

2.27.1.7 UsageContext

See also Base Definition, Detailed Descriptions and Mappings.

For UsageContext examples, see the examples in Library, ServiceDefinition, Measure, and PlanDefinition.

2.27.2 Metadata Types - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/metadatatypes-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.27.2 Metadata Types - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | UsageContext

2.27.2.1 ContactDetail

See also Base Definition, Examples and Mappings.

ContactDetail
Definition | Specifies contact information for a person or organization.
Control | 1..1
Requirements | Need to track contact information in the same way across multiple resources.
Summary | true
ContactDetail.name
Definition | The name of an individual to contact.
Control | 0..1
Type | string
Summary | true
Comments | If there is no named individual, the telecom information is for the organization as a whole.
ContactDetail.telecom
Definition | The contact details for the individual (if a name was provided) or the organization.
Control | 0..*
Type | ContactPoint
Summary | true

2.27.2.2 Contributor

See also Base Definition, Examples and Mappings.

Contributor
Definition | A contributor to the content of a knowledge asset, including authors, editors, reviewers, and endorsers.
Control | 1..1
Requirements | Need to track contributor information in the same way across multiple resources.
Summary | true
Contributor.type
Definition | The type of contributor.
Control | 1..1
Terminology Binding | ContributorType (Required)
Type | code
Summary | true
Contributor.name
Definition | The name of the individual or organization responsible for the contribution.
Control | 1..1
Type | string
Summary | true

Contributor.contact
Definition | Contact details to assist a user in finding and communicating with the contributor.
Control | 0..*
Type | ContactDetail
Summary | true

2.27.2.3 DataRequirement

See also Base Definition, Examples and Mappings.

DataRequirement
Definition | Describes a required data item for evaluation in terms of the type of data, and optional code or date-based filters of the data.
Control | 1..1
Summary | true
DataRequirement.type
Definition | The type of the required data, specified as the type name of a resource. For profiles, this value is set to the type of the base resource of the profile.
Control | 1..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Summary | true
DataRequirement.profile
Definition | The profile of the required data, specified as the uri of the profile definition.
Control | 0..*
Type | uri
Summary | true
DataRequirement.mustSupport
Definition | Indicates that specific elements of the type are referenced by the knowledge module and must be supported by the consumer in order to obtain an effective evaluation. This does not mean that a value is required for this element, only that the consuming system must understand the element and be able to provide values for it if they are available. Note that the value for this element can be a path to allow references to nested elements. In that case, all the elements along the path must be supported.
Control | 0..*
Type | string
Summary | true
DataRequirement.codeFilter
Definition | Code filters specify additional constraints on the data, specifying the value set of interest for a particular element of the data.
Control | 0..*
Summary | true
DataRequirement.codeFilter.path
Definition | The code-valued attribute of the filter. The specified path must be resolvable from the type of the required data. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements. Note that the index must be an integer constant. The path must resolve to an element of type code, Coding, or CodeableConcept.
Control | 1..1
Type | string
Summary | true
DataRequirement.codeFilter.valueSet[x]
Definition | The valueset for the code filter. The valueSet and value elements are exclusive. If valueSet is specified, the filter will return only those data items for which the value of the code-valued element specified in the path is a member of the specified valueset.
Control | 0..1
Type | string|Reference(ValueSet)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DataRequirement.codeFilter.valueCode
Definition | The codes for the code filter. Only one of valueSet, valueCode, valueCoding, or valueCodeableConcept may be specified. If values are given, the filter will return only those data items for which the code-valued attribute specified by the path has a value that is one of the specified codes.
Control | 0..*
Type | code
Summary | true
DataRequirement.codeFilter.valueCoding
Definition | The Codings for the code filter. Only one of valueSet, valueCode, valueConding, or valueCodeableConcept may be specified. If values are given, the filter will return only those data items for which the code-valued attribute specified by the path has a value that is one of the specified Codings.
Control | 0..*
Type | Coding
Summary | true
DataRequirement.codeFilter.valueCodeableConcept
Definition | The CodeableConcepts for the code filter. Only one of valueSet, valueCode, valueConding, or valueCodeableConcept may be specified. If values are given, the filter will return only those data items for which the code-valued attribute specified by the path has a value that is one of the specified CodeableConcepts.
Control | 0..*
Type | CodeableConcept
Summary | true
DataRequirement.dateFilter
Definition | Date filters specify additional constraints on the data in terms of the applicable date range for specific elements.

Control | 0..*
Summary | true
DataRequirement.dateFilter.path
Definition | The date-valued attribute of the filter. The specified path must be resolvable from the type of the required data. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements. Note that the index must be an integer constant. The path must resolve to an element of type dateTime, Period, Schedule, or Timing.
Control | 1..1
Type | string
Summary | true
DataRequirement.dateFilter.value[x]
Definition | The value of the filter. If period is specified, the filter will return only those data items that fall within the bounds determined by the Period, inclusive of the period boundaries. If dateTime is specified, the filter will return only those data items that are equal to the specified dateTime. If a Duration is specified, the filter will return only those data items that fall within Duration from now.
Control | 0..1
Type | dateTime|Period|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

2.27.2.4 ParameterDefinition

See also Base Definition, Examples and Mappings.

ParameterDefinition
Definition | The parameters to the module. This collection specifies both the input and output parameters. Input parameters are provided by the caller as part of the $evaluate operation. Output parameters are included in the GuidanceResponse.
Control | 1..1
Summary | true
ParameterDefinition.name
Definition | The name of the parameter used to allow access to the value of the parameter in evaluation contexts.
Control | 0..1
Type | code
Summary | true
ParameterDefinition.use
Definition | Whether the parameter is input or output for the module.
Control | 1..1
Terminology Binding | OperationParameterUse (Required)
Type | code
Summary | true
ParameterDefinition.min
Definition | The minimum number of times this parameter SHALL appear in the request or response.
Control | 0..1
Type | integer
Summary | true
ParameterDefinition.max
Definition | The maximum number of times this element is permitted to appear in the request or response.
Control | 0..1
Type | string
Summary | true
ParameterDefinition.documentation
Definition | A brief discussion of what the parameter is for and how it is used by the module.
Control | 0..1
Type | string
Summary | true
ParameterDefinition.type
Definition | The type of the parameter.
Control | 1..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Summary | true
ParameterDefinition.profile
Definition | If specified, this indicates a profile that the input data must conform to, or that the output data will conform to.
Control | 0..1
Type | Reference(StructureDefinition)
Summary | true

2.27.2.5 RelatedArtifact

See also Base Definition, Examples and Mappings.

RelatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Control | 1..1
Requirements | Knowledge resources must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Summary | true

Comments | Each related artifact is either an attachment, or a reference to another knowledge resource, but not both.
RelatedArtifact.type
Definition | The type of relationship to the related artifact.
Control | 1..1
Terminology Binding | RelatedArtifactType (Required)
Type | code
Summary | true
RelatedArtifact.display
Definition | A brief description of the document or knowledge resource being referenced, suitable for display to a consumer.
Control | 0..1
Type | string
Summary | true
RelatedArtifact.citation
Definition | A bibliographic citation for the related artifact. This text SHOULD be formatted according to an accepted citation format.
Control | 0..1
Type | string
Summary | true
Comments | Additional structured information about citations should be captured as extensions.
RelatedArtifact.url
Definition | A url for the artifact that can be followed to access the actual content.
Control | 0..1
Type | uri
Summary | true
Comments | If a document or resource element is present, this element SHALL NOT be provided (use the url or reference in the Attachment or resource reference).
RelatedArtifact.document
Definition | The document being referenced, represented as an attachment. This is exclusive with the resource element.
Control | 0..1
Type | Attachment
Summary | true
RelatedArtifact.resource
Definition | The related resource, such as a library, value set, profile, or other knowledge resource.
Control | 0..1
Type | Reference(Any)
Summary | true
Comments | If the type is predecessor, this is a reference to the succeeding knowledge resource. If the type is successor, this is a reference to the prior knowledge resource.

2.27.2.6 TriggerDefinition

See also Base Definition, Examples and Mappings.

TriggerDefinition
Definition | A description of a triggering event.
Control | 1..1
Summary | true
TriggerDefinition.type
Definition | The type of triggering event.
Control | 1..1
Terminology Binding | TriggerType (Required)
Type | code
Summary | true
TriggerDefinition.eventName
Definition | The name of the event (if this is a named-event trigger).
Control | 0..1
Type | string
Summary | true
TriggerDefinition.eventTiming[x]
Definition | The timing of the event (if this is a period trigger).
Control | 0..1
Type | Timing|Reference(Schedule)|date|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
TriggerDefinition.eventData
Definition | The triggering data of the event (if this is a data trigger).
Control | 0..1
Type | DataRequirement
Summary | true

2.27.2.7 UsageContext

See also Base Definition, Examples and Mappings.

UsageContext

Definition | Specifies clinical/business/etc metadata that can be used to retrieve, index and/or categorize an artifact. This metadata can either be specific to the applicable population (e.g., age category, DRG) or the specific context of care (e.g., venue, care setting, provider of care).
Control | 1..1
Requirements | Consumers of the resource must be able to determine the intended applicability for the resource. Ideally, this information would be used programmatically to determine when and how it should be incorporated or exposed.
Summary | true
UsageContext.code
Definition | A code that identifies the type of context being specified by this usage context.
Control | 1..1
Terminology Binding | UsageContextType (Extensible)
Type | Coding
Summary | true
UsageContext.value[x]
Definition | A value that defines the context specified in this context of use. The interpretation of the value is defined by the code.
Control | 1..1
Terminology Binding | Context of Use ValueSet (Example)
Type | CodeableConcept|Quantity|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

2.27.3 Metadata Types - Mappings

URL: https://hl7.org/fhir/STU3/metadatatypes-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.27.3 Metadata Types - Mappings

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

This page provides mappings for the data types. There are mappings to HL7 v2 , HL7 v3 , and (where appropriate) vCard.

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | UsageContext

2.27.3.1 ContactDetail

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.2 Contributor

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.3 DataRequirement

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.4 ParameterDefinition

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.5 RelatedArtifact

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.6 TriggerDefinition

See also Base Definition, Examples and Detailed Descriptions.

2.27.3.7 UsageContext

See also Base Definition, Examples and Detailed Descriptions.

2.28 Base Resource Definitions

URL: https://hl7.org/fhir/STU3/resource.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: 5 | Compartments: Not linked to any defined compartments

2.28.1 Scope and Usage

This specification defines a series of different types of resource that can be used to exchange and/or store data in order to solve a wide range of healthcare related problems, both clinical and administrative. In addition, this specification defines several different ways of exchanging the resources.

A resource is an entity that:

* has a known identity (a URL) by which it can be addressed

* identifies itself as one of the types of resource defined in this specification

* contains a set of structured data items as described by the definition of the resource type

* has an identified version that changes if the contents of the resource change

Resources have multiple representations.

2.28.2 Boundaries and Relationships

The following optional elements and properties are defined for all resources:

* An identity

* Meta data

* A base language

* A reference to "Implicit Rules"

Most resources are derived from Domain Resources - so they also can contain text, contained resources, extensions, and data elements specific to the particular domain of the resource. There is a special type of resource called Bundle for collections of resources.

Note: there is documentation for the Structure, UML, XML, and JSON representations of the resource structure.

2.28.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Resource | | | n/a | Base Resource
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Common Languages (Extensible but limited to All Languages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content -->
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  "id" : "<id>", // Logical id of this artifact
  "meta" : { Meta }, // Metadata about the resource
  "implicitRules" : "<uri>", // A set of rules under which this content was created
  "language" : "<code>" // Language of the resource content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Resource.id [ id ]; # 0..1 Logical id of this artifact
  fhir:Resource.meta [ Meta ]; # 0..1 Metadata about the resource
  fhir:Resource.implicitRules [ uri ]; # 0..1 A set of rules under which this content was created
  fhir:Resource.language [ code ]; # 0..1 Language of the resource content
]
```

Structure

Name | Flags | Card. | Type | Description & Constraints
Resource | | | n/a | Base Resource
id | ? | 0..1 | id | Logical id of this artifact
meta | ? | 0..1 | Meta | Metadata about the resource
implicitRules | ?!? | 0..1 | uri | A set of rules under which this content was created

language | | 0..1 | code | Language of the resource content Common Languages (Extensible but limited to All Languages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content -->
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  "id" : "<id>", // Logical id of this artifact
  "meta" : { Meta }, // Metadata about the resource
  "implicitRules" : "<uri>", // A set of rules under which this content was created
  "language" : "<code>" // Language of the resource content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Resource.id [ id ]; # 0..1 Logical id of this artifact
  fhir:Resource.meta [ Meta ]; # 0..1 Metadata about the resource
  fhir:Resource.implicitRules [ uri ]; # 0..1 A set of rules under which this content was created
  fhir:Resource.language [ code ]; # 0..1 Language of the resource content
]
```

2.28.3.1 Terminology Bindings

Path | Definition | Type | Reference
Resource.language | A human language. | Extensible, but limited to All Languages | Common Languages

2.28.3.2 Resource Identity

Each resource has an "id" element which contains the logical identity of the resource assigned by the server responsible for storing it. Resources always have a known identity except for the special case when a new resource is being sent to a server to assign an identity (create interaction). The logical identity is unique within the space of all resources of the same type on the same server. Once assigned, the identity is never changed. Note that if the resource is copied to another server, the copy might not be able to retain the same logical identity.

The unique identifier of a resource instance is an absolute URI constructed from the server base address at which the instance is found, the resource type and the Logical ID, such as http://test.fhir.org/rest/Patient/123 (where 123 is the Logical Id). When the literal identity is an HTTP address, this address can generally be used to retrieve or manipulate the resource. Note that implementations SHOULD NOT assume that the identity of a resource is always resolvable to a literal server - it may be temporarily unavailable, or not available by policy (e.g. firewalls) or in some cases, it may not actually exist (e.g. use of resource outside a RESTful environment). Resources reference each other by their identity. These references are allowed to be absolute or relative (see Resource References for further discussion). Copying or moving resources from one server to another means that resources acquire a new identity. For further details, see Managing Resource Identity.

Logical ids (and therefore literal identities) are case sensitive. Logical Ids are always opaque, and external systems need not and should not attempt to determine their internal structure. A logical id SHALL always be represented in the same way in resource references and URLs. Ids can be up to 64 characters long, and contain any combination of upper and lowercase ASCII letters, numerals, "-" and ".".

2.28.3.2.1 "Business" Identifiers

In addition to the logical id and literal identity discussed above, many resources contain an element named "identifier", which, if populated, contains a different kind of identifier. As resources are copied from server to server, their literal identity will change, and their logical id may change.

However, all copies of the resource refer to the same underlying concept, and this concept may also be represented in other formats (variously, HL7 v2 , CDA , XDS, and many more). Each representation carries the same identifier that identifies it consistently across all contexts of use. This is known as the business identifier, and is found in the identifier element. In a few resources, there is a url element that serves a similar purpose, but is constrained to be a literal URL for implementation reasons.

All resources that have an identifier element support searching by the identifier, so that records can be located by that method. So if an HL7 v2 message has the following OBR:

```
OBR|1|845439^GHH OE|1045813^GHH LAB|1554-5^GLUCOSE^LN|||200202150730|...
```

Then the DiagnosticReport it represents can be located using the following query:

```
 GET [base]/DiagnosticReport?identifier=1045813
```

If a FHIR server is a stable server that is the canonical master source for the definition of a concept, the business identifier for all systems may be the same as the literal identity of the resource on the master server.

2.28.3.3 Resource Metadata

Each resource contains an element "meta", of type "Meta", which is a set of metadata that provides technical and workflow context to the resource. The metadata items are all optional, though some or all of them may be required in particular implementations or contexts of use.

Metadata Item | Type | Usage

versionId (0..1) | id | Changes each time the content of the resource changes. Can be referenced in a resource reference. Can be used to ensure that updates are based on the latest version of the resource. The version can be globally unique, or scoped by the Logical Id of the resource. Version identifiers are generally either a serially incrementing id scoped by the logical id, or a uuid, though neither of these approaches is required. There is no fixed order for version ids - clients cannot assume that a versionId that comes after another one either numerically or alphabetically represents a later version. The same versionId can never be used for more than one version of the same resource. Note that servers SHOULD support versions, but some are unable to

lastUpdated (0..1) | instant | If populated, this value changes each time the content of the resource changes. it can be used by a system or a human to judge the currency of the resource content. Note that version aware updates do not use this element

profile (0..*) | uri | An assertion that the content conforms to a resource profile (a StructureDefinition). See Extending and Restricting Resources for further discussion. Can be changed as profiles and value sets change or the system rechecks conformance

security (0..*) | Coding | Security labels applied to this resource. These tags connect resources in specific ways to the overall security policy and infrastructure. Security tags can be updated when the resource changes, or whenever the security sub-system chooses to

tag (0..*) | Coding | Tags applied to this resource. Tags are used to relate resources to process and workflow. Applications are not required to consider the tags when interpreting the meaning of a resource.

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Meta | ? | | Element | Metadata about a resource Elements defined in Ancestors: id, extension

versionId | ? | 0..1 | id | Version specific identifier

lastUpdated | ? | 0..1 | instant | When the resource version last changed

profile | ? | 0..* | uri | Profiles this resource claims to conform to

security | ? | 0..* | Coding | Security Labels applied to this resource All Security Labels (Extensible)

tag | ? | 0..* | Coding | Tags applied to this resource Common Tags (Example)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<meta xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <versionId value="[id]"/><!-- 0..1 Version specific identifier -->
 <lastUpdated value="[instant]"/><!-- 0..1 When the resource version last changed -->
 <profile value="[uri]"/><!-- 0..* Profiles this resource claims to conform to -->
 <security><!-- 0..* Coding Security Labels applied to this resource --></security>
 <tag><!-- 0..* Coding Tags applied to this resource --></tag>
</meta>
```

JSON Template

```
{
  // from Element: extension
  "versionId" : "<id>", // Version specific identifier
  "lastUpdated" : "<instant>", // When the resource version last changed
  "profile" : ["<uri>"], // Profiles this resource claims to conform to
  "security" : [{ Coding }], // Security Labels applied to this resource
  "tag" : [{ Coding }] // Tags applied to this resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Meta.versionId [ id ]; # 0..1 Version specific identifier
  fhir:Meta.lastUpdated [ instant ]; # 0..1 When the resource version last changed
  fhir:Meta.profile [ uri ], ... ; # 0..* Profiles this resource claims to conform to
  fhir:Meta.security [ Coding ], ... ; # 0..* Security Labels applied to this resource
  fhir:Meta.tag [ Coding ], ... ; # 0..* Tags applied to this resource
]
```

Changes since DSTU2

Meta

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Meta | ? | | Element | Metadata about a resource Elements defined in Ancestors: id, extension
versionId | ? | 0..1 | id | Version specific identifier
lastUpdated | ? | 0..1 | instant | When the resource version last changed
profile | ? | 0..* | uri | Profiles this resource claims to conform to
security | ? | 0..* | Coding | Security Labels applied to this resource All Security Labels (Extensible)
tag | ? | 0..* | Coding | Tags applied to this resource Common Tags (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<meta xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <versionId value="[id]"/><!-- 0..1 Version specific identifier -->
 <lastUpdated value="[instant]"/><!-- 0..1 When the resource version last changed -->
 <profile value="[uri]"/><!-- 0..* Profiles this resource claims to conform to -->
 <security><!-- 0..* Coding Security Labels applied to this resource --></security>
 <tag><!-- 0..* Coding Tags applied to this resource --></tag>
</meta>
```

JSON Template

```
{
  // from Element: extension
  "versionId" : "<id>", // Version specific identifier
  "lastUpdated" : "<instant>", // When the resource version last changed
  "profile" : ["<uri>"], // Profiles this resource claims to conform to
  "security" : [{ Coding }], // Security Labels applied to this resource
  "tag" : [{ Coding }] // Tags applied to this resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Meta.versionId [ id ]; # 0..1 Version specific identifier
  fhir:Meta.lastUpdated [ instant ]; # 0..1 When the resource version last changed
  fhir:Meta.profile [ uri ], ... ; # 0..* Profiles this resource claims to conform to
  fhir:Meta.security [ Coding ], ... ; # 0..* Security Labels applied to this resource
  fhir:Meta.tag [ Coding ], ... ; # 0..* Tags applied to this resource
]
```

Changes since DSTU2

Meta

See the Full Difference for further information

Note that the RESTful API defines some Operations that provide direct read and write access to the meta element.

2.28.3.3.1 Technical vs Business Versions

All resources are conceptually versioned, and each resource sits at the head of a linear list of past versions. The past versions are superseded by the current version, and only available for audit/integrity purposes. The current version is e.g. http://acme.org/fhir/ResourceType/id123, and a past version would be http://acme.org/fhir/ResourceType/id123/_history/v2.

Note that there's no requirement for servers to keep a history. The history interaction is provided for where this is an appropriate service to provide. However, whether a server keeps them or not, past versions are dead and gone. The current version of the resource is in the Resource.meta.versionId. For a value set this would be:

```
<ValueSet>
  <meta>
    <versionId value="v2"/>
  </meta>
</ValueSet>
```

That version changes every time the server updates the resource and writes a new version over the top of an existing one.

Some resources have another version marker in them. For instance, ValueSet has another version in it:

```
<ValueSet>
  <url value="http://acme.com/fhir/ValueSet/example"/>
  <version value="2.0"/>
</ValueSet>
```

This says that this is version 2.0 of the 'example' value set. This is the business version of the value set, the one that humans get involved with. These 2 versions elements have quite different lifecycles. To illustrate, take these cases:

* A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Later, another user creates a revised version of the value set, and this is called version 2. It is committed to the server as an update (PUT [base]/ValueSet/x1). Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 2 and ValueSet.meta.versionId = 2

* A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Then a typo is found in the definition, so this is fixed, but it's still v1 of the value set. This is PUT to [base]/ValueSet/x1. Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 1 and ValueSet.meta.versionId = 2. Later, another user creates a revised version of the value set, and this is called version 2. It is commmited to the server as an update (PUT [base]/ValueSet/x1). Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 2 and ValueSet.meta.versionId = 3

* A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Later, another user creates a revised version of the value set, and this is called version 2. This time, as well as supporting this new version 2, there are production systems still using version 1, and both need to be valid on the value set server. So a new value set is created on the server (POST [base]/ValueSet) and is assigned the identiity 'x2'. Now, there are two different value sets, both with URL "http://acme.com/valuesets/example". One has ValueSet.id = x1, ValueSet.version = 1 and ValueSet.meta.versionId = 1 and the other has ValueSet.id = x2, ValueSet.version = 2 and ValueSet.meta.versionId = 1.

2.28.3.4 Implicit Rules

A reference to a custom agreement that describes how the resource is being used (e.g. an implementation guide) that was followed when the resource was constructed, where the implemenation guide must be known and understood in order to safely processing the content.

Asserting this rule set restricts the content to be only understood by a limited set of trading partners. This inherently limits the usefulness of the data in the long term, and should be avoided where possible. However, the existing health eco-system is highly fractured, and not yet ready to define, collect, and exchange data in a generally exchangeable sense.

Note that resources are almost always constructed following some custom agreement. Best practice - which is recommended through out the conformance resources - is that such agreements make all knowledge about the content of the resource explicit in the content of the resource (e.g. no default values in profiles); if custom agreements follow this advice, and declare their extensions as required, then it is not necessary to understand the agreement in order to safely process the resource content. For this reason, use of implicitRules is rare.

2.28.3.5 Language

Each resource may have a language element that specifies the base language of the content using a code defined in BCP 47 . The language element is provided to support indexing and accessibility (e.g. text-to-speech use the language tag).

There is no default language, though one may be inferred from the context of use. Not all of the content of the resource has to be in the specified language.

If a language is specified, it should also be specified on the Narrative Text. The html language tag in the narrative is used when processing the narrative. The language tag on the resource is provided so that applications processing the data in the resource can specify the language of any alternate presentations generated from the data.

2.28.3.6 Tags, Profiles, and Security Labels

These 3 metadata attributes are part of the resource, but are never used to keep information that needs to be understood when interpreting the content of a resource; their function is limited to finding and controlling access to the resource, and connecting resources to technical or clinical workflow.

2.28.3.6.1 Tags

Tags are used to associate additional operational information with the Resources, including such as workflow management. A typical use of tagging would be to maintain an informal list of resources that need review.

In a general tag, the coded concept may be a reference to a healthcare vocabulary, including ones defined in this specification, or vocabularies such as those defined by HL7 for other purposes (e.g. HL7 v2 and HL7 v3 /CDA), LOINC, or SNOMED CT. Alternatively, the concept may be one defined by the implementation in the local context.

The list of tags on a resource is a set, where each tag is unique based on the system+code combination.

2.28.3.6.2 Profile Tags

A profile assertion represents a claim that the resource conforms to the identified StructureDefinition, which makes rules about what content is allowed to be in a resource. In a profile tag, the term is a URL that references an identified StructureDefinition resource.

It's always possible to determine whether a resource conforms to any profile simply by testing it against the profile (the validation tools provide the functionality to perform this test in a variety of contexts). However there are several circumstances where simply examining whether a resource conforms to a particular profile as needed is impractical:

* A server searching a set of resources for ones that conform to a particular profile

* A receiver that has many profiles to choose when validating resource

Profile Tags serve these use cases - a client/creator of a resource can tag the resource with an assertion that the resource conforms to a particular structure definition. The server/receiver of the resource can choose to take this assertion at face value, or to assist in locating the correct StructureDefinition against which to validate the resource.

Note: resources can conform to multiple profiles at once. A resource can conform to a profile without ever being labeled that it does, or a resource may falsely claim to conform to a profile. For this reason, applications processing resources SHOULD always depend on the contents of the resource when processing them, and/or check resources against the StructureDefinitions directly rather than relying the existence of profile tags for meaning. Profile Tags are provided as a method of finding resources that conform to a particular StructureDefinition, not statements of meaning about the resource.

Many trading partner agreements will make rules about what claims can be made and when they must be tested, which will make the profile assertion more reliable.

The list of profiles on a resource is a set, where each profile is unique based on the value of the URI.

2.28.3.6.3 Security Labels

A security label is attached to a resource to provide specific security metadata about the information in the resource. For further information, see Security Labels.

The list of security on a resource is a set, where each tag is unique based on the system+code combination.

2.28.3.7 Updates to Tags, Profiles, and Security Labels

When a resource is updated (e.g. on the RESTful interface), servers generally follow this pattern:

* Merge existing and new tags

* Replace existing profile tags with new profile tags

* Merge existing and new security labels

However, in some cases, an update may invalidate existing tags. Servers may update or remove previously existing recognized tags if this is known to be appropriate.

2.28.3.8 Further Information

* Conformance Rules

* Resource Definitions

* References between Resources

* Narratives

* Formats: XML, JSON

* Extensibility (Examples)

* Detailed Descriptions

* Inter-version Compatibility

2.28.3.9 Maturity Levels

The Maturity level documentation has moved.

2.28.4 Search Parameters

Common search parameters defined by this resource. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths
_content | string | Search on the entire content of the resource
_id | token | Logical id of this artifact | Resource.id
_lastUpdated | date | When the resource version last changed | Resource.meta.lastUpdated
_profile | uri | Profiles this resource claims to conform to | Resource.meta.profile
_query | token | A custom search profile that describes a specific defined query operation
_security | token | Security Labels applied to this resource | Resource.meta.security
_tag | token | Tags applied to this resource | Resource.meta.tag

2.28.5 Resource Resource - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/resource-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.28.5 Resource Resource - Detailed Descriptions

Detailed Descriptions for the elements in the Base Resource Definitions.

Resource
Definition | This is the base resource type for everything.
Control | 1..1
Resource.id
Definition | The logical id of the resource, as used in the URL for the resource. Once assigned, this value never changes.
Control | 0..1
Type | id
Summary | true

Comments | The only time that a resource does not have an id is when it is being submitted to the server using a create operation.

Resource.meta

Definition | The metadata about the resource. This is content that is maintained by the infrastructure. Changes to the content may not always be associated with version changes to the resource.

Control | 0..1

Type | Meta

Summary | true

Resource.implicitRules

Definition | A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content.

Control | 0..1

Type | uri

Is Modifier | true

Summary | true

Comments | Asserting this rule set restricts the content to be only understood by a limited set of trading partners. This inherently limits the usefulness of the data in the long term. However, the existing health eco-system is highly fractured, and not yet ready to define, collect, and exchange data in a generally computable sense. Wherever possible, implementers and/or specification writers should avoid using this element. This element is labelled as a modifier because the implicit rules may provide additional knowledge about the resource that modifies it's meaning or interpretation.

Resource.language

Definition | The base language in which the resource is written.

Control | 0..1

Terminology Binding | Common Languages (Extensible but limited to ??)

Type | code

Comments | Language is provided to support indexing and accessibility (typically, services such as text to speech use the language tag). The html language tag in the narrative applies to the narrative. The language tag on the resource may be used to specify the language of other presentations generated from the data in the resource Not all the content has to be in the base language. The Resource.language should not be assumed to apply to the narrative automatically. If a language is specified, it should it also be specified on the div element in the html (see rules in HTML5 for information about the relationship between xml:lang and the html lang attribute).

Meta

Definition | The metadata about a resource. This is content in the resource that is maintained by the infrastructure. Changes to the content may not always be associated with version changes to the resource.

Control | 1..1

Summary | true

Meta.versionId

Definition | The version specific identifier, as it appears in the version portion of the URL. This values changes when the resource is created, updated, or deleted.

Control | 0..1

Type | id

Summary | true

Comments | The server assigns this value, and ignores what the client specifies, except in the case that the server is imposing version integrity on updates/deletes.

Meta.lastUpdated

Definition | When the resource last changed - e.g. when the version changed.

Control | 0..1

Type | instant

Summary | true

Comments | This value is always populated except when the resource is first being created. The server / resource manager sets this value; what a client provides is irrelevant.

Meta.profile

Definition | A list of profiles (references to StructureDefinition resources) that this resource claims to conform to. The URL is a reference to StructureDefinition.url.

Control | 0..*

Type | uri

Summary | true

Comments | It is up to the server and/or other infrastructure of policy to determine whether/how these claims are verified and/or updated over time. The list of profile URLs is a set.

Meta.security

Definition | Security labels applied to this resource. These tags connect specific resources to the overall security policy and infrastructure.

Control | 0..*

Terminology Binding | All Security Labels (Extensible)

Type | Coding

Summary | true

Comments | The security labels can be updated without changing the stated version of the resource The list of security labels is a set. Uniqueness is based the system/code, and version and display are ignored.

Meta.tag

Definition | Tags applied to this resource. Tags are intended to be used to identify and relate resources to process and workflow, and applications are not required to consider the tags when interpreting the meaning of a resource.

Control | 0..*

Terminology Binding | Common Tags (Example)

Type | Coding

Summary | true

Comments | The tags can be updated without changing the stated version of the resource. The list of tags is a set. Uniqueness is based the system/code, and version and display are ignored.

2.28.6 Resource HL7 Extensions

URL: https://hl7.org/fhir/STU3/resource-extensions.html

Defines common extensions used across multiple different resources

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.28.6 Resource HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

2.28.6.1 Content

Extensions:

resource-approvalDate | approvalDate : The date on which the asset content was approved by the publisher. Approval happens once when the content is officially approved for usage.

resource-lastReviewDate | lastReviewDate : The date on which the asset content was last reviewed. Review happens periodically after that, but doesn't change the original approval date.

resource-effectivePeriod | effectivePeriod : The period during which the resource content was or is planned to be effective.

2.28.7 Resource Resource - Operations

URL: https://hl7.org/fhir/STU3/resource-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.28.7 Resource Resource - Operations

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 4 operations associated with it:

$validate | Validate a resource

$meta | Access a list of profiles, tags, and security labels

$meta-add | Add profiles, tags, and security labels to a resource

$meta-delete | Delete profiles, tags, and security labels for a resource

For more information about operations, including how they are invoked, see Operations.

2.28.7.1 Validate a resource

The validate operation checks whether the attached content would be acceptable either generally, as a create, an update or as a delete to an existing resource. The action the server takes depends on the mode parameter:

* [mode not provided]: The server checks the content of the resource against any schema, constraint rules, and other general terminology rules

* create: The server checks the content, and then checks that the content would be acceptable as a create (e.g. that the content would not violate any uniqueness constraints)

* update: The server checks the content, and then checks that it would accept it as an update against the nominated specific resource (e.g. that there are no changes to immutable fields the server does not allow to change, and checking version integrity if appropriate)

* delete: The server ignores the content, and checks that the nominated resource is allowed to be deleted (e.g. checking referential integrity rules)

Modes update and delete can only be used when the operation is invoked at the resource instance level. The return from this operation is an OperationOutcome

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-validate
```

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/$validate

URL: [base]/[Resource]/[id]/$validate

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

resource | 0..1 | Resource | | | Must be present unless the mode is "delete"

mode | 0..1 | code | ResourceValidationMode (Required) | | Default is 'no action'; (e.g. general validation)

profile | 0..1 | uri | | | If this is nominated, then the resource is validated against this specific profile. If a profile is nominated, and the server cannot validate against the nominated profile, it SHALL return an error

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | OperationOutcome | | | If the operation outcome does not list any errors, and a mode was specified, then this is an indication that the operation would be expected to succeed (excepting for transactional integrity issues, see below) Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

This operation may be used during design and development to validate application design. It can also be used at run-time. One possible use might be that a client asks the server whether a proposed update is valid as the user is editing a dialog and displays an updated error to the user. The operation can be used as part of a light-weight two phase commit protocol but there is no expectation that the server will hold the content of the resource after this operation is used, or that the server guarantees to successfully perform an actual create, update or delete after the validation operation completes.

This operation returns a 200 OK whether or not the resource is valid. A 4xx or 5xx error means that the validation itself could not be performed, and it is unknown whether the resource is valid or not.

2.28.7.1.1 Examples

Validate a Patient against the DAF patient profile using POST directly. (Request):

```
POST /open/Patient/$validate?profile=http://hl7.org/fhir/StructureDefinition/daf-patient
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<Patient xmlns="http://hl7.org/fhir">
  <id value="us01"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>
        <b>Generated Narrative with Details</b>
      </p>
      <p>
        <b>id</b>: us01</p>
      <p>
        <b>active</b>: true</p>
        <p>
        <b>birthsex</b>: Male</p>
      <p>
        <b>name</b>: Peter James Chalmers (OFFICIAL)</p>
      <p>
        <b>telecom</b>: ph: (03) 5555 6473(WORK), person@example.org(WORK)</p>
      <p>
        <b>address</b>: 534 Erewhon St PleasantVille Vic 3999 (HOME)</p>
      <p>
        <b>race</b>: White, American Indian or Alaska Native, Asian, Shoshone, Filipino</p>
      <p>
        <b>ethnicity</b>: Hispanic or Latino, Dominican, Mexican</p>
    </div>
  </text>
  <extension url="http://hl7.org/fhir/StructureDefinition/us-core-race">
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Race"/>
        <code value="2106-3"/>
        <display value="White"/>
      </valueCoding>
    </extension>
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Race"/>
        <code value="1002-5"/>
        <display value="American Indian or Alaska Native"/>
      </valueCoding>
    </extension>
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Race"/>
        <code value="2028-9"/>
        <display value="Asian"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Race"/>
        <code value="1586-7"/>
        <display value="Shoshone"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Race"/>
        <code value="2036-2"/>
        <display value="Filipino"/>
      </valueCoding>
    </extension>
  </extension>
  <extension url="http://hl7.org/fhir/StructureDefinition/us-core-ethnicity">
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Ethnicity"/>
        <code value="2135-2"/>
        <display value="Hispanic or Latino"/>
      </valueCoding>
    </extension>
```

```
    <extension url="detailed">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Ethnicity"/>
        <code value="2184-0"/>
        <display value="Dominican"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://hl7.org/fhir/v3/Ethnicity"/>
        <code value="2148-5"/>
        <display value="Mexican"/>
      </valueCoding>
    </extension>
  </extension>
  <extension url="http://hl7.org/fhir/StructureDefinition/us-core-birthsex">
    <valueCode value="M"/>
  </extension>
  <active value="true"/>
  <!--  Peter James Chalmers, but called "Jim"  -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 6473"/>
    <use value="work"/>
  </telecom>
  <telecom>
    <extension url="http://hl7.org/fhir/StructureDefinition/us-core-direct">
      <valueBoolean value="true"/>
    </extension>
    <system value="email"/>
    <value value="person@example.org"/>
    <use value="work"/>
  </telecom>
  <address>
    <use value="home"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
    <district value="Orange County"/>
    <state value="Vic"/>
    <postalCode value="3999"/>
  </address>
</Patient>
```

Validate a proposed patient update - and for this example, using the Parameters syntax (Request):

```
POST /open/Patient/example/$validate
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="mode"/>
    <valueCode value="update"/>
  <parameter>
  <parameter>
    <name value="mode"/>
    <resource>
      <?xml version="1.0" encoding="UTF-8"?>
      <Patient xmlns="http://hl7.org/fhir">
        <id value="example"/>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">
            <table>
              <tbody>
                <tr>
                  <td>Name</td>
                  <td>Peter James
                    <b>Chalmers</b> (&amp;quot;Jim&amp;quot;)
                  </td>
                </tr>
                <tr>
                  <td>Address</td>
                  <td>534 Erewhon, Pleasantville, Vic, 3999</td>
                </tr>
                <tr>
                  <td>Contacts</td>
                  <td>Home: unknown. Work: (03) 5555 6473</td>
                </tr>
                <tr>
                  <td>Id</td>
                  <td>MRN: 12345 (Acme Healthcare)</td>
                </tr>
              </tbody>
```

```
      </table>
    </div>
  </text>
  <!--  MRN assigned by ACME healthcare on 6-May 2001  -->
  <identifier>
    <use value="usual"/>
    <type>
      <coding>
        <system value="http://hl7.org/fhir/v2/0203"/>
        <code value="MR"/>
      </coding>
    </type>
    <system value="urn:oid:1.2.36.146.595.217.0.1"/>
    <value value="12345"/>
    <period>
      <start value="2001-05-06"/>
    </period>
    <assigner>
      <display value="Acme Healthcare"/>
    </assigner>
  </identifier>
  <active value="true"/>
  <!--  Peter James Chalmers, but called "Jim"  -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <name>
    <use value="usual"/>
    <given value="Jim"/>
  </name>
  <name>
    <!-- Maiden names apply for anyone whose name changes as a result of marriage - irrespective of gender -->
    <use value="maiden"/>
    <family value="Windsor"/>
    <given value="Peter"/>
    <given value="James"/>
    <period>
      <end value="2002"/>
    </period>
  </name>
  <telecom>
    <use value="home"/>
    <!--  home communication details aren't known  -->
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 6473"/>
    <use value="work"/>
    <rank value="1"/>
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 3410 5613"/>
    <use value="mobile"/>
    <rank value="2"/>
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 8834"/>
    <use value="old"/>
    <period>
      <end value="2014"/>
    </period>
  </telecom>
  <!--  use FHIR code system for male / female  -->
  <gender value="male"/>
  <birthDate value="1974-12-25">
    <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
      <valueDateTime value="1974-12-25T14:35:45-05:00"/>
    </extension>
  </birthDate>
  <deceasedBoolean value="false"/>
  <address>
    <use value="home"/>
    <type value="both"/>
    <text value="534 Erewhon St
PeasantVille, Rainbow, Vic  3999"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
    <district value="Rainbow"/>
    <state value="Vic"/>
    <postalCode value="3999"/>
    <period>
      <start value="1974-12-25"/>
    </period>
  </address>
```

```
        <contact>
          <relationship>
            <coding>
              <system value="http://hl7.org/fhir/v2/0131"/>
              <code value="N"/>
            </coding>
          </relationship>
          <name>
            <family value="du Marché">
              <!--  the "du" part is a family name prefix (VV in iso 21090)  -->
              <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix">
                <valueString value="VV"/>
              </extension>
            </family>
            <given value="Bénédicte"/>
          </name>
          <telecom>
            <system value="phone"/>
            <value value="+33 (237) 998327"/>
          </telecom>
          <address>
            <use value="home"/>
            <type value="both"/>
            <line value="534 Erewhon St"/>
            <city value="PleasantVille"/>
            <district value="Rainbow"/>
            <state value="Vic"/>
            <postalCode value="3999"/>
            <period>
              <start value="1974-12-25"/>
            </period>
          </address>
          <gender value="female"/>
          <period>
            <!--  The contact relationship started in 2012  -->
            <start value="2012"/>
          </period>
        </contact>
        <managingOrganization>
          <reference value="Organization/1"/>
        </managingOrganization>
      </Patient>
    </resource>
  <parameter>
</Parameters>
```

Returned when the resource is valid (Response):

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="allok"/>
  <text>
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>All OK</p>
    </div>
  </text>
  <issue>
    <severity value="information"/>
    <code value="informational"/>
    <details>
      <text value="All OK"/>
    </details>
 </issue>
</OperationOutcome>
```

Returned when the resource is not valid (Response):

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="validationfail"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Error parsing resource XML (Unknown Content &amp;quot;label&amp;quot; @ /Patient/identifier/label)</p>
    </div>
  </text>
  <issue>
    <severity value="error"/>
    <code value="structure"/>
    <details>
      <text value="Error parsing resource XML (Unknown Content &amp;quot;label&amp;quot;"/>
    </details>
    <location value="/f:Patient/f:identifier"/>
    <expression value="Patient.identifier"/>
```

```
  </issue>
</OperationOutcome>
```

2.28.7.2 Access a list of profiles, tags, and security labels

This operation retrieves a summary of the profiles, tags, and security labels for the given scope; e.g. for each scope:

* system-wide: a list of all profiles, tags and security labels in use by the system

* resource-type level: A list of all profiles, tags, and security labels for the resource type

* individual resource level: A list of all profiles, tags, and security labels for the current version of the resource. Also, as a special case, this operation (and other meta operations) can be performed on a historical version of a resource)

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-meta
```

Formal Definition (as a OperationDefinition).

URL: [base]/$meta

URL: [base]/[Resource]/$meta

URL: [base]/[Resource]/[id]/$meta

This is an idempotent operation

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
return	1..1	Meta			The meta returned by the operation

At the system and type levels, the $meta operation is used to get a summary of all the labels that are in use across the system. The principle use for this operation is to support search e.g. what tags can be searched for. At these levels, the meta will not contain versionId, lastUpdated etc. Systems are not obligated to implement the operation at this level (and should return a 4xx error if they don't). At the resource and historical entry level, the $meta operation returns the same meta as would be returned by accessing the resource directly. This can be used to allow a system to get access to the meta-information for the resource without accessing the resource itself, e.g. for security reasons

2.28.7.2.1 Examples

Get all the tags, profiles, and security labels used on patient resources (Request):

```
GET /fhir/Patient/$meta
```

All the tags, profiles, and security labels used on patient resources (Response):

```
HTTP/1.1 200 OK
[other headers]

<Parameters>
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <profile value="http://hl7.org/fhir/StructureDefinition/uslab-patient"/>
      <security>
        <system value="http://hl7.org/fhir/v3/ActCode"/>
        <code value="EMP"/>
        <display value="employee information sensitivity"/>
      </security>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
        <display value="Current Inpatient"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

2.28.7.3 Add profiles, tags, and security labels to a resource

This operation takes a meta, and adds the profiles, tags, and security labels found in it to the nominated resource. This operation can also be used on historical entries - to update them without creating a different historical version

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-meta-add
```

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/[id]/$meta-add

This is not an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
meta	1..1	Meta			Profiles, tags, and security labels to add to the existing resource. Note that profiles, tags, and security labels are sets, and duplicates are not created. The identity of a tag or security label is the system+code. When matching existing tags during adding, version and display are ignored. For profiles, matching is based on the full URL

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
return	1..1	Meta			Resulting meta for the resource

This operation is special in that executing this operation does not cause a new version of the resource to be created. The meta is updated directly. This is because the content in meta does not affect the meaning of the resource, and the security labels (in particular) are used to apply access rules to existing versions of resources

2.28.7.3.1 Examples

Add a 'paper record lost' tag to a resource (Request):

```
POST /fhir/Patient/example/$meta-add
[Other headers]

<Parameters>
  <parameter>
    <name value="meta"/>
    <valueMeta>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

All the tags, profiles, and security labels used on patient resources (Response):

```
HTTP/1.1 200 OK
[other headers]

<Parameters>
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
        <display value="Current Inpatient"/>
      </tag>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

2.28.7.4 Delete profiles, tags, and security labels for a resource

This operation takes a meta, and deletes the profiles, tags, and security labels found in it from the nominated resource. This operation can also be used on historical entries

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Resource-meta-delete`

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/[id]/$meta-delete

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

meta | 1..1 | Meta | | | Profiles, tags, and security labels to delete from the existing resource. It is not an error if these tags, profiles, and labels do not exist. The identity of a tag or security label is the system+code. When matching existing tags during deletion, version and display are ignored. For profiles, matching is based on the full URL

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Meta | | | Resulting meta for the resource

This operation is special in that executing this operation does not cause a new version of the resource to be created. The meta is updated directly. This is because the content in meta does not affect the meaning of the resource, and the security labels (in particular) are used to apply access rules to existing versions of resources

2.28.7.4.1 Examples

Remove the 'current patient' tag from a resource (Request):

```
POST /fhir/Patient/example/$meta-delete
[Other headers]

<Parameters>
  <parameter>
    <name value="meta"/>
    <valueMeta>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
```

```
        <display value="Current Inpatient"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

All the tags, profiles, and security labels used on patient resources (Response):

```
HTTP/1.1 200 OK
[other headers]

<Parameters>
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

2.28.8 Resource Resource - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/resource-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.28.8 Resource Resource - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Resource.

Functional status for this map: Not Mapped (see documentation)

2.28.8.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Resource2to3" = "R2 to R3 Conversions for Resource"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Resource" alias ResourceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Resource" alias Resource as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/Meta2to3"

group for type+types  Resource
  input src : ResourceR2 as source
  input tgt : Resource as target

  "Resource-id" : for src.id make tgt.id
  "Resource-meta" : for src.meta make tgt.meta
  "Resource-implicitRules" : for src.implicitRules make tgt.implicitRules
  "Resource-language" : for src.language make tgt.language

endgroup
```

2.28.8.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Resource3to2" = "R3 to R2 Conversion for Resource"

uses "http://hl7.org/fhir/StructureDefinition/Resource" alias Resource as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Resource" alias ResourceR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/Meta3to2"

group for type+types  Resource
  input src : Resource as source
  input tgt : ResourceR2 as target

  "Resource-id" : for src.id make tgt.id
  "Resource-meta" : for src.meta make tgt.meta
  "Resource-implicitRules" : for src.implicitRules make tgt.implicitRules
  "Resource-language" : for src.language make tgt.language

endgroup
```

2.28.8.3 R3 Validation Errors

n/a

2.29 DomainResource Resource

URL: https://hl7.org/fhir/STU3/domainresource.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: 5 | Compartments: Not linked to any defined compartments

2.29.1 Scope and Usage

A domain resource is an resource that:

* has a human-readable XHTML representation of the content of the resource (see Human Narrative in resources)

* can contain additional related resources inside the resource (see Contained Resources)

* can have additional extensions and modifierExtensions as well as the defined data (See Extensibility)

As an abstract resource, this resource is never created directly; instead, one of its descendent resources (see List of Resources) is created.

2.29.2 Boundaries and Relationships

This resource extends the base resource. All of the listed Resources except Bundle, Parameters and Binary extend this resource.

2.29.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

DomainResource | I | | Resource | A resource with narrative, extensions, and contained resources + If the resource is contained in another resource, it SHALL NOT contain nested Resources + If the resource is contained in another resource, it SHALL NOT contain any narrative + If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated + If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource Elements defined in Ancestors: id, meta, implicitRules, language

text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation

contained | | 0..* | Resource | Contained, inline Resources

extension | | 0..* | Extension | Additional Content defined by implementations

modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <extension><!-- 0..* Extension Additional Content defined by implementations --></extension>
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  // from Resource: id, meta, implicitRules, and language
  "text" : { Narrative }, // C? Text summary of the resource, for human interpretation
  "contained" : [{ Resource }], // Contained, inline Resources
  (Extensions - see JSON page)
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:DomainResource.text [ Narrative ]; # 0..1 Text summary of the resource, for human interpretation
  fhir:DomainResource.contained [ Resource ], ... ; # 0..* Contained, inline Resources
  fhir:DomainResource.extension [ Extension ], ... ; # 0..* Additional Content defined by implementations
  fhir:DomainResource.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

Structure

Name | Flags | Card. | Type | Description & Constraints

DomainResource | I | | Resource | A resource with narrative, extensions, and contained resources + If the resource is contained in another resource, it SHALL NOT contain nested Resources + If the resource is contained in another resource, it SHALL NOT contain any narrative + If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated + If the resource is contained in another resource, it

SHALL be referred to from elsewhere in the resource Elements defined in Ancestors: id, meta, implicitRules, language
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional Content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <text><!--  0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <extension><!-- 0..* Extension Additional Content defined by implementations --></extension>
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  // from Resource: id, meta, implicitRules, and language
  "text" : { Narrative }, // C? Text summary of the resource, for human interpretation
  "contained" : [{ Resource }], // Contained, inline Resources
  (Extensions - see JSON page)
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:DomainResource.text [ Narrative ]; # 0..1 Text summary of the resource, for human interpretation
  fhir:DomainResource.contained [ Resource ], ... ; # 0..* Contained, inline Resources
  fhir:DomainResource.extension [ Extension ], ... ; # 0..* Additional Content defined by implementations
  fhir:DomainResource.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

2.29.3.1 Constraints

* dom-1: If the resource is contained in another resource, it SHALL NOT contain any narrative (expression : contained.text.empty())

* dom-2: If the resource is contained in another resource, it SHALL NOT contain nested Resources (expression : contained.contained.empty())

* dom-3: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource (expression : contained.where(('#'+id in %resource.descendants().reference).not()).empty())

* dom-4: If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated (expression : contained.meta.versionId.empty() and contained.meta.lastUpdated.empty())

2.29.4 Searching Extensions

To search for extensions, define a search parameter for the extension. All other search parameters are named aliases for existing content in the resource. In some cases, though not all, the search parameter name is the same as the element that it searches, but this is not required. Searching for extensions is the same - define a name that identifies the value extension by its URL, and then searches can filter based on the value of the extension.

2.29.5 Search Parameters

Common search parameters defined by this resource. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths
_text | string | Search on the narrative of the resource

2.29.6 Resource DomainResource - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/domainresource-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.29.6 Resource DomainResource - Detailed Descriptions

Detailed Descriptions for the elements in the DomainResource Resource.

DomainResource
Definition | A resource that includes narrative, extensions, and contained resources.
Control | 1..1

Invariants | Defined on this element dom-1: If the resource is contained in another resource, it SHALL NOT contain any narrative (expression : contained.text.empty(), xpath: not(parent::f:contained and f:text)) dom-2: If the resource is contained in another resource, it SHALL NOT contain nested Resources (expression : contained.contained.empty(), xpath: not(parent::f:contained and f:contained)) dom-3: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource (expression : contained.where(('#'+id in %resource.descendants().reference).not()).empty(), xpath: not(exists(for $id in f:contained/*/@id return $id[not(ancestor::f:contained/parent::*/descendant::f:reference/@value=concat('#', $id))])) dom-4: If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated (expression : contained.meta.versionId.empty() and contained.meta.lastUpdated.empty(), xpath: not(exists(f:contained/*/f:meta/f:versionId)) and not(exists(f:contained/*/f:meta/f:lastUpdated)))

DomainResource.text

Definition | A human-readable narrative that contains a summary of the resource, and may be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it "clinically safe" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.

Control | 0..1

Type | Narrative

Alternate Names | narrative; html; xhtml; display

Comments | Contained resources do not have narrative. Resources that are not contained SHOULD have a narrative. In some cases, a resource may only have text with little or no additional discrete data (as long as all minOccurs=1 elements are satisfied). This may be necessary for data from legacy systems where information is captured as a "text blob" or where text is additionally entered raw or narrated and encoded in formation is added later.

Invariants | Affect this element dom-1: If the resource is contained in another resource, it SHALL NOT contain any narrative (expression : contained.text.empty(), xpath: not(parent::f:contained and f:text))

DomainResource.contained

Definition | These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.

Control | 0..*

Type | Resource

Alternate Names | inline resources; anonymous resources; contained resources

Comments | This should never be done when the content can be identified properly, as once identification is lost, it is extremely difficult (and context dependent) to restore it again.

DomainResource.extension

Definition | May be used to represent additional information that is not part of the basic definition of the resource. In order to make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.

Control | 0..*

Type | Extension

Alternate Names | extensions; user content

Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

DomainResource.modifierExtension

Definition | May be used to represent additional information that is not part of the basic definition of the resource, and that modifies the understanding of the element that contains it. Usually modifier elements provide negation or qualification. In order to make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.

Control | 0..*

Type | Extension

Is Modifier | true

Alternate Names | extensions; user content

Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.29.7 Resource DomainResource - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/domainresource-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.29.7 Resource DomainResource - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for DomainResource.

Functional status for this map: Not Mapped (see documentation)

2.29.7.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DomainResource2to3" = "R2 to R3 Conversions for DomainResource"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DomainResource" alias DomainResourceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DomainResource" alias DomainResource as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DomainResource extends Resource
  input src : DomainResourceR2 as source
```

```
  input tgt : DomainResource as target

  "DomainResource-contained" : for src.contained make tgt.contained
  "DomainResource-text" : for src.text make tgt.text
  "DomainResource-extension" : for src.extension make tgt.extension
  "DomainResource-modifierExtension" : for src.modifierExtension make tgt.modifierExtension

endgroup
```

2.29.7.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DomainResource3to2" = "R3 to R2 Conversion for DomainResource"

uses "http://hl7.org/fhir/StructureDefinition/DomainResource" alias DomainResource as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DomainResource" alias DomainResourceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DomainResource extends Resource
  input src : DomainResource as source
  input tgt : DomainResourceR2 as target

  "DomainResource-contained" : for src.contained make tgt.contained
  "DomainResource-text" : for src.text make tgt.text
  "DomainResource-extension" : for src.extension make tgt.extension
  "DomainResource-modifierExtension" : for src.modifierExtension make tgt.modifierExtension

endgroup
```

2.29.7.3 R3 Validation Errors

n/a

2.30 Element

URL: https://hl7.org/fhir/STU3/element.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The base definition for all elements contained inside a resource. All elements, whether defined as a Data Type (including primitives) or as part of a resource structure, have this base content:

* Extensions

* An internal id

There are 3 kinds of descendent types that specialize Element:

* Primitive data types, that add a primitive value property of the specified type

* Complex data types, that add their own children (all of which are also elements)

* BackboneElement, A specialization that adds modifierExtension, which is the super-type of all the element types defined in resource definitions (e.g. Patient.contact)

Note that resources themselves all specialize the base type Resource.

2.30.0.1 Content

* Structure

* UML

* XML

* JSON

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Element | I | | n/a | Base for all elements All FHIR elements must have a @value or children
id | | 0..1 | string | xml:id (or equivalent in JSON)
extension | | 0..* | Extension | Additional Content defined by implementations
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir" id="Internal id (e.g. like xml:id) (string)">
 <extension><!-- 0..* Extension Additional Content defined by implementations --></extension>
</[name]>
```

JSON Template

```
// complex types:

{
  "id" : "<string>", // Internal Id
```

```
  "extension" : [{//  Additional Content defined by implementations
    // from Element: extension (recursive)
    "url" : "<uri>", // R!  identifies the meaning of the extension
    "value[x]" : <*> // identifies the meaning of the extension
  }]
}
// primitive types:

{
  "{name}" : "value",  // The primitive value
  "{name}_" : { // Special syntax for the id and extensions
    "id" : "<string>", // Internal Id
    "extension" : [{//  Additional Content defined by implementations
      // from Element: extension (recursive)
      "url" : "<uri>", // R!  identifies the meaning of the extension
      "value[x]" : <*> // identifies the meaning of the extension
    }]
  }
}
```

Structure

Name | Flags | Card. | Type | Description & Constraints

Element | I | | Element | Base for all elements All FHIR elements must have a @value or children

id | | 0..1 | string | xml:id (or equivalent in JSON)

extension | | 0..* | Extension | Additional Content defined by implementations

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir" id="Internal id (e.g. like xml:id) (string)">
 <extension><!-- 0..* Extension Additional Content defined by implementations --></extension>
</[name]>
```

JSON Template

```
// complex types:

{
  "id" : "<string>", // Internal Id
  "extension" : [{//  Additional Content defined by implementations
    // from Element: extension (recursive)
    "url" : "<uri>", // R!  identifies the meaning of the extension
    "value[x]" : <*> // identifies the meaning of the extension
  }]
}
// primitive types:

{
  "{name}" : "value",  // The primitive value
  "{name}_" : { // Special syntax for the id and extensions
    "id" : "<string>", // Internal Id
    "extension" : [{//  Additional Content defined by implementations
      // from Element: extension (recursive)
      "url" : "<uri>", // R!  identifies the meaning of the extension
      "value[x]" : <*> // identifies the meaning of the extension
    }]
  }
}
```

Constraints

* ele-1: All FHIR elements must have a @value or children (expression : hasValue() | (children().count() > id.count()))

This constraint exists to reduce syntactical variation in resource contents. If an element has no children, then it is always omitted from the resource, as opposed to optionally present without any content.

2.30.0.2 Representation of Element

As the base type for all elements included in a resource, Element is an important structural element of FHIR. Even the primitive types inherit the base features and representation rules that apply to the Element type.

2.30.0.3 XML Representation

Elements are represented by an XML element. The name of the element comes from the context in which it is used, not from the type. The internal id is represented as an attribute (similar to xml:id, but see below about scope). Extensions are represented as XML elements. Here is the representation for an element 'code' of type Coding:

```
  <code id="[internal id]">
    <extension url="..."/>
      ... if there are any extensions
    <extension>
    .. elements of Coding type...
  </code>
```

Primitive types have the same representation; the actual primitive value appears as an XML attribute named value on the XML element. For example, a string property named "name" will be represented like this:

```
<code id="[internal id] value="[value of string]"">
  <extension url="..."/>
    ... if there are any extensions
  <extension>
</code>
```

2.30.0.4 JSON Representation

Elements (except for primitive types, see below) are represented by a JSON object property. The name of the property comes from the context in which it is used, not from the type. The internal id is represented as a JSON string property named "_id". Extensions are represented in a JSON array of objects named "extension". Here is the representation for a property 'code' of type Coding:

```
{
  "code" : {
    "id" : "[internal id]",
    "extension" : [
      ..extensions, if present...
    ],
    .. properties of Coding type...
  }
}
```

Primitive types are represented differently; the actual primitive value appears as a JSON string or number property. If an internal id or extensions are present, they appear in a JSON object with the name of the primitive value property with "_" prepended. For example, a string property named "name" will be represented like this:

```
{
  "name" : "[value of string]",
  "_name" : {
    "id" : "[internal id]",
    "extension" : [
      ..extensions, if present...
    ]
  }
}
```

The exact use of this pattern is described here.

2.30.0.5 Internal Id Scope

The id property of the element is defined to allow implementers to build implementation functionality that makes use of internal references inside the resource. This specification does not use the internal id on the element in any way.

The internal id is unique within the scope of the resource that contains it. Specifically, this means:

* The id SHALL be unique within a given resource

* The uniqueness boundary extends into contained resources. i.e. a contained resource cannot have the same id as any element in the resource that contains it or any other contained resource

* The uniqueness boundary is broken at Bundle.entry.resource and Parameters.parameter.resource, since these are elements that aggregate different resources

These rules ensure that there is no need to change internal identifiers while exchanging resources.

2.30.1 Detailed Descriptions

URL: https://hl7.org/fhir/STU3/element-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.30.1 Element - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

2.30.1.1 Element

Element

Definition | Base definition for all elements in a resource.

Control | 1..1

Invariants | Defined on this element ele-1: All FHIR elements must have a @value or children (expression : hasValue() | (children().count() > id.count()), xpath: @value|f:*|h:div) Affect this element

Element.id

Definition | unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.

Control | 0..1

Type | string

Element.extension

Definition | May be used to represent additional information that is not part of the basic definition of the element. In order to make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.

Control | 0..*

Type | Extension

Alternate Names | extensions; user content

Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.31 BackboneElement

URL: https://hl7.org/fhir/STU3/backboneelement.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The base definition for elements defined as part of a resource definition. Data Type elements do not use this type. For instance, Patient.contact is an element that is defined as part of the patient resource, so it automatically has the type BackboneElement.

Note that the descendent types of BackboneElement are all declared implicitly as part of the definitions of the resources.

2.31.0.1 Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

BackboneElement | | | Element | Base for elements defined inside a resource Elements defined in Ancestors: id, extension

modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:BackboneElement.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

Changes since DSTU2

BackboneElement

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

BackboneElement | | | Element | Base for elements defined inside a resource Elements defined in Ancestors: id, extension

modifierExtension | ?!? | 0..* | Extension | Extensions that cannot be ignored

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  (Modifier Extensions - see JSON page)
```

```
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:BackboneElement.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

Changes since DSTU2

BackboneElement

See the Full Difference for further information

2.31.0.2 Specializations

This table lists all the specializations of BackboneElement that are defined, and the path at which they are defined.

Path
Account
Account.coverage
Account.guarantor
ActivityDefinition
ActivityDefinition.participant
ActivityDefinition.dynamicValue
AdverseEvent
AdverseEvent.suspectEntity
AllergyIntolerance
AllergyIntolerance.reaction
Appointment
Appointment.participant
AuditEvent
AuditEvent.agent
AuditEvent.agent.network
AuditEvent.source
AuditEvent.entity
AuditEvent.entity.detail
Bundle
Bundle.link
Bundle.entry
Bundle.entry.search
Bundle.entry.request
Bundle.entry.response
CapabilityStatement
CapabilityStatement.software
CapabilityStatement.implementation
CapabilityStatement.rest
CapabilityStatement.rest.security
CapabilityStatement.rest.security.certificate
CapabilityStatement.rest.resource
CapabilityStatement.rest.resource.interaction
CapabilityStatement.rest.resource.searchParam
CapabilityStatement.rest.interaction
CapabilityStatement.rest.operation
CapabilityStatement.messaging
CapabilityStatement.messaging.endpoint
CapabilityStatement.messaging.supportedMessage
CapabilityStatement.messaging.event
CapabilityStatement.document
CarePlan
CarePlan.activity
CarePlan.activity.detail
CareTeam
CareTeam.participant
ChargeItem
ChargeItem.participant
Claim
Claim.related
Claim.payee
Claim.careTeam
Claim.information
Claim.diagnosis
Claim.procedure
Claim.insurance
Claim.accident

Claim.item
Claim.item.detail
Claim.item.detail.subDetail
ClaimResponse
ClaimResponse.item
ClaimResponse.item.adjudication
ClaimResponse.item.detail
ClaimResponse.item.detail.subDetail
ClaimResponse.addItem
ClaimResponse.addItem.detail
ClaimResponse.error
ClaimResponse.payment
ClaimResponse.processNote
ClaimResponse.insurance
ClinicalImpression
ClinicalImpression.investigation
ClinicalImpression.finding
CodeSystem
CodeSystem.filter
CodeSystem.property
CodeSystem.concept
CodeSystem.concept.designation
CodeSystem.concept.property
Communication
Communication.payload
CommunicationRequest
CommunicationRequest.payload
CommunicationRequest.requester
CompartmentDefinition
CompartmentDefinition.resource
Composition
Composition.attester
Composition.relatesTo
Composition.event
Composition.section
ConceptMap
ConceptMap.group
ConceptMap.group.element
ConceptMap.group.element.target
ConceptMap.group.element.target.dependsOn
ConceptMap.group.unmapped
Condition
Condition.stage
Condition.evidence
Consent
Consent.actor
Consent.policy
Consent.data
Consent.except
Consent.except.actor
Consent.except.data
Contract
Contract.agent
Contract.signer
Contract.valuedItem
Contract.term
Contract.term.agent
Contract.term.valuedItem
Contract.friendly
Contract.legal
Contract.rule
Coverage
Coverage.grouping
DataElement
DataElement.mapping
DetectedIssue
DetectedIssue.mitigation
Device
Device.udi
DeviceComponent
DeviceComponent.productionSpecification
DeviceMetric

DeviceMetric.calibration
DeviceRequest
DeviceRequest.requester
DiagnosticReport
DiagnosticReport.performer
DiagnosticReport.image
DocumentManifest
DocumentManifest.content
DocumentManifest.related
DocumentReference
DocumentReference.relatesTo
DocumentReference.content
DocumentReference.context
DocumentReference.context.related
EligibilityResponse
EligibilityResponse.insurance
EligibilityResponse.insurance.benefitBalance
EligibilityResponse.insurance.benefitBalance.financial
EligibilityResponse.error
Encounter
Encounter.statusHistory
Encounter.classHistory
Encounter.participant
Encounter.diagnosis
Encounter.hospitalization
Encounter.location
EpisodeOfCare
EpisodeOfCare.statusHistory
EpisodeOfCare.diagnosis
ExpansionProfile
ExpansionProfile.fixedVersion
ExpansionProfile.excludedSystem
ExpansionProfile.designation
ExpansionProfile.designation.include
ExpansionProfile.designation.include.designation
ExpansionProfile.designation.exclude
ExpansionProfile.designation.exclude.designation
ExplanationOfBenefit
ExplanationOfBenefit.related
ExplanationOfBenefit.payee
ExplanationOfBenefit.information
ExplanationOfBenefit.careTeam
ExplanationOfBenefit.diagnosis
ExplanationOfBenefit.procedure
ExplanationOfBenefit.insurance
ExplanationOfBenefit.accident
ExplanationOfBenefit.item
ExplanationOfBenefit.item.adjudication
ExplanationOfBenefit.item.detail
ExplanationOfBenefit.item.detail.subDetail
ExplanationOfBenefit.addItem
ExplanationOfBenefit.addItem.detail
ExplanationOfBenefit.payment
ExplanationOfBenefit.processNote
ExplanationOfBenefit.benefitBalance
ExplanationOfBenefit.benefitBalance.financial
FamilyMemberHistory
FamilyMemberHistory.condition
Goal
Goal.target
GraphDefinition
GraphDefinition.link
GraphDefinition.link.target
GraphDefinition.link.target.compartment
Group
Group.characteristic
Group.member
HealthcareService
HealthcareService.availableTime
HealthcareService.notAvailable
ImagingManifest
ImagingManifest.study

ImagingManifest.study.series
ImagingManifest.study.series.instance
ImagingStudy
ImagingStudy.series
ImagingStudy.series.instance
Immunization
Immunization.practitioner
Immunization.explanation
Immunization.reaction
Immunization.vaccinationProtocol
ImmunizationRecommendation
ImmunizationRecommendation.recommendation
ImmunizationRecommendation.recommendation.dateCriterion
ImmunizationRecommendation.recommendation.protocol
ImplementationGuide
ImplementationGuide.dependency
ImplementationGuide.package
ImplementationGuide.package.resource
ImplementationGuide.global
ImplementationGuide.page
Linkage
Linkage.item
List
List.entry
Location
Location.position
Measure
Measure.group
Measure.group.population
Measure.group.stratifier
Measure.supplementalData
MeasureReport
MeasureReport.group
MeasureReport.group.population
MeasureReport.group.stratifier
MeasureReport.group.stratifier.stratum
MeasureReport.group.stratifier.stratum.population
Medication
Medication.ingredient
Medication.package
Medication.package.content
Medication.package.batch
MedicationAdministration
MedicationAdministration.performer
MedicationAdministration.dosage
MedicationDispense
MedicationDispense.performer
MedicationDispense.substitution
MedicationRequest
MedicationRequest.requester
MedicationRequest.dispenseRequest
MedicationRequest.substitution
MessageDefinition
MessageDefinition.focus
MessageDefinition.allowedResponse
MessageHeader
MessageHeader.destination
MessageHeader.source
MessageHeader.response
NamingSystem
NamingSystem.uniqueId
NutritionOrder
NutritionOrder.oralDiet
NutritionOrder.oralDiet.nutrient
NutritionOrder.oralDiet.texture
NutritionOrder.supplement
NutritionOrder.enteralFormula
NutritionOrder.enteralFormula.administration
Observation
Observation.referenceRange
Observation.related
Observation.component

OperationDefinition
OperationDefinition.parameter
OperationDefinition.parameter.binding
OperationDefinition.overload
OperationOutcome
OperationOutcome.issue
Organization
Organization.contact
Patient
Patient.contact
Patient.animal
Patient.communication
Patient.link
PaymentReconciliation
PaymentReconciliation.detail
PaymentReconciliation.processNote
Person
Person.link
PlanDefinition
PlanDefinition.goal
PlanDefinition.goal.target
PlanDefinition.action
PlanDefinition.action.condition
PlanDefinition.action.relatedAction
PlanDefinition.action.participant
PlanDefinition.action.dynamicValue
Practitioner
Practitioner.qualification
PractitionerRole
PractitionerRole.availableTime
PractitionerRole.notAvailable
Procedure
Procedure.performer
Procedure.focalDevice
ProcedureRequest
ProcedureRequest.requester
ProcessRequest
ProcessRequest.item
ProcessResponse
ProcessResponse.processNote
Provenance
Provenance.agent
Provenance.entity
Questionnaire
Questionnaire.item
Questionnaire.item.enableWhen
Questionnaire.item.option
QuestionnaireResponse
QuestionnaireResponse.item
QuestionnaireResponse.item.answer
ReferralRequest
ReferralRequest.requester
RequestGroup
RequestGroup.action
RequestGroup.action.condition
RequestGroup.action.relatedAction
ResearchStudy
ResearchStudy.arm
RiskAssessment
RiskAssessment.prediction
SearchParameter
SearchParameter.component
Sequence
Sequence.referenceSeq
Sequence.variant
Sequence.quality
Sequence.repository
Specimen
Specimen.collection
Specimen.processing
Specimen.container
StructureDefinition

StructureDefinition.mapping
StructureDefinition.snapshot
StructureDefinition.differential
StructureMap
StructureMap.structure
StructureMap.group
StructureMap.group.input
StructureMap.group.rule
StructureMap.group.rule.source
StructureMap.group.rule.target
StructureMap.group.rule.target.parameter
StructureMap.group.rule.dependent
Subscription
Subscription.channel
Substance
Substance.instance
Substance.ingredient
SupplyDelivery
SupplyDelivery.suppliedItem
SupplyRequest
SupplyRequest.orderedItem
SupplyRequest.requester
Task
Task.requester
Task.restriction
Task.input
Task.output
TestReport
TestReport.participant
TestReport.setup
TestReport.setup.action
TestReport.setup.action.operation
TestReport.setup.action.assert
TestReport.test
TestReport.test.action
TestReport.teardown
TestReport.teardown.action
TestScript
TestScript.origin
TestScript.destination
TestScript.metadata
TestScript.metadata.link
TestScript.metadata.capability
TestScript.fixture
TestScript.variable
TestScript.rule
TestScript.rule.param
TestScript.ruleset
TestScript.ruleset.rule
TestScript.ruleset.rule.param
TestScript.setup
TestScript.setup.action
TestScript.setup.action.operation
TestScript.setup.action.operation.requestHeader
TestScript.setup.action.assert
TestScript.setup.action.assert.rule
TestScript.setup.action.assert.rule.param
TestScript.setup.action.assert.ruleset
TestScript.setup.action.assert.ruleset.rule
TestScript.setup.action.assert.ruleset.rule.param
TestScript.test
TestScript.test.action
TestScript.teardown
TestScript.teardown.action
ValueSet
ValueSet.compose
ValueSet.compose.include
ValueSet.compose.include.concept
ValueSet.compose.include.concept.designation
ValueSet.compose.include.filter
ValueSet.expansion
ValueSet.expansion.parameter

ValueSet.expansion.contains
VisionPrescription
VisionPrescription.dispense

2.31.1 Detailed Descriptions

URL: https://hl7.org/fhir/STU3/backboneelement-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.31.1 BackboneElement - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

2.31.1.1 BackboneElement

Element
Definition | Base definition for all elements in a resource.
Control | 1..1
Invariants | Defined on this element ele-1: All FHIR elements must have a @value or children (expression : hasValue() | (children().count() > id.count()), xpath: @value|f:*|h:div) Affect this element
Element.id
Definition | unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.
Control | 0..1
Type | string
Element.extension
Definition | May be used to represent additional information that is not part of the basic definition of the element. In order to make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.
Control | 0..*
Type | Extension
Alternate Names | extensions; user content
Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.32 Element Definition

URL: https://hl7.org/fhir/STU3/elementdefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

The definition of an element in a resource or an extension. The definition includes:

* Path (name), Cardinality, and data type

* Definitions, usage notes, and requirements

* Default or fixed values

* Constraints, Length limits, and other usage rules

* Terminology Binding

* Mappings to other specifications

* Structural Usage Information such as Slicing

The ElementDefinition type is the core of the FHIR metadata layer, and is closely (conceptually) aligned to ISO 11179. The DataElement resource is a packaging wrapper around the ElementDefinition type. All the data elements defined in this specification are published as a collection of data elements (XML or JSON).

2.32.0.1 Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

ElementDefinition | ?I | | Element | Definition of an element in a resource or extension + Min <= Max + if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding + Pattern may only be specified if there is one type + Fixed value may only be specified if there is one type + Binding can only be present for coded elements, string, and uri + Pattern and value are mutually exclusive + Constraints must be unique by key + Types must be unique by the combination of code and profile + sliceName must be composed of proper tokens separated by "/" + default value and meaningWhenMissing are mutually exclusive Elements defined in Ancestors: id, extension
path | ? | 1..1 | string | Path of the element in the hierarchy of elements
representation | ? | 0..* | code | xmlAttr | xmlText | typeAttr | cdaText | xhtml PropertyRepresentation (Required)
sliceName | ? | 0..1 | string | Name for this particular element (in a set of slices)
label | ? | 0..1 | string | Name for element to display with or prompt for element
code | ? | 0..* | Coding | Corresponding codes in terminologies LOINC Codes (Example)
slicing | ?I | 0..1 | Element | This element is sliced - slices follow + If there are no discriminators, there must be a definition
discriminator | ? | 0..* | Element | Element values that are used to distinguish the slices
type | ? | 1..1 | code | value | exists | pattern | type | profile DiscriminatorType (Required)
path | ? | 1..1 | string | Path to element value
description | ?I | 0..1 | string | Text description of how slicing works (or not)
ordered | ? | 0..1 | boolean | If elements must be in same order as slices
rules | ? | 1..1 | code | closed | open | openAtEnd SlicingRules (Required)
short | ? | 0..1 | string | Concise definition for space-constrained presentation
definition | ? | 0..1 | markdown | Full formal definition as narrative text
comment | ? | 0..1 | markdown | Comments about the use of this element
requirements | ? | 0..1 | markdown | Why this resource has been created
alias | ? | 0..* | string | Other names
min | ?I | 0..1 | unsignedInt | Minimum Cardinality
max | ?I | 0..1 | string | Maximum Cardinality (a number or *) + Max SHALL be a number or "*"
base | ? | 0..1 | Element | Base definition information for tools
path | ? | 1..1 | string | Path that identifies the base element
min | ? | 1..1 | unsignedInt | Min cardinality of the base element
max | ? | 1..1 | string | Max cardinality of the base element
contentReference | ?I | 0..1 | uri | Reference to definition of content for the element
type | ?I | 0..* | Element | Data type and Profile for this element + Aggregation may only be specified if one of the allowed types for the element is a resource
code | ? | 1..1 | uri | Data type or Resource (reference to definition) FHIRDefinedType (Extensible)
profile | ? | 0..1 | uri | Profile (StructureDefinition) to apply (or IG)
targetProfile | ? | 0..1 | uri | Profile (StructureDefinition) to apply to reference target (or IG)
aggregation | ?I | 0..* | code | contained | referenced | bundled - how aggregated AggregationMode (Required)
versioning | ? | 0..1 | code | either | independent | specific ReferenceVersionRules (Required)
defaultValue[x] | ?I | 0..1 | * | Specified value if missing from instance
meaningWhenMissing | ?I | 0..1 | markdown | Implicit meaning when this element is missing
orderMeaning | ? | 0..1 | string | What the order of the elements means
fixed[x] | ?I | 0..1 | * | Value must be exactly this
pattern[x] | ?I | 0..1 | * | Value must have at least these property values
example | ? | 0..* | Element | Example value (as defined for type)
label | ? | 1..1 | string | Describes the purpose of this example
value[x] | ? | 1..1 | * | Value of Example (one of allowed types)
minValue[x] | ? | 0..1 | | Minimum Allowed Value (for some types)
minValueDate | | | date
minValueDateTime | | | dateTime
minValueInstant | | | instant
minValueTime | | | time
minValueDecimal | | | decimal
minValueInteger | | | integer
minValuePositiveInt | | | positiveInt
minValueUnsignedInt | | | unsignedInt
minValueQuantity | | | Quantity
maxValue[x] | ? | 0..1 | | Maximum Allowed Value (for some types)
maxValueDate | | | date
maxValueDateTime | | | dateTime
maxValueInstant | | | instant
maxValueTime | | | time
maxValueDecimal | | | decimal
maxValueInteger | | | integer
maxValuePositiveInt | | | positiveInt
maxValueUnsignedInt | | | unsignedInt
maxValueQuantity | | | Quantity
maxLength | ? | 0..1 | integer | Max length for strings
condition | ? | 0..* | id | Reference to invariant about presence
constraint | ? | 0..* | Element | Condition that must evaluate to true
key | ?I | 1..1 | id | Target of 'condition' reference above
requirements | ? | 0..1 | string | Why this constraint is necessary or appropriate
severity | ? | 1..1 | code | error | warning ConstraintSeverity (Required)
human | ? | 1..1 | string | Human description of constraint
expression | ? | 1..1 | string | FHIRPath expression of constraint

xpath | ? | 0..1 | string | XPath expression of constraint
source | ? | 0..1 | uri | Reference to original source of constraint
mustSupport | ? | 0..1 | boolean | If the element must supported
isModifier | ? | 0..1 | boolean | If this modifies the meaning of other elements
isSummary | ? | 0..1 | boolean | Include when _summary = true?
binding | ?I | 0..1 | Element | ValueSet details if this is coded + ValueSet as a URI SHALL start with http:// or https:// or urn:
strength | ? | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
description | ?I | 0..1 | string | Human explanation of the value set
valueSet[x] | ?I | 0..1 | | Source of value set
valueSetUri | | | uri
valueSetReference | | | Reference(ValueSet)
mapping | ? | 0..* | Element | Map element to another set of definitions
identity | ? | 1..1 | id | Reference to mapping declaration
language | ? | 0..1 | code | Computable language of mapping MimeType (Required)
map | ? | 1..1 | string | Details of the mapping
comment | ? | 0..1 | string | Comments about the mapping or its use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <path value="[string]"/><!-- 1..1 Path of the element in the hierarchy of elements -->
 <representation value="[code]"/><!-- 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml -->
 <sliceName value="[string]"/><!-- 0..1 Name for this particular element (in a set of slices) -->
 <label value="[string]"/><!-- 0..1 Name for element to display with or prompt for element -->
 <code><!-- 0..* Coding Corresponding codes in terminologies --></code>
 <slicing>  <!-- 0..1 This element is sliced - slices follow -->
  <discriminator>  <!-- 0..* Element values that are used to distinguish the slices -->
   <type value="[code]"/><!-- 1..1 value | exists | pattern | type | profile -->
   <path value="[string]"/><!-- 1..1 Path to element value -->
  </discriminator>
  <description value="[string]"/><!--  0..1 Text description of how slicing works (or not) -->
  <ordered value="[boolean]"/><!-- 0..1 If elements must be in same order as slices -->
  <rules value="[code]"/><!-- 1..1 closed | open | openAtEnd -->
 </slicing>
 <short value="[string]"/><!-- 0..1 Concise definition for space-constrained presentation -->
 <definition value="[markdown]"/><!-- 0..1 Full formal definition as narrative text -->
 <comment value="[markdown]"/><!-- 0..1 Comments about the use of this element -->
 <requirements value="[markdown]"/><!-- 0..1 Why this resource has been created -->
 <alias value="[string]"/><!-- 0..* Other names -->
 <min value="[unsignedInt]"/><!--  0..1 Minimum Cardinality -->
 <max value="[string]"/><!--  0..1 Maximum Cardinality (a number or *) -->
 <base>  <!-- 0..1 Base definition information for tools -->
  <path value="[string]"/><!-- 1..1 Path that identifies the base element -->
  <min value="[unsignedInt]"/><!-- 1..1 Min cardinality of the base element -->
  <max value="[string]"/><!-- 1..1 Max cardinality of the base element -->
 </base>
 <contentReference value="[uri]"/><!--  0..1 Reference to definition of content for the element -->
 <type>  <!--  0..* Data type and Profile for this element -->
  <code value="[uri]"/><!-- 1..1 Data type or Resource (reference to definition) -->
  <profile value="[uri]"/><!-- 0..1 Profile (StructureDefinition) to apply (or IG) -->
  <targetProfile value="[uri]"/><!-- 0..1 Profile (StructureDefinition) to apply to reference target (or IG) -->
  <aggregation value="[code]"/><!--  0..* contained | referenced | bundled - how aggregated -->
  <versioning value="[code]"/><!-- 0..1 either | independent | specific -->
 </type>
 <defaultValue[x]><!--  0..1 * Specified value if missing from instance --></defaultValue[x]>
 <meaningWhenMissing value="[markdown]"/><!--  0..1 Implicit meaning when this element is missing -->
 <orderMeaning value="[string]"/><!-- 0..1 What the order of the elements means -->
 <fixed[x]><!--  0..1 * Value must be exactly this --></fixed[x]>
 <pattern[x]><!--  0..1 * Value must have at least these property values --></pattern[x]>
 <example>  <!-- 0..* Example value (as defined for type) -->
  <label value="[string]"/><!-- 1..1 Describes the purpose of this example -->
  <value[x]><!-- 1..1 * Value of Example (one of allowed types) --></value[x]>
 </example>
 <minValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Minimum Allowed Value (for some types) --></minValue[x]>
 <maxValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Maximum Allowed Value (for some types) --></maxValue[x]>
 <maxLength value="[integer]"/><!-- 0..1 Max length for strings -->
 <condition value="[id]"/><!-- 0..* Reference to invariant about presence -->
 <constraint>  <!-- 0..* Condition that must evaluate to true -->
  <key value="[id]"/><!--  1..1 Target of 'condition' reference above -->
  <requirements value="[string]"/><!-- 0..1 Why this constraint is necessary or appropriate -->
  <severity value="[code]"/><!-- 1..1 error | warning -->
  <human value="[string]"/><!-- 1..1 Human description of constraint -->
  <expression value="[string]"/><!-- 1..1 FHIRPath expression of constraint -->
  <xpath value="[string]"/><!-- 0..1 XPath expression of constraint -->
  <source value="[uri]"/><!-- 0..1 Reference to original source of constraint -->
 </constraint>
 <mustSupport value="[boolean]"/><!-- 0..1 If the element must supported -->
 <isModifier value="[boolean]"/><!-- 0..1 If this modifies the meaning of other elements -->
 <isSummary value="[boolean]"/><!-- 0..1 Include when _summary = true? -->
 <binding>  <!--  0..1 ValueSet details if this is coded -->
```

```
  <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
  <description value="[string]"/><!--  0..1 Human explanation of the value set -->
  <valueSet[x]><!--  0..1 uri|Reference(ValueSet) Source of value set --></valueSet[x]>
 </binding>
 <mapping>  <!-- 0..* Map element to another set of definitions -->
  <identity value="[id]"/><!-- 1..1 Reference to mapping declaration -->
  <language value="[code]"/><!-- 0..1 Computable language of mapping  -->
  <map value="[string]"/><!-- 1..1 Details of the mapping -->
  <comment value="[string]"/><!-- 0..1 Comments about the mapping or its use -->
 </mapping>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "path" : "<string>", // R!  Path of the element in the hierarchy of elements
  "representation" : ["<code>"], // xmlAttr | xmlText | typeAttr | cdaText | xhtml
  "sliceName" : "<string>", // Name for this particular element (in a set of slices)
  "label" : "<string>", // Name for element to display with or prompt for element
  "code" : [{ Coding }], // Corresponding codes in terminologies
  "slicing" : { // This element is sliced - slices follow
    "discriminator" : [{ // Element values that are used to distinguish the slices
      "type" : "<code>", // R!  value | exists | pattern | type | profile
      "path" : "<string>" // R!  Path to element value
    }],
    "description" : "<string>", // C? Text description of how slicing works (or not)
    "ordered" : <boolean>, // If elements must be in same order as slices
    "rules" : "<code>" // R!  closed | open | openAtEnd
  },
  "short" : "<string>", // Concise definition for space-constrained presentation
  "definition" : "<markdown>", // Full formal definition as narrative text
  "comment" : "<markdown>", // Comments about the use of this element
  "requirements" : "<markdown>", // Why this resource has been created
  "alias" : ["<string>"], // Other names
  "min" : "<unsignedInt>", // C? Minimum Cardinality
  "max" : "<string>", // C? Maximum Cardinality (a number or *)
  "base" : { // Base definition information for tools
    "path" : "<string>", // R!  Path that identifies the base element
    "min" : "<unsignedInt>", // R!  Min cardinality of the base element
    "max" : "<string>" // R!  Max cardinality of the base element
  },
  "contentReference" : "<uri>", // C? Reference to definition of content for the element
  "type" : [{ // C? Data type and Profile for this element
    "code" : "<uri>", // R!  Data type or Resource (reference to definition)
    "profile" : "<uri>", // Profile (StructureDefinition) to apply (or IG)
    "targetProfile" : "<uri>", // Profile (StructureDefinition) to apply to reference target (or IG)
    "aggregation" : ["<code>"], // C? contained | referenced | bundled - how aggregated
    "versioning" : "<code>" // either | independent | specific
  }],
  // defaultValue[x]: Specified value if missing from instance. One of these 23:
  "defaultValueInteger" : <integer>,
  "defaultValueDecimal" : <decimal>,
  "defaultValueDateTime" : "<dateTime>",
  "defaultValueDate" : "<date>",
  "defaultValueInstant" : "<instant>",
  "defaultValueString" : "<string>",
  "defaultValueUri" : "<uri>",
  "defaultValueBoolean" : <boolean>,
  "defaultValueCode" : "<code>",
  "defaultValueBase64Binary" : "<base64Binary>",
  "defaultValueCoding" : { Coding },
  "defaultValueCodeableConcept" : { CodeableConcept },
  "defaultValueAttachment" : { Attachment },
  "defaultValueIdentifier" : { Identifier },
  "defaultValueQuantity" : { Quantity },
  "defaultValueRange" : { Range },
  "defaultValuePeriod" : { Period },
  "defaultValueRatio" : { Ratio },
  "defaultValueHumanName" : { HumanName },
  "defaultValueAddress" : { Address },
  "defaultValueContactPoint" : { ContactPoint },
  "defaultValueSchedule" : { Schedule },
  "defaultValueReference" : { Reference },
  "meaningWhenMissing" : "<markdown>", // C? Implicit meaning when this element is missing
  "orderMeaning" : "<string>", // What the order of the elements means
  // fixed[x]: Value must be exactly this. One of these 23:
  "fixedInteger" : <integer>,
  "fixedDecimal" : <decimal>,
  "fixedDateTime" : "<dateTime>",
  "fixedDate" : "<date>",
  "fixedInstant" : "<instant>",
  "fixedString" : "<string>",
  "fixedUri" : "<uri>",
  "fixedBoolean" : <boolean>,
  "fixedCode" : "<code>",
  "fixedBase64Binary" : "<base64Binary>",
  "fixedCoding" : { Coding },
  "fixedCodeableConcept" : { CodeableConcept },
  "fixedAttachment" : { Attachment },
```

```
"fixedIdentifier" : { Identifier },
"fixedQuantity" : { Quantity },
"fixedRange" : { Range },
"fixedPeriod" : { Period },
"fixedRatio" : { Ratio },
"fixedHumanName" : { HumanName },
"fixedAddress" : { Address },
"fixedContactPoint" : { ContactPoint },
"fixedSchedule" : { Schedule },
"fixedReference" : { Reference },
// pattern[x]: Value must have at least these property values. One of these 23:
"patternInteger" : <integer>,
"patternDecimal" : <decimal>,
"patternDateTime" : "<dateTime>",
"patternDate" : "<date>",
"patternInstant" : "<instant>",
"patternString" : "<string>",
"patternUri" : "<uri>",
"patternBoolean" : <boolean>,
"patternCode" : "<code>",
"patternBase64Binary" : "<base64Binary>",
"patternCoding" : { Coding },
"patternCodeableConcept" : { CodeableConcept },
"patternAttachment" : { Attachment },
"patternIdentifier" : { Identifier },
"patternQuantity" : { Quantity },
"patternRange" : { Range },
"patternPeriod" : { Period },
"patternRatio" : { Ratio },
"patternHumanName" : { HumanName },
"patternAddress" : { Address },
"patternContactPoint" : { ContactPoint },
"patternSchedule" : { Schedule },
"patternReference" : { Reference },
"example" : [{ // Example value (as defined for type)
  "label" : "<string>", // R!  Describes the purpose of this example
  // value[x]: Value of Example (one of allowed types). One of these 23:
  "valueInteger" : <integer>
  "valueDecimal" : <decimal>
  "valueDateTime" : "<dateTime>"
  "valueDate" : "<date>"
  "valueInstant" : "<instant>"
  "valueString" : "<string>"
  "valueUri" : "<uri>"
  "valueBoolean" : <boolean>
  "valueCode" : "<code>"
  "valueBase64Binary" : "<base64Binary>"
  "valueCoding" : { Coding }
  "valueCodeableConcept" : { CodeableConcept }
  "valueAttachment" : { Attachment }
  "valueIdentifier" : { Identifier }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valuePeriod" : { Period }
  "valueRatio" : { Ratio }
  "valueHumanName" : { HumanName }
  "valueAddress" : { Address }
  "valueContactPoint" : { ContactPoint }
  "valueSchedule" : { Schedule }
  "valueReference" : { Reference }
}],
// minValue[x]: Minimum Allowed Value (for some types). One of these 9:
"minValueDate" : "<date>",
"minValueDateTime" : "<dateTime>",
"minValueInstant" : "<instant>",
"minValueTime" : "<time>",
"minValueDecimal" : <decimal>,
"minValueInteger" : <integer>,
"minValuePositiveInt" : "<positiveInt>",
"minValueUnsignedInt" : "<unsignedInt>",
"minValueQuantity" : { Quantity },
// maxValue[x]: Maximum Allowed Value (for some types). One of these 9:
"maxValueDate" : "<date>",
"maxValueDateTime" : "<dateTime>",
"maxValueInstant" : "<instant>",
"maxValueTime" : "<time>",
"maxValueDecimal" : <decimal>,
"maxValueInteger" : <integer>,
"maxValuePositiveInt" : "<positiveInt>",
"maxValueUnsignedInt" : "<unsignedInt>",
"maxValueQuantity" : { Quantity },
"maxLength" : <integer>, // Max length for strings
"condition" : ["<id>"], // Reference to invariant about presence
"constraint" : [{ // Condition that must evaluate to true
  "key" : "<id>", // C? R!  Target of 'condition' reference above
  "requirements" : "<string>", // Why this constraint is necessary or appropriate
  "severity" : "<code>", // R!  error | warning
  "human" : "<string>", // R!  Human description of constraint
  "expression" : "<string>", // R!  FHIRPath expression of constraint
```

```
    "xpath" : "<string>", // XPath expression of constraint
    "source" : "<uri>" // Reference to original source of constraint
  }],
  "mustSupport" : <boolean>, // If the element must supported
  "isModifier" : <boolean>, // If this modifies the meaning of other elements
  "isSummary" : <boolean>, // Include when _summary = true?
  "binding" : { // C? ValueSet details if this is coded
    "strength" : "<code>", // R!  required | extensible | preferred | example
    "description" : "<string>", // Human explanation of the value set
    // valueSet[x]: Source of value set. One of these 2:
    "valueSetUri" : "<uri>"
    "valueSetReference" : { Reference(ValueSet) }
  },
  "mapping" : [{ // Map element to another set of definitions
    "identity" : "<id>", // R!  Reference to mapping declaration
    "language" : "<code>", // Computable language of mapping
    "map" : "<string>", // R!  Details of the mapping
    "comment" : "<string>" // Comments about the mapping or its use
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ElementDefinition.path [ string ]; # 1..1 Path of the element in the hierarchy of elements
  fhir:ElementDefinition.representation [ code ], ... ; # 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml
  fhir:ElementDefinition.sliceName [ string ]; # 0..1 Name for this particular element (in a set of slices)
  fhir:ElementDefinition.label [ string ]; # 0..1 Name for element to display with or prompt for element
  fhir:ElementDefinition.code [ Coding ], ... ; # 0..* Corresponding codes in terminologies
  fhir:ElementDefinition.slicing [ # 0..1 This element is sliced - slices follow
    fhir:ElementDefinition.slicing.discriminator [ # 0..* Element values that are used to distinguish the slices
      fhir:ElementDefinition.slicing.discriminator.type [ code ]; # 1..1 value | exists | pattern | type | profile
      fhir:ElementDefinition.slicing.discriminator.path [ string ]; # 1..1 Path to element value
    ], ...;
    fhir:ElementDefinition.slicing.description [ string ]; # 0..1 Text description of how slicing works (or not)
    fhir:ElementDefinition.slicing.ordered [ boolean ]; # 0..1 If elements must be in same order as slices
    fhir:ElementDefinition.slicing.rules [ code ]; # 1..1 closed | open | openAtEnd
  ];
  fhir:ElementDefinition.short [ string ]; # 0..1 Concise definition for space-constrained presentation
  fhir:ElementDefinition.definition [ markdown ]; # 0..1 Full formal definition as narrative text
  fhir:ElementDefinition.comment [ markdown ]; # 0..1 Comments about the use of this element
  fhir:ElementDefinition.requirements [ markdown ]; # 0..1 Why this resource has been created
  fhir:ElementDefinition.alias [ string ], ... ; # 0..* Other names
  fhir:ElementDefinition.min [ unsignedInt ]; # 0..1 Minimum Cardinality
  fhir:ElementDefinition.max [ string ]; # 0..1 Maximum Cardinality (a number or *)
  fhir:ElementDefinition.base [ # 0..1 Base definition information for tools
    fhir:ElementDefinition.base.path [ string ]; # 1..1 Path that identifies the base element
    fhir:ElementDefinition.base.min [ unsignedInt ]; # 1..1 Min cardinality of the base element
    fhir:ElementDefinition.base.max [ string ]; # 1..1 Max cardinality of the base element
  ];
  fhir:ElementDefinition.contentReference [ uri ]; # 0..1 Reference to definition of content for the element
  fhir:ElementDefinition.type [ # 0..* Data type and Profile for this element
    fhir:ElementDefinition.type.code [ uri ]; # 1..1 Data type or Resource (reference to definition)
    fhir:ElementDefinition.type.profile [ uri ]; # 0..1 Profile (StructureDefinition) to apply (or IG)
    fhir:ElementDefinition.type.targetProfile [ uri ]; # 0..1 Profile (StructureDefinition) to apply to reference target
     (or IG)
    fhir:ElementDefinition.type.aggregation [ code ], ... ; # 0..* contained | referenced | bundled - how aggregated
    fhir:ElementDefinition.type.versioning [ code ]; # 0..1 either | independent | specific
  ], ...;
  # ElementDefinition.defaultValue[x] : 0..1 Specified value if missing from instance. One of these 38
    fhir:ElementDefinition.defaultValueBase64Binary [ base64Binary ]
    fhir:ElementDefinition.defaultValueBoolean [ boolean ]
    fhir:ElementDefinition.defaultValueCode [ code ]
    fhir:ElementDefinition.defaultValueDate [ date ]
    fhir:ElementDefinition.defaultValueDateTime [ dateTime ]
    fhir:ElementDefinition.defaultValueDecimal [ decimal ]
    fhir:ElementDefinition.defaultValueId [ id ]
    fhir:ElementDefinition.defaultValueInstant [ instant ]
    fhir:ElementDefinition.defaultValueInteger [ integer ]
    fhir:ElementDefinition.defaultValueMarkdown [ markdown ]
    fhir:ElementDefinition.defaultValueOid [ oid ]
    fhir:ElementDefinition.defaultValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.defaultValueString [ string ]
    fhir:ElementDefinition.defaultValueTime [ time ]
    fhir:ElementDefinition.defaultValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.defaultValueUri [ uri ]
    fhir:ElementDefinition.defaultValueAddress [ Address ]
    fhir:ElementDefinition.defaultValueAge [ Age ]
    fhir:ElementDefinition.defaultValueAnnotation [ Annotation ]
    fhir:ElementDefinition.defaultValueAttachment [ Attachment ]
    fhir:ElementDefinition.defaultValueCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.defaultValueCoding [ Coding ]
    fhir:ElementDefinition.defaultValueContactPoint [ ContactPoint ]
    fhir:ElementDefinition.defaultValueCount [ Count ]
    fhir:ElementDefinition.defaultValueDistance [ Distance ]
    fhir:ElementDefinition.defaultValueDuration [ Duration ]
    fhir:ElementDefinition.defaultValueHumanName [ HumanName ]
```

```
    fhir:ElementDefinition.defaultValueIdentifier [ Identifier ]
    fhir:ElementDefinition.defaultValueMoney [ Money ]
    fhir:ElementDefinition.defaultValuePeriod [ Period ]
    fhir:ElementDefinition.defaultValueQuantity [ Quantity ]
    fhir:ElementDefinition.defaultValueRange [ Range ]
    fhir:ElementDefinition.defaultValueRatio [ Ratio ]
    fhir:ElementDefinition.defaultValueReference [ Reference ]
    fhir:ElementDefinition.defaultValueSampledData [ SampledData ]
    fhir:ElementDefinition.defaultValueSignature [ Signature ]
    fhir:ElementDefinition.defaultValueTiming [ Timing ]
    fhir:ElementDefinition.defaultValueMeta [ Meta ]
  fhir:ElementDefinition.meaningWhenMissing [ markdown ]; # 0..1 Implicit meaning when this element is missing
  fhir:ElementDefinition.orderMeaning [ string ]; # 0..1 What the order of the elements means
  # ElementDefinition.fixed[x] : 0..1 Value must be exactly this. One of these 38
    fhir:ElementDefinition.fixedBase64Binary [ base64Binary ]
    fhir:ElementDefinition.fixedBoolean [ boolean ]
    fhir:ElementDefinition.fixedCode [ code ]
    fhir:ElementDefinition.fixedDate [ date ]
    fhir:ElementDefinition.fixedDateTime [ dateTime ]
    fhir:ElementDefinition.fixedDecimal [ decimal ]
    fhir:ElementDefinition.fixedId [ id ]
    fhir:ElementDefinition.fixedInstant [ instant ]
    fhir:ElementDefinition.fixedInteger [ integer ]
    fhir:ElementDefinition.fixedMarkdown [ markdown ]
    fhir:ElementDefinition.fixedOid [ oid ]
    fhir:ElementDefinition.fixedPositiveInt [ positiveInt ]
    fhir:ElementDefinition.fixedString [ string ]
    fhir:ElementDefinition.fixedTime [ time ]
    fhir:ElementDefinition.fixedUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.fixedUri [ uri ]
    fhir:ElementDefinition.fixedAddress [ Address ]
    fhir:ElementDefinition.fixedAge [ Age ]
    fhir:ElementDefinition.fixedAnnotation [ Annotation ]
    fhir:ElementDefinition.fixedAttachment [ Attachment ]
    fhir:ElementDefinition.fixedCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.fixedCoding [ Coding ]
    fhir:ElementDefinition.fixedContactPoint [ ContactPoint ]
    fhir:ElementDefinition.fixedCount [ Count ]
    fhir:ElementDefinition.fixedDistance [ Distance ]
    fhir:ElementDefinition.fixedDuration [ Duration ]
    fhir:ElementDefinition.fixedHumanName [ HumanName ]
    fhir:ElementDefinition.fixedIdentifier [ Identifier ]
    fhir:ElementDefinition.fixedMoney [ Money ]
    fhir:ElementDefinition.fixedPeriod [ Period ]
    fhir:ElementDefinition.fixedQuantity [ Quantity ]
    fhir:ElementDefinition.fixedRange [ Range ]
    fhir:ElementDefinition.fixedRatio [ Ratio ]
    fhir:ElementDefinition.fixedReference [ Reference ]
    fhir:ElementDefinition.fixedSampledData [ SampledData ]
    fhir:ElementDefinition.fixedSignature [ Signature ]
    fhir:ElementDefinition.fixedTiming [ Timing ]
    fhir:ElementDefinition.fixedMeta [ Meta ]
  # ElementDefinition.pattern[x] : 0..1 Value must have at least these property values. One of these 38
    fhir:ElementDefinition.patternBase64Binary [ base64Binary ]
    fhir:ElementDefinition.patternBoolean [ boolean ]
    fhir:ElementDefinition.patternCode [ code ]
    fhir:ElementDefinition.patternDate [ date ]
    fhir:ElementDefinition.patternDateTime [ dateTime ]
    fhir:ElementDefinition.patternDecimal [ decimal ]
    fhir:ElementDefinition.patternId [ id ]
    fhir:ElementDefinition.patternInstant [ instant ]
    fhir:ElementDefinition.patternInteger [ integer ]
    fhir:ElementDefinition.patternMarkdown [ markdown ]
    fhir:ElementDefinition.patternOid [ oid ]
    fhir:ElementDefinition.patternPositiveInt [ positiveInt ]
    fhir:ElementDefinition.patternString [ string ]
    fhir:ElementDefinition.patternTime [ time ]
    fhir:ElementDefinition.patternUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.patternUri [ uri ]
    fhir:ElementDefinition.patternAddress [ Address ]
    fhir:ElementDefinition.patternAge [ Age ]
    fhir:ElementDefinition.patternAnnotation [ Annotation ]
    fhir:ElementDefinition.patternAttachment [ Attachment ]
    fhir:ElementDefinition.patternCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.patternCoding [ Coding ]
    fhir:ElementDefinition.patternContactPoint [ ContactPoint ]
    fhir:ElementDefinition.patternCount [ Count ]
    fhir:ElementDefinition.patternDistance [ Distance ]
    fhir:ElementDefinition.patternDuration [ Duration ]
    fhir:ElementDefinition.patternHumanName [ HumanName ]
    fhir:ElementDefinition.patternIdentifier [ Identifier ]
    fhir:ElementDefinition.patternMoney [ Money ]
    fhir:ElementDefinition.patternPeriod [ Period ]
    fhir:ElementDefinition.patternQuantity [ Quantity ]
    fhir:ElementDefinition.patternRange [ Range ]
    fhir:ElementDefinition.patternRatio [ Ratio ]
    fhir:ElementDefinition.patternReference [ Reference ]
    fhir:ElementDefinition.patternSampledData [ SampledData ]
    fhir:ElementDefinition.patternSignature [ Signature ]
```

```
  fhir:ElementDefinition.patternTiming [ Timing ]
  fhir:ElementDefinition.patternMeta [ Meta ]
fhir:ElementDefinition.example [ # 0..* Example value (as defined for type)
  fhir:ElementDefinition.example.label [ string ]; # 1..1 Describes the purpose of this example
  # ElementDefinition.example.value[x] : 1..1 Value of Example (one of allowed types). One of these 38
    fhir:ElementDefinition.example.valueBase64Binary [ base64Binary ]
    fhir:ElementDefinition.example.valueBoolean [ boolean ]
    fhir:ElementDefinition.example.valueCode [ code ]
    fhir:ElementDefinition.example.valueDate [ date ]
    fhir:ElementDefinition.example.valueDateTime [ dateTime ]
    fhir:ElementDefinition.example.valueDecimal [ decimal ]
    fhir:ElementDefinition.example.valueId [ id ]
    fhir:ElementDefinition.example.valueInstant [ instant ]
    fhir:ElementDefinition.example.valueInteger [ integer ]
    fhir:ElementDefinition.example.valueMarkdown [ markdown ]
    fhir:ElementDefinition.example.valueOid [ oid ]
    fhir:ElementDefinition.example.valuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.example.valueString [ string ]
    fhir:ElementDefinition.example.valueTime [ time ]
    fhir:ElementDefinition.example.valueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.example.valueUri [ uri ]
    fhir:ElementDefinition.example.valueAddress [ Address ]
    fhir:ElementDefinition.example.valueAge [ Age ]
    fhir:ElementDefinition.example.valueAnnotation [ Annotation ]
    fhir:ElementDefinition.example.valueAttachment [ Attachment ]
    fhir:ElementDefinition.example.valueCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.example.valueCoding [ Coding ]
    fhir:ElementDefinition.example.valueContactPoint [ ContactPoint ]
    fhir:ElementDefinition.example.valueCount [ Count ]
    fhir:ElementDefinition.example.valueDistance [ Distance ]
    fhir:ElementDefinition.example.valueDuration [ Duration ]
    fhir:ElementDefinition.example.valueHumanName [ HumanName ]
    fhir:ElementDefinition.example.valueIdentifier [ Identifier ]
    fhir:ElementDefinition.example.valueMoney [ Money ]
    fhir:ElementDefinition.example.valuePeriod [ Period ]
    fhir:ElementDefinition.example.valueQuantity [ Quantity ]
    fhir:ElementDefinition.example.valueRange [ Range ]
    fhir:ElementDefinition.example.valueRatio [ Ratio ]
    fhir:ElementDefinition.example.valueReference [ Reference ]
    fhir:ElementDefinition.example.valueSampledData [ SampledData ]
    fhir:ElementDefinition.example.valueSignature [ Signature ]
    fhir:ElementDefinition.example.valueTiming [ Timing ]
    fhir:ElementDefinition.example.valueMeta [ Meta ]
], ...;
# ElementDefinition.minValue[x] : 0..1 Minimum Allowed Value (for some types). One of these 9
  fhir:ElementDefinition.minValueDate [ date ]
  fhir:ElementDefinition.minValueDateTime [ dateTime ]
  fhir:ElementDefinition.minValueInstant [ instant ]
  fhir:ElementDefinition.minValueTime [ time ]
  fhir:ElementDefinition.minValueDecimal [ decimal ]
  fhir:ElementDefinition.minValueInteger [ integer ]
  fhir:ElementDefinition.minValuePositiveInt [ positiveInt ]
  fhir:ElementDefinition.minValueUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.minValueQuantity [ Quantity ]
# ElementDefinition.maxValue[x] : 0..1 Maximum Allowed Value (for some types). One of these 9
  fhir:ElementDefinition.maxValueDate [ date ]
  fhir:ElementDefinition.maxValueDateTime [ dateTime ]
  fhir:ElementDefinition.maxValueInstant [ instant ]
  fhir:ElementDefinition.maxValueTime [ time ]
  fhir:ElementDefinition.maxValueDecimal [ decimal ]
  fhir:ElementDefinition.maxValueInteger [ integer ]
  fhir:ElementDefinition.maxValuePositiveInt [ positiveInt ]
  fhir:ElementDefinition.maxValueUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.maxValueQuantity [ Quantity ]
fhir:ElementDefinition.maxLength [ integer ]; # 0..1 Max length for strings
fhir:ElementDefinition.condition [ id ], ... ; # 0..* Reference to invariant about presence
fhir:ElementDefinition.constraint [ # 0..* Condition that must evaluate to true
  fhir:ElementDefinition.constraint.key [ id ]; # 1..1 Target of 'condition' reference above
  fhir:ElementDefinition.constraint.requirements [ string ]; # 0..1 Why this constraint is necessary or appropriate
  fhir:ElementDefinition.constraint.severity [ code ]; # 1..1 error | warning
  fhir:ElementDefinition.constraint.human [ string ]; # 1..1 Human description of constraint
  fhir:ElementDefinition.constraint.expression [ string ]; # 1..1 FHIRPath expression of constraint
  fhir:ElementDefinition.constraint.xpath [ string ]; # 0..1 XPath expression of constraint
  fhir:ElementDefinition.constraint.source [ uri ]; # 0..1 Reference to original source of constraint
], ...;
fhir:ElementDefinition.mustSupport [ boolean ]; # 0..1 If the element must supported
fhir:ElementDefinition.isModifier [ boolean ]; # 0..1 If this modifies the meaning of other elements
fhir:ElementDefinition.isSummary [ boolean ]; # 0..1 Include when _summary = true?
fhir:ElementDefinition.binding [ # 0..1 ValueSet details if this is coded
  fhir:ElementDefinition.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
  fhir:ElementDefinition.binding.description [ string ]; # 0..1 Human explanation of the value set
  # ElementDefinition.binding.valueSet[x] : 0..1 Source of value set. One of these 2
    fhir:ElementDefinition.binding.valueSetUri [ uri ]
    fhir:ElementDefinition.binding.valueSetReference [ Reference(ValueSet) ]
];
fhir:ElementDefinition.mapping [ # 0..* Map element to another set of definitions
  fhir:ElementDefinition.mapping.identity [ id ]; # 1..1 Reference to mapping declaration
  fhir:ElementDefinition.mapping.language [ code ]; # 0..1 Computable language of mapping
  fhir:ElementDefinition.mapping.map [ string ]; # 1..1 Details of the mapping
```

```
    fhir:ElementDefinition.mapping.comment [ string ]; # 0..1 Comments about the mapping or its use
  ], ...;
]
```

Changes since DSTU2

ElementDefinition
ElementDefinition.representation | Add Codes xmlText, typeAttr, cdaText, xhtml
ElementDefinition.sliceName | Renamed from name to sliceName
ElementDefinition.slicing.discriminator | Type changed from string to Element
ElementDefinition.slicing.discriminator.type | Added Element
ElementDefinition.slicing.discriminator.path | Added Element
ElementDefinition.comment | Renamed from comments to comment
ElementDefinition.min | Type changed from integer to unsignedInt
ElementDefinition.base.min | Type changed from integer to unsignedInt
ElementDefinition.contentReference | Renamed from nameReference to contentReference Type changed from string to uri
ElementDefinition.type.code | Type changed from code to uri Change binding strength from required to extensible
ElementDefinition.type.profile | Max Cardinality changed from * to 1
ElementDefinition.type.targetProfile | Added Element
ElementDefinition.type.versioning | Added Element
ElementDefinition.defaultValue[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.orderMeaning | Added Element
ElementDefinition.fixed[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.pattern[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money
ElementDefinition.example | Renamed from example[x] to example Max Cardinality changed from 1 to * Remove boolean, Remove integer, Remove decimal, Remove base64Binary, Remove instant, Remove string, Remove uri, Remove date, Remove dateTime, Remove time, Remove code, Remove oid, Remove id, Remove unsignedInt, Remove positiveInt, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Quantity, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.example.label | Added Element
ElementDefinition.example.value[x] | Added Element
ElementDefinition.minValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.maxValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.constraint.expression | Added Element
ElementDefinition.constraint.xpath | Min Cardinality changed from 1 to 0
ElementDefinition.constraint.source | Added Element
ElementDefinition.mapping.comment | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
ElementDefinition | ?I | | Element | Definition of an element in a resource or extension + Min <= Max + if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding + Pattern may only be specified if there is one type + Fixed value may only be specified if there is one type + Binding can only be present for coded elements, string, and uri + Pattern and value are mutually exclusive + Constraints must be unique by key + Types must be unique by the combination of code and profile + sliceName must be composed of proper tokens separated by "/" + default value and meaningWhenMissing are mutually exclusive Elements defined in Ancestors: id, extension
path | ? | 1..1 | string | Path of the element in the hierarchy of elements
representation | ? | 0..* | code | xmlAttr | xmlText | typeAttr | cdaText | xhtml PropertyRepresentation (Required)
sliceName | ? | 0..1 | string | Name for this particular element (in a set of slices)
label | ? | 0..1 | string | Name for element to display with or prompt for element
code | ? | 0..* | Coding | Corresponding codes in terminologies LOINC Codes (Example)
slicing | ?I | 0..1 | Element | This element is sliced - slices follow + If there are no discriminators, there must be a definition
discriminator | ? | 0..* | Element | Element values that are used to distinguish the slices
type | ? | 1..1 | code | value | exists | pattern | type | profile DiscriminatorType (Required)
path | ? | 1..1 | string | Path to element value
description | ?I | 0..1 | string | Text description of how slicing works (or not)
ordered | ? | 0..1 | boolean | If elements must be in same order as slices
rules | ? | 1..1 | code | closed | open | openAtEnd SlicingRules (Required)
short | ? | 0..1 | string | Concise definition for space-constrained presentation
definition | ? | 0..1 | markdown | Full formal definition as narrative text
comment | ? | 0..1 | markdown | Comments about the use of this element
requirements | ? | 0..1 | markdown | Why this resource has been created
alias | ? | 0..* | string | Other names
min | ?I | 0..1 | unsignedInt | Minimum Cardinality
max | ?I | 0..1 | string | Maximum Cardinality (a number or *) + Max SHALL be a number or "*"
base | ? | 0..1 | Element | Base definition information for tools
path | ? | 1..1 | string | Path that identifies the base element
min | ? | 1..1 | unsignedInt | Min cardinality of the base element

max | ? | 1..1 | string | Max cardinality of the base element
contentReference | ?I | 0..1 | uri | Reference to definition of content for the element
type | ?I | 0..* | Element | Data type and Profile for this element + Aggregation may only be specified if one of the allowed types for the element is a resource
code | ? | 1..1 | uri | Data type or Resource (reference to definition) FHIRDefinedType (Extensible)
profile | ? | 0..1 | uri | Profile (StructureDefinition) to apply (or IG)
targetProfile | ? | 0..1 | uri | Profile (StructureDefinition) to apply to reference target (or IG)
aggregation | ?I | 0..* | code | contained | referenced | bundled - how aggregated AggregationMode (Required)
versioning | ? | 0..1 | code | either | independent | specific ReferenceVersionRules (Required)
defaultValue[x] | ?I | 0..1 | * | Specified value if missing from instance
meaningWhenMissing | ?I | 0..1 | markdown | Implicit meaning when this element is missing
orderMeaning | ? | 0..1 | string | What the order of the elements means
fixed[x] | ?I | 0..1 | * | Value must be exactly this
pattern[x] | ?I | 0..1 | * | Value must have at least these property values
example | ? | 0..* | Element | Example value (as defined for type)
label | ? | 1..1 | string | Describes the purpose of this example
value[x] | ? | 1..1 | * | Value of Example (one of allowed types)
minValue[x] | ? | 0..1 | | Minimum Allowed Value (for some types)
minValueDate | | | date
minValueDateTime | | | dateTime
minValueInstant | | | instant
minValueTime | | | time
minValueDecimal | | | decimal
minValueInteger | | | integer
minValuePositiveInt | | | positiveInt
minValueUnsignedInt | | | unsignedInt
minValueQuantity | | | Quantity
maxValue[x] | ? | 0..1 | | Maximum Allowed Value (for some types)
maxValueDate | | | date
maxValueDateTime | | | dateTime
maxValueInstant | | | instant
maxValueTime | | | time
maxValueDecimal | | | decimal
maxValueInteger | | | integer
maxValuePositiveInt | | | positiveInt
maxValueUnsignedInt | | | unsignedInt
maxValueQuantity | | | Quantity
maxLength | ? | 0..1 | integer | Max length for strings
condition | ? | 0..* | id | Reference to invariant about presence
constraint | ? | 0..* | Element | Condition that must evaluate to true
key | ?I | 1..1 | id | Target of 'condition' reference above
requirements | ? | 0..1 | string | Why this constraint is necessary or appropriate
severity | ? | 1..1 | code | error | warning ConstraintSeverity (Required)
human | ? | 1..1 | string | Human description of constraint
expression | ? | 1..1 | string | FHIRPath expression of constraint
xpath | ? | 0..1 | string | XPath expression of constraint
source | ? | 0..1 | uri | Reference to original source of constraint
mustSupport | ? | 0..1 | boolean | If the element must supported
isModifier | ? | 0..1 | boolean | If this modifies the meaning of other elements
isSummary | ? | 0..1 | boolean | Include when _summary = true?
binding | ?I | 0..1 | Element | ValueSet details if this is coded + ValueSet as a URI SHALL start with http:// or https:// or urn:
strength | ? | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
description | ?I | 0..1 | string | Human explanation of the value set
valueSet[x] | ?I | 0..1 | | Source of value set
valueSetUri | | | uri
valueSetReference | | | Reference(ValueSet)
mapping | ? | 0..* | Element | Map element to another set of definitions
identity | ? | 1..1 | id | Reference to mapping declaration
language | ? | 0..1 | code | Computable language of mapping MimeType (Required)
map | ? | 1..1 | string | Details of the mapping
comment | ? | 0..1 | string | Comments about the mapping or its use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <path value="[string]"/><!-- 1..1 Path of the element in the hierarchy of elements -->
 <representation value="[code]"/><!-- 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml -->
 <sliceName value="[string]"/><!-- 0..1 Name for this particular element (in a set of slices) -->
 <label value="[string]"/><!-- 0..1 Name for element to display with or prompt for element -->
 <code><!-- 0..* Coding Corresponding codes in terminologies --></code>
 <slicing>  <!-- 0..1 This element is sliced - slices follow -->
  <discriminator>  <!-- 0..* Element values that are used to distinguish the slices -->
```

```
   <type value="[code]"/><!-- 1..1 value | exists | pattern | type | profile -->
   <path value="[string]"/><!-- 1..1 Path to element value -->
  </discriminator>
  <description value="[string]"/><!--  0..1 Text description of how slicing works (or not) -->
  <ordered value="[boolean]"/><!-- 0..1 If elements must be in same order as slices -->
  <rules value="[code]"/><!-- 1..1 closed | open | openAtEnd -->
 </slicing>
 <short value="[string]"/><!-- 0..1 Concise definition for space-constrained presentation -->
 <definition value="[markdown]"/><!-- 0..1 Full formal definition as narrative text -->
 <comment value="[markdown]"/><!-- 0..1 Comments about the use of this element -->
 <requirements value="[markdown]"/><!-- 0..1 Why this resource has been created -->
 <alias value="[string]"/><!-- 0..* Other names -->
 <min value="[unsignedInt]"/><!--  0..1 Minimum Cardinality -->
 <max value="[string]"/><!--  0..1 Maximum Cardinality (a number or *) -->
 <base>  <!-- 0..1 Base definition information for tools -->
  <path value="[string]"/><!-- 1..1 Path that identifies the base element -->
  <min value="[unsignedInt]"/><!-- 1..1 Min cardinality of the base element -->
  <max value="[string]"/><!-- 1..1 Max cardinality of the base element -->
 </base>
 <contentReference value="[uri]"/><!--  0..1 Reference to definition of content for the element -->
 <type>  <!--  0..* Data type and Profile for this element -->
  <code value="[uri]"/><!-- 1..1 Data type or Resource (reference to definition) -->
  <profile value="[uri]"/><!-- 0..1 Profile (StructureDefinition) to apply (or IG) -->
  <targetProfile value="[uri]"/><!-- 0..1 Profile (StructureDefinition) to apply to reference target (or IG) -->
  <aggregation value="[code]"/><!--  0..* contained | referenced | bundled - how aggregated -->
  <versioning value="[code]"/><!-- 0..1 either | independent | specific -->
 </type>
 <defaultValue[x]><!--  0..1 * Specified value if missing from instance --></defaultValue[x]>
 <meaningWhenMissing value="[markdown]"/><!--  0..1 Implicit meaning when this element is missing -->
 <orderMeaning value="[string]"/><!-- 0..1 What the order of the elements means -->
 <fixed[x]><!--  0..1 * Value must be exactly this --></fixed[x]>
 <pattern[x]><!--  0..1 * Value must have at least these property values --></pattern[x]>
 <example>  <!-- 0..* Example value (as defined for type) -->
  <label value="[string]"/><!-- 1..1 Describes the purpose of this example -->
  <value[x]><!-- 1..1 * Value of Example (one of allowed types) --></value[x]>
 </example>
 <minValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Minimum Allowed Value (for some types) --></minValue[x]>
 <maxValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Maximum Allowed Value (for some types) --></maxValue[x]>
 <maxLength value="[integer]"/><!-- 0..1 Max length for strings -->
 <condition value="[id]"/><!-- 0..* Reference to invariant about presence -->
 <constraint>  <!-- 0..* Condition that must evaluate to true -->
  <key value="[id]"/><!--  1..1 Target of 'condition' reference above -->
  <requirements value="[string]"/><!-- 0..1 Why this constraint is necessary or appropriate -->
  <severity value="[code]"/><!-- 1..1 error | warning -->
  <human value="[string]"/><!-- 1..1 Human description of constraint -->
  <expression value="[string]"/><!-- 1..1 FHIRPath expression of constraint -->
  <xpath value="[string]"/><!-- 0..1 XPath expression of constraint -->
  <source value="[uri]"/><!-- 0..1 Reference to original source of constraint -->
 </constraint>
 <mustSupport value="[boolean]"/><!-- 0..1 If the element must supported -->
 <isModifier value="[boolean]"/><!-- 0..1 If this modifies the meaning of other elements -->
 <isSummary value="[boolean]"/><!-- 0..1 Include when _summary = true? -->
 <binding>  <!--  0..1 ValueSet details if this is coded -->
  <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
  <description value="[string]"/><!--  0..1 Human explanation of the value set -->
  <valueSet[x]><!--  0..1 uri|Reference(ValueSet) Source of value set --></valueSet[x]>
 </binding>
 <mapping>  <!-- 0..* Map element to another set of definitions -->
  <identity value="[id]"/><!-- 1..1 Reference to mapping declaration -->
  <language value="[code]"/><!-- 0..1 Computable language of mapping  -->
  <map value="[string]"/><!-- 1..1 Details of the mapping -->
  <comment value="[string]"/><!-- 0..1 Comments about the mapping or its use -->
 </mapping>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "path" : "<string>", // R!  Path of the element in the hierarchy of elements
  "representation" : ["<code>"], // xmlAttr | xmlText | typeAttr | cdaText | xhtml
  "sliceName" : "<string>", // Name for this particular element (in a set of slices)
  "label" : "<string>", // Name for element to display with or prompt for element
  "code" : [{ Coding }], // Corresponding codes in terminologies
  "slicing" : { // This element is sliced - slices follow
    "discriminator" : [{ // Element values that are used to distinguish the slices
      "type" : "<code>", // R!  value | exists | pattern | type | profile
      "path" : "<string>" // R!  Path to element value
    }],
    "description" : "<string>", // C? Text description of how slicing works (or not)
    "ordered" : <boolean>, // If elements must be in same order as slices
    "rules" : "<code>" // R!  closed | open | openAtEnd
  },
  "short" : "<string>", // Concise definition for space-constrained presentation
  "definition" : "<markdown>", // Full formal definition as narrative text
  "comment" : "<markdown>", // Comments about the use of this element
  "requirements" : "<markdown>", // Why this resource has been created
  "alias" : ["<string>"], // Other names
```

```
"min" : "<unsignedInt>", // C? Minimum Cardinality
"max" : "<string>", // C? Maximum Cardinality (a number or *)
"base" : { // Base definition information for tools
  "path" : "<string>", // R!  Path that identifies the base element
  "min" : "<unsignedInt>", // R!  Min cardinality of the base element
  "max" : "<string>" // R!  Max cardinality of the base element
},
"contentReference" : "<uri>", // C? Reference to definition of content for the element
"type" : [{ // C? Data type and Profile for this element
  "code" : "<uri>", // R!  Data type or Resource (reference to definition)
  "profile" : "<uri>", // Profile (StructureDefinition) to apply (or IG)
  "targetProfile" : "<uri>", // Profile (StructureDefinition) to apply to reference target (or IG)
  "aggregation" : ["<code>"], // C? contained | referenced | bundled - how aggregated
  "versioning" : "<code>" // either | independent | specific
}],
// defaultValue[x]: Specified value if missing from instance. One of these 23:
"defaultValueInteger" : <integer>,
"defaultValueDecimal" : <decimal>,
"defaultValueDateTime" : "<dateTime>",
"defaultValueDate" : "<date>",
"defaultValueInstant" : "<instant>",
"defaultValueString" : "<string>",
"defaultValueUri" : "<uri>",
"defaultValueBoolean" : <boolean>,
"defaultValueCode" : "<code>",
"defaultValueBase64Binary" : "<base64Binary>",
"defaultValueCoding" : { Coding },
"defaultValueCodeableConcept" : { CodeableConcept },
"defaultValueAttachment" : { Attachment },
"defaultValueIdentifier" : { Identifier },
"defaultValueQuantity" : { Quantity },
"defaultValueRange" : { Range },
"defaultValuePeriod" : { Period },
"defaultValueRatio" : { Ratio },
"defaultValueHumanName" : { HumanName },
"defaultValueAddress" : { Address },
"defaultValueContactPoint" : { ContactPoint },
"defaultValueSchedule" : { Schedule },
"defaultValueReference" : { Reference },
"meaningWhenMissing" : "<markdown>", // C? Implicit meaning when this element is missing
"orderMeaning" : "<string>", // What the order of the elements means
// fixed[x]: Value must be exactly this. One of these 23:
"fixedInteger" : <integer>,
"fixedDecimal" : <decimal>,
"fixedDateTime" : "<dateTime>",
"fixedDate" : "<date>",
"fixedInstant" : "<instant>",
"fixedString" : "<string>",
"fixedUri" : "<uri>",
"fixedBoolean" : <boolean>,
"fixedCode" : "<code>",
"fixedBase64Binary" : "<base64Binary>",
"fixedCoding" : { Coding },
"fixedCodeableConcept" : { CodeableConcept },
"fixedAttachment" : { Attachment },
"fixedIdentifier" : { Identifier },
"fixedQuantity" : { Quantity },
"fixedRange" : { Range },
"fixedPeriod" : { Period },
"fixedRatio" : { Ratio },
"fixedHumanName" : { HumanName },
"fixedAddress" : { Address },
"fixedContactPoint" : { ContactPoint },
"fixedSchedule" : { Schedule },
"fixedReference" : { Reference },
// pattern[x]: Value must have at least these property values. One of these 23:
"patternInteger" : <integer>,
"patternDecimal" : <decimal>,
"patternDateTime" : "<dateTime>",
"patternDate" : "<date>",
"patternInstant" : "<instant>",
"patternString" : "<string>",
"patternUri" : "<uri>",
"patternBoolean" : <boolean>,
"patternCode" : "<code>",
"patternBase64Binary" : "<base64Binary>",
"patternCoding" : { Coding },
"patternCodeableConcept" : { CodeableConcept },
"patternAttachment" : { Attachment },
"patternIdentifier" : { Identifier },
"patternQuantity" : { Quantity },
"patternRange" : { Range },
"patternPeriod" : { Period },
"patternRatio" : { Ratio },
"patternHumanName" : { HumanName },
"patternAddress" : { Address },
"patternContactPoint" : { ContactPoint },
"patternSchedule" : { Schedule },
"patternReference" : { Reference },
```

```
  "example" : [{ // Example value (as defined for type)
    "label" : "<string>", // R!  Describes the purpose of this example
    // value[x]: Value of Example (one of allowed types). One of these 23:
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueDateTime" : "<dateTime>"
    "valueDate" : "<date>"
    "valueInstant" : "<instant>"
    "valueString" : "<string>"
    "valueUri" : "<uri>"
    "valueBoolean" : <boolean>
    "valueCode" : "<code>"
    "valueBase64Binary" : "<base64Binary>"
    "valueCoding" : { Coding }
    "valueCodeableConcept" : { CodeableConcept }
    "valueAttachment" : { Attachment }
    "valueIdentifier" : { Identifier }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valuePeriod" : { Period }
    "valueRatio" : { Ratio }
    "valueHumanName" : { HumanName }
    "valueAddress" : { Address }
    "valueContactPoint" : { ContactPoint }
    "valueSchedule" : { Schedule }
    "valueReference" : { Reference }
  }],
  // minValue[x]: Minimum Allowed Value (for some types). One of these 9:
  "minValueDate" : "<date>",
  "minValueDateTime" : "<dateTime>",
  "minValueInstant" : "<instant>",
  "minValueTime" : "<time>",
  "minValueDecimal" : <decimal>,
  "minValueInteger" : <integer>,
  "minValuePositiveInt" : "<positiveInt>",
  "minValueUnsignedInt" : "<unsignedInt>",
  "minValueQuantity" : { Quantity },
  // maxValue[x]: Maximum Allowed Value (for some types). One of these 9:
  "maxValueDate" : "<date>",
  "maxValueDateTime" : "<dateTime>",
  "maxValueInstant" : "<instant>",
  "maxValueTime" : "<time>",
  "maxValueDecimal" : <decimal>,
  "maxValueInteger" : <integer>,
  "maxValuePositiveInt" : "<positiveInt>",
  "maxValueUnsignedInt" : "<unsignedInt>",
  "maxValueQuantity" : { Quantity },
  "maxLength" : <integer>, // Max length for strings
  "condition" : ["<id>"], // Reference to invariant about presence
  "constraint" : [{ // Condition that must evaluate to true
    "key" : "<id>", // C? R!  Target of 'condition' reference above
    "requirements" : "<string>", // Why this constraint is necessary or appropriate
    "severity" : "<code>", // R!  error | warning
    "human" : "<string>", // R!  Human description of constraint
    "expression" : "<string>", // R!  FHIRPath expression of constraint
    "xpath" : "<string>", // XPath expression of constraint
    "source" : "<uri>" // Reference to original source of constraint
  }],
  "mustSupport" : <boolean>, // If the element must supported
  "isModifier" : <boolean>, // If this modifies the meaning of other elements
  "isSummary" : <boolean>, // Include when _summary = true?
  "binding" : { // C? ValueSet details if this is coded
    "strength" : "<code>", // R!  required | extensible | preferred | example
    "description" : "<string>", // Human explanation of the value set
    // valueSet[x]: Source of value set. One of these 2:
    "valueSetUri" : "<uri>"
    "valueSetReference" : { Reference(ValueSet) }
  },
  "mapping" : [{ // Map element to another set of definitions
    "identity" : "<id>", // R!  Reference to mapping declaration
    "language" : "<code>", // Computable language of mapping
    "map" : "<string>", // R!  Details of the mapping
    "comment" : "<string>" // Comments about the mapping or its use
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ElementDefinition.path [ string ]; # 1..1 Path of the element in the hierarchy of elements
  fhir:ElementDefinition.representation [ code ], ... ; # 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml
  fhir:ElementDefinition.sliceName [ string ]; # 0..1 Name for this particular element (in a set of slices)
  fhir:ElementDefinition.label [ string ]; # 0..1 Name for element to display with or prompt for element
  fhir:ElementDefinition.code [ Coding ], ... ; # 0..* Corresponding codes in terminologies
  fhir:ElementDefinition.slicing [ # 0..1 This element is sliced - slices follow
    fhir:ElementDefinition.slicing.discriminator [ # 0..* Element values that are used to distinguish the slices
```

```
    fhir:ElementDefinition.slicing.discriminator.type [ code ]; # 1..1 value | exists | pattern | type | profile
    fhir:ElementDefinition.slicing.discriminator.path [ string ]; # 1..1 Path to element value
  ], ...;
  fhir:ElementDefinition.slicing.description [ string ]; # 0..1 Text description of how slicing works (or not)
  fhir:ElementDefinition.slicing.ordered [ boolean ]; # 0..1 If elements must be in same order as slices
  fhir:ElementDefinition.slicing.rules [ code ]; # 1..1 closed | open | openAtEnd
];
fhir:ElementDefinition.short [ string ]; # 0..1 Concise definition for space-constrained presentation
fhir:ElementDefinition.definition [ markdown ]; # 0..1 Full formal definition as narrative text
fhir:ElementDefinition.comment [ markdown ]; # 0..1 Comments about the use of this element
fhir:ElementDefinition.requirements [ markdown ]; # 0..1 Why this resource has been created
fhir:ElementDefinition.alias [ string ], ... ; # 0..* Other names
fhir:ElementDefinition.min [ unsignedInt ]; # 0..1 Minimum Cardinality
fhir:ElementDefinition.max [ string ]; # 0..1 Maximum Cardinality (a number or *)
fhir:ElementDefinition.base [ # 0..1 Base definition information for tools
  fhir:ElementDefinition.base.path [ string ]; # 1..1 Path that identifies the base element
  fhir:ElementDefinition.base.min [ unsignedInt ]; # 1..1 Min cardinality of the base element
  fhir:ElementDefinition.base.max [ string ]; # 1..1 Max cardinality of the base element
];
fhir:ElementDefinition.contentReference [ uri ]; # 0..1 Reference to definition of content for the element
fhir:ElementDefinition.type [ # 0..* Data type and Profile for this element
  fhir:ElementDefinition.type.code [ uri ]; # 1..1 Data type or Resource (reference to definition)
  fhir:ElementDefinition.type.profile [ uri ]; # 0..1 Profile (StructureDefinition) to apply (or IG)
  fhir:ElementDefinition.type.targetProfile [ uri ]; # 0..1 Profile (StructureDefinition) to apply to reference target
   (or IG)
  fhir:ElementDefinition.type.aggregation [ code ], ... ; # 0..* contained | referenced | bundled - how aggregated
  fhir:ElementDefinition.type.versioning [ code ]; # 0..1 either | independent | specific
], ...;
# ElementDefinition.defaultValue[x] : 0..1 Specified value if missing from instance. One of these 38
  fhir:ElementDefinition.defaultValueBase64Binary [ base64Binary ]
  fhir:ElementDefinition.defaultValueBoolean [ boolean ]
  fhir:ElementDefinition.defaultValueCode [ code ]
  fhir:ElementDefinition.defaultValueDate [ date ]
  fhir:ElementDefinition.defaultValueDateTime [ dateTime ]
  fhir:ElementDefinition.defaultValueDecimal [ decimal ]
  fhir:ElementDefinition.defaultValueId [ id ]
  fhir:ElementDefinition.defaultValueInstant [ instant ]
  fhir:ElementDefinition.defaultValueInteger [ integer ]
  fhir:ElementDefinition.defaultValueMarkdown [ markdown ]
  fhir:ElementDefinition.defaultValueOid [ oid ]
  fhir:ElementDefinition.defaultValuePositiveInt [ positiveInt ]
  fhir:ElementDefinition.defaultValueString [ string ]
  fhir:ElementDefinition.defaultValueTime [ time ]
  fhir:ElementDefinition.defaultValueUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.defaultValueUri [ uri ]
  fhir:ElementDefinition.defaultValueAddress [ Address ]
  fhir:ElementDefinition.defaultValueAge [ Age ]
  fhir:ElementDefinition.defaultValueAnnotation [ Annotation ]
  fhir:ElementDefinition.defaultValueAttachment [ Attachment ]
  fhir:ElementDefinition.defaultValueCodeableConcept [ CodeableConcept ]
  fhir:ElementDefinition.defaultValueCoding [ Coding ]
  fhir:ElementDefinition.defaultValueContactPoint [ ContactPoint ]
  fhir:ElementDefinition.defaultValueCount [ Count ]
  fhir:ElementDefinition.defaultValueDistance [ Distance ]
  fhir:ElementDefinition.defaultValueDuration [ Duration ]
  fhir:ElementDefinition.defaultValueHumanName [ HumanName ]
  fhir:ElementDefinition.defaultValueIdentifier [ Identifier ]
  fhir:ElementDefinition.defaultValueMoney [ Money ]
  fhir:ElementDefinition.defaultValuePeriod [ Period ]
  fhir:ElementDefinition.defaultValueQuantity [ Quantity ]
  fhir:ElementDefinition.defaultValueRange [ Range ]
  fhir:ElementDefinition.defaultValueRatio [ Ratio ]
  fhir:ElementDefinition.defaultValueReference [ Reference ]
  fhir:ElementDefinition.defaultValueSampledData [ SampledData ]
  fhir:ElementDefinition.defaultValueSignature [ Signature ]
  fhir:ElementDefinition.defaultValueTiming [ Timing ]
  fhir:ElementDefinition.defaultValueMeta [ Meta ]
fhir:ElementDefinition.meaningWhenMissing [ markdown ]; # 0..1 Implicit meaning when this element is missing
fhir:ElementDefinition.orderMeaning [ string ]; # 0..1 What the order of the elements means
# ElementDefinition.fixed[x] : 0..1 Value must be exactly this. One of these 38
  fhir:ElementDefinition.fixedBase64Binary [ base64Binary ]
  fhir:ElementDefinition.fixedBoolean [ boolean ]
  fhir:ElementDefinition.fixedCode [ code ]
  fhir:ElementDefinition.fixedDate [ date ]
  fhir:ElementDefinition.fixedDateTime [ dateTime ]
  fhir:ElementDefinition.fixedDecimal [ decimal ]
  fhir:ElementDefinition.fixedId [ id ]
  fhir:ElementDefinition.fixedInstant [ instant ]
  fhir:ElementDefinition.fixedInteger [ integer ]
  fhir:ElementDefinition.fixedMarkdown [ markdown ]
  fhir:ElementDefinition.fixedOid [ oid ]
  fhir:ElementDefinition.fixedPositiveInt [ positiveInt ]
  fhir:ElementDefinition.fixedString [ string ]
  fhir:ElementDefinition.fixedTime [ time ]
  fhir:ElementDefinition.fixedUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.fixedUri [ uri ]
  fhir:ElementDefinition.fixedAddress [ Address ]
  fhir:ElementDefinition.fixedAge [ Age ]
  fhir:ElementDefinition.fixedAnnotation [ Annotation ]
```

```
    fhir:ElementDefinition.fixedAttachment [ Attachment ]
    fhir:ElementDefinition.fixedCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.fixedCoding [ Coding ]
    fhir:ElementDefinition.fixedContactPoint [ ContactPoint ]
    fhir:ElementDefinition.fixedCount [ Count ]
    fhir:ElementDefinition.fixedDistance [ Distance ]
    fhir:ElementDefinition.fixedDuration [ Duration ]
    fhir:ElementDefinition.fixedHumanName [ HumanName ]
    fhir:ElementDefinition.fixedIdentifier [ Identifier ]
    fhir:ElementDefinition.fixedMoney [ Money ]
    fhir:ElementDefinition.fixedPeriod [ Period ]
    fhir:ElementDefinition.fixedQuantity [ Quantity ]
    fhir:ElementDefinition.fixedRange [ Range ]
    fhir:ElementDefinition.fixedRatio [ Ratio ]
    fhir:ElementDefinition.fixedReference [ Reference ]
    fhir:ElementDefinition.fixedSampledData [ SampledData ]
    fhir:ElementDefinition.fixedSignature [ Signature ]
    fhir:ElementDefinition.fixedTiming [ Timing ]
    fhir:ElementDefinition.fixedMeta [ Meta ]
  # ElementDefinition.pattern[x] : 0..1 Value must have at least these property values. One of these 38
    fhir:ElementDefinition.patternBase64Binary [ base64Binary ]
    fhir:ElementDefinition.patternBoolean [ boolean ]
    fhir:ElementDefinition.patternCode [ code ]
    fhir:ElementDefinition.patternDate [ date ]
    fhir:ElementDefinition.patternDateTime [ dateTime ]
    fhir:ElementDefinition.patternDecimal [ decimal ]
    fhir:ElementDefinition.patternId [ id ]
    fhir:ElementDefinition.patternInstant [ instant ]
    fhir:ElementDefinition.patternInteger [ integer ]
    fhir:ElementDefinition.patternMarkdown [ markdown ]
    fhir:ElementDefinition.patternOid [ oid ]
    fhir:ElementDefinition.patternPositiveInt [ positiveInt ]
    fhir:ElementDefinition.patternString [ string ]
    fhir:ElementDefinition.patternTime [ time ]
    fhir:ElementDefinition.patternUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.patternUri [ uri ]
    fhir:ElementDefinition.patternAddress [ Address ]
    fhir:ElementDefinition.patternAge [ Age ]
    fhir:ElementDefinition.patternAnnotation [ Annotation ]
    fhir:ElementDefinition.patternAttachment [ Attachment ]
    fhir:ElementDefinition.patternCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.patternCoding [ Coding ]
    fhir:ElementDefinition.patternContactPoint [ ContactPoint ]
    fhir:ElementDefinition.patternCount [ Count ]
    fhir:ElementDefinition.patternDistance [ Distance ]
    fhir:ElementDefinition.patternDuration [ Duration ]
    fhir:ElementDefinition.patternHumanName [ HumanName ]
    fhir:ElementDefinition.patternIdentifier [ Identifier ]
    fhir:ElementDefinition.patternMoney [ Money ]
    fhir:ElementDefinition.patternPeriod [ Period ]
    fhir:ElementDefinition.patternQuantity [ Quantity ]
    fhir:ElementDefinition.patternRange [ Range ]
    fhir:ElementDefinition.patternRatio [ Ratio ]
    fhir:ElementDefinition.patternReference [ Reference ]
    fhir:ElementDefinition.patternSampledData [ SampledData ]
    fhir:ElementDefinition.patternSignature [ Signature ]
    fhir:ElementDefinition.patternTiming [ Timing ]
    fhir:ElementDefinition.patternMeta [ Meta ]
  fhir:ElementDefinition.example [ # 0..* Example value (as defined for type)
    fhir:ElementDefinition.example.label [ string ]; # 1..1 Describes the purpose of this example
    # ElementDefinition.example.value[x] : 1..1 Value of Example (one of allowed types). One of these 38
      fhir:ElementDefinition.example.valueBase64Binary [ base64Binary ]
      fhir:ElementDefinition.example.valueBoolean [ boolean ]
      fhir:ElementDefinition.example.valueCode [ code ]
      fhir:ElementDefinition.example.valueDate [ date ]
      fhir:ElementDefinition.example.valueDateTime [ dateTime ]
      fhir:ElementDefinition.example.valueDecimal [ decimal ]
      fhir:ElementDefinition.example.valueId [ id ]
      fhir:ElementDefinition.example.valueInstant [ instant ]
      fhir:ElementDefinition.example.valueInteger [ integer ]
      fhir:ElementDefinition.example.valueMarkdown [ markdown ]
      fhir:ElementDefinition.example.valueOid [ oid ]
      fhir:ElementDefinition.example.valuePositiveInt [ positiveInt ]
      fhir:ElementDefinition.example.valueString [ string ]
      fhir:ElementDefinition.example.valueTime [ time ]
      fhir:ElementDefinition.example.valueUnsignedInt [ unsignedInt ]
      fhir:ElementDefinition.example.valueUri [ uri ]
      fhir:ElementDefinition.example.valueAddress [ Address ]
      fhir:ElementDefinition.example.valueAge [ Age ]
      fhir:ElementDefinition.example.valueAnnotation [ Annotation ]
      fhir:ElementDefinition.example.valueAttachment [ Attachment ]
      fhir:ElementDefinition.example.valueCodeableConcept [ CodeableConcept ]
      fhir:ElementDefinition.example.valueCoding [ Coding ]
      fhir:ElementDefinition.example.valueContactPoint [ ContactPoint ]
      fhir:ElementDefinition.example.valueCount [ Count ]
      fhir:ElementDefinition.example.valueDistance [ Distance ]
      fhir:ElementDefinition.example.valueDuration [ Duration ]
      fhir:ElementDefinition.example.valueHumanName [ HumanName ]
      fhir:ElementDefinition.example.valueIdentifier [ Identifier ]
```

```
      fhir:ElementDefinition.example.valueMoney [ Money ]
      fhir:ElementDefinition.example.valuePeriod [ Period ]
      fhir:ElementDefinition.example.valueQuantity [ Quantity ]
      fhir:ElementDefinition.example.valueRange [ Range ]
      fhir:ElementDefinition.example.valueRatio [ Ratio ]
      fhir:ElementDefinition.example.valueReference [ Reference ]
      fhir:ElementDefinition.example.valueSampledData [ SampledData ]
      fhir:ElementDefinition.example.valueSignature [ Signature ]
      fhir:ElementDefinition.example.valueTiming [ Timing ]
      fhir:ElementDefinition.example.valueMeta [ Meta ]
  ], ...;
  # ElementDefinition.minValue[x] : 0..1 Minimum Allowed Value (for some types). One of these 9
    fhir:ElementDefinition.minValueDate [ date ]
    fhir:ElementDefinition.minValueDateTime [ dateTime ]
    fhir:ElementDefinition.minValueInstant [ instant ]
    fhir:ElementDefinition.minValueTime [ time ]
    fhir:ElementDefinition.minValueDecimal [ decimal ]
    fhir:ElementDefinition.minValueInteger [ integer ]
    fhir:ElementDefinition.minValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.minValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.minValueQuantity [ Quantity ]
  # ElementDefinition.maxValue[x] : 0..1 Maximum Allowed Value (for some types). One of these 9
    fhir:ElementDefinition.maxValueDate [ date ]
    fhir:ElementDefinition.maxValueDateTime [ dateTime ]
    fhir:ElementDefinition.maxValueInstant [ instant ]
    fhir:ElementDefinition.maxValueTime [ time ]
    fhir:ElementDefinition.maxValueDecimal [ decimal ]
    fhir:ElementDefinition.maxValueInteger [ integer ]
    fhir:ElementDefinition.maxValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.maxValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.maxValueQuantity [ Quantity ]
  fhir:ElementDefinition.maxLength [ integer ]; # 0..1 Max length for strings
  fhir:ElementDefinition.condition [ id ], ... ; # 0..* Reference to invariant about presence
  fhir:ElementDefinition.constraint [ # 0..* Condition that must evaluate to true
    fhir:ElementDefinition.constraint.key [ id ]; # 1..1 Target of 'condition' reference above
    fhir:ElementDefinition.constraint.requirements [ string ]; # 0..1 Why this constraint is necessary or appropriate
    fhir:ElementDefinition.constraint.severity [ code ]; # 1..1 error | warning
    fhir:ElementDefinition.constraint.human [ string ]; # 1..1 Human description of constraint
    fhir:ElementDefinition.constraint.expression [ string ]; # 1..1 FHIRPath expression of constraint
    fhir:ElementDefinition.constraint.xpath [ string ]; # 0..1 XPath expression of constraint
    fhir:ElementDefinition.constraint.source [ uri ]; # 0..1 Reference to original source of constraint
  ], ...;
  fhir:ElementDefinition.mustSupport [ boolean ]; # 0..1 If the element must supported
  fhir:ElementDefinition.isModifier [ boolean ]; # 0..1 If this modifies the meaning of other elements
  fhir:ElementDefinition.isSummary [ boolean ]; # 0..1 Include when _summary = true?
  fhir:ElementDefinition.binding [ # 0..1 ValueSet details if this is coded
    fhir:ElementDefinition.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
    fhir:ElementDefinition.binding.description [ string ]; # 0..1 Human explanation of the value set
    # ElementDefinition.binding.valueSet[x] : 0..1 Source of value set. One of these 2
      fhir:ElementDefinition.binding.valueSetUri [ uri ]
      fhir:ElementDefinition.binding.valueSetReference [ Reference(ValueSet) ]
  ];
  fhir:ElementDefinition.mapping [ # 0..* Map element to another set of definitions
    fhir:ElementDefinition.mapping.identity [ id ]; # 1..1 Reference to mapping declaration
    fhir:ElementDefinition.mapping.language [ code ]; # 0..1 Computable language of mapping
    fhir:ElementDefinition.mapping.map [ string ]; # 1..1 Details of the mapping
    fhir:ElementDefinition.mapping.comment [ string ]; # 0..1 Comments about the mapping or its use
  ], ...;
]
```

Changes since DSTU2

ElementDefinition

ElementDefinition.representation | Add Codes xmlText, typeAttr, cdaText, xhtml

ElementDefinition.sliceName | Renamed from name to sliceName

ElementDefinition.slicing.discriminator | Type changed from string to Element

ElementDefinition.slicing.discriminator.type | Added Element

ElementDefinition.slicing.discriminator.path | Added Element

ElementDefinition.comment | Renamed from comments to comment

ElementDefinition.min | Type changed from integer to unsignedInt

ElementDefinition.base.min | Type changed from integer to unsignedInt

ElementDefinition.contentReference | Renamed from nameReference to contentReference Type changed from string to uri

ElementDefinition.type.code | Type changed from code to uri Change binding strength from required to extensible

ElementDefinition.type.profile | Max Cardinality changed from * to 1

ElementDefinition.type.targetProfile | Added Element

ElementDefinition.type.versioning | Added Element

ElementDefinition.defaultValue[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money

ElementDefinition.orderMeaning | Added Element

ElementDefinition.fixed[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money

ElementDefinition.pattern[x] | Add Age, Add Count, Add Distance, Add Duration, Add Money

ElementDefinition.example | Renamed from example[x] to example Max Cardinality changed from 1 to * Remove boolean, Remove integer, Remove decimal, Remove base64Binary, Remove instant, Remove string, Remove uri, Remove date, Remove dateTime, Remove time, Remove code, Remove oid, Remove id, Remove unsignedInt, Remove positiveInt, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Quantity, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta

ElementDefinition.example.label | Added Element
ElementDefinition.example.value[x] | Added Element
ElementDefinition.minValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.maxValue[x] | Remove boolean, Remove base64Binary, Remove string, Remove uri, Remove code, Remove oid, Remove id, Remove markdown, Remove Annotation, Remove Attachment, Remove Identifier, Remove CodeableConcept, Remove Coding, Remove Range, Remove Period, Remove Ratio, Remove SampledData, Remove Signature, Remove HumanName, Remove Address, Remove ContactPoint, Remove Timing, Remove Reference, Remove Meta
ElementDefinition.constraint.expression | Added Element
ElementDefinition.constraint.xpath | Min Cardinality changed from 1 to 0
ElementDefinition.constraint.source | Added Element
ElementDefinition.mapping.comment | Added Element

See the Full Difference for further information

Constraints

* eld-1: On ElementDefinition.slicing: If there are no discriminators, there must be a definition (expression on ElementDefinition.slicing: discriminator.exists() or description.exists())

* eld-2: Min <= Max (expression : min.empty() or max.empty() or (max = '*') or (min <= max.toInteger()))

* eld-3: On ElementDefinition.max: Max SHALL be a number or "*" (expression on ElementDefinition.max: empty() or ($this = '*') or (toInteger() >= 0))

* eld-4: On ElementDefinition.type: Aggregation may only be specified if one of the allowed types for the element is a resource (expression on ElementDefinition.type: aggregation.empty() or (code = 'Reference'))

* eld-5: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding (expression : contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()))

* eld-6: Fixed value may only be specified if there is one type (expression : fixed.empty() or (type.count() <= 1))

* eld-7: Pattern may only be specified if there is one type (expression : pattern.empty() or (type.count() <= 1))

* eld-8: Pattern and value are mutually exclusive (expression : pattern.empty() or fixed.empty())

* eld-11: Binding can only be present for coded elements, string, and uri (expression : binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'Extension') or (code = 'string') or (code = 'uri')).exists())

* eld-12: On ElementDefinition.binding: ValueSet as a URI SHALL start with http:// or https:// or urn: (expression on ElementDefinition.binding: valueSet.is(uri).not() or valueSet.as(uri).startsWith('http:') or valueSet.as(uri).startsWith('https') or valueSet.as(uri).startsWith('urn:'))

* eld-13: Types must be unique by the combination of code and profile (expression : type.select(code&profile&targetProfile).isDistinct())

* eld-14: Constraints must be unique by key (expression : constraint.select(key).isDistinct())

* eld-15: default value and meaningWhenMissing are mutually exclusive (expression : defaultValue.empty() or meaningWhenMissing.empty())

* eld-16: sliceName must be composed of proper tokens separated by "/" (expression : sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\/\\-_]+$'))

2.32.0.2 Use of ElementDefinition.path

The path element is the most important property of the element definition. It both names the element, and locates the element within a hierarchy defined within a particular context. Within the FHIR specification, there is only one original definition for each path (that is, an element definition with no base element). This is the master definition to which all the other definitions with the same path must conform.

All elements defined within the FHIR specification itself are defined within a StructureDefinition that defines a resource, or a datatype. This defines the identity of the element and provides the context in which the meaning of the element is understood. When Elements are defined, the following rules apply:

* Element names (the parts of a path delineated by the '.' character) SHALL not contain whitespace (i.e. Unicode characters marked as whitespace)

* Element names SHALL not contain the characters ,:;'"/|?|!@#$%^&*()[]{}

* Element names SHOULD not contain non-ASCII characters

* Element paths SHALL not exceed 64 characters in length

* Element paths cannot imply elements that are not explicitly defined i.e. a.b.c.d cannot be defined unless a.b.c is explicitly defined

* By convention, each path start with an uppercase letter (type) but all the element names that follow this are lowercase (not type names). All resources and data types (except for Primitive data types) follow this convention, but logical models are not

required to do so

Elements may be defined in:

* StructureDefinitions of kind = resource, complex-type or primitive-type, where derivation = specialization. These are either Resources or Data types defined in the specification, or in custom resource definitions (see Custom Resource Policy)

* StructureDefinitions of kind = logical

* Data Elements

StructureDefinitions with derivation = constraint (i.e. Resource and Data Type profiles) are not allowed to define or include ElementDefinitions with a path not defined within the base resource definition from which they derive (e.g. in the FHIR specification).

2.32.0.3 ElementDefinition.id

In addition to the path, every ElementDefinition SHALL have a populated id, and the id SHALL have a unique value populated by following this algorithm:

* The id will be constructed as a dot separated string, each part corresponding to a token in the path

* For each token in the path, use the syntax pathpart:slicename/reslicename

Note that in a profile with no slices, this id will match the path exactly and entirely. id values constructed in this fashion are unique, and persistent, and may be used as the target of external references into the definition, where necessary.

2.32.0.4 Interpretation of ElementDefinition in different contexts

The data type ElementDefinition is used both in StructureDefinition and DataElement, and within those resources, the way its elements are to be used and interpreted depends on the context where ElementDefinition used:

ElementDefinition field	DataElement	Type definition, first element	Type definition, following elements	Constraint Definition, first element	Constraint Definition, following elements
sliceName	prohibited	prohibited	prohibited	prohibited	required for slices, else prohibited
label	optional	optional	optional	recommended	recommended
code	optional	optional	optional	optional	optional
slicing	prohibited	prohibited	prohibited	prohibited	optional
short/definition	optional	required	required	required‡	required‡
requirements/ comments/alias	optional	prohibited	optional	prohibited‡	optional‡
base	prohibited	prohibited	prohibited	required	required
type	required	required	required	optional	optional
nameReference	prohibited	prohibited	optional	prohibited	optional
min/max	optional	optional (irrelevant)	required	optional	optional†
defaultValue[x]	optional	prohibited	optional	prohibited	optional†
meaningWhenMissing	optional	prohibited	optional	prohibited	optional†
fixed[x]	prohibited	prohibited	prohibited	prohibited	optional
pattern[x]	optional	prohibited	prohibited	prohibited	optional
example[x]	optional	prohibited	optional	prohibited	optional
minValue[x]	optional	prohibited	prohibited	prohibited	optional
maxValue[x]	optional	prohibited	prohibited	prohibited	optional
maxLength	optional	prohibited	prohibited	prohibited	optional
mustSupport	optional	prohibited	prohibited	optional	optional
isModifier	prohibited	prohibited	optional	prohibited	optional†
isSummary	prohibited	prohibited	optional	prohibited	optional†
binding	optional	prohibited	optional	prohibited	optional
constraint	optional	optional	optional	optional?	optional?
condition	optional	prohibited	optional	prohibited	optional?
mapping	optional	optional	optional	optional?	optional?

Notes:

* Type definition: A StructureDefinition without a base element, or where the derivation type is 'specialization'

* Constraint definition: A StructureDefinition with a base element and a derivation of 'constraint' - e.g. a definition of a structure that constrains another base structure, referring to the differential portion

* †: The element's presence, and value, must match the definition in the base structure

* ‡: The element content must be consistent with that matching element in the base structure

* ?: Additional constraints and mappings can be defined, but they do not replace the ones in the base structure

The use of Path and type depends more deeply on the context where the ElementDefinition is used:

Context	path (1st element)	path (following elements)	type (1st element)
Data Element	Any token	Any token (distinct in the DataElement)	Any type
Base definition of a data type (example: Quantity - XML, JSON)	Name of the type	Path inside the datatype	Element
A constrained data type (example: Money - XML, JSON)	Name of the base type	Path inside the datatype	Name of the base type
Base definition of a resource (example: Patient - XML, JSON)	The name of the resource	Path inside the resource	DomainResource or sometimes Resource
Constraint on a resource (example: DAF Patient - XML, JSON)	The name of the resource	Path inside the resource (including into the data types)	The name of the resource

Base Extension (a standard data type) (example: Extension - XML, JSON) | Extension | Extension.value[x] or Extension.extension | Extension
A defined Extension (example: Extension - XML, JSON) | Extension | Extension.value[x] or Extension.extension (for complex extensions) | Extension

There are additional notes about the use of ElementDefinition when defining Extensions on the Defining Extensions page.

2.32.0.4.1 Rules about Slicing

For a description of slicing, see Slicing

* Slicing is only allowed when constraining an existing structure

* slicing can only be used on the first repeat of an element, this element is considered to be the slicing entry

* All elements following the first repeat that containing a slicing SHALL have a sliceName

* The first entry (the one having the information) is understood to be the set of constraints to be used for the "open" slice, i.e., when the slice is open, this definition is used as a constraint on elements in the instance that are not part of a slice and belong to the open portion of the slice. Its use follows the "normal case", except:

slicing

* slicing must be present

* min and max govern the number of total occurrences of the sliced element including the number of occurrences in the open portion of the slice.

* An element with a cardinality of 0..1 and a choice of multiple types can be sliced by type. This is to specify different constraints for different types. In this case, the discriminator SHALL be "@type"

2.32.0.4.2 Rules about min and max

In a DataElement, min and max are always required.

In a StructureDefinition:

* If there is no StructureDefinition.base: min and max are always required

* Otherwise, in StructureDefinition.differential: min and max are always optional; if they are not present, they default to the base min and max

* In StructureDefinition.snapshot: min and max are always required

2.32.0.4.3 Missing Elements

Most elements have a minimum cardinality of 0, which means that they may be missing from a resource when it is exchanged between systems. Generally, when an element is missing, all that an application processing the resource can say about the element is that the value is unknown - it may have a correct value, but it has not been provided for security or workflow reasons. On the other hand, it may not have a value at all. All the application can say is that the value is unknown.

However, for some elements, this specification makes specific rules about what it means if the element is missing. For some elements, this takes the form of a default value, a value that should be assumed if an element is present. For example, here are some default values:

Element | Default Value
ElementDefinition.mustSupport | false
SampledData.factor | 1
ValueSet.codeSystem.concept.abstract | false
List.entry.deleted | false

Most default values are mathematical / numerical non-operations e.g. false, 0 for addition or 1 for multiplication. Default values are rarely used because:

* The value must be known by all implementations

* When an element has a default value, it can never be unknown - e.g. it is implicitly mandatory

* The default value can never be changed.

For other elements, the resource explicitly documents the meaning of a missing element, and that meaning is not represented by a default value. Some examples:

Element | Meaning when missing
Period.end | If the end of the period is missing, it means that the period is ongoing
Quantity.comparator | If there is no comparator, then there is no modification of the value
MedicationAdministration.wasNotGiven | If this is missing, then the medication was administered
Substance.instance | If this element is not present, then the substance resource describes a kind of substance

Like default values, and for the same reasons, assigning a meaning when an element is missing is something that is not done very often. Note that Profiles cannot change the default value or missing meaning of an element, since this cannot be done in the base specification either.

Here is a list of all elements with a default value or a missing meaning:

* Period.end: If the end of the period is missing, it means that the period is ongoing

* Quantity.comparator: If there is no comparator, then there is no modification of the value

* SampledData.factor: 1

* ElementDefinition.slicing.ordered: false

* ElementDefinition.type.versioning: either
* ElementDefinition.mustSupport: false
* ElementDefinition.isModifier: false
* ElementDefinition.isSummary: false
* Timing.repeat.frequency: 1
* BodySite.active: true
* CarePlan.activity.detail.prohibited: false
* CommunicationRequest.priority: If missing, this task should be performed with normal priority
* ConceptMap.group.element.target.equivalence: equivalent
* ConceptMap.group.element.target.dependsOn.system: The code is in the source (dependsOn) or target (product) system
* Condition.verificationStatus: unknown
* Consent.period: all date ranges included
* Consent.actor: There is no specific actor associated with the consent
* Consent.action: all actions
* Consent.data: all data
* Consent.except.actor: There is no specific actor associated with the exception
* Consent.except.action: all actions
* Consent.except.data: all data
* DeviceRequest.priority: If missing, normal priority
* ExpansionProfile.excludedSystem.version: Exclusion is not version specific
* ExpansionProfile.includeDefinition: false
* ExpansionProfile.activeOnly: false
* FamilyMemberHistory.notDone: false
* FamilyMemberHistory.estimatedAge: It is unknown whether the age is an estimate or not
* Group.active: true
* Group.member.period: The member is in the group at this time
* Group.member.inactive: false
* HealthcareService.active: true
* Linkage.active: true
* List.entry.deleted: false
* Media.frames: 1
* MedicationAdministration.notGiven: If this is missing, then the medication was administered
* MessageDefinition.focus.min: 0
* MessageDefinition.responseRequired: false
* NamingSystem.uniqueId.preferred: If there are multiple ids, and one is labeled "preferred", then the assumption is that the others are not preferred. In the absence of any id marked as preferred, no inference can be drawn
* Organization.active: true
* Patient.active: true
* Practitioner.active: true
* PractitionerRole.active: true
* Procedure.notDone: false
* ProcedureRequest.priority: If missing, this task should be performed with normal priority
* ProcedureRequest.doNotPerform: false
* Questionnaire.item.required: false
* Questionnaire.item.repeats: false
* RelatedPerson.active: true
* RequestGroup.priority: If missing, this request should be performed with normal priority
* Schedule.active: true
* Slot.overbooked: If overbooked is missing, systems may assume that there are still appointments available
* Substance.instance: If this element is not present, then the substance resource describes a kind of substance
* SupplyRequest.priority: If missing, this task should be performed with normal priority
* Task.priority: If missing, this task should be performed with normal priority

* TestScript.origin.profile: FHIR-Client
* TestScript.destination.profile: FHIR-Server
* TestScript.metadata.capability.required: false
* TestScript.metadata.capability.validated: false
* TestScript.fixture.autocreate: false
* TestScript.fixture.autodelete: false
* TestScript.setup.action.operation.accept: xml
* TestScript.setup.action.operation.contentType: xml
* TestScript.setup.action.operation.encodeRequestUrl: true
* TestScript.setup.action.assert.contentType: xml
* TestScript.setup.action.assert.warningOnly: false
* ValueSet.expansion.offset: Paging is not being used
* ValueSet.expansion.contains.abstract: false
* ValueSet.expansion.contains.inactive: false

2.32.1 Examples

URL: https://hl7.org/fhir/STU3/elementdefinition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.32.1 Element Definition Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

For ElementDefinition examples, see the extensive StructureDefinition example list.

2.32.2 Detailed Descriptions

URL: https://hl7.org/fhir/STU3/elementdefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

2.32.2 Element Definition Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 5 | Ballot Status: Trial Use

2.32.2.1 ElementDefinition

ElementDefinition

Definition | Captures constraints on each element within the resource, profile, or extension.

Control | 1..1

Summary | true

Invariants | Defined on this element eld-11: Binding can only be present for coded elements, string, and uri (expression : binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'Extension') or (code = 'string') or (code = 'uri')).exists(), xpath: not(exists(f:binding)) or (count(f:type/f:code) = 0) or f:type/f:code/@value=('code','Coding','CodeableConcept','Quantity','Extension', 'string', 'uri')) eld-13: Types must be unique by the combination of code and profile (expression : type.select(code&profile&targetProfile).isDistinct(), xpath: not(exists(for $type in f:type return $type/preceding-sibling::f:type[f:code/@value=$type/f:code/@value and f:profile/@value = $type/f:profile/@value]))) eld-14: Constraints must be unique by key (expression : constraint.select(key).isDistinct(), xpath: count(f:constraint) = count(distinct-values(f:constraint/f:key/@value))) eld-15: default value and meaningWhenMissing are mutually exclusive (expression : defaultValue.empty() or meaningWhenMissing.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'fixed')])) or not(exists(f:meaningWhenMissing))) eld-16: sliceName must be composed of proper tokens separated by "/" (expression : sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\/\\-_]+$'), xpath: not(exists(f:sliceName/@value)) or matches(f:sliceName/@value, '^[a-zA-Z0-9\\/\\-_]+$')) eld-2: Min <= Max (expression : min.empty() or max.empty() or (max = '*') or (min <= max.toInteger()), xpath: not(exists(f:min)) or not(exists(f:max)) or (not(f:max/@value) and not(f:min/@value)) or (f:max/@value = '*') or (number(f:max/@value) >= f:min/@value)) eld-5: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding (expression : contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()), xpath: not(exists(f:contentReference) and (exists(f:type) or exists(f:*[starts-with(local-name(.), 'value')]) or exists(f:*[starts-with(local-name(.), 'defaultValue')]) or exists(f:*[starts-with(local-name(.), 'fixed')]) or exists(f:*[starts-with(local-name(.), 'pattern')]) or exists(f:*[starts-with(local-name(.), 'example')]) or exists(f:*[starts-with(local-name(.), 'f:minValue')]) or exists(f:*[starts-with(local-name(.), 'f:maxValue')]) or exists(f:maxLength) or exists(f:binding)))) eld-6: Fixed value may only be specified if there is one type (expression : fixed.empty() or (type.count() <= 1), xpath: not(exists(f:*[starts-with(local-name(.), 'fixed')])) or (count(f:type)<=1)) eld-7: Pattern may only be specified if there is one type (expression : pattern.empty() or (type.count() <= 1), xpath: not(exists(f:*[starts-with(local-name(.), 'pattern')])) or (count(f:type)<=1)) eld-8: Pattern and value are mutually exclusive (expression : pattern.empty() or fixed.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'pattern')])) or not(exists(f:*[starts-with(local-name(.), 'fixed')])))

ElementDefinition.path

Definition | The path identifies the element and is expressed as a "."-separated list of ancestor elements, beginning with the name of the resource or extension.

Control | 1..1

Type | string

Summary | true
ElementDefinition.representation
Definition | Codes that define how this element is represented in instances, when the deviation varies from the normal case.
Control | 0..*
Terminology Binding | PropertyRepresentation (Required)
Type | code
Summary | true
Comments | In resources, this is rarely used except for special cases where the representation deviates from the normal, and can only be done in the base standard (and profiles must reproduce what the base standard does). This element is used quite commonly in Logical models when the logical models represent a specific serialization format (e.g. CDA, v2 etc).
ElementDefinition.sliceName
Definition | The name of this element definition slice, when slicing is working. The name must be a token with no dots or spaces. This is a unique name referring to a specific set of constraints applied to this element, used to provide a name to different slices of the same element.
Control | 0..1
Type | string
Requirements | May also be used for code generation purposes.
Summary | true
Comments | The name SHALL be unique within the structure within the context of the constrained resource element. (Though to avoid confusion, uniqueness across all elements is recommended.).
ElementDefinition.label
Definition | A single preferred label which is the text to display beside the element indicating its meaning or to use to prompt for the element in a user display or form.
Control | 0..1
Type | string
Summary | true
Comments | See also the extension (http://hl7.org/fhir/StructureDefinition/elementdefinition-question)[extension-elementdefinition-question.html].
ElementDefinition.code
Definition | A code that has the same meaning as the element in a particular terminology.
Control | 0..*
Terminology Binding | LOINC Codes (Example)
Type | Coding
Requirements | Links the meaning of an element to an external terminology, and is very useful for searching and indexing.
Summary | true
Comments | The concept SHALL be properly aligned with the data element definition and other constraints, as defined in the code system, including relationships, of any code listed here. Where multiple codes exist in a terminology that could correspond to the data element, the most granular code(s) should be selected, so long as they are not more restrictive than the data element itself. The mappings may be used to provide more or less granular or structured equivalences in the code system.
LOINC Code | LOINC_NUM
ElementDefinition.slicing
Definition | Indicates that the element is sliced into a set of alternative definitions (i.e. in a structure definition, there are multiple different constraints on a single element in the base resource). Slicing can be used in any resource that has cardinality ..* on the base resource, or any resource with a choice of types. The set of slices is any elements that come after this in the element sequence that have the same path, until a shorter path occurs (the shorter path terminates the set).
Control | 0..1
Summary | true
Comments | The first element in the sequence, the one that carries the slicing, is the definition that applies to all the slices. This is based on the unconstrained element, but can apply any constraints as appropriate. This may include the common constraints on the children of the element.
Invariants | Defined on this element eld-1: If there are no discriminators, there must be a definition (expression : discriminator.exists() or description.exists(), xpath: (f:discriminator) or (f:description))
ElementDefinition.slicing.discriminator
Definition | Designates which child elements are used to discriminate between the slices when processing an instance. If one or more discriminators are provided, the value of the child elements in the instance data SHALL completely distinguish which slice the element in the resource matches based on the allowed values for those elements in each of the slices.
Control | 0..*
Summary | true
Comments | If there is no discriminator, the content is hard to process, so this should be avoided. If the base element has a cardinality of ..1, and there is a choice of types, the discriminator must be "@type".
ElementDefinition.slicing.discriminator.type
Definition | How the element value is interpreted when discrimination is evaluated.
Control | 1..1
Terminology Binding | DiscriminatorType (Required)
Type | code
Summary | true
ElementDefinition.slicing.discriminator.path
Definition | A FHIRPath expression, using a restricted subset of FHIRPath, that is used to identify the element on which discrimination is based.
Control | 1..1
Type | string
Summary | true
Comments | The only FHIRPath functions that are allowed are resolve(), and extension(url).
ElementDefinition.slicing.description
Definition | A human-readable text description of how the slicing works. If there is no discriminator, this is required to be present to provide whatever information is possible about how the slices can be differentiated.
Control | 0..1
Type | string

Summary | true
Comments | If it's really not possible to differentiate them, the design should be re-evaluated to make the content usable.
Invariants | Affect this element eld-1: If there are no discriminators, there must be a definition (expression : discriminator.exists() or description.exists(), xpath: (f:discriminator) or (f:description))
ElementDefinition.slicing.ordered
Definition | If the matching elements have to occur in the same order as defined in the profile.
Control | 0..1
Type | boolean
Default Value | false
Summary | true
Comments | Order should only be required when it is a pressing concern for presentation. Profile authors should consider making the order a feature of the rules about the narrative, not the rules about the data - requiring ordered data makes the profile much less re-usable.
ElementDefinition.slicing.rules
Definition | Whether additional slices are allowed or not. When the slices are ordered, profile authors can also say that additional slices are only allowed at the end.
Control | 1..1
Terminology Binding | SlicingRules (Required)
Type | code
Summary | true
Comments | Allowing additional elements makes for a much for flexible template - it's open for use in wider contexts, but also means that the content of the resource is not closed, and applications have to decide how to handle content not described by the profile.
ElementDefinition.short
Definition | A concise description of what this element means (e.g. for use in autogenerated summaries).
Control | 0..1
Type | string
Summary | true
Comments | May change the term to provide language more appropriate to the context of the profile or to reflect slicing.
ElementDefinition.definition
Definition | Provides a complete explanation of the meaning of the data element for human readability. For the case of elements derived from existing elements (e.g. constraints), the definition SHALL be consistent with the base definition, but convey the meaning of the element in the particular context of use of the resource.
Control | 0..1
Type | markdown
Requirements | To allow a user to state the usage of an element in a particular context.
Alternate Names | Description
Summary | true
Comments | It is easy for a different definition to change the meaning of an element and this can have nasty downstream consequences. Please be careful when providing definitions.
LOINC Code | TERM DEFINITION/DESCRIPTION(S)
ElementDefinition.comment
Definition | Explanatory notes and implementation guidance about the data element, including notes about how to use the data properly, exceptions to proper use, etc.
Control | 0..1
Type | markdown
Summary | true
Comments | If it is possible to capture usage rules using constraints, that mechanism should be used in preference to this element.
LOINC Code | COMMENTS
ElementDefinition.requirements
Definition | This element is for traceability of why the element was created and why the constraints exist as they do. This may be used to point to source materials or specifications that drove the structure of this element.
Control | 0..1
Type | markdown
Summary | true
Comments | This element does not describe the usage of the element (that's done in comments), rather it's for traceability of why the element is either needed or why the constraints exist as they do. This may be used to point to source materials or specifications that drove the structure of this data element.
LOINC Code | R/O/C
ElementDefinition.alias
Definition | Identifies additional names by which this element might also be known.
Control | 0..*
Type | string
Requirements | Allows for better easier recognition of the element by multiple communities, including international communities.
Alternate Names | synonym; other name
Summary | true
LOINC Code | RELATED NAMES (only some of these apply)
ElementDefinition.min
Definition | The minimum number of times this element SHALL appear in the instance.
Control | 0..1
Type | unsignedInt
Summary | true
Invariants | Affect this element eld-2: Min <= Max (expression : min.empty() or max.empty() or (max = '*') or (min <= max.toInteger()), xpath: not(exists(f:min)) or not(exists(f:max)) or (not(f:max/@value) and not(f:min/@value)) or (f:max/@value = '*') or (number(f:max/@value) >= f:min/@value))

ElementDefinition.max
Definition | The maximum number of times this element is permitted to appear in the instance.
Control | 0..1
Type | string
Summary | true
Invariants | Defined on this element eld-3: Max SHALL be a number or "*" (expression : empty() or ($this = '*') or (toInteger() >= 0), xpath: @value='*' or (normalize-space(@value)!='' and normalize-space(translate(@value, '0123456789',''))='')) Affect this element eld-2: Min <= Max (expression : min.empty() or max.empty() or (max = '*') or (min <= max.toInteger()), xpath: not(exists(f:min)) or not(exists(f:max)) or (not(f:max/@value) and not(f:min/@value)) or (f:max/@value = '*') or (number(f:max/@value) >= f:min/@value))
ElementDefinition.base
Definition | Information about the base definition of the element, provided to make it unnecessary for tools to trace the deviation of the element through the derived and related profiles. This information is provided when the element definition is not the original definition of an element - i.g. either in a constraint on another type, or for elements from a super type in a snap shot.
Control | 0..1
Summary | true
Comments | The base information does not carry any information that could not be determined from the path and related profiles, but making this determination requires both that the related profiles are available, and that the algorithm to determine them be available. So they are deformalised into this location for tooling convenience, and to ensure that the base information is available without dependencies.
ElementDefinition.base.path
Definition | The Path that identifies the base element - this matches the ElementDefinition.path for that element. Across FHIR, there is only one base definition of any element - that is, an element definition on a StructureDefinition without a StructureDefinition.base.
Control | 1..1
Type | string
Summary | true
ElementDefinition.base.min
Definition | Minimum cardinality of the base element identified by the path.
Control | 1..1
Type | unsignedInt
Summary | true
Comments | This is provided for consistency with max, and may affect code generation of mandatory elements of the base resource are generated differently (some reference implementations have done this).
ElementDefinition.base.max
Definition | Maximum cardinality of the base element identified by the path.
Control | 1..1
Type | string
Summary | true
Comments | This is provided to code generation, since the serialization representation in JSON differs depending on whether the base element has max > 1. Also, some forms of code generation may differ.
ElementDefinition.contentReference
Definition | Identifies the identity of an element defined elsewhere in the profile whose content rules should be applied to the current element.
Control | 0..1
Type | uri
Summary | true
Invariants | Affect this element eld-5: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding (expression : contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()), xpath: not(exists(f:contentReference) and (exists(f:type) or exists(f:*[starts-with(local-name(.), 'value')]) or exists(f:*[starts-with(local-name(.), 'defaultValue')]) or exists(f:*[starts-with(local-name(.), 'fixed')]) or exists(f:*[starts-with(local-name(.), 'pattern')]) or exists(f:*[starts-with(local-name(.), 'example')]) or exists(f:*[starts-with(local-name(.), 'f:minValue')]) or exists(f:*[starts-with(local-name(.), 'f:maxValue')]) or exists(f:maxLength) or exists(f:binding))))
ElementDefinition.type
Definition | The data type or resource that the value of this element is permitted to be.
Control | 0..*
Summary | true
Comments | The Type of the element can be left blank in a differential constraint, in which case the type is inherited from the resource. Abstract types are not permitted to appear as a type when multiple types are listed. (I.e. Abstract types cannot be part of a choice).
Invariants | Defined on this element eld-4: Aggregation may only be specified if one of the allowed types for the element is a resource (expression : aggregation.empty() or (code = 'Reference'), xpath: not(exists(f:aggregation)) or exists(f:code[@value = 'Reference'])) Affect this element eld-13: Types must be unique by the combination of code and profile (expression : type.select(code&profile&targetProfile).isDistinct(), xpath: not(exists(for $type in f:type return $type/preceding-sibling::f:type[f:code/@value=$type/f:code/@value and f:profile/@value = $type/f:profile/@value])))
LOINC Code | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
ElementDefinition.type.code
Definition | URL of Data type or Resource that is a(or the) type used for this element. References are URLs that are relative to http://hl7.org/fhir/StructureDefinition e.g. "string" is a reference to http://hl7.org/fhir/StructureDefinition/string. Absolute URLs are only allowed in logical models.
Control | 1..1
Terminology Binding | FHIRDefinedType (Extensible)
Type | uri
Summary | true
Comments | If the element is a reference to another resource, this element contains "Reference", and the targetProfile element defines what resources can be referenced. The targetProfile may be a reference to the general definition of a resource (e.g. http://hl7.org/fhir/StructureDefinition/Patient). There would be one pair of type/code for each allowed target resource type.
LOINC Code | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
ElementDefinition.type.profile

Definition | Identifies a profile structure or implementation Guide that SHALL hold for the datatype this element refers to. Can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the resource SHALL conform to at least one profile defined in the implementation guide.
Control | 0..1
Type | uri
Summary | true
ElementDefinition.type.targetProfile
Definition | Identifies a profile structure or implementation Guide that SHALL hold for the target of the reference this element refers to. Can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the resource SHALL conform to at least one profile defined in the implementation guide.
Control | 0..1
Type | uri
Summary | true
ElementDefinition.type.aggregation
Definition | If the type is a reference to another resource, how the resource is or can be aggregated - is it a contained resource, or a reference, and if the context is a bundle, is it included in the bundle.
Control | 0..*
Terminology Binding | AggregationMode (Required)
Type | code
Summary | true
Invariants | Affect this element eld-4: Aggregation may only be specified if one of the allowed types for the element is a resource (expression : aggregation.empty() or (code = 'Reference'), xpath: not(exists(f:aggregation)) or exists(f:code[@value = 'Reference']))
ElementDefinition.type.versioning
Definition | Whether this reference needs to be version specific or version independent, or whether either can be used.
Control | 0..1
Terminology Binding | ReferenceVersionRules (Required)
Type | code
Default Value | either
Summary | true
Comments | The base specification never makes a rule as to which form is allowed, but implementation guides may do this.
ElementDefinition.defaultValue[x]
Definition | The value that should be used if there is no value stated in the instance (e.g. 'if not otherwise specified, the abstract is false').
Control | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Default values can only be specified on a resource, data type, or extension definition, and never in a profile that applies to one of these. Specifying a default value means that the property can never been unknown - it must always have a value. Further, the default value can never be changed. For these reasons, default values are (and should be) used extremely sparingly.
Invariants | Affect this element eld-15: default value and meaningWhenMissing are mutually exclusive (expression : defaultValue.empty() or meaningWhenMissing.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'fixed')])) or not(exists(f:meaningWhenMissing)))
ElementDefinition.meaningWhenMissing
Definition | The Implicit meaning that is to be understood when this element is missing (e.g. 'when this element is missing, the period is ongoing'.
Control | 0..1
Type | markdown
Summary | true
Comments | Implicit meanings for missing values can only be specified on a resource, data type, or extension definition, and never in a profile that applies to one of these. An implicit meaning for a missing value can never be changed, and specifying one has the consequence that constraining its use in profiles eliminates use cases as possibilities, not merely moving them out of scope.
Invariants | Affect this element eld-15: default value and meaningWhenMissing are mutually exclusive (expression : defaultValue.empty() or meaningWhenMissing.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'fixed')])) or not(exists(f:meaningWhenMissing)))
ElementDefinition.orderMeaning
Definition | If present, indicates that the order of the repeating element has meaning and describes what that meaning is. If absent, it means that the order of the element has no meaning.
Control | 0..1
Type | string
Summary | true
Comments | This element can only be asserted on repeating elements and can only be introduced when defining resources or data types. It can be further refined profiled elements but if absent in the base type, a profile cannot assert meaning.
ElementDefinition.fixed[x]
Definition | Specifies a value that SHALL be exactly the value for this element in the instance. For purposes of comparison, non-significant whitespace is ignored, and all values must be an exact match (case and accent sensitive). Missing elements/attributes must also be missing.
Control | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This is not recommended for Coding and CodeableConcept since these often have highly contextual properties such as version or display.
Invariants | Affect this element eld-5: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding (expression : contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()), xpath: not(exists(f:contentReference) and (exists(f:type) or exists(f:*[starts-with(local-name(.), 'value')]) or exists(f:*[starts-with(local-name(.), 'defaultValue')]) or exists(f:*[starts-with(local-name(.), 'fixed')]) or exists(f:*[starts-with(local-name(.), 'pattern')]) or exists(f:*[starts-with(local-name(.), 'example')]) or exists(f:*[starts-with(local-name(.), 'f:minValue')]) or exists(f:*[starts-with(local-name(.), 'f:maxValue')]) or exists(f:maxLength) or

exists(f:binding)))) eld-6: Fixed value may only be specified if there is one type (expression : fixed.empty() or (type.count() <= 1), xpath: not(exists(f:*[starts-with(local-name(.), 'fixed')])) or (count(f:type)<=1)) eld-8: Pattern and value are mutually exclusive (expression : pattern.empty() or fixed.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'pattern')])) or not(exists(f:*[starts-with(local-name(.), 'fixed')])))

ElementDefinition.pattern[x]

Definition | Specifies a value that the value in the instance SHALL follow - that is, any value in the pattern must be found in the instance. Other additional values may be found too. This is effectively constraint by example. The values of elements present in the pattern must match exactly (case-sensitive, accent-sensitive, etc.).

Control | 0..1

Type | *

[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true

Comments | Mostly used for fixing values of CodeableConcept. At present, pattern[x] is not recommended as a basis for slicing while issues related to this are investigated during the STU period.

Invariants | Affect this element eld-5: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding (expression : contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()), xpath: not(exists(f:contentReference) and (exists(f:type) or exists(f:*[starts-with(local-name(.), 'value')]) or exists(f:*[starts-with(local-name(.), 'defaultValue')]) or exists(f:*[starts-with(local-name(.), 'fixed')]) or exists(f:*[starts-with(local-name(.), 'pattern')]) or exists(f:*[starts-with(local-name(.), 'example')]) or exists(f:*[starts-with(local-name(.), 'f:minValue')]) or exists(f:*[starts-with(local-name(.), 'f:maxValue')]) or exists(f:maxLength) or exists(f:binding)))) eld-7: Pattern may only be specified if there is one type (expression : pattern.empty() or (type.count() <= 1), xpath: not(exists(f:*[starts-with(local-name(.), 'pattern')])) or (count(f:type)<=1)) eld-8: Pattern and value are mutually exclusive (expression : pattern.empty() or fixed.empty(), xpath: not(exists(f:*[starts-with(local-name(.), 'pattern')])) or not(exists(f:*[starts-with(local-name(.), 'fixed')])))

ElementDefinition.example

Definition | A sample value for this element demonstrating the type of information that would typically be found in the element.

Control | 0..*

Summary | true

Comments | Examples will most commonly be present for data where it's not implicitly obvious from either the data type or value set what the values might be. (I.e. Example values for dates or quantities would generally be unnecessary.) If the example value is fully populated, the publication tool can generate an instance automatically.

ElementDefinition.example.label

Definition | Describes the purpose of this example amoung the set of examples.

Control | 1..1

Type | string

Summary | true

ElementDefinition.example.value[x]

Definition | The actual value for the element, which must be one of the types allowed for this element.

Control | 1..1

Type | *

[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true

LOINC Code | EXMPL_ANSWERS

ElementDefinition.minValue[x]

Definition | The minimum allowed value for the element. The value is inclusive. This is allowed for the types date, dateTime, instant, time, decimal, integer, and Quantity.

Control | 0..1

Type | date|dateTime|instant|time|decimal|integer|positiveInt|unsignedInt|Quantity

[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true

Comments | Except for date/date/instant, the type of the minValue[x] SHALL be the same as the specified type of the element. For the date/dateTime/instant values, the type of minValue[x] SHALL be either the same, or a a Duration which specifies a relative time limit to the current time. The duration value is positive, and is subtracted from the current clock to determine the minimum allowable value. A minimum value for a Quantity is interpreted as an canonical minimum - e.g. you cannot provide 100mg if the minimum value is 10g.

ElementDefinition.maxValue[x]

Definition | The maximum allowed value for the element. The value is inclusive. This is allowed for the types date, dateTime, instant, time, decimal, integer, and Quantity.

Control | 0..1

Type | date|dateTime|instant|time|decimal|integer|positiveInt|unsignedInt|Quantity

[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true

Comments | Except for date/date/instant, the type of the maxValue[x] SHALL be the same as the specified type of the element. For the date/dateTime/instant values, the type of maxValue[x] SHALL be either the same, or a a Duration which specifies a relative time limit to the current time. The duration value is positive, and is added to the current clock to determine the maximum allowable value. A maximum value for a Quantity is interpreted as an canonical maximum - e.g. you cannot provide 10g if the maximum value is 50mg.

ElementDefinition.maxLength

Definition | Indicates the maximum length in characters that is permitted to be present in conformant instances and which is expected to be supported by conformant consumers that support the element.

Control | 0..1

Type | integer

Summary | true

Comments | Receivers are not required to reject instances that exceed the maximum length. The full length could be stored. In some cases, data might be truncated, though truncation should be undertaken with care and an understanding of the consequences of doing so. If not specified, there is no conformance expectation for length support.

ElementDefinition.condition

Definition | A reference to an invariant that may make additional statements about the cardinality or value in the instance.

Control | 0..*

Type | id
Summary | true
ElementDefinition.constraint
Definition | Formal constraints such as co-occurrence and other constraints that can be computationally evaluated within the context of the instance.
Control | 0..*
Summary | true
Comments | Constraints should be declared on the "context" element - the lowest element in the hierarchy that is common to all nodes referenced by the constraint.
ElementDefinition.constraint.key
Definition | Allows identification of which elements have their cardinalities impacted by the constraint. Will not be referenced for constraints that do not affect cardinality.
Control | 1..1
Type | id
Summary | true
Invariants | Affect this element eld-14: Constraints must be unique by key (expression : constraint.select(key).isDistinct(), xpath: count(f:constraint) = count(distinct-values(f:constraint/f:key/@value)))
ElementDefinition.constraint.requirements
Definition | Description of why this constraint is necessary or appropriate.
Control | 0..1
Type | string
Summary | true
Comments | To be used if the reason for the constraint may not be intuitive to all implementers.
ElementDefinition.constraint.severity
Definition | Identifies the impact constraint violation has on the conformance of the instance.
Control | 1..1
Terminology Binding | ConstraintSeverity (Required)
Type | code
Summary | true
Comments | This allows constraints to be asserted as "shall" (error) and "should" (warning).
ElementDefinition.constraint.human
Definition | Text that can be used to describe the constraint in messages identifying that the constraint has been violated.
Control | 1..1
Type | string
Summary | true
Comments | Should be expressed in business terms as much as possible.
ElementDefinition.constraint.expression
Definition | A FHIRPath expression of constraint that can be executed to see if this constraint is met.
Control | 1..1
Type | string
Requirements | Used by validation tooling tests of the validity of the resource.
Summary | true
ElementDefinition.constraint.xpath
Definition | An XPath expression of constraint that can be executed to see if this constraint is met.
Control | 0..1
Type | string
Requirements | Used in Schematron tests of the validity of the resource.
Summary | true
Comments | Elements SHALL use "f" as the namespace prefix for the FHIR namespace, and "x" for the xhtml namespace, and SHALL NOT use any other prefixes. Note: we are considering deprecating the xpath element. Implementer feedback is welcome.
ElementDefinition.constraint.source
Definition | A reference to the original source of the constraint, for traceability purposes.
Control | 0..1
Type | uri
Summary | true
Comments | This is used when, e.g. rendering, where it is not useful to present inherited constraints when rendering the snapshot.
ElementDefinition.mustSupport
Definition | If true, implementations that produce or consume resources SHALL provide "support" for the element in some meaningful way. If false, the element may be ignored and not supported.
Control | 0..1
Type | boolean
Default Value | false
Requirements | Allows a profile to set expectations for system capabilities beyond merely respecting cardinality constraints.
Summary | true
Comments | "Something useful" is context dependent and impossible to describe in the base FHIR specification. For this reason, tue mustSupport flag is never set to true by the FHIR specification itself - it is only set to true in profiles. This is done in Resource Profiles, where the profile labels an element as mustSupport=true. When a profile does this, it SHALL also make clear exactly what kind of "support" is required, as this can mean many things. Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.
ElementDefinition.isModifier
Definition | If true, the value of this element affects the interpretation of the element or resource that contains it, and the value of the element cannot be ignored. Typically, this is used for status, negation and qualification codes. The effect of this is that the element cannot be ignored by systems: they SHALL either recognize the element and process it, and/or a pre-determination has been made that it is not relevant to their particular system.

Control | 0..1
Type | boolean
Default Value | false
Requirements | Allows elements to be introduced into a specification that can't safely be ignored by applications that don't recognize them.
Summary | true
Comments | Only the definition of an element can set IsModifier true - either the specification itself or where an extension is originally defined. Once set, it cannot be changed in derived profiles. An element/extension that has isModifier=true SHOULD also have a minimum cardinality of 1, so that there is no lack of clarity about what to do if it is missing. If it can be missing, the definition SHALL make the meaning of a missing element clear.
ElementDefinition.isSummary
Definition | Whether the element should be included if a client requests a search with the parameter _summary=true.
Control | 0..1
Type | boolean
Default Value | false
Requirements | Allow clients to search through large resources quickly.
Summary | true
Comments | Some resources include a set of simple metadata, and some very large data. This element is used to reduce the quantity of data returned in searches. Note that servers may pre-cache summarized resources for optimal performance, so servers may not support per-profile use of the isSummary flag. When a request is made with _summary=true, serailisers only include elements marked as 'isSummary = true'. Other than Attachment.data, all data type properties are included in the summary form. Modifier elements or elements with minimum cardinality = 1 must be marked as summary elements.
ElementDefinition.binding
Definition | Binds to a value set if this element is coded (code, Coding, CodeableConcept, Quantity), or the data types (string, uri).
Control | 0..1
Summary | true
Comments | For a CodeableConcept, when no codes are allowed - only text, use a binding of strength "required" with a description explaining that no coded values are allowed and what sort of information to put in the "text" element.
Invariants | Defined on this element eld-11: Binding can only be present for coded elements, string, and uri (expression : binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'Extension') or (code = 'string') or (code = 'uri')).exists(), xpath: not(exists(f:binding)) or (count(f:type/f:code) = 0) or f:type/f:code/@value=('code','Coding','CodeableConcept','Quantity','Extension', 'string', 'uri'))
LOINC Code | ANSWER LIST, NORMATIVE ANSWER LIST
ElementDefinition.binding.strength
Definition | Indicates the degree of conformance expectations associated with this binding - that is, the degree to which the provided value set must be adhered to in the instances.
Control | 1..1
Terminology Binding | BindingStrength (Required)
Type | code
Alternate Names | conformance; extensibility
Summary | true
Comments | For further discussion, see Using Terminologies.
LOINC Code | N/A
ElementDefinition.binding.description
Definition | Describes the intended use of this particular set of codes.
Control | 0..1
Type | string
Summary | true
ElementDefinition.binding.valueSet[x]
Definition | Points to the value set or external definition (e.g. implicit value set) that identifies the set of codes to be used. If the binding refers to an explicit value set - the normal case - then use a Reference(ValueSet) preferably containing the canonical URL for the value set. If the reference is to an implicit value set - usually, an IETF RFC that defines a grammar, such as mime types - then use a uri.
Control | 0..1
Type | uri|Reference(ValueSet)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | For value sets with a referenceResource, the display can contain the value set description. The reference may be version-specific or not.
Invariants | Affect this element eld-12: ValueSet as a URI SHALL start with http:// or https:// or urn: (expression : valueSet.is(uri).not() or valueSet.as(uri).startsWith('http:') or valueSet.as(uri).startsWith('https') or valueSet.as(uri).startsWith('urn:'), xpath: not(exists(f:valueSetUri)) or (starts-with(string(f:valueSetUri/@value), 'http:') or starts-with(string(f:valueSetUri/@value), 'https:') or starts-with(string(f:valueSetUri/@value), 'urn:')))
LOINC Code | CODE_TABLE
ElementDefinition.mapping
Definition | Identifies a concept from an external specification that roughly corresponds to this element.
Control | 0..*
Requirements | Provides guidance to implementers familiar with or converting content from other specifications.
Summary | true
Comments | Mappings are not necessarily specific enough for safe translation.
ElementDefinition.mapping.identity
Definition | An internal reference to the definition of a mapping.
Control | 1..1
Type | id
Summary | true
ElementDefinition.mapping.language
Definition | Identifies the computable language in which mapping.map is expressed.

Control | 0..1
Terminology Binding | MimeType :
Type | code
Summary | true
Comments | If omitted, then there can be no expectation of computational interpretation of the mapping.
ElementDefinition.mapping.map
Definition | Expresses what part of the target specification corresponds to this element.
Control | 1..1
Type | string
Summary | true
Comments | For most mappings, the syntax is undefined. Syntax will be provided for mappings to the RIM. Multiple mappings may be possible and may include constraints on other resource elements that identify when a particular mapping applies.
ElementDefinition.mapping.comment
Definition | Comments that provide information about the mapping or its use.
Control | 0..1
Type | string
Summary | true